Command and Control Interface Volume

For

A-GRA

6.0a

31 JULY 2026

Revision History

Version	Description	Release Date
5.0a	Initial Public ASK 5.0a Release	21 APR 2026
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Table of Contents

Table of Figures

Table of Tables

1 C2 INTERFACE

C2 for "Autonomous" Collaborative Platforms may seem like an oxymoron–the platforms for Mission "Autonomy" are not RPAs–however, the A-GRA is currently intended for semi-autonomous capabilities and platforms. The semi-autonomous designation keeps humans in the loop as much as is necessary for mission execution. The A-GRA C2 interface documentation should be approached with that context in mind.

The C2 L1 interface includes all interactions between a C2 "entity," e.g., a user, and MA. Compliant MA solutions must appropriately process and respond to all required C2 L1 interactions. C2 is typically accomplished utilizing a C2 system (C2 application + C2 node) with a Human Machine Interface (HMI), however the A-GRA is method agnostic. Compliant C2 applications must only support all interactions required by their users. Compliance for C2 applications means implementing interactions according to the A-GRA definitions. C2 applications do not need to support every A-GRA defined C2 L1 interaction to meet compliance.

1.1 Interface Description

TEST C2 is accomplished via three distinct paradigms, each designed for different operational situations and intents: Direct, Planned (Indirect), and Responsive.

Direct C2

This paradigm involves the issuance of live commands directly to executing MA(s) to address immediate needs. It allows C2 entities to dynamically manage assets and supersede pre-planned actions without manually creating or updating plans.

Planned C2

This paradigm refers to C2 performed by interacting with MA via the on-board mission plan(s) and data. Thus, it can be thought of as indirect C2. C2 entities can activate, deactivate, update, or send new mission plans and/or data to affect actions and responses throughout a mission as needed to achieve desired mission outcomes.

Responsive C2

This paradigm relies upon a trigger-and-response model where automated services monitor the environment for predefined events. When a trigger condition is met, the service automatically initiates a predetermined response.

These paradigms should be kept in mind when consuming C2 ICD content. Future versions of the ASK will present a clearer organization of the content.

1.2 Interface Interactions

The C2 - MA interface assumes "off-board" or "Over-The-Air" (OTA) communications are necessary for C2 entities to reach MA and does not include the interaction with MS in diagrams. In a platform implementation, the Decentralized Messaging Service (DMS), defined in the Mission Systems section, is the current approach to accomplish back and forth MA and C2 communication OTA. Both C2 and MA must utilize compliant DMS software and aligned configurations. The DMS utilizes the Data Distribution Service (DDS) middleware for communication which is also specified by the A-GRA. The core method for messages to transfer between C2 and MA utilizes a UCI message wrapper (*MA_TxDataPayloadCommandMT*) to sends messages to the DMS and another UCI message wrapper (*MA_RxDataPayloadMT*) the DMS publishes when messages are received. These wrapper messages are specified in more detail in the Mission Systems section. The "Additional Comms Capability" MUC specifies a more holistic approach to managing the OTA communications with a Data Dissemination Plan. The Data Dissemination Plan specifies how often and whether ingress/egress messages will be subscribed/published directly on the ASB or if the UCI wrappers can be used. MA is responsible for authentication of the received messages before processing.

The DMS interaction must be implemented for both C2 and MA, either directly or via platform or enclave capabilities, for a compliant end-to-end system. See the appropriate Mission Systems content for details.

Figure 1-1: C2 Interactions

1.2.1 C2 Acknowledgment Configurations Package

1.2.1.1 C2 Message Acknowledgment Configuration Status

This interaction provides the current Message Acknowledgment Configuration defining what messages C2 will respond to with an acknowledgment upon receipt. C2 will report the current Message Acknowledgment Configuration on change by sending a *MA_MessageAcknowledgmentConfigurationMT* message.

The *DefaultGroupMessageFreq* field defines the frequency at which the Grouped Acknowledgment message is set. If the field is not set, there is no grouping. The *DefaultMessagesToAck* field sets the list of messages that an Acknowledgment message will be sent for when received. If the field is not set, all messages listed in *MA_AckMessageEnum* will receive a response. The *ReceiverSpecificSettings* field specifies how MA will respond to individual C2s and/or Peers if they do not fall under the default settings. If MA does not support Message Acknowledgment Configuration, the *ConfigurationSupported* field will be set to "FALSE" and it will respond with an ungrouped acknowledgment for all possible messages listed in *MA_AckMessageEnum*.

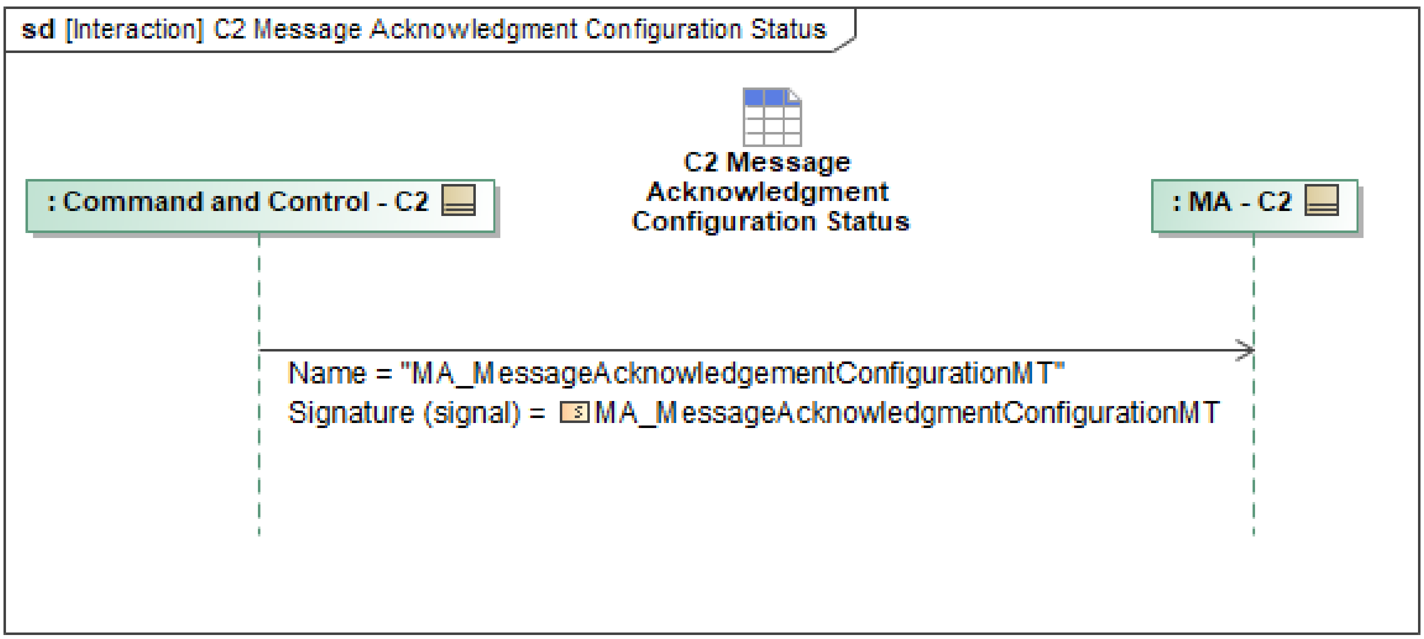


Figure 1-2: C2 Message Acknowledgment Configuration Status

Table 1-1: C2 Message Acknowledgment Configuration Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessageAcknowledgementConfigurationMT: MA_MessageAcknowledgmentConfigurationMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.2 C2 Updates Message Acknowledgment Configuration

This interaction specifies when C2 wants to set or update the C2 Message Acknowledgment Configuration. C2 can update default responses by filling out the *DefaultGroupMessageFreq* and

DefaultMessagesToAck. C2 can also use the *RequestedReceiverSpecificSetting* field to define the messages that C2 expects MA to respond with an acknowledgment for and if the messages will be grouped. C2 should only request *RequestedReceiverSpecificSetting* for itself and not other systems.

C2 sends the *MA_MessageAcknowledgmentConfigurationRequestMT* message to MA. MA may accept or reject the configuration request. MA accepts the request by sending a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_1* with the *RequestProcessingState* set to "COMPLETED", followed by a *MA_MessageAcknowledgmentConfigurationMT* that reflects the updated configuration.

MA may reject the configuration request for the following reasons:

- It does not support configuration. MA sends a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_2* message and sets the *RequestProcessingStateReason* to "SYSTEM_UNAVAILABLE."
- There is limited bandwidth. MA sends a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_2* message, sets the *RequestProcessingStateReason* to "INSUFFICIENT_RESOURCES", and fills out the description field.
- The system receives an unsupported input. MA sends a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_2* message and sets the *RequestProcessingStateReason* to "INVALID_INPUT_PARAMETER."
- MA requests *RequestedReceiverSpecificSetting* for another system. MA sends a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_2* message and sets the *RequestProcessingStateReason* to "INELIGIBLE_CONTROL_SOURCE."

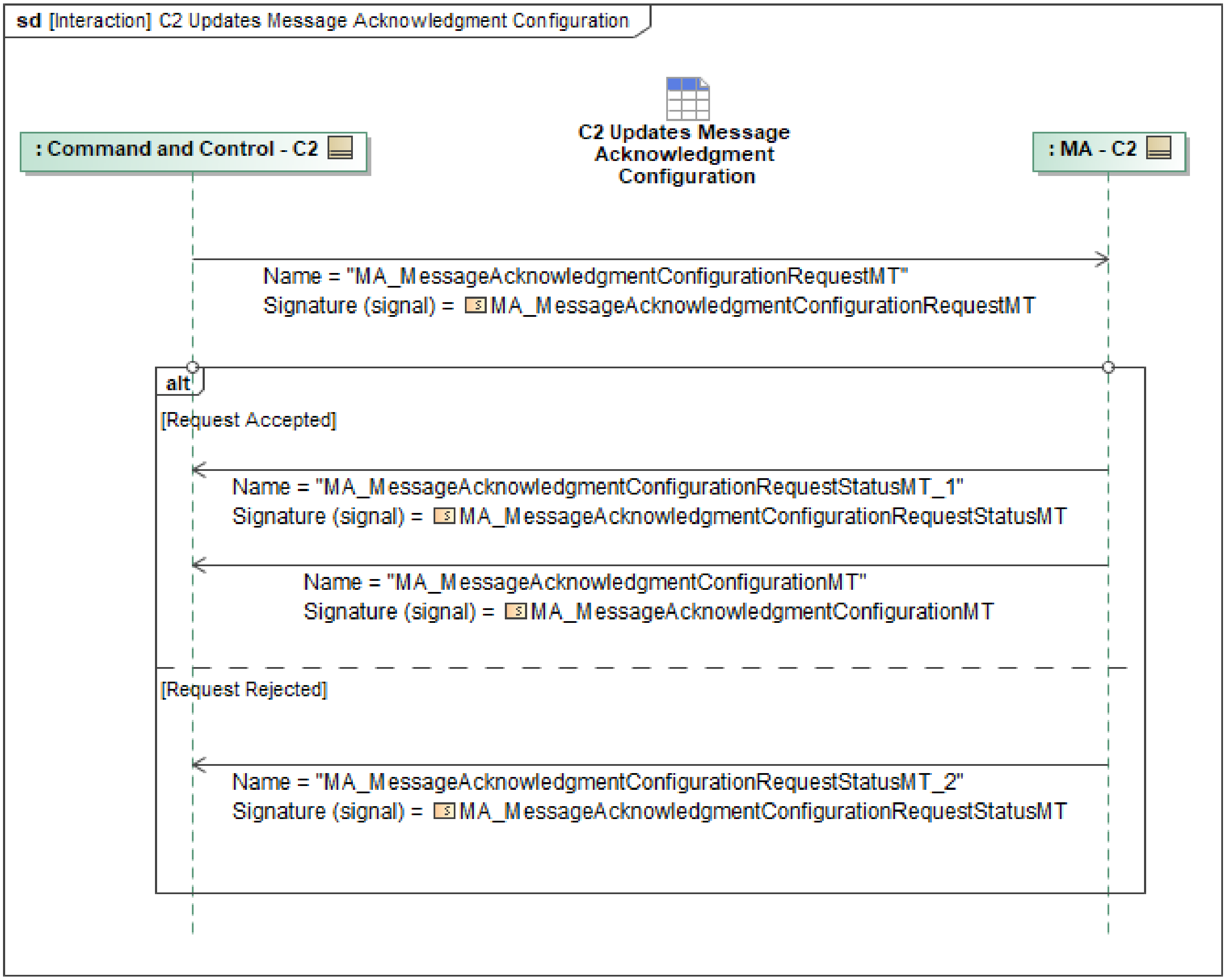


Figure 1-3: C2 Updates Message Acknowledgment Configuration

Table 1-2: C2 Updates Message Acknowledgment Configuration

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessageAcknowledgmentConfigurationMT : MA_MessageAcknowledgmentConfigurationMT	None
MA_MessageAcknowledgmentConfigurationRequestMT: MA_MessageAcknowledgmentConfigurationRequestMT	None
MA_MessageAcknowledgmentConfigurationRequestStatusMT_1: MA_MessageAcknowledgmentConfigurationRequestStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessageAcknowledgmentConfigurationRequestStatusMT_2: MA_MessageAcknowledgmentConfigurationRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.3 Message Acknowledgment Configuration Status

This interaction provides the current Message Acknowledgment Configuration defining what messages MA will respond to with an acknowledgment upon receipt. MA will report the current Message Acknowledgment Configuration on change by sending a *MA_MessageAcknowledgmentConfigurationMT* message.

The *DefaultGroupMessageFreq* field defines the frequency at which the Grouped Acknowledgment message is set. If the field is not set, there is no grouping. The *DefaultMessagesToAck* field sets the list of messages that an Acknowledgment message will be sent for when received. If the field is not set, all messages listed in *MA_AckMessageEnum* will receive a response. The *ReceiverSpecificSettings* field specifies how MA will respond to individual C2s and/or Peers if they do not fall under the default settings. If MA does not support Message Acknowledgment Configuration, the *ConfigurationSupported* field will be set to "FALSE" and it will respond with an ungrouped acknowledgment for all possible messages listed in *MA_AckMessageEnum*.

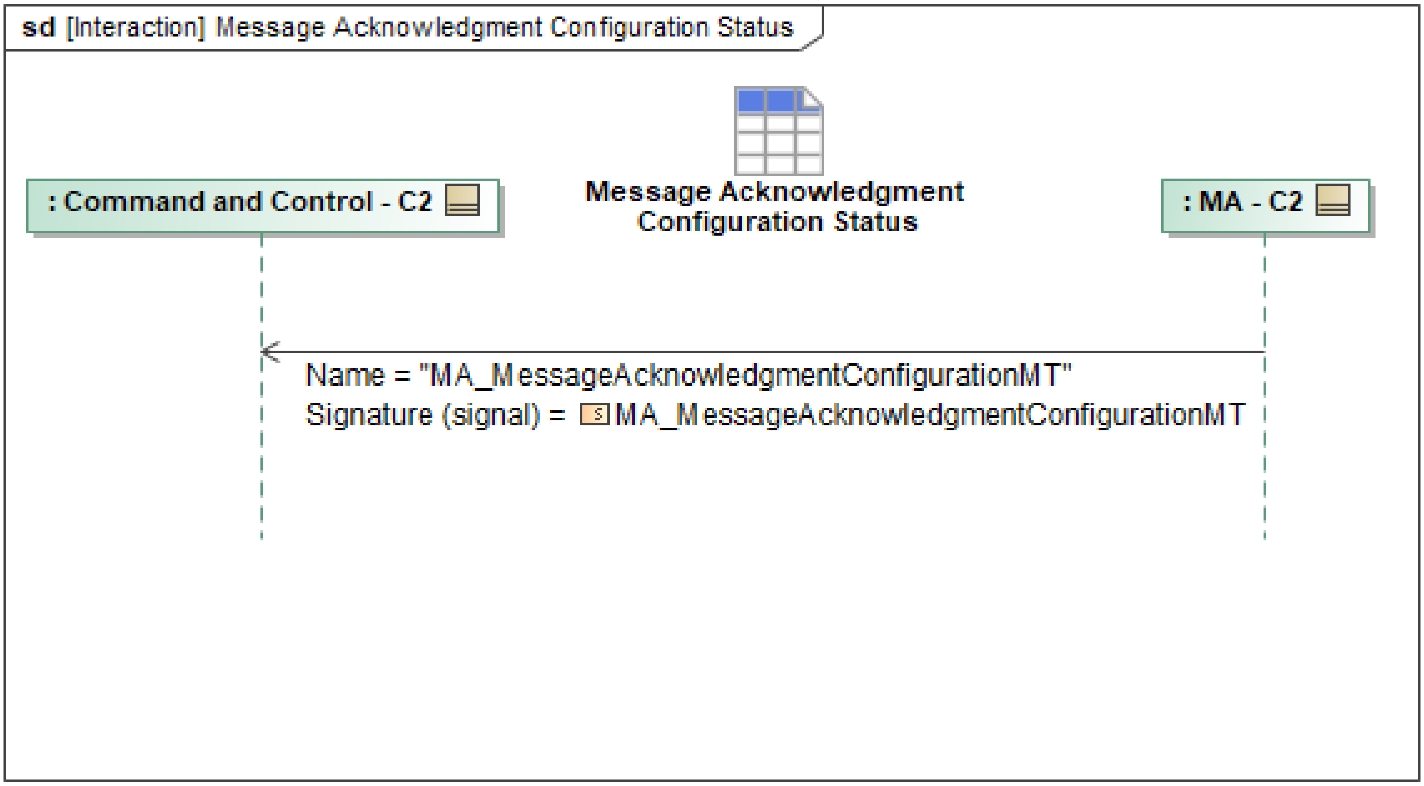


Figure 1-4: Message Acknowledgment Configuration Status

Table 1-3: Message Acknowledgment Configuration Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessageAcknowledgmentConfigurationMT : MA_MessageAcknowledgmentConfigurationMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.1.4 Update Message Acknowledgment Configuration

This interaction specifies when MA wants to set or update the C2 Message Acknowledgment Configuration. MA can update default responses by filling out the *DefaultGroupMessageFreq* and *DefaultMessagesToAck.* MA can also use the *RequestedReceiverSpecificSetting* field to define the messages that MA expects C2 to respond with an acknowledgment for and if the messages will be grouped. MA should only request *RequestedReceiverSpecificSetting* for itself and not other systems.

MA sends the *MA_MessageAcknowledgmentConfigurationRequestMT* message to C2. C2 may accept or reject the configuration request. C2 accepts the request by sending a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_1* with the *RequestProcessingState* set to "COMPLETED", followed by a *MA_MessageAcknowledgmentConfigurationMT* that reflects the updated configuration.

C2 may reject the configuration request for the following reasons:

- It does not support configuration. C2 sends a

MA_MessageAcknowledgmentConfigurationRequestStatusMT_2 message and sets the *RequestProcessingStateReason* to "SYSTEM_UNAVAILABLE."

- There is limited bandwidth. C2 sends a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_2* message, sets the *RequestProcessingStateReason* to "INSUFFICIENT_RESOURCES", and fills out the description field.
- The system receives an unsupported input. C2 sends a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_2* message and sets the *RequestProcessingStateReason* to "INVALID_INPUT_PARAMETER."
- MA requests *RequestedReceiverSpecificSetting* for another system. C2 sends a *MA_MessageAcknowledgmentConfigurationRequestStatusMT_2* message and sets the *RequestProcessingStateReason* to "INELIGIBLE_CONTROL_SOURCE."

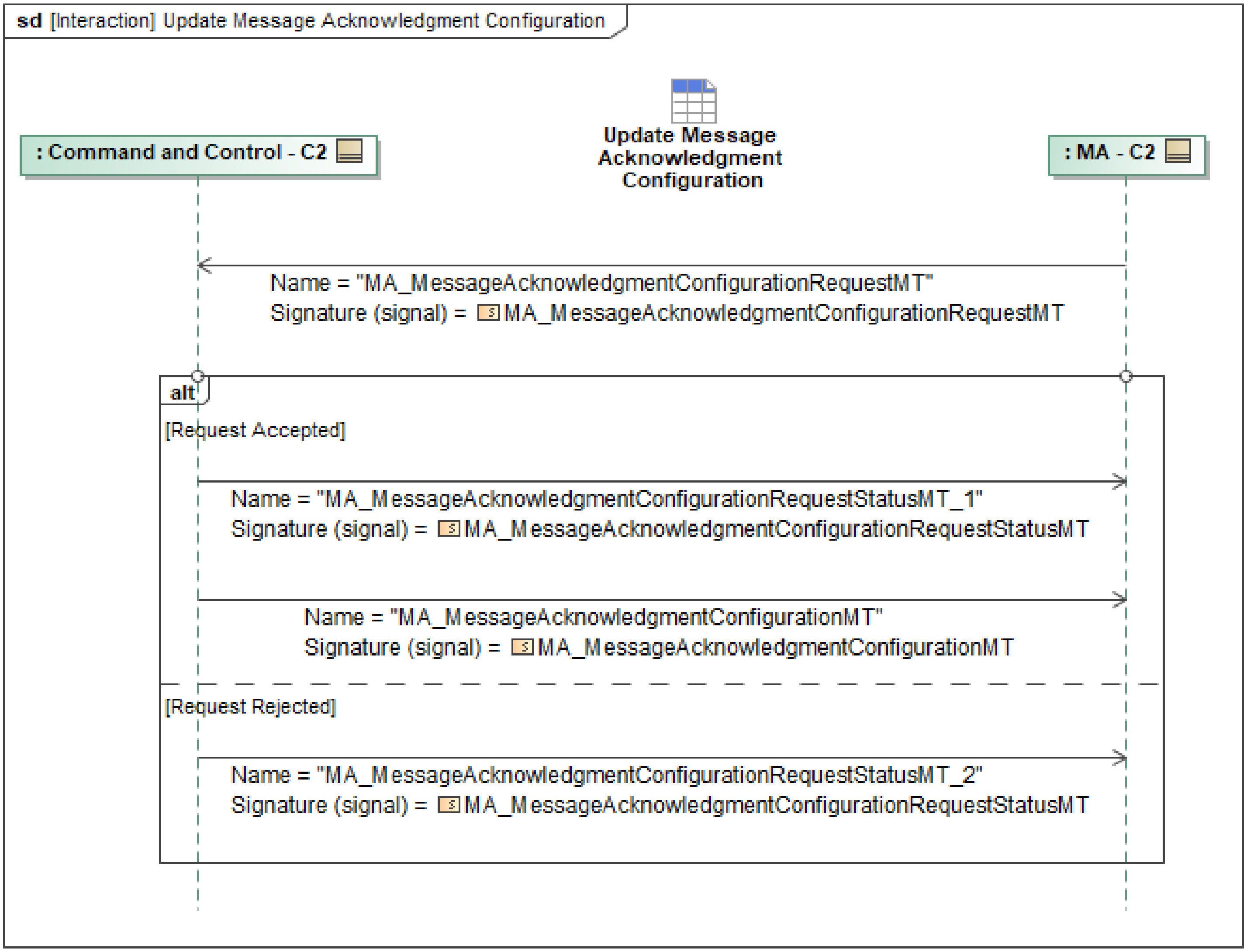


Figure 1-5: Update Message Acknowledgment Configuration

Table 1-4: Update Message Acknowledgment Configuration

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessageAcknowledgmentConfigurationMT : MA_MessageAcknowledgmentConfigurationMT	None
MA_MessageAcknowledgmentConfigurationRequestMT: MA_MessageAcknowledgmentConfigurationRequestMT	None
MA_MessageAcknowledgmentConfigurationRequestStatusMT_1: MA_MessageAcknowledgmentConfigurationRequestStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MessageAcknowledgmentConfigurationRequestStatusMT_2: MA_MessageAcknowledgmentConfigurationRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.2 C2 CAP Behaviors Package

1.2.2.1 Share CAP Execution Status

This interaction specifies the CAP Execution Status message which is sent from MA to C2. This message is expected to be sent periodically to C2 when one or multiple ACPs are executing a CAP (commanded by either an Action or a Task). When multiple ACPs are executing a CAP, the ACPs are assumed to be within a Package. A single ACP performing CAP may be within a Package but is not required to be within a Package for a single-ship mission.

If the ACP is the Package Leader, then the Leader MA sends *MA_CAP_ExecutionStatusReportMT periodically* to C2 after receiving an individual execution status report from all Followers (See "Receive CAP Execution Status"). *MA_CAP_ExecutionStatusReportMT* defines fields to capture both package-level and individual ACP data. At the package-level, the *PackageID* shall be specified, and the *PackageRequirementID* field shall include the unique ID of the package-level CAP Task (if the Package is commanded via a Task). If the CAP was commanded as a result of an Action, the *PackageRequirementID* field shall include the unique ID of the package-level CAP Action. Additionally, the Leader MA should include the *AOI_Coverage* and *PlanMetrics* fields when reporting CAP Execution Status to C2. The leader will compute these fields prior to this interaction. The *PlanMetrics* are computed by the leader when computing the CAP routes, and the *AOI_Coverage* is computed by the leader upon receiving threat zone information and Field of Regard reports from package members. The *ExecutionStatus* field shall contain the status of each individual ACP executing CAP within the package. The *SystemIDs* shall be set to the IDs of the ACPs, *RequirementIDs* shall be set to the IDs of the vehicle-level CAP Tasks, and the *CAP_Status* object shall be defined. Included in the *CAP_Status* object is the required *CAP_State* field and optional status fields.

If the ACP is executing a CAP for a single-ship mission (ACP is not part of a Package), MA will send *MA_CAP_ExecutionStatusReportMT periodically* to C2 at a constant rate in addition to anytime its *CAP_State* is updated. At the package-level, no fields are required. At the vehicle level, the *ExecutionStatus* shall be defined for just the single ACP where the *SystemID*, *RequirementID,* and *CAP_Status* object are required.

MA will send a *TaskStatusMT* message to C2 as part of the COP and therefore is not specified as part of this interaction. If the ACP is the package leader, it will send a *TaskStatusMT* message for reach vehicle performing a CAP. Within the *TaskStatusMT* message, the *TimeWindow* field may be defined to specify the remaining time-on-station for performing the CAP Task.

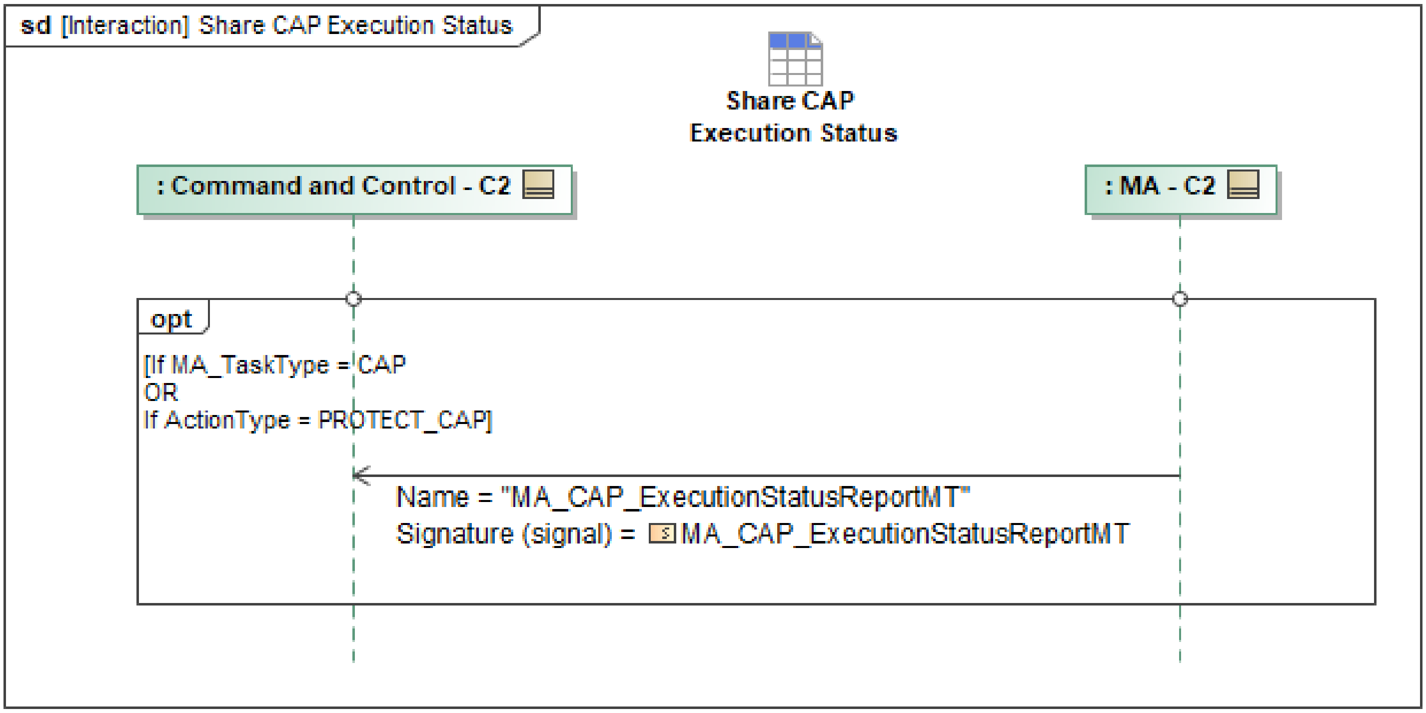


Figure 1-6: Share CAP Execution Status

Table 1-5: Share CAP Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_CAP_ExecutionStatusReportMT: MA_CAP_ExecutionStatusReportMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3 C2 Communications Package

1.2.3.1 C2 Overrides Data Dissemination Plan

Mission Use Case: Additional Comms Capabilities

In some situations a modification to the Data Dissemination Plan is required that are short term in nature and a complete update to the *MA_DataDisseminationPlanMT* message is not required. The *MA_DataDisseminationPlanOverrideRequestMT* can be used for these sorts of modifications. C2 can send the *MA_DataDisseminationPlanOverrideRequestMT* message to MA which forwards it to MS. MS can then accept or reject the request(s). If the request(s) are accepted, MS sends the *MA_DataDisseminationPlanOverrideRequestStatusMT_1* message to MA who forwards it to C2 with the Request(s) having the *Request.RequestProcessingState* set to "QUEUED", "PROCESSING", "CANCELED", or "COMPLETED" as needed. Depending on the original request, the *EffectivityActivationStatus* or *OperatorModificationStatus* fields will be filled out as needed. The interaction "MA Updates Data Dissemination Plan Activation Status" is initiated which will provide an update to the active status and include the *OverrideModificationState* associated with the override request status. For each applicable Gateway Activity, MS sends the *MA_GatewayActivityMT* message(s) to MA who forwards it to C2 which will include any modifications that were made. If the request is NOT accepted, MS sends the *MA_DataDisseminationPlanOverrideRequestStatusMT_2* message to MA who forwards it to C2 with the *RequestProcessingState* set to "REJECTED" or "FAILED", and the *RequestProcessingStateReason* filled out as needed. The *EffectivityActivationStatus* or *OperatorModificationStatus* fields will be filled out outlining the reason for the rejections/failures.

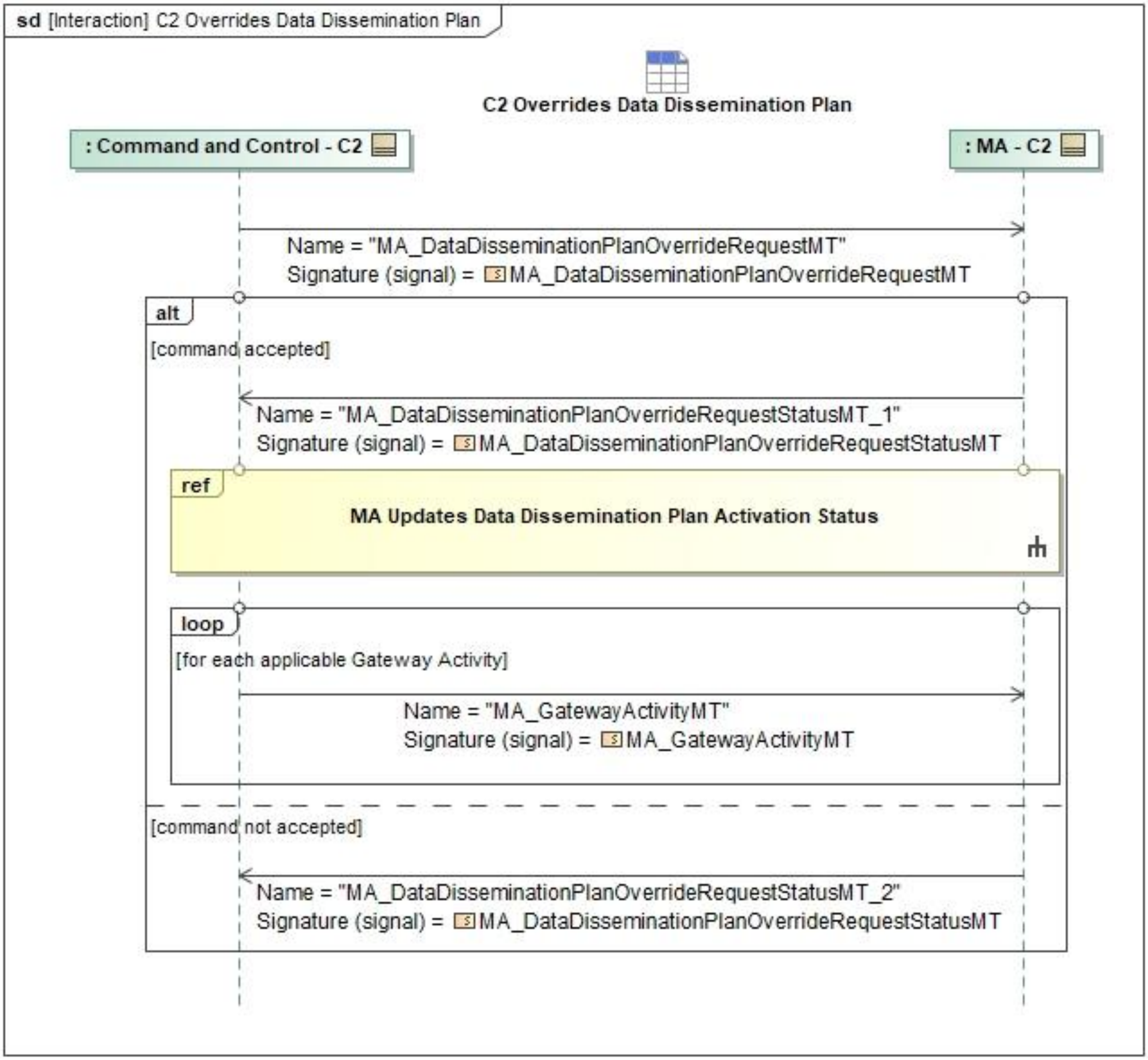


Figure 1-7: C2 Overrides Data Dissemination Plan

Table 1-6: C2 Overrides Data Dissemination Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanOverrideRequestMT: MA_DataDisseminationPlanOverrideRequestMT	AllocatedToSystemIDs: MA_PackageSystemType
MA_DataDisseminationPlanOverrideRequestStatusMT_1: MA_DataDisseminationPlanOverrideRequestStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanOverrideRequestStatusMT_2: MA_DataDisseminationPlanOverrideRequestStatusMT	RequestProcessingStateReason: CannotComplyType
MA_GatewayActivityMT: MA_GatewayActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3.2 C2 Updates Data Dissemination Plan

Mission Use Case: Additional Comms Capabilities

The *MA_DataDisseminationPlanMT* is used to configure how messages ingress and egress over the air. MS manages/implements the Data Dissemination Plan and Mission Autonomy acts as a pass through if C2 needs to update and/or activate the plan. The MS section "Update Data Dissemination Plan" details the contents of the *MA_DataDisseminationPlanMT* message and how it can be configured.

If the Data Dissemination Plan needs update or is a new plan, C2 sends the *MA_DataDisseminationPlanMT* message and the *MA_DataDisseminationPlanActivationCommandMT* message to MA who forwards them to MS. If the command is accepted by MS, MS sends the *MA_DataDisseminationPlanActivationCommandStatusMT_1* message to MA who forwards it to C2 with the *CommandProcessingStateEnum* set to "ACCEPTED", the interaction "MA Updates Data Dissemination Plan Activation Status" is initiated, and for each applicable Gateway Activity MS sends the *MA_GatewayActivityMT* message(s) to MA who forwards them to C2. If the command is NOT accepted, MS send the *MA_DataDisseminationPlanActivationCommandStatusMT_2* message to MA who forwards it to C2 with the *CommandProcessingStateEnum* set to "REJECTED" and the *PlanActivationCommandStatus* is set according to the reason it was rejected.

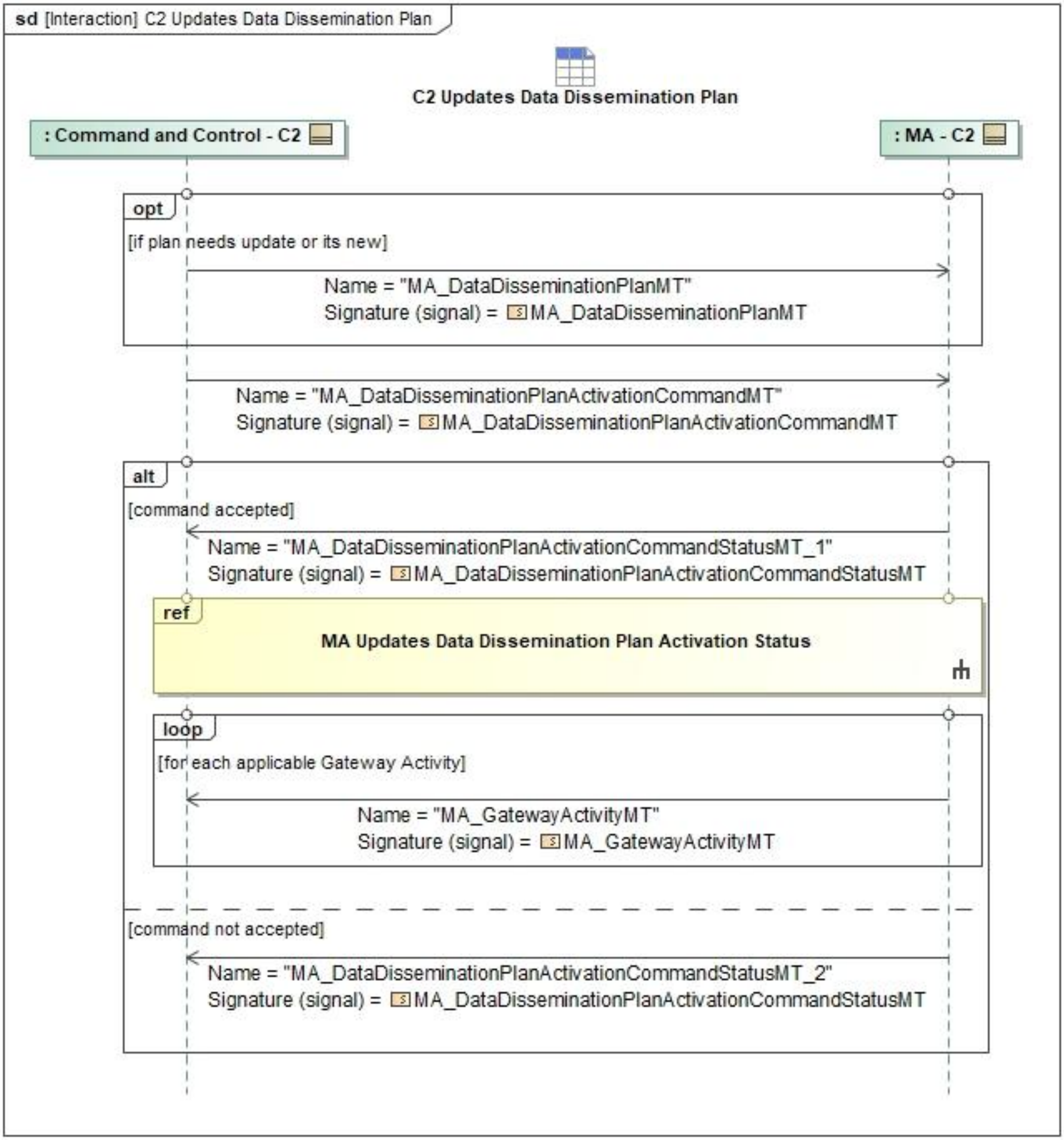


Figure 1-8: C2 Updates Data Dissemination Plan

Table 1-7: C2 Updates Data Dissemination Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanActivationCommandMT: MA_DataDisseminationPlanActivationCommandMT	Applicability: MA_PackageSystemType
MA_DataDisseminationPlanActivationCommandStatusMT_1: MA_DataDisseminationPlanActivationCommandStatusMT	None
MA_DataDisseminationPlanActivationCommandStatusMT_2: MA_DataDisseminationPlanActivationCommandStatusMT	RequestProcessingStateReason: CannotComplyType
MA_DataDisseminationPlanMT: MA_DataDisseminationPlanMT	None
MA_GatewayActivityMT: MA_GatewayActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3.3 MA Updates Comms Gateway Status

Mission Use Case: Additional Comms Capabilities

The Gateway Capability in MS describes how messages egress or ingress over the air. To provide information about each of the active gateways available for use, MA receives and forwards the *GatewayCapabilityMT* and *GatewayCapabilityStatusMT* messages from MS to C2. Then, for each active Gateway Activity, MA also forwards the *MA_GatewayActivityMT* message(s) to C2. These messages provide the information needed to understand the state of the over the air messaging, they will provide context for how things are currently configured, and provide information for how to configure over the air messaging. MS will typically configure the gateways internally, but the information can be useful when creating the plans that MS uses.

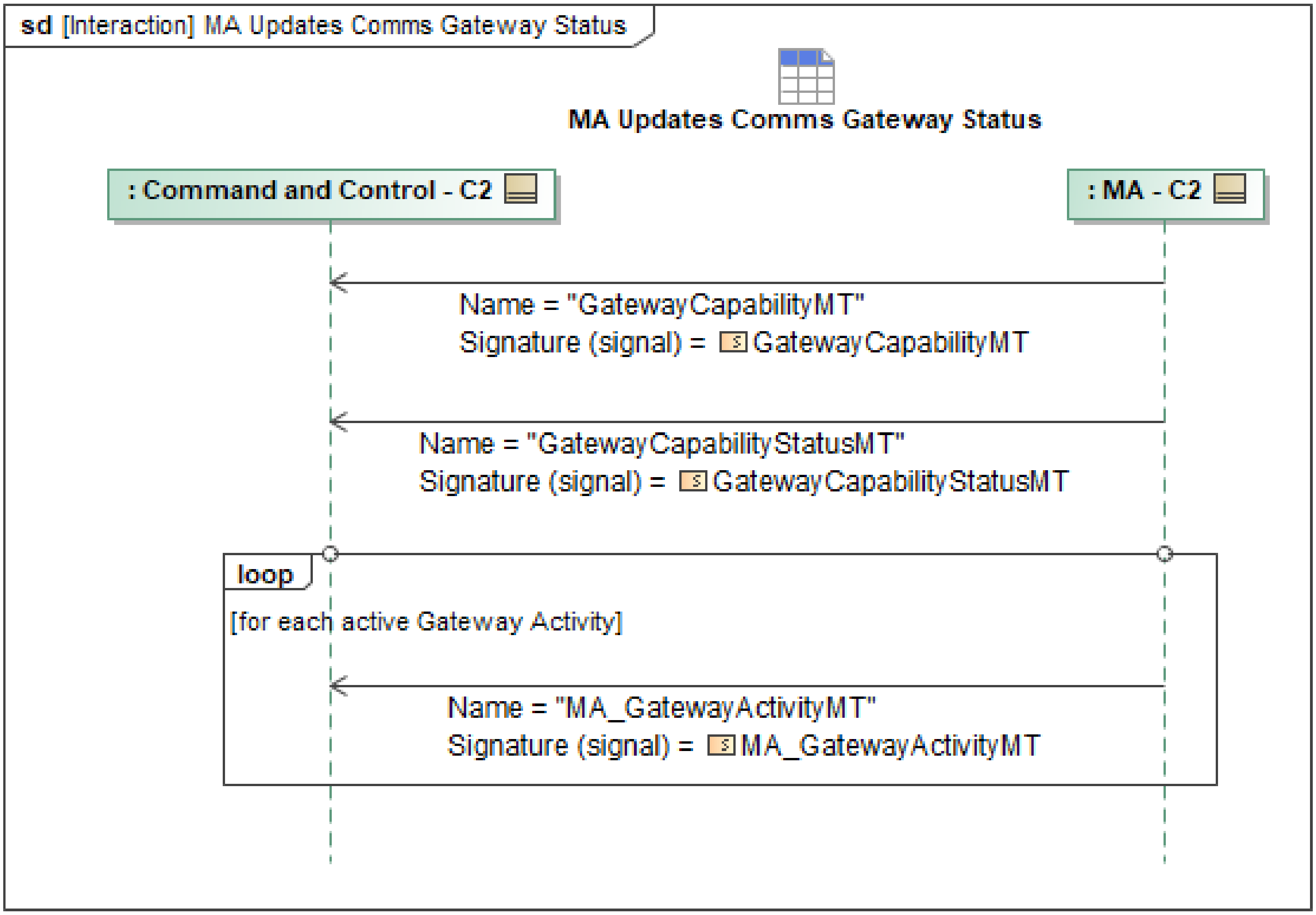


Figure 1-9: MA Updates Comms Gateway Status

Table 1-8: MA Updates Comms Gateway Status

Message Name (Name:Type)	Required Fields (Name:Type)
GatewayCapabilityMT: GatewayCapabilityMT	None
GatewayCapabilityStatusMT: GatewayCapabilityStatusMT	None
MA_GatewayActivityMT: MA_GatewayActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.3.4 MA Updates Data Dissemination Plan Activation Status

Mission Use Case: Additional Comms Capabilities

The MS will send the *MA_DataDisseminationPlanActivationStatusMT* message to MA who forwards it to C2 that will reflect the active state of Data Dissemination Plan. This can be sent periodically, but will always be sent upon change to the active state. The message will reflect the active Effectivities and Override Modifications of the Data Dissemination Plan.

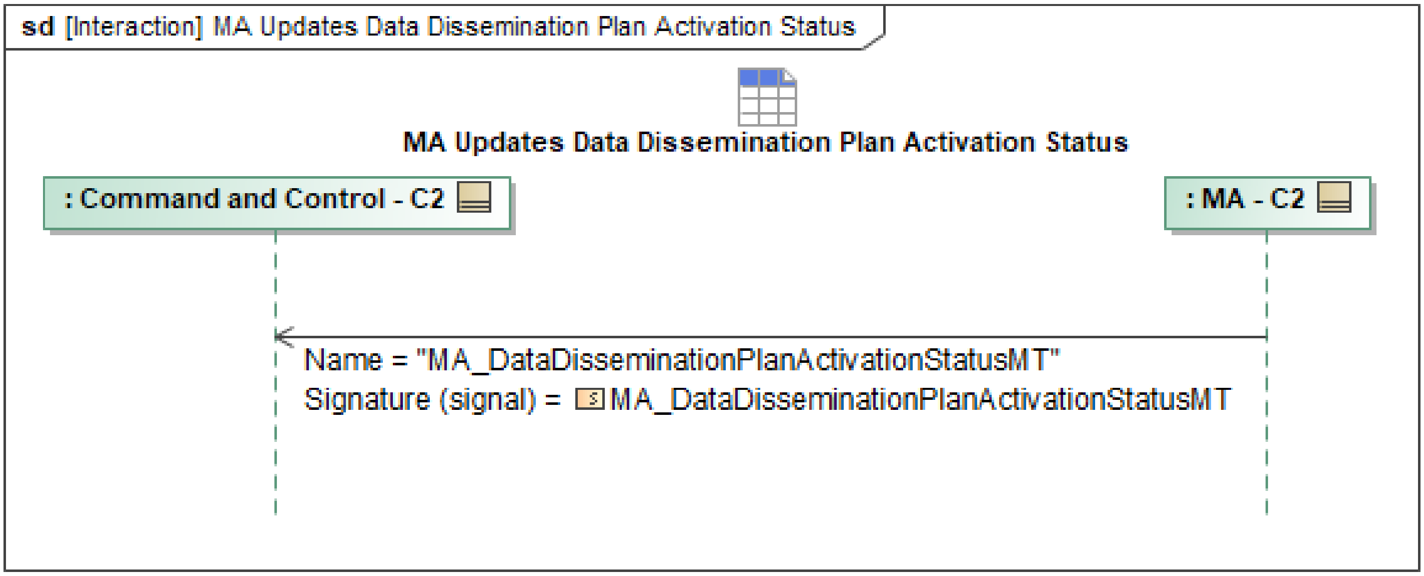


Figure 1-10: MA Updates Data Dissemination Plan Activation Status

Table 1-9: MA Updates Data Dissemination Plan Activation Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DataDisseminationPlanActivationStatusMT: MA_DataDisseminationPlanActivationStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4 C2 Contingency Behaviors Package

C2 systems have the ability to send contingency plans to MA in the form of a Response. Responses are sets of “Trigger - Response” pairs generated by C2 and sent to MA. The Trigger can be any sort of message generated internally by MA or another onboard system, or it can be sent from C2 to command the desired Response. The Response can generate new requirements/tasks, activate a previously developed Mission Plan or part of a Mission Plan, generate a contingency alert, or do nothing at all.

In the event of a contingency, MA can be configured to react in one of three different ways: alert generation only (all will generate an alert), activate a specific mission plan, or activate autonomous replanning. A C2 system will use the *PlanningFunction** family of UCI messages along with Response Requirements to achieve these three contingency reactions. The sections below break down these use cases further. The first defines sequences for setting up the three autonomous reactions, the second defines the sequences for the message flow when a trigger is tripped, and the third defines how to command these responses directly.

In order to set up a Trigger-Response pair, a C2 system will send a *PlanningFunctionSettingsCommandMT* message to define a trigger for MA to monitor. This message also specifies what sort of action MA should take once its trigger is tripped. If a C2 System wishes to specify an existing plan to execute as a reaction to a contingency, it will also need to construct a

Response Requirement containing the desired plan. The specifics of these messages and fields are discussed in the following sections.

MA alerts the C2 systems when a contingency is triggered and notifies it of the status of the contingency response. The corresponding sequence to create a contingency trigger and optionally define a response action must have been completed by either C2 or during Mission Planning before these triggered contingency sequences can occur.

For lost communication contingencies, MA assumes that the loss is reported by the Mission Systems Communications Manager or is determined by the absence of a heartbeat message past a configurable timeout interval. During planned communication outages, MA depends on MS for lost comms notification.

1.2.4.1 C2 Activate a Contingency

C2 sends a *MA_ResponseCommandMT* to activate or deactivate a previously uploaded *MA_ResponseMT*. The *MA_ResponseCommandMT* will set the *ConstraintOverride.Allocation.RequiredAllocation.Allocation* field to indicate the target *SystemIDs* or *PackageID* to activate the *MA_ResponseMT*. Optionally, the *AssociatedPlan* field can be used to link back to a plan which contains the *MA_ResponseMT*. MA responds with a *MA_ResponseCommandStatusMT* to indicate if the command was successful or not.

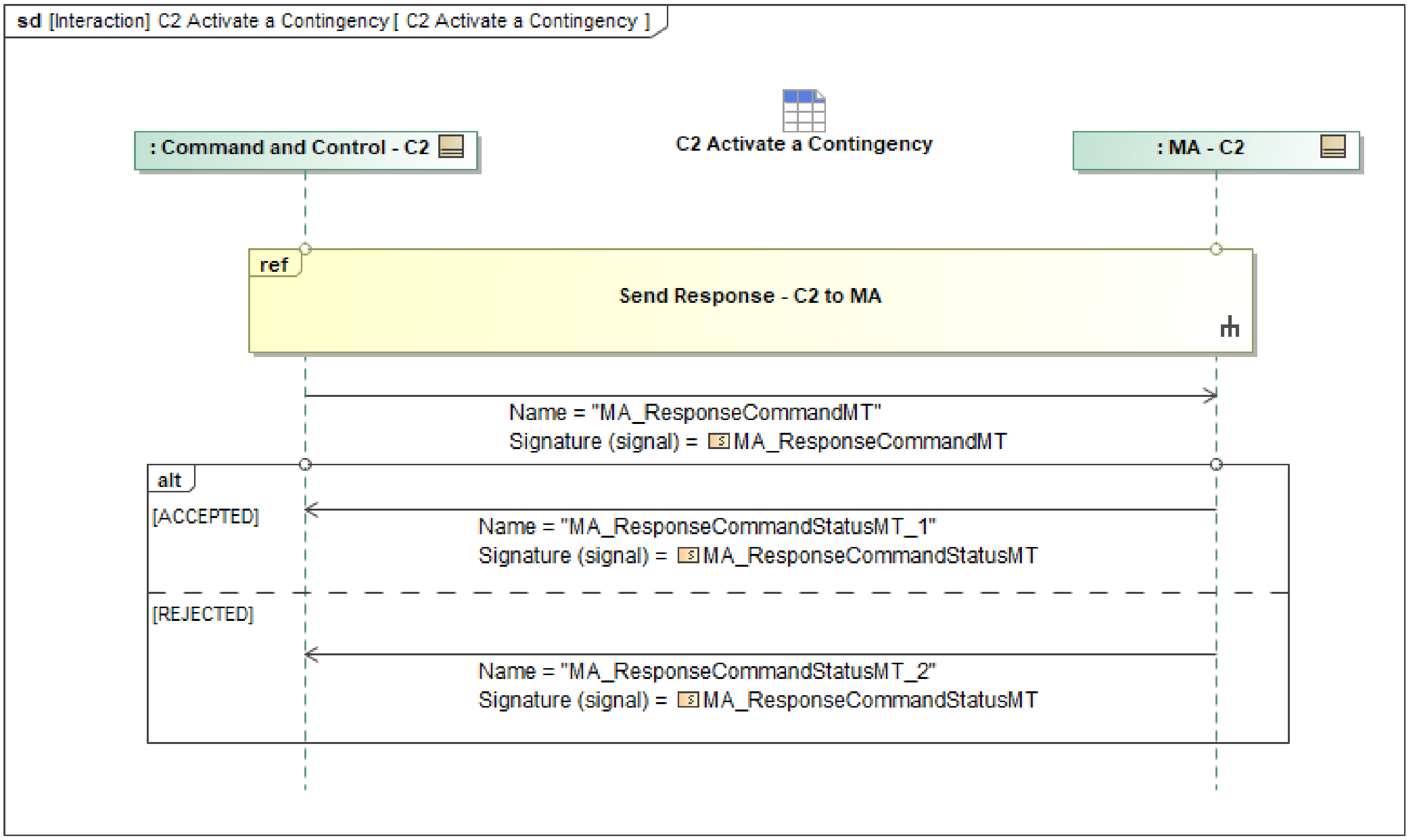


Figure 1-11: C2 Activate a Contingency

Table 1-10: C2 Activate a Contingency

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ResponseCommandMT: MA_ResponseCommandMT	None
MA_ResponseCommandStatusMT_1: MA_ResponseCommandStatusMT	None
MA_ResponseCommandStatusMT_2: MA_ResponseCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.2 C2 Incorrect Messaging Sequence Error Handling

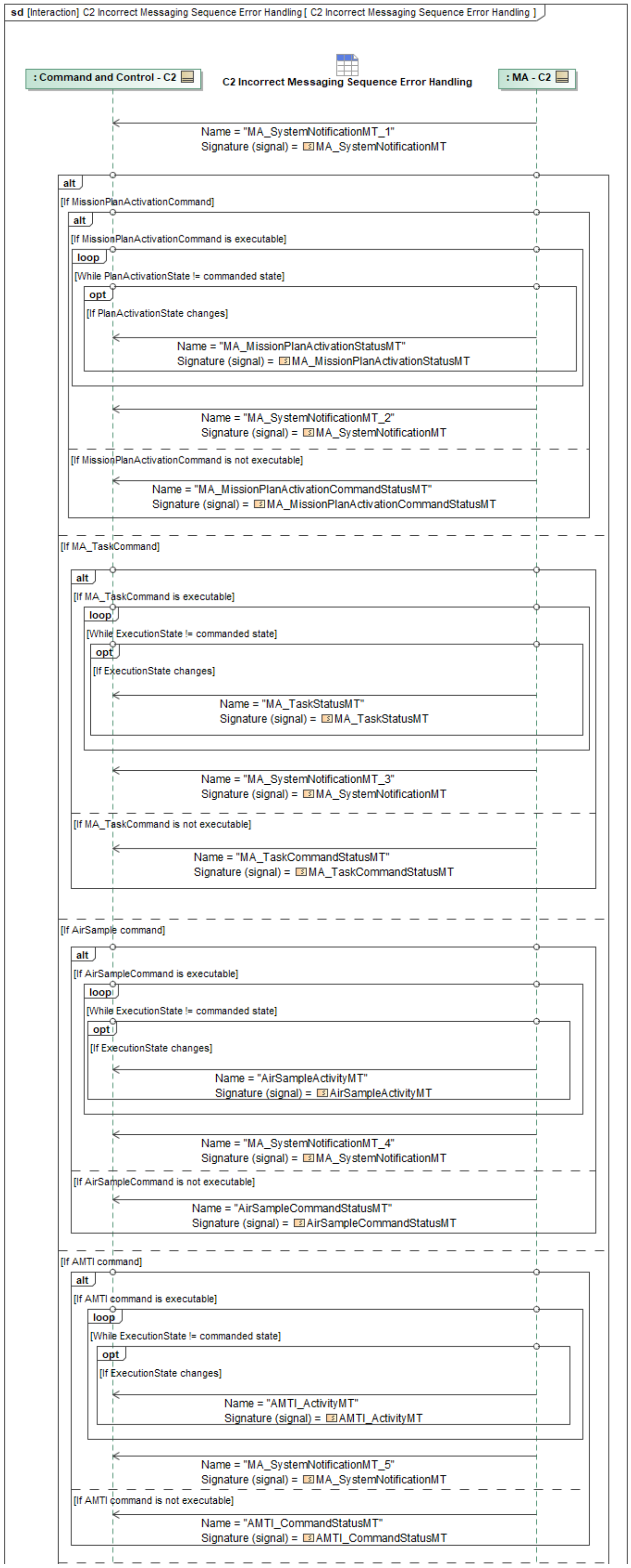
When MA receives a command, it is possible that the commanded state of an object (e.g., *MissionPlan, Task, EA_Capability*) is more than one state away from the object's current state. If this is the case, an Incorrect Messaging Sequence error is triggered. Upon identification of an Incorrect Messaging Sequence error, MA will send *MA_SystemNotificationMT* to C2 to indicate that the commanded state is out of sequence and that corrective state transitions will be performed. Within the message, MA sets *NotificationState* to *ACTIVE* and sets *NotificationDetail.Reason* to *STATE_OR_SETTINGS*. *NotificationState* communicates that the error exists and *NotificationDetail.Reason* indicates that the error exists due to the object's current state.

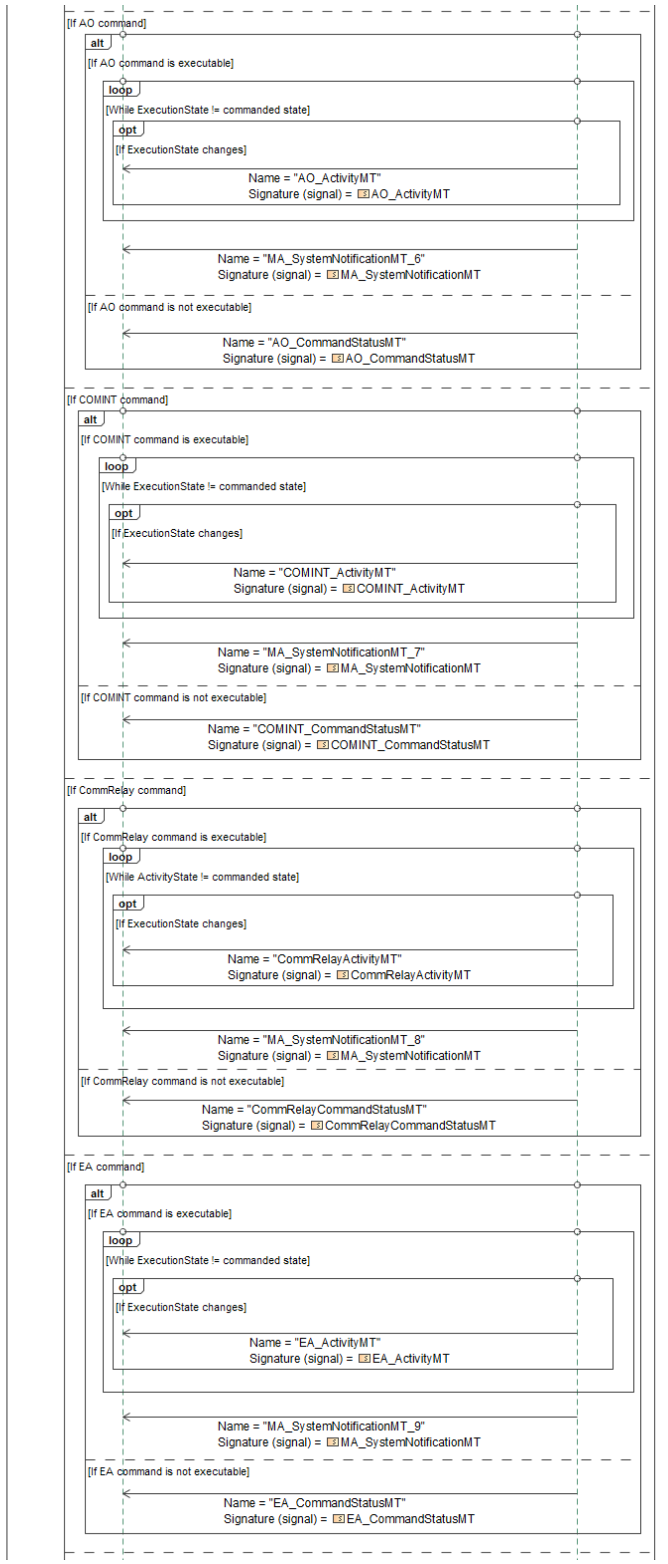
AssociatedMessage.MessageType identifies the command associated with the error, and *AssociatedMessage.AssociatedID* will reference the ID of the associated command.

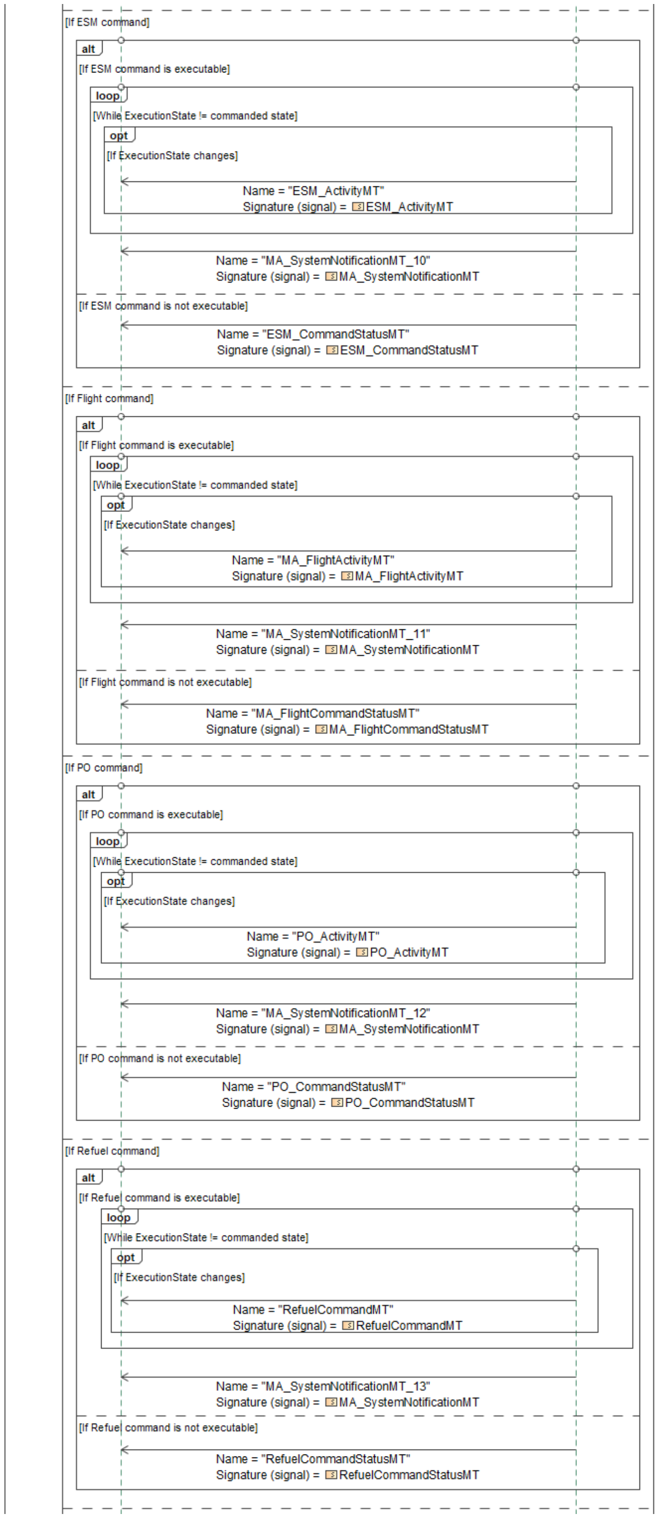
If MA determines that the command cannot be carried out for any reason, MA will publish the corresponding **CommandStatus* message with *CommandProcessingState* set to *REJECTED.* If MA determines that the command can be carried out, MA will publish the corresponding **CommandStatus* message with *CommandProcessingState* set to *ACCEPTED*, and will progress the object's status towards the commanded state.

MA communicates the object's status when the object's status changes during corrective actions. Status updates are not intended to be periodic; they are sent when MA transitions the object to a different state while progressing toward the commanded state. Object status reflects the object's actual state at the time the status is reported and provides visibility into the intermediate transitions performed by MA during resolution of the Incorrect Messaging Sequence error. The applicable status message depends on the affected object. For mission plan activation commands, MA uses *MA_MissionPlanActivationStatus* to communicate the current *PlanActivationState* (e.g., *READY_FOR_ACTIVATION, ACTIVATING, ACTIVATED, ACTIVE_DEGRADED, DEACTIVATING, and DEACTIVATED*). For a TaskCommand, MA uses *MA_TaskStatusMT* to communicate the current *ExecutionState* (e.g. *AWAITING_PLAN_ACTIVATION, EXECUTING, COMPLETED, DROPPED, FAILED, CANCELED*). For UCI capability commands, MA uses the capability's corresponding **Activity message* (e.g., *EA_Activity, AMTI_Activity, AO_Activity*) to communicate the current *ActivityState* (e.g., *ENABLED, ACTIVE_PARTIALLY_CONSTRAINED, ACTIVE_UNCONSTRAINED, COMPLETED, FAILED*).

The commanded state represents the state requested by the originating command and serves as the target state against which MA evaluates progress during executing corrective actions. When the current object state matches the commanded state, MA considers the Incorrect Messaging Sequence Error to be resolved and will send updated *MA_SystemNotificationMT* with *NotificationState* set to *CLOSED*.


sd [Interaction] C2 Incorrect Messaging Sequence Error Handling [C2 Incorrect Messaging Sequence Error Handling]
: Command and Control - C2
C2 Incorrect Messaging Sequence Error Handling
: MA - C2
Name = "MA_SystemNotificationMT_1"
Signature (signal) = MA_SystemNotificationMT
alt
[If MissionPlanActivationCommand]
alt
[If MissionPlanActivationCommand is executable]
loop
[While PlanActivationState != commanded state]
opt
[If PlanActivationState changes]
Name = "MA_MissionPlanActivationStatusMT"
Signature (signal) = MA_MissionPlanActivationStatusMT
Name = "MA_SystemNotificationMT_2"
Signature (signal) = MA_SystemNotificationMT
[If MissionPlanActivationCommand is not executable]
Name = "MA_MissionPlanActivationCommandStatusMT"
Signature (signal) = MA_MissionPlanActivationCommandStatusMT
[If MA_TaskCommand]
alt
[If MA_TaskCommand is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "MA_TaskStatusMT"
Signature (signal) = MA_TaskStatusMT
Name = "MA_SystemNotificationMT_3"
Signature (signal) = MA_SystemNotificationMT
[If MA_TaskCommand is not executable]
Name = "MA_TaskCommandStatusMT"
Signature (signal) = MA_TaskCommandStatusMT
[If AirSample command]
alt
[If AirSampleCommand is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "AirSampleActivityMT"
Signature (signal) = AirSampleActivityMT
Name = "MA_SystemNotificationMT_4"
Signature (signal) = MA_SystemNotificationMT
[If AirSampleCommand is not executable]
Name = "AirSampleCommandStatusMT"
Signature (signal) = AirSampleCommandStatusMT
[If AMTI command]
alt
[If AMTI command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "AMTI_ActivityMT"
Signature (signal) = AMTI_ActivityMT
Name = "MA_SystemNotificationMT_5"
Signature (signal) = MA_SystemNotificationMT
[If AMTI command is not executable]
Name = "AMTI_CommandStatusMT"
Signature (signal) = AMTI_CommandStatusMT


[If AO command]
alt
[If AO command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "AO_ActivityMT"
Signature (signal) = AO_ActivityMT
Name = "MA_SystemNotificationMT_6"
Signature (signal) = MA_SystemNotificationMT
[If AO command is not executable]
Name = "AO_CommandStatusMT"
Signature (signal) = AO_CommandStatusMT
[If COMINT command]
alt
[If COMINT command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "COMINT_ActivityMT"
Signature (signal) = COMINT_ActivityMT
Name = "MA_SystemNotificationMT_7"
Signature (signal) = MA_SystemNotificationMT
[If COMINT command is not executable]
Name = "COMINT_CommandStatusMT"
Signature (signal) = COMINT_CommandStatusMT
[If CommRelay command]
alt
[If CommRelay command is executable]
loop
[While ActivityState != commanded state]
opt
[If ExecutionState changes]
Name = "CommRelayActivityMT"
Signature (signal) = CommRelayActivityMT
Name = "MA_SystemNotificationMT_8"
Signature (signal) = MA_SystemNotificationMT
[If CommRelay command is not executable]
Name = "CommRelayCommandStatusMT"
Signature (signal) = CommRelayCommandStatusMT
[If EA command]
alt
[If EA command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "EA_ActivityMT"
Signature (signal) = EA_ActivityMT
Name = "MA_SystemNotificationMT_9"
Signature (signal) = MA_SystemNotificationMT
[If EA command is not executable]
Name = "EA_CommandStatusMT"
Signature (signal) = EA_CommandStatusMT


[If ESM command]
alt
[If ESM command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "ESM_ActivityMT"
Signature (signal) = ESM_ActivityMT
Name = "MA_SystemNotificationMT_10"
Signature (signal) = MA_SystemNotificationMT
[If ESM command is not executable]
Name = "ESM_CommandStatusMT"
Signature (signal) = ESM_CommandStatusMT
[If Flight command]
alt
[If Flight command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "MA_FlightActivityMT"
Signature (signal) = MA_FlightActivityMT
Name = "MA_SystemNotificationMT_11"
Signature (signal) = MA_SystemNotificationMT
[If Flight command is not executable]
Name = "MA_FlightCommandStatusMT"
Signature (signal) = MA_FlightCommandStatusMT
[If PO command]
alt
[If PO command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "PO_ActivityMT"
Signature (signal) = PO_ActivityMT
Name = "MA_SystemNotificationMT_12"
Signature (signal) = MA_SystemNotificationMT
[If PO command is not executable]
Name = "PO_CommandStatusMT"
Signature (signal) = PO_CommandStatusMT
[If Refuel command]
alt
[If Refuel command is executable]
loop
[While ExecutionState != commanded state]
opt
[If ExecutionState changes]
Name = "RefuelCommandMT"
Signature (signal) = RefuelCommandMT
Name = "MA_SystemNotificationMT_13"
Signature (signal) = MA_SystemNotificationMT
[If Refuel command is not executable]
Name = "RefuelCommandStatusMT"
Signature (signal) = RefuelCommandStatusMT

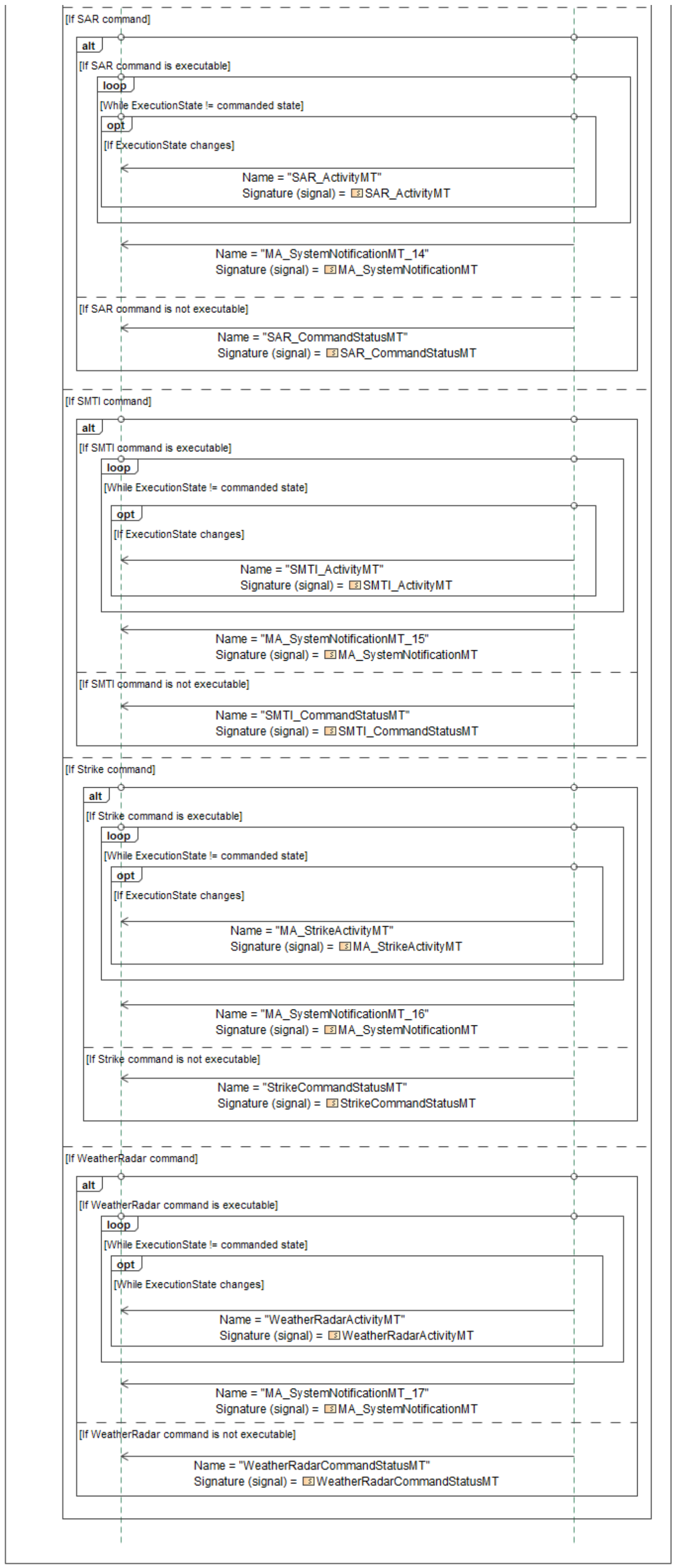


Figure 1-12: C2 Incorrect Messaging Sequence Error Handling

Table 1-11: C2 Incorrect Messaging Sequence Error Handling

Message Name (Name:Type)	Required Fields (Name:Type)
AirSampleActivityMT: AirSampleActivityMT	None
AirSampleCommandStatusMT: AirSampleCommandStatusMT	None
AMTI_ActivityMT: AMTI_ActivityMT	None
AMTI_CommandStatusMT: AMTI_CommandStatusMT	None
AO_ActivityMT: AO_ActivityMT	None
AO_CommandStatusMT: AO_CommandStatusMT	None
COMINT_ActivityMT: COMINT_ActivityMT	None
COMINT_CommandStatusMT: COMINT_CommandStatusMT	None
CommRelayActivityMT: CommRelayActivityMT	None
CommRelayCommandStatusMT: CommRelayCommandStatusMT	None
EA_ActivityMT: EA_ActivityMT	None
EA_CommandStatusMT: EA_CommandStatusMT	None
ESM_ActivityMT: ESM_ActivityMT	None
ESM_CommandStatusMT: ESM_CommandStatusMT	None
MA_FlightActivityMT: MA_FlightActivityMT	None
MA_FlightCommandStatusMT: MA_FlightCommandStatusMT	None
MA_MissionPlanActivationCommandStatusMT: MA_MissionPlanActivationCommandStatusMT	None
MA_MissionPlanActivationStatusMT: MA_MissionPlanActivationStatusMT	None
MA_StrikeActivityMT: MA_StrikeActivityMT	None
MA_SystemNotificationMT_1: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_2: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_3: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_4: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_5: MA_SystemNotificationMT	AssociatedID: ID_Type

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemNotificationMT_6: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_7: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_8: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_9: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_10: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_11: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_12: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_13: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_14: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_15: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_16: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_SystemNotificationMT_17: MA_SystemNotificationMT	AssociatedID: ID_Type
MA_TaskCommandStatusMT: MA_TaskCommandStatusMT	None
MA_TaskStatusMT: MA_TaskStatusMT	None
PO_ActivityMT: PO_ActivityMT	None
PO_CommandStatusMT: PO_CommandStatusMT	None
RefuelCommandMT: RefuelCommandMT	None
RefuelCommandStatusMT: RefuelCommandStatusMT	None
SAR_ActivityMT: SAR_ActivityMT	None
SAR_CommandStatusMT: SAR_CommandStatusMT	None
SMTI_ActivityMT: SMTI_ActivityMT	None
SMTI_CommandStatusMT: SMTI_CommandStatusMT	None
StrikeCommandStatusMT: StrikeCommandStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
WeatherRadarActivityMT: WeatherRadarActivityMT	None
WeatherRadarCommandStatusMT: WeatherRadarCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.3 C2 Lost Communication

A C2 system can configure MA to generate a *MissionContingencyAlertMT* and activate an existing plan when losing communications with C2. This sequence will follow the sequence laid out in section *Define Contingency Plan*, with some specifics. To define the contingency settings, C2 System will send a *MA_PlanningFunctionSettingsCommandMT* to MA with the *SystemID* field set to the unique ID of the system whose planning functions settings are being commanded or changed, the *MissionPlanningAutonomy.Trigger.CommsLost* set with the *MissionPlanningAutonomy.Trigger.CommsLost.Threshold* set to the amount of time in seconds to wait with no communications before triggering the lost communications contingency, and the *MissionPlaningAutonomy.AutonomousPlanningResponseChoice.AlertOnly* field set to "*FALSE*" indicating that MA should not initiate autonomous replanning. When MA receives a *MA_PlanningFunctionSettingsCommandMT*, if C2 is authorized it will reply with a *MA_PlanningFunctionSettingsCommandStatusMT* with the *CommandProcessingState* set to either "*ACCEPTED*" if the new autonomy settings were accepted or "*REJECTED*" if the autonomy settings were not able to be applied. The C2 system will send the MA a *MA_ResponseMT* with the *ResponseType* field set to "*PLAN_ACTIVATION*", the *Option.Trigger.AnyMessage.MessageType* field set to "*MISSION_CONTINGENCY_ALERT*", and the *Option.Response.ActivatePlan.MissionPlanID* field set to the desired *MissionPlanID* in order to configure the plan that should be activated in the event of lost communication with C2. C2 Systems can further specify what should be activated by using *Option.Response.ActivatePlan.ActivationDetails.ByMissionPlan, Option.Response.ActivatePlan.ActivationDetails.BySubPlan, and Option.Response.ActivatePlan.ActivationDetails.ByExecutionPlanSet*. *ByMissionPlan* indicates that the response should activate all sub-*Plans included in the mission plan. The *BySubPlan* field offers the ability to select specific sub-*Plan to activate, which is recommended for this specific contingency scenario. The *ByExecutionPlanSet* field offers the ability to select a specific ExecutionPlanSet to activate, which is recommended when using a mission plan operating with an *ExecutionSequence*.

Following the *MA_ResponseMT* messages, C2 systems will send a *ResponsePlanCommandMT* message with the *CommandState* set to "*NEW*" or "*UPDATE*" to tell Mission Autonomy to generate or edit a *ResponsePlanMT* based on the *MA_ResponseMTs* that were sent. A *CommandState* of "*NEW*" indicates that the Response is new and doesn't currently exist. *CommandState* of "*UPDATE*" indicates that C2 systems wish to perform an update to an existing *ResponsePlanMT*. Once a Plan has been successfully generated, Mission Autonomy will send out a second *ResponsePlanCommandStatusMT* with *PlanningStatus.CommandProcessingState* to "*ACCEPTED*", *PlanningStatus.CommandStatus* to "*COMPLETED*" and the *ResultingPlanID* field filled out with ID of the new Response Plan. Mission Autonomy will also send a *ResponsePlanMT* containing the newly generated Response Plan to C2 systems with corresponding *PlanCommandID,* specifying the originating *ResponsePlanCommandID,* and *Plan:AllocatedResponse* to indicate that a response should be assigned to the system that the plan is allocated to.

Mission Autonomy will then request the C2 systems to approve the Response Plan via a *MA_ApprovalRequestMT* message with the *RequirementID* of the Response Plan set in *ApprovalReferences.ApprovalItem.PlanApproval*. The C2 system will send back an *MA_ApprovalRequestStatusMT* message with *OperatorID* populated with the approving operator's UUID, and *ApprovalRequestProcessingState* set as "*APPROVED*" if the Response Plan is deemed acceptable or "*REJECTED*" if the Response Plan is not acceptable. If the Response Plan is rejected, Mission Autonomy will initiate replanning and propose a new Response Plan to C2. If the Response Plan is accepted, the plan is ready to be executed. MA sends *ResponsePlanApprovalStatusMT* to verify that the Response Plan was successfully approved. To begin execution of the new Response Plan, C2 systems will send a *MA_MissionPlanActivationCommandMT* message with *Command.MissionPlanID* set to the ID of the approved Response Plan and the *ActivationDetails:BySubPlan:ResponsePlan* populated with the applicable Response Plan. MA will respond with a *MA_MissionPlanActivationCommandStatusMT* notifying C2 of the command's status. Once activated, response monitoring will begin and send *ResponseActivityMT_1* message to the C2 system. An *ActivityStatus* of "*ENABLED*" means the trigger is currently being monitored. An *ActivityStatus* of "*ACTIVE*" indicates the response trigger was tripped and the response is currently executing the desired *Plan specified in *Option.Response.ActivatePlan.ActivationDetails*. An *ActivityStatus* of "*COMPLETED*" means that the desired *Plan has finished execution.

In the event that there are lost communications, which is defined by no *SystemStatusMT* received for the duration of the *MissionPlanningAutonomy.Trigger.CommsLost.Threshold*, MA will activate the contingency plan and execute it. Though the messages will not deliver to C2 due to the comms loss, MA generates *MissionContingencyAlertMT* in response to the contingency being activated. Additionally, MA sends *MA_ResponseActivityMT_2, _3*, and *_4* with information on the *ActivatedPlans* of the contingency response plan and the *AlertID* to specify the MissionContingencyAlert for the Response.

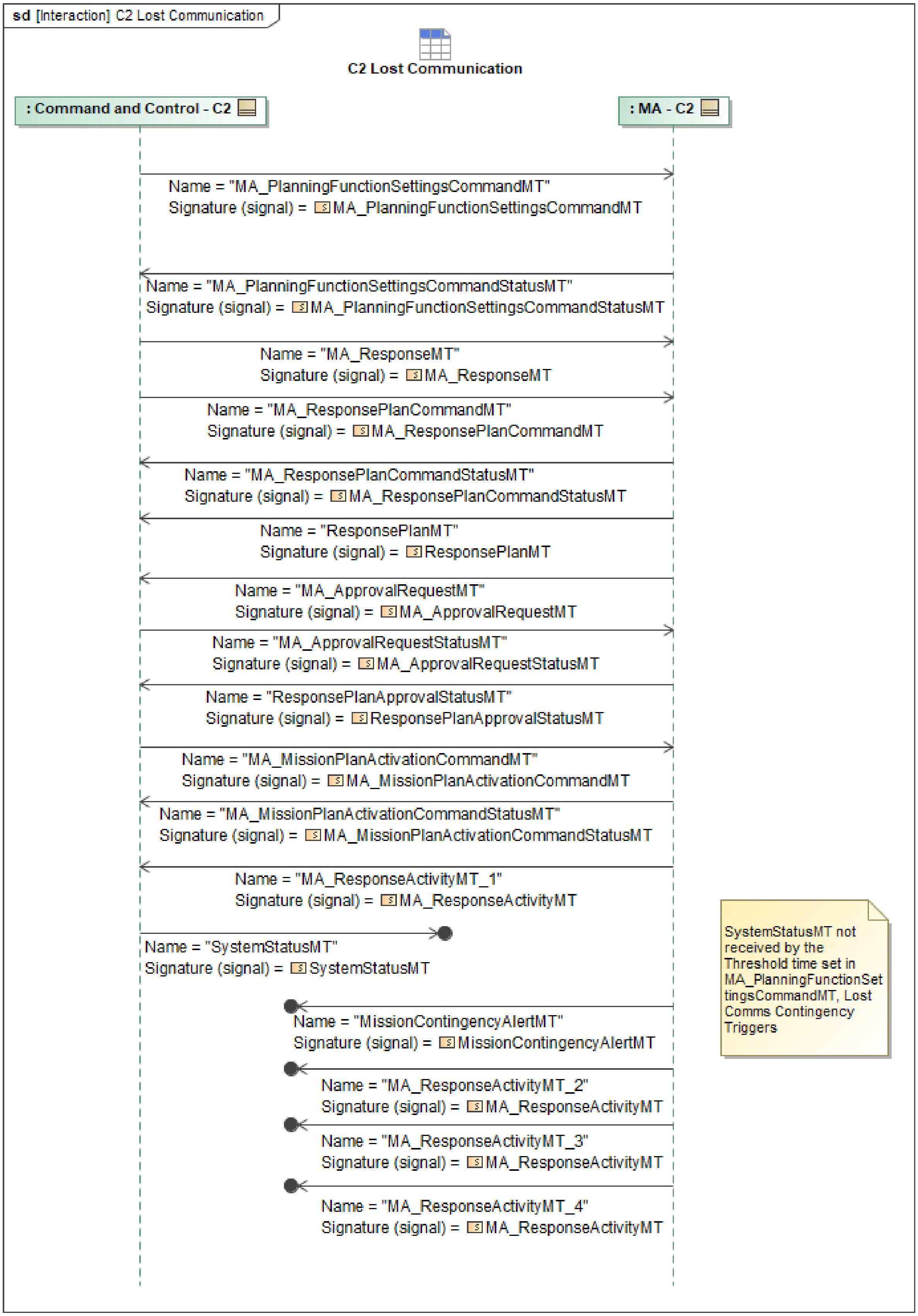


Figure 1-13: C2 Lost Communication

Table 1-12: C2 Lost Communication

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalRequestMT: MA_ApprovalRequestMT	None
MA_ApprovalRequestStatusMT: MA_ApprovalRequestStatusMT	OperatorID: OperatorID_Type
MA_MissionPlanActivationCommandMT: MA_MissionPlanActivationCommandMT	ResponsePlan: ResponsePlanActivationType
MA_MissionPlanActivationCommandStatusMT: MA_MissionPlanActivationCommandStatusMT	None
MA_PlanningFunctionSettingsCommandMT: MA_PlanningFunctionSettingsCommandMT	Threshold: DurationType AlertOnly: EmptyType MissionPlanningAutonomy: MA_MissionPlanningAutonomySettingType CommsLost: CommsLostTriggerDataType CommsLost: CommsLostTriggerDataType AlertOnly: EmptyType
MA_PlanningFunctionSettingsCommandStatusMT: MA_PlanningFunctionSettingsCommandStatusMT	None
MA_ResponseActivityMT_1: MA_ResponseActivityMT	MissionPlanID: MissionPlanID_Type
MA_ResponseActivityMT_2: MA_ResponseActivityMT	AlertID: MissionContingencyAlertID_Type ActivatedPlans: PlansReferenceType
MA_ResponseActivityMT_3: MA_ResponseActivityMT	AlertID: MissionContingencyAlertID_Type ActivatedPlans: PlansReferenceType
MA_ResponseActivityMT_4: MA_ResponseActivityMT	AlertID: MissionContingencyAlertID_Type ActivatedPlans: MA_PlansReferenceType
MA_ResponseMT: MA_ResponseMT	AnyMessage: QueryMessageType MessageType: MessageTypeEnum ActivatePlan: MA_MissionPlanActivationCommandType
MA_ResponsePlanCommandMT: MA_ResponsePlanCommandMT	None
MA_ResponsePlanCommandStatusMT: MA_ResponsePlanCommandStatusMT	ResultingPlanID: ResponsePlanID_Type
MissionContingencyAlertMT: MissionContingencyAlertMT	None
ResponsePlanApprovalStatusMT: ResponsePlanApprovalStatusMT	None
ResponsePlanMT: ResponsePlanMT	PlanCommandID: ResponsePlanCommandID_ChoiceType

Message Name (Name:Type)	Required Fields (Name:Type)
	AllocatedResponse: ResponseAllocationType
SystemStatusMT: SystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.4 C2 Malformed Message/Precondition Not Met Handling

Upon receipt of a malformed message, MA will send *MA_SystemNotificationMT* to C2, indicating that the message could not be successfully processed. Within the message, MA sets *NotificationState* to ACTIVE and populates *NotificationDetail.Reason* with *INVALID_INPUT_PARAMETER*. *NotificationDetail.Description* will contain a human-readable string stating which fields are malformed. *AssociatedMessage* will reference the malformed message. The *MessageType* will be set to the same type as the malformed message. *AssociatedID* will be populated with a reference ID from the malformed message if applicable, the *UUID* of that message. If additional data is required to resolve the condition and the fault remains active, MA will issue *QueryDataRequestMT* with the *Query* field containing the ID of the specific message. C2 will respond with *QueryDataRequestStatusMT*, specifically the *Message* field within the *Result* field will contain the corrected message containing the originally missing data. If the fault has been addressed, then MA will only send a new *MA_SystemNotificationMT* with the same fields as above and *NotificationState* set to CLOSED.

Upon receipt of a command from C2 for which the required inputs are not present on the platform/data store, MA will send *MA_SystemNotificationMT* to C2, indicating that the command could not be successfully executed. Within the message, MA sets *NotificationState* to ACTIVE and populates *NotificationDetail.Reason* with *UNKNOWN_ID*. *NotificationDetail.Description* will contain a human-readable string stating which data is not present. *AssociatedMessage* will reference the attempted command. The *MessageType* will be set to the same type as the command. *AssociatedID* will be populated with a reference ID from the command if applicable, the *UUID* of that command. If additional data is required to resolve the condition and the fault remains active, MA can issue *QueryDataRequestMT* with the Query field containing the ID of the specific command. C2 would then respond with *QueryDataRequestStatusMT*, specifically the Message field within the Result field will contain the corrected command containing the originally missing data. If the fault has been addressed, then MA will only send a new *MA_SystemNotificationMT* with the same fields as above and *NotificationState* set to CLOSED.

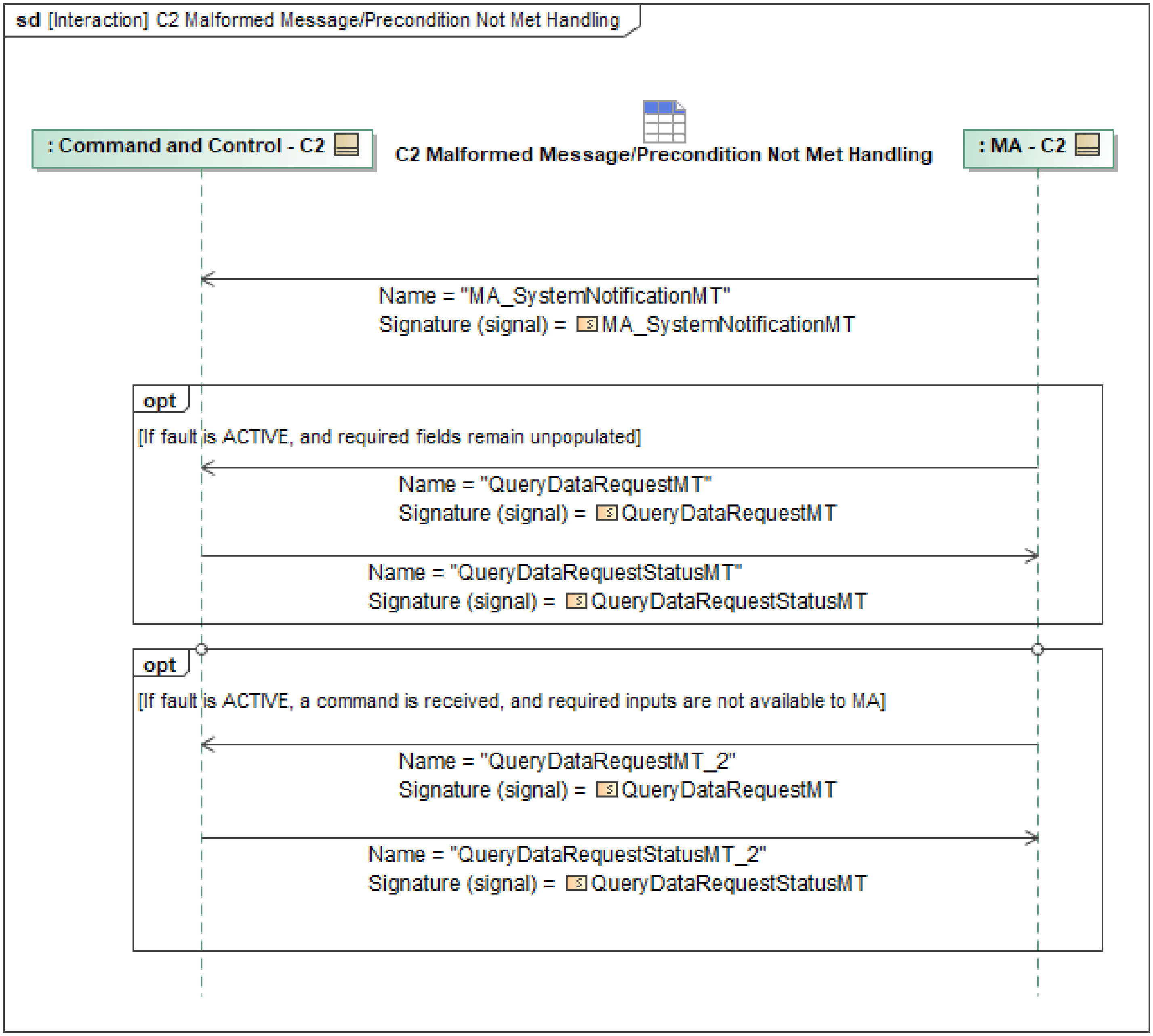


Figure 1-14: C2 Malformed Message/Precondition Not Met Handling

Table 1-13: C2 Malformed Message/Precondition Not Met Handling

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemNotificationMT: MA_SystemNotificationMT	Description: VisibleString1024Type AssociatedID: ID_Type
QueryDataRequestMT: QueryDataRequestMT	Query: QueryPET
QueryDataRequestMT_2: QueryDataRequestMT	None
QueryDataRequestStatusMT: QueryDataRequestStatusMT	Message: MessageType

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.5 C2 Receive Contingency Status

MA will periodically publish *MA_ResponseActivityMT* to convey monitoring and triggered status for all active contingencies. An *ActivityState* of *ENABLED* indicates MA is actively monitoring the trigger condition for the *Response* while *DISABLED* means monitoring has stopped. When the trigger condition has been satisfied, the *ActivityState* will change to *COMPLETED* with the *RequirementInstantiation* or *ActivatedPlans* fields filled out if the *Response* was configured to perform a response action, otherwise only *AlertID* will be set.

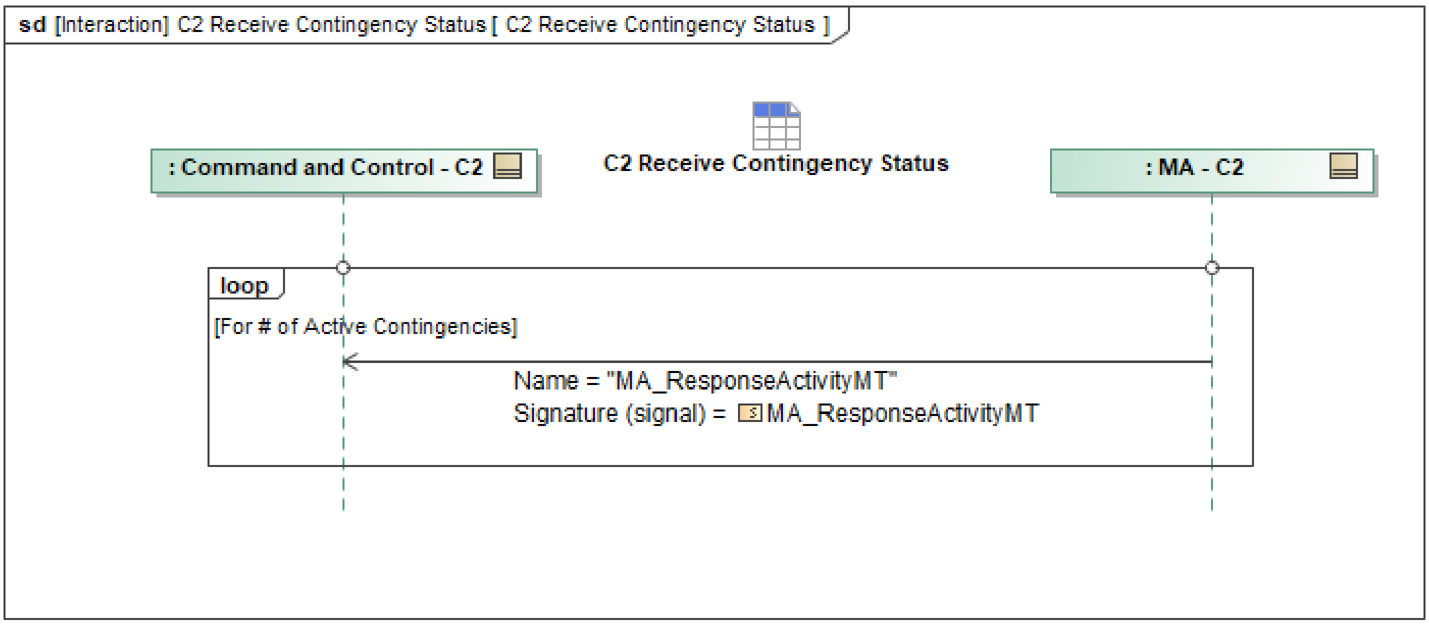


Figure 1-15: C2 Receive Contingency Status

Table 1-14: C2 Receive Contingency Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ResponseActivityMT: MA_ResponseActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.6 Define Autonomous Contingency Re-Plan

C2 systems can configure MA to initiate autonomous replanning in the event of a contingency. C2 systems will send a *MA_PlanningFunctionSettingsCommandMT* to set the contingency trigger and indicate the desired replanning that should occur if the contingency condition is met.

MissionPlanningAutonomy.AutonomousPlanningResponseChoice field should use *AutonomousPlanningAction* or *AlertOnly* field should be "*FALSE*". *MissionPlanningAutonomy.AutonomousPlanningResponseChoice.AutonomousPlanningAction* is used to configure what type of replanning MA should initiate. *MissionPlanningAutonomy.AutonomousPlanningResponseChoice.AutonomousPlanningAction* should include a *AutonomousPlanningActionID* to be used to reference this planning action, as well as *AllowedType* field which will configure the specifics of the planning process. *AllowedType.PlanningProcessID* should be set to the desired ID for this planning process and will be used to reference this process in future messages. *AllowedType.ForPlanningUseOnly* should be set to "*FALSE*" as this plan must be executed to resolve the contingency.

When MA receives *MA_PlanningFunctionSettingsCommandMT*, and if C2 is authorized, it replies with *MA_PlanningFunctionSettingsCommandStatusMT_2*, with *CommandedProcessingState* set to "*ACCEPTED*" if the system is able to apply the autonomy settings or "*REJECTED*" is the system is not able to apply these settings.

In order for MA to execute a generated plan, the plan must first be approved per the *ApprovalPolicy*. Mission Planning will set the criteria for what plans can be approved with the use of *MA_ApprovalPolicyMT* by an operator vs. which plans can be automatically activated by MA.

In the event C2 wishes to modify an existing approval policy, C2 sends an *MA_ApprovalPolicyMT* message. For contingencies, the approval policy will have an *ApprovalPolicyID* set to the desired ID of the *ApprovalPolicy*. The Plans field should have *Plans.ApplicablePlanningActivity* set to **_PLAN*, where the asterisk indicates the appropriate requirement for this planning service. Additionally, the approval policy should have *ByTriggerPolicy.TriggerSource* set to "*AUTONOMOUS_TRIGGER*" to indicate this policy applies to replanning from autonomous triggers and not operator-commanded replanning. *ByTriggerPolicy.Trigger* can be used to specify that this policy is only for a specific type of trigger. The *MA_ApprovalPolicyMT* will also have *PlanActivation.Policy.ApprovalRequirement* set to either "*AUTO_APPROVE*" if C2 wishes the contingency replan to be executed immediately, or "*APPROVAL_REQUIRED*" which will ask a C2 operator for approval of the proposed plan immediately before execution. *MA_ApprovalPolicyMT* with the desired settings, specifically *Plans.ByTriggerPolicy.Policy* can be set to *"APPROVAL_REQUIRED"* to require MA to wait for operator approval before activating an autonomous replan in response to the defined contingency trigger.

For more information on execution sequences with respect to autonomous replanning, see section Triggered Autonomous Contingency Replan.

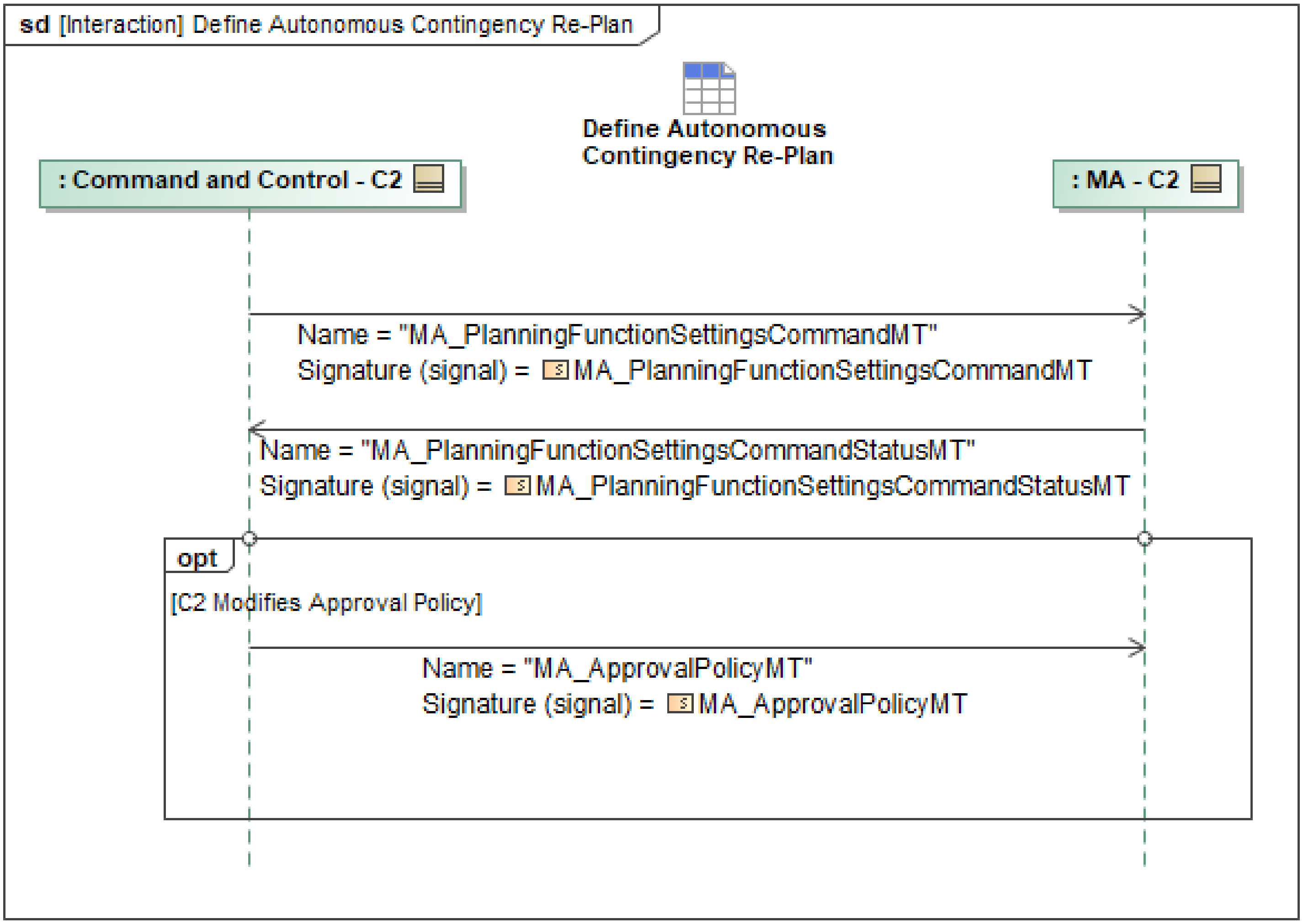


Figure 1-16: Define Autonomous Contingency Re-Plan

Table 1-15: Define Autonomous Contingency Re-Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalPolicyMT: MA_ApprovalPolicyMT	PlanActivation: PlanActivationPolicyType Plans: PlanPolicyType ByTriggerPolicy: ByTriggerPolicyType DataRecordInstanceID: DataRecordInstanceID_Type
MA_PlanningFunctionSettingsCommandMT: MA_PlanningFunctionSettingsCommandMT	MissionPlanningAutonomy: MissionPlanningAutonomySettingType
MA_PlanningFunctionSettingsCommandStatusMT: MA_PlanningFunctionSettingsCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.7 Define Contingency Alert

C2 systems can configure MA to generate a *MissionContingencyAlertMT* when a contingency is triggered. C2 Systems will send a *MA_PlanningFunctionSettingsComamndMT* message to MA. *MissionPlanningAutonomy.Trigger* field is used to configure what type of contingency to monitor for. The *PlanningByCaseTriggerType* offers a wide range of possible contingency conditions that can be set. It offers typical contingency conditions like "*CommsLost*", "*EnduranceLow*", "*OffRoute*" etc, as well as other triggers more specific to UCI like "*CapabilityAdded*" or "*RequirementFailed*". The *PlanningByCaseTriggerType* also includes types for predicting future contingencies; for example, the "*RouteVulnerability*" type can be used to set vulnerability thresholds that can predict the likelihood of a future contingency. Depending on the type of trigger chosen, more information may need to be provided. As an example, A "*Lost Comms*" trigger requires a duration value to measure how long communications have been interrupted for (see section C2 Lost Communications where the lost communications example is walked through in more detail). *MissionPlanningAutonomy.AutonomousPlanningResponseChoice* field must be set to "*Alert Only*" to indicate to MA that an alert should be generated for the chosen Trigger when its conditions are met. This field also indicates that MA should not initiate autonomous contingency replanning. When MA receives *MA_PlanningFunctionSettingsCommandMT* and C2 is authorized, it will reply with a *MA_PlanningFunctionSettingsCommandStatusMT* with the *CommandProcessingState* field set to either "*ACCEPTED*" if the new autonomy settings were accepted or "*REJECTED*" if the autonomy settings were rejected. The autonomy settings can be rejected if the system is already constrained by a different set of autonomy settings or has been tasked with other requirements that would interfere with the system's ability to effectively perform the autonomous processes described in the message. If MA rejects the autonomy settings, it will set the *CommandProcessingStateReason* field in *MA_PlanningFunctionSettingsCommandStatusMT* to indicate why the autonomy settings were rejected. C2 systems can use *CommandProcessingStateReason* field in the returned status message to reconfigure the *MissionPlanningAutonomy* field in *MA_PlanningFunctionSettingsCommandMT* and resend to MA.

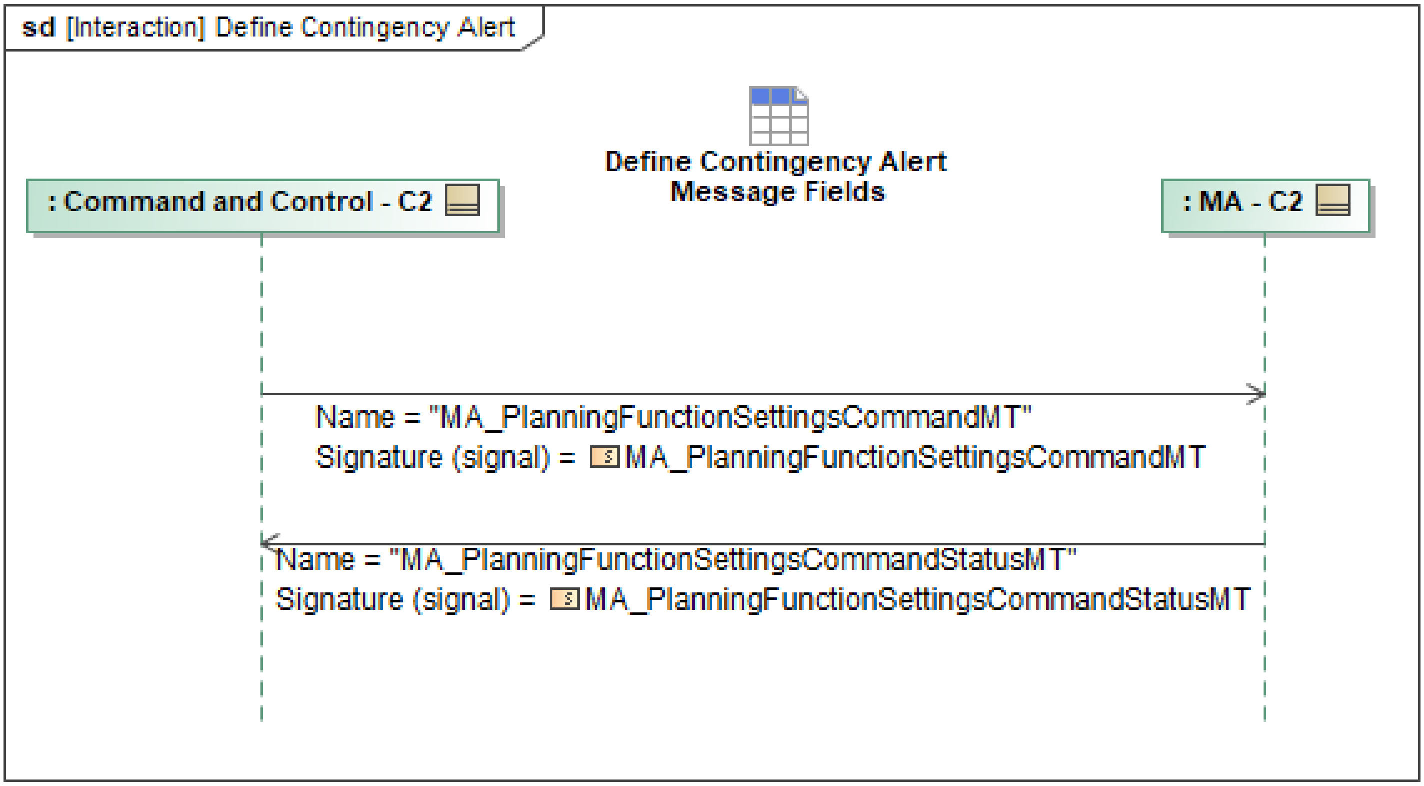


Figure 1-17: Define Contingency Alert

Table 1-16: Define Contingency Alert

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PlanningFunctionSettingsCommandMT: MA_PlanningFunctionSettingsCommandMT	None
MA_PlanningFunctionSettingsCommandStatusMT: MA_PlanningFunctionSettingsCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.8 Define Contingency Plan

C2 systems will send an *MA_ApprovalPolicyMT* message for each unique contingency plan and will be sent alongside a corresponding *MA_ResponseMT*. Information such as timeout policies and approval durations will be addressed within *ApprovalPolicy.Plans.DefaultPolicy*. Following approval, C2 Systems will use *MA_ResponseMT* in order to activate a specific plan based on a contingency trigger. Refer to section *Define Contingency Alert* for details on setting up an alert with *PlanningFunctionSettingsCommandMT*. C2 systems next step will be to configure a *MA_ResponseMT* to activate a plan when this alert is generated.

The *MA_ResponseMT* will specify the mission plan to be activated when the specified trigger is tripped. The *ResponseType* field should be set to "*PLAN_ACTIVATION*". The *MA_ResponseMT Option.Trigger* should contain a "*QueryMessageType*" that will be filtering for

MissionContingencyAlertMTs from the system configured with the previous *PlanningFunctionSettingsCommandMT*. The *QueryMessage* should match *SourceSystemID*, *ContingencyCondition.Trigger*, and *ContingencyCondition.AutonomousActionStatus.AlertOnly* fields in the *MissionContingencyAlertMT*. C2 systems will set *Option.Response.ActivatePlan.MissionPlanID* field to the desired *MissionPlanID* in order to configure the plan that should be activated in the event of the contingency. C2 Systems can further specify what should be activated by using *Option.Response.ActivatePlan.ActivationDetails.ByMissionPlan, Option.Response.ActivatePlan.ActivationDetails.BySubPlan,* and *Option.Response.ActivatePlan.ActivationDetails.ByExecutionPlanSet* fields. *ByMissionPlan* indicates that the response should activate all sub-*Plans included in the mission plan. The *BySubPlan* field offers the ability to select specific sub-*Plans to activate, like for example a RTB route plan in response to a "Endurance Low" contingency alert. The *ByExecutionPlanSet* field offers the ability to select a specific ExecutionPlanSet to activate, such as one corresponding to a 'Transit' portion of the mission. C2 systems can send one or multiple *ResponseMTs* depending on the number of desired plans that need to be activated to complete the mission.

Following the *MA_ResponseMT* messages, C2 systems will send a *ResponsePlanCommandMT* message with the *CommandState* set to "*NEW*" or "*UPDATE*" to tell Mission Autonomy to generate or edit a *ResponsePlanMT* based on the *MA_ResponseMTs* that were sent. A *CommandState* of "*NEW*" indicates that the Response is new and doesn't currently exist. *CommandState* of "*UPDATE*" indicates that C2 systems wish to perform an update to an existing *ResponsePlanMT*. Once a Plan has been successfully generated, Mission Autonomy will send out a second *ResponsePlanCommandStatusMT* with *PlanningStatus.CommandProcessingState* to "*ACCEPTED*", *PlanningStatus.CommandStatus* to "*COMPLETED*" and the *ResultingPlanID* field filled out with ID of the new Response Plan. Mission Autonomy will also send a *ResponsePlanMT* containing the newly generated Response Plan with corresponding *PlanCommandID,* specifying the originating *ResponsePlanCommandID,* and *Plan:AllocatedResponse* to indicate that a response should be assigned to the system that the plan is allocated to.

Mission Autonomy will then request the C2 systems to approve the Response Plan via a *MA_ApprovalRequestMT* message with the *RequirementID* of the Response Plan set in *ApprovalReferences.ApprovalItem.PlanApproval*. The C2 system will send back an *MA_ApprovalRequestStatusMT* message with *ApprovalRequestProcessingState* set as "*APPROVED*" if the Response Plan is deemed acceptable or "*REJECTED*" if the Response Plan is not acceptable. If the Response Plan is rejected, Mission Autonomy will initiate replanning and propose a new Response Plan to C2. If the Response Plan is accepted, the plan is ready to be executed. MA sends *ResponsePlanApprovalStatusMT* to verify that the Response Plan was successfully approved. To begin execution of the new Response Plan, C2 systems will send a *MA_MissionPlanActivationCommandMT* message with *Command.MissionPlanID* set to the ID of the approved Response Plan. MA will respond with a *MA_MissionPlanActivationCommandStatusMT* notifying C2 of the command's status. Once activated, response monitoring will begin and send *ResponseActivityMT* messages to the C2 system. An *ActivityStatus* of "*ENABLED*" means the trigger is currently being monitored. An *ActivityStatus* of "*ACTIVE*" indicates the response trigger was tripped and the response is currently executing the desired *Plan specified in *Option.Response.ActivatePlan.ActivationDetails*. An *ActivityStatus* of "*COMPLETED*" means that the desired *Plan has finished execution.

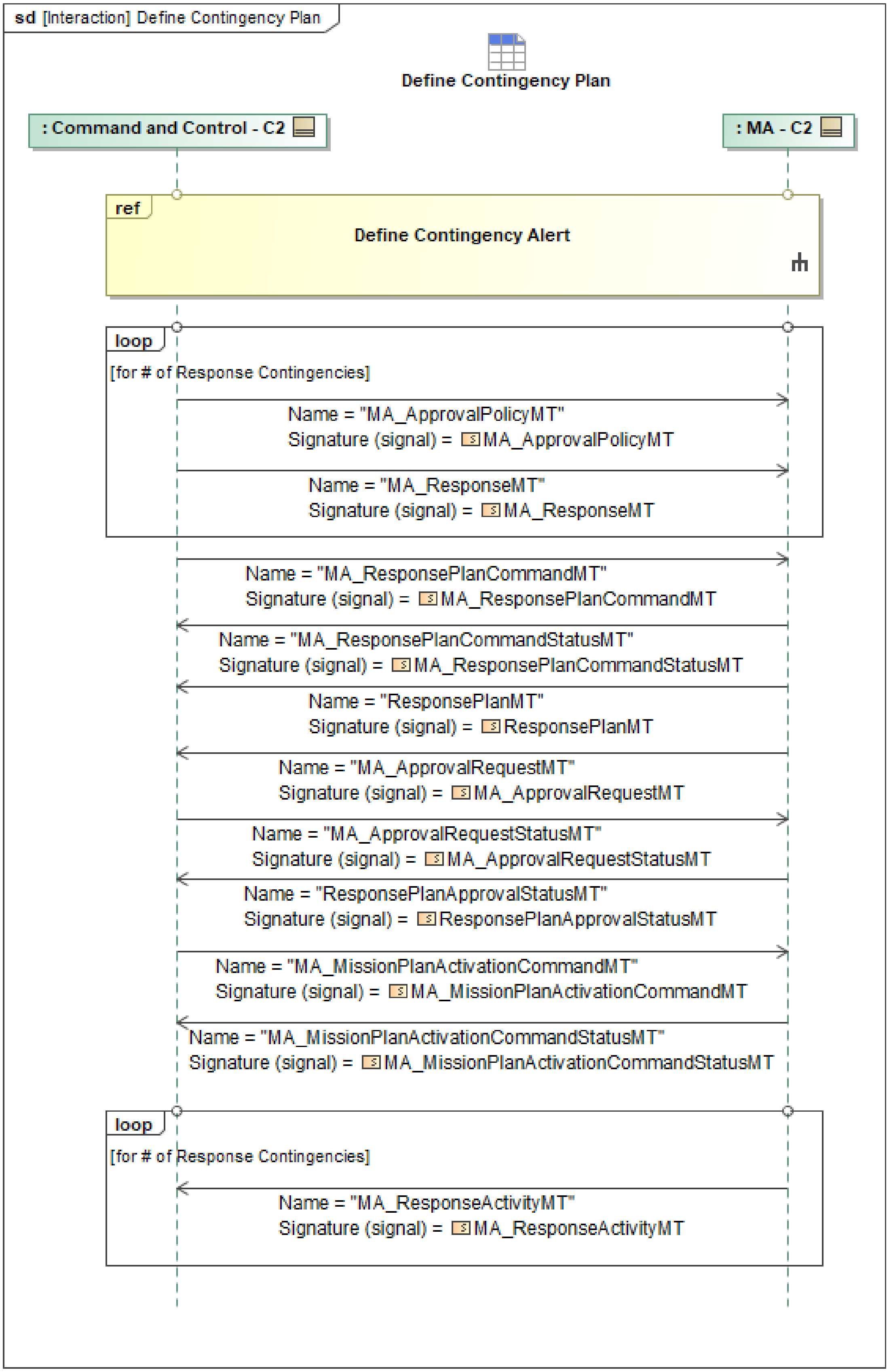


Figure 1-18: Define Contingency Plan

Table 1-17: Define Contingency Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalPolicyMT: MA_ApprovalPolicyMT	DefaultPolicy: ApprovalPolicyBaseType Plans: PlanPolicyType
MA_ApprovalRequestMT: MA_ApprovalRequestMT	None
MA_ApprovalRequestStatusMT: MA_ApprovalRequestStatusMT	OperatorID: OperatorID_Type
MA_MissionPlanActivationCommandMT: MA_MissionPlanActivationCommandMT	None
MA_MissionPlanActivationCommandStatusMT: MA_MissionPlanActivationCommandStatusMT	None
MA_ResponseActivityMT: MA_ResponseActivityMT	MissionPlanID: MissionPlanID_Type
MA_ResponseMT: MA_ResponseMT	None
MA_ResponsePlanCommandMT: MA_ResponsePlanCommandMT	MissionPlanID: MissionPlanID_Type
MA_ResponsePlanCommandStatusMT: MA_ResponsePlanCommandStatusMT	ResultingPlanID: ResponsePlanID_Type
ResponsePlanApprovalStatusMT: ResponsePlanApprovalStatusMT	None
ResponsePlanMT: ResponsePlanMT	PlanCommandID: ResponsePlanCommandID_ChoiceType AllocatedResponse: ResponseAllocationType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.9 Triggered Autonomous Contingency Re-Plan

When a trigger for a contingency is tripped, the MA will send the C2 systems a *MissionContingencyAlertMT_1* with the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detects the contingency and is generating the alert, the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field set to "*FALSE"* to indicate that an action will be taken, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that has the contingency condition, the *ContingencyCondition.ConflictState* field set to "*CONFLICTED*" to indicate that the conflict that caused the trigger has not been resolved, the *ContingencyCondition.Conflict.ConflictID* set to the unique ID for this conflict, the *ContingencyCondition.Trigger.Source* field set to "*AUTONOMOUS_TRIGGER*", the *ContingencyCondition.Trigger.ByCaseTrigger* field set to the initial by case trigger for autonomous mission planning functions, the *ContingencyCondition.AutonomousActionStatus.AutonomousPlanningActionStatus.AutonomousPlanningActionID* field set to the ID of the *AutonomousPlanningAction* and the *ContingencyCondition.AutonomousActionStatus.AutonomousPlanningActionStatus.Status* field set to "*IN_PROGRESS*".

Next, MA must check the approval required to execute the appropriate plan. In a previous sequence (Update Contingency Plan), the contingency plans were loaded and *ApprovalPolicyMT* sent. *ApprovalPolicyMT* contains an *ApprovalPolicyID*, *PlanID*, the plan itself, and approval requirements under the *ApprovalPolicyID, Plans.ApplicablePlanningActivity, and ApprovalPolicy.Plans.ByTriggerPolicy.Policy* fields, respectively. If *ApprovalPolicy.Plans.ByTriggerPolicy.Policy = Approval_Required,* MA will send the plan to be executed to the C2 systems. The plan could be any type of plan (i.e *MA_TaskPlanMT*, *MA_RoutePlanMT*, *ActionPlanMT*, etc.) depending on the constraints that were given to Mission Autonomy when the trigger was defined. The sequence diagram is using an *MA_RoutePlanMT* as an example.

If the *ApprovalPolicy* was previously set to "operator" when the contingency was setup, then after the *MA_RoutePlanMT_1* is sent, the MA will send the C2 systems an *MA_ApprovalRequestMT* with the *RequestID* field set to the unique ID of the request message, the *Approver.NonOperatorIdentifier.SystemID* set to the unique ID of the system, the *Approver.NonOperatorIdentifier.ServiceID* set to the unique ID of the service, the *ApprovalReferences.ApprovalPolicyID* set to the unique ID of the *ApprovalPolicy* that resulted in the *MA_ApprovalRequestMT*, the *ApprovalReferences.ApprovalItem.PlanApproval* set to the the unique ID of the Plan that is under review for approval, and the *RespondByField* set to the date and time that the request must be approved by. The C2 systems will respond with an *MA_ApprovalRequestStatusMT* with the *RequestID* set to the unique ID of the request message corresponding to this status message, the *RequestProcessingState* set to "*COMPLETED*", the *OperatorID* field filled out so MA has awareness of who approved the request, and the *ApprovalRequestProcessingState* set to "*APPROVED*". If it is not approved, then MA will send another *MA_RoutePlanMT_1* message until the operator approves it. If the *ApprovalPolicy* was not set to "*operator*", the MA will use the plan it generated without any external approval via MA_RoutePlanMT_2.

The execution status of the activated *Plan will be sent via **PlanExecutionStatusMT*. The example in the sequence diagram is for a *RoutePlanExecutionStatusMT to be sent periodically*. When the contingency has been cleared, the MA will send the C2 systems a *MissionContingencyAlertMT_2* with the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detected the contingency and generated the alert, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that had the contingency condition, the *ContingencyCondition.ConflictState* field set to "*CLEARED*" to indicate that the conflict that caused the trigger has been resolved, the *ContingencyCondition.Conflict.ConflictID* set to the unique id for this conflict, and the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field set to "*FALSE*" and the *ContingencyCondition.AutonomousActionStatus.AutonomousPlanningActionStatus.Status* field set to "*SUCCESSFUL*" to indicate successful action to clear the contingency.

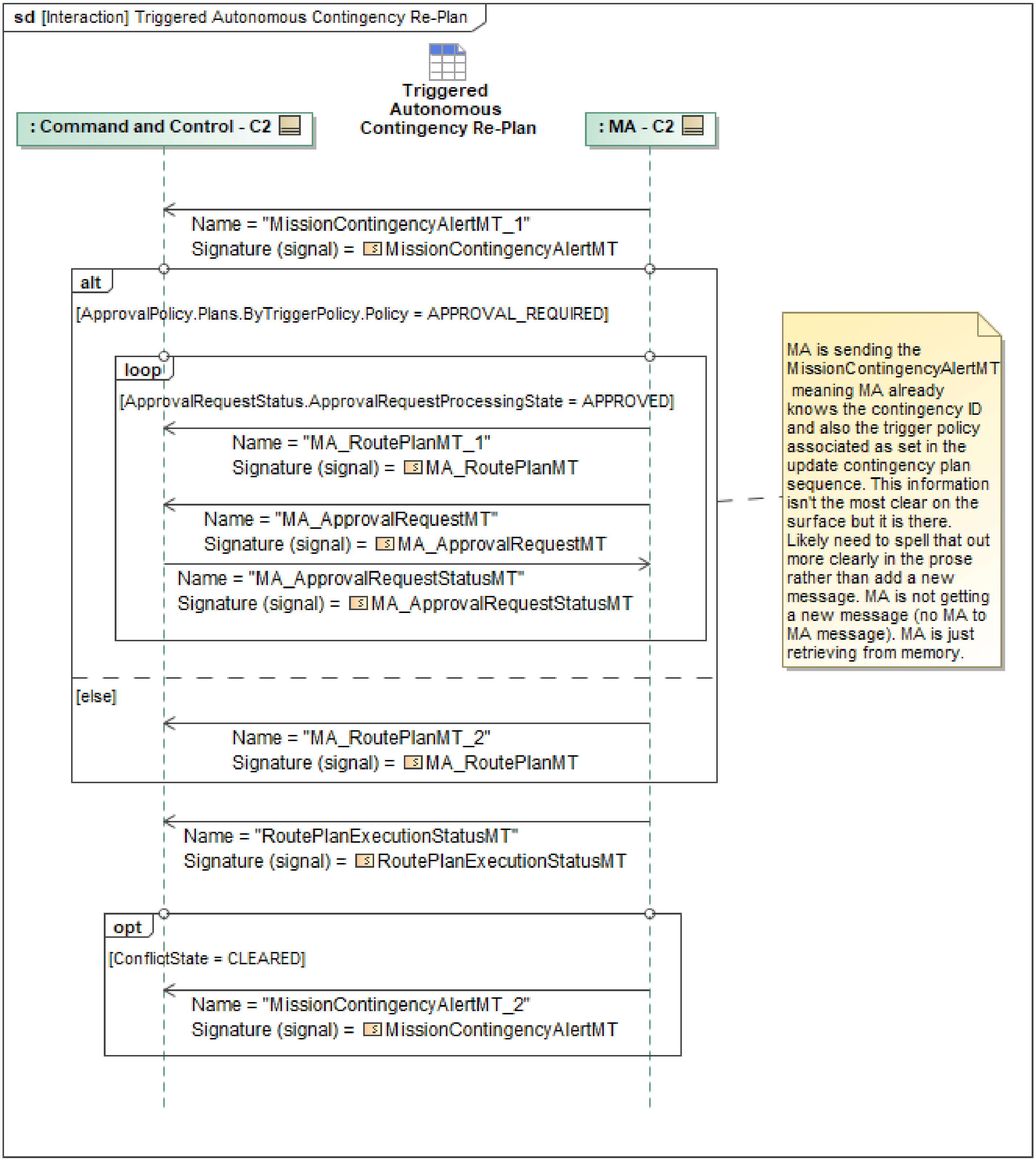


Figure 1-19: Triggered Autonomous Contingency Re-Plan

Table 1-18: Triggered Autonomous Contingency Re-Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalRequestMT: MA_ApprovalRequestMT	None
MA_ApprovalRequestStatusMT: MA_ApprovalRequestStatusMT	OperatorID: OperatorID_Type
MA_RoutePlanMT_1: MA_RoutePlanMT	None
MA_RoutePlanMT_2: MA_RoutePlanMT	None
MissionContingencyAlertMT_1: MissionContingencyAlertMT	ByCaseTrigger: PlanningByCaseTriggerType
MissionContingencyAlertMT_2: MissionContingencyAlertMT	None
RoutePlanExecutionStatusMT: RoutePlanExecutionStatusMT	PlanExecutionStatus: RoutePlanExecutionStateType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.10 Triggered Contingency Alert

When a trigger for a contingency is tripped, the MA will send the C2 systems a *MissionContingencyAlertMT_1* with the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detects the contingency and is generating the alert, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that has the contingency condition, the *ContingencyCondition.ConflictState* field set to "*CONFLICTED*" to indicate that the conflict that caused the trigger has not been resolved, the *ContingencyCondition.Conflict.ConflictID* set to the unique ID for this conflict, the *ContingencyCondition.Trigger.Source* field set to "*AUTONOMOUS_TRIGGER*", the *ContingencyCondition.Trigger.ByCaseTrigger* field set to the initial by case trigger for autonomous mission planning functions, and the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field set to "*TRUE*" to indicate that there will be no action taken.

When the contingency has been cleared, the MA will send the C2 systems a *MissionContingencyAlertMT_2* with the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detected the contingency and generated the alert, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that had the contingency condition, the *ContingencyCondition.ConflictState* field set to "*CLEARED*" to indicate that the conflict that caused the trigger has been resolved, the *ContingencyCondition.Conflict.ConflictID* set to the unique id for this conflict, the *ContingencyCondition.Trigger.Source* field set to "*AUTONOMOUS_TRIGGER*", the *ContingencyCondition.Trigger.ByCaseTrigger* field set to the initial by case trigger for autonomous mission planning functions, and the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field set to "*TRUE*" to indicate that no was action was taken.

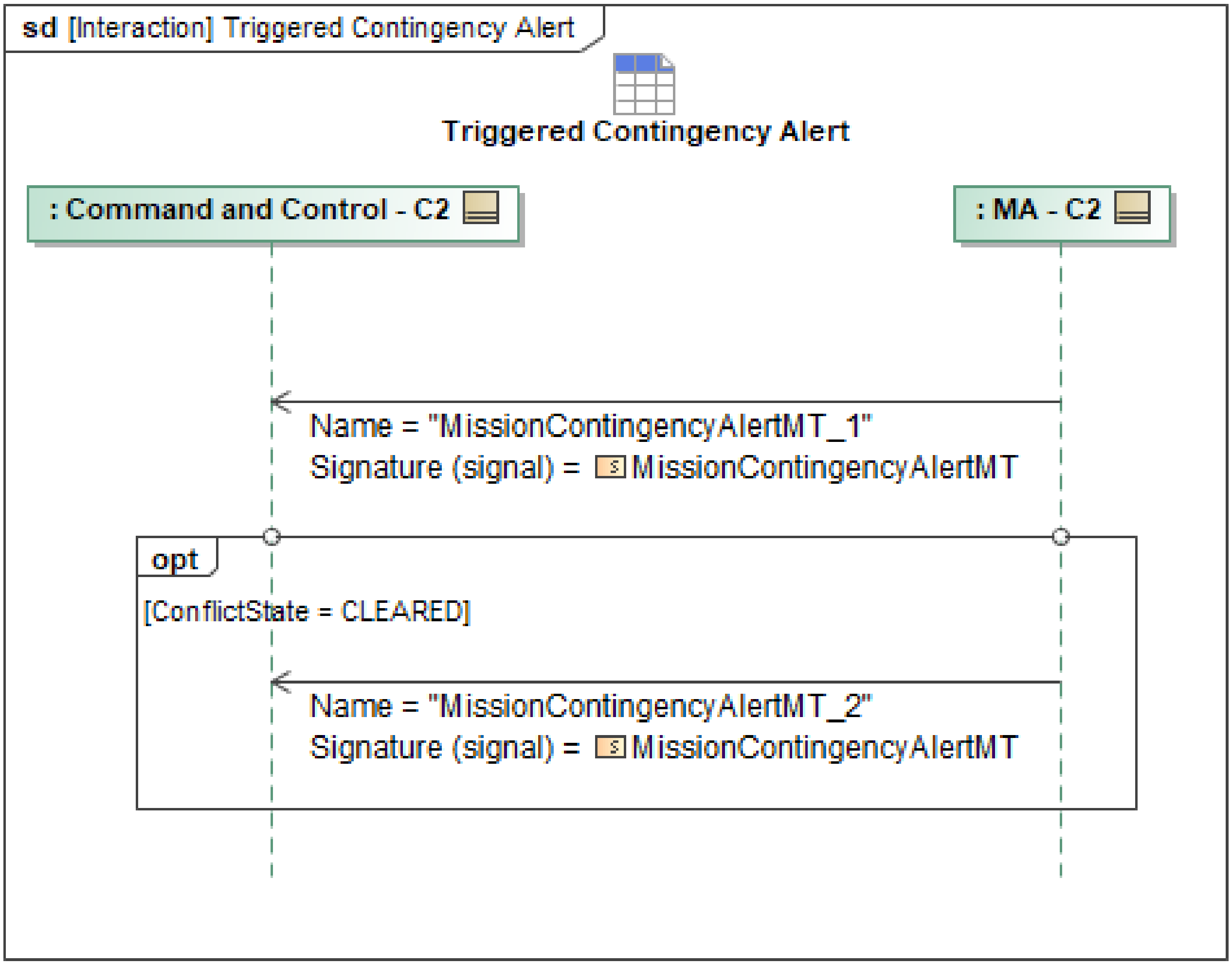


Figure 1-20: Triggered Contingency Alert

Table 1-19: Triggered Contingency Alert

Message Name (Name:Type)	Required Fields (Name:Type)
MissionContingencyAlertMT_1: MissionContingencyAlertMT	ByCaseTrigger: PlanningByCaseTriggerType
MissionContingencyAlertMT_2: MissionContingencyAlertMT	ByCaseTrigger: PlanningByCaseTriggerType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.11 Triggered Contingency Plan

When a trigger for a contingency is tripped, the MA will send the C2 systems a *MissionContingencyAlertMT_1* with the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detects the contingency and is generating the alert, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that has the contingency condition, the *ContingencyCondition.ConflictState* field set to "*CONFLICTED*" to indicate that the conflict that caused the trigger has not been resolved, the

ContingencyCondition.Conflict.ConflictID set to the unique ID for this conflict, the *ContingencyCondition.Trigger.Source* field set to "*AUTONOMOUS_TRIGGER*", the *ContingencyCondition.Trigger.ByCaseTrigger* field set to the initial by case trigger for autonomous mission planning functions, and the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field set to "*TRUE*" to indicate that the autonomous planning will not be happening because this contingency was linked to the activation of a specific plan in the ResponsePlan (see Section Define Contingency Plan).

MA sends a *MA_ResponseActivityMT_1* with the *Activity.ActivityState* field set to "*ACTIVE_**" (depending on if the Activity is able to be performed based on resource limitations), the *Activity.ActivatedPlans.MissionPlanID* field set to the ID of the MissionPlan if a MissionPlan is being activated, and the *Activity.ActivatedPlans.*PlanID*(s) field(s) set to indicate activation of desired sub-*Plans into the same activation state. A seperate *Activity* must be used to set plans to a different *ActivityState*. Once the response activity has been completed, MA sends a *MA_ResponseActivityMT_2* with the ActivityState field set to "*COMPLETED*" and the *Activity.ActivatedPlans.MissionPlanID* and/or *Activity.ActivatedPlans.*PlanID* field set to the ID of the *Plan(s) that was/were executed. MA sends another *MA_ResponseActivityMT_3* with the *ActivityState* field set to "*ENABLED*" to indicate that the trigger is enabled, the *Activity.ActivatedPlans.MissionPlanID* field set to the ID of the MissionPlan that was activated when the contingency was tripped, and the *Activity.AlertID* field set to the *MissionContingencyAlertID* of the *MissionContingencyAlertMT*.

When the contingency has been cleared, the MA will send the C2 systems a *MissionContingencyAlertMT_2* with the *MissionContingencyAlertID* field set to the ID of the alert, the *SourceSystemID* field set to the ID of the system that detected the contingency and generated the alert, the *ContingencyCondition.ConflictedSystemID* field set to the ID of the system that had the contingency condition, the *ContingencyCondition.ConflictState* field set to "*CLEARED*" to indicate that the conflict that caused the trigger has been resolved, the *ContingencyCondition.Conflict.ConflictID* set to the unique id for this conflict, the *ContingencyCondition.Trigger.Source* field set to "*AUTONOMOUS_TRIGGER*", the *ContingencyCondition.Trigger.ByCaseTrigger* field set to the initial by case trigger for autonomous mission planning functions, and the *ContingencyCondition.AutonomousActionStatus.AlertOnly* field set to "*TRUE*" to indicate that no was action taken.

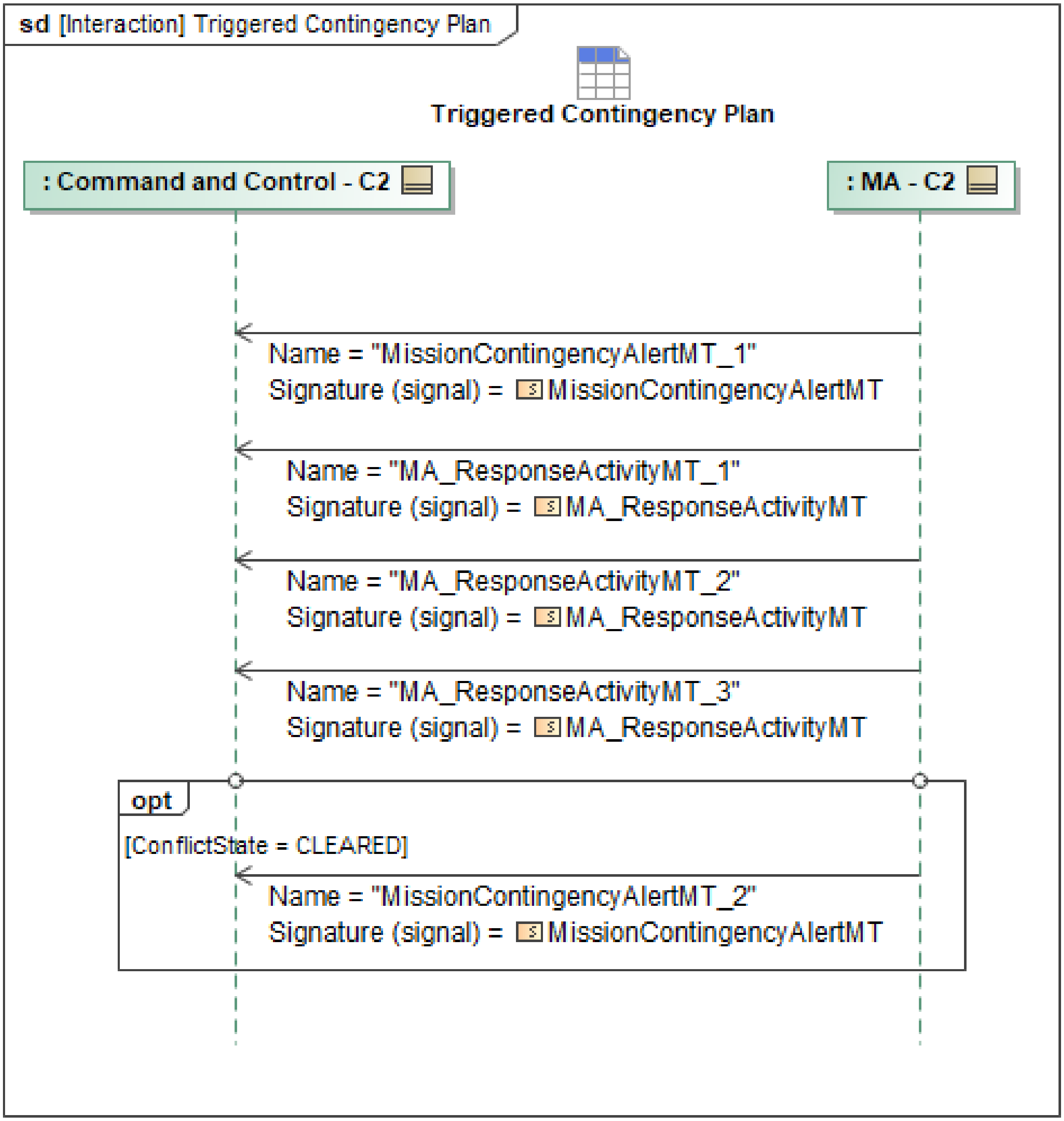


Figure 1-21: Triggered Contingency Plan

Table 1-20: Triggered Contingency Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ResponseActivityMT_1: MA_ResponseActivityMT	AlertID: MissionContingencyAlertID_Type ActivatedPlans: PlansReferenceType
MA_ResponseActivityMT_2: MA_ResponseActivityMT	AlertID: MissionContingencyAlertID_Type ActivatedPlans: PlansReferenceType

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ResponseActivityMT_3: MA_ResponseActivityMT	MissionPlanID: MissionPlanID_Type
MissionContingencyAlertMT_1: MissionContingencyAlertMT	ByCaseTrigger: PlanningByCaseTriggerType
MissionContingencyAlertMT_2: MissionContingencyAlertMT	ByCaseTrigger: PlanningByCaseTriggerType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.12 Team Contingencies

1.2.4.12.1 C2 Lost Comms with Team Leader

In the event that a Team leader loses connection with C2, the lost comms contingency response behavior is activated (see *Define Team Lost Comms Contingency)*. In normal operations, C2 sends SystemStatusMT_1 to MA periodically, and MA sends SystemStatusMT_2 to C2 periodically. The lost comms condition occurs when SystemStatusMT_1 and SystemStatusMT_2 fail to reach their intended recipients. If the Team Leader regains communication with C2 via receipt of a *SystemStatusMT_3 message* (returning to normal periodic behavior), the Team Leader will send C2 a fresh *PackageStatusMT* updating it of any changes that may have occurred while the team could not communicate with C2. See section *Team Leader Lost Comms with C2* in the peer section for more information.

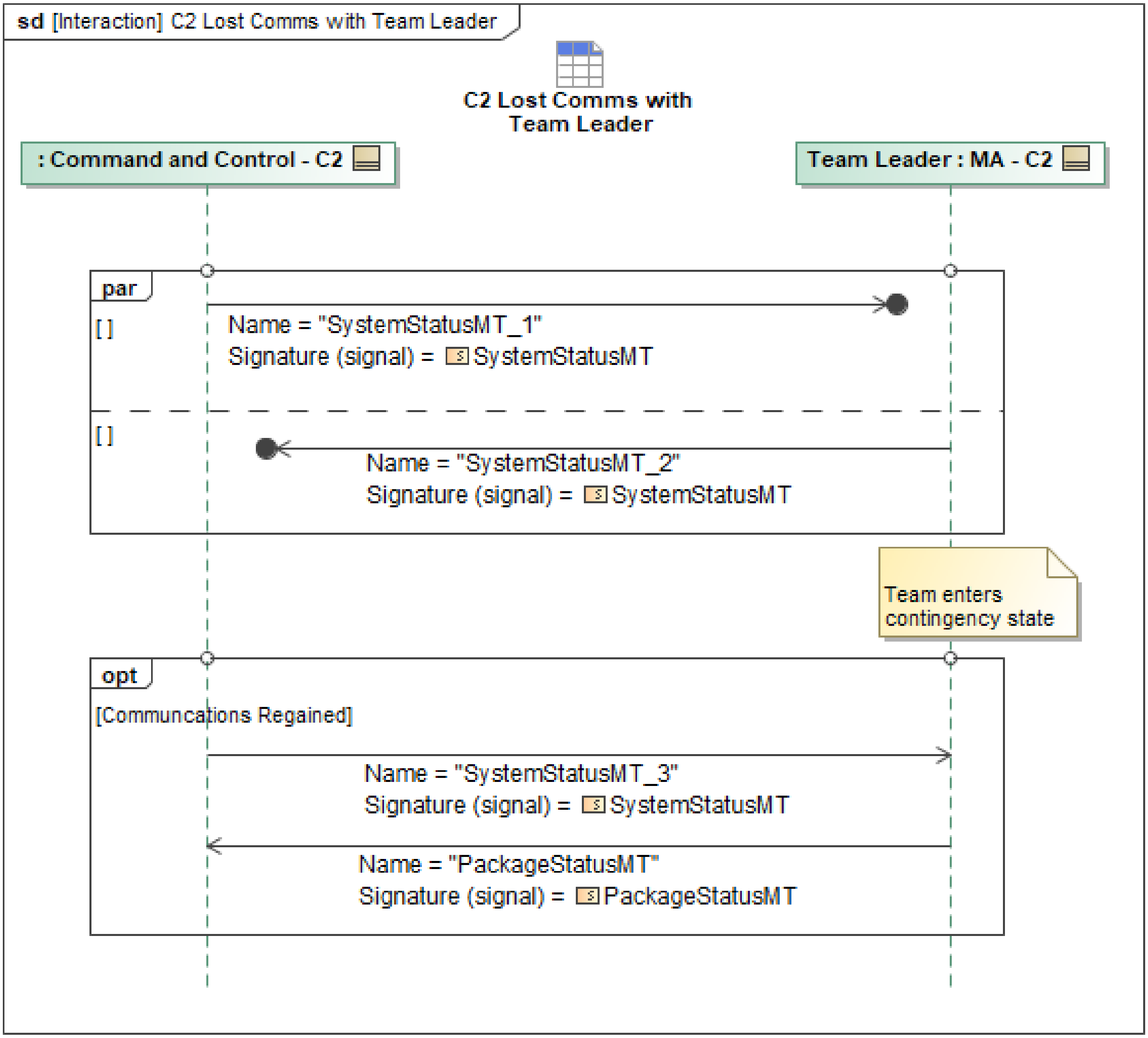


Figure 1-22: C2 Lost Comms with Team Leader

Table 1-21: C2 Lost Comms with Team Leader

Message Name (Name:Type)	Required Fields (Name:Type)
PackageStatusMT: PackageStatusMT	None
SystemStatusMT_1: SystemStatusMT	None
SystemStatusMT_2: SystemStatusMT	None
SystemStatusMT_3: SystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set

Tables section)

1.2.4.12.2 Define Team Lost Comms Contingency

For C2 to define a contingency for a team, a C2 system will send a *MA_SystemManagementRequestMT* to the Team Leader. The *ReferenceSystemID* field will be set to the system ID of the Team Leader. The *RequestType* will be *VehicleSettings*, with *VehicleSettings.LostCommTimeout* set to the maximum time allowed between *SystemStatusMT*'s before triggering a lost comms contingency. The Team Leader will then communicate the contingency information to its team, see section *Define Teams Lost Comms Contingency* in the Peer Section for more information. Once the Team Leader has gotten confirmation that all team members have been configured, it will send a *MA_SystemManagementRequestStatusMT* back to C2. The *MA_SystemManagementRequestStatusMT* will have the *RequestProcessingState* field set to "*COMPLETED".* If the *MA_SystemManagementRequestMT* was not able to be applied to the entire team, then the Team Leader will send C2 systems a *MA_SystemManagementRequestStatusMT* with the *RequestProcessingState* set to "*FAILED*" and the *RequestProcessingStateReason* set to the reason the *MA_SystemManagementRequestMT* couldn't be applied. Additionally, a contingency could also be triggered by an alert generated by the Comms Manager when it detects a communication outage. See section *Mission Systems* for more information on the Comms Manager.

C2 will then continue to communicate with the Team Leader, following the contingency set up as described in section *Define Contingency Plan.* The Team Leader will route *ResponsePlanMTs* and other contingency messages as needed. See section *Define Team Lost Comms Contingency* in the Peer section for more information on how a Team Leader defines a contingency for its team members.

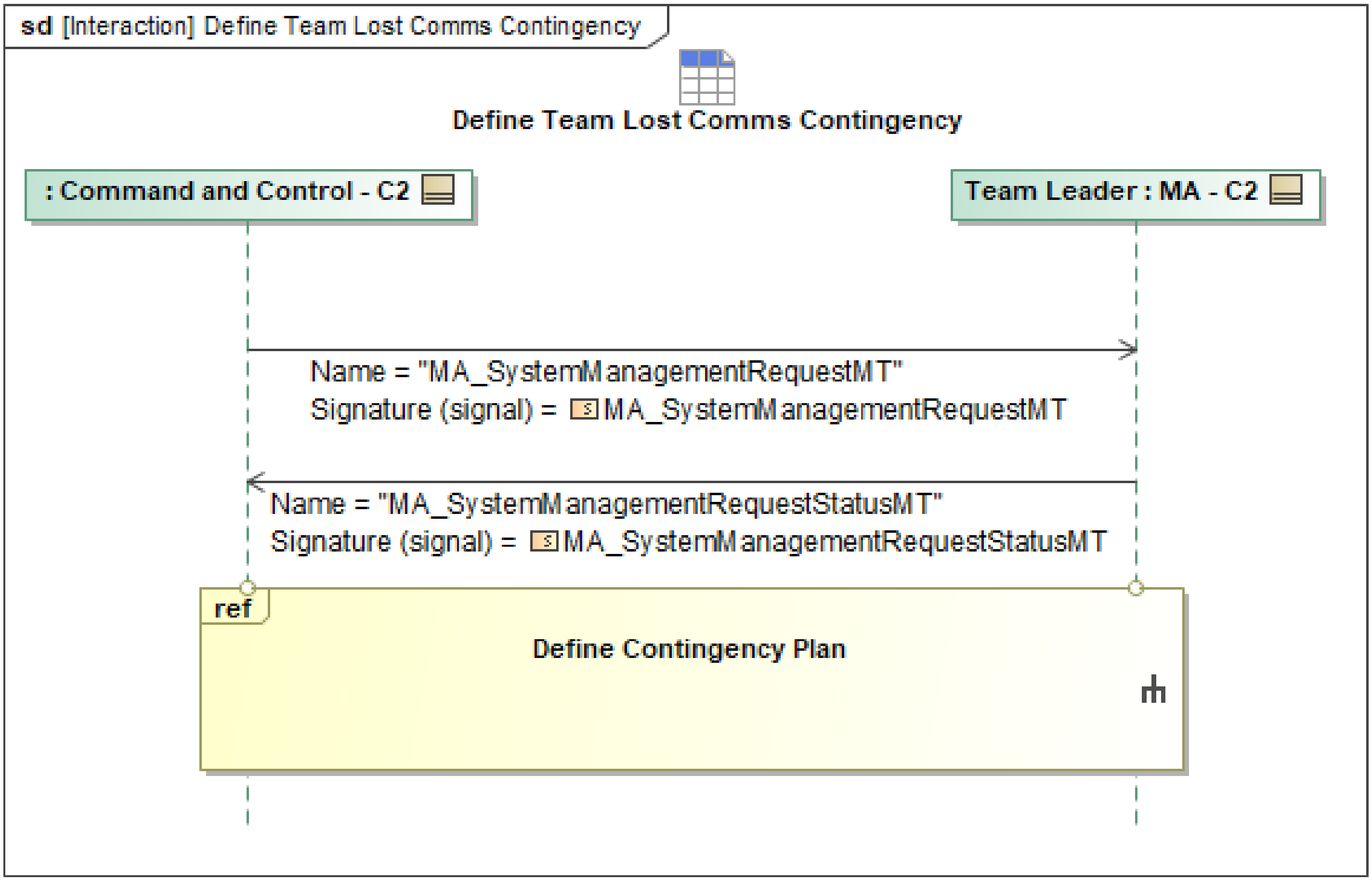


Figure 1-23: Define Team Lost Comms Contingency

Table 1-22: Define Team Lost Comms Contingency

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemManagementRequestMT: MA_SystemManagementRequestMT	LostCommTimeout: DurationType
MA_SystemManagementRequestStatusMT: MA_SystemManagementRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.12.3 Team Leader Lost Comms with Multiple Team Members

In the event that multiple peers are disconnected from the Team Leader, the team will split into two separate teams. It is assumed that if both halves of the segmented team can communicate with the same C2 system, they will communicate via that C2 system so there will be no need to split the team. If a segment of a team can no longer communicate with its team leader or its original C2 system, but can connect to a new C2 system, then it is expected that the disconnected segment will begin communicating with that C2 system as described below. An example of such a situation is if a team experiences a geographical split, where one side loses communication with the rest of the team and C2, but there is another C2 node currently in range. If the disconnected segment cannot connect to any other C2 systems, it should enter a contingency state as described in section *C2 Lost Comms with Team Leader.*

If the side of the split team that has the Team Leader can contact a C2 system, it will notify C2 with a *MissionContingencyAlertMT* followed by sending each *PackageStatusMT* periodically which will give C2 an update on the team as a whole. In the event that Team Leader cannot communicate with C2, see section *C2 Lost Comms With Team Leader.* The *MissionContingencyAlertMT* will have a *MissionContingencyAlertID* field set to the alert ID generated for this contingency instance. The *SourceSystemID* will be set to the system ID of the Team Leader. The *MissionContingencyAlertMT* will have a *ContingencyCondition* element for each team member it can no longer communicate with. The field *ContingencyCondition.ConflictedSystemID* will correspond to the system whose lost communication is being reported. The field *ContingencyCondition.ConflictState* will be set to "*CONFLICTED*" to indicate that the contingency is still ongoing. The field *ContingencyCondition.Conflict.ConflictID* will hold an ID for this particular contingency. The *ContingencyCondition.Trigger.Source* should be set to "*AUTONOMOUS_TRIGGER*" to indicate this alert was generated from MA. The *ContingencyCondition.Trigger.ByCaseTrigger* should be set to "*CommsLost*". The *ContingencyCondition.AutonomousActionStatus* should be set to *AlertOnly*. The *PackageStatusMT* will have a *PackageID* field to specify which package this status update pertains to. It will also set the *PackageState* field to "*DEGRADED*" indicating there has been an issue within the package, but it is still operating with some functionality. The *PackageSource* field will be set to "*ACTUAL*", indicating that there is a known issue with the package. There will also be *PackagePartnerStatus* element for each peer that is part of the team. Each *PackagePartnerStatus* element will have the *SystemID* field set to the system ID of the team member that they are describing. For fully operational peers, the *PackagePartnerStatus.PartnerState* will be set to "*OPERATIONAL*" (indicating the system is healthy) and the *PackagePartnerState.PartnerSource* will be set to "*ACTUAL*", meaning the Team Leader has up to date information that this system is healthy. For team members in the package experiencing lost communications, the *PackagePartnerStatus.PartnerState* field will be set to "*FAILED*", indicating that the lost team members are not expected to rejoin this team if they are able to regain communication. The *PackagePartnerStatus.PartnerSource* field will be set to "*ESTIMATED*" as the Team Leader does not have up to date information about the state of the peer experiencing the lost communications.

After notifying C2 of the systems that lost communication, the Team Leader will send a *PackageMT_1* to indicate to C2 that the team's makeup has been altered and that this message reflects the currently operational team members. The *PackageID* will be set to the original team's package ID, as the side of the split with the Team Leader will retain the original team's package ID. The *LeadSystemID* field will be set to the system ID of the current Team Leader. The *PackageMT_1* will only contain *PackagePartner* elements for the team members that the Team Leader can still communicate with. The *PackagePartner.SystemID* should be set to the system ID of each member in this team and can list as many *PackagePartner.CapabilityID* elements as needed for the corresponding system ID. Right after notifying C2 of this change to the team configuration, the Team Leader will send a *PackageStatusMT_1* periodically with the fields filled out to showcase a fully operational team for this package ID.

The side of the split that doesn't have a Team Leader will immediately go about the process of electing a new Leader (see section Create a Full Package Amongst Peers). If the new Team Leader can connect to a C2 system, it will send its own *PackageMT_2* to C2. This *PackageMT_2* will indicate the new makeup for this side of the split team. The *PackageMT_2* will have a new package ID set in the *PackageID* field, the newly elected Team Leader's system ID will be set in the *LeadSystemID* field, and there will be a *PackagePartner* element for each team member on this side of the split. The *PackagePartner.SystemID* should be set to the system ID of each member in this team, and can list as many *PackagePartner.CapabilityID* elements as needed for the corresponding system ID. After the new Team Leader has let C2 know that its team exists, it will give a status update by sending a *PackageStatusMT_2* to C2, indicating that all team members are operational. If the new Team Leader cannot establish communication with C2, it should behave as described in the section *C2 Lost comms with Team Leader.*

In the event that a split team is able to reconnect, team leads need to communicate status between each other and their respective C2 nodes. First, each team Leader sends the other Team Leader a system status notification using *SystemStatusMT_1* and *SystemStatusMT_2*, periodically and respectively. Then the two teams will each send an *OperatorNotificationMT* to let C2 know that they have regained communication with each other. The original team leader sends *OperatorNotificationMT_1* to the original C2, and the new team leader sends *OperatorNotificationMT_2* to the other C2. The *OperatorNotificationMT* will have an *OperatorNotificationID* set to the ID for this notification. The *Title* field will be set to a string that reads "*Split Team Regained Communication"*, the *Category* field will be set to "SYSTEM", the *Severity* field will be set to "*INFORMATIONAL*" and the *Source* field will be set to each Team Leader's system ID. An operator can then decide if it wants to keep them as two separate teams or merge them together after being notified. (*See Creating or Changing a Package* section for more information on managing teams).

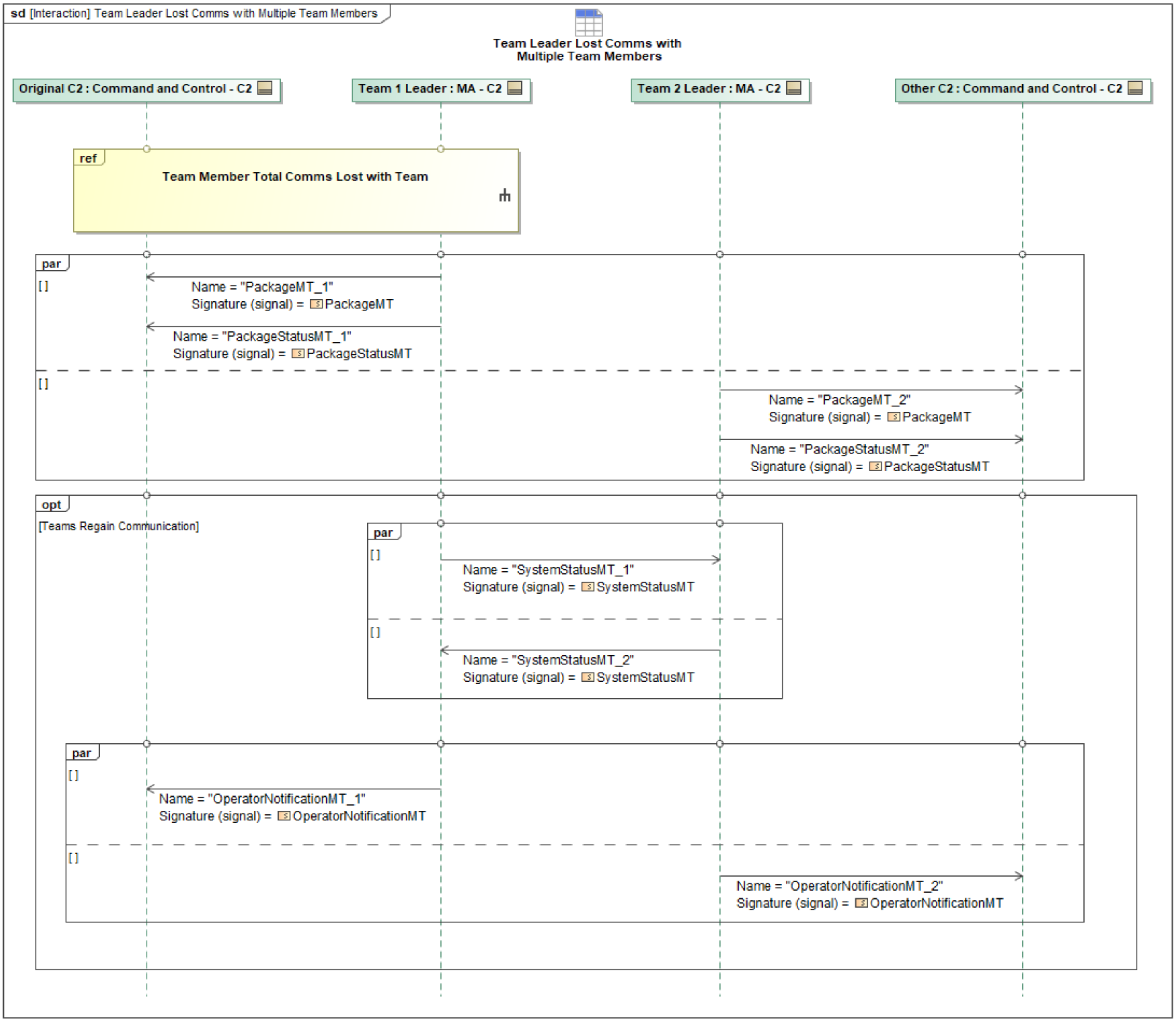


Figure 1-24: Team Leader Lost Comms with Multiple Team Members

Table 1-23: Team Leader Lost Comms with Multiple Team Members

Message Name (Name:Type)	Required Fields (Name:Type)
OperatorNotificationMT_1: OperatorNotificationMT	Source: NotificationSourceType
OperatorNotificationMT_2: OperatorNotificationMT	Source: NotificationSourceType
PackageMT_1: PackageMT	LeadSystemID: SystemID_Type
PackageMT_2: PackageMT	LeadSystemID: SystemID_Type
PackageStatusMT_1: PackageStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
PackageStatusMT_2: PackageStatusMT	None
SystemStatusMT_1: SystemStatusMT	None
SystemStatusMT_2: SystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.4.12.4 Team Member Total Comms Lost with Team

When the Team Leader loses communication with one of its team members, it will alert the C2 by sending a *MissionContingencyAlertMT_1* with the *MissionContingencyAlertID* field set to the alert ID generated for this contingency, the *Source.SystemID* field set to the system that detected the communications outage, the *ContingencyCondition.ConflictedSystemID* field set to the system ID of the system that has lost its communications, the *ContingencyCondition,ConflictState* field set to *"CONFLICTED"* to indicate that the contingency has not been resolved, the *ContingencyCondition.Conflict.ConflictID* field set to the ID for this contingency, the *ConflictState.Trigger.Source* field set to *"AUTONOMOUS_TRIGGER"* to indicated that this alert was generated from MA, the *ContingencyCondition.Trigger.ByCaseTrigger* field set to *"CommsLost",* the *ContingencyCondition.AutonomousActionStatus.AlertOnly.* It will also update the C2 on the package by sending a *PackageStatusMT_1* with the *PackageID* field set to the package that this status update pertains to, the *PackageState* field set to "*DEGRADED*" indicating there has been an issue within the package, but it is still operating with some functionality, the *PackageSource* field will be set to "*ACTUAL*" indicating that there is a known issue with the package. There will also be *PackagePartnerStatus* element for each peer that is part of the team. Each *PackagePartnerStatus* element will have the *SystemID* field set to the system ID of the team member that they are describing. For fully operational peers, the *PackagePartnerStatus.PartnerState* will be set to "*OPERATIONAL*" (indicating the system is healthy) and the *PackagePartnerState.PartnerSource* will be set to "*ACTUAL*", meaning the Team Leader has up to date information that this system is healthy. For team members in the package experiencing lost communications, the *PackagePartnerStatus.PartnerState* field will be set to either "*DEGRADED*" if the Team Leader thinks it will regain communications or "*FAILED*" if it thinks the communication loss is permanent. In either case, the *PackagePartnerStatus.PartnerSource* field will be set for "*ESTIMATED*" as the Team Leader does not have up to date information about the state of the peer experiencing the lost communications.

If the team member is able to communicate again, the Team Leader will send the C2 a *MissionContingencyAlertMT_2* with all the same field values except the *ContingencyCondition.ConflictState* field set to *"CLEARED"* to indicate that the contingency has been resolved. The Team Leader will also send the C2 an updated *PackageStatusMT_2* with the *PackagePartnerStatus.PartnerState* field that was previously set to *"DEGRADED"* or *"FAILED"* for this team member set to "*OPERATIONAL*".

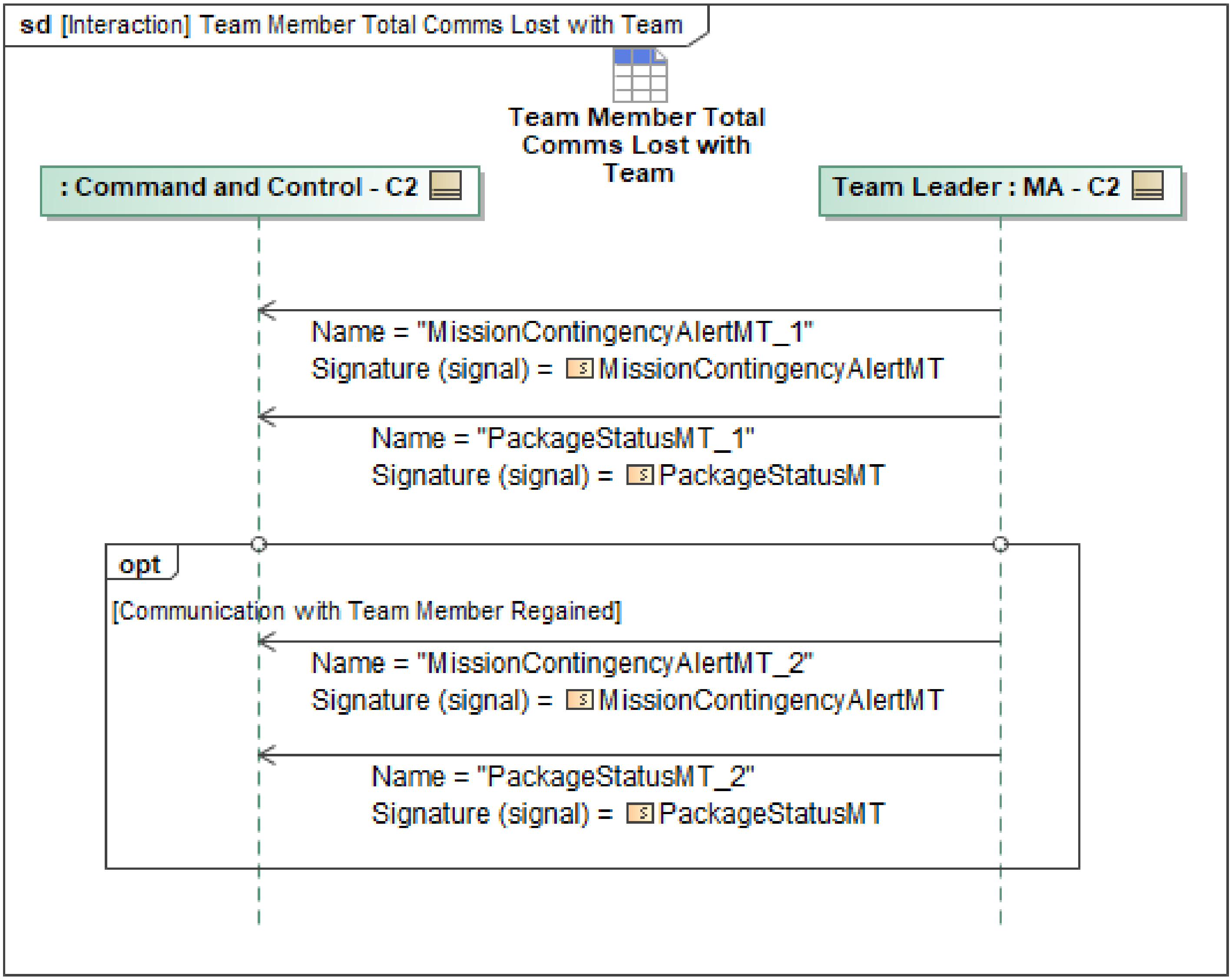


Figure 1-25: Team Member Total Comms Lost with Team

Table 1-24: Team Member Total Comms Lost with Team

Message Name (Name:Type)	Required Fields (Name:Type)
MissionContingencyAlertMT_1: MissionContingencyAlertMT	ByCaseTrigger: PlanningByCaseTriggerType
MissionContingencyAlertMT_2: MissionContingencyAlertMT	ByCaseTrigger: PlanningByCaseTriggerType
PackageStatusMT_1: PackageStatusMT	None
PackageStatusMT_2: PackageStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5 C2 COP Behaviors Package

1.2.5.1 C2 Requests the COP Configuration Settings

To request an update of the current configuration settings, C2 may send *MA_COP_ConfigurationSettingsRequestMT*, requesting a status update on the COP Configuration Settings from one or more packages. While processing the request, the COP Lead may optionally send a *MA_COP_ConfigurationSettingsRequestStatusMT_1* with *RequestProcessingState* set to *PROCESSING*. Once the request is completed, the COP Lead will send *MA_COP_ConfigurationSettingsRequestStatusMT_2* with the *RequestProcessingState* set to *COMPLETED* and the current settings in the ConfigurationSettings. If unable to complete the command, the COP Lead will send *MA_COP_ConfigurationSettingsRequestStatusMT_3* with the *RequestProcessingState* set to *FAILED*.

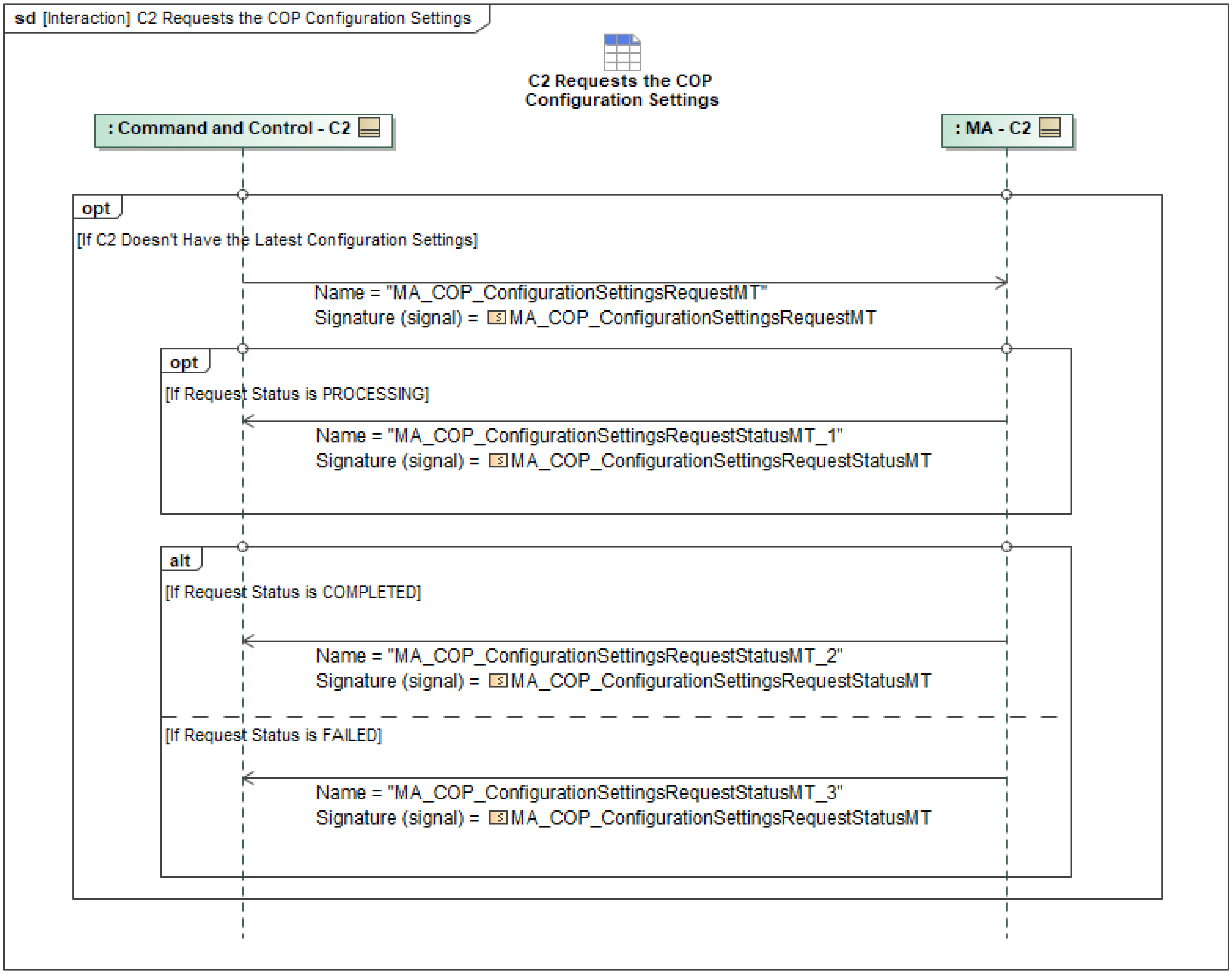


Figure 1-26: C2 Requests the COP Configuration Settings

Table 1-25: C2 Requests the COP Configuration Settings

Message Name (Name:Type)	Required Fields (Name:Type)
MA_COP_ConfigurationSettingsRequestMT: MA_COP_ConfigurationSettingsRequestMT	None
MA_COP_ConfigurationSettingsRequestStatusMT_1: MA_COP_ConfigurationSettingsRequestStatusMT	None
MA_COP_ConfigurationSettingsRequestStatusMT_2: MA_COP_ConfigurationSettingsRequestStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_COP_ConfigurationSettingsRequestStatusMT_3: MA_COP_ConfigurationSettingsRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5.2 C2 Updates the COP Configuration Settings

This interaction describes the process of C2 configuring the Common Operating Picture (COP). If C2 configures or updates the COP configuration settings, the *MA_COP_ConfigurationSettingsCommandMT* will be sent to MA on the Recipient ACPs, with the *PackageID*, *COP_LeaderID*, and *RequestAuthority* field set to *ORDER*, to update their COP configuration settings accordingly. To update a Configuration item, set *ConfigurationID* to the UUID of the item to be updated. To add a Configuration item, the *ConfigurationID* is optional. To delete a Configuration item, set the *ConfigurationID* to the UUID of the item to be removed and omit the remaining fields in the Configuration. Once received, MA on the recipient ACP can optionally respond with *MA_COP_ConfigurationSettingsCommandStatusMT* with *CommandProcessingState* set to *RECEIVED*. If both C2 and the Lead ACP send *MA_COP_ConfigurationSettingsCommandMT* to Recipient ACPs, the C2 command will be prioritized over the Lead ACP command. If the command is completed, the COP Lead will send *MA_COP_ConfigurationSettingsCommandStatusMT* with *CommandProcessingState* set to *ACCEPTED* and *MA_COP_ConfigurationSettingsMT* with the updated settings. If unable to execute the command, the COP Lead will send *MA_COP_ConfigurationSettingsCommandStatusMT* with *CommandProcessingState* set to *REJECTED*.

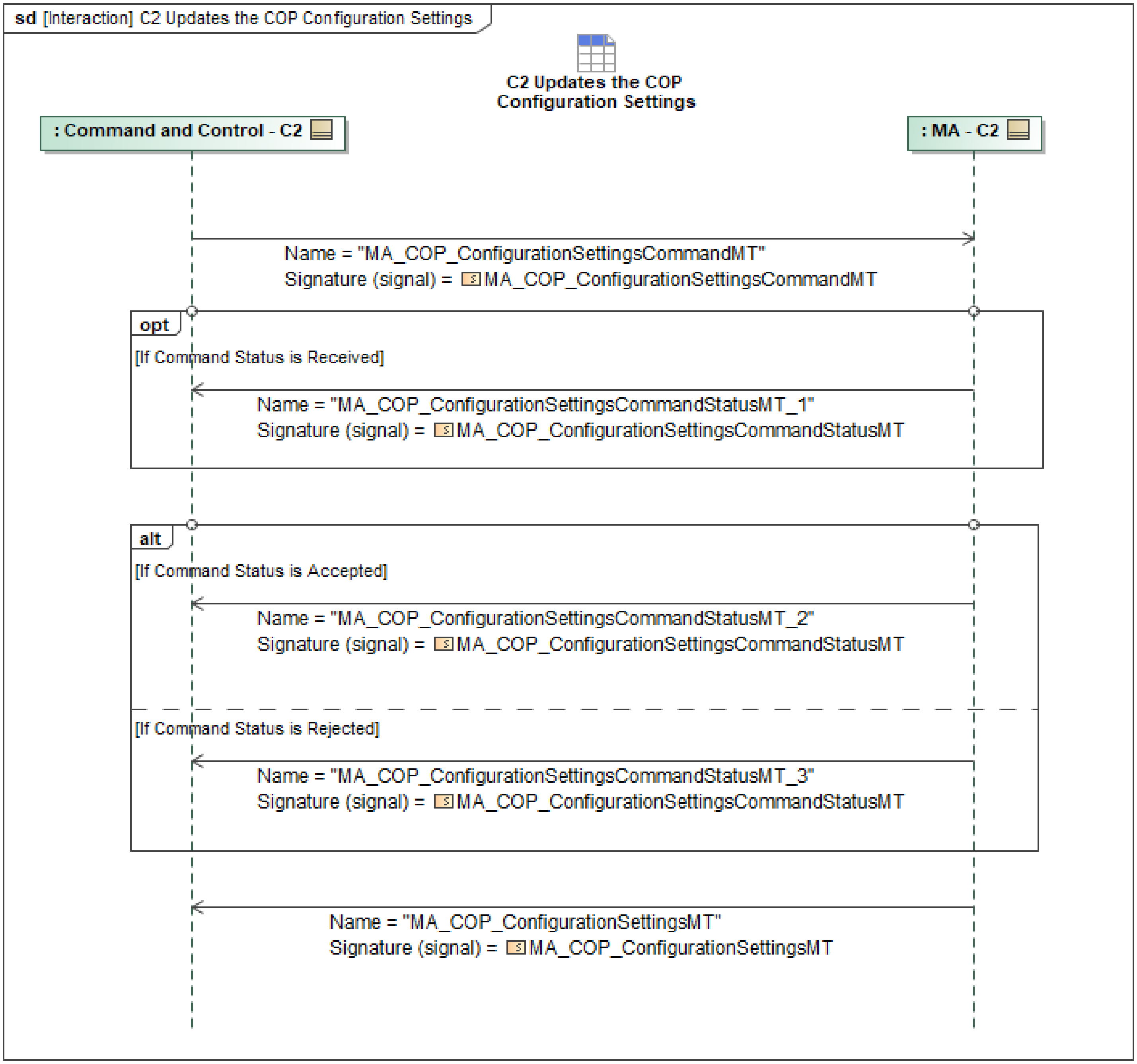


Figure 1-27: C2 Updates the COP Configuration Settings

Table 1-26: C2 Updates the COP Configuration Settings

Message Name (Name:Type)	Required Fields (Name:Type)
MA_COP_ConfigurationSettingsCommandMT: MA_COP_ConfigurationSettingsCommandMT	None
MA_COP_ConfigurationSettingsCommandStatusMT_1: MA_COP_ConfigurationSettingsCommandStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_COP_ConfigurationSettingsCommandStatusMT_2: MA_COP_ConfigurationSettingsCommandStatusMT	None
MA_COP_ConfigurationSettingsCommandStatusMT_3: MA_COP_ConfigurationSettingsCommandStatusMT	CommandProcessingStateReason: CannotComplyType
MA_COP_ConfigurationSettingsMT: MA_COP_ConfigurationSettingsMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.5.3 Synchronize Global COP to C2

The Synchronize Global COP to C2 interaction supports the use case in which the COP Leader has generated a new or updated Global COP and needs to distribute the data to C2. The output of the Global COP includes updated Entity data and updated Team States. This information needs to be synchronized to C2 each time the Global COP is updated. The COP Leader sends *EntityMT* and *PackageStatusMT* to C2. *EntityMT* includes the *MeasurementID* and the *SourceTypeIdentifier* fields to provide context for the updated measurements and originating sources, respectively. MA sends the *PackageStatusMT* periodically which communicates the package status, as well as Team States of each member in the package using *PackagePartnerStatus*.

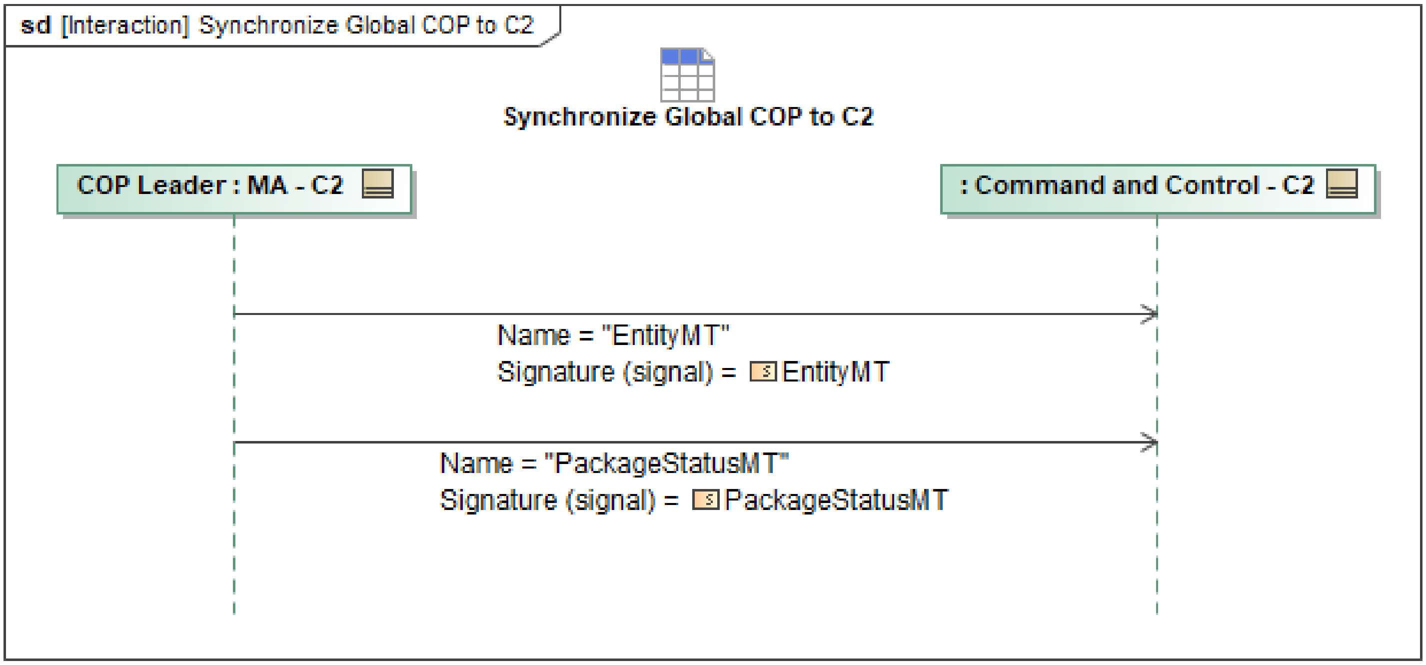


Figure 1-28: Synchronize Global COP to C2

Table 1-27: Synchronize Global COP to C2

Message Name (Name:Type)	Required Fields (Name:Type)
EntityMT: EntityMT	MeasurementID: MeasurementID_Type SourceTypeIdentifier: EntitySourceIdentifierType
PackageStatusMT: PackageStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.6 C2 Engagement Criteria Behaviors Package

The Engagement Criteria schema allows an authorized operator to set specific criteria that an ACP must use while engaging a Target. This includes defining where an ACP can engage a Target, how it can identify *EntityMTs*, what is considered a Target, what a Targeting Act is, who can authorize a Targeting Act against a Target, and how it can keep Custody of that Target. There can only be a single engagement criteria active at any given time. An engagement criteria message (*MA_EngagementCriteriaMDT*) can be activated by using the command message *MA_EngagementCriteriaCommandMDT*. If the *MA_EngagementCriteriaCommandMDT.ApplicableRegion* field is left empty, the engagement criteria specified is active everywhere. When the field is not empty, it restricts the engagement criteria to only be active in specific geozone(s). The Target and the actual weapons release must occur within that geozone(s). The ACP may start engagement activities while not inside the geozone(s), but cannot perform that actual weapons release outside of that area. For example, while the ACP is outside of the geozone(s), C2 can designate a Target and the Requirements can start to execute, but the Strike Activity must not fire on the Target unless both the ACP and Target are inside the geozone(s).

<u>Identity Matrix:</u>

The Engagement Criteria Identity Matrix (*MA_EngagementCriteriaMDT.IdMatrixCriteria*) can be used to classify the identity of *EntitiyMT* (*EntityMDT.Identity.Standard.StandardIdentity* field set to *FRIEND*, *NEUTRAL*, *SUSPECT*, *HOSTILE*, or *UNKNOWN* if unable to classify). If this field is not populated, then ACPs will use the Entity classified identity or populated by Fusion and not attempt to classify the identity (or confirm identity of entities). If populated, Fusion will apply the identity criteria combined with onboard/shared sensor data to identify/classify entities and possibly alter their identity as appropriate. Any identity updates will be shared via existing COP/fusion processes. An authorized C2 can classify the identity of *EntityMTs* which would take precedence.

<u>Targeting:</u>

Definitions:

- An "Enemy" is a track that lacks positive Friend markers and has been positively identified as an enemy by Enemy Electromagnetic Indication (EEI), by Visual Identification (VID), or other approved indications (*EntityMDT.Identity.Standard.StandardIdentity* is set to *SUSPECT*).
- A "Threat" is an Enemy that has shown hostile intent or demonstrated a hostile act, or other approved indications (*EntityMDT.Identity.Standard.StandardIdentity* is set to *HOSTILE*).
- A “Targeting Act” is any Requirement (e.g. *MA_TaskMT, ActionMT*, etc.) which requires approval under the *ApprovalPolicyMT* referenced by the *ApprovalAuthorityMT* referenced by *MA_EngagementCriteriaMDT.TargetAuthorityCriteria.ApprovalAuthorityID* or within the *ApprovalPolicyMT* referenced by *MA_EngagementCriteriaMDT.TargetAuthorityCriteria.ApprovalPolicyID*. The Requirements listed in this approval policy will typically include *EffectMTs* of type *ATTRIT* or *DEFEAT**, *ActionMTs* of type *ATTACK** or *TARGET*, *MA_TaskMTs* of type *STRIKE* or *TACTICAL_ORDER* and possibly other Requirements that may result in a hostile act against an enemy.
- A “Target” is a *TargetType* entity that has been designated for engagement via kinetic or non-kinetic means by a Target Authority and is the subject of a Targeting Act.
- A “Target Authority” is an operator/system/service listed in the *ApprovalAuthorityMT* message referenced by *MA_EngagementCriteriaMDT.TargetAuthorityCriteria.ApprovalAuthorityID* or *MA_EngagementCriteriaMDT.TargetAuthorityCriteria.AuthorizedOperatorRoles*.
- A "Self Promoted Target" is when MA uses the Engagement Criteria Target Designation Criteria (*MA_EngagementCriteriaMDT.TargetDesignationCriteria*) to trigger a *MA_DesignationRequestMT* to be sent to a Target Authority.

An ACP may execute a Targeting Act against an entity (thereby making that entity a Target) if and only if at least one of the following is true:

1. A Target Authority has sent a *MA_DesignationMT* message that designates the entity as a *TARGET*. MA may request such a designation message via a *MA_DesignationRequestMT*.
2. A Target Authority has commanded (i.e. via *MissionPlanActivationCommandMT*, *TaskCommandMT*, *ActionCommandMT*, etc.) a Targeting Act via direct C2.
3. A Target Authority approves a Targeting Act via a standard UCI *ApprovalRequestMT* / *ApprovalRequestStatusMT* interaction.

If a Target Authority has commanded a Targeting Act or has sent a *MA_DesignationMT* that triggers a *ResponseMT* that starts executing a Targeting Act, an *ApprovalRequestMT* / *ApprovalRequestStatusMT* interaction is not required even if the Targeting Authority linked *ApprovalPolicy* specifies it is required. This would only be needed if a C2 that is not a Target Authority commanded a Targeting Act or has sent a *MA_DesignationMT* that triggers a *ResponseMT* that starts executing a Targeting Act.

Target Custody:

The Engagement Criteria Target Custody Criteria (*MA_EngagementCriteriaMDT.TargetCustodyCriteria*) details how the ACP will ensure that it has custody of the target during the weapons engagement process. If this criteria/field is populated, the Strike Activity can use this to determine if it does or does not have custody of the Target. If custody is lost this would pause or cancel the Strike Activity on that Target. If the criteria specifies that the *EntityMT* must be *HOSTILE* and the Engagement Criteria Identity Matrix is populated, the Identity Matrix Criteria can be used to continually confirm that the *EntityMT* is *HOSTILE* given the provided criteria and not just the set *EntityMT* identity setting.

1.2.6.1 Activate Engagement Criteria

The Activate Engagement Criteria sequence diagram specifies the interaction of C2 commanding MA to activate one of its stored engagement criteria activation schema. First, C2 sends a *MA_EngagementCriteriaCommandMT* message to MA. The *AppliesToID* field can have multiple entries of *PackageID* or *SystemID*. This allows a C2 to send a Lead MA the Package ID (or multiple sub Package IDs) or a set of System IDs. The *MA_EngagementCriteriaCommandMT* message shall specify the *EngagementCriteriaID* field containing the ID of the Engagement Criteria message that C2 wishes to be active. Any new *MA_EngagementCriteriaCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA_EngagementCriteriaCommandMT* is sent with the *CommandedState* set to *"CANCEL"*, this will cancel the active command and active engagement criteria. In response to receiving the command, MA shall send *MA_EngagementCriteriaCommandStatusMT_1* to C2. The *CommandProcessingState* will be set to *"RECEIVED"* to indicate receipt of the command. For this status message and all other status messages (*MA_EngagementCriteriaCommandStatus* & *MA_EngagementCriteriaStatus*), the *AppliesToID* field shall be set to the Package ID if MA is the leader and has received a status message from all members of the Package ID, otherwise it shall be set to System ID(s) and/or sub Package ID(s) that the status message applies to. The MA shall then use the *EngagementCriteriaID* to query its onboard MP to retrieve the *MA_EngagementCriteriaMT*. If the MA is a non-package ACP, it shall send *MA_EngagementCriteriaCommandStatusMT_2* to C2 with the *CommandProcessingState* set to *"ACCEPTED"*. The MA will send *MA_EngagementCriteriaStatusMT_1* to C2 containing the *EngagementCriteriaID* and other configured data as needed. Optionally, if MA is the lead ACP, then MA shall message its follower ACPs with the *MA_EngagementCriteriaCommandMT* (See "Activate Peer Engagement Criteria"). After receiving the respective responses from the follower ACPs, MA (the lead ACP) would send C2 the *MA_EngagementCriteriaCommandStatusMT_3* with the *CommandProcessingState* set to *"ACCEPTED"* and the *MA_EngagementCriteriaStatusMT_1* both reflecting the *AppliesToID* as needed and other configured data.

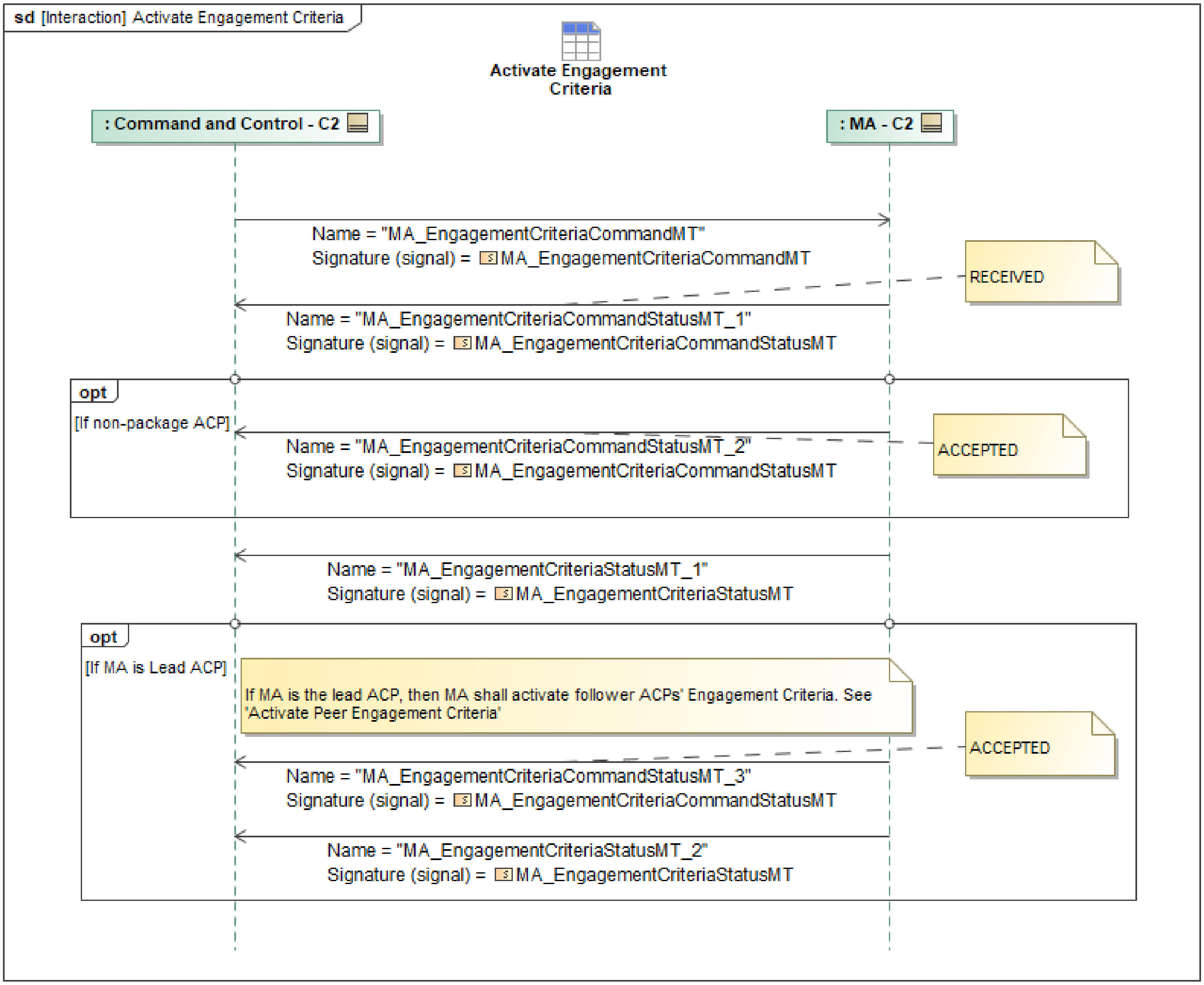


Figure 1-29: Activate Engagement Criteria

Table 1-28: Activate Engagement Criteria

Message Name (Name:Type)	Required Fields (Name:Type)
MA_EngagementCriteriaCommandMT: MA_EngagementCriteriaCommandMT	None
MA_EngagementCriteriaCommandStatusMT_1: MA_EngagementCriteriaCommandStatusMT	None
MA_EngagementCriteriaCommandStatusMT_2: MA_EngagementCriteriaCommandStatusMT	None
MA_EngagementCriteriaCommandStatusMT_3: MA_EngagementCriteriaCommandStatusMT	None
MA_EngagementCriteriaStatusMT_1: MA_EngagementCriteriaStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_EngagementCriteriaStatusMT_2: MA_EngagementCriteriaStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7 C2 Generic Behaviors Package

1.2.7.1 C2 Commanded Geozone

This diagram contains the collection of data messages used when C2 commands the state of any Op* message (e.g., *MA_OpZoneMT*, *MA_OpVolumeMT*, etc.) currently stored on MA, independent of mission plan. When the state of a given Op* message is "*ACTIVE*", the constraints imposed by said message must be considered by MA.

Upon receipt of an *MA_OpCommandMT*, MA shall store the command locally. The *MA_OpCommandMT* includes a reference to an existing Op* message (*OpID*), an indication of whether the command is to activate or de-activate (*CommandedState*), and any relevant conditions that must be met in order for the command to execute (*ConstraintsOverride*). Additionally, the *Ranking* field allows MA to resolve conflicting command messages through comparative ranking. MA acknowledges a received command by sending *MA_OpCommandStatusMT_1* with *CommandProcessingState* of "*RECEIVED*". MA then evaluates the conditions and ranking of the received *MA_OpCommandMT* and sends an *MA_OpCommandStatusMT_2* message with *CommandProcessingState* of "*ACCEPTED*" if successful. Otherwise, *MA_OpCommandStatusMT_3* is sent with the *CommandProcessingState* of "*REJECTED*", and the accompanying *CommandProcessingStateReason* populated with the reason for the rejection (e.g., "*RANKING*").

MA continuously monitors the conditions specified in ConstraintsOverride for all stored *MA_OpCommandMT* messages. If there are no conditions specified, or all conditions are met, MA changes the state of the referenced Op* message and sends an *MA_OpStatusMT* notifying C2 of the updated status (*CurrentState*) of the Op* message and the command that triggered the change (*CommandID*).

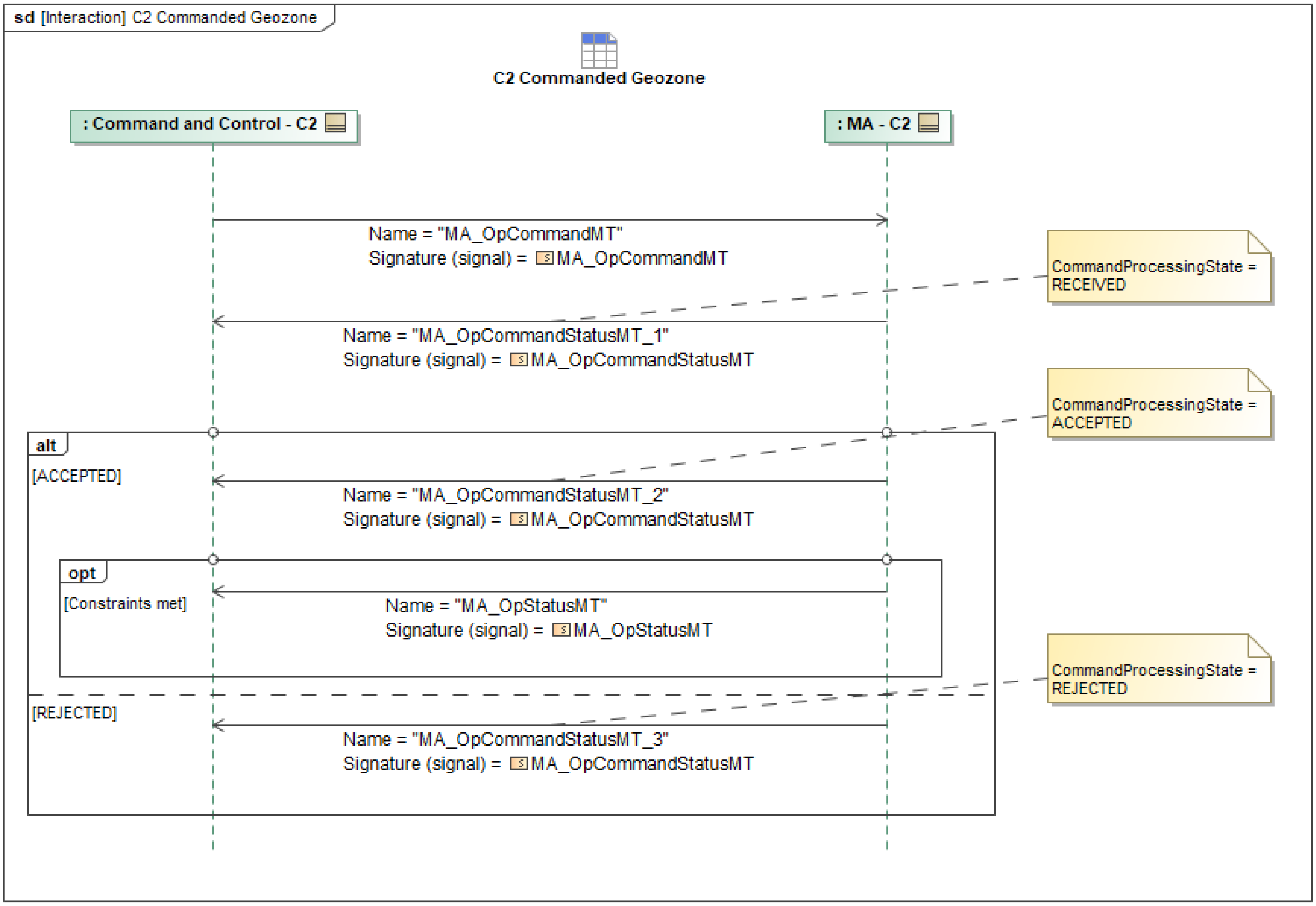


Figure 1-30: C2 Commanded Geozone

Table 1-29: C2 Commanded Geozone

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpCommandMT: MA_OpCommandMT	OpID: OpID_ChoiceType
MA_OpCommandStatusMT_1: MA_OpCommandStatusMT	None
MA_OpCommandStatusMT_2: MA_OpCommandStatusMT	None
MA_OpCommandStatusMT_3: MA_OpCommandStatusMT	CommandProcessingStateReason: CannotComplyType
MA_OpStatusMT: MA_OpStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.2 C2 Plan Approval Status

The C2 Plan Approval Status sequence diagram specifies the interaction of an ACP and the C2 authority where the ACP has received approval from C2 already for the plans indicated, and the ACP is responding with the confirmation that the plans have now been designated as approved. While the Approval Status specifically statuses the response of the Approval Request, the plan approval status messages are a status of the approval state of the plan itself, and they are utilized by the system to confirm individual plans approval or rejection status. As each executable plan is approved, the ACP sends the plan approval status to C2 via the *TaskPlanApprovalStatusMT, MissionPlanApprovalStatusMT, RoutePlanApprovalStatusMT*, and *ActivityPlanApprovalStatusMT* messages. For each of the messages sent, the *PlanID* and *PlanApprovalState* is populated with the applicable plan UUID and approval state, respectively.

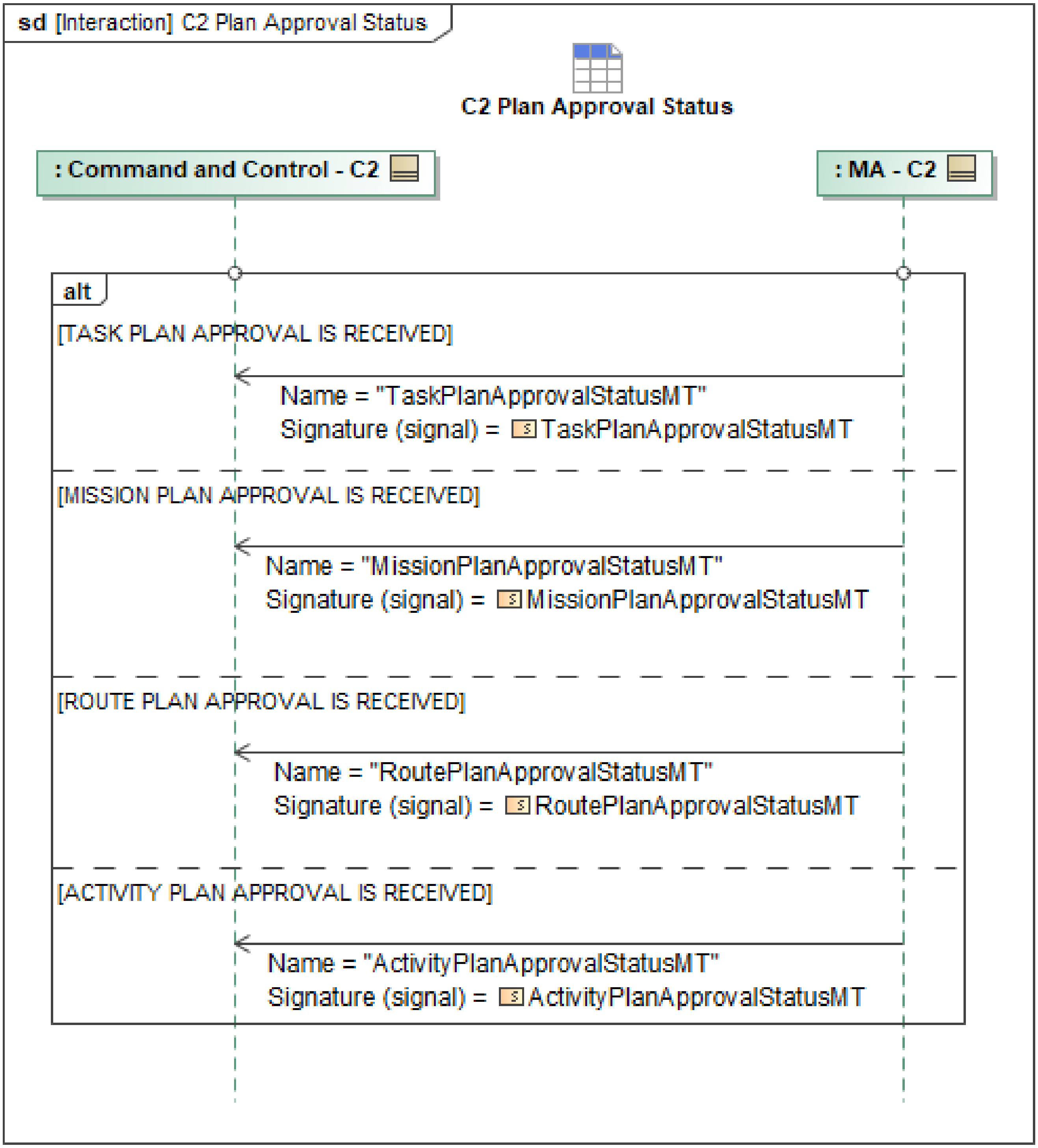


Figure 1-31: C2 Plan Approval Status

Table 1-30: C2 Plan Approval Status

Message Name (Name:Type)	Required Fields (Name:Type)
ActivityPlanApprovalStatusMT: ActivityPlanApprovalStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MissionPlanApprovalStatusMT: MissionPlanApprovalStatusMT	None
RoutePlanApprovalStatusMT: RoutePlanApprovalStatusMT	None
TaskPlanApprovalStatusMT: TaskPlanApprovalStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.3 C2 Publish Control Status

Flight Autonomy (FA), through the Vehicle Interface (VI), will periodically publish the *ControlStatusMT* message to Mission Autonomy (MA), indicating the current controlling system and service. MA sends *ControlStatusMT* periodically to relay the *PRIMARY_CONTROLLER and SECONDARY*_CONTROLLER designations to C2 for monitoring. This is depicted through the C2 Publish Control Status interaction, where the status is indicated by specifying through *ControlStatusMT/MessageData/ControlType/CapabilityControl/PrimaryController* and *ControlStatusMT/MessageData/ControlType/CapabilityControl/SecondaryController*.

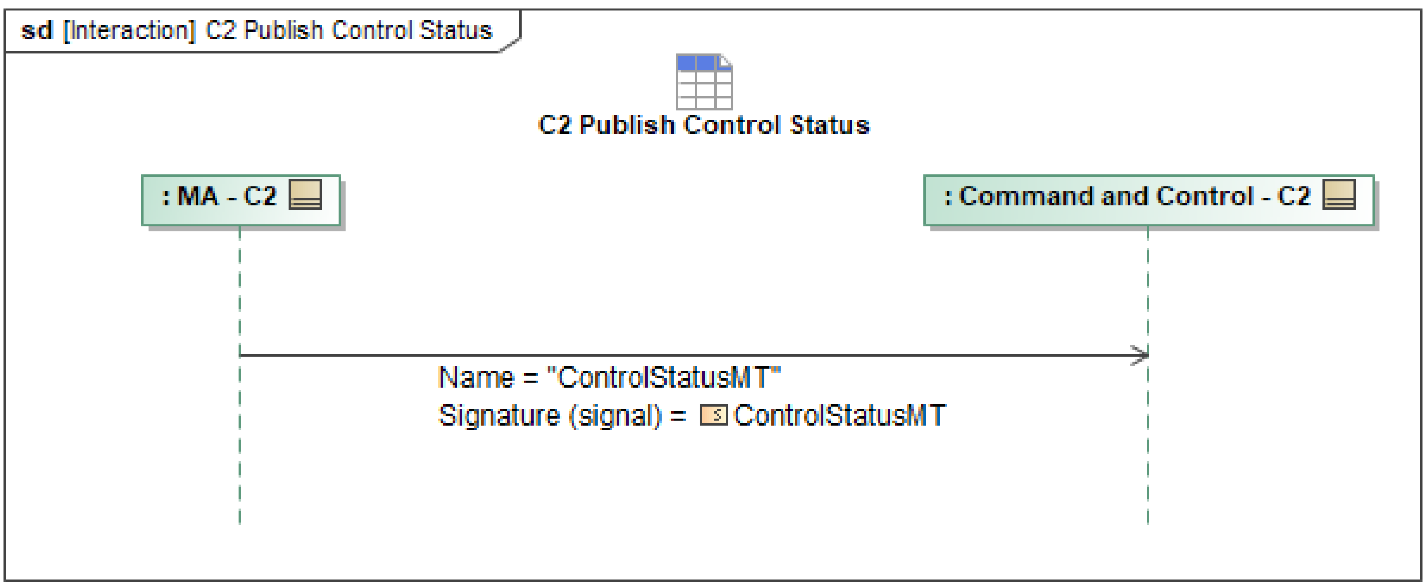


Figure 1-32: C2 Publish Control Status

Table 1-31: C2 Publish Control Status

Message Name (Name:Type)	Required Fields (Name:Type)
ControlStatusMT: ControlStatusMT	PrimaryController: PrimaryControllerType SecondaryController: SecondaryControllerType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.4 C2 Receives Geozone Status

This diagram contains the data message used when the state of any Op* message (e.g., *MA_OpZoneMT*, *MA_OpVolumeMT*, etc.) is changed. This may occur as a result of an *MA_OpCommandMT* or *MA_OpPlanMT* once the constraints on the message are met.

The *MA_OpStatusMT* shall include a reference to the Op* message that changed state (*OpID*), the updated state of the Op* message (*CurrentState*), and a reference to the command or plan that caused the state change (*CommandID*).

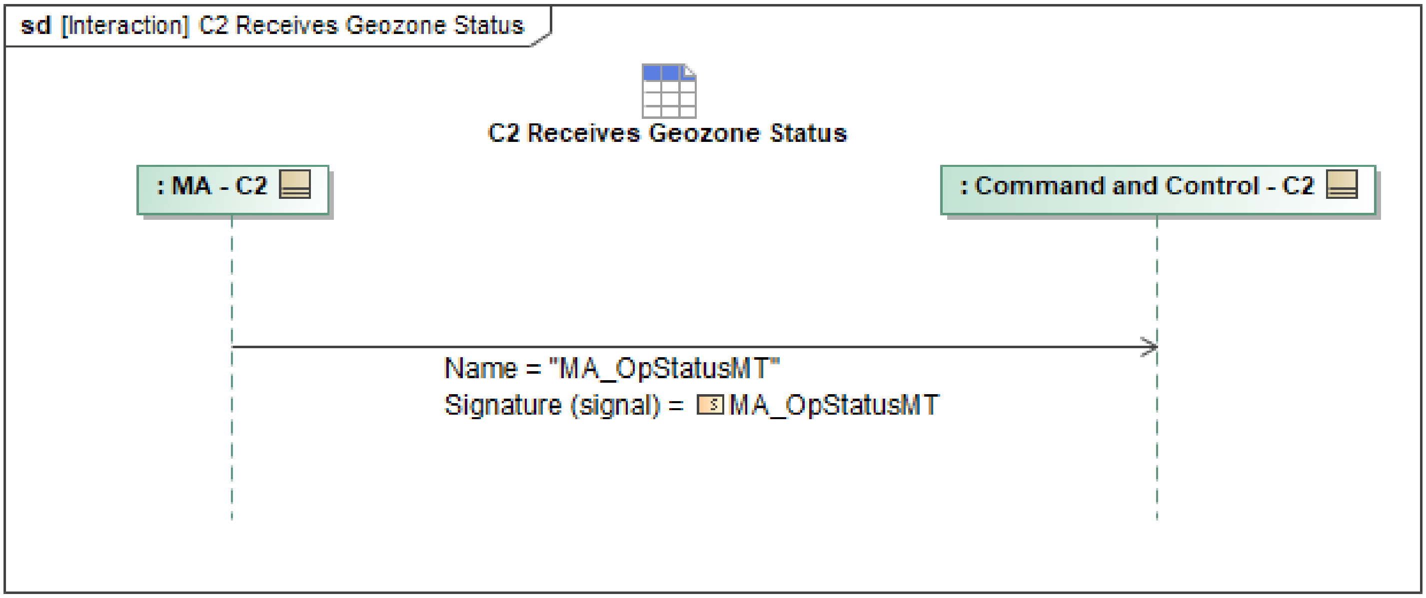


Figure 1-33: C2 Receives Geozone Status

Table 1-32: C2 Receives Geozone Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpStatusMT: MA_OpStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.5 C2 Receives Message Acknowledgment

C2 Acknowledgment Configurations sequences defines which messages C2 would expect MA to respond with an acknowledgment of receipt. If MA receives a configured message, MA will send a *MA_SystemNotificationMT* with the *NotificationState* set to *CONFIRMED*, *Source* set to its own SystemID, *Severity* set to *INFORMATIONAL*, *SystemSubjectIDs* set to its own SystemID, *SystemPerspective* set to *SOURCE*, and *AssociatedMessage* set to the *MessageType* of the received message and the *AssociatedID* of the applicable ID in the received message. The *MA_SystemNotificationMT* can also be sent at limited periodic rates for acknowledgments based on the Message Acknowledgment Configuration where it would group together all received *AssociatedMessages* over the periodic time that was configured. This is done by adding multiple entries to the *AssociatedMessage* field for each message. The *MA_SystemNotificationMT* would only be sent at that periodic rate if there was a received message during that time.

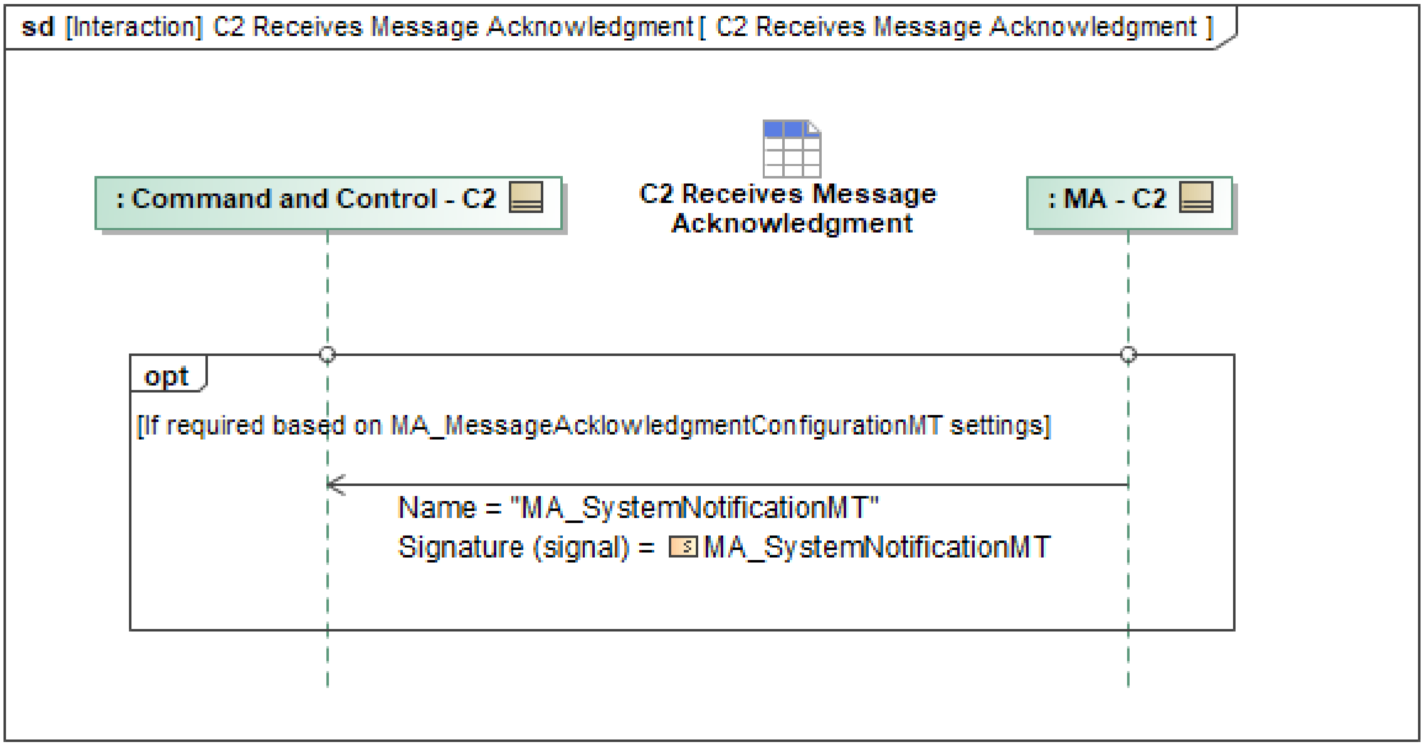


Figure 1-34: C2 Receives Message Acknowledgment

Table 1-33: C2 Receives Message Acknowledgment

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemNotificationMT: MA_SystemNotificationMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.6 C2 Sends Message Acknowledgment

C2 Acknowledgment Configurations sequences defines which messages MA would expect C2 to respond with an acknowledgment of receipt. If C2 receives a message which it is configured to acknowledge receipt of, C2 will send a *MA_SystemNotificationMT* with the *NotificationState* set to *CONFIRMED*, *Source* set to its own SystemID, Severity set to *INFORMATIONAL*, *SystemSubjectIDs* set to its own *SystemID*, SystemPerspective set to *SOURCE*, and *AssociatedMessage* set to the *MessageType* of the received message and the *AssociatedID* of the applicable ID in the received message. The *MA_SystemNotificationMT* can also be sent at limited periodic rates for acknowledgments based on the Message Acknowledgment Configuration where it would group together all received *AssociatedMessages* over the periodic time that was configured. This is done by adding multiple entries to the *AssociatedMessage* field for each message. The *MA_SystemNotificationMT* would only be sent at that periodic rate if there was a received message during that time.

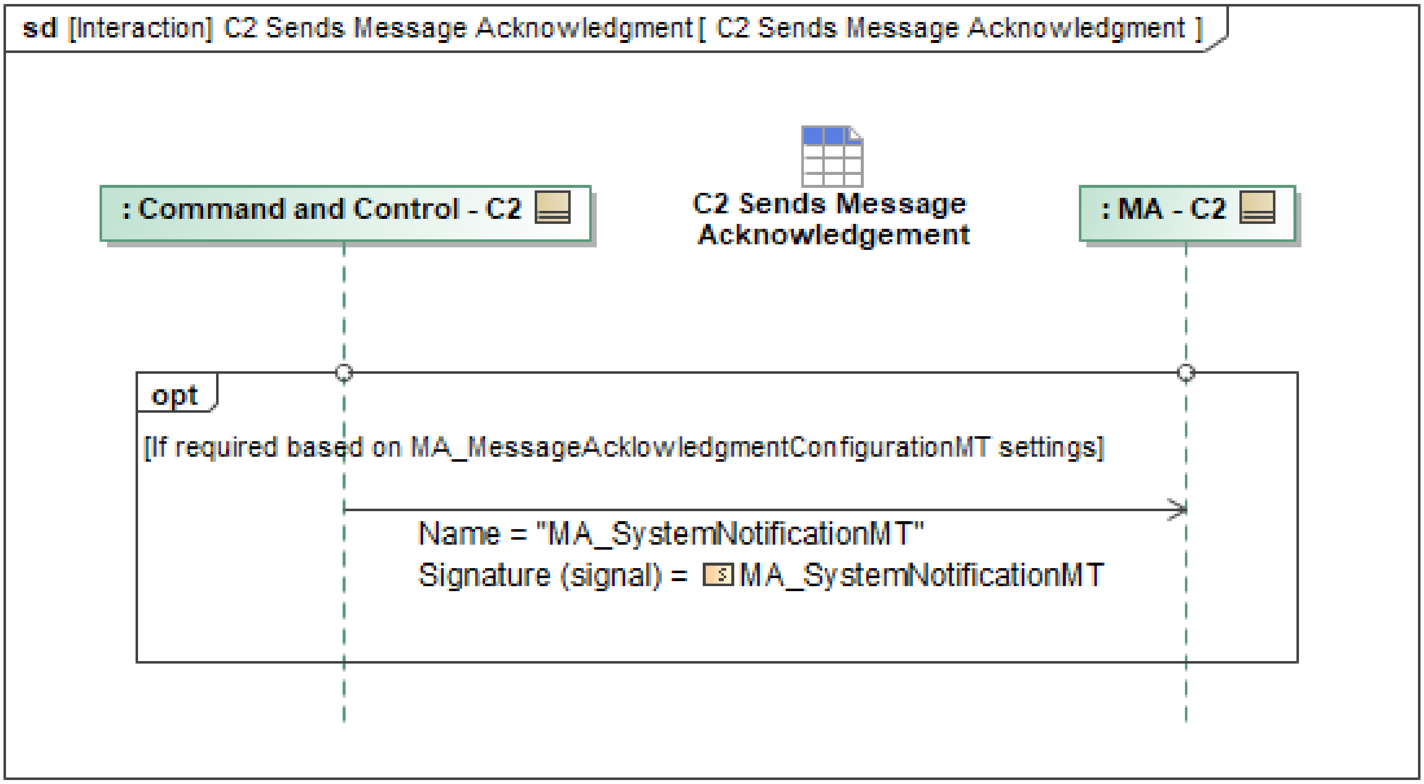


Figure 1-35: C2 Sends Message Acknowledgment

Table 1-34: C2 Sends Message Acknowledgment

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemNotificationMT: MA_SystemNotificationMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.7 Control Mode Authorization

C2 should be made aware of the capabilities that MA has been granted by Flight Autonomy. For information on the VI-MA interaction, see Control Authorization in the Vehicle Interface for more details. MA communicates with C2 how MA can control the vehicle as MA sends *MA_FlightCapabilityMT periodically*. The *MA_FlightCapabilityMT* message requires the *FlightCapabilityPerformanceProfile* field be populated with the appropriate subtypes of the capabilities advertised and any parameters need by MA for commanding each type present**.** MA sends the *MA_FlightCapabilityStatusMT* message periodically to communicate the availability of control modes. Both of these messages should be passed to C2 nodes when connecting to MA and *MA_FlightCapabilityStatusMT* should be forwarded along to C2 nodes whenever there is a change in flight capabilities.

Once the required capabilities are available, C2 can issue commands to the MA via the Direct HSA/CSA Command use case or the Direct Waypoint Command interaction. The *MA_FlightCommandMT* message within these use cases carries control mode-specific data, as described in later sections. ***MA_FlightCommandMT*** **is not used to set or command a formation on the C2–MA interface; a formation is planned and commanded via the Task path (*MA_TaskMT* and *MA_TaskCommandMT*).**

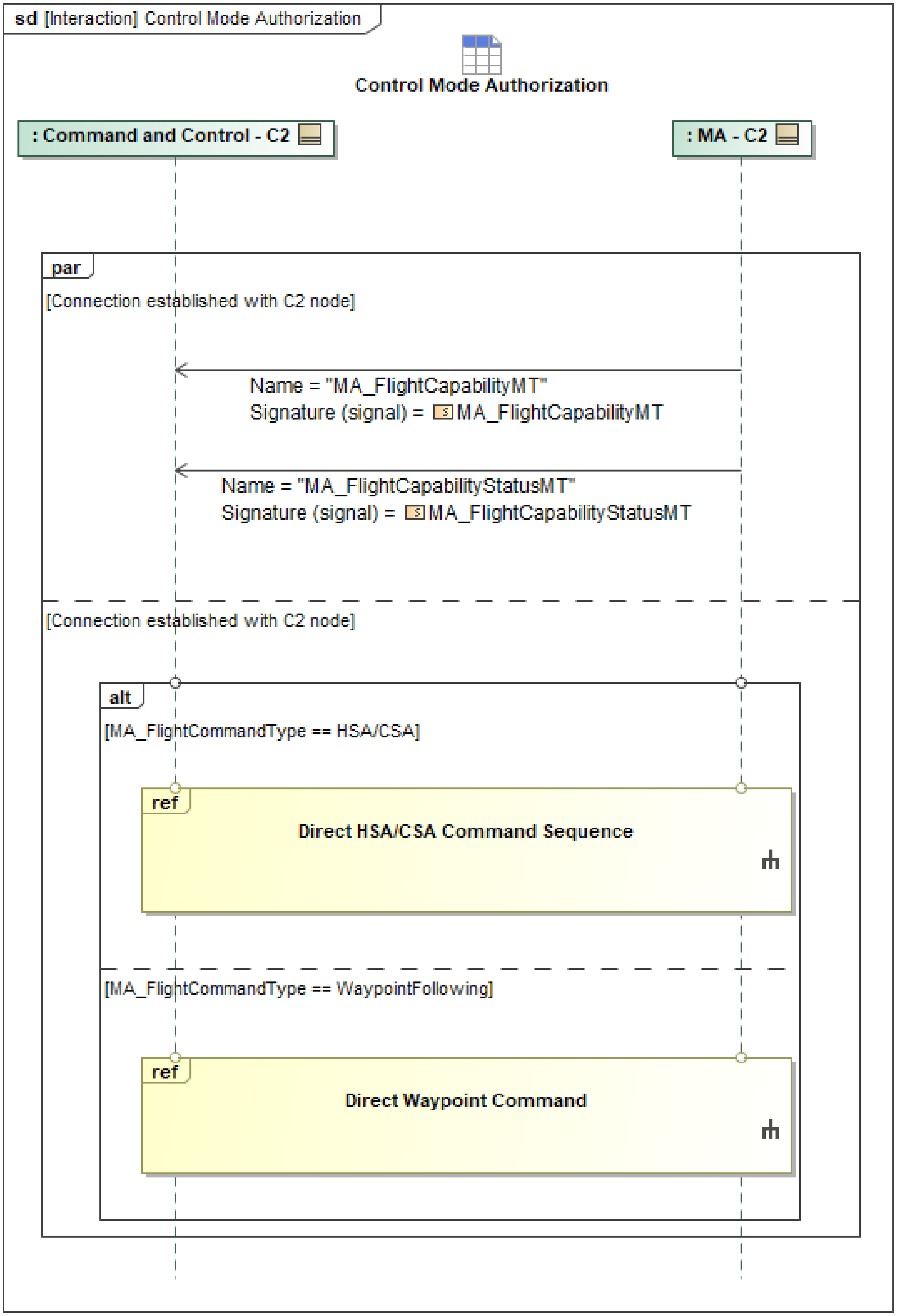


Figure 1-36: Control Mode Authorization

Table 1-35: Control Mode Authorization

Message Name (Name:Type)	Required Fields (Name:Type)
MA_FlightCapabilityMT: MA_FlightCapabilityMT	None
MA_FlightCapabilityStatusMT: MA_FlightCapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.8 Direct HSA/CSA Command Sequence

The C2 system can send heading/speed/altitude or course/speed/altitude (HSA/CSA) commands to MA via the *MA_FlightCommandMT* message. The C2 system sends *MA_FlightCommandMT* to command the system and invokes a previously advertised capability. Note: MA_FlightCommandMT is not used to set or command a formation on the C2–MA interface; a formation is planned and commanded via the Task path (MA_TaskMT and MA_TaskCommandMT). MA responds with *MA_FlightCommandStatusMT*, either approving or rejecting the command. If the command is *REJECTED*, the loop in the sequence will break and the *MA_FlightCommandStatusMT* will specify the *CannotComplyDetails* field to provide further details. VI may optionally specify a suggested flight task with adjusted parameters that constitute a "best-effort" using the *MA_TaskMT* message and Flight field. *MA_TaskMT* is used to encapsulate a suggested flight task from VI and gives VI the ability to define the parameters of the command that could be accepted. MA can restart the sequence with the VI's "best-effort" suggestion or choose an entirely different flight command. The break may also occur if the *Activity State* is *COMPLETED* or *FAILED*.

If the command is *ACCEPTED*, the VI will then begin publishing the *MA_FlightActivityMT* report, which includes data on the commanded state of the ownship, and the execution state of the command described initially in *MA_FlightCommandMT*. The cycle of *MA_FlightCommandMT > MA_FlightCommandStatusMT > MA_FlightActivityMT* repeats until either control is taken away from MA or MA relinquishes control.

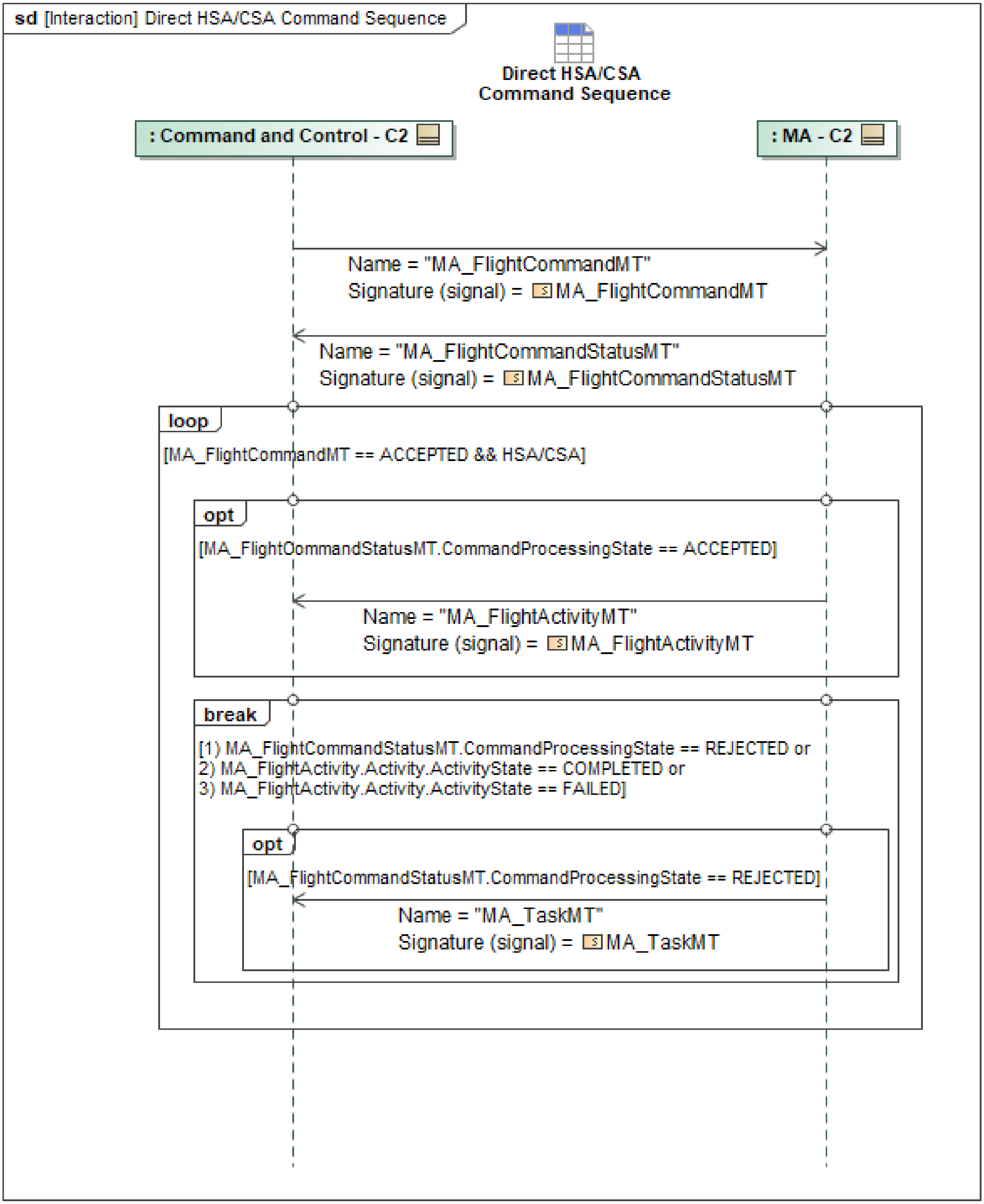


Figure 1-37: Direct HSA/CSA Command Sequence

Table 1-36: Direct HSA/CSA Command Sequence

Message Name (Name:Type)	Required Fields (Name:Type)
MA_FlightActivityMT: MA_FlightActivityMT	None
MA_FlightCommandMT: MA_FlightCommandMT	HSA_CSA: MA_HSA_CSA_Type
MA_FlightCommandStatusMT: MA_FlightCommandStatusMT	None
MA_TaskMT: MA_TaskMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.9 Direct Waypoint Command

For Direct C2, before the communication of new waypoints begins, MA needs to transmit to C2 its capabilities through the Capability Publication sequence so that MA can let C2 know what it can control. These capabilities can then be invoked via the *MA_FlightCommandMT* message to execute a waypoint command. Note: *MA_FlightCommandMT* is not used to set or command a formation on the C2–MA interface; a formation is planned and commanded via the Task path (*MA_TaskMT* and *MA_TaskCommandMT*). MA then responds with an *MA_FlightCommandStatusMT* accepting or rejecting the command. If the command is *REJECTED*, the loop in the sequence will break and the *MA_FlightCommandStatusMT* message will specify the *CannotComplyDetails* field to provide further details. VI may optionally specify a suggested flight task with adjusted parameters that constitute a "best-effort" using the *MA_TaskMT* message and *Flight* field. *MA_TaskMT* is used to encapsulate a suggested flight task from VI and gives VI the ability to define the parameters of the command that could be accepted. MA can restart the sequence with the VI's "best-effort" suggestion or choose an entirely different flight command. The break may also occur if the *Activity State* is *COMPLETED* or *FAILED*.

If the command is *ACCEPTED*, the VI will then begin publishing the *MA_FlightActivityMT* report, which includes data on the commanded state of the ownship, and the execution state of the command described initially in *MA_FlightCommandMT*. The cycle of *MA_FlightCommandMT* > *MA_FlightCommandStatusMT* > *MA_FlightActivityMT* repeats until either control is taken away from MA, or MA relinquishes control.

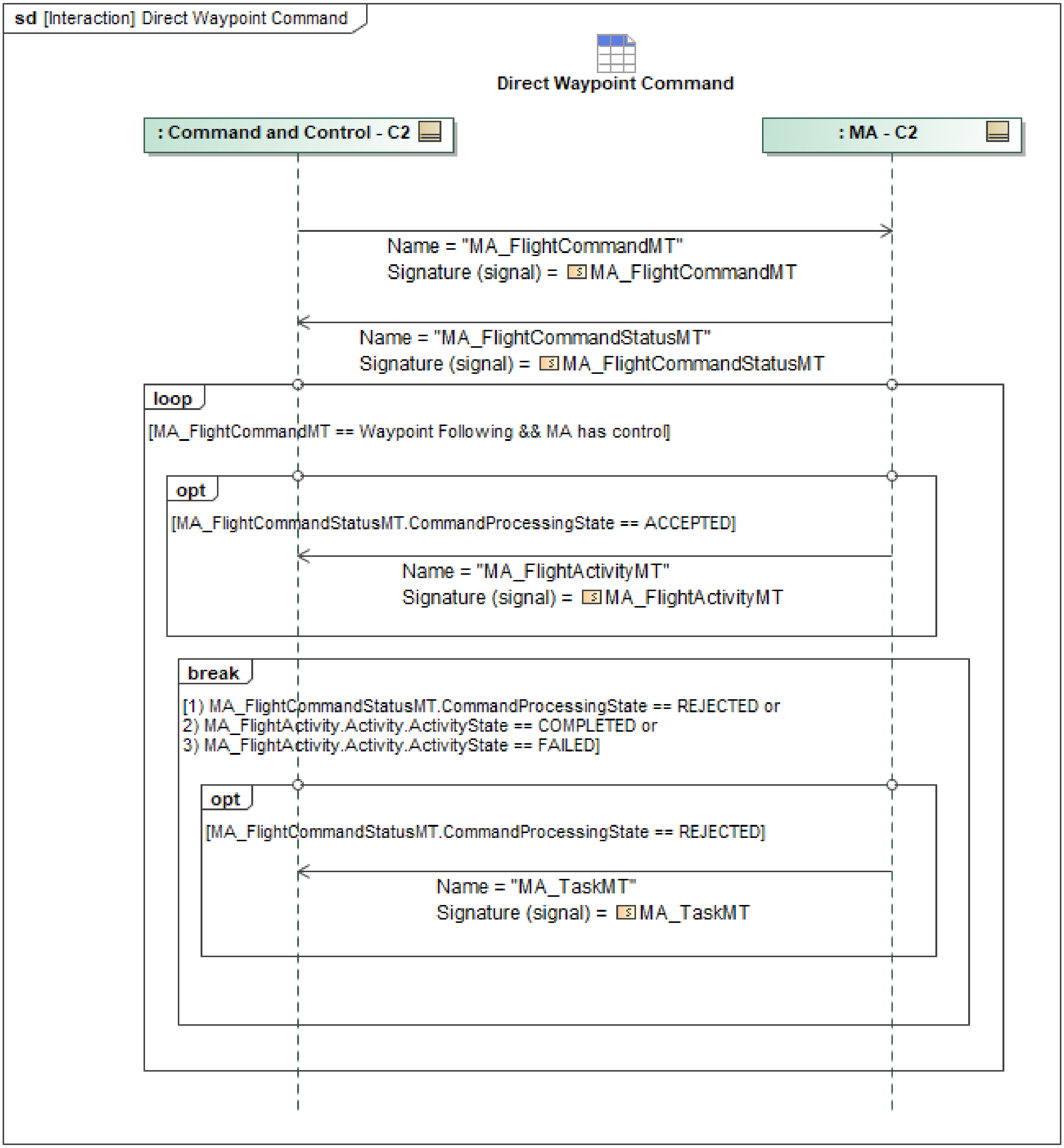


Figure 1-38: Direct Waypoint Command

Table 1-37: Direct Waypoint Command

Message Name (Name:Type)	Required Fields (Name:Type)
MA_FlightActivityMT: MA_FlightActivityMT	None
MA_FlightCommandMT: MA_FlightCommandMT	WaypointFollowing: MA_WaypointFollowingType
MA_FlightCommandStatusMT: MA_FlightCommandStatusMT	None
MA_TaskMT: MA_TaskMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.10 Planned HSA/CSA Command Sequence

To direct a vehicle to follow an HSA/CSA flight control mode via Planned C2, *MA_TaskMT* should be used with *FlightTaskType* populated to define the *HSA_CSA* flight control mode. C2 first sends out a *MA_MissionPlanMT*, followed by an *MA_TaskPlanMT* and *MA_TaskMT*.

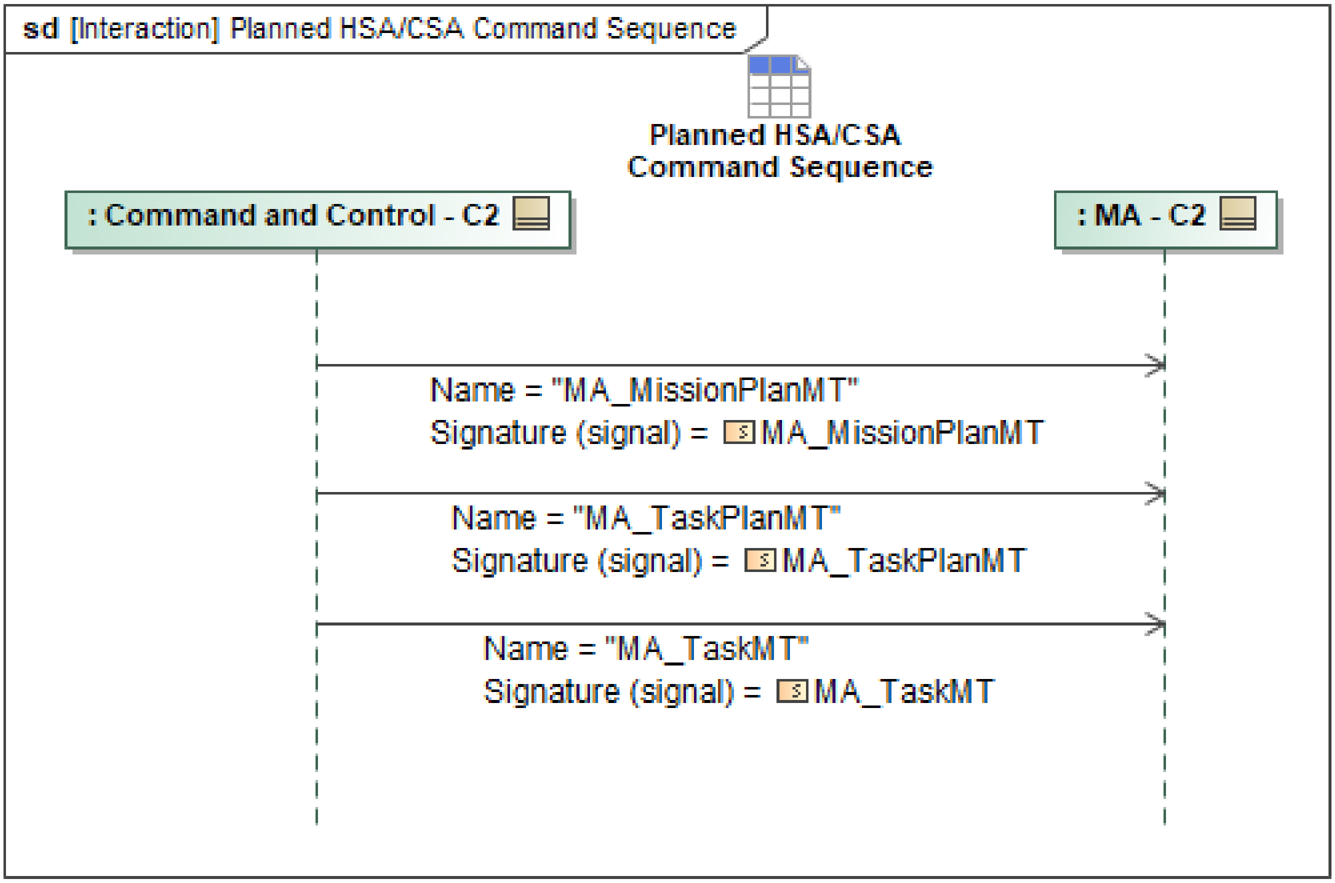


Figure 1-39: Planned HSA/CSA Command Sequence

Table 1-38: Planned HSA/CSA Command Sequence

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanMT: MA_MissionPlanMT	None
MA_TaskMT: MA_TaskMT	HSA_CSA: MA_HSA_CSA_Type
MA_TaskPlanMT: MA_TaskPlanMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.11 Planned Waypoint Command

Waypoint following, alternatively referred to as route following, is a control mode where a C2 node provides MA a series of waypoints describing a desired flight path. Waypoint behavior may include simple commands such as fly-by or fly-through; complex commands including payload action and loiter/hold; or bounded by constraints such as vehicle attitude or timing.

For Planned C2, system can upload multiple routes to MA before selecting one to execute. C2 starts by sending the *MA_MissionPlanMT* with the route referenced in the associated *SubPlans*. A well-conditioned *MA_RoutePlanMT* message will simply include an *MA_RoutePlanType* for MA to execute.

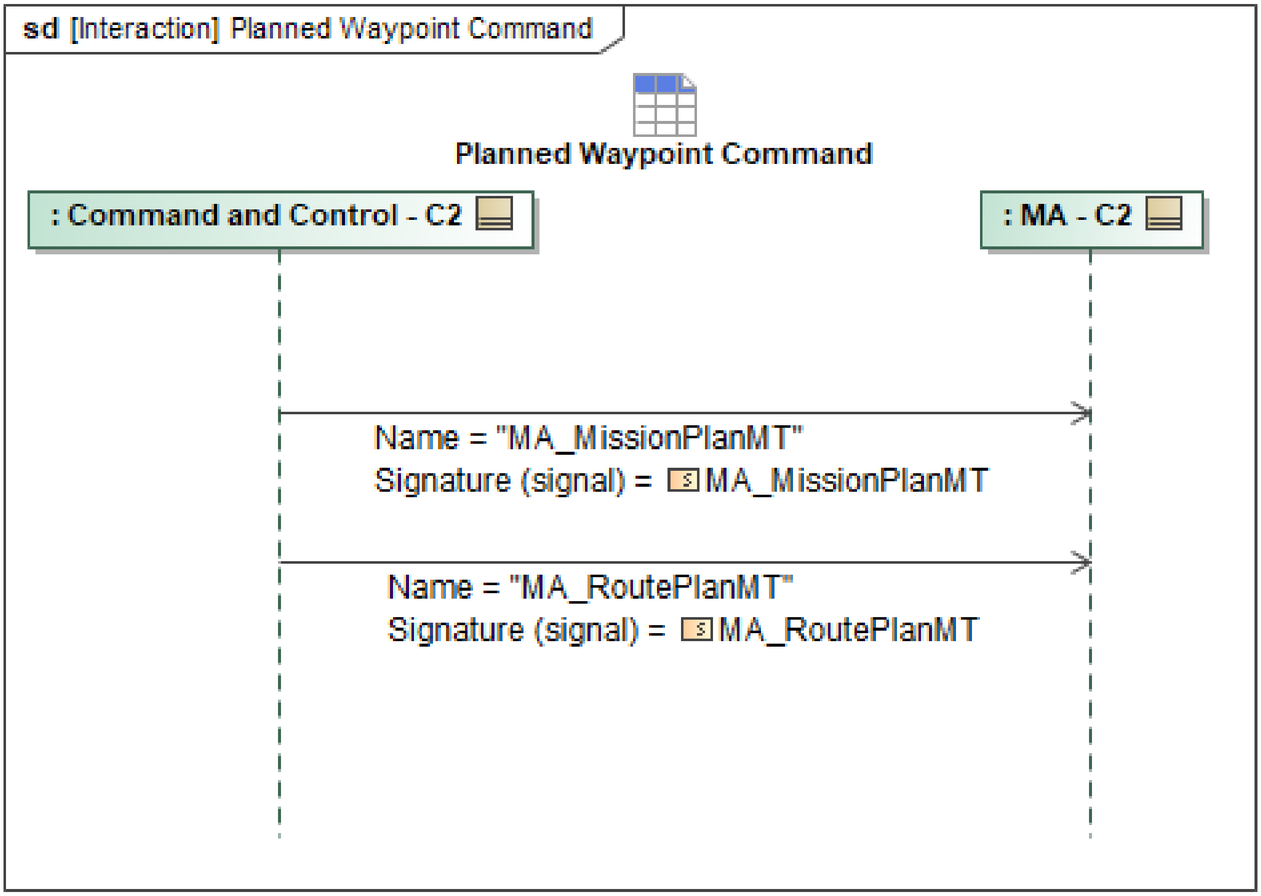


Figure 1-40: Planned Waypoint Command

Table 1-39: Planned Waypoint Command

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanMT: MA_MissionPlanMT	None
MA_RoutePlanMT: MA_RoutePlanMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.12 Receive Execution Status

MA will periodically provide updates to C2 on the status of actively executing or queued actions, tasks, plans, routes, and missions.

MA will send *MA_ActionStatusMT* and *MA_TaskStatusMT* periodically to convey the status of the ongoing Action or Task. Both *MA_ActionStatusMT* and *MA_TaskStatusMT* contain the required *ExecutionState* field denoting the executing state. When a task or action is allocated to an entire Package, the Lead ACP shall aggregate status reporting by populating the *SystemID* or *ExecutingSystemID* field within *MA_TaskStatusMT* and *MA_ActionStatusMT* messages with the *PackageID* UUID. This represents the collective state of the team. In an off-nominal condition where 1 or more Package members have an off-nominal *Status, the Lead ACP shall revert to sending individual *StatusMT messages strictly for the subset of platforms that failed.

If C2 allocates a requirement to a specific subset of platforms (via discrete *SystemID* allocations rather than *PackageID*), the Lead ACP shall bypass package-level aggregation and return individual status messages for those tasked members.

If MA is executing a route, it will send *RoutePlanExecutionStatusMT* and *RouteActivityPlanExecutionStatusMT* periodically containing the required field *PlanExecutionStatus.PlanExecutionState* and *PlanExecutionStatus.RouteActivityPlanExecutionStateType*, respectively, denoting the executing state.

If MA is executing a mission, MA will send *MissionPlanExecutionStatusMT* and *MA_MissionPlanActivationStatusMT* periodically containing the required field *PlanExecutionStatus.MissionPlanExecutionStateType* and *PlanActivationState.PlanActivationStateEnum*, respectively, denoting the executing state.

If MA is executing an activity plan or task plan, it will send *ActivityPlanExecutionStatusMT* and *TaskPlanExecutionStatusMT* periodically containing the required field *PlanExecutionStatus.ActivityPlanExecutionStateType* and *PlanExecutionStatus.TaskPlanExecutionStateType*, respectively, denoting the executing state.

If MA is executing a response, it will send *ResponsePlanExecutionStatusMT* periodically containing the required field *PlanExecutionStatus.ResponsePlanExecutionStateType* denoting the executing state of the response plan.

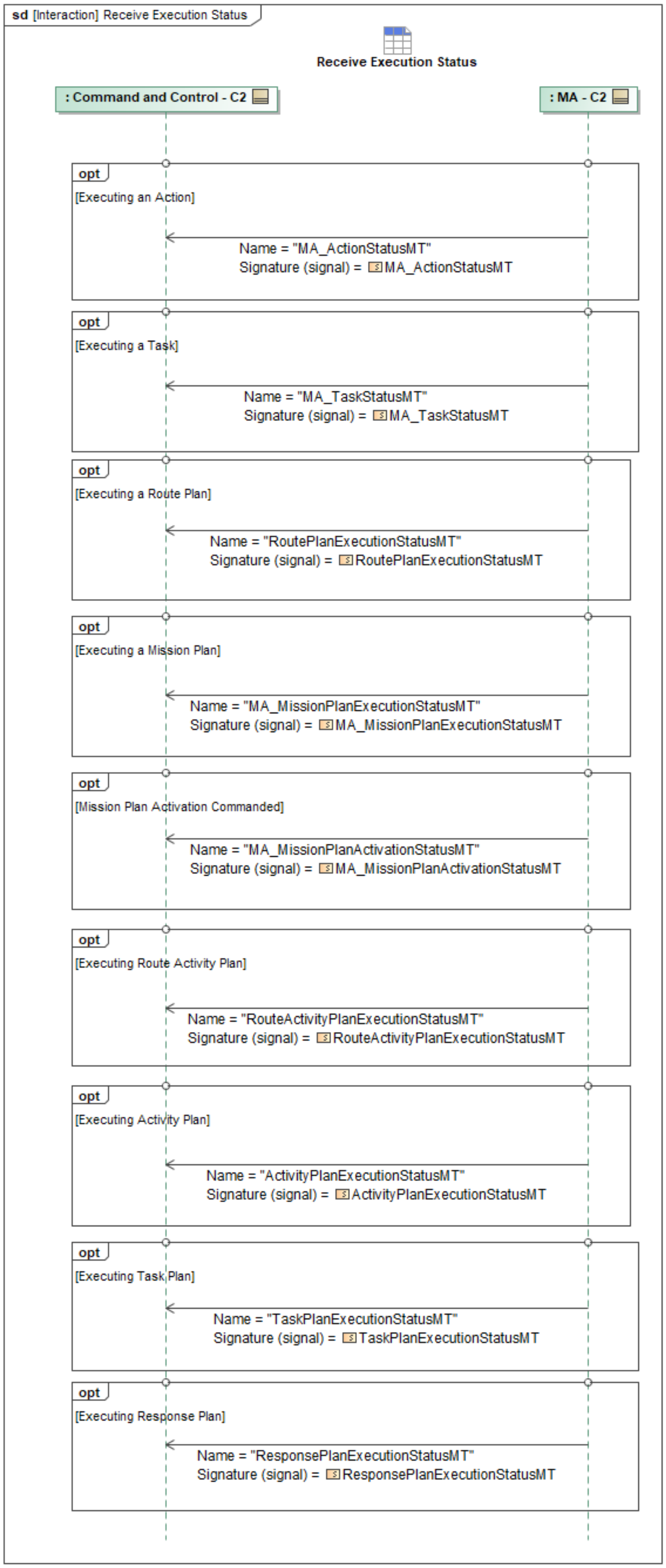


Figure 1-41: Receive Execution Status

Table 1-40: Receive Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
ActivityPlanExecutionStatusMT: ActivityPlanExecutionStatusMT	None
MA_ActionStatusMT: MA_ActionStatusMT	None
MA_MissionPlanActivationStatusMT: MA_MissionPlanActivationStatusMT	None
MA_MissionPlanExecutionStatusMT: MA_MissionPlanExecutionStatusMT	None
MA_TaskStatusMT: MA_TaskStatusMT	None
ResponsePlanExecutionStatusMT: ResponsePlanExecutionStatusMT	None
RouteActivityPlanExecutionStatusMT: RouteActivityPlanExecutionStatusMT	None
RoutePlanExecutionStatusMT: RoutePlanExecutionStatusMT	None
TaskPlanExecutionStatusMT: TaskPlanExecutionStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.13 Report Departure Checkpoints to C2

MA will report departure checkpoints to C2 using an *OperatorNotificationMT* message. The diagram below shows all the *OperatorNotificationMT* signals associated with reporting departure checkpoints and the notes will indicate the details of the Title and any associations in the *OperatorNotificationMT* signals:

Runway Advancement

As an ACP releases the brakes and starts down the runway MA will report advancing down the runway to C2 using an *OperatorNotificationMT* message.

MA will associate this with the *MA_LaunchTask* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType* to *TASK* and the *AssociatedID* to the *MA_TaskMT:TaskID*.

Airborne

When an ACP has met the criteria for being airborne MA will report that to C2 using an *OperatorNotificationMT* message.

MA will associate this with the *MA_LaunchTask* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType* to *TASK* and the *AssociatedID* to the *MA_TaskMT:TaskID*.

FA-MA Handoff

When an ACP has completed the FA-MA handoff MA will report that to C2 using an

OperatorNotificationMT message.

MA will associate *OperatorNotificationMT* with the *ControlStatusMT* message that indicates the handoff completed by setting *AssociatedMessageType* to *CONTROL_STATUS* and *AssociatedID* to the *ControlType:MissionControl:ControllerSystemID* which is the C2 that currently has mission control.

Departure Route Complete on Route Path

When an ACP has completed the departure route MA will report that to C2 using an *OperatorNotificationMT* message.

MA will associate this with the *MA_RoutePlanMT* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType* to *ROUTE_PLAN* and the *AssociatedID* to the *MA_RoutePlan:RoutePlanID.*

Note: this sequence diagram could be used to report the completion of other route paths that are not departure routes

Departure Route Complete on Path Segment

When an ACP has completed the departure route segments, MA will report that to C2 using an *OperatorNotificationMT message*.

MA will associate this with the *MA_RoutePlanMT* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType* to *ROUTE_PLAN* and the *AssociatedID* to the *MA_RoutePlan:PathSegmentID.*

Note: this sequence diagram could be used to report the completion of other route path segments that are not departure routes

Departure Orbit Join

When an ACP has started the departure orbit MA will report that to C2 using an *OperatorNotificationMT* message.

MA will associate this with the *Orbit MA_TaskMT* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType* to *TASK* and the *AssociatedID* to the *MA_TaskMT:TaskID.*

Departure Orbit Completion

When an ACP has completed the departure orbit MA will report that to C2 using an *OperatorNotificationMT* message.

MA will associate this with the *Orbit MA_TaskMT* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType* to *TASK* and the *AssociatedID* to the *MA_TaskMT:TaskID*.

Transition to Transit Completion on Route

When an ACP has completed the departure route MA will report that to C2 using an *OperatorNotificationMT message*.

MA will associate this with the *MA_RoutePlanMT* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType* to *ROUTE_PLAN* and the *AssociatedID* to the *MA_RoutePlanMT:RoutePlanID*

Transition to Transit Completion on Route Path Segment

When an ACP has completed the route segment for Departure MA will report that to C2 using an *OperatorNotificationMT message*.

MA will associate this with the *MA_RoutePlanMT* by setting the *OperatorNotificationMT:AssociatedMessage:MessageType to ROUTE_PLAN* and the *AssociatedID* to the *MA_RoutePlanMT:RoutePlanID.*

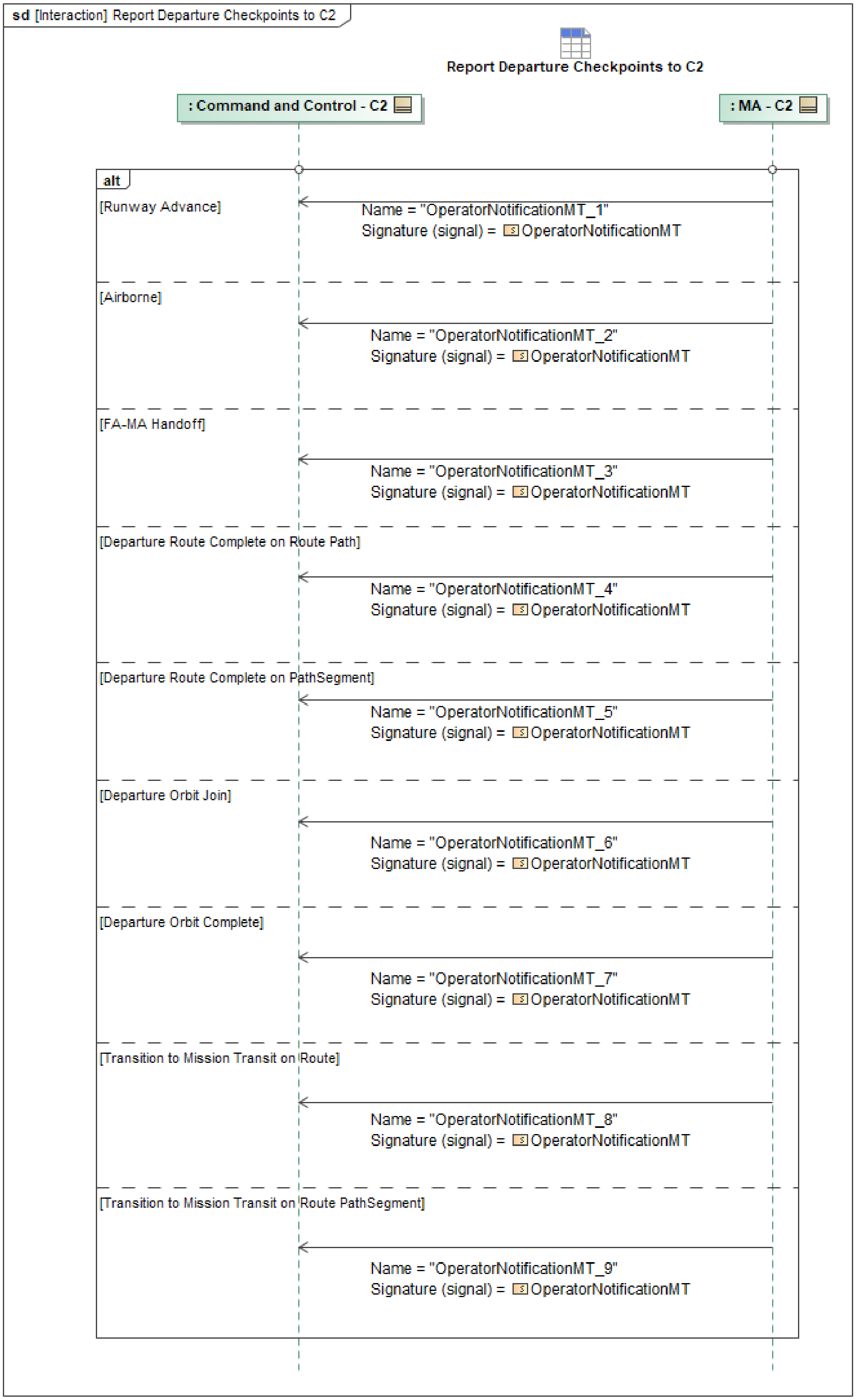


Figure 1-42: Report Departure Checkpoints to C2

Table 1-41: Report Departure Checkpoints to C2

Message Name (Name:Type)	Required Fields (Name:Type)
OperatorNotificationMT_1: OperatorNotificationMT	TASK: AssociatedMessage: AssociatedMessageType AssociatedID: ID_Type
OperatorNotificationMT_2: OperatorNotificationMT	AssociatedID: ID_Type AssociatedMessage: AssociatedMessageType TASK:
OperatorNotificationMT_3: OperatorNotificationMT	AssociatedMessage: AssociatedMessageType CONTROL_STATUS: ControllerSystemID: SystemID_Type
OperatorNotificationMT_4: OperatorNotificationMT	ROUTE_PLAN: AssociatedMessage: AssociatedMessageType AssociatedID: ID_Type
OperatorNotificationMT_5: OperatorNotificationMT	AssociatedID: ID_Type AssociatedMessage: AssociatedMessageType ROUTE_PLAN:
OperatorNotificationMT_6: OperatorNotificationMT	TASK: AssociatedMessage: AssociatedMessageType AssociatedID: ID_Type
OperatorNotificationMT_7: OperatorNotificationMT	AssociatedID: ID_Type TASK: AssociatedMessage: AssociatedMessageType
OperatorNotificationMT_8: OperatorNotificationMT	AssociatedID: ID_Type AssociatedMessage: AssociatedMessageType ROUTE_PLAN:
OperatorNotificationMT_9: OperatorNotificationMT	AssociatedID: ID_Type AssociatedMessage: AssociatedMessageType ROUTE_PLAN:

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.14 Request Generation of Route Metrics

RouteMetricsMT contains Route Metric Data and is stored or cached in data manager services which may have a database backend. This request to generate data is sent via the *RouteMetricsRequestMT* and responded to with *RouteMetricsRequestStatusMT* messages. Data management services respond with intermediate *RouteMetricsRequestStatusMT* messages with the *RequestProcessingStateEnum* set to indicate additional in-process states, such as *"QUEUED"* and

"PROCESSING". When complete, the *RequestProcessingStateEnum* is set to *"COMPLETED"* and contains the requested data. When the original *RouteMetricsRequestStatusMT*'s optional field *ResultsInNative* is populated with the matching message type, the underlying data management services instead publishes the requested data in its original or raw format as a separate message and indicates *RouteDataRequestStatusMT* message's *RequestProcessingStateEnum* to *"COMPLETED"*.

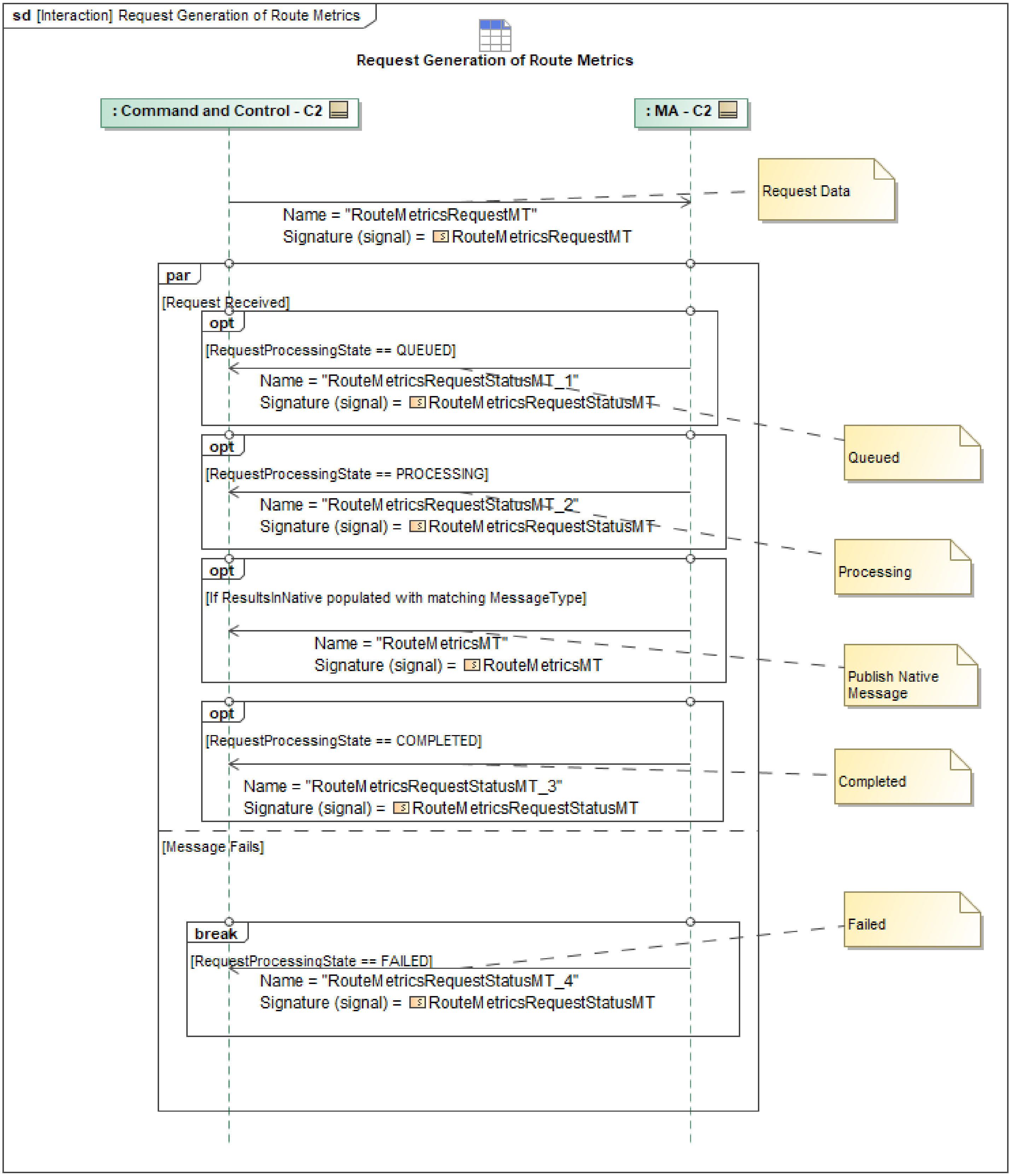


Figure 1-43: Request Generation of Route Metrics

Table 1-42: Request Generation of Route Metrics

Message Name (Name:Type)	Required Fields (Name:Type)
RouteMetricsMT: RouteMetricsMT	None
RouteMetricsRequestMT: RouteMetricsRequestMT	None
RouteMetricsRequestStatusMT_1: RouteMetricsRequestStatusMT	None
RouteMetricsRequestStatusMT_2: RouteMetricsRequestStatusMT	None
RouteMetricsRequestStatusMT_3: RouteMetricsRequestStatusMT	None
RouteMetricsRequestStatusMT_4: RouteMetricsRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.15 Respond to Approval Request

In the event that MA requires external approval, C2 can respond to the approval request sent by MA. MA publishes an *MA_ApprovalRequestMT*, and on receipt of this message, C2 will make an approval determination and respond to MA with *MA_ApprovalRequestStatusMT*.

MA does not require external approval in all situations. One example of a situation where external approval is required is the case where an *ApprovalPolicyMT* contains a *Policy* field that is set to *APPROVAL_REQUIRED*.

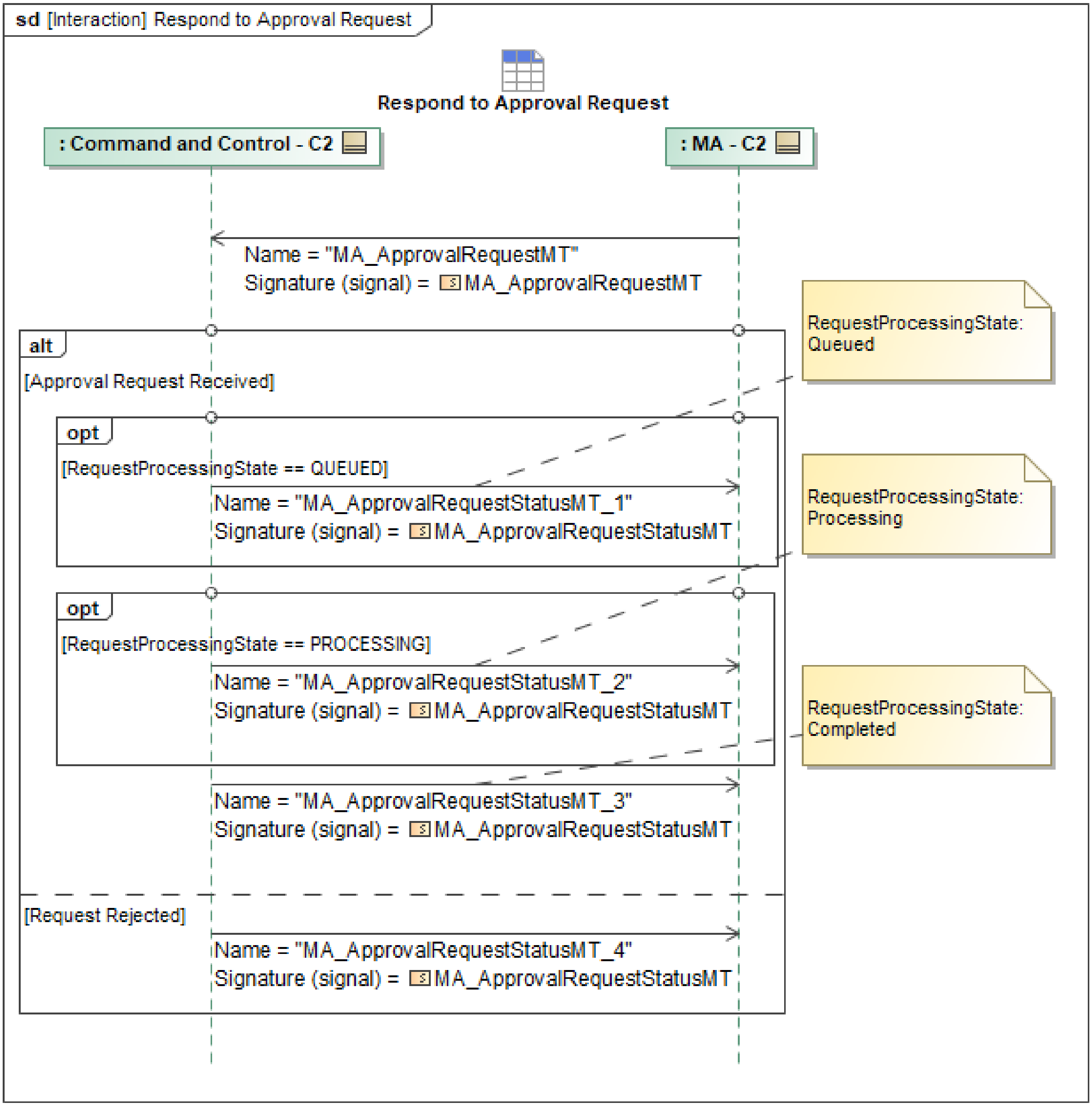


Figure 1-44: Respond to Approval Request

Table 1-43: Respond to Approval Request

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalRequestMT: MA_ApprovalRequestMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalRequestStatusMT_1: MA_ApprovalRequestStatusMT	None
MA_ApprovalRequestStatusMT_2: MA_ApprovalRequestStatusMT	None
MA_ApprovalRequestStatusMT_3: MA_ApprovalRequestStatusMT	None
MA_ApprovalRequestStatusMT_4: MA_ApprovalRequestStatusMT	ApprovalRequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.16 Send Flight Phase Status

FA must establish a clear mechanism to publish the operational phase of flight to MA. This will also inform downstream systems, such as C2 and Peer, for monitoring. MA sends *MA_FlightActivityMT* periodically as it serves as the primary mechanism for FA to assert the current phase to MA, and then from MA to C2. The *FlightPhase* Boolean indicates the current phase of flight identified by Flight Autonomy, specifically its safety criticality. True indicates that the current flight phase is safety critical; False indicates a non-safety critical phase.

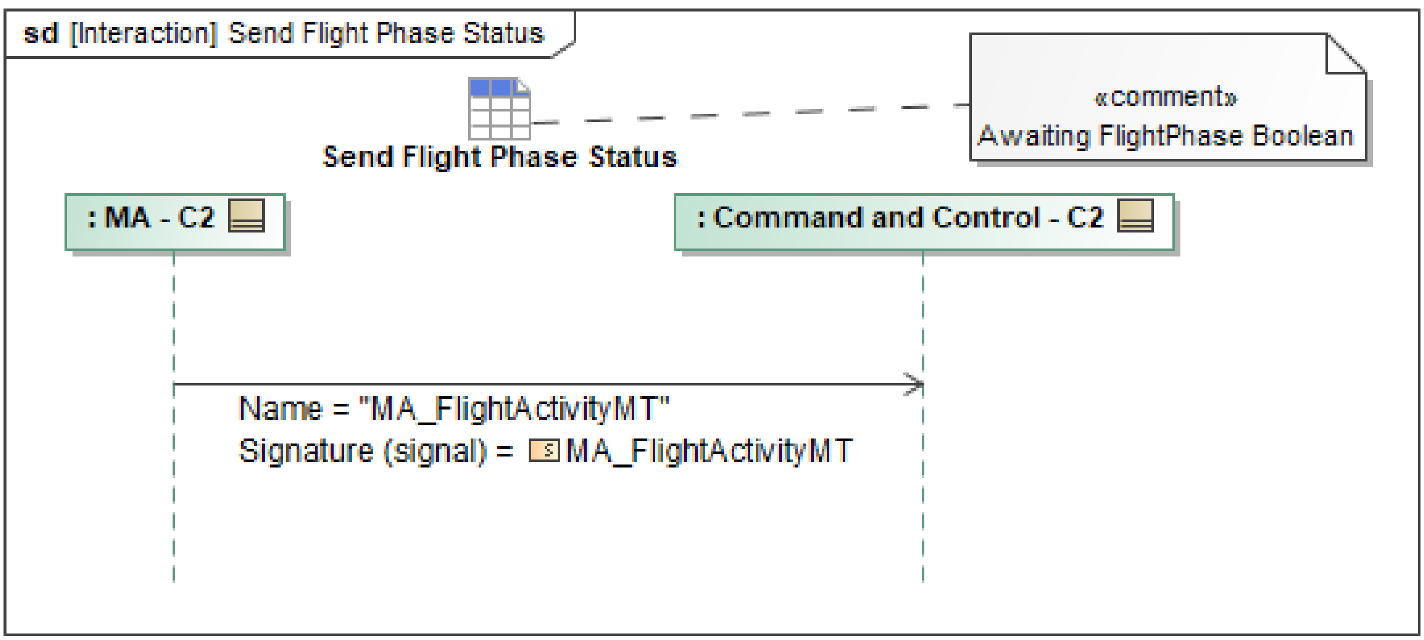


Figure 1-45: Send Flight Phase Status

Table 1-44: Send Flight Phase Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_FlightActivityMT: MA_FlightActivityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set

Tables section)

1.2.7.17 Send Heartbeat

Both MA and C2 exchange SystemStatusMT periodically to convey a regular heartbeat. The C2 system should identify itself as such by setting *SystemStatusMT.PlatformStatus.CommandAndControlIndicator* to *True*. No response is required for heartbeat messages issued.

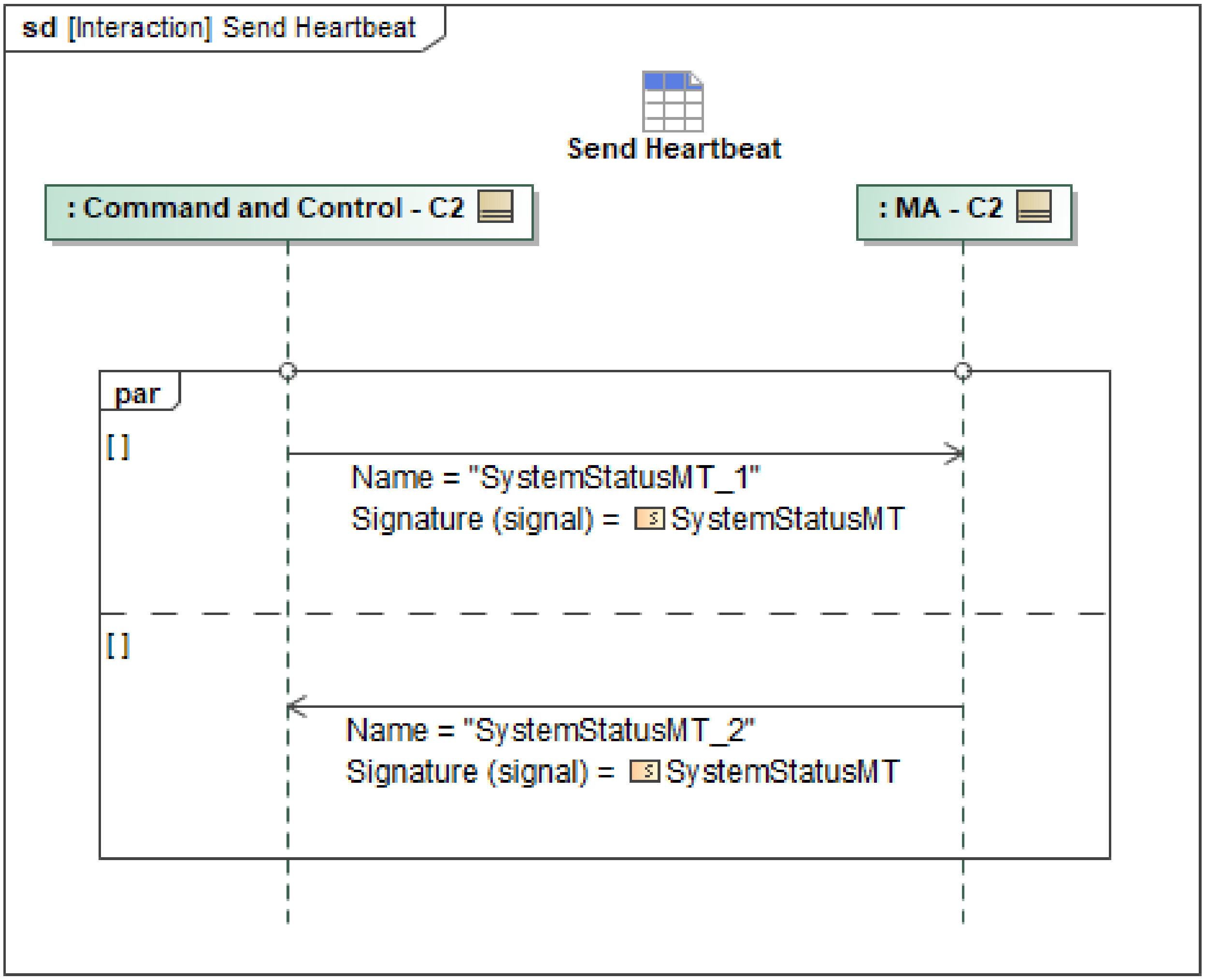


Figure 1-46: Send Heartbeat

Table 1-45: Send Heartbeat

Message Name (Name:Type)	Required Fields (Name:Type)
SystemStatusMT_1: SystemStatusMT	PlatformStatus: PlatformStatusType OperatorID: OperatorID_Type
SystemStatusMT_2: SystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.18 Send Subsystem Status to C2

MA can send *SubsystemStatusMT* periodically to inform C2 of the status of any subsystem on the platform.

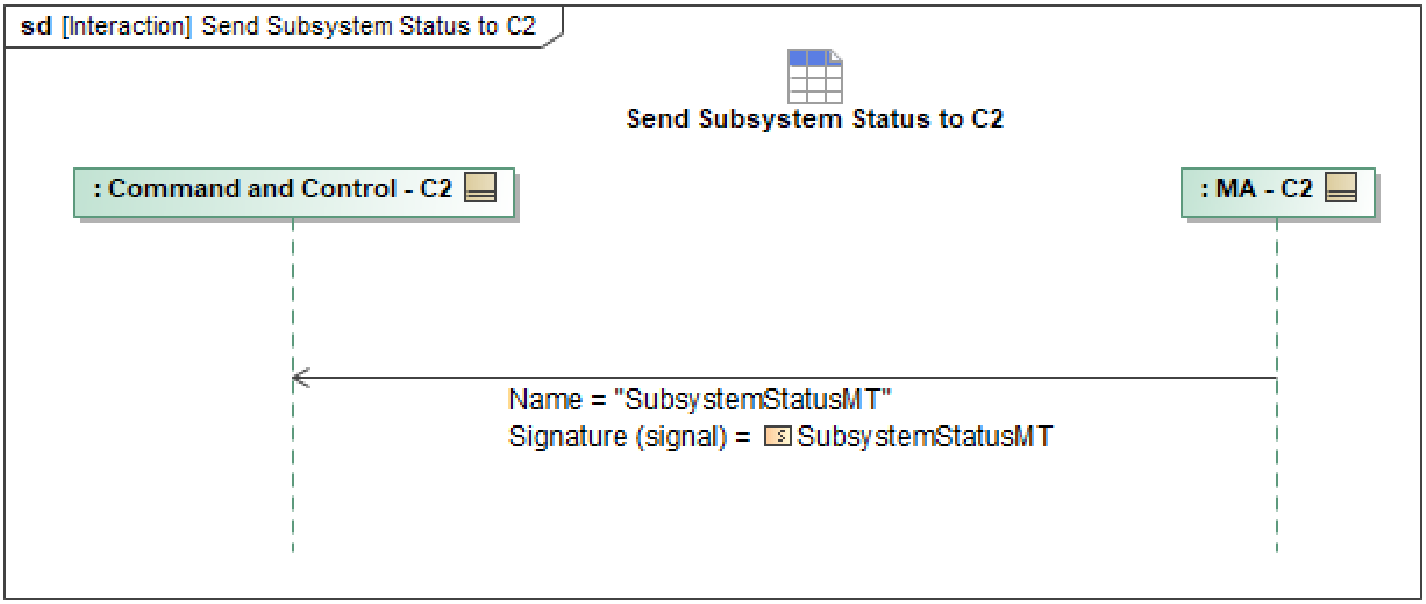


Figure 1-47: Send Subsystem Status to C2

Table 1-46: Send Subsystem Status to C2

Message Name (Name:Type)	Required Fields (Name:Type)
SubsystemStatusMT: SubsystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.19 Share Vehicle State Data

MA reports vehicle state data to C2 by sending *MA_FlightActivityMT*, *ComponentStatusMT, MA_PositionReportMT, and MA_NavigationReportMT periodically*. *MA_PositionReportMT* includes *InertialState*, defining *Position*, *DomainVelocity*, *GroundVelocity*, *CalibratedAirspeed*, and *Mach*. *DomainVelocity* represents the True Airspeed of the vehicle. *ComponentStatusMT* is optionally included if additional status data is required by C2. The transmission of these four messages being sent from MA to C2 will allow for the C2 instance(s) to gain a full picture of the state and status of the ACP.

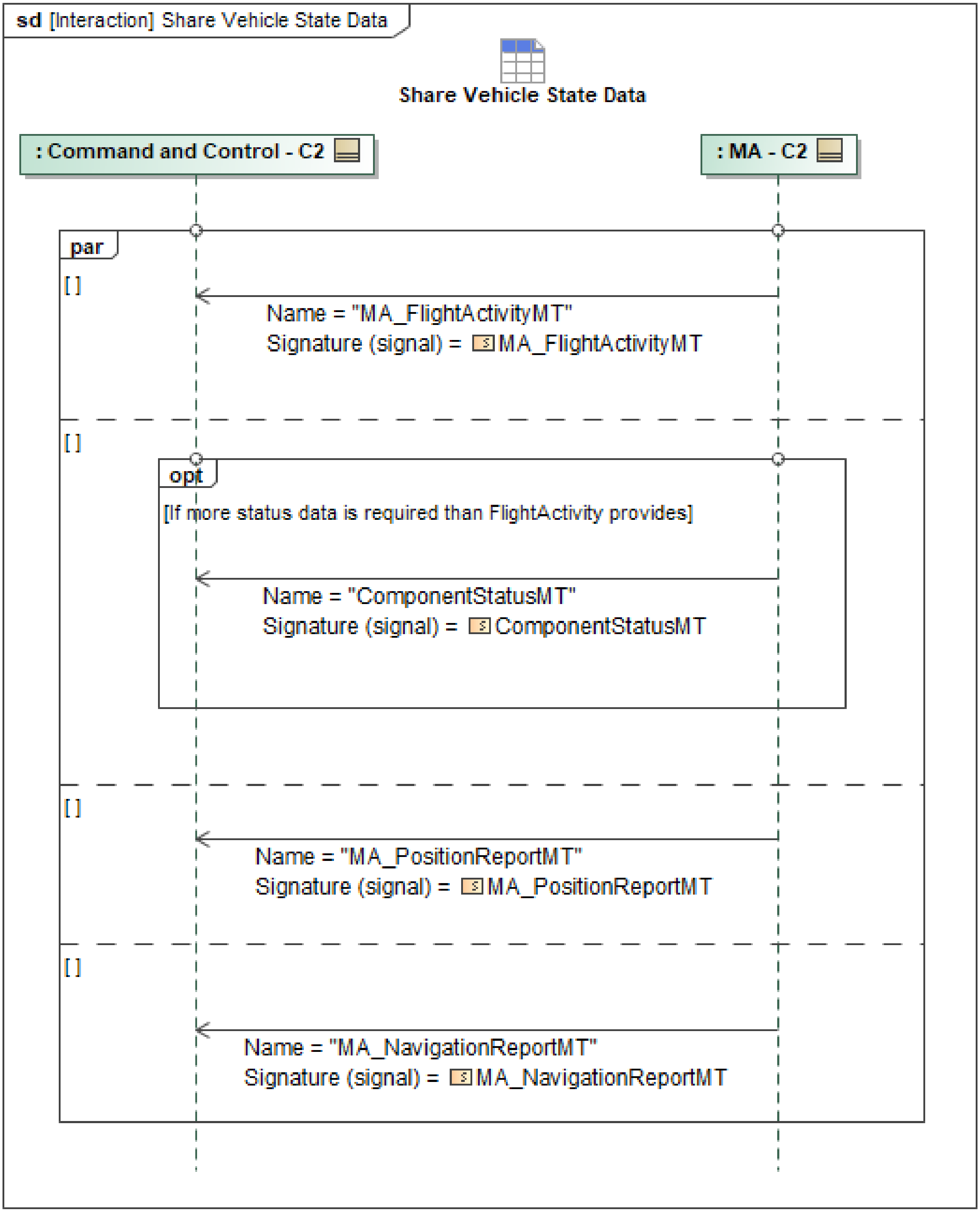


Figure 1-48: Share Vehicle State Data

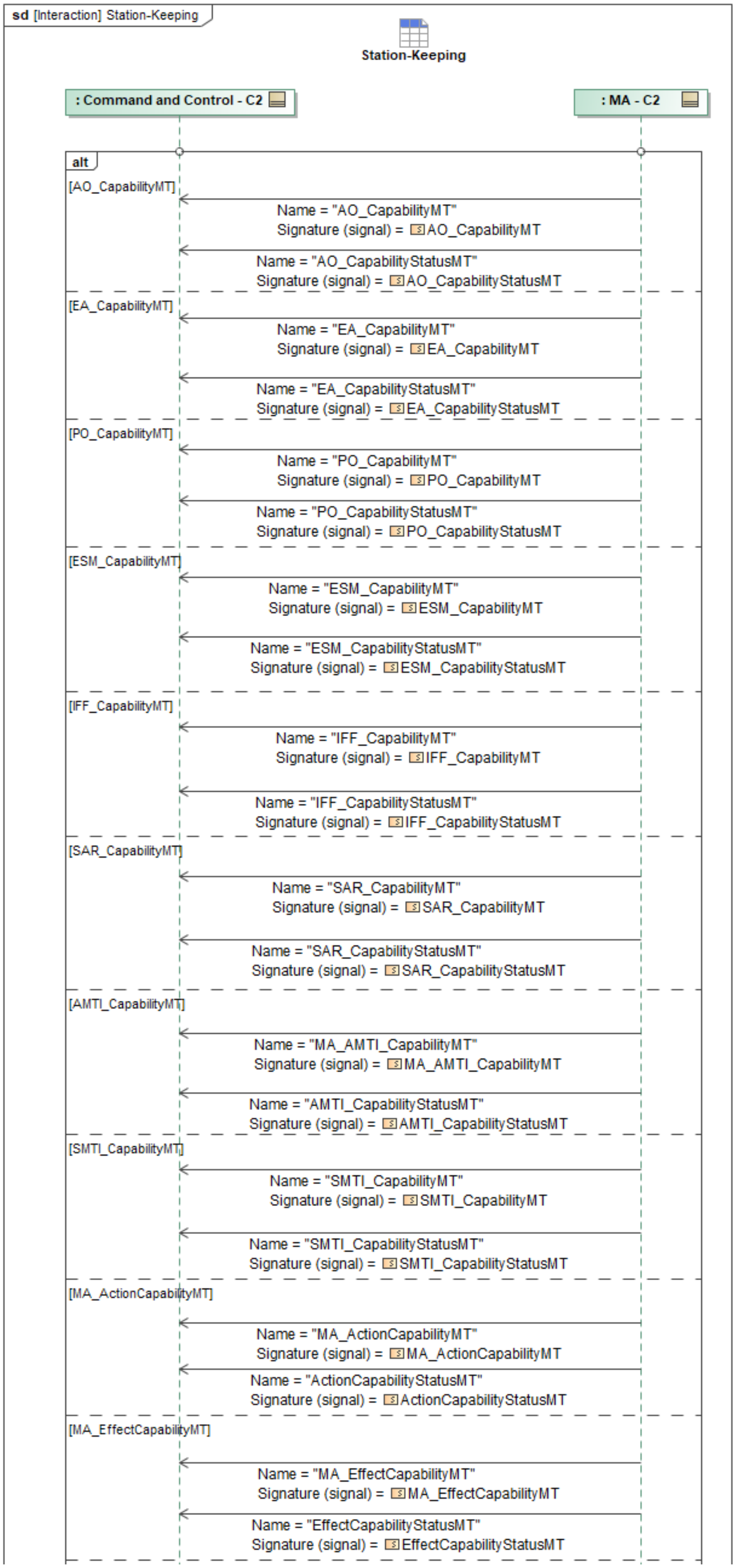
Table 1-47: Share Vehicle State Data

Message Name (Name:Type)	Required Fields (Name:Type)
ComponentStatusMT: ComponentStatusMT	None
MA_FlightActivityMT: MA_FlightActivityMT	None
MA_NavigationReportMT: MA_NavigationReportMT	None
MA_PositionReportMT: MA_PositionReportMT	DomainVelocity: Velocity3D_Type GroundVelocity: Velocity2D_Type CalibratedAirspeed: SpeedType Mach: MachType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.20 Station-Keeping

The known set of station-keeping capabilities is presented in the following sequence diagram. A platform may advertise multiple of these capabilities (*[Capability]Activity* and *[Capability]Statuses*) based on platform configuration and each is sent periodically as a separate instance of this sequence.


sd [Interaction] Station-Keeping
Station-Keeping
: Command and Control - C2
: MA - C2
alt
[AO_CapabilityMT]
Name = "AO_CapabilityMT"
Signature (signal) = AO_CapabilityMT
Name = "AO_CapabilityStatusMT"
Signature (signal) = AO_CapabilityStatusMT
[EA_CapabilityMT]
Name = "EA_CapabilityMT"
Signature (signal) = EA_CapabilityMT
Name = "EA_CapabilityStatusMT"
Signature (signal) = EA_CapabilityStatusMT
[PO_CapabilityMT]
Name = "PO_CapabilityMT"
Signature (signal) = PO_CapabilityMT
Name = "PO_CapabilityStatusMT"
Signature (signal) = PO_CapabilityStatusMT
[ESM_CapabilityMT]
Name = "ESM_CapabilityMT"
Signature (signal) = ESM_CapabilityMT
Name = "ESM_CapabilityStatusMT"
Signature (signal) = ESM_CapabilityStatusMT
[IFF_CapabilityMT]
Name = "IFF_CapabilityMT"
Signature (signal) = IFF_CapabilityMT
Name = "IFF_CapabilityStatusMT"
Signature (signal) = IFF_CapabilityStatusMT
[SAR_CapabilityMT]
Name = "SAR_CapabilityMT"
Signature (signal) = SAR_CapabilityMT
Name = "SAR_CapabilityStatusMT"
Signature (signal) = SAR_CapabilityStatusMT
[AMTI_CapabilityMT]
Name = "MA_AMTI_CapabilityMT"
Signature (signal) = MA_AMTI_CapabilityMT
Name = "AMTI_CapabilityStatusMT"
Signature (signal) = AMTI_CapabilityStatusMT
[SMTI_CapabilityMT]
Name = "SMTI_CapabilityMT"
Signature (signal) = SMTI_CapabilityMT
Name = "SMTI_CapabilityStatusMT"
Signature (signal) = SMTI_CapabilityStatusMT
[MA_ActionCapabilityMT]
Name = "MA_ActionCapabilityMT"
Signature (signal) = MA_ActionCapabilityMT
Name = "ActionCapabilityStatusMT"
Signature (signal) = ActionCapabilityStatusMT
[MA_EffectCapabilityMT]
Name = "MA_EffectCapabilityMT"
Signature (signal) = MA_EffectCapabilityMT
Name = "EffectCapabilityStatusMT"
Signature (signal) = EffectCapabilityStatusMT

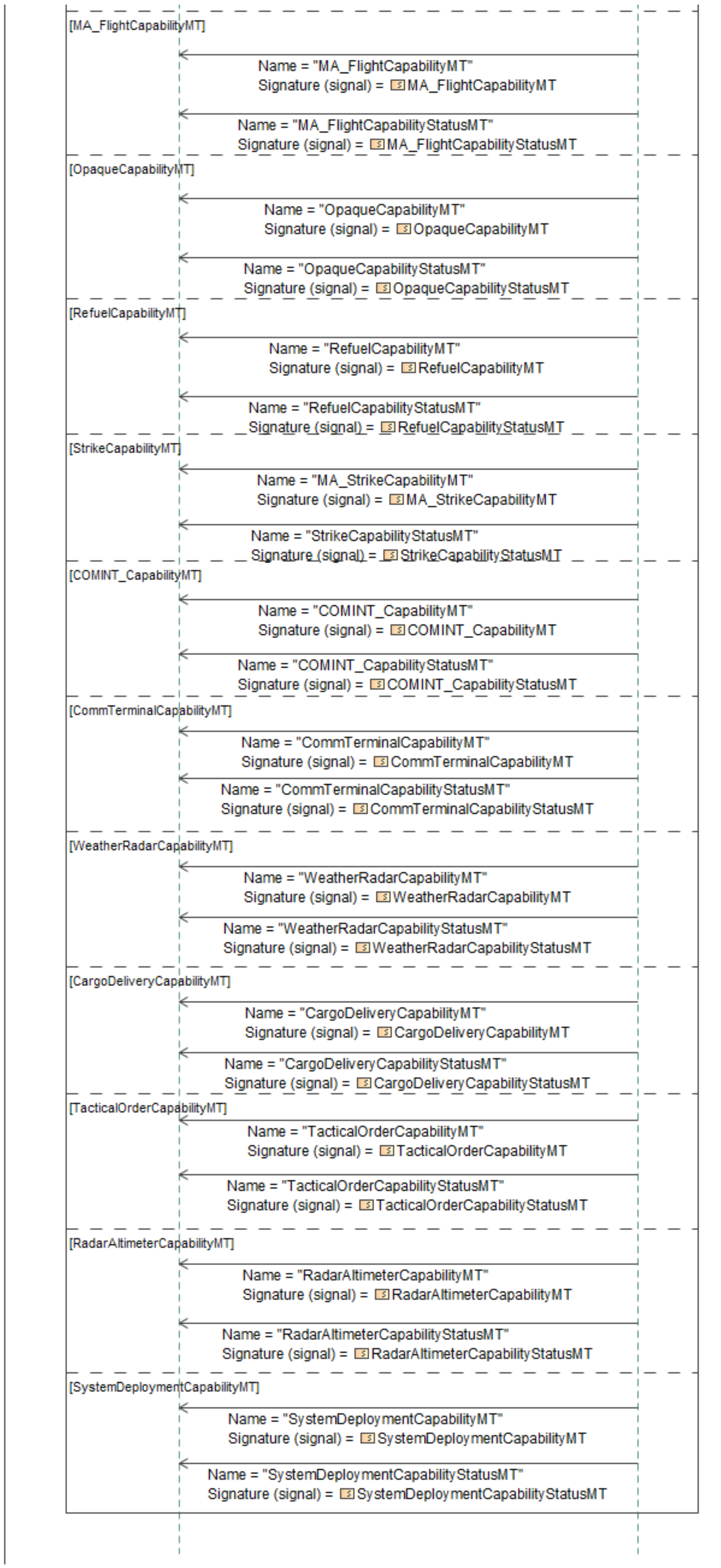


Figure 1-49: Station-Keeping

Table 1-48: Station-Keeping

Message Name (Name:Type)	Required Fields (Name:Type)
ActionCapabilityStatusMT: ActionCapabilityStatusMT	None
AMTI_CapabilityStatusMT: AMTI_CapabilityStatusMT	None
AO_CapabilityMT: AO_CapabilityMT	None
AO_CapabilityStatusMT: AO_CapabilityStatusMT	None
CargoDeliveryCapabilityMT: CargoDeliveryCapabilityMT	None
CargoDeliveryCapabilityStatusMT: CargoDeliveryCapabilityStatusMT	None
COMINT_CapabilityMT: COMINT_CapabilityMT	None
COMINT_CapabilityStatusMT: COMINT_CapabilityStatusMT	None
CommTerminalCapabilityMT: CommTerminalCapabilityMT	None
CommTerminalCapabilityStatusMT: CommTerminalCapabilityStatusMT	None
EA_CapabilityMT: EA_CapabilityMT	None
EA_CapabilityStatusMT: EA_CapabilityStatusMT	None
EffectCapabilityStatusMT: EffectCapabilityStatusMT	None
ESM_CapabilityMT: ESM_CapabilityMT	None
ESM_CapabilityStatusMT: ESM_CapabilityStatusMT	None
IFF_CapabilityMT: IFF_CapabilityMT	None
IFF_CapabilityStatusMT: IFF_CapabilityStatusMT	None
MA_ActionCapabilityMT: MA_ActionCapabilityMT	None
MA_AMTI_CapabilityMT: MA_AMTI_CapabilityMT	None
MA_EffectCapabilityMT: MA_EffectCapabilityMT	None
MA_FlightCapabilityMT: MA_FlightCapabilityMT	None
MA_FlightCapabilityStatusMT: MA_FlightCapabilityStatusMT	None
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
OpaqueCapabilityMT: OpaqueCapabilityMT	None
OpaqueCapabilityStatusMT: OpaqueCapabilityStatusMT	None
PO_CapabilityMT: PO_CapabilityMT	None
PO_CapabilityStatusMT: PO_CapabilityStatusMT	None
RadarAltimeterCapabilityMT: RadarAltimeterCapabilityMT	None
RadarAltimeterCapabilityStatusMT: RadarAltimeterCapabilityStatusMT	None
RefuelCapabilityMT: RefuelCapabilityMT	None
RefuelCapabilityStatusMT: RefuelCapabilityStatusMT	None
SAR_CapabilityMT: SAR_CapabilityMT	None
SAR_CapabilityStatusMT: SAR_CapabilityStatusMT	None
SMTI_CapabilityMT: SMTI_CapabilityMT	None
SMTI_CapabilityStatusMT: SMTI_CapabilityStatusMT	None
StrikeCapabilityStatusMT: StrikeCapabilityStatusMT	None
SystemDeploymentCapabilityMT: SystemDeploymentCapabilityMT	None
SystemDeploymentCapabilityStatusMT: SystemDeploymentCapabilityStatusMT	None
TacticalOrderCapabilityMT: TacticalOrderCapabilityMT	None
TacticalOrderCapabilityStatusMT: TacticalOrderCapabilityStatusMT	None
WeatherRadarCapabilityMT: WeatherRadarCapabilityMT	None
WeatherRadarCapabilityStatusMT: WeatherRadarCapabilityStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.21 Systems Needed Request

The Systems Needed Request sequence diagram specifies the interaction of a Lead ACP and the Command and Control (C2) authority where the Lead ACP has identified a need for an additional package asset and is requesting support from C2 to identify which asset to add. This supports scenarios in which the Lead ACP identified that it needs an additional asset in order to fulfill its current package level action or tasking. The Lead ACP doesn't necessarily have insight into the assets currently available, or what other assets' capabilities are, therefore the Lead ACP needs support from C2 in order to determine the best asset to add to the package. This request can be utilized in tandem with the Creating or Changing a package interaction to complete the formal join command.

Once the Lead ACP has determined the need for a new asset and a need for C2 support in selecting and allocating that asset, the Lead ACP sends the *MA_SystemNeededRequestMT* message to C2. The *MA_SystemsNeededRequestMT* fields are used to communicate the needed capabilities or tasking which the new ACP will fulfill. *RequestState* is utilized to communicate if the request is *NEW* or *UPDATED*, depending on if the request if being sent for the first time or not. *Candidate* is used as a *ChoiceType* to populate either a list of candidate systems via *SystemCandidate*, or an enumeration of *SystemCapabilityDescription* that the new asset needs to have. *ProposedRequirements* are used to communicate a specific Task, Effect, Action, or Response via an existing ID for that data object.

After C2 has processed the request, it responds to the Lead ACP with the *MA_SystemsNeededRequestStatusMT* message. The *RequestProcessingState reports the progress towards processing the request.* Once a *RequestProcessingState* of *COMPLETED* is reported, the *Option* field with *SystemsNeeded* and *SystemID* is populated to specify the asset selected to fulfill the request.

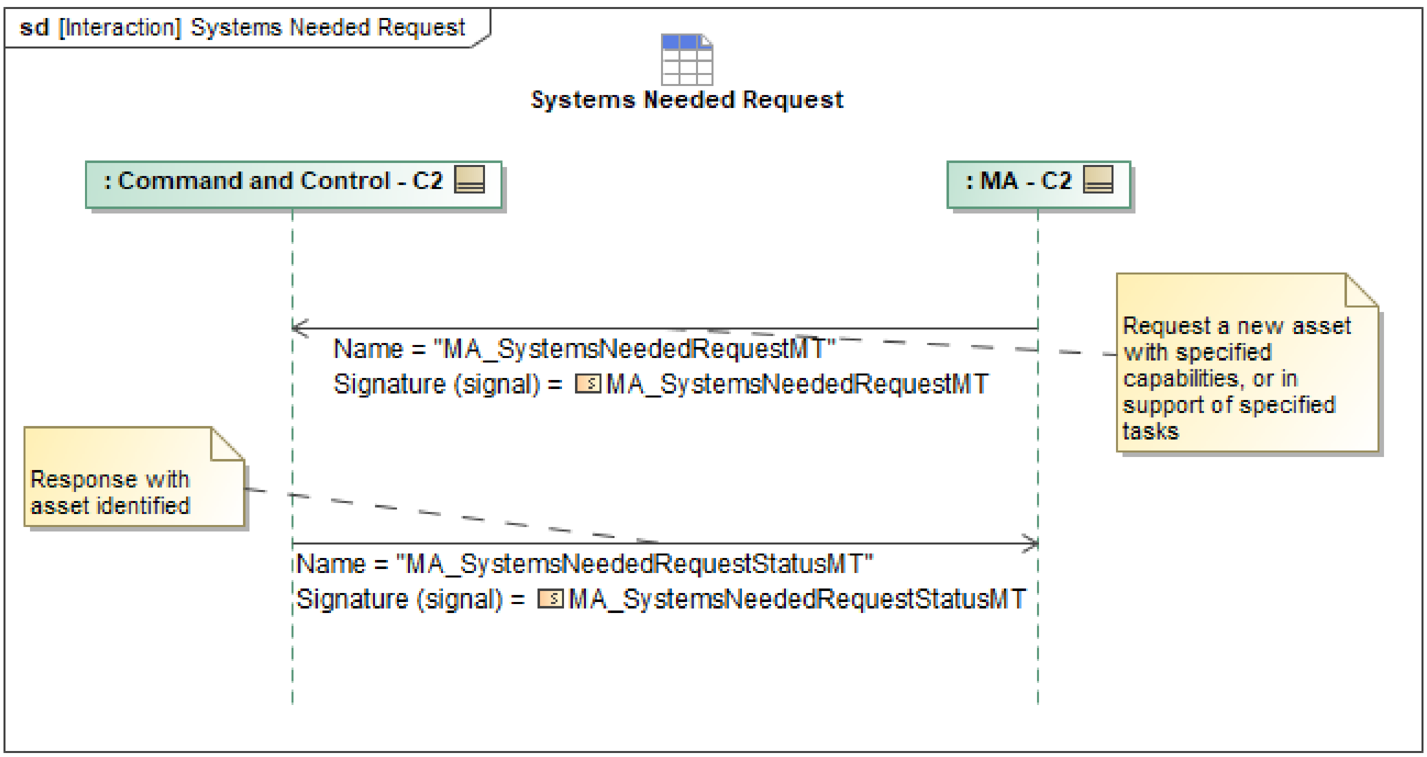


Figure 1-50: Systems Needed Request

Table 1-49: Systems Needed Request

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemsNeededRequestMT: MA_SystemsNeededRequestMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemsNeededRequestStatusMT: MA_SystemsNeededRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.22 Update Barometric Pressure

The C2 system can send the barometric pressure reference command to MA via *MA_SystemManagementRequestMT*. This value is the QNH pressure and this interaction allows the value to be changed in order to accurately calculate the vehicle's altitude above mean sea level. The *RequestType* will be *VehicleSettings*, with *VehicleSettings.QNH_Setting* set to the barometric pressure in kilopascals (kPa).

In response, MA sends C2 the *MA_SystemManagementRequestStatusMT* message, accepting or rejecting the command. The *MA_SystemManagementRequestStatusMT* will have the *RequestProcessingState* field set to "COMPLETED". If the *MA_SystemManagementRequestMT* was not able to be applied, then the *RequestProcessingState* will be set to "FAILED" and the *RequestProcessingStateReason* set to the reason the *MA_SystemManagementRequestMT* couldn't be applied.

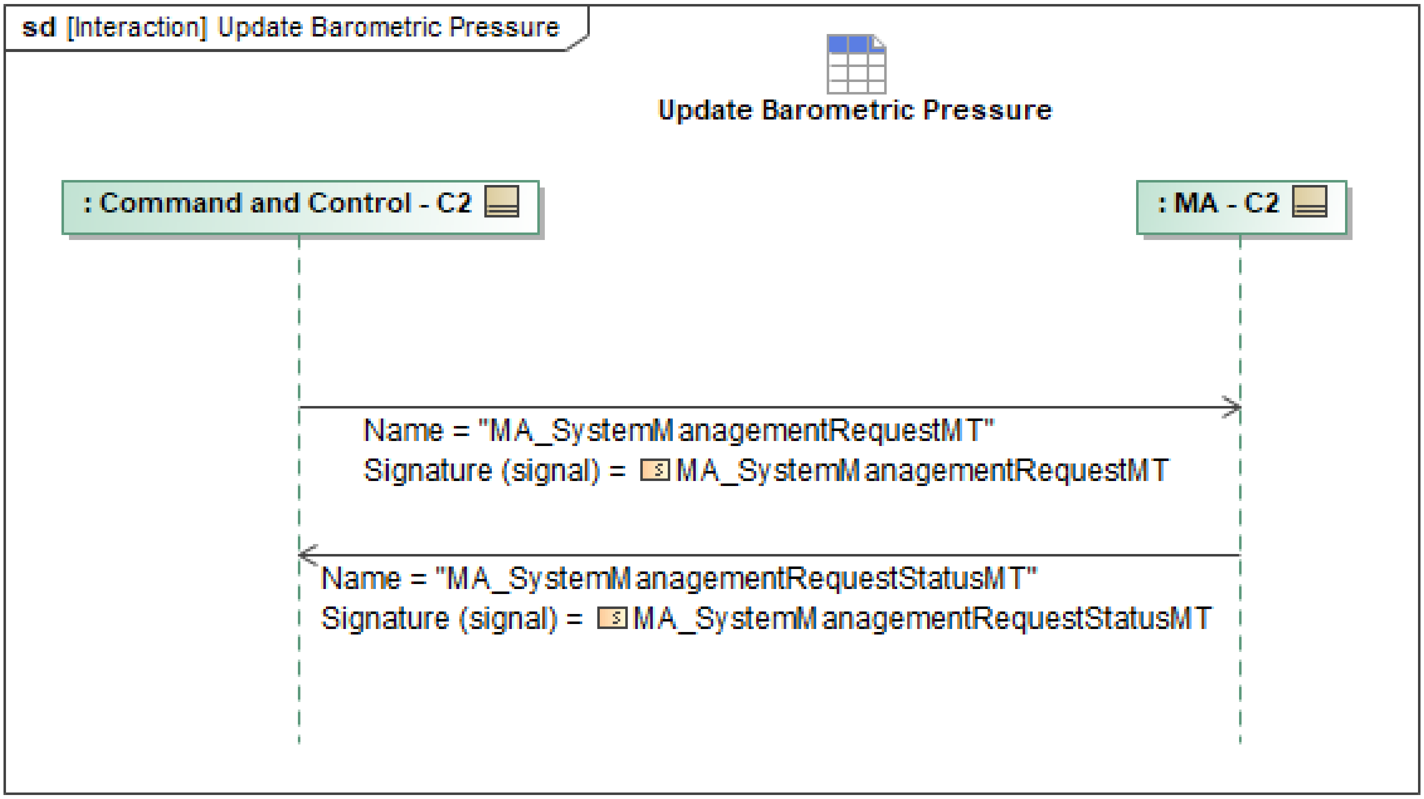


Figure 1-51: Update Barometric Pressure

Table 1-50: Update Barometric Pressure

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemManagementRequestMT: MA_SystemManagementRequestMT	None
MA_SystemManagementRequestStatusMT: MA_SystemManagementRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.7.23 Update MA-FA Control Designations

C2 may command the aircraft control paradigm including revoking or authorizing MA to command the flight of the aircraft. C2 will revoke or authorize MA to command specific flight capabilities of the aircraft. To communicate the authorized flight capabilities, C2 will send MA a *MA_FlightCapabilityMT* populating the *CapabilityType.MA_FlightCapabilityEnum* field with the allowable control modes that MA can command. To revoke flight capabilities, C2 will not include them in the *MA_FlightCapabilityEnum* and to authorize flight capabilities C2 will select them as a part of the *MA_FlightCapabilityEnum.*

To enact the control paradigm, C2 will then send a *ControlRequestMT* populating the *Controller* as the MA System ID and the *Controllee* as the FA system ID as MA will be controlling FA through flight commands. C2 will populate the *ControlType* for MA as *CAPABILITY_SECONDARY*, as the *CAPABILITY_PRIMARY* control type is reserved for FA as the safety critical system on the aircraft.

Lastly, the *CapabilityID* will be set as the *CapabilityID* of the *MA_FlightCapabilityMT* that proceeded the *ControlRequestMT.*

Both the *MA_FlightCapabilityMT* and *ControlRequestMT* will be forwarded from MA to FA to inform FA of the C2 desired control paradigm. This sequence is outlined in the VI. MA will then respond with the *ControlRequestStatusMT* to acknowledge the commanded control paradigm.

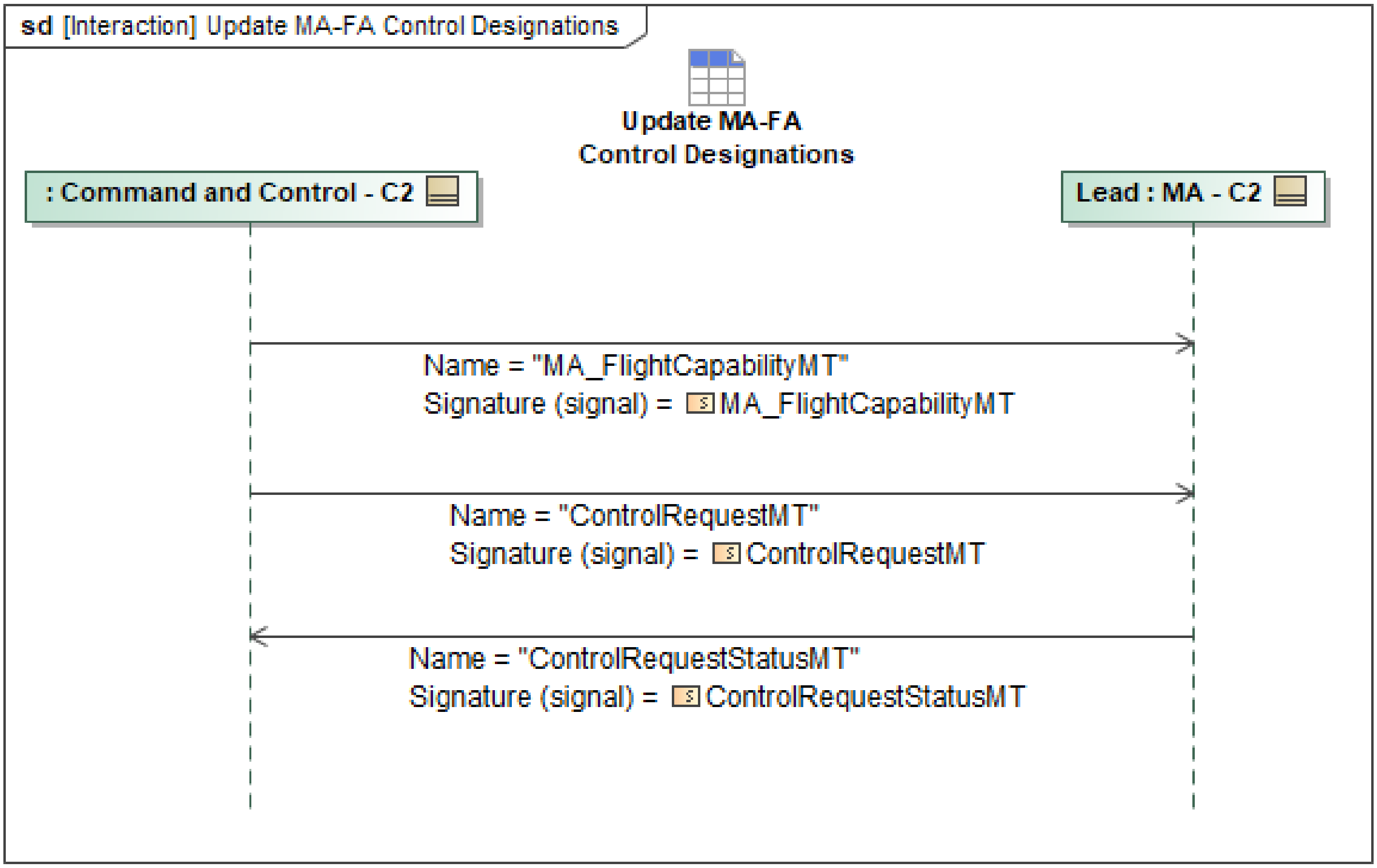


Figure 1-52: Update MA-FA Control Designations

Table 1-51: Update MA-FA Control Designations

Message Name (Name:Type)	Required Fields (Name:Type)
ControlRequestMT: ControlRequestMT	CapabilityID: CapabilityID_Type
ControlRequestStatusMT: ControlRequestStatusMT	None
MA_FlightCapabilityMT: MA_FlightCapabilityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8 C2 Mission Plan Behaviors Package

1.2.8.1 Activate or Deactivate Mission Plan

A C2 node can activate or deactivate a mission plan by sending a

MA_MissionPlanActivationCommandMT message. The *MA_MissionPlanActivationCommandMT's CommandState* should be *NEW*, the *Command* field should specify which mission plan to activate or deactivate, and *CommandID* should be a unique identifier for the command. To activate or deactivate a mission plan and all its sub-plans, send *Command.ActivationDetails.ByMissionPlan*. To activate or deactivate only a subset of subplans belonging to a mission plan, send *Command.ActivationDetails.BySubPlan* detailing which subplans should be activated or deactivated. To activate or deactivate only a specific ExecutionPlanSet belonging to a mission plan, send *Command.ActivationDetails.ByExecutionPlanSet*.

MA will send MA_MissionPlanActivationStatusMT and MA_MissionPlanExecutionStatusMT messages periodically.

MA will send at least one *MA_MissionPlanActivationCommandStatusMT* message. While the command is being initiated by MA, it can optionally send status updates to C2 via *MA_MissionPlanActivationCommandStatusMT_1* with *CommandStatus set to QUEUED or PROCESSING.* Once the command status has reached terminal status, *MA_MissionPlanActivationCommandStatusMT_2* is sent with *CommandProcessingState ACCEPTED* or *REJECTED* and *CommandStatus COMPLETED* or *FAILED*.

If a mission plan or subplan was activated or deactivated successfully, MA will send C2 a *MA_MissionPlanActivationStatusMT* detailing the current state of the mission plan. If the mission plan has subplans, *MA_MissionPlanActivationStatusMT_1* is sent and it must include *SubPlanActivationState.* If there are no SubPlans, then *MA_MissionPlanActivationStatusMT_2* is sent. If plan execution status has changed, MA will send C2 a *MA_MissionPlanExecutionStatusMT* with *Source* set to *ACTUAL* and *MA_PlanExecutionStatus* detailing the change in execution status. For a mission plan operating with an *ExecutionSequence*, *PlanExecutionStatus.ExecutionPlanSetStatus* can be included to provide statuses on the *ExecutionPlanSet*s within the mission plan.

See the behaviors "Update RoutePlan" and "Update Mission Plan" on how the different plans interact in sequence

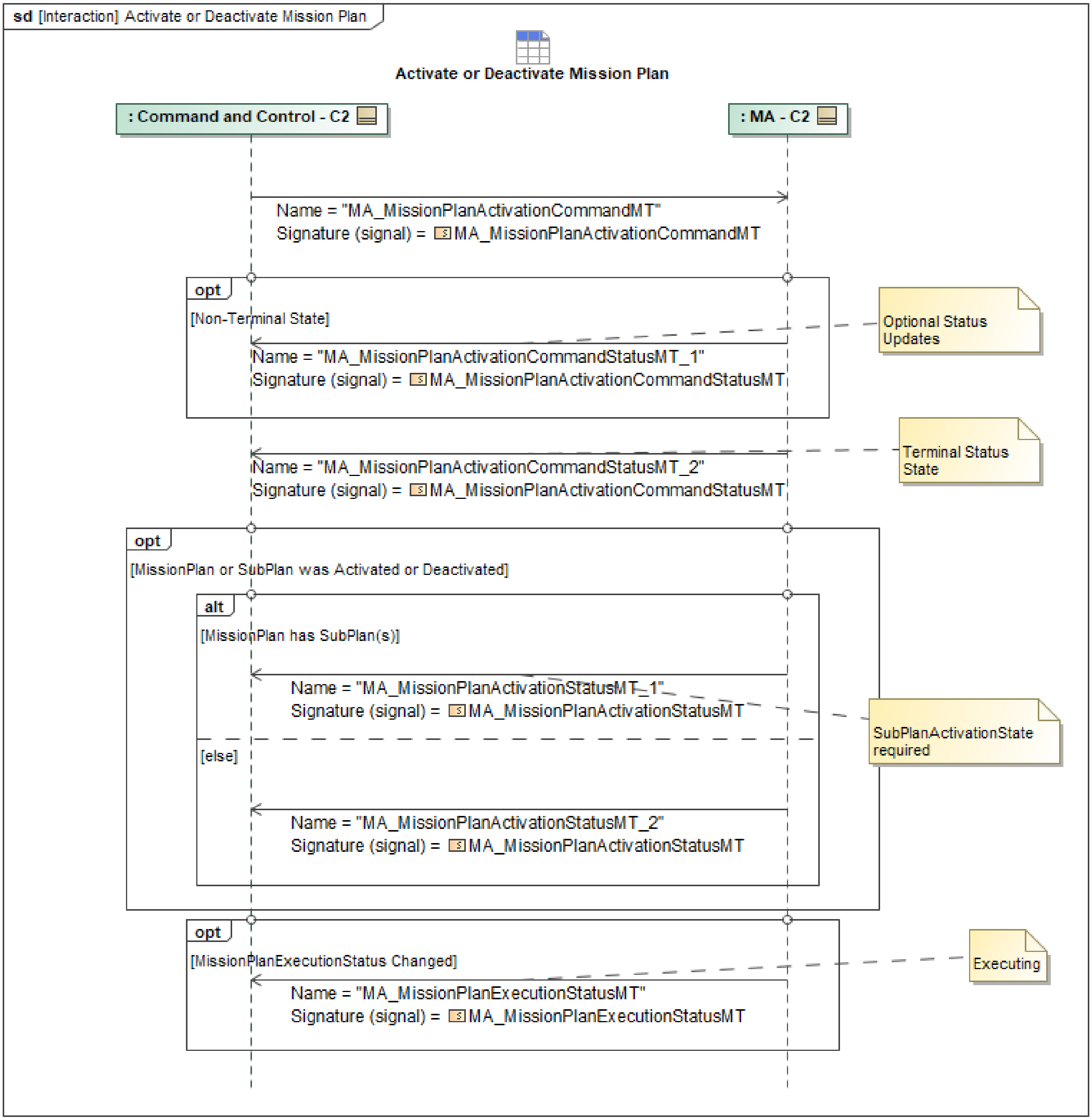


Figure 1-53: Activate or Deactivate Mission Plan

Table 1-52: Activate or Deactivate Mission Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanActivationCommandMT: MA_MissionPlanActivationCommandMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanActivationCommandStatusMT_1: MA_MissionPlanActivationCommandStatusMT	None
MA_MissionPlanActivationCommandStatusMT_2: MA_MissionPlanActivationCommandStatusMT	None
MA_MissionPlanActivationStatusMT_1: MA_MissionPlanActivationStatusMT	SubPlanActivationState: PlanActivationStateType
MA_MissionPlanActivationStatusMT_2: MA_MissionPlanActivationStatusMT	None
MA_MissionPlanExecutionStatusMT: MA_MissionPlanExecutionStatusMT	PlanExecutionStatus: MissionPlanExecutionStateType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.2 C2 Commanded Abort

A C2 system may also wish to command MA to abort its current behavior. This sequence begins with the *MA_MissionPlanActivationCommandMT* sent from the C2 system to MA, which includes the following set fields: *MissionPlanActivationCommandID* specifying the *UUID* of the abort command, the *CommandState* field denoting the abort command by the enum value "*CANCEL*", and the *Command* field containing the MissionPlanID and optional SubPlan(s) or ExecutionPlanSet(s) to cancel executing. Within the plan to abort, the *CommandType* field is set to the *PlanActivationCommandEnum* value of "*DEACTIVATE*."

MA shall first respond with a *MA_MissionPlanActivationCommandStatusMT* reflecting the acceptance and active processing of the abort command. Specifically, this message confirms the command through a crossed referenced *CommandID* matching the UUID of the abort command, *ActivationStatus.CommandProcessingState* enumerated value specifying "*ACCEPTED*", and *ActivationStatus.CommandStatus* set to "*COMPLETED*". *ActivationCommandByState.PlanActivationCommandState* will be set to the *PlanActivationStateEnum* value of "*DEACTIVATED*", and *ActivationCommandByState.Plans* will be set to the specific plan to be deactivated (can be any type), as was specified in the *MA_MissionPlanActivationCommandMT*. Should MA encounter any issues during the processing phase for this command, then MA shall inform the C2 system through an alternate *ActivationStatus.CommandStatus* of "*FAILED*" and *ActivationStatus.CommandProcessingState* of "*REJECTED*".

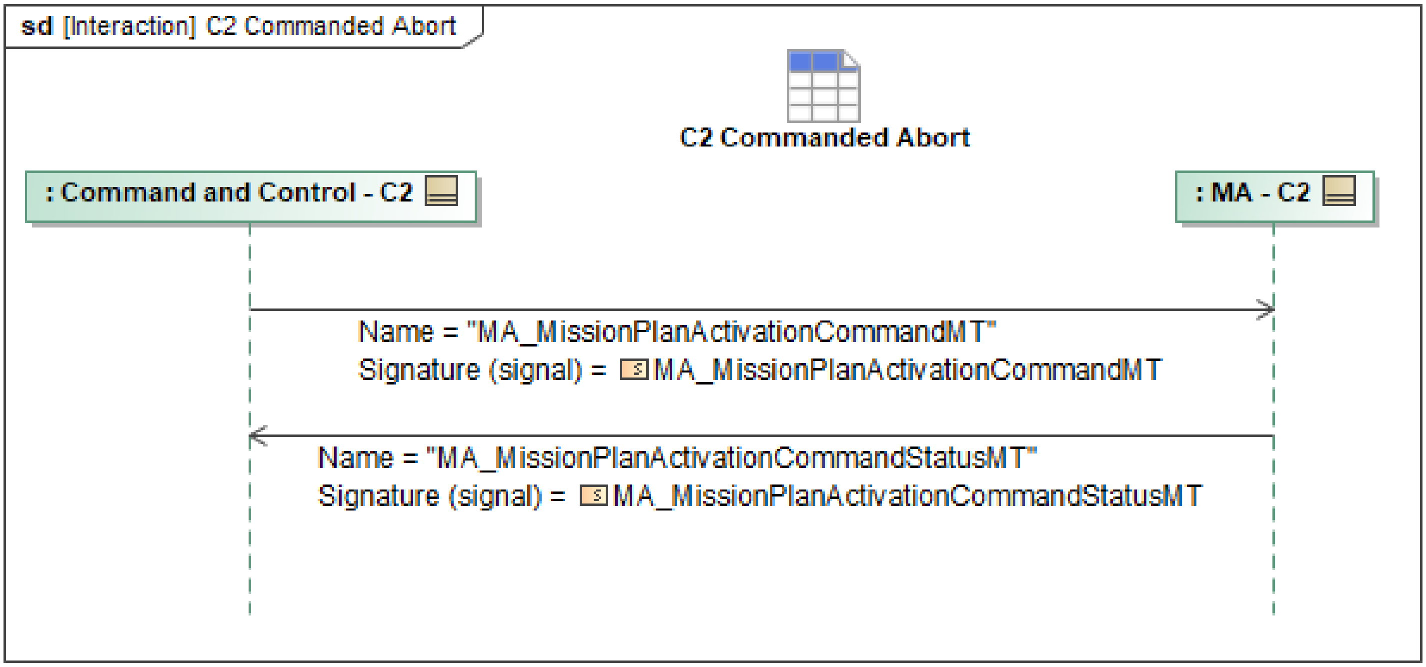


Figure 1-54: C2 Commanded Abort

Table 1-53: C2 Commanded Abort

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanActivationCommandMT: MA_MissionPlanActivationCommandMT	BySubPlan: MA_MissionPlanSubplanActivationType
MA_MissionPlanActivationCommandStatusMT: MA_MissionPlanActivationCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.3 C2 Detect Mission Plan Route Discontinuity Response

Upon identification of a Mission Plan Route Discontinuity Error, MA will send *MA_SystemNotificationMT* to C2 and P2P, to indicate that a Mission Plan Route Discontinuity condition exists. Within the message, MA sets *NotificationState* to ACTIVE and populates *NotificationDetail.Reason* with INFEASIBLE_ROUTE. This indicates that the ACP can no longer continue along the nominal Mission Plan path from its current or projected position. *AssociatedMessage.MessageType* will have the ROUTE_PLAN enumeration to indicate to C2 that the route is no longer feasible. *AssociatedMessage.AssociatedID* may reference the affected *PathSegmentID* that precedes the discontinuity, or alternatively, the *FlightCommandID* can be used if a *PathSegmentID* does not exist or cannot be referenced.

At the time of the discontinuity, MA also sends *MA_PositionReportDetailed* so that C2 has access to the current and projected position data relevant to the error condition. The *Kinematics* field communicates the ACP's current position and velocity associated with the discontinuity. Resolution actions may include MA executing a plan modification or receiving a resolution from C2 or P2P. Upon resolution of the error, MA will send *MA_SystemNotificationMT* to C2 and P2P with *NotificationState* set to CLOSED and the same other fields mentioned in the initial *MA_SystemNotificationMT*, including *NotificationDetail.Reason* and AssociatedMessage details, to provide a consistent update on the resolved condition.

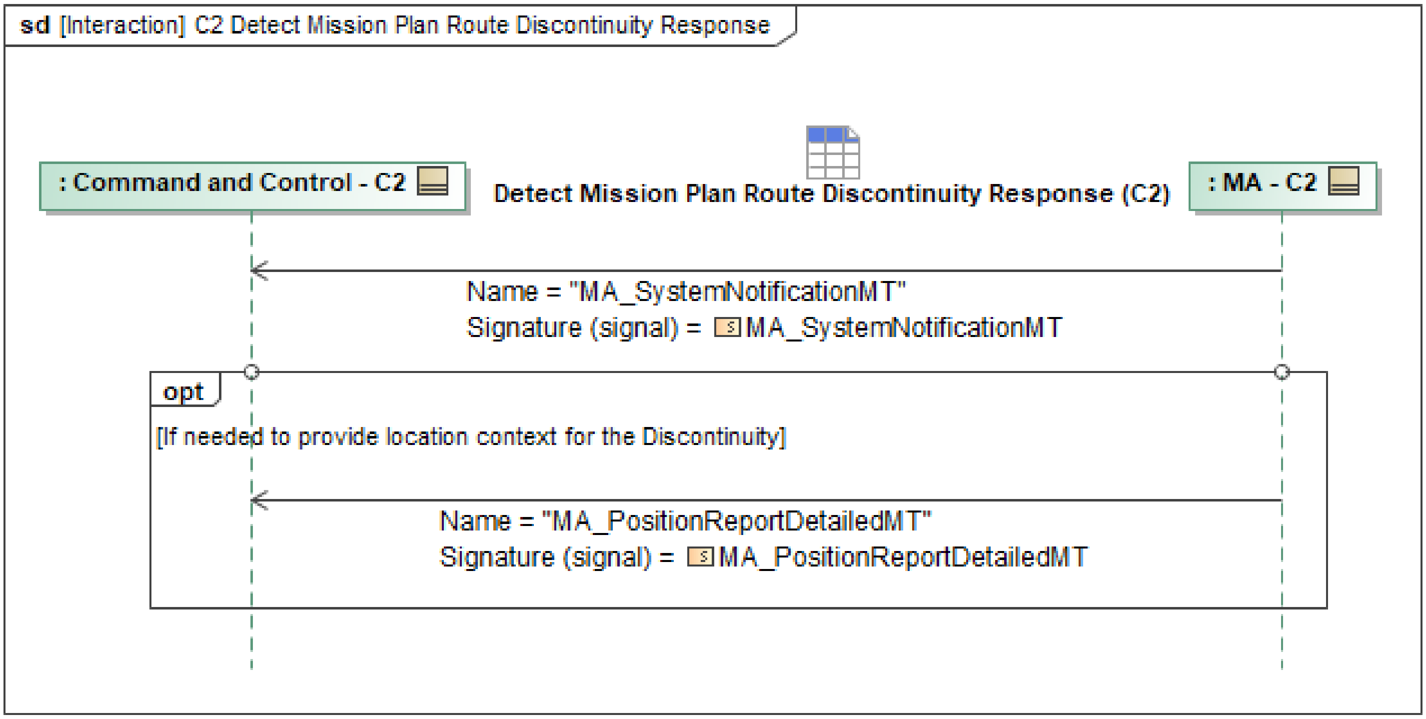


Figure 1-55: C2 Detect Mission Plan Route Discontinuity Response

Table 1-54: C2 Detect Mission Plan Route Discontinuity Response

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PositionReportDetailedMT: MA_PositionReportDetailedMT	None
MA_SystemNotificationMT: MA_SystemNotificationMT	AssociatedID: ID_Type Description: VisibleString1024Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.4 C2 Receive Abort Status

This interaction between MA and C2 periodically forwards the *MA_MissionPlanActivationStatus* and *RoutePlanExecutionStatus* from MA to C2. *MA_MissionPlanActivationStatusMT* is sent with information populated for *RoutePlanID* of the route plan that has been aborted and *MissionPlanActivationStatus:SubPlanActivationState:ActivationState* is *set to DEACTIVATED.*

Then, RoutePlanExecutionStatusMT is sent with information populated for PlanID of the route plan that has been aborted and a *PlanExecutionState set to Canceled.*

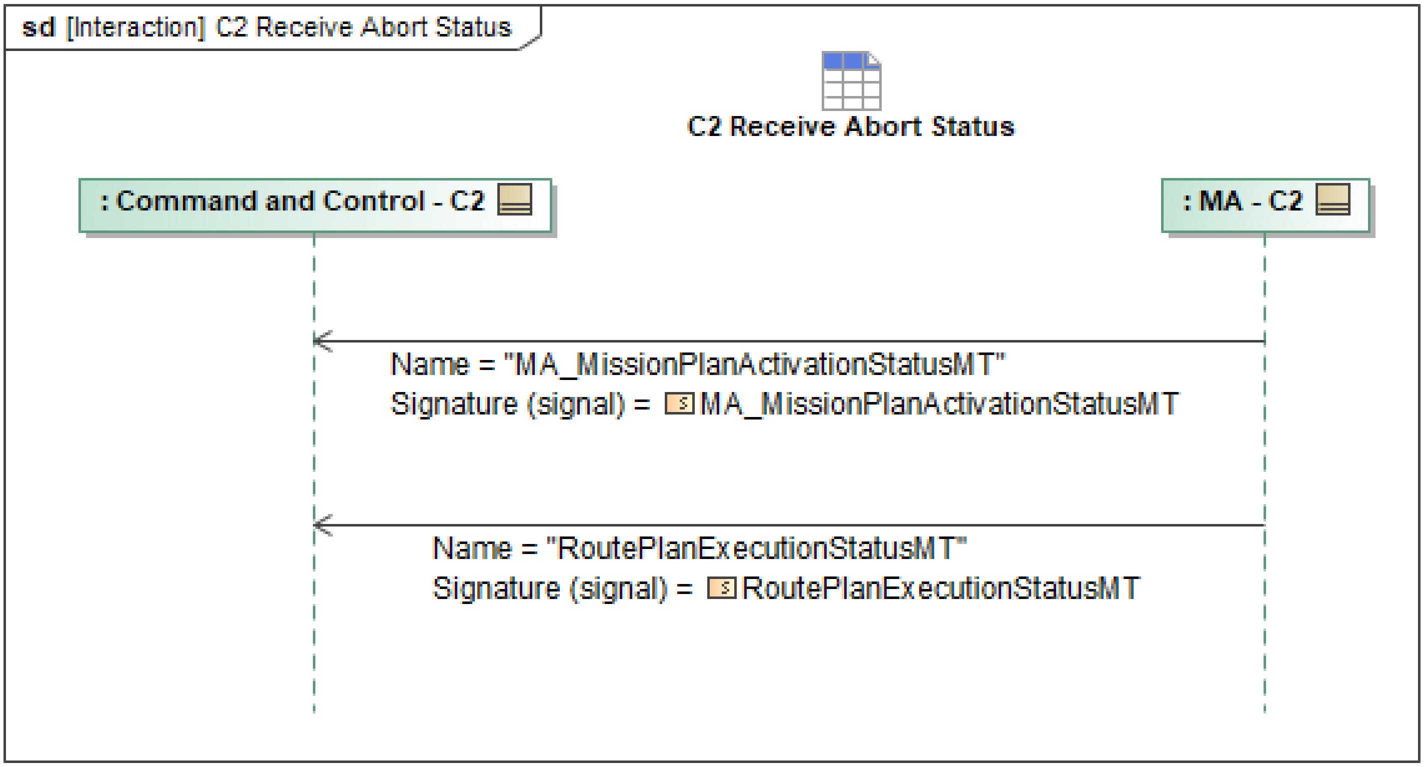


Figure 1-56: C2 Receive Abort Status

Table 1-55: C2 Receive Abort Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanActivationStatusMT: MA_MissionPlanActivationStatusMT	RoutePlanID: RoutePlanID_Type DEACTIVATED: ActivationState: PlanActivationStateEnum
RoutePlanExecutionStatusMT: RoutePlanExecutionStatusMT	PlanExecutionState: PlanExecutionStateEnum PlanID: RoutePlanID_Type CANCELED:

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.5 C2 Receive Plan Modification Notification

When MA updates a Plan, there may be C2 elements that need to be informed for situational awareness or synchronization reasons. This sequence describes the messaging that MA can use to broadcast the fact that MA has changed a Plan. If MA has modified a *MissionPlan* or a *SubPlan*, MA will send a *MA_PlanModificationRequestMT* to C2, requesting that C2 updates the Plan in question. Likewise, if MA has only modified a *RoutePlan*, MA will send a *RouteModificationRequestMT* to the same effect. Contained in the *MessageData* fields of these messages are fields that describe the Plan modifications which are populated as required to mirror the change that MA has already made to the Plan. Upon receipt of request, C2 will respond to MA with a *MA_PlanModificationRequestStatusMT* or *RouteModificationStatusMT* describing the status of the

Plan update.

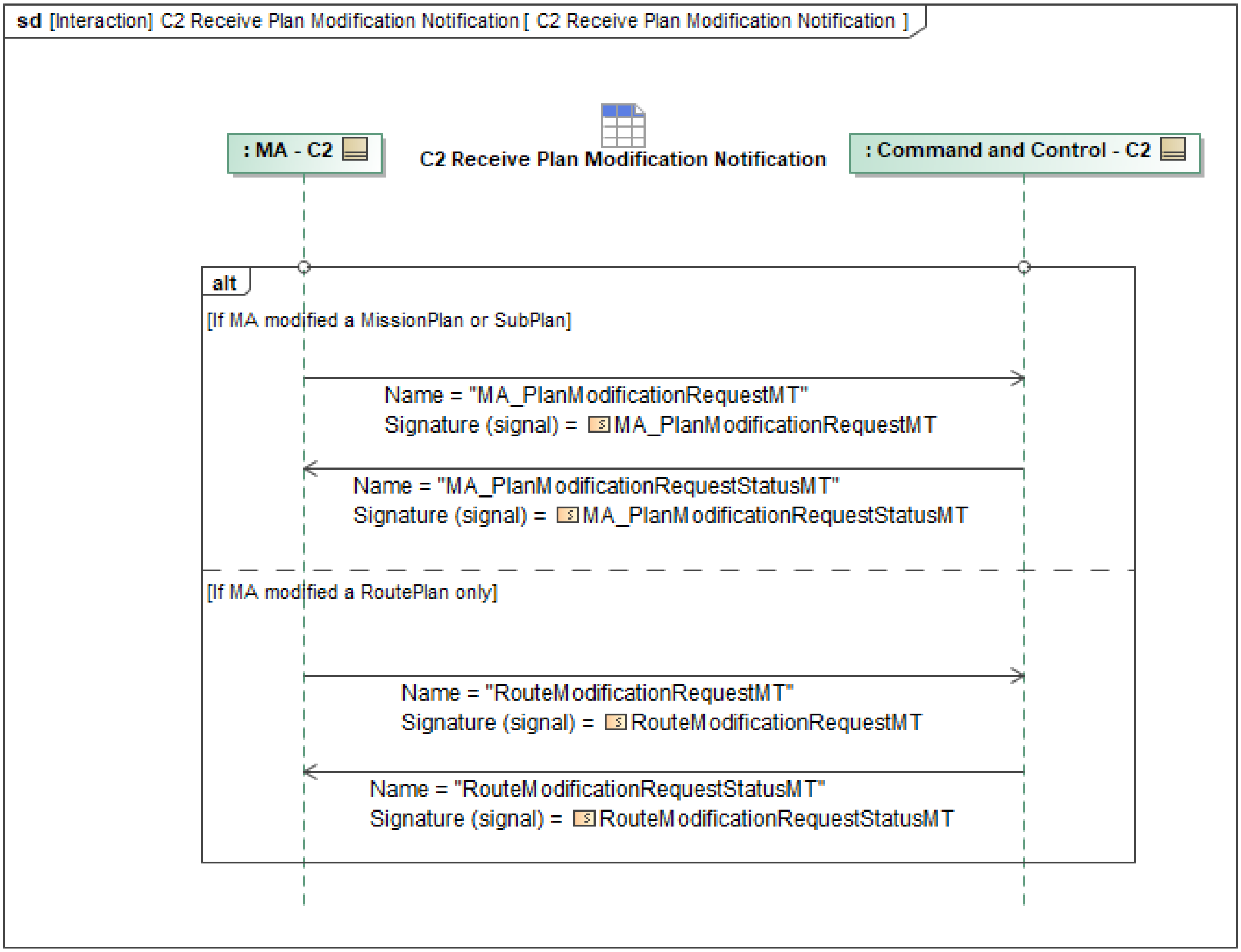


Figure 1-57: C2 Receive Plan Modification Notification

Table 1-56: C2 Receive Plan Modification Notification

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PlanModificationRequestMT: MA_PlanModificationRequestMT	None
MA_PlanModificationRequestStatusMT: MA_PlanModificationRequestStatusMT	None
RouteModificationRequestMT: RouteModificationRequestMT	None
RouteModificationRequestStatusMT: RouteModificationRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.6 C2 Request Plan Patch

In the event that C2 determines that a Plan Patch is necessary, C2 can request that MA perform a Plan Patch. If the desired Plan Patch is applicable to a Route Plan, C2 sends MA a *RouteModificationRequestMT.* The *RouteModificationRequestMT* contains a *Path* field and a *PathSegment* field that communicate one of three possible outcomes: the addition of a path, removal of a path, or update a path, through the *Add, RemoveID, or Update* fields*,* respectively. Contained in the *PathSegment* field, the *PathID* field identifies the Path which the segment changes apply to.

If the desired Plan Patch is applicable to any other type of Plan, C2 sends MA a *MA_PlanModificationRequestMT.* The *MA_PlanModificationRequest* contains a *PlanModificationDetails* field which allows for a number of potential plan modification methods. For Plan Patching, the *SubPlansModification* field will be used to either add a *SubPlan* (*SubPlanAdditions* field) or retract a *SubPlan (SubPlanRetractions* field*).* MA then responds with the corresponding **StatusMT message (RouteModificationRequestStatusMT or MA_PlanModificationRequestStatusMT),* which contains the *RequestID* and the *RequestProcessingState* which indicates the status of the request.

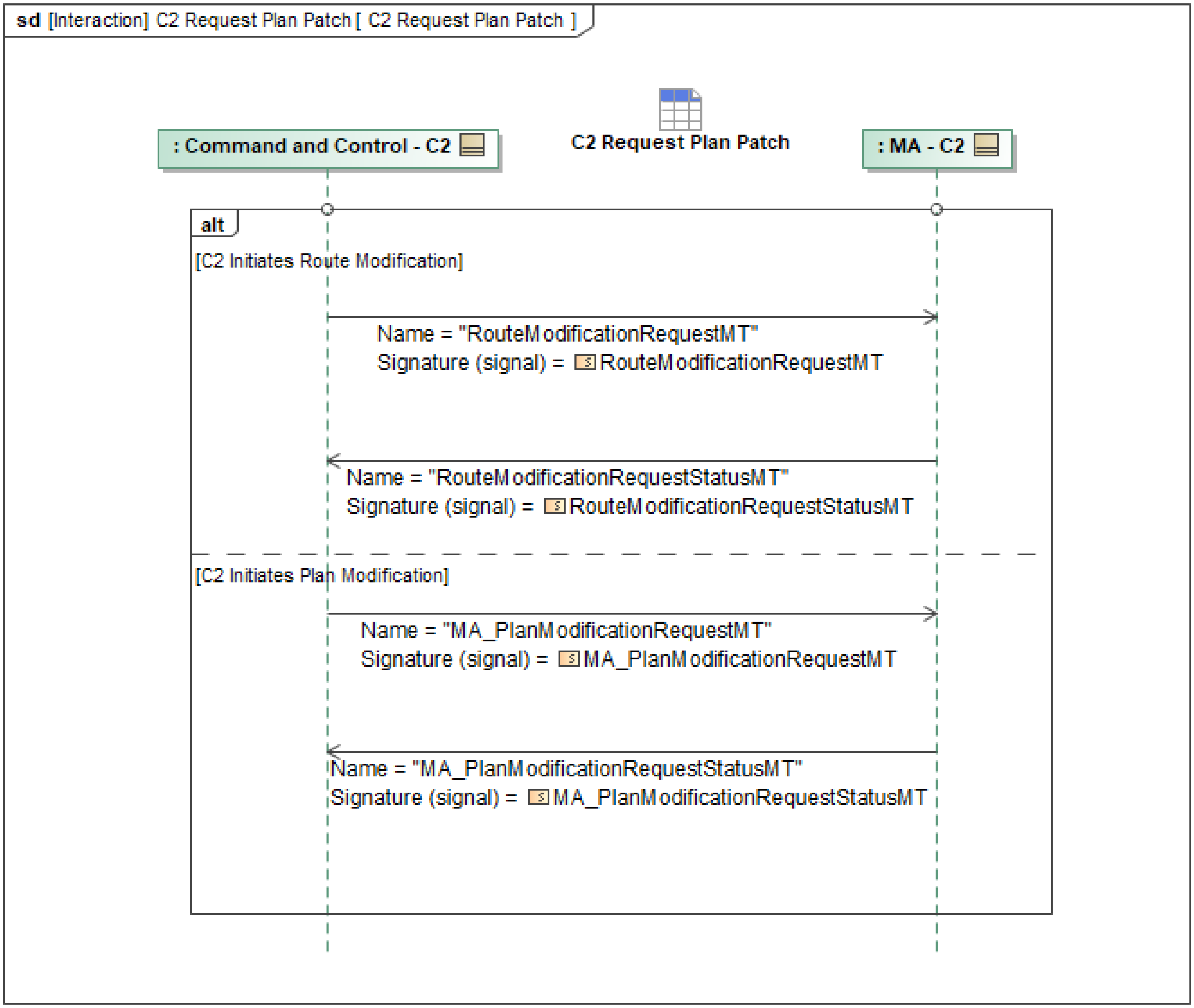


Figure 1-58: C2 Request Plan Patch

Table 1-57: C2 Request Plan Patch

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PlanModificationRequestMT: MA_PlanModificationRequestMT	None
MA_PlanModificationRequestStatusMT: MA_PlanModificationRequestStatusMT	None
RouteModificationRequestMT: RouteModificationRequestMT	None
RouteModificationRequestStatusMT: RouteModificationRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy

Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.7 Command Mission Replan

MA can be commanded by C2 to generate a new mission plan. C2 sends a *MA_MissionPlanCommandMT* which may contain inputs to indicate data and references to other messages and their data to be used to create the *MA_MissionPlanMT*. Upon receipt of the *MA_MissionPlanCommandMT*, if C2 is authorized, MA begins generating a new mission plan, responding with *MA_MissionPlanCommandStatusMT* as required by the command. When MA is done generating the mission plan, it will publish a *MA_MissionPlanMT* message for approval. *MA_MissionPlanMT* will define the *ObjectState*, *ExecutionSequence*, *MissionPlanCommandID*, and optionally *SubPlans* if relevant. MA will include the *ResultingPlanIdentifier* field in the *MA_MissionPlanCommandStatusMT_2* to allow C2 to identify which *MA_MissionPlanMT* message has been generated as a result of its original *MA_MissionPlanCommandMT.*

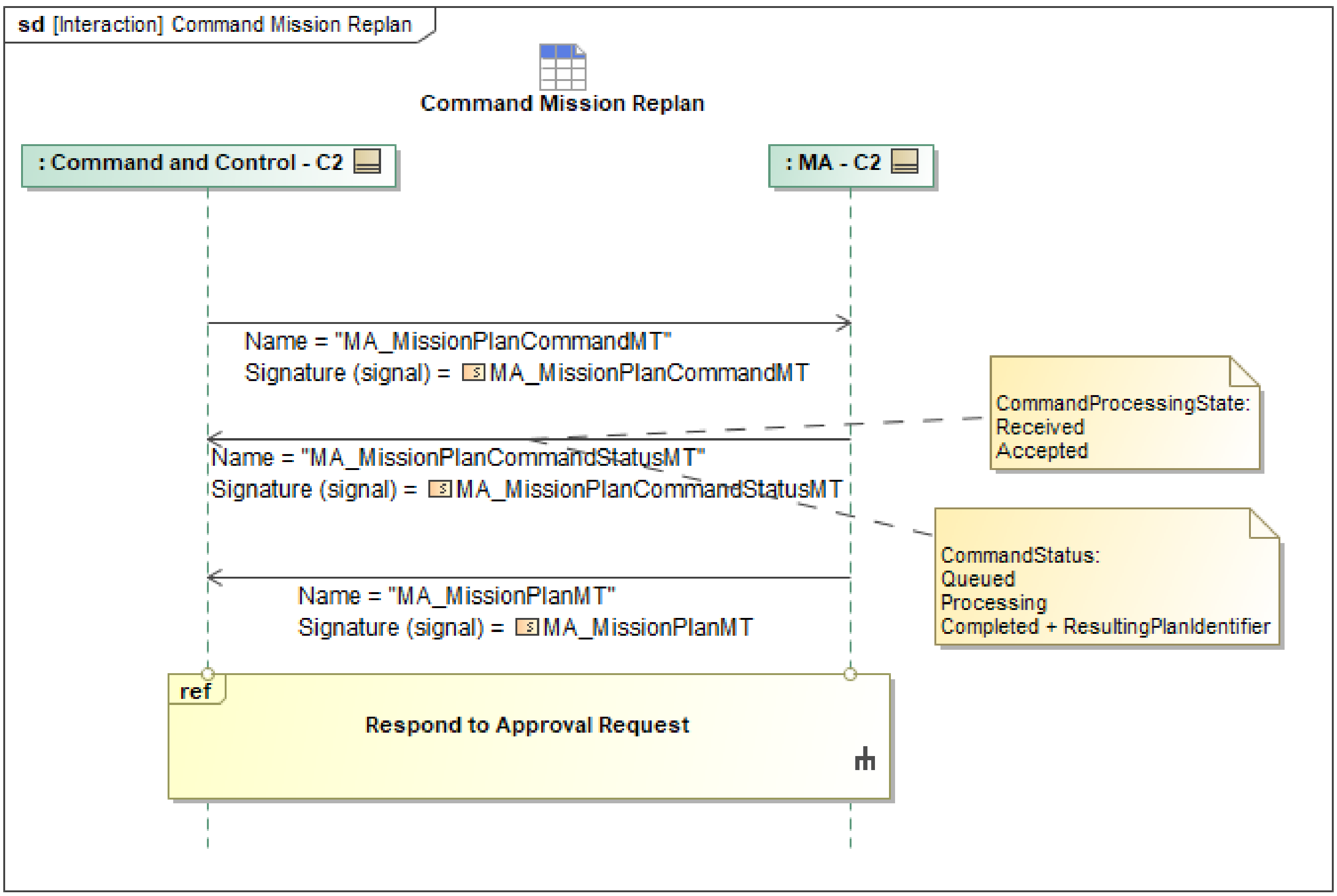


Figure 1-59: Command Mission Replan

Table 1-58: Command Mission Replan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanCommandMT: MA_MissionPlanCommandMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanCommandStatusMT: MA_MissionPlanCommandStatusMT	ResultingPlanIdentifier: PlansReferenceType
MA_MissionPlanMT: MA_MissionPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.8 MA Notifies C2 of Replanning

MA sends C2 an *MissionPlanStatusMT, RoutePlanStatusMT*, T*askPlanStatusMT*, *RouteActivityPlanStatusMT*, *ActivityPlanStatusMT*, or *ActionPlanStatusMT* to indicate replanning is required and has begun. The message used is based on the type of replanning that needs to be done (MissionPlan, RoutePlan, TaskPlan, RouteActivityPlan, ActivityPlan, or ActionPlan respectively). Within the message, *PlanningState* is set to *PLANNING* and *ValidationState* is set to *INVALID* to indicate replanning has begun due to an issue with the current plan.

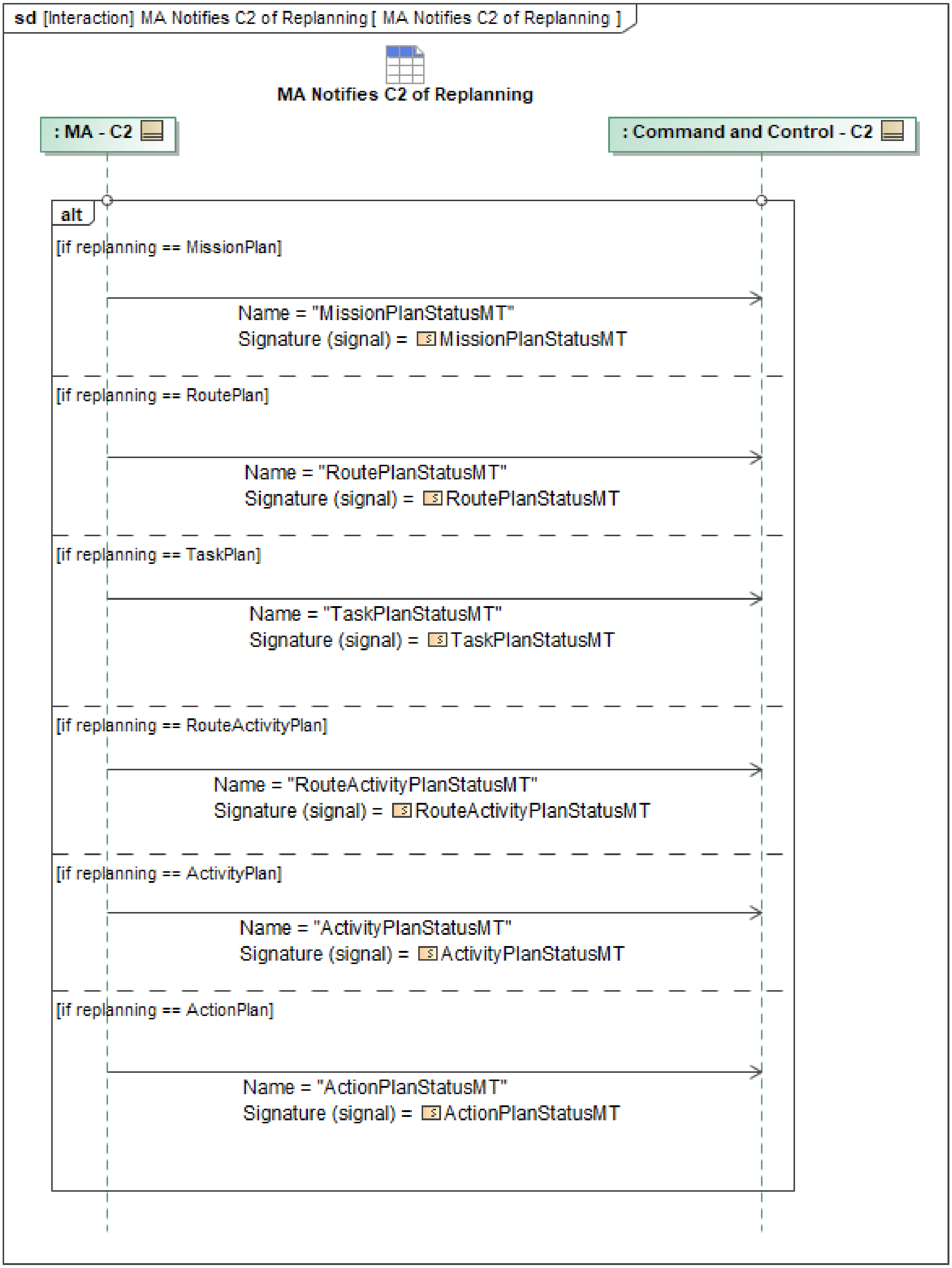


Figure 1-60: MA Notifies C2 of Replanning

Table 1-59: MA Notifies C2 of Replanning

Message Name (Name:Type)	Required Fields (Name:Type)
ActionPlanStatusMT: ActionPlanStatusMT	None
ActivityPlanStatusMT: ActivityPlanStatusMT	None
MissionPlanStatusMT: MissionPlanStatusMT	None
RouteActivityPlanStatusMT: RouteActivityPlanStatusMT	None
RoutePlanStatusMT: RoutePlanStatusMT	None
TaskPlanStatusMT: TaskPlanStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.9 MA Provides Replan Status to C2

This interaction enables MA to send additional *MA_MissionPlanCommandStatusMT* messages to C2 to report replanning status. This could be triggered by a contingency or other event that changed the planning or tasking status.

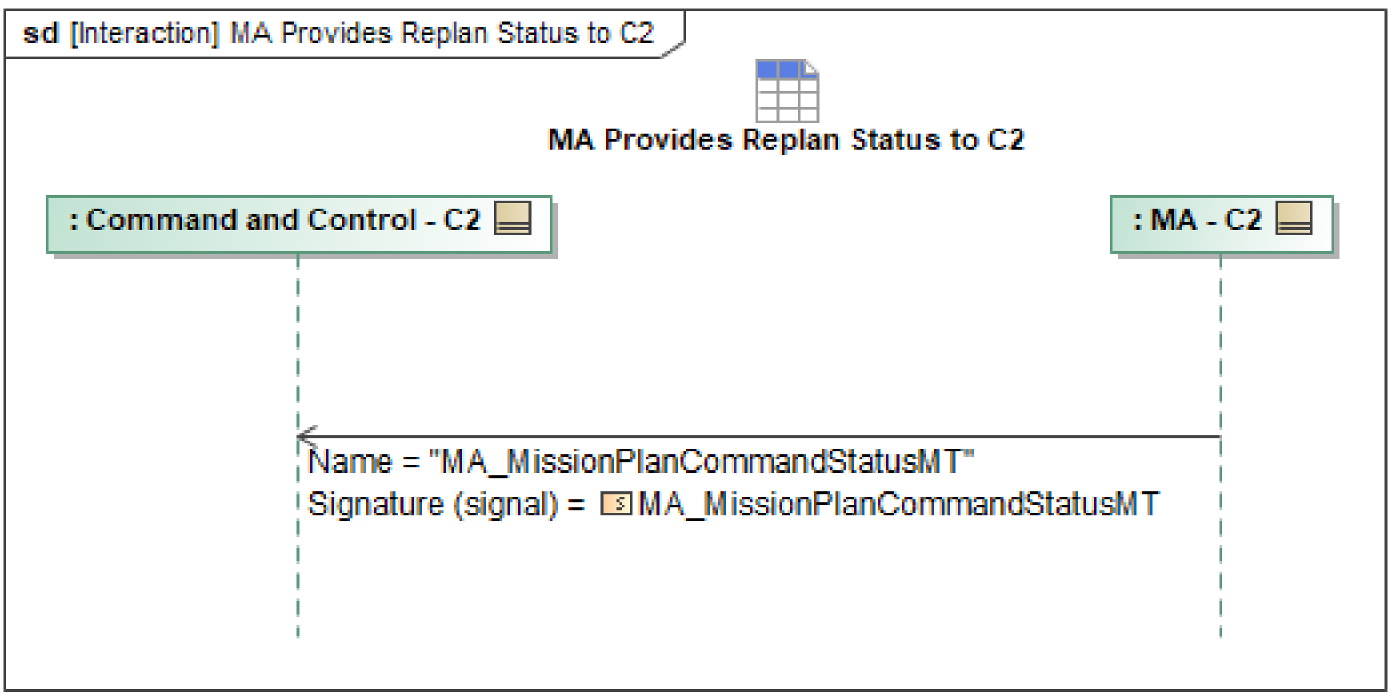


Figure 1-61: MA Provides Replan Status to C2

Table 1-60: MA Provides Replan Status to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanCommandStatusMT: MA_MissionPlanCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set

Tables section)

1.2.8.10 QB Commanded RTB

At any point during an active connection with MA, a C2 system shall be capable of issuing commanded control over MA Contingencies, assuming the C2 system is authorized under the appropriate RBAC level. The appropriate RBAC level shall indicate the ability for C2 to specify or command a contingency response at or when a qualifying triggering event occurs. In the case for a commanded "*Return to Base*" (RTB) contingency, an authorized C2 system can either activate an existing RTB Action by sending MA_ActionCommandMT with the associated CapabilityID, or it can send MA_ActionMT with ActionType set to RETURN_TO_BASE which is then activated by MA_ActionCommandMT. MA shall acknowledge this message sequence with a *MA_MissionPlanActivationCommandStatusMT*, with *MissionPlanActivationCommandID* confirming the command's identity, *ActivationStatus.CommandProcessingState* as "*ACCEPTED*", and *ActivationStatus.CommandStatus* set to "*COMPLETED*", *ActivationCommandByState.PlanActivationCommandState* set to "*ACTIVATED*", and *ActivationCommandByState.Plans* will be set to the RTB *Plan that was specified in the *MA_MissionPlanActivationCommandMT* to be activated.

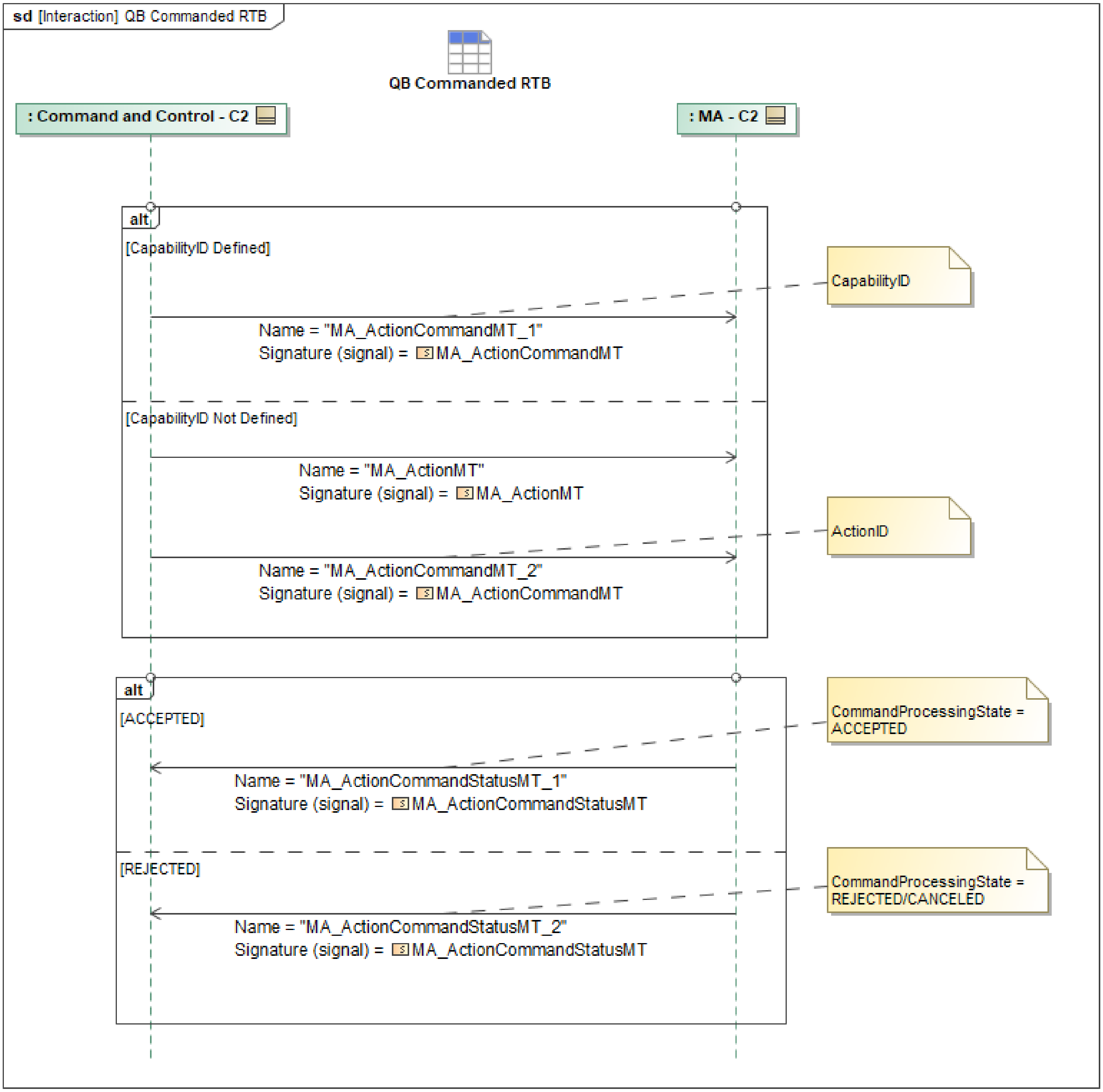


Figure 1-62: QB Commanded RTB

Table 1-61: QB Commanded RTB

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandMT_1: MA_ActionCommandMT	None
MA_ActionCommandMT_2: MA_ActionCommandMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandStatusMT_1: MA_ActionCommandStatusMT	None
MA_ActionCommandStatusMT_2: MA_ActionCommandStatusMT	None
MA_ActionMT: MA_ActionMT	ActionType: MA_ActionTypeEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.11 Query C2 for Missing Data

If MA identifies that there is Mission Plan data that is missing or inaccessible, most likely as part of pre-flight startup checks, it can request that C2 send the data.

This data is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may also respond with intermediate *QueryDataRequestStatusMT_2 and QueryDataRequestStatusMT_3* messages with *RequestProcessingStateEnum* set to indicate additional in-process states, such as “*QUEUED*” and “*PROCESSING*”, respectively.

When complete, the *QueryDataRequestStatusMT_3* message's *RequestProcessingStateEnum* is set to "*COMPLETED*" and contains the requested data.

When the original *QueryDataRequestMT*’s optional field *ResultsInNative* is populated with the matching message type, the underlying data management services instead publishes the requested data in its original or raw format as a separate message and indicates *QueryDataRequestStatusMT_3* message's *RequestProcessingStateEnum* to "*COMPLETED*".

If C2 is unable to complete the query it will ignore the *QueryDataRequestMT* command and reply with *QueryDataRequestStatusMT_4* where *RequestProcessingState* is set to "REJECTED". If the query is rejected because C2 does not have the requested information, the *Reason* field within *RequestProcessingStateReason* will be set to "UNKNOWN_ID". If the query is rejected because the missing requested information cannot be transmitted via UCI, the *Reason* field will be set instead to "INVALID_INPUT_PARAMETER".

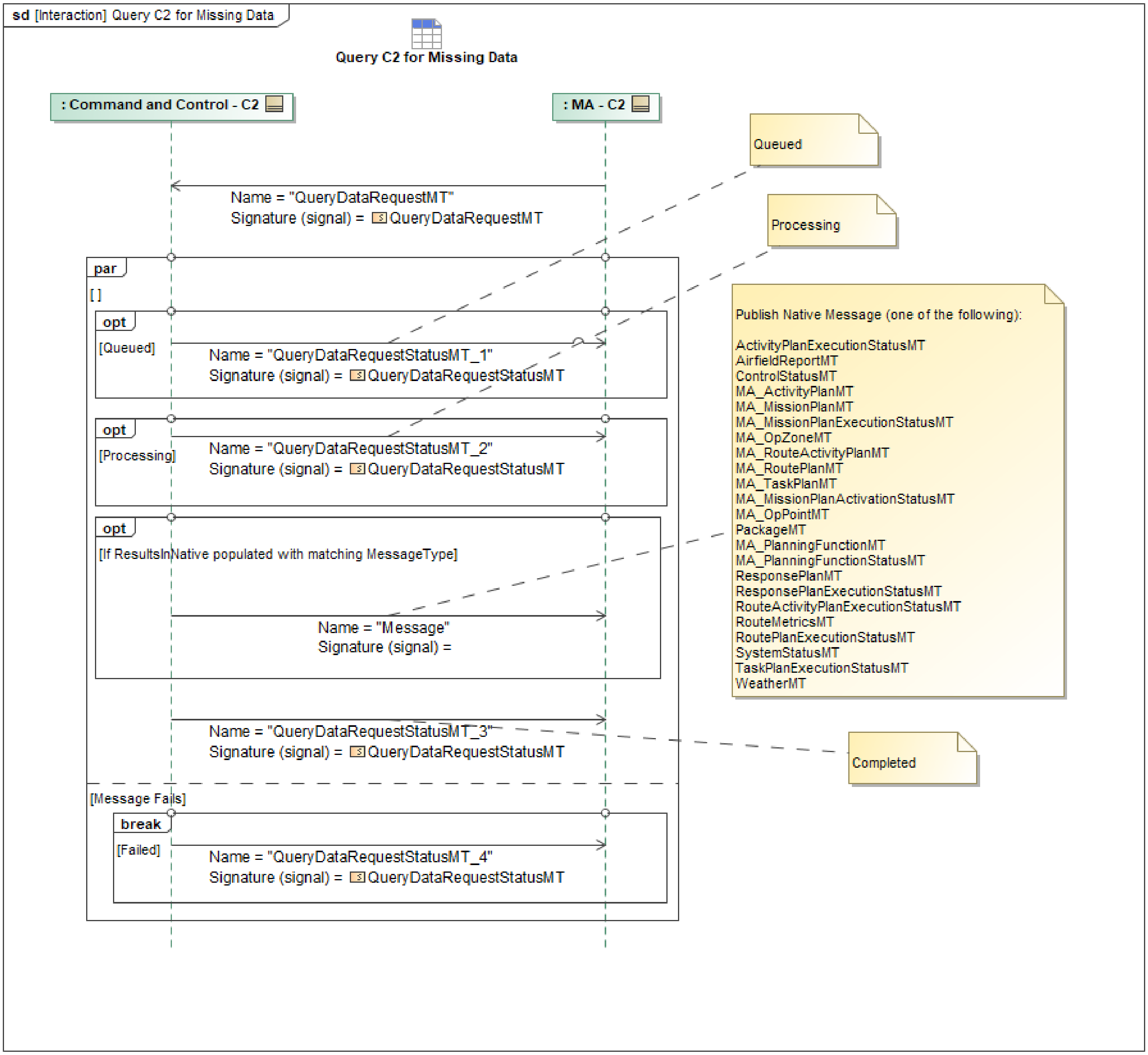


Figure 1-63: Query C2 for Missing Data

Table 1-62: Query C2 for Missing Data

Message Name (Name:Type)	Required Fields (Name:Type)
Message:	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.12 Query Deconfliction Data

Asset Deconfliction data, used to prevent the ACP from colliding with another platform, is defined by *MA_OpLineMT*, *MA_OpZoneMT*, and *MA_OpRoutingMT*.

MA_OpLineMT contains data used to convey a Corridor: Line with width along the designated Route, describing the passageway designated for Mission planning deconfliction for each ACP. Using the *Category* field set to *CORRIDOR*. The *LeftWidth* and *RightWidth* fields are required to describe the overall width of the corridor.

The defined data is to be used when deconflicting entities routes or generating routes to ensure proper separation. These parameters are applied at the vehicle level.

MA_OpZoneMT contains data used to convey Altitude. Data in the *Zone* field captures the shape of the region, along with the Max and Min altitudes. Describing the region designated for mission planning deconfliction for each ACP. The defined data is to be used when deconflicting entities routes or generating routes to ensure proper Altitude block compliance. These parameters are applied at the vehicle level.

MA_OpRoutingMT contains data used to convey a Bubble: The *SpecificBlueSeparation* and *SpecificRedSeparation* define parameters to use when deconflicting entity routes or generating routes to ensure compliance of the designated keep in and keep out zones. These parameters are applied at the vehicle type level.

Data is stored or cached in data manager services which may have a database backend. C2 can query this data via *QueryDataRequestMT* by including three instances of *MessageType* with their enumerations set to *OP_LINE*, *OP_ROUTING*, and *OP_ZONE* respectively. This will be responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may also respond with intermediate *QueryDataRequestStatusMT_1 and QueryDataRequestStatusMT_2* messages with *RequestProcessingStateEnum* set to indicate additional in-process states, such as "*QUEUED*" and "*PROCESSING*", respectively. When complete, the *QueryDataRequestStatusMT_3* message's *RequestProcessingState* field is set to "*COMPLETED*" and contains the requested data. When the original *QueryDataRequestMT*'s optional field *ResultsInNative* is populated, the underlying data management services publishes the requested data in its original or raw format as a separate message from the *QueryDataRequestStatusMT_3* message at completion.

If C2 is unable to complete the query it will ignore the *QueryDataRequestMT* command and reply with *QueryDataRequestStatusMT_4* where *RequestProcessingState* is set to "REJECTED". If the query is rejected because C2 does not have the requested information, the *Reason* field within *RequestProcessingStateReason* will be set to "UNKNOWN_ID". If the query is rejected because the missing requested information cannot be transmitted via UCI, the *Reason* field will be set instead to "INVALID_INPUT_PARAMETER".

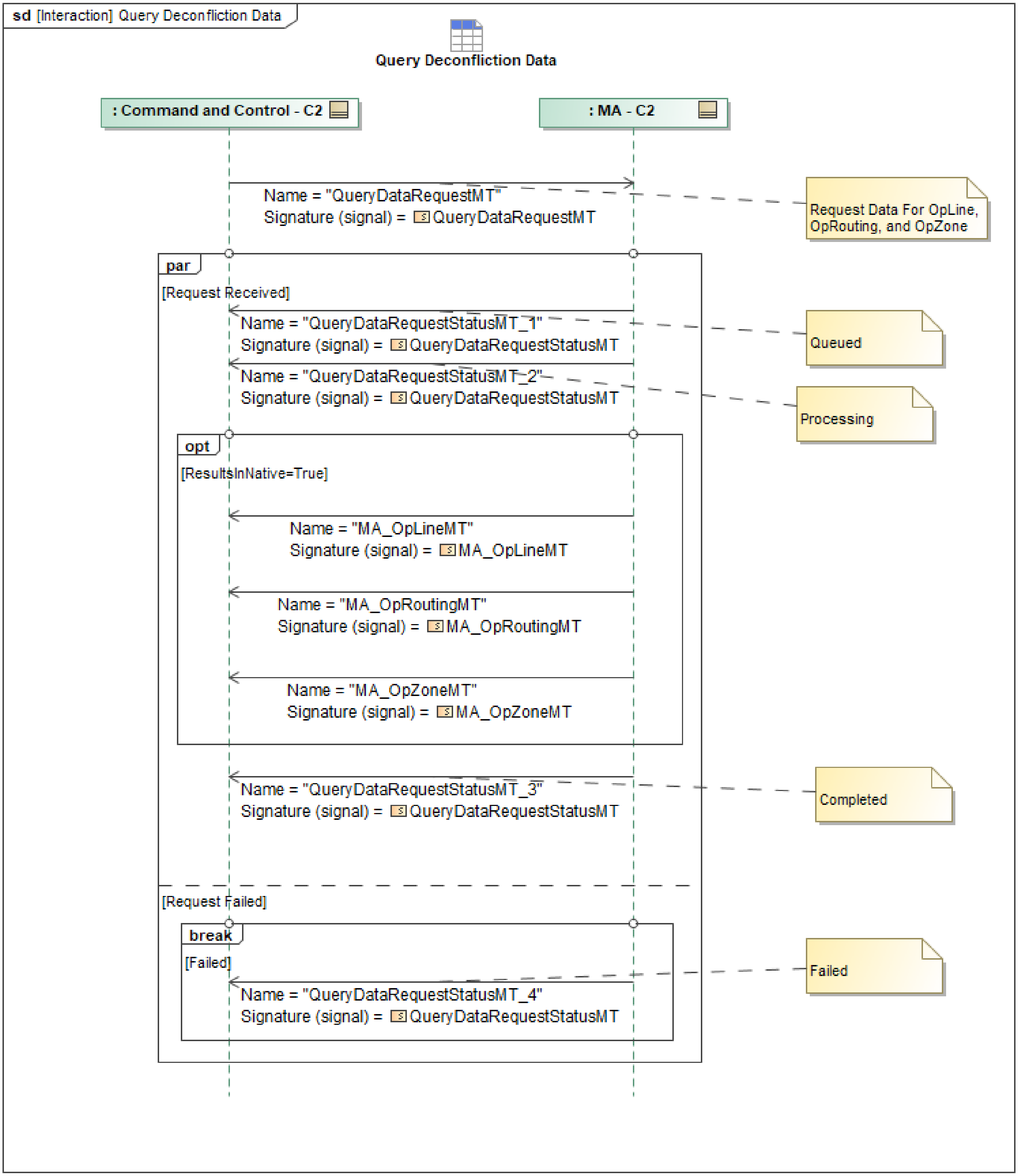


Figure 1-64: Query Deconfliction Data

Table 1-63: Query Deconfliction Data

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpLineMT: MA_OpLineMT	LeftWidth: DistanceType RightWidth: DistanceType
MA_OpRoutingMT: MA_OpRoutingMT	SpecificRedSeparation: SpecificRedSeparationType SpecificBlueSeparation: SpecificBlueSeparationType
MA_OpZoneMT: MA_OpZoneMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.13 Receive Autonomy Settings

MA sends *MA_PlanningFunctionMT* and *MA_PlanningFunctionStatusMT* periodically to allow C2 to programmatically and dynamically discover how autonomous a system is and how its associated planning services are configured.

MA_PlanningFunctionStatusMT allows a system to advertise up to 10 *PlanningInterface*s, one per UCI *PlanType*. Furthermore, the system can advertise its ability to create *Plans (*PlanInterface* set to "*PLAN_COMMAND*"), validate *Plans (PlanInterface set to "*PLAN_VALIDATION*"), automated *Plan activation (*PlanInterface* set to "*AUTOMATED_PLANNING*"), planning of *[Requirements]Plans* (*PlanningInterface* set to *"REQUIREMENT_MANAGEMENT"*), scoring plans (*PlanningInterface* set to *"PLAN_SCORES"*), computing metrics for plans (*PlanningInterface* set to *"PLAN_METRICS"*), and assessing plans (*PlanInterface* set to *"PLAN_ASSESSMENT"*). The *PlanActivationAutonomy* is similar to the Command field of *MissionPlanActivationCommandMT* and can be thought of as a template for any automated plan activations as to how far to advance the plan state machine upon activation. These abilities, when onboard, are critical for automated planning and plan activation. Beyond these settings, *MA_PlanningFunctionStatusMT* provides subscribers with high level details on how plans will be generated. For example, *ResultsInMissionPlan* specifies that the generated *Plans will be composed in a generated *MissionPlanMT*. Plan generation can be constrained by parts with *OutputPlanParts* or by other plans via *ConstrainingPlans* or *OtherSystemConstrainingPlans*.

MA_PlanningFunctionMT is an even more detailed representation into a system's autonomous planning functions. Its main difference is it exposes individual planning functions by providing their *PlanningProcessID*. This is useful if the desire is to issue a **PlanCommandMT* directly to an individual planning process. *PlanningFunctionSettingsCommandMT* can be paired with information discovered in *MA_PlanningFunctionMT* to reconfigure an individual planning function.

MA_PlanningFunctionMT's fields are similar to *MA_PlanningFunctionStatusMT* but with more detail by virtue of exposing settings of individual planners rather than an aggregation of a platform's planning abilities.

The interaction starts with each planning service publishing *MA_PlanningFunctionMT* with the (required) supported *PlanType* (*InterfacePlanTypeEnum*) and (required) *PlanInterface* (*PlanningInterfaceEnum*). It is optional for C2 to consume these initial *MA_PlanningFunctionMT* messages as MA will support subsequent querying for this content. MA can also optionally publish *MA_PlanningFunctionStatusMT* to indicate a high level summary of all *PlanType* (*InterfacePlanTypeEnum*) and *PlanInterface* (*PlanningInterfaceEnum*). These messages should be published shortly after the corresponding planning services are ready to consume **PlanCommandMTs*, **PlanValidationCommandMT*, *AssessmentRequest*, *PlanScoresRequest*, **MetricsRequest*, etc depending on the interface supported by that particular service.

For best information on a given planner's availability, C2 should consider the last published *MA_PlanningFunctionMT* by that planner's unique *PlanningFunctionID* or consider using the C2 - Query Autonomy Settings message exchange.

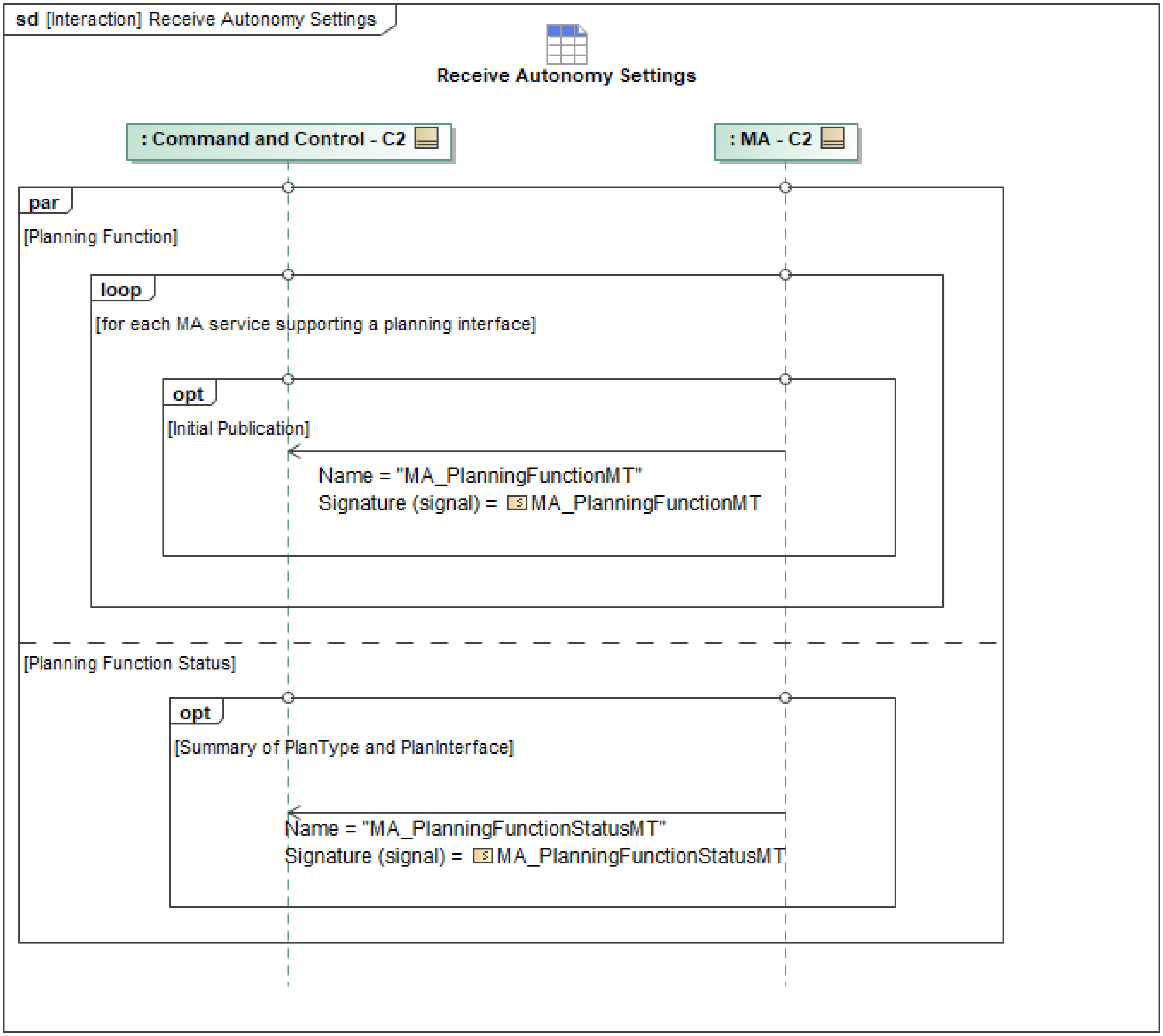


Figure 1-65: Receive Autonomy Settings

Table 1-64: Receive Autonomy Settings

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PlanningFunctionMT: MA_PlanningFunctionMT	None
MA_PlanningFunctionStatusMT: MA_PlanningFunctionStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.14 Receive MA Plan Status

This interaction describes the process of MA providing status on known plans back to C2, enabling C2 to monitor the completeness and validity of all stored plan data. Unlike periodic status updates, this interaction is strictly event-driven and is only initiated when a change in the local plan store results in a status change, aiding in bookkeeping, error detection, and synchronization between MA and C2. For example, a MissionPlan links to a specific RoutePlan, and if that RoutePlan is modified or inadvertently removed by C2, MA automatically runs a consistency check, invalidates the parent MissionPlan, and issues a status message indicating the plan is no longer valid. After an updated is completed, MA sends C2 a corresponding status message based on the *Plan that was updated via *MissionPlanStatusMT*, *TaskPlanStatusMT*, *ActivityPlanStatusMT*, *RouteActivityPlanStatusMT*, *RoutePlanStatusMT* or *ActionPlanStatusMT.* These messages convey the current completeness of the plan using the *PlanningState* field and the plan's integrity using the *ValidationState* field.

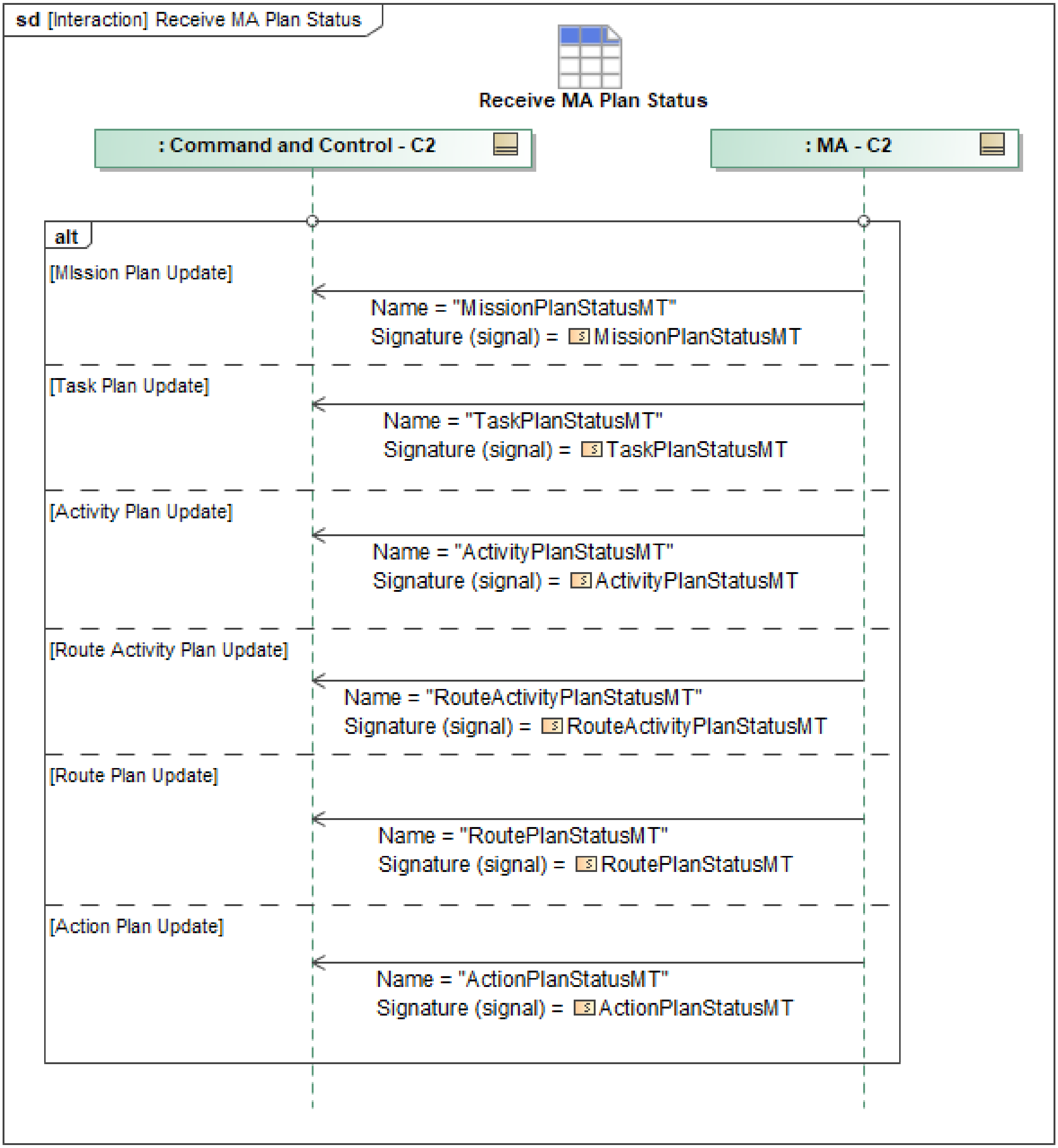


Figure 1-66: Receive MA Plan Status

Table 1-65: Receive MA Plan Status

Message Name (Name:Type)	Required Fields (Name:Type)
ActionPlanStatusMT: ActionPlanStatusMT	None
ActivityPlanStatusMT: ActivityPlanStatusMT	None
MissionPlanStatusMT: MissionPlanStatusMT	None
RouteActivityPlanStatusMT: RouteActivityPlanStatusMT	None
RoutePlanStatusMT: RoutePlanStatusMT	None
TaskPlanStatusMT: TaskPlanStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.15 Respond to Airfield Report Query

AirfieldReportMT contains Airfield Data and is stored or cached in data manager services which may have a database backend. This data is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may also respond with intermediate *QueryDataRequestStatusMT_1 and QueryDataRequestStatusMT_2* messages with *RequestProcessingStateEnum* set to indicate additional in-process states, such as "*QUEUED*" and "*PROCESSING*", respectively.

When complete, the *QueryDataRequestStatusMT_3* message's *RequestProcessingStateEnum* is set to "*COMPLETED*" and contains the requested data in the *Result* field.

When the original *QueryDataRequestMT*'s optional field *ResultsInNative* is populated with the matching message type, the underlying data management services instead publishes the requested data in its original or raw format as a separate message and indicates *QueryDataRequestStatusMT_3* message's *RequestProcessingStateEnum* to "*COMPLETED*".

If C2 is unable to complete the query it will ignore the *QueryDataRequestMT* command and reply with *QueryDataRequestStatusMT_4* where *RequestProcessingState* is set to "REJECTED". If the query is rejected because C2 does not have the requested information, the *Reason* field within *RequestProcessingStateReason* will be set to "UNKNOWN_ID". If the query is rejected because the missing requested information cannot be transmitted via UCI, the *Reason* field will be set instead to "INVALID_INPUT_PARAMETER".

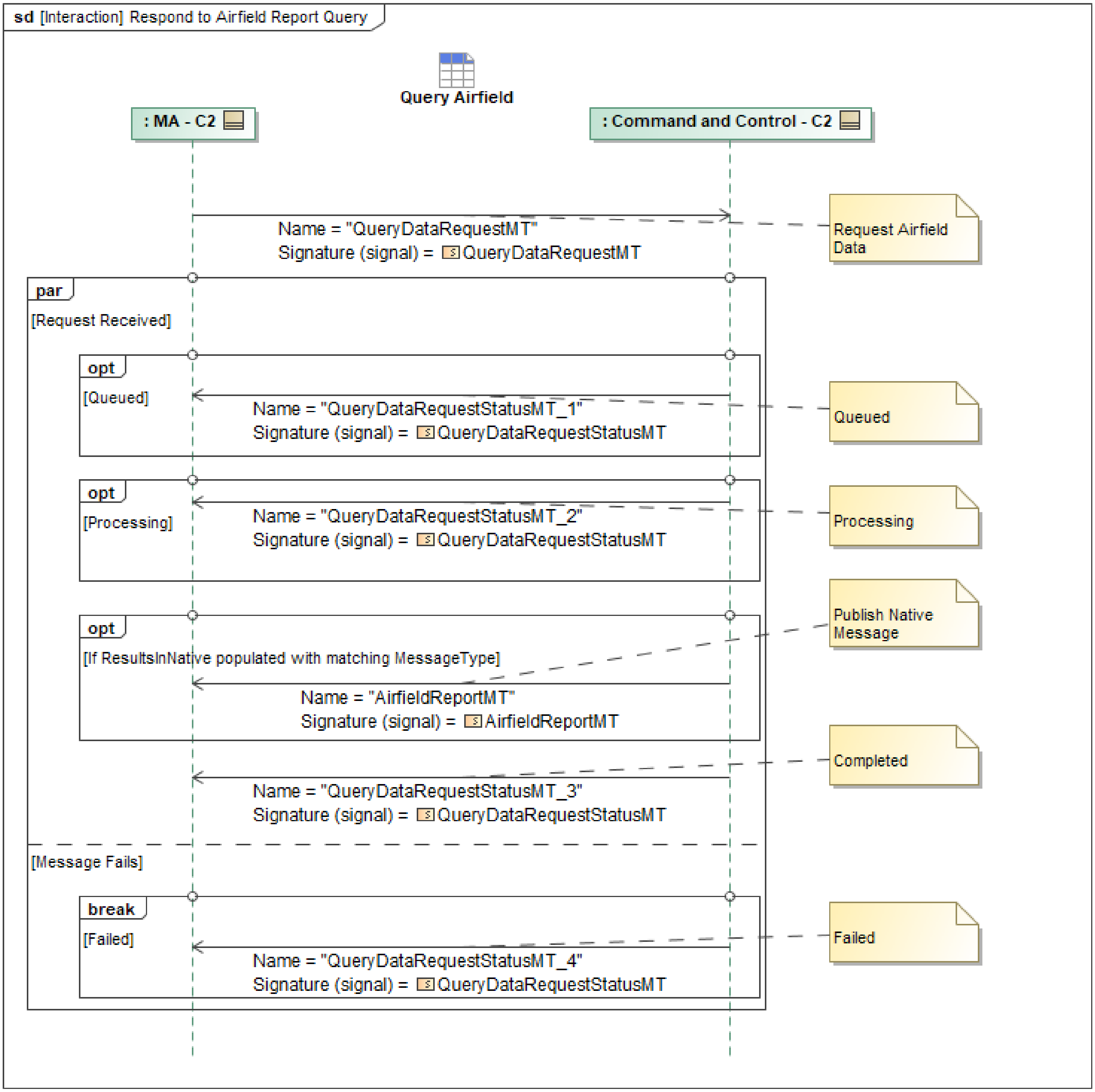


Figure 1-67: Respond to Airfield Report Query

Table 1-66: Respond to Airfield Report Query

Message Name (Name:Type)	Required Fields (Name:Type)
AirfieldReportMT: AirfieldReportMT	None
QueryDataRequestMT: QueryDataRequestMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.16 Send Response Plan

The diagram contains the collection of unique plan messages sent to Mission Autonomy following the logic described below for a contingency plan represented in the diagram. C2 systems will send an *MA_ApprovalPolicyMT* message for each unique contingency plan. The *DataRecordInstanceID* is included in the message to include tracking ID for the approval policy.

Response Plan Data is sent the via a *ResponsePlanMT* message for the plan that is being shared with Mission Autonomy. If the message being sent represents the first instance of the plan, the *ObjectState* field is set to "*NEW*". When sending an update to revise an existing plan, *ObjectState* is set to “*UPDATED*”. The response plan message is populated with the definition to detail the plan being conveyed. MA will determine whether C2 is authorized to command any plans.

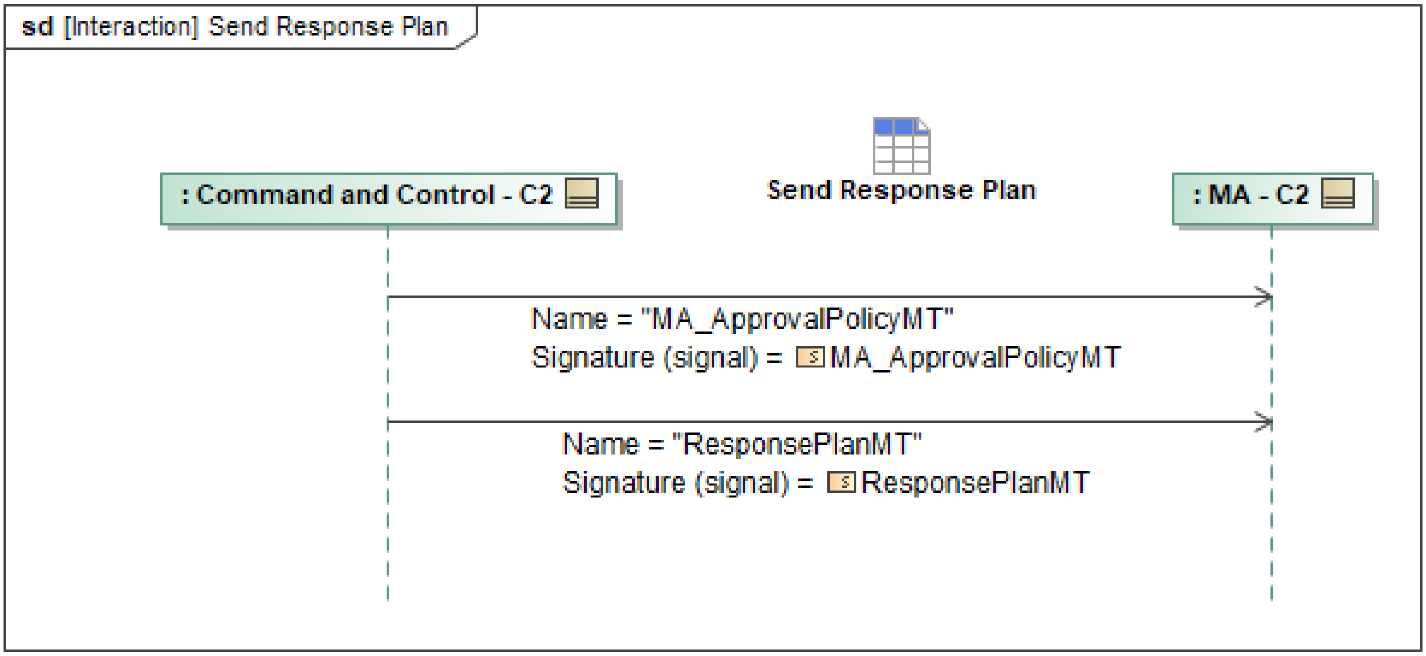


Figure 1-68: Send Response Plan

Table 1-67: Send Response Plan

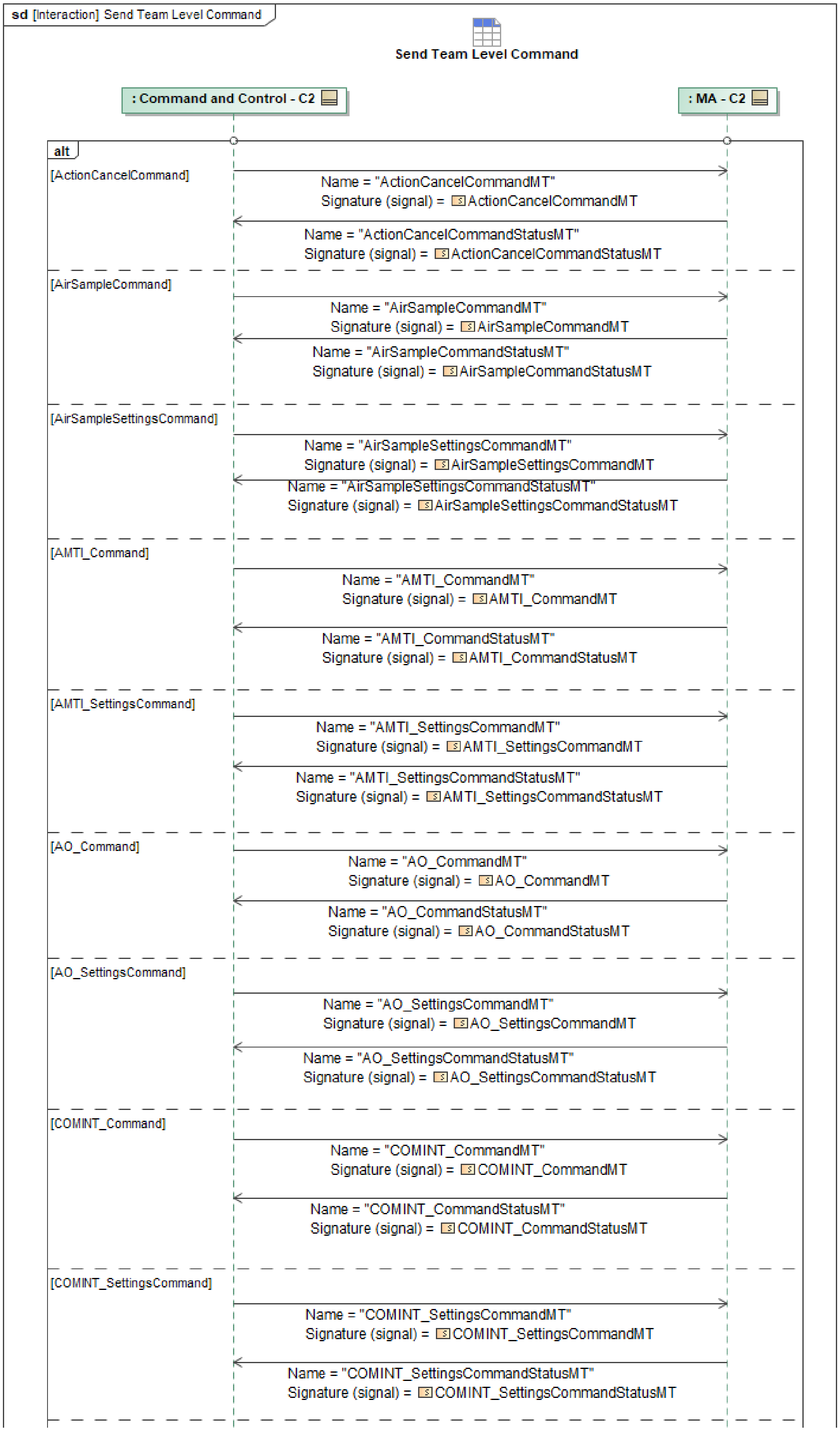
Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalPolicyMT: MA_ApprovalPolicyMT	DataRecordInstanceID: DataRecordInstanceID_Type
ResponsePlanMT: ResponsePlanMT	ObjectState: ObjectStateEnum

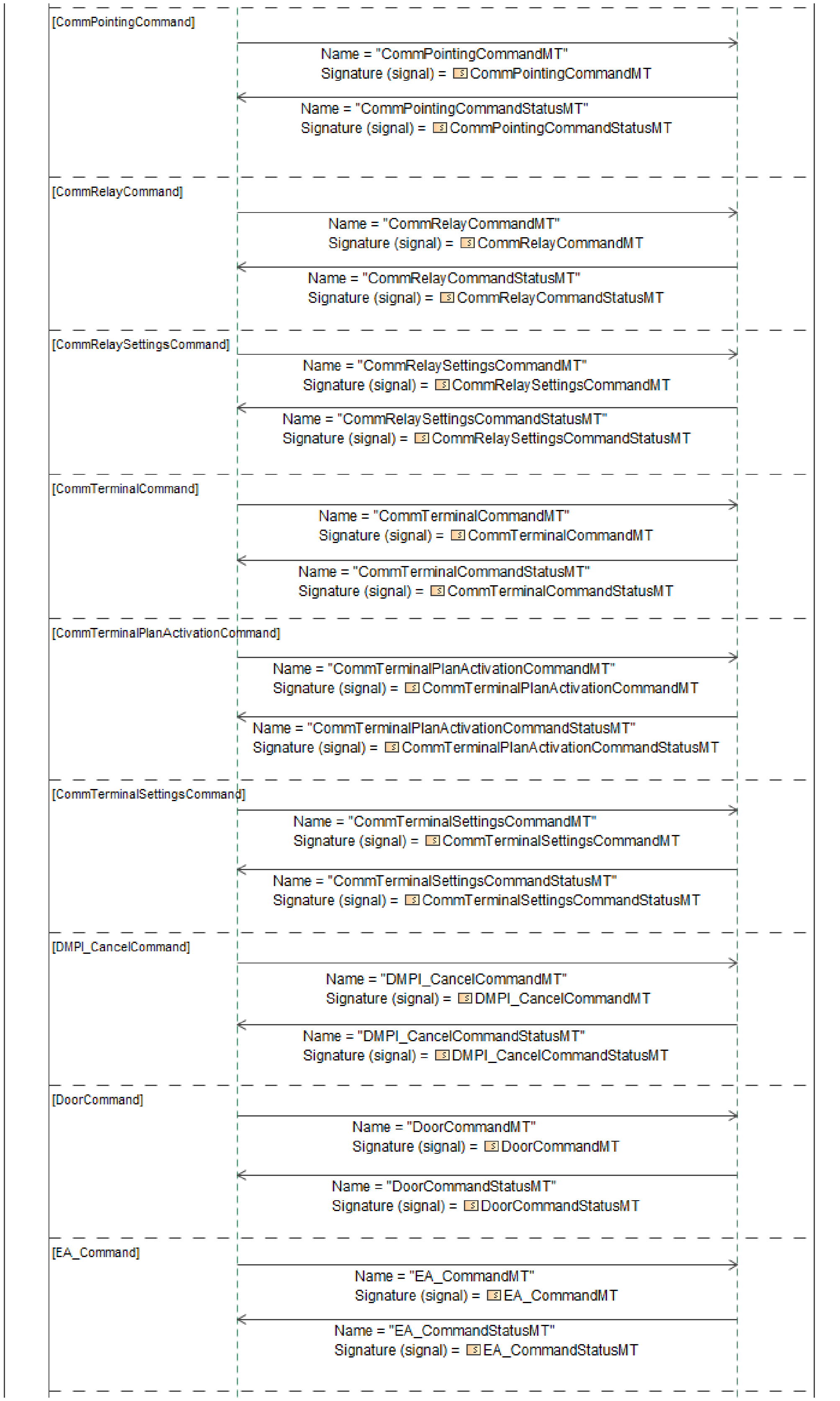
Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

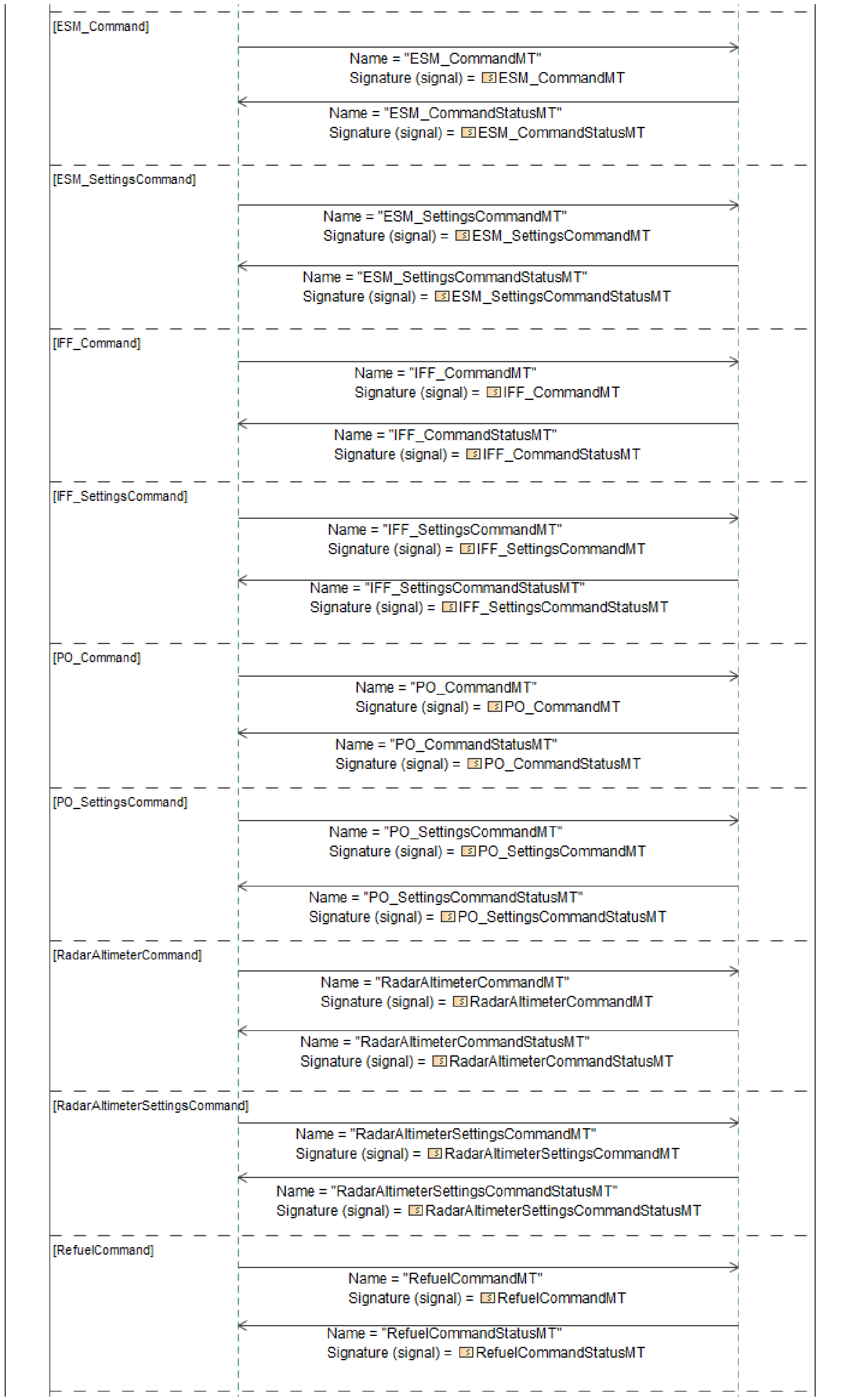
1.2.8.17 Send Team Level Command

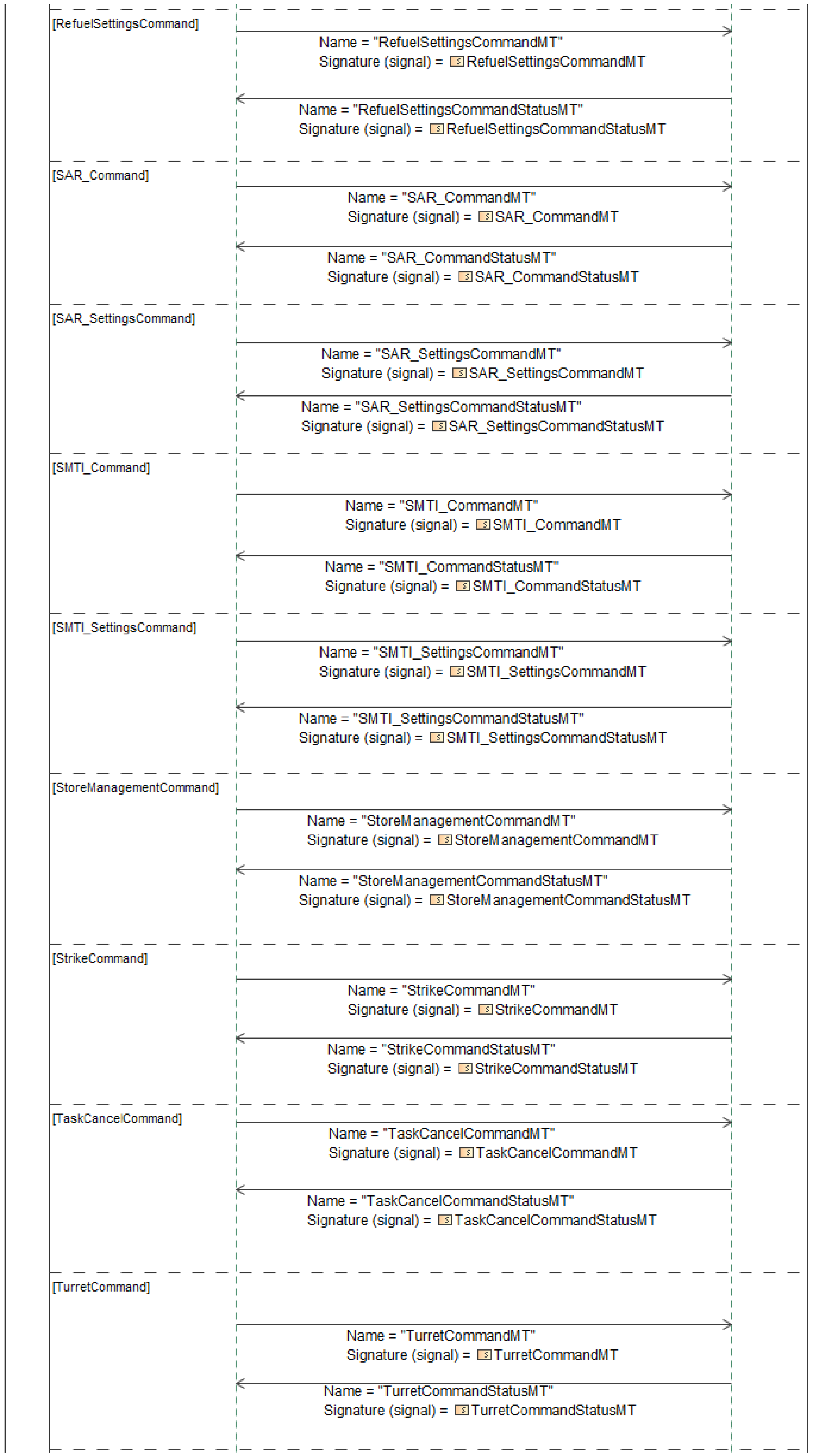
Package level tasking can be commanded to a team by C2. C2 sends a package level command (any UCI **CommandMT*) to the team lead MA, where the command can be decomposed or forwarded to the appropriate team members. Team lead MA responds to the command with a **CommandStatusMT* message that corresponds to the original command.

MA will respond to the original **CommandMT* message with a **CommandStatusMT* message response.


sd [Interaction] Send Team Level Command
Send Team Level Command
: Command and Control - C2
: MA - C2
alt
[ActionCancelCommand]
Name = "ActionCancelCommandMT"
Signature (signal) = ActionCancelCommandMT
Name = "ActionCancelCommandStatusMT"
Signature (signal) = ActionCancelCommandStatusMT
[AirSampleCommand]
Name = "AirSampleCommandMT"
Signature (signal) = AirSampleCommandMT
Name = "AirSampleCommandStatusMT"
Signature (signal) = AirSampleCommandStatusMT
[AirSampleSettingsCommand]
Name = "AirSampleSettingsCommandMT"
Signature (signal) = AirSampleSettingsCommandMT
Name = "AirSampleSettingsCommandStatusMT"
Signature (signal) = AirSampleSettingsCommandStatusMT
[AMTI_Command]
Name = "AMTI_CommandMT"
Signature (signal) = AMTI_CommandMT
Name = "AMTI_CommandStatusMT"
Signature (signal) = AMTI_CommandStatusMT
[AMTI_SettingsCommand]
Name = "AMTI_SettingsCommandMT"
Signature (signal) = AMTI_SettingsCommandMT
Name = "AMTI_SettingsCommandStatusMT"
Signature (signal) = AMTI_SettingsCommandStatusMT
[AO_Command]
Name = "AO_CommandMT"
Signature (signal) = AO_CommandMT
Name = "AO_CommandStatusMT"
Signature (signal) = AO_CommandStatusMT
[AO_SettingsCommand]
Name = "AO_SettingsCommandMT"
Signature (signal) = AO_SettingsCommandMT
Name = "AO_SettingsCommandStatusMT"
Signature (signal) = AO_SettingsCommandStatusMT
[COMINT_Command]
Name = "COMINT_CommandMT"
Signature (signal) = COMINT_CommandMT
Name = "COMINT_CommandStatusMT"
Signature (signal) = COMINT_CommandStatusMT
[COMINT_SettingsCommand]
Name = "COMINT_SettingsCommandMT"
Signature (signal) = COMINT_SettingsCommandMT
Name = "COMINT_SettingsCommandStatusMT"
Signature (signal) = COMINT_SettingsCommandStatusMT


[CommPointingCommand]
Name = "CommPointingCommandMT"
Signature (signal) = CommPointingCommandMT
Name = "CommPointingCommandStatusMT"
Signature (signal) = CommPointingCommandStatusMT
[CommRelayCommand]
Name = "CommRelayCommandMT"
Signature (signal) = CommRelayCommandMT
Name = "CommRelayCommandStatusMT"
Signature (signal) = CommRelayCommandStatusMT
[CommRelaySettingsCommand]
Name = "CommRelaySettingsCommandMT"
Signature (signal) = CommRelaySettingsCommandMT
Name = "CommRelaySettingsCommandStatusMT"
Signature (signal) = CommRelaySettingsCommandStatusMT
[CommTerminalCommand]
Name = "CommTerminalCommandMT"
Signature (signal) = CommTerminalCommandMT
Name = "CommTerminalCommandStatusMT"
Signature (signal) = CommTerminalCommandStatusMT
[CommTerminalPlanActivationCommand]
Name = "CommTerminalPlanActivationCommandMT"
Signature (signal) = CommTerminalPlanActivationCommandMT
Name = "CommTerminalPlanActivationCommandStatusMT"
Signature (signal) = CommTerminalPlanActivationCommandStatusMT
[CommTerminalSettingsCommand]
Name = "CommTerminalSettingsCommandMT"
Signature (signal) = CommTerminalSettingsCommandMT
Name = "CommTerminalSettingsCommandStatusMT"
Signature (signal) = CommTerminalSettingsCommandStatusMT
[DMPI_CancelCommand]
Name = "DMPI_CancelCommandMT"
Signature (signal) = DMPI_CancelCommandMT
Name = "DMPI_CancelCommandStatusMT"
Signature (signal) = DMPI_CancelCommandStatusMT
[DoorCommand]
Name = "DoorCommandMT"
Signature (signal) = DoorCommandMT
Name = "DoorCommandStatusMT"
Signature (signal) = DoorCommandStatusMT
[EA_Command]
Name = "EA_CommandMT"
Signature (signal) = EA_CommandMT
Name = "EA_CommandStatusMT"
Signature (signal) = EA_CommandStatusMT


[ESM_Command]
Name = "ESM_CommandMT"
Signature (signal) = ESM_CommandMT
Name = "ESM_CommandStatusMT"
Signature (signal) = ESM_CommandStatusMT
[ESM_SettingsCommand]
Name = "ESM_SettingsCommandMT"
Signature (signal) = ESM_SettingsCommandMT
Name = "ESM_SettingsCommandStatusMT"
Signature (signal) = ESM_SettingsCommandStatusMT
[IFF_Command]
Name = "IFF_CommandMT"
Signature (signal) = IFF_CommandMT
Name = "IFF_CommandStatusMT"
Signature (signal) = IFF_CommandStatusMT
[IFF_SettingsCommand]
Name = "IFF_SettingsCommandMT"
Signature (signal) = IFF_SettingsCommandMT
Name = "IFF_SettingsCommandStatusMT"
Signature (signal) = IFF_SettingsCommandStatusMT
[PO_Command]
Name = "PO_CommandMT"
Signature (signal) = PO_CommandMT
Name = "PO_CommandStatusMT"
Signature (signal) = PO_CommandStatusMT
[PO_SettingsCommand]
Name = "PO_SettingsCommandMT"
Signature (signal) = PO_SettingsCommandMT
Name = "PO_SettingsCommandStatusMT"
Signature (signal) = PO_SettingsCommandStatusMT
[RadarAltimeterCommand]
Name = "RadarAltimeterCommandMT"
Signature (signal) = RadarAltimeterCommandMT
Name = "RadarAltimeterCommandStatusMT"
Signature (signal) = RadarAltimeterCommandStatusMT
[RadarAltimeterSettingsCommand]
Name = "RadarAltimeterSettingsCommandMT"
Signature (signal) = RadarAltimeterSettingsCommandMT
Name = "RadarAltimeterSettingsCommandStatusMT"
Signature (signal) = RadarAltimeterSettingsCommandStatusMT
[RefuelCommand]
Name = "RefuelCommandMT"
Signature (signal) = RefuelCommandMT
Name = "RefuelCommandStatusMT"
Signature (signal) = RefuelCommandStatusMT


[RefuelSettingsCommand]
Name = "RefuelSettingsCommandMT"
Signature (signal) = RefuelSettingsCommandMT
Name = "RefuelSettingsCommandStatusMT"
Signature (signal) = RefuelSettingsCommandStatusMT
[SAR_Command]
Name = "SAR_CommandMT"
Signature (signal) = SAR_CommandMT
Name = "SAR_CommandStatusMT"
Signature (signal) = SAR_CommandStatusMT
[SAR_SettingsCommand]
Name = "SAR_SettingsCommandMT"
Signature (signal) = SAR_SettingsCommandMT
Name = "SAR_SettingsCommandStatusMT"
Signature (signal) = SAR_SettingsCommandStatusMT
[SMTI_Command]
Name = "SMTI_CommandMT"
Signature (signal) = SMTI_CommandMT
Name = "SMTI_CommandStatusMT"
Signature (signal) = SMTI_CommandStatusMT
[SMTI_SettingsCommand]
Name = "SMTI_SettingsCommandMT"
Signature (signal) = SMTI_SettingsCommandMT
Name = "SMTI_SettingsCommandStatusMT"
Signature (signal) = SMTI_SettingsCommandStatusMT
[StoreManagementCommand]
Name = "StoreManagementCommandMT"
Signature (signal) = StoreManagementCommandMT
Name = "StoreManagementCommandStatusMT"
Signature (signal) = StoreManagementCommandStatusMT
[StrikeCommand]
Name = "StrikeCommandMT"
Signature (signal) = StrikeCommandMT
Name = "StrikeCommandStatusMT"
Signature (signal) = StrikeCommandStatusMT
[TaskCancelCommand]
Name = "TaskCancelCommandMT"
Signature (signal) = TaskCancelCommandMT
Name = "TaskCancelCommandStatusMT"
Signature (signal) = TaskCancelCommandStatusMT
[TurretCommand]
Name = "TurretCommandMT"
Signature (signal) = TurretCommandMT
Name = "TurretCommandStatusMT"
Signature (signal) = TurretCommandStatusMT

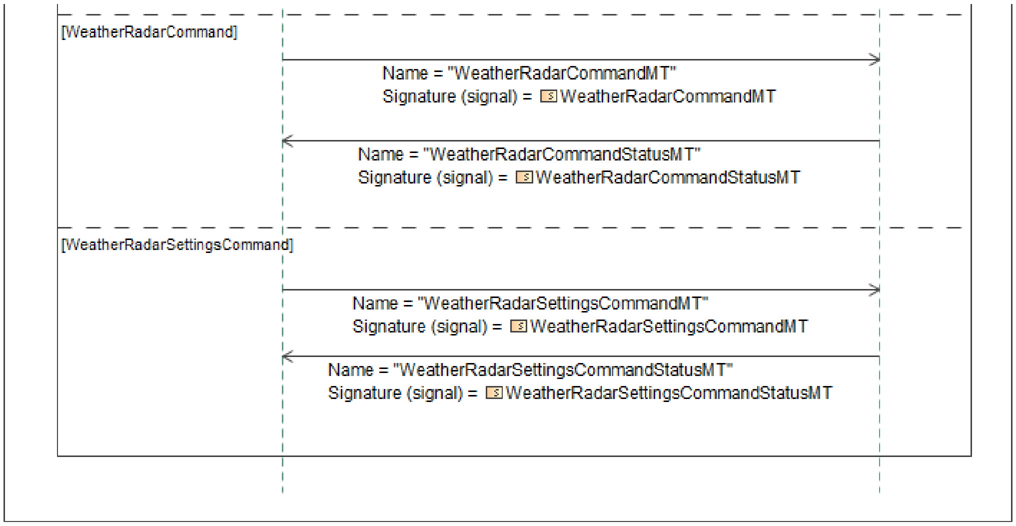


Figure 1-69: Send Team Level Command

Table 1-68: Send Team Level Command

Message Name (Name:Type)	Required Fields (Name:Type)
ActionCancelCommandMT: ActionCancelCommandMT	None
ActionCancelCommandStatusMT: ActionCancelCommandStatusMT	None
AirSampleCommandMT: AirSampleCommandMT	None
AirSampleCommandStatusMT: AirSampleCommandStatusMT	None
AirSampleSettingsCommandMT: AirSampleSettingsCommandMT	None
AirSampleSettingsCommandStatusMT: AirSampleSettingsCommandStatusMT	None
AMTI_CommandMT: AMTI_CommandMT	None
AMTI_CommandStatusMT: AMTI_CommandStatusMT	None
AMTI_SettingsCommandMT: AMTI_SettingsCommandMT	None
AMTI_SettingsCommandStatusMT: AMTI_SettingsCommandStatusMT	None
AO_CommandMT: AO_CommandMT	None
AO_CommandStatusMT: AO_CommandStatusMT	None
AO_SettingsCommandMT: AO_SettingsCommandMT	None
AO_SettingsCommandStatusMT: AO_SettingsCommandStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
COMINT_CommandMT: COMINT_CommandMT	None
COMINT_CommandStatusMT: COMINT_CommandStatusMT	None
COMINT_SettingsCommandMT: COMINT_SettingsCommandMT	None
COMINT_SettingsCommandStatusMT: COMINT_SettingsCommandStatusMT	None
CommPointingCommandMT: CommPointingCommandMT	None
CommPointingCommandStatusMT: CommPointingCommandStatusMT	None
CommRelayCommandMT: CommRelayCommandMT	None
CommRelayCommandStatusMT: CommRelayCommandStatusMT	None
CommRelaySettingsCommandMT: CommRelaySettingsCommandMT	None
CommRelaySettingsCommandStatusMT: CommRelaySettingsCommandStatusMT	None
CommTerminalCommandMT: CommTerminalCommandMT	None
CommTerminalCommandStatusMT: CommTerminalCommandStatusMT	None
CommTerminalPlanActivationCommandMT: CommTerminalPlanActivationCommandMT	None
CommTerminalPlanActivationCommandStatusMT: CommTerminalPlanActivationCommandStatusMT	None
CommTerminalSettingsCommandMT: CommTerminalSettingsCommandMT	None
CommTerminalSettingsCommandStatusMT: CommTerminalSettingsCommandStatusMT	None
DMPI_CancelCommandMT: DMPI_CancelCommandMT	None
DMPI_CancelCommandStatusMT: DMPI_CancelCommandStatusMT	None
DoorCommandMT: DoorCommandMT	None
DoorCommandStatusMT: DoorCommandStatusMT	None
EA_CommandMT: EA_CommandMT	None
EA_CommandStatusMT: EA_CommandStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
ESM_CommandMT: ESM_CommandMT	None
ESM_CommandStatusMT: ESM_CommandStatusMT	None
ESM_SettingsCommandMT: ESM_SettingsCommandMT	None
ESM_SettingsCommandStatusMT: ESM_SettingsCommandStatusMT	None
IFF_CommandMT: IFF_CommandMT	None
IFF_CommandStatusMT: IFF_CommandStatusMT	None
IFF_SettingsCommandMT: IFF_SettingsCommandMT	None
IFF_SettingsCommandStatusMT: IFF_SettingsCommandStatusMT	None
PO_CommandMT: PO_CommandMT	None
PO_CommandStatusMT: PO_CommandStatusMT	None
PO_SettingsCommandMT: PO_SettingsCommandMT	None
PO_SettingsCommandStatusMT: PO_SettingsCommandStatusMT	None
RadarAltimeterCommandMT: RadarAltimeterCommandMT	None
RadarAltimeterCommandStatusMT: RadarAltimeterCommandStatusMT	None
RadarAltimeterSettingsCommandMT: RadarAltimeterSettingsCommandMT	None
RadarAltimeterSettingsCommandStatusMT: RadarAltimeterSettingsCommandStatusMT	None
RefuelCommandMT: RefuelCommandMT	None
RefuelCommandStatusMT: RefuelCommandStatusMT	None
RefuelSettingsCommandMT: RefuelSettingsCommandMT	None
RefuelSettingsCommandStatusMT: RefuelSettingsCommandStatusMT	None
SAR_CommandMT: SAR_CommandMT	None
SAR_CommandStatusMT: SAR_CommandStatusMT	None
SAR_SettingsCommandMT: SAR_SettingsCommandMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
SAR_SettingsCommandStatusMT: SAR_SettingsCommandStatusMT	None
SMTI_CommandMT: SMTI_CommandMT	None
SMTI_CommandStatusMT: SMTI_CommandStatusMT	None
SMTI_SettingsCommandMT: SMTI_SettingsCommandMT	None
SMTI_SettingsCommandStatusMT: SMTI_SettingsCommandStatusMT	None
StoreManagementCommandMT: StoreManagementCommandMT	None
StoreManagementCommandStatusMT: StoreManagementCommandStatusMT	None
StrikeCommandMT: StrikeCommandMT	None
StrikeCommandStatusMT: StrikeCommandStatusMT	None
TaskCancelCommandMT: TaskCancelCommandMT	None
TaskCancelCommandStatusMT: TaskCancelCommandStatusMT	None
TurretCommandMT: TurretCommandMT	None
TurretCommandStatusMT: TurretCommandStatusMT	None
WeatherRadarCommandMT: WeatherRadarCommandMT	None
WeatherRadarCommandStatusMT: WeatherRadarCommandStatusMT	None
WeatherRadarSettingsCommandMT: WeatherRadarSettingsCommandMT	None
WeatherRadarSettingsCommandStatusMT: WeatherRadarSettingsCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.18 Synchronize Contingency Plan

This diagram contains the messages sent between MA and C2 to synchronize an updated contingency plan. The *MA_ApprovalPolicyMT* and *ResponsePlanMT* will be sent with an *ObjectState* enumeration set to *UPDATE*.

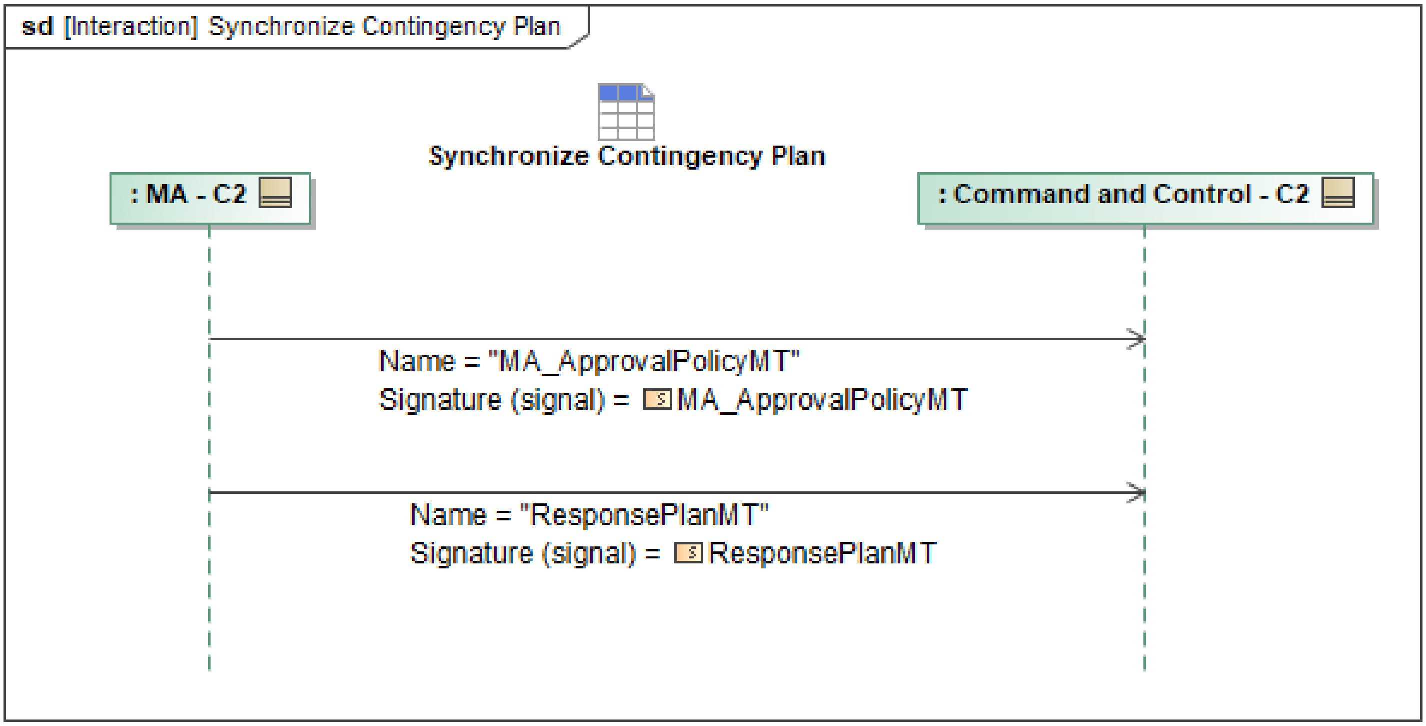


Figure 1-70: Synchronize Contingency Plan

Table 1-69: Synchronize Contingency Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalPolicyMT: MA_ApprovalPolicyMT	DataRecordInstanceID: DataRecordInstanceID_Type
ResponsePlanMT: ResponsePlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.19 Update C2 with Package Mission Plan Synchronization Status

This interaction describes the process of the MA sending the C2 an update on the package mission plan synchronization status. This is done through a *MA_SystemNotificationMT* message with the *ApplicableToID* field containing either the relevant package or system ID. This allows a reference to the mission plan's ID in order for the recipient to understand which plan was synchronized.

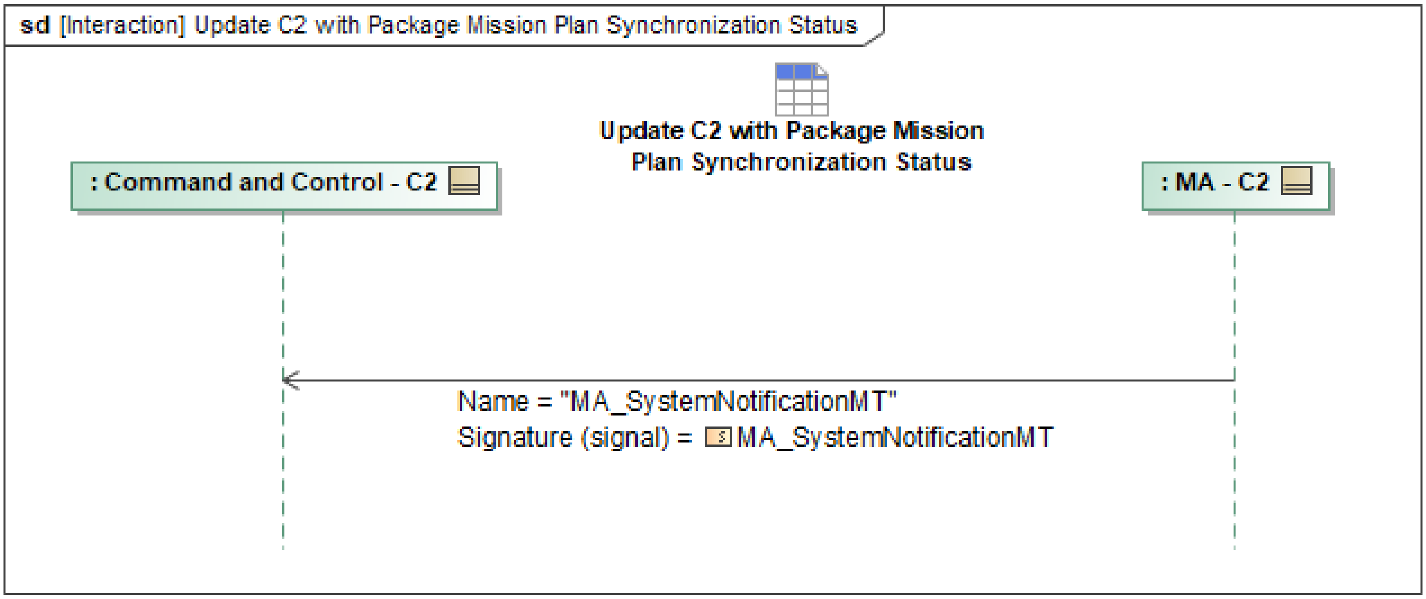


Figure 1-71: Update C2 with Package Mission Plan Synchronization Status

Table 1-70: Update C2 with Package Mission Plan Synchronization Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemNotificationMT: MA_SystemNotificationMT	AppliesTo: SubjectType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.20 Zone Alerts

When MA detects a conflict between a geozone and a plan/requirement (for example, a RoutePlan into a KOZ). MA sends an *OpNotificationMT* to alert C2 of the conflict. The geozone with the conflict is set within the *OpZoneID* field inside the *OpSubject* field. *AssociatedMessage* fields contain each conflicted Route, Plan, Requirement, etc., with the respective *AssociatedID* filled. *NotificationNarrative* contains a human-readable version of the conflict.

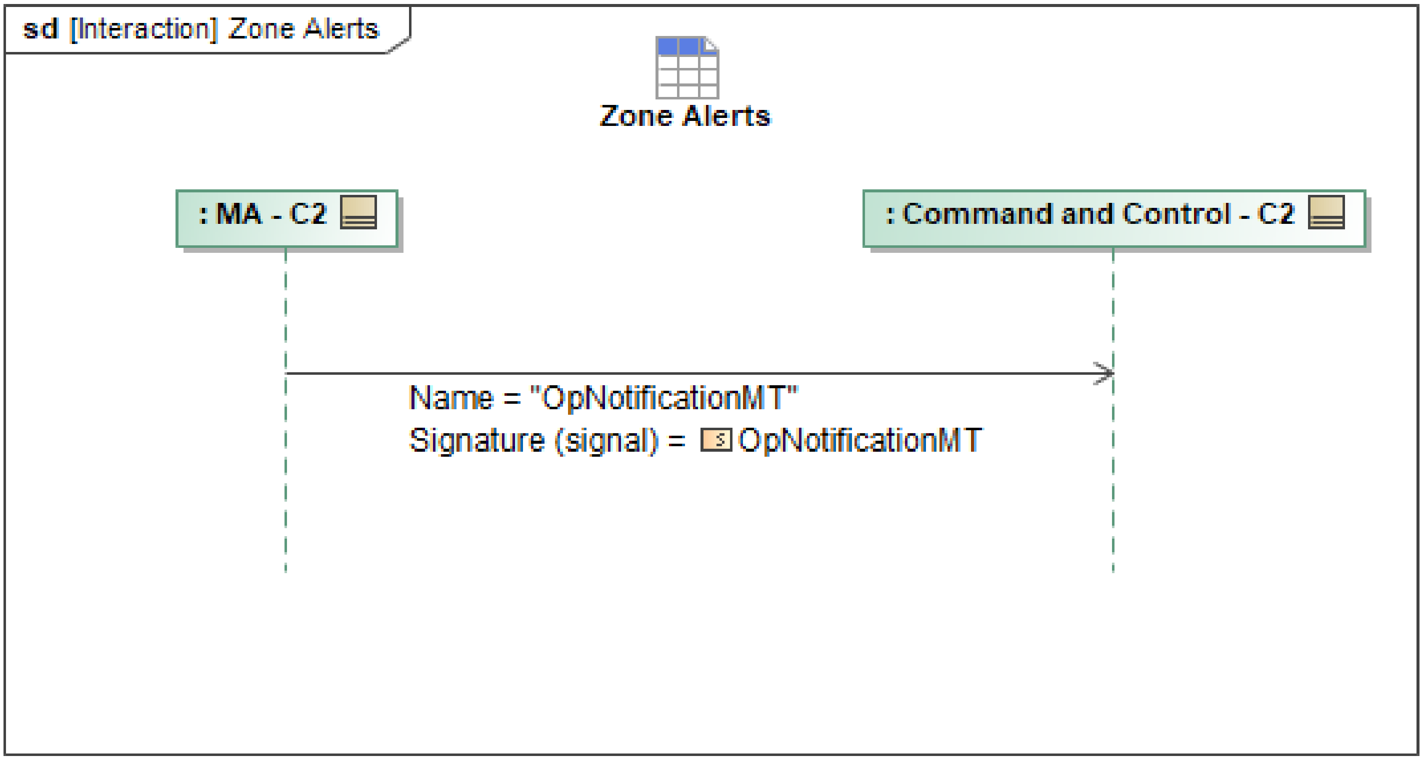


Figure 1-72: Zone Alerts

Table 1-71: Zone Alerts

Message Name (Name:Type)	Required Fields (Name:Type)
OpNotificationMT: OpNotificationMT	AssociatedID: ID_Type NotificationNarrative: VisibleString1024Type AssociatedMessage: AssociatedMessageType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21 Query Data

1.2.8.21.1 Query Activity Plan

MA_ActivityPlanMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_ActivityPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_ActivityPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_ActivityPlanMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_ActivityPlanMT*.

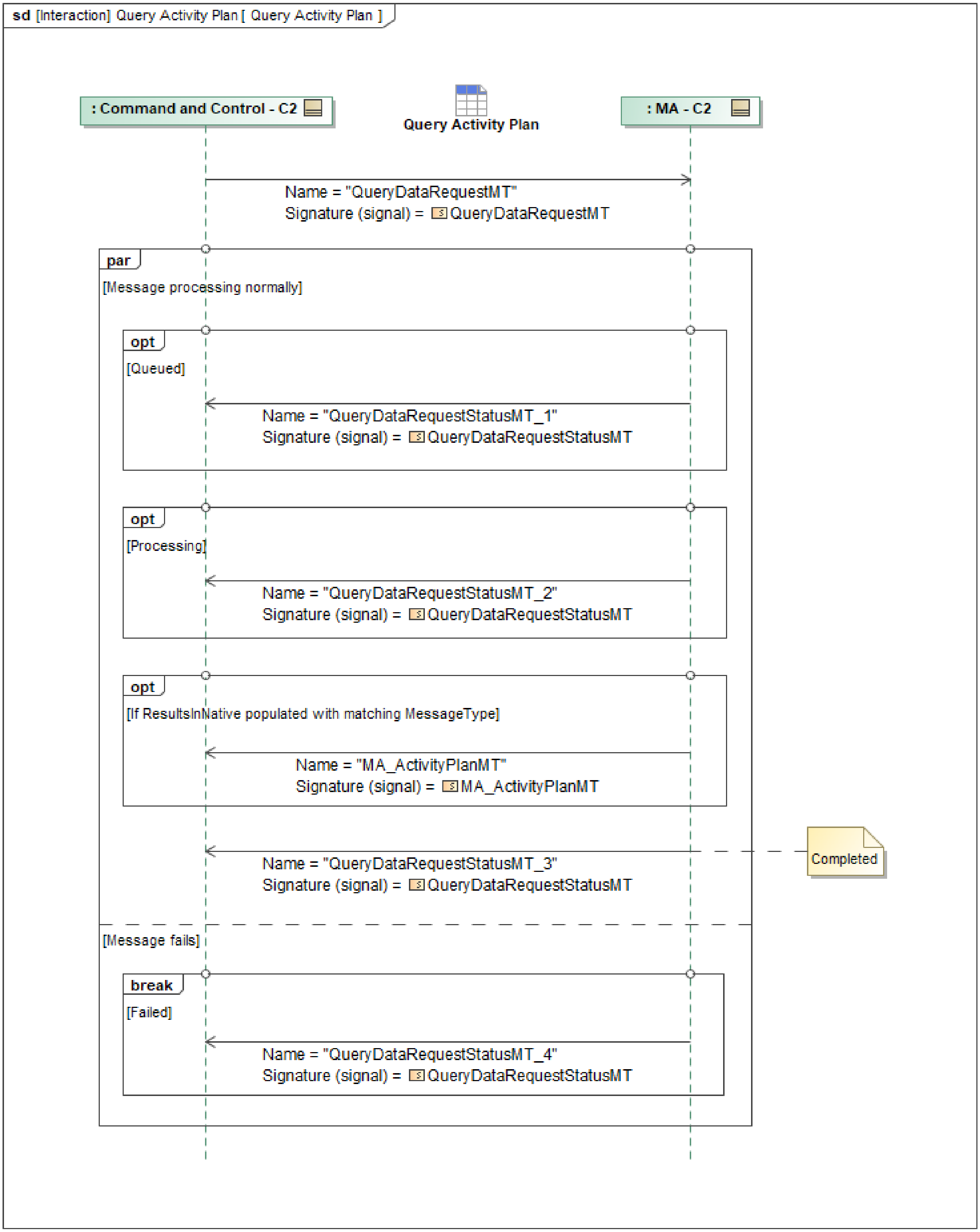


Figure 1-73: Query Activity Plan

Table 1-72: Query Activity Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActivityPlanMT: MA_ActivityPlanMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.2 Query Activity Plan Execution Status

ActivityPlanExecutionStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *ActivityPlanExecutionStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *ActivityPlanExecutionStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *ActivityPlanExecutionStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *ActivityPlanExecutionStatusMT*.

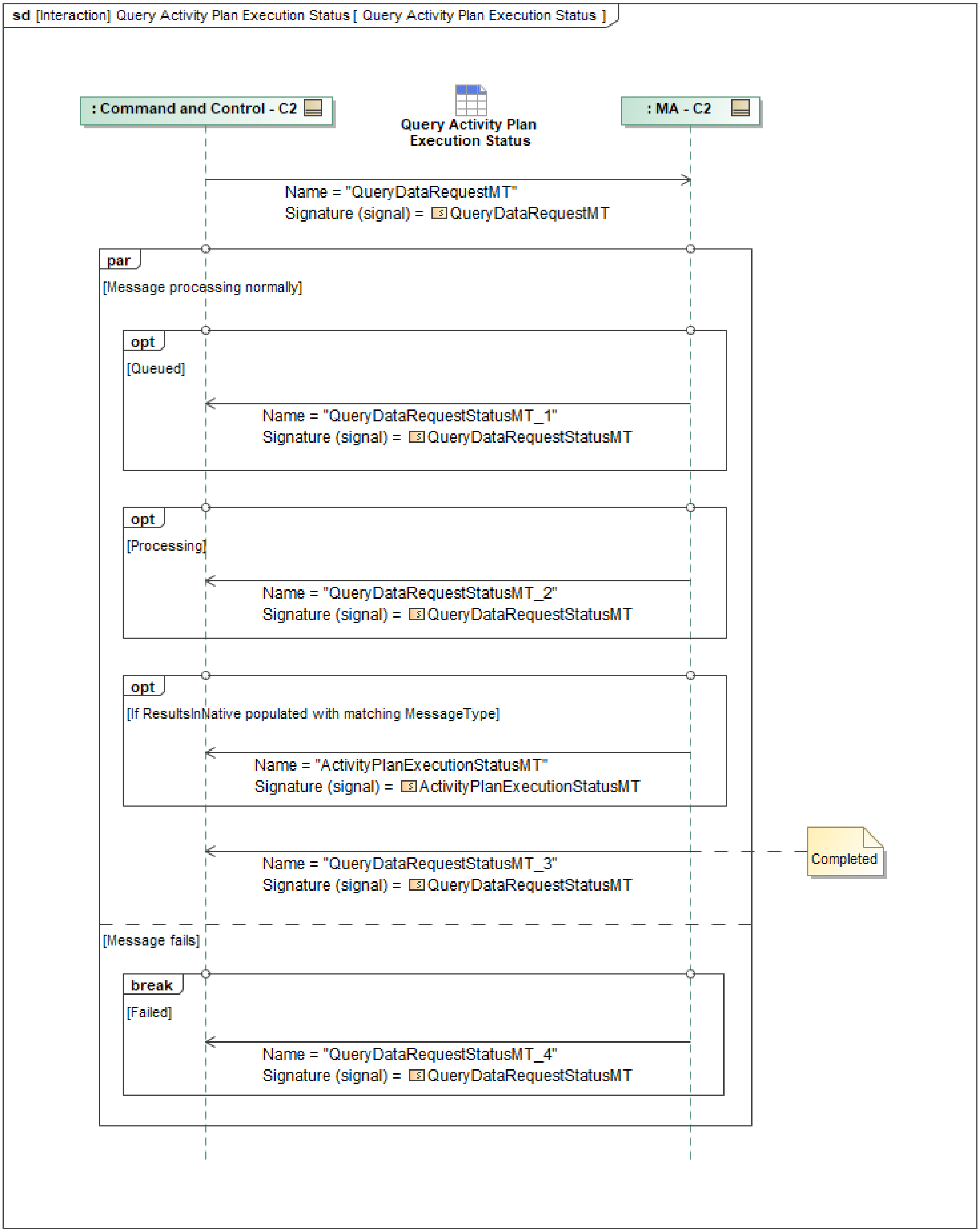


Figure 1-74: Query Activity Plan Execution Status

Table 1-73: Query Activity Plan Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
ActivityPlanExecutionStatusMT: ActivityPlanExecutionStatusMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.3 Query Airfield Report

AirfieldReportMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *AirfieldReportMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *AirfieldReportMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *AirfieldReportMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *AirfieldReportMT*.

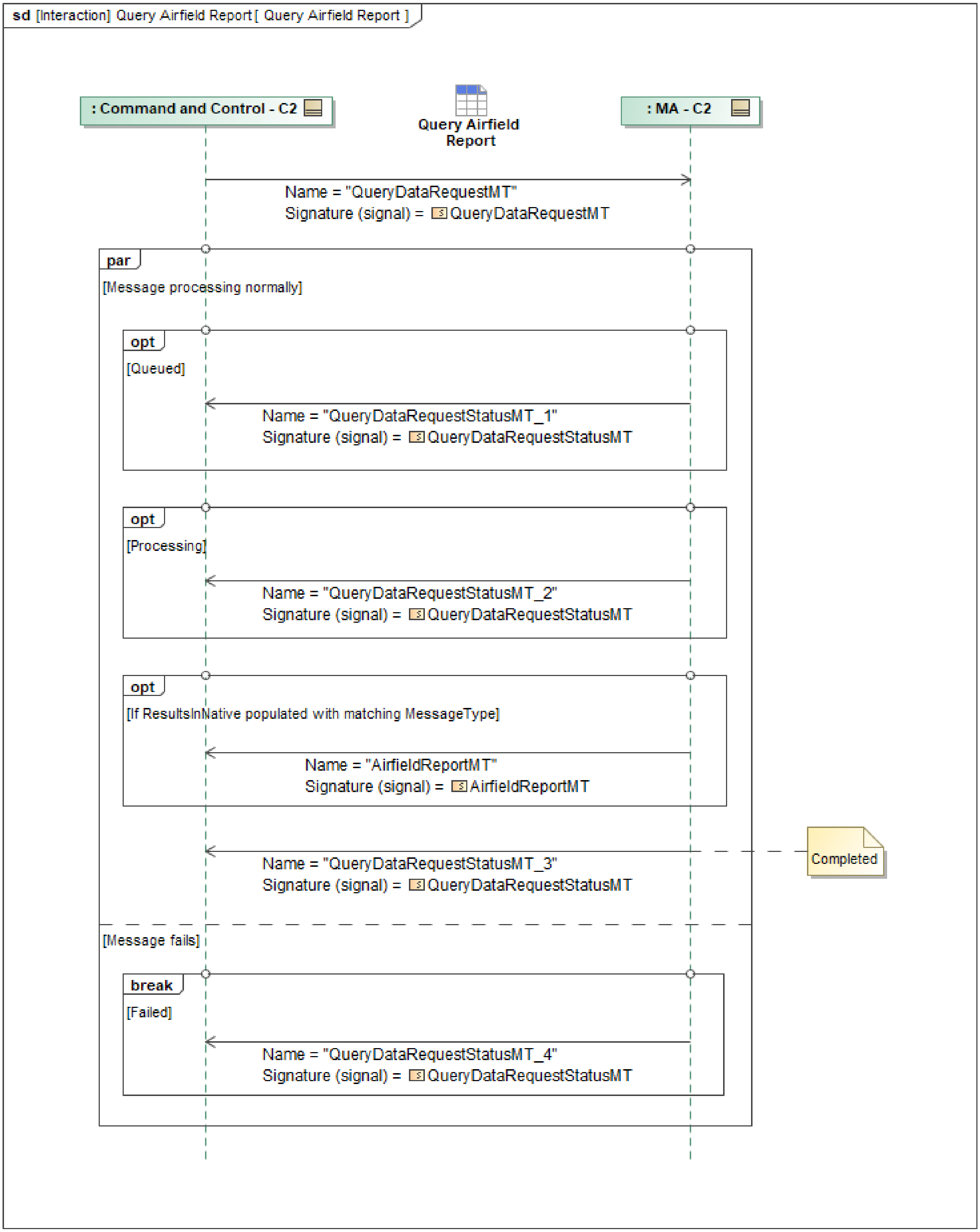


Figure 1-75: Query Airfield Report

Table 1-74: Query Airfield Report

Message Name (Name:Type)	Required Fields (Name:Type)
AirfieldReportMT: AirfieldReportMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.4 Query Control Status

ControlStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *ControlStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *ControlStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *ControlStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *ControlStatusMT*.

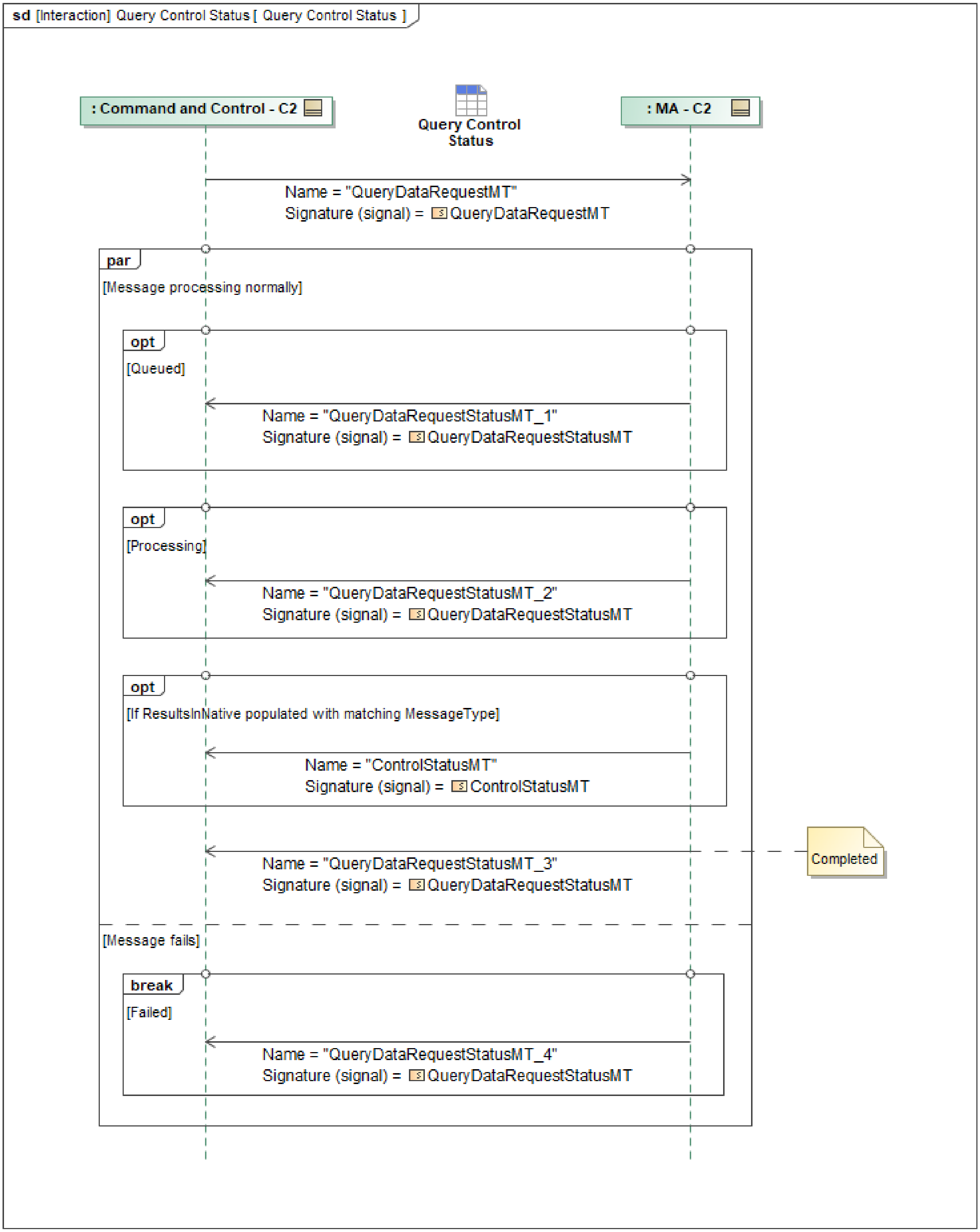


Figure 1-76: Query Control Status

Table 1-75: Query Control Status

Message Name (Name:Type)	Required Fields (Name:Type)
ControlStatusMT: ControlStatusMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.5 Query Mission Plan

MA_MissionPlanMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_MissionPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_MissionPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_MissionPlanMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_MissionPlanMT*.

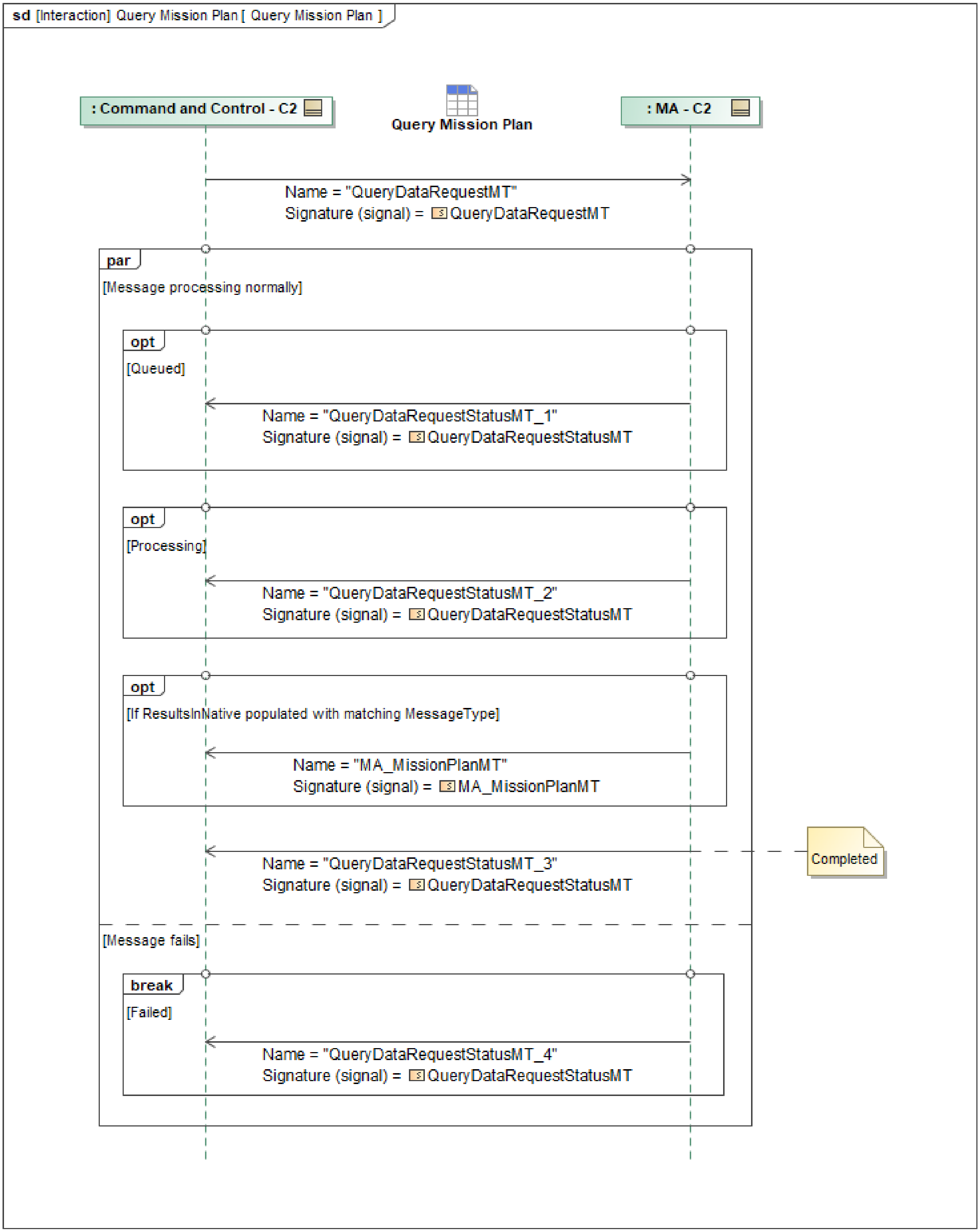


Figure 1-77: Query Mission Plan

Table 1-76: Query Mission Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanMT: MA_MissionPlanMT	SubPlans: MA_PlansReferenceBaseType
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.6 Query Mission Plan Activation Status

MA_MissionPlanActivationStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_MissionPlanActivationStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_MissionPlanActivationStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_MissionPlanActivationStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_MissionPlanActivationStatusMT*.

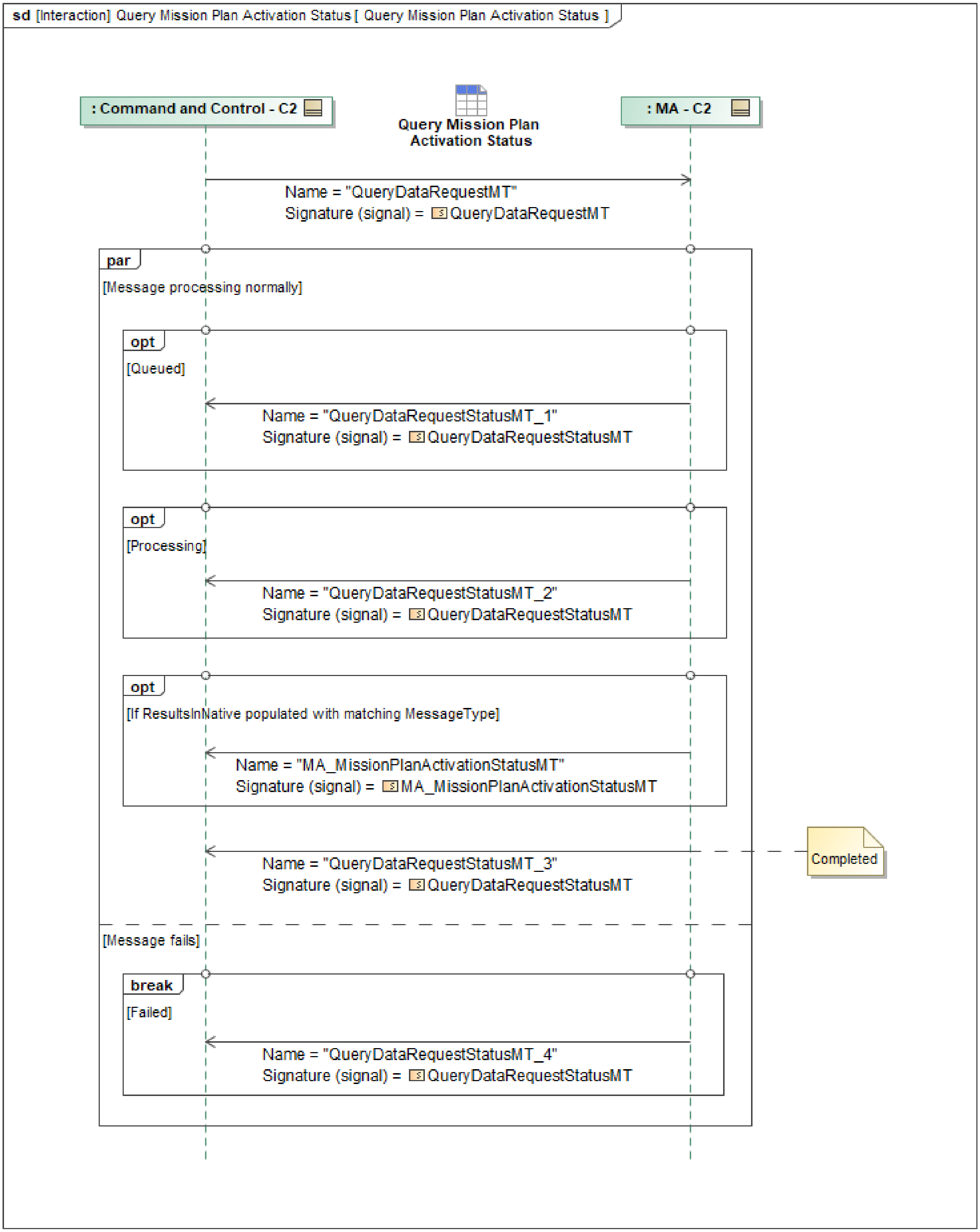


Figure 1-78: Query Mission Plan Activation Status

Table 1-77: Query Mission Plan Activation Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanActivationStatusMT: MA_MissionPlanActivationStatusMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.7 Query Mission Plan Execution Status

MA_MissionPlanExecutionStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_MissionPlanExecutionStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_MissionPlanExecutionStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_MissionPlanExecutionStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_MissionPlanExecutionStatusMT*.

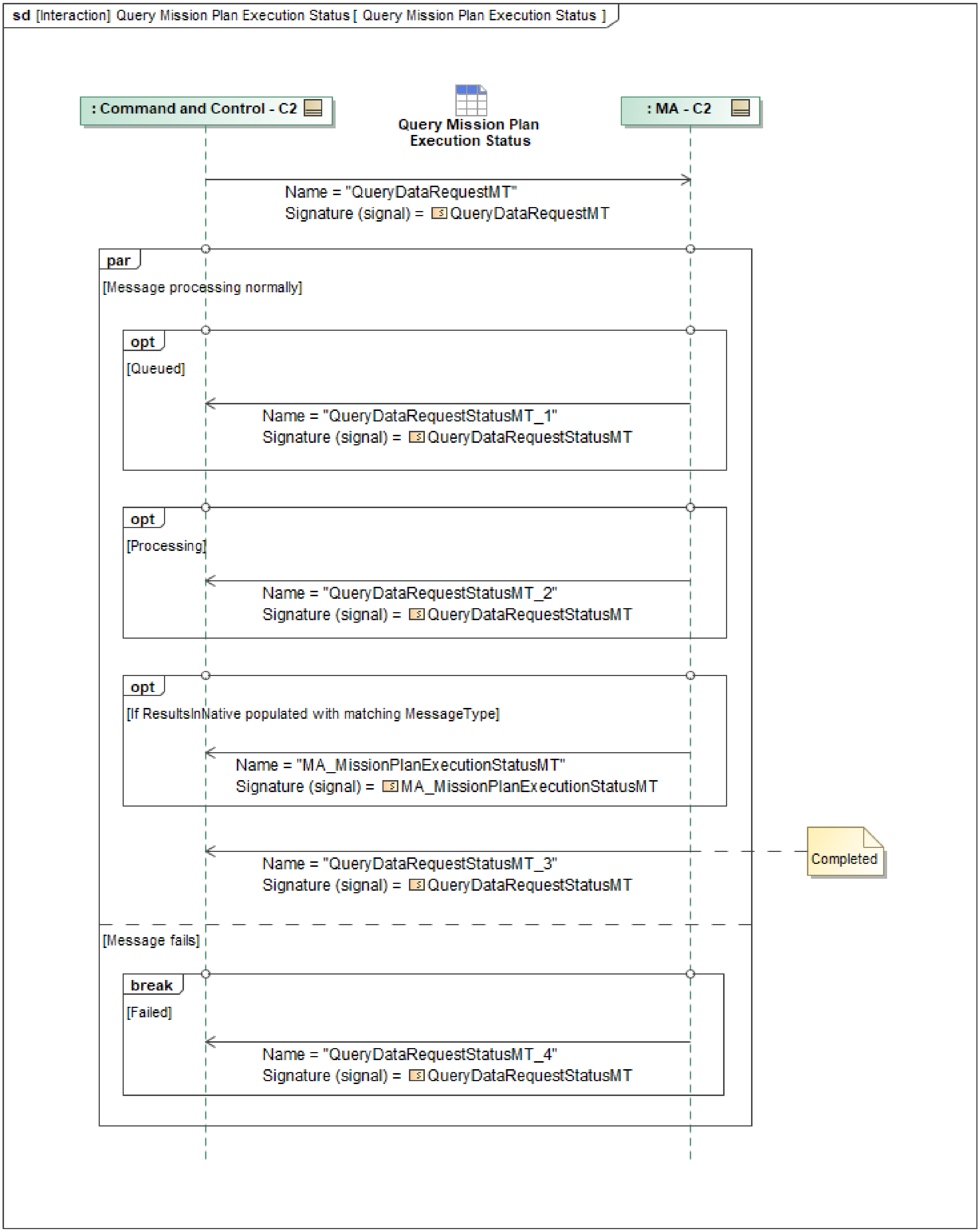


Figure 1-79: Query Mission Plan Execution Status

Table 1-78: Query Mission Plan Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanExecutionStatusMT: MA_MissionPlanExecutionStatusMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.8 Query OpPoint

MA_OpPointMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_OpPointMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_OpPointMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_OpPointMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_OpPointMT*.

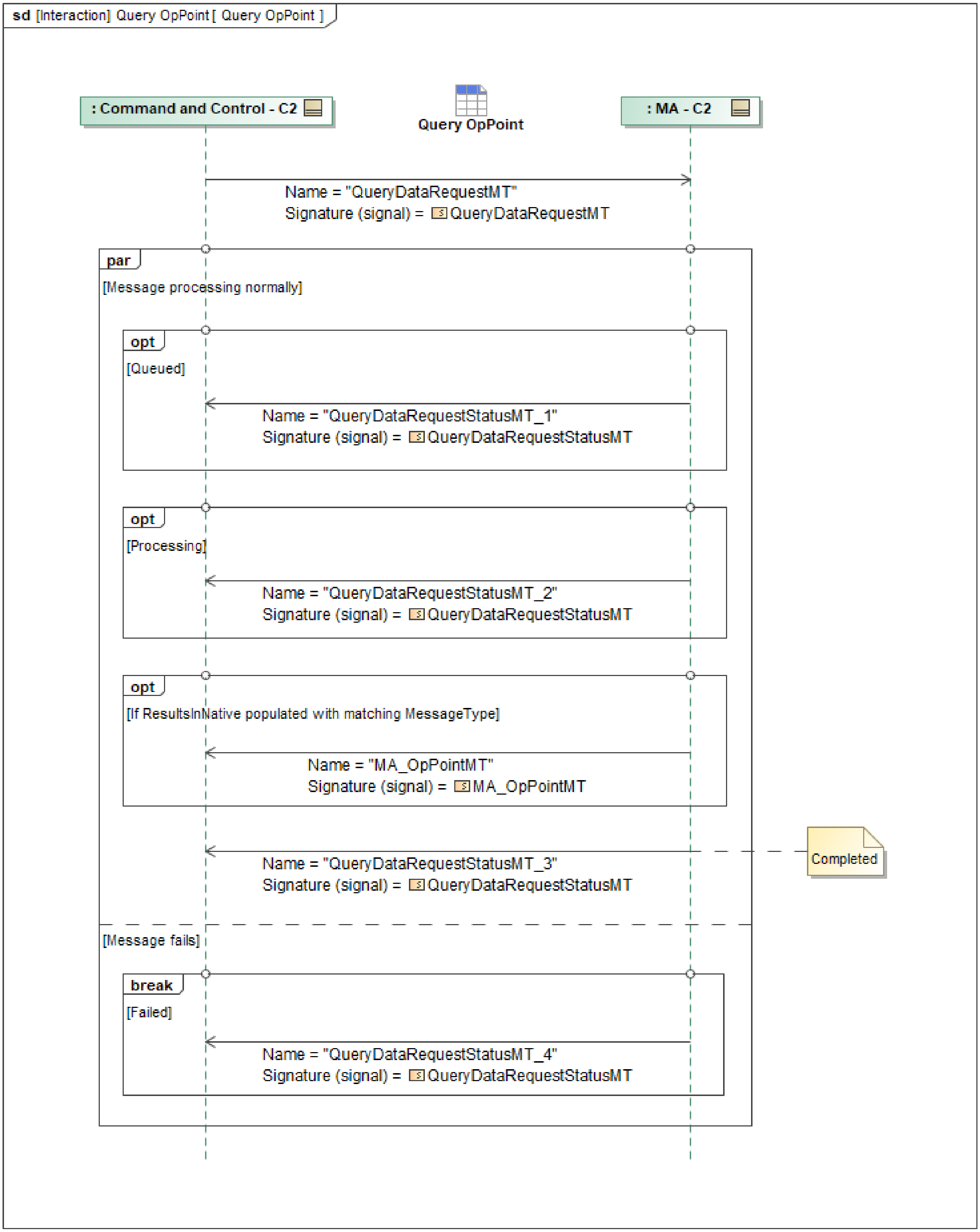


Figure 1-80: Query OpPoint

Table 1-79: Query OpPoint

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpPointMT: MA_OpPointMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.9 Query OpZone

MA_OpZoneMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_OpZoneMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_OpZoneMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_OpZoneMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_OpZoneMT*.

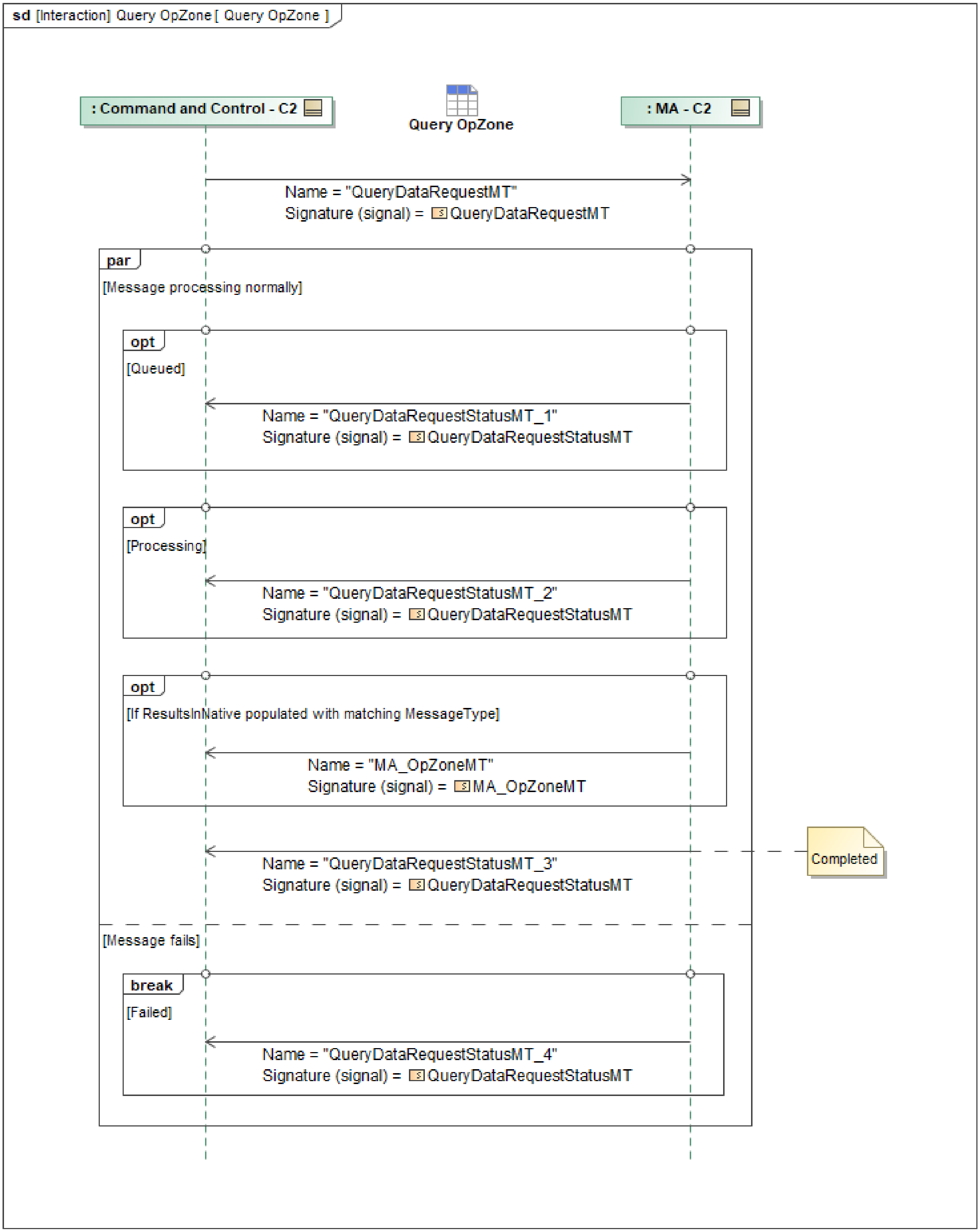


Figure 1-81: Query OpZone

Table 1-80: Query OpZone

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpZoneMT: MA_OpZoneMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.10 Query Package

PackageMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *PackageMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *PackageMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *PackageMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *PackageMT*.

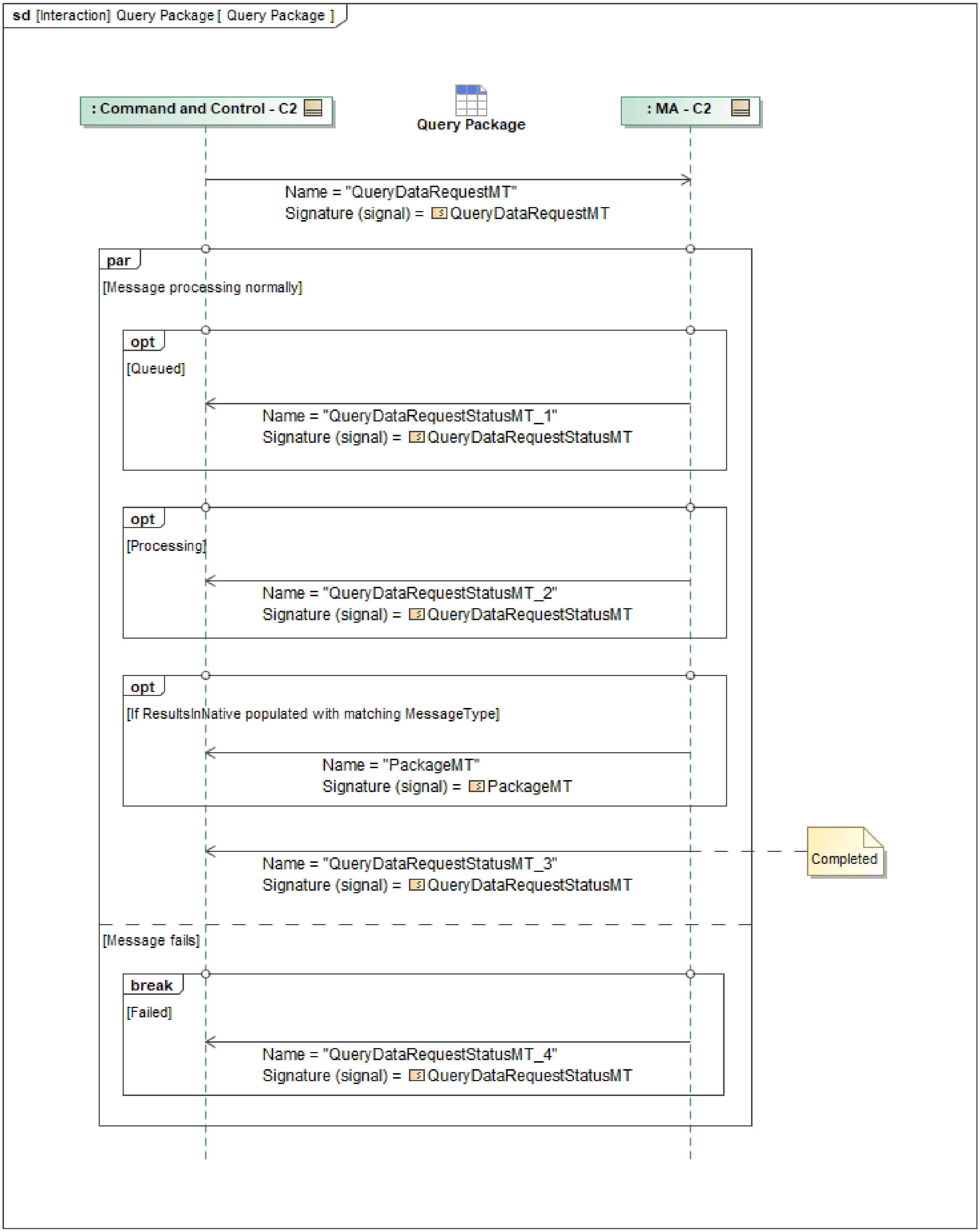


Figure 1-82: Query Package

Table 1-81: Query Package

Message Name (Name:Type)	Required Fields (Name:Type)
PackageMT: PackageMT	ObjectState: ObjectStateEnum
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.11 Query Planning Function

MA_PlanningFunctionMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_PlanningFunctionMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_PlanningFunctionMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_PlanningFunctionMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_PlanningFunctionMT*.

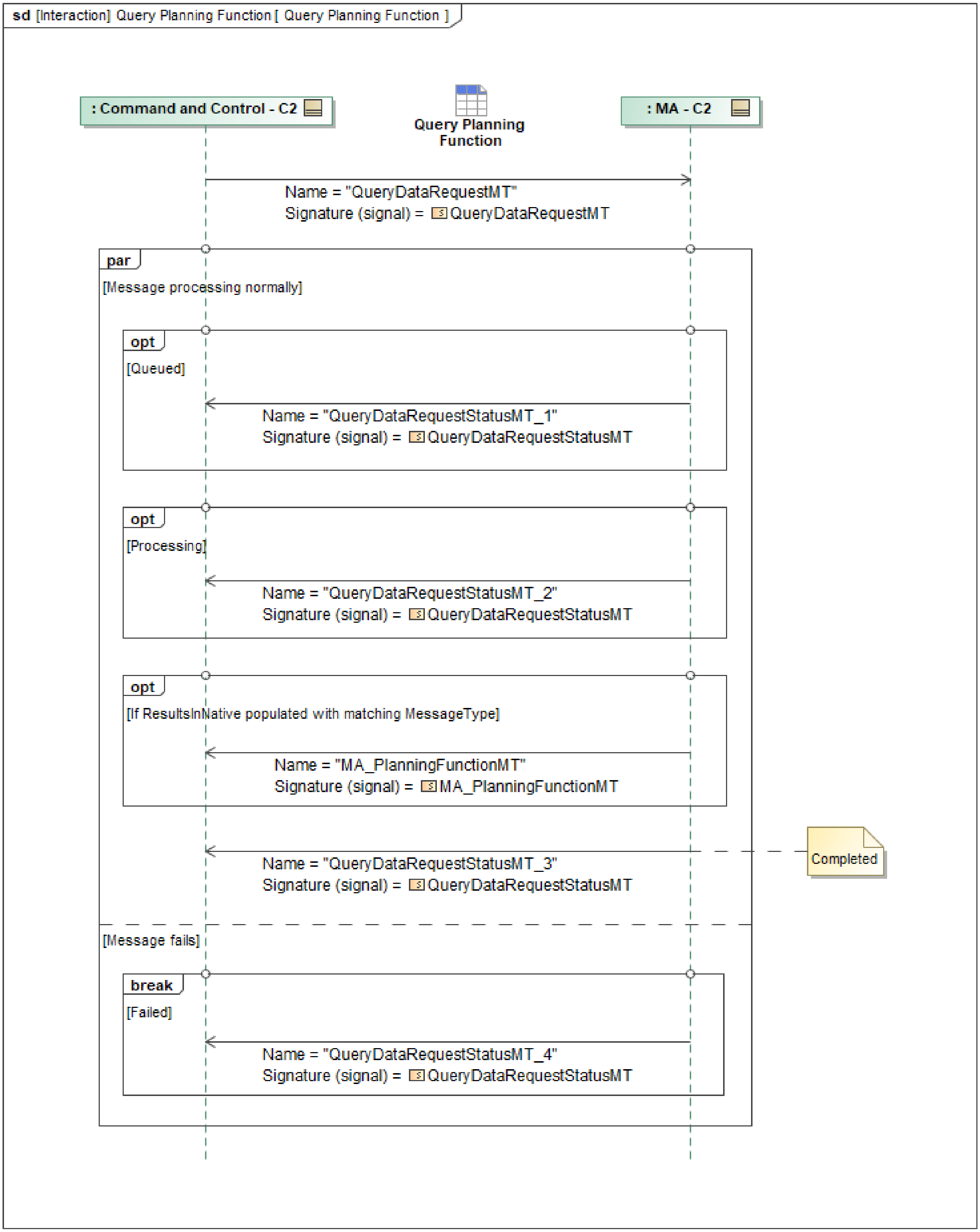


Figure 1-83: Query Planning Function

Table 1-82: Query Planning Function

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PlanningFunctionMT: MA_PlanningFunctionMT	PlanningInterfaces: PlanningInterfacesType PlanningInterfaces: PlanningInterfacesType
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.12 Query Planning Function Status

MA_PlanningFunctionStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_PlanningFunctionStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_PlanningFunctionStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_PlanningFunctionStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_PlanningFunctionStatusMT*.

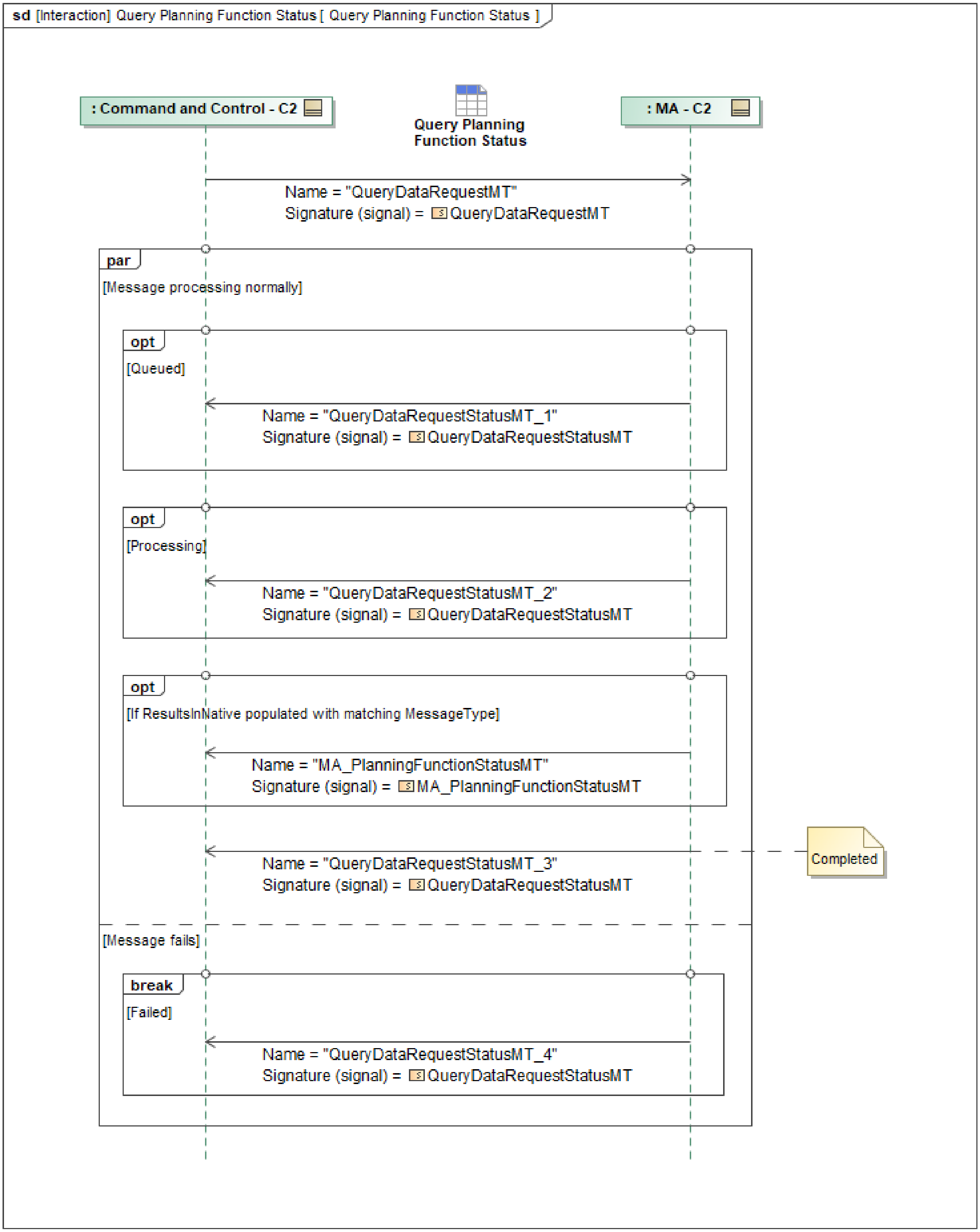


Figure 1-84: Query Planning Function Status

Table 1-83: Query Planning Function Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PlanningFunctionStatusMT: MA_PlanningFunctionStatusMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.13 Query Response Plan

ResponsePlanMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *ResponsePlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *ResponsePlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *ResponsePlanMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *ResponsePlanMT*.

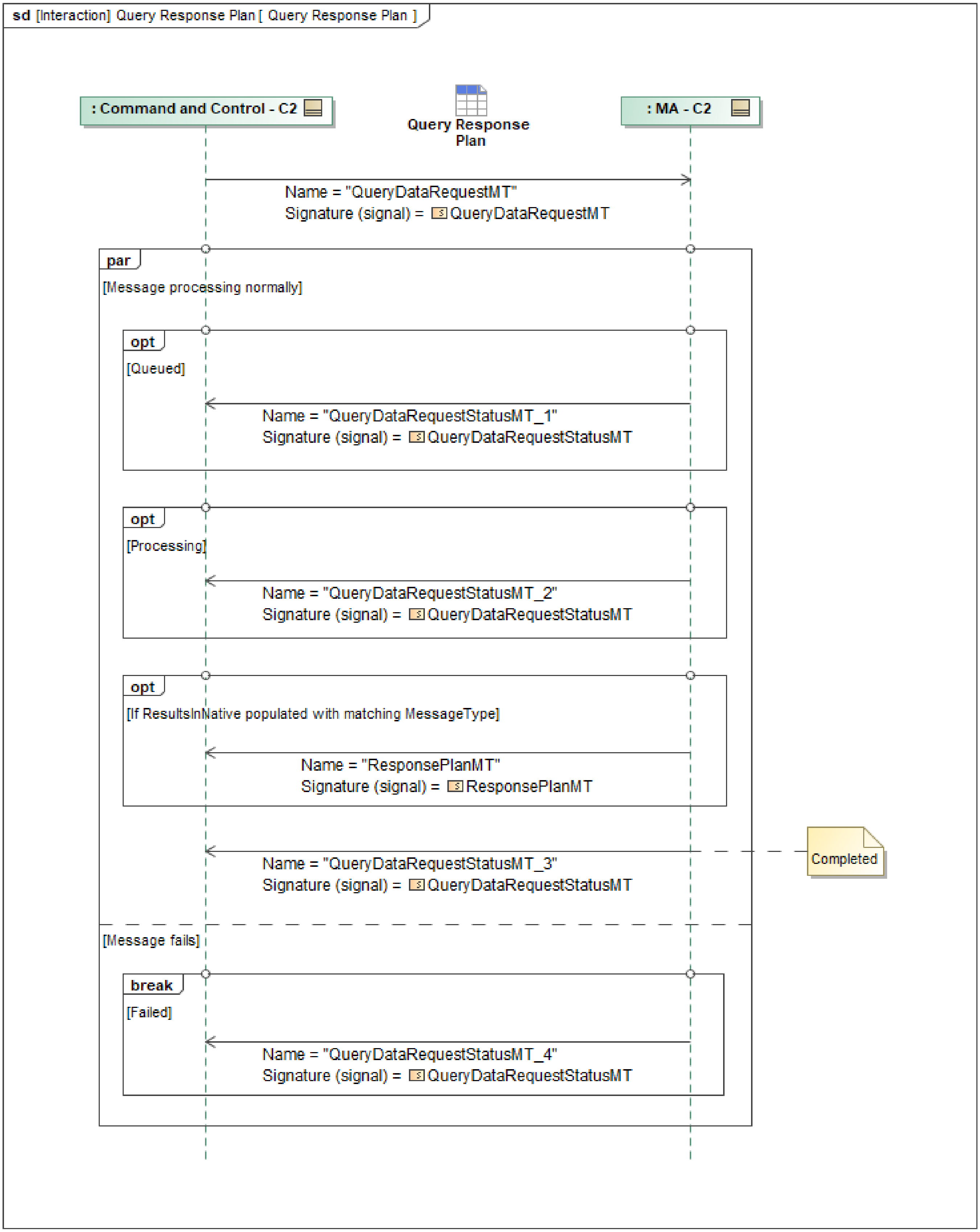


Figure 1-85: Query Response Plan

Table 1-84: Query Response Plan

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
ResponsePlanMT: ResponsePlanMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.14 Query Response Plan Execution Status

ResponsePlanExecutionStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *ResponsePlanExecutionStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *ResponsePlanExecutionStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *ResponsePlanExecutionStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *ResponsePlanExecutionStatusMT*.

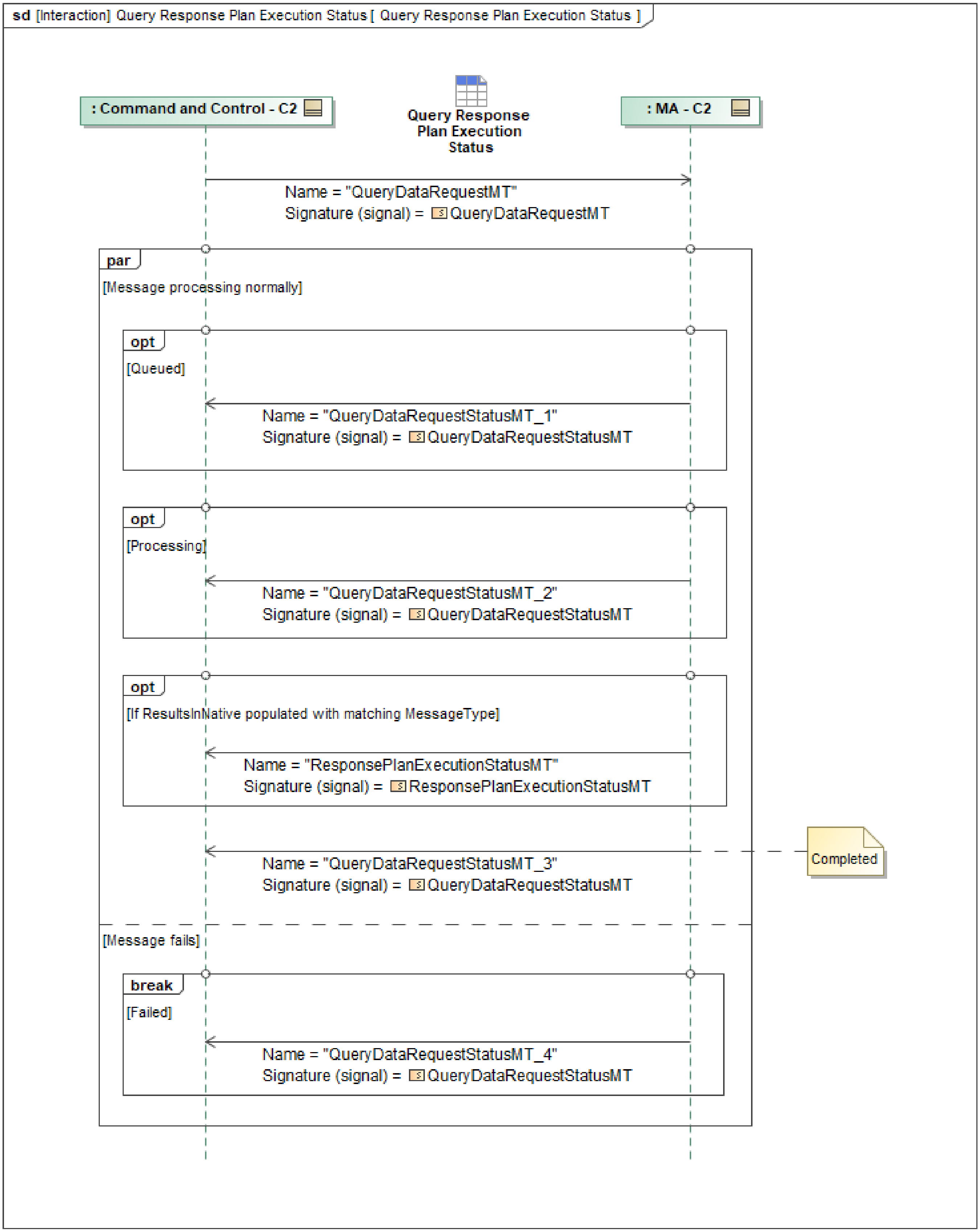


Figure 1-86: Query Response Plan Execution Status

Table 1-85: Query Response Plan Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
ResponsePlanExecutionStatusMT: ResponsePlanExecutionStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.15 Query Route Activity Plan

MA_RouteActivityPlanMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_RouteActivityPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_RouteActivityPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_RouteActivityPlanMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_RouteActivityPlanMT*.

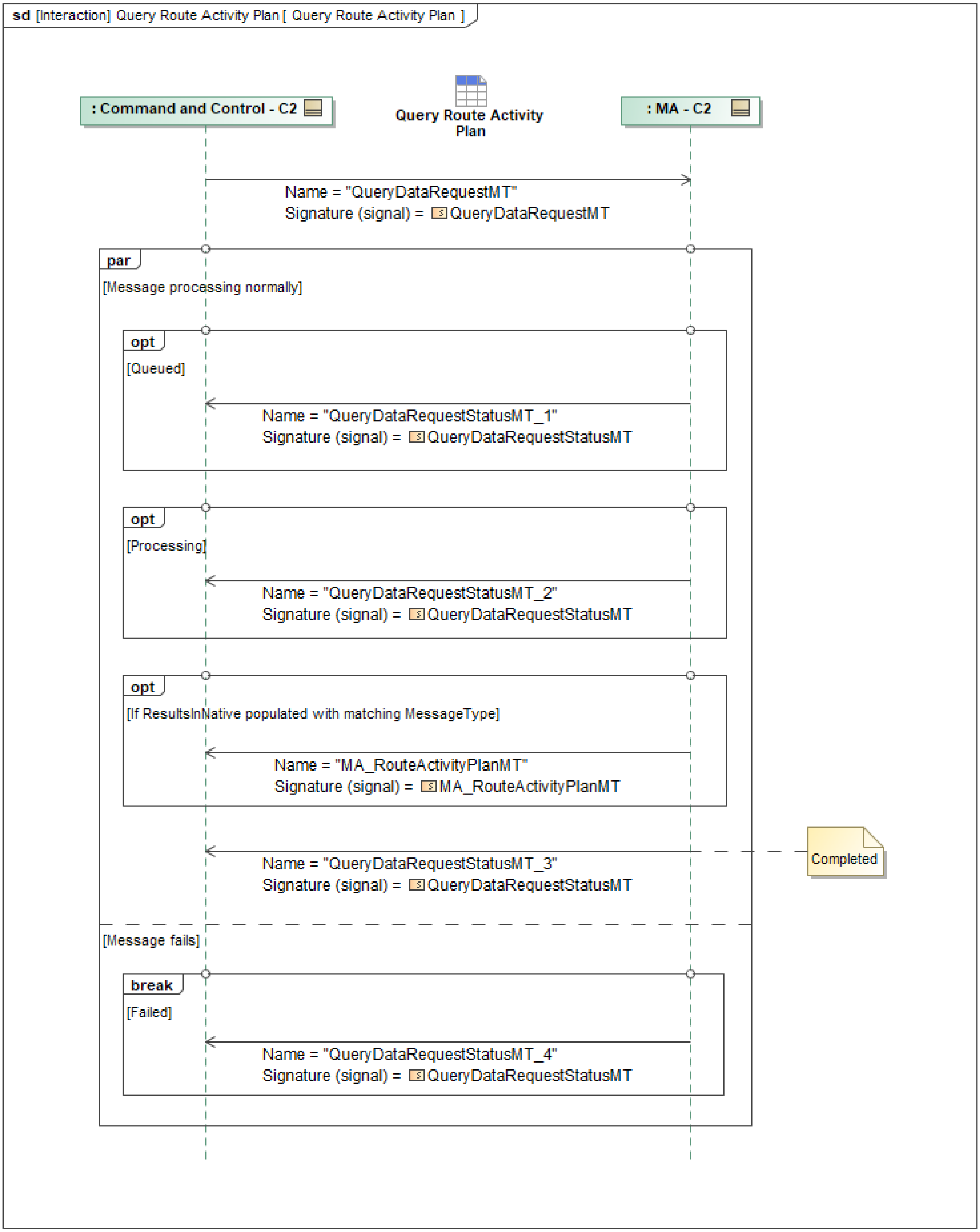


Figure 1-87: Query Route Activity Plan

Table 1-86: Query Route Activity Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RouteActivityPlanMT: MA_RouteActivityPlanMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.16 Query Route Activity Plan Execution Status

RouteActivityPlanExecutionStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *RouteActivityPlanExecutionStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *RouteActivityPlanExecutionStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *RouteActivityPlanExecutionStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *RouteActivityPlanExecutionStatusMT*.

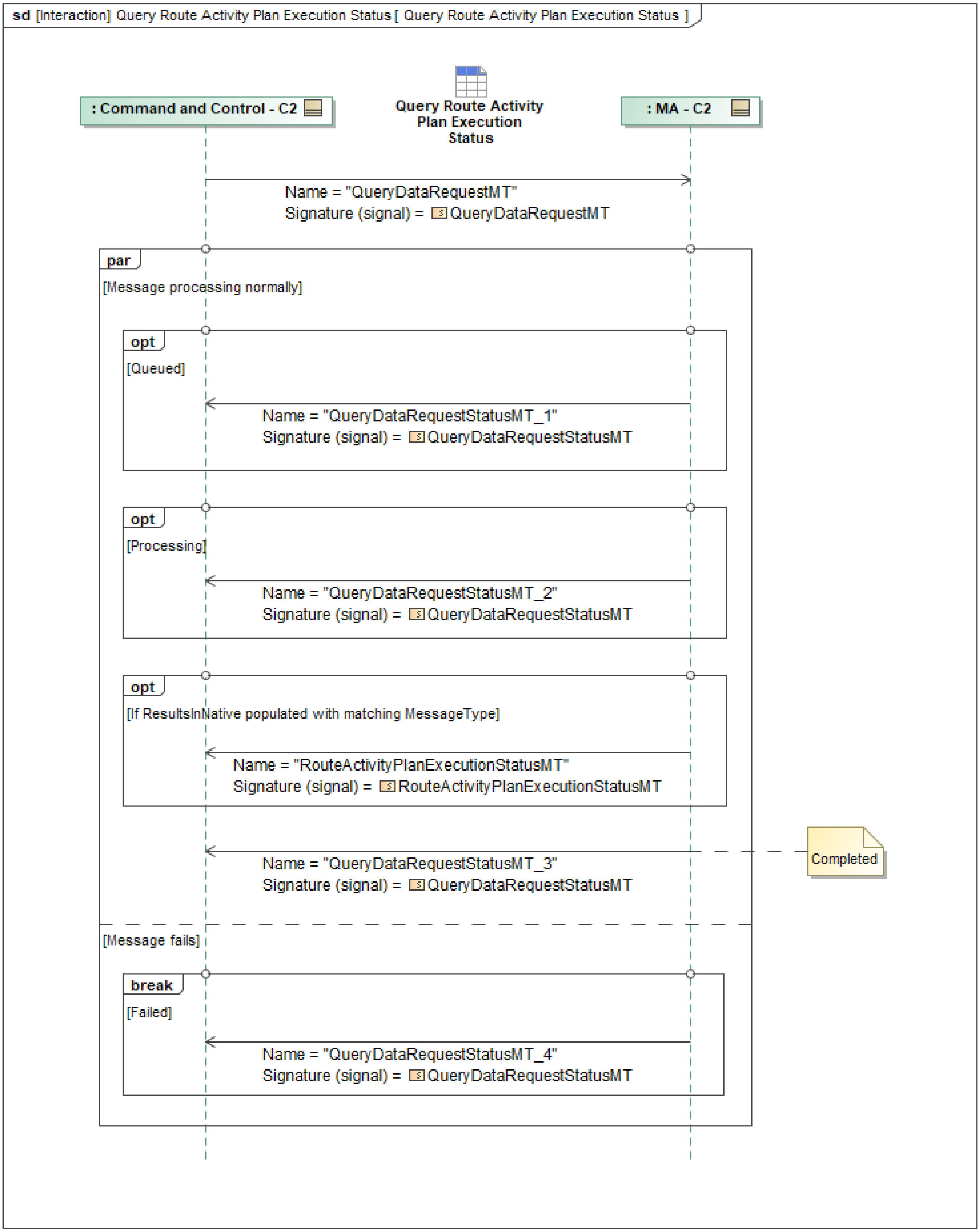


Figure 1-88: Query Route Activity Plan Execution Status

Table 1-87: Query Route Activity Plan Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
RouteActivityPlanExecutionStatusMT: RouteActivityPlanExecutionStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.17 Query Route Metrics

RouteMetricsMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *RouteMetricsMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *RouteMetricsMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *RouteMetricsMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *RouteMetricsMT*.

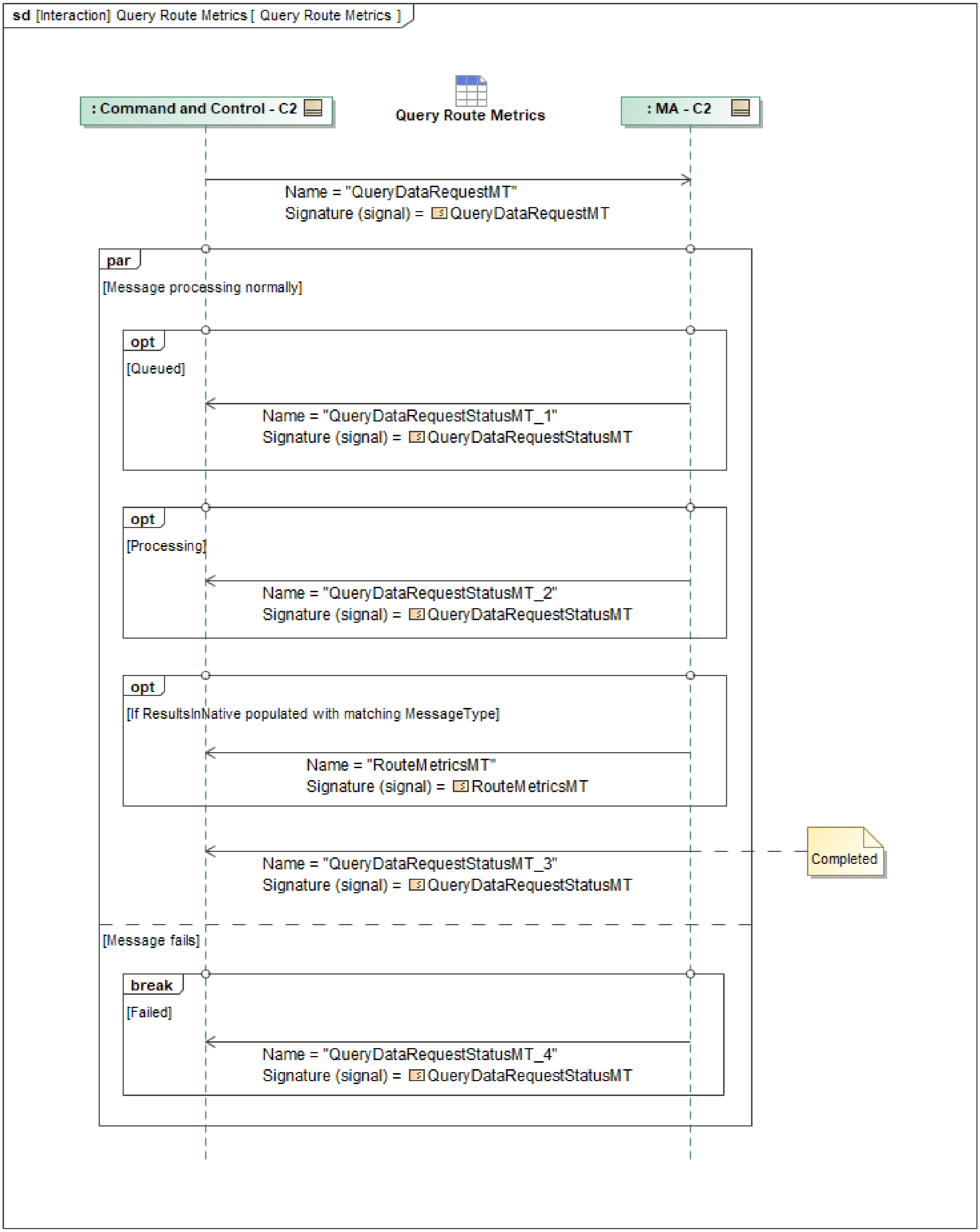


Figure 1-89: Query Route Metrics

Table 1-88: Query Route Metrics

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
RouteMetricsMT: RouteMetricsMT	EnduranceUsage: EnduranceType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.18 Query Route Plan

MA_RoutePlanMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_RoutePlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_RoutePlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_RoutePlanMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_RoutePlanMT*.

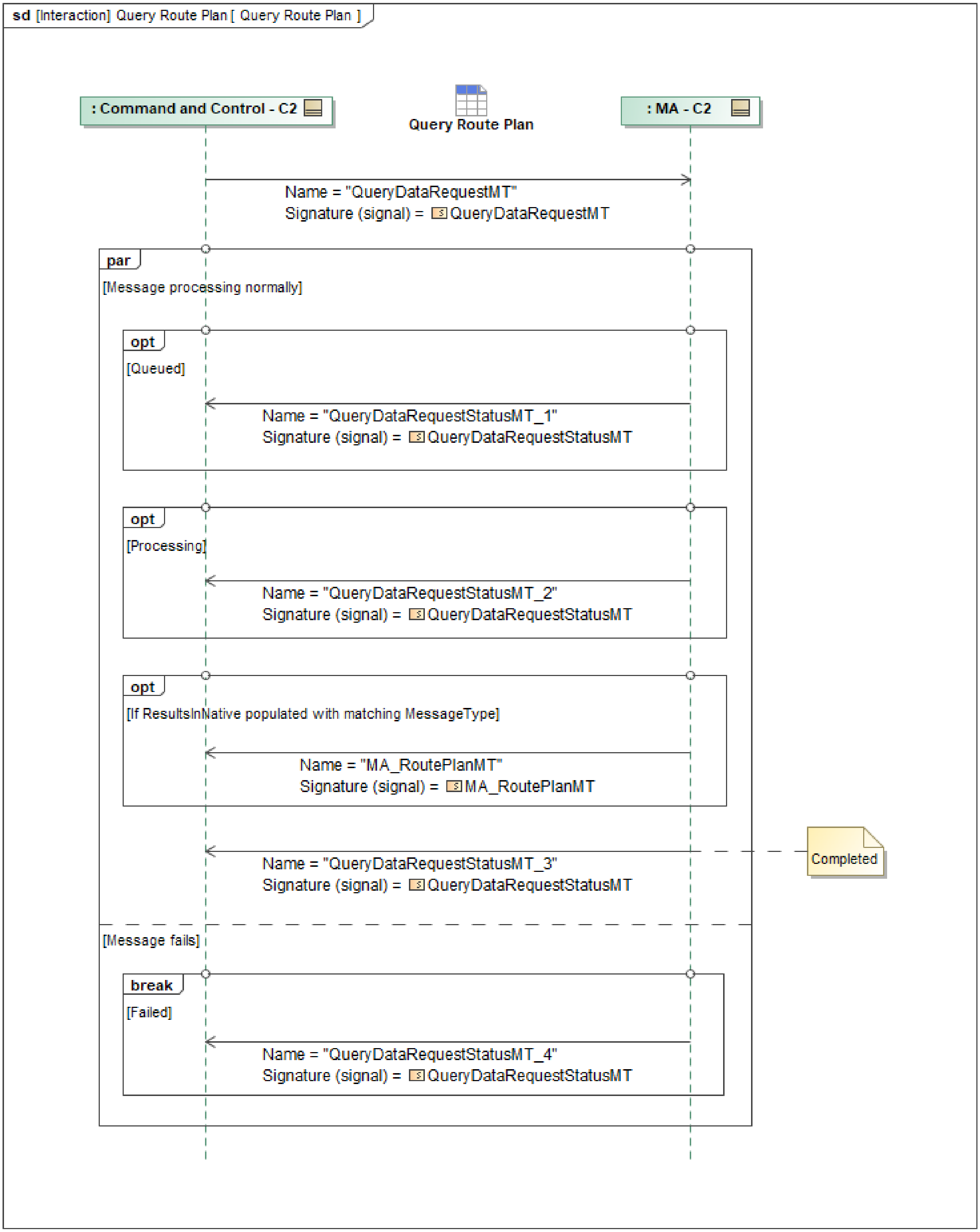


Figure 1-90: Query Route Plan

Table 1-89: Query Route Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RoutePlanMT: MA_RoutePlanMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.19 Query Route Plan Execution Status

RoutePlanExecutionStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *RoutePlanExecutionStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *RoutePlanExecutionStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *RoutePlanExecutionStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *RoutePlanExecutionStatusMT*.

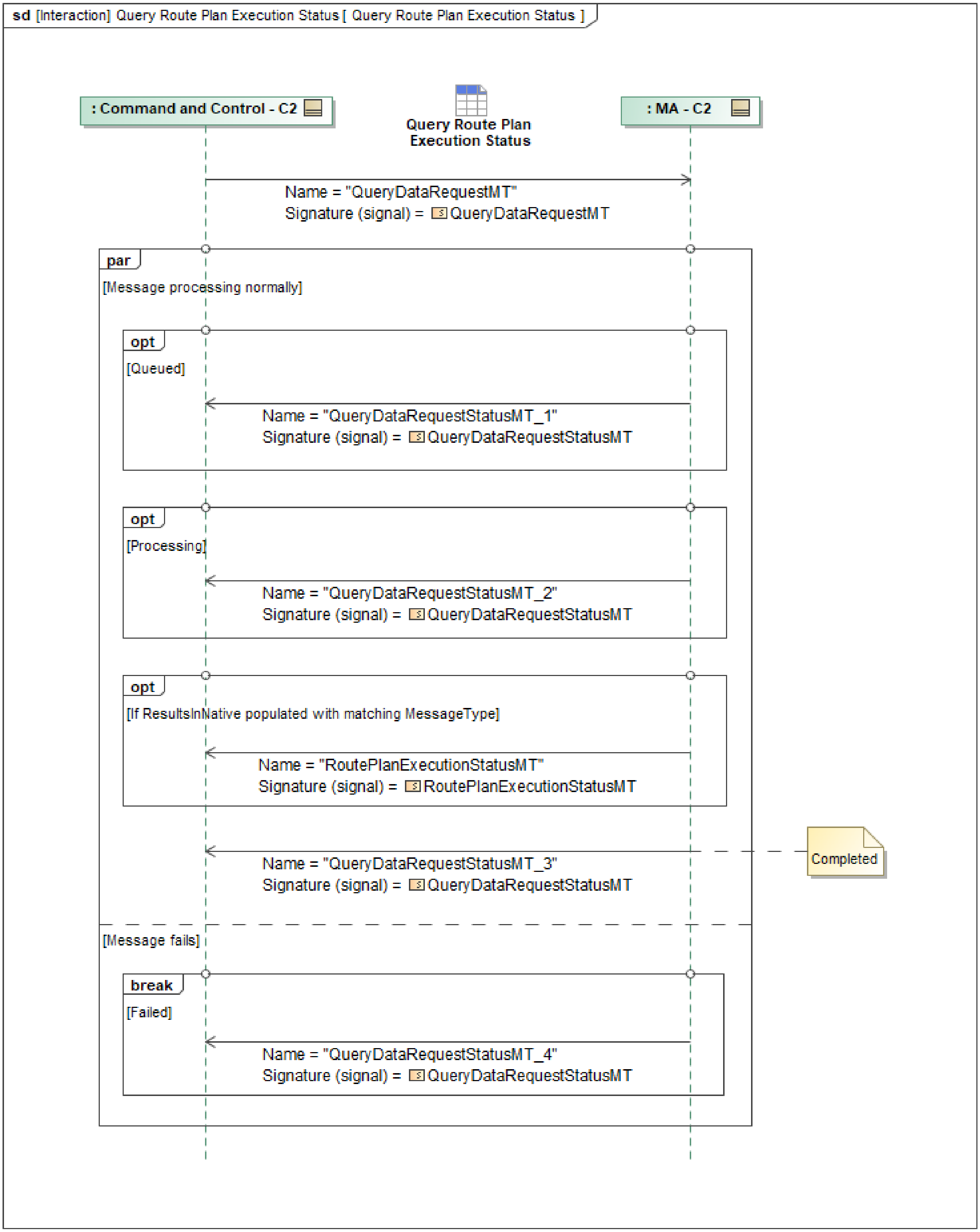


Figure 1-91: Query Route Plan Execution Status

Table 1-90: Query Route Plan Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
RoutePlanExecutionStatusMT: RoutePlanExecutionStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.20 Query System Status

SystemStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *SystemStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *SystemStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *SystemStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *SystemStatusMT*.

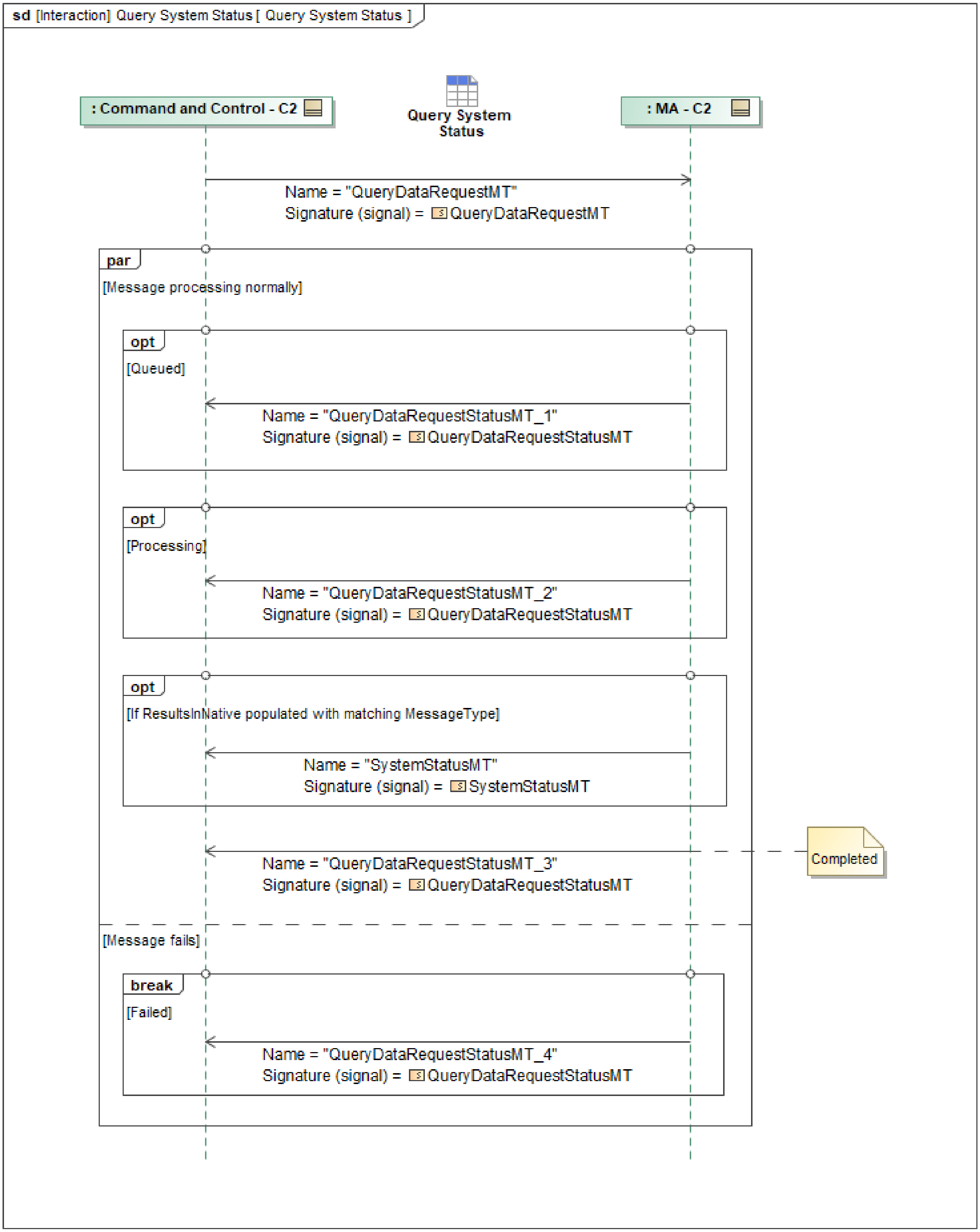


Figure 1-92: Query System Status

Table 1-91: Query System Status

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
SystemStatusMT: SystemStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.21 Query Task Plan

MA_TaskPlanMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *MA_TaskPlanMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *MA_TaskPlanMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *MA_TaskPlanMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *MA_TaskPlanMT*.

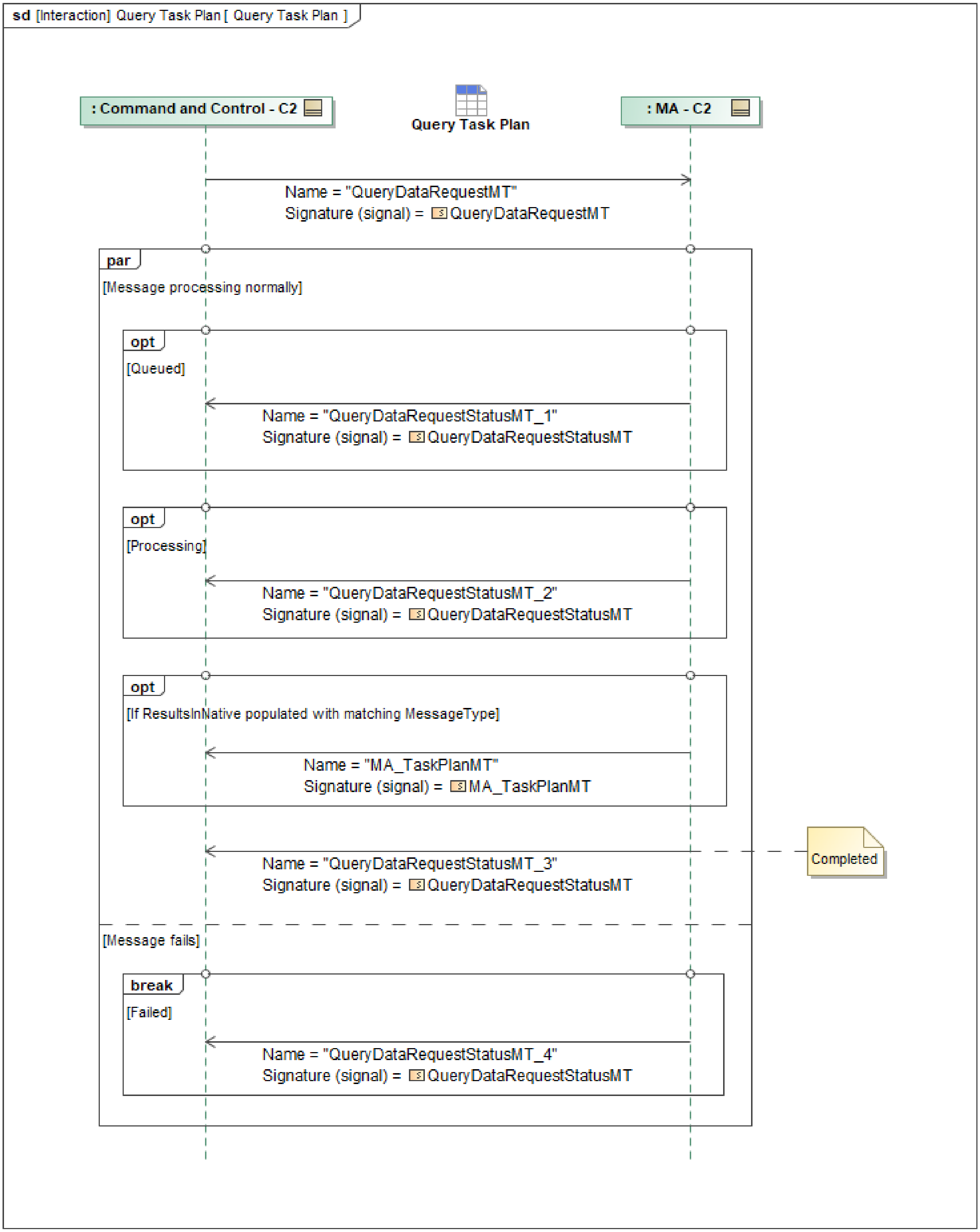


Figure 1-93: Query Task Plan

Table 1-92: Query Task Plan

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskPlanMT: MA_TaskPlanMT	None
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.22 Query Task Plan Execution Status

TaskPlanExecutionStatusMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *TaskPlanExecutionStatusMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *TaskPlanExecutionStatusMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *TaskPlanExecutionStatusMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *TaskPlanExecutionStatusMT*.

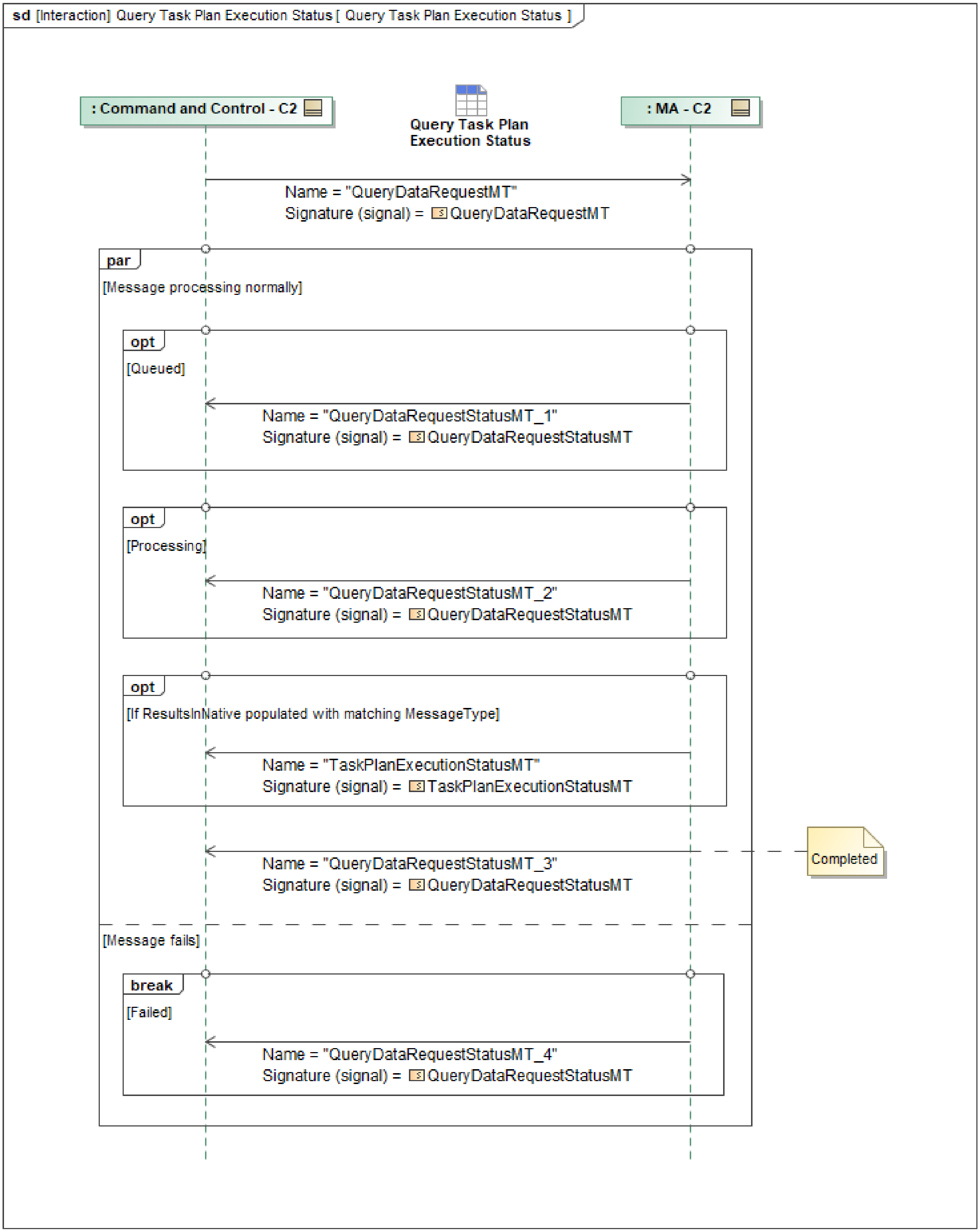


Figure 1-94: Query Task Plan Execution Status

Table 1-93: Query Task Plan Execution Status

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
TaskPlanExecutionStatusMT: TaskPlanExecutionStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.21.23 Query Weather

WeatherMT is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate status messages, beginning with *QueryDataRequestStatusMT_1*, which has *RequestProcessingStateEnum* set to *QUEUED* to indicate that the request has been queued for processing. The service also may send *QueryDataRequestStatusMT_2* with *RequestProcessingStateEnum* set to *PROCESSING*, indicating that the request is actively being processed.

When processing is complete and the requested data is returned within the status message itself, the service sends *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED*, and this message contains the requested *WeatherMT* payload. When the original *QueryDataRequestMT* message's optional field *ResultsInNative* is populated with the matching *WeatherMT* message type, the underlying data management services instead publish the requested data in its original or raw format as a separate *WeatherMT* message and still send *QueryDataRequestStatusMT_3* with *RequestProcessingStateEnum* set to *COMPLETED* to indicate that requested data has been published.

If the request fails, the service sends *QueryDataRequestStatusMT_4* with *RequestProcessingStateEnum* set to *FAILED*, indicating that processing has terminated without successfully providing the requested *WeatherMT*.

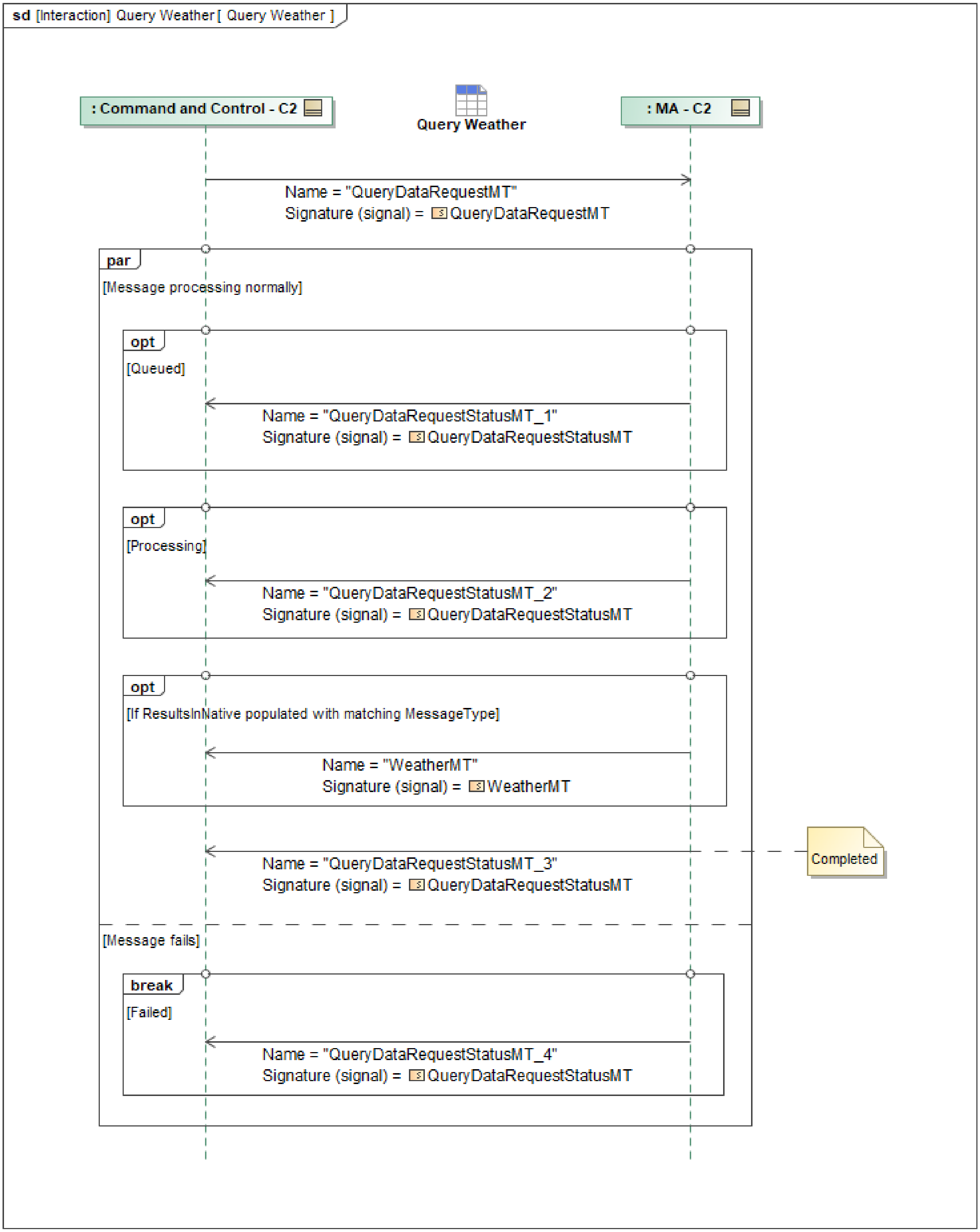


Figure 1-95: Query Weather

Table 1-94: Query Weather

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	None
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType
WeatherMT: WeatherMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22 Send Data - C2 to MA

1.2.8.22.1 Send Action - C2 to MA

This interaction has C2 send *MA_ActionMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

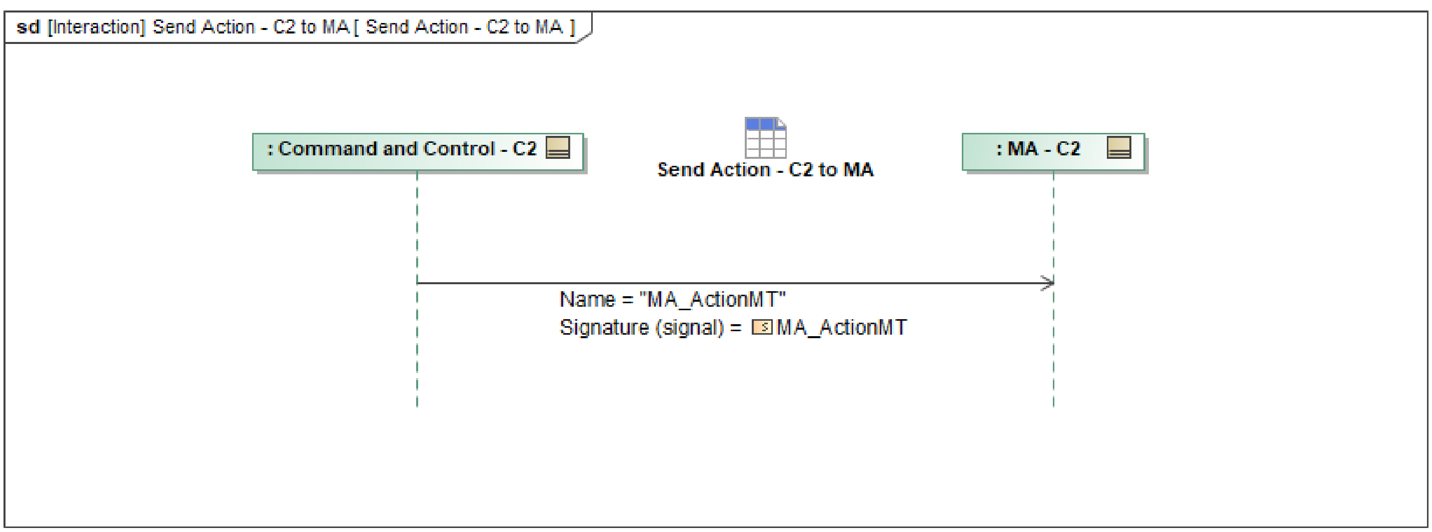


Figure 1-96: Send Action - C2 to MA

Table 1-95: Send Action - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionMT: MA_ActionMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set

Tables section)

1.2.8.22.2 Send Action Plan - C2 to MA

This interaction has C2 send *MA_ActionPlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

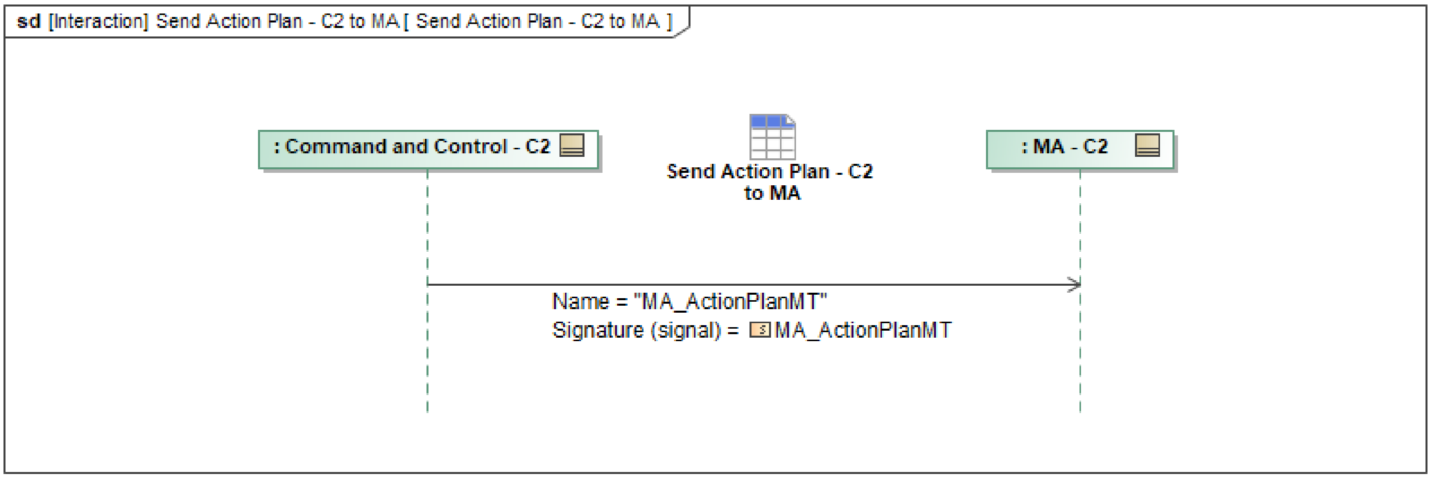


Figure 1-97: Send Action Plan - C2 to MA

Table 1-96: Send Action Plan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionPlanMT: MA_ActionPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.3 Send Activity Plan - C2 to MA

This interaction has C2 send *MA_ActivityPlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

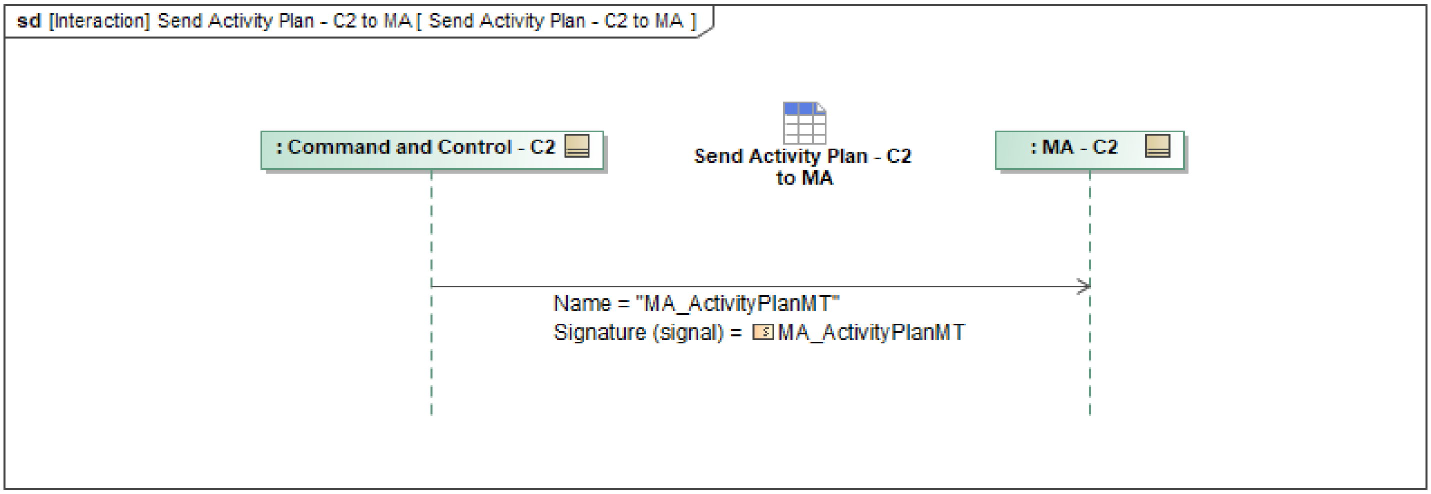


Figure 1-98: Send Activity Plan - C2 to MA

Table 1-97: Send Activity Plan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActivityPlanMT: MA_ActivityPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.4 Send Airfield Report - C2 to MA

This interaction has C2 send *AirfieldReportMT* to MA .

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

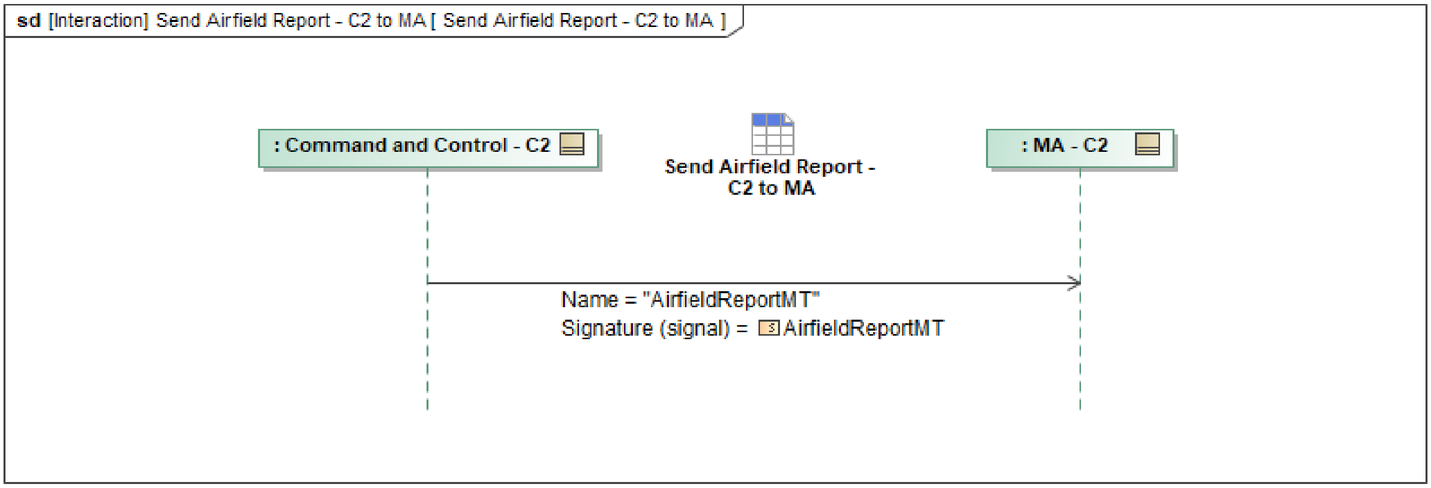


Figure 1-99: Send Airfield Report - C2 to MA

Table 1-98: Send Airfield Report - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
AirfieldReportMT: AirfieldReportMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.5 Send Approval Authority - C2 to MA

This interaction has C2 send *MA_ApprovalAuthorityMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

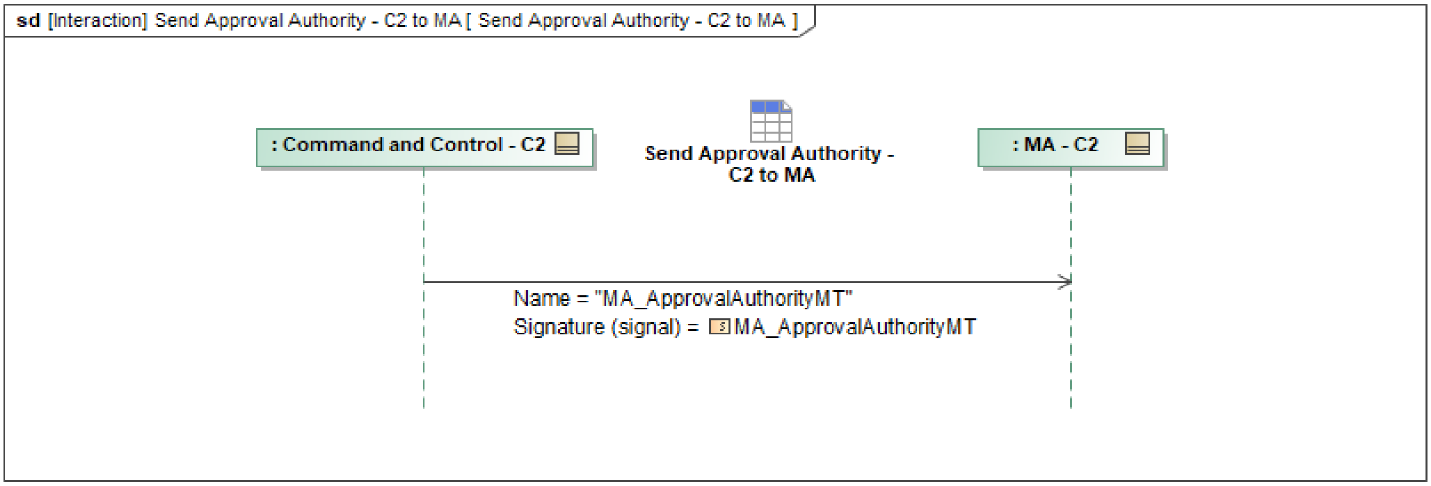


Figure 1-100: Send Approval Authority - C2 to MA

Table 1-99: Send Approval Authority - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalAuthorityMT: MA_ApprovalAuthorityMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.6 Send Approval Policy - C2 to MA

This interaction has C2 send *MA_ApprovalPolicyMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

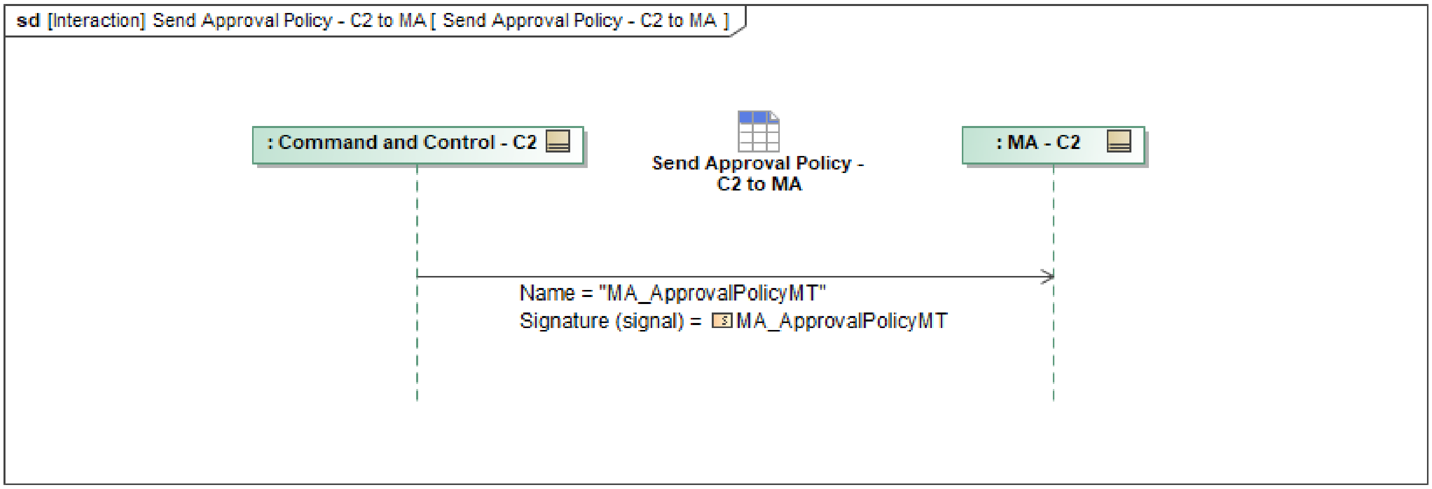


Figure 1-101: Send Approval Policy - C2 to MA

Table 1-100: Send Approval Policy - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalPolicyMT: MA_ApprovalPolicyMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.7 Send Capability Set - C2 to MA

This interaction has C2 send *MA_CapabilitySetMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

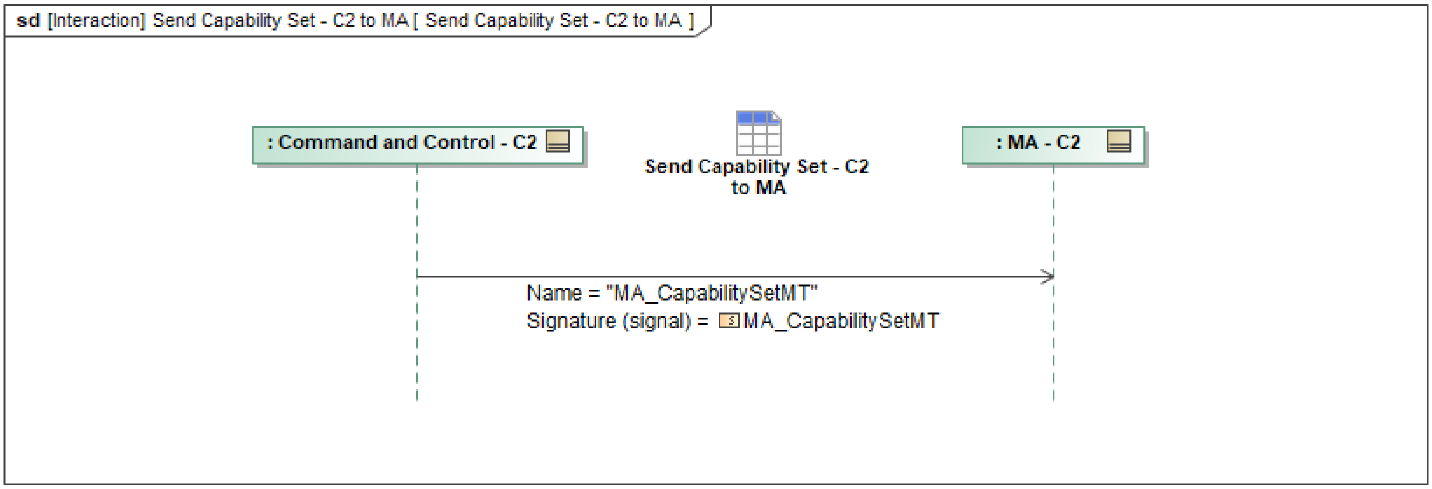


Figure 1-102: Send Capability Set - C2 to MA

Table 1-101: Send Capability Set - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_CapabilitySetMT: MA_CapabilitySetMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.8 Send Constraint Priority List - C2 to MA

This interaction has C2 send *MA_ConstraintPriorityListMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

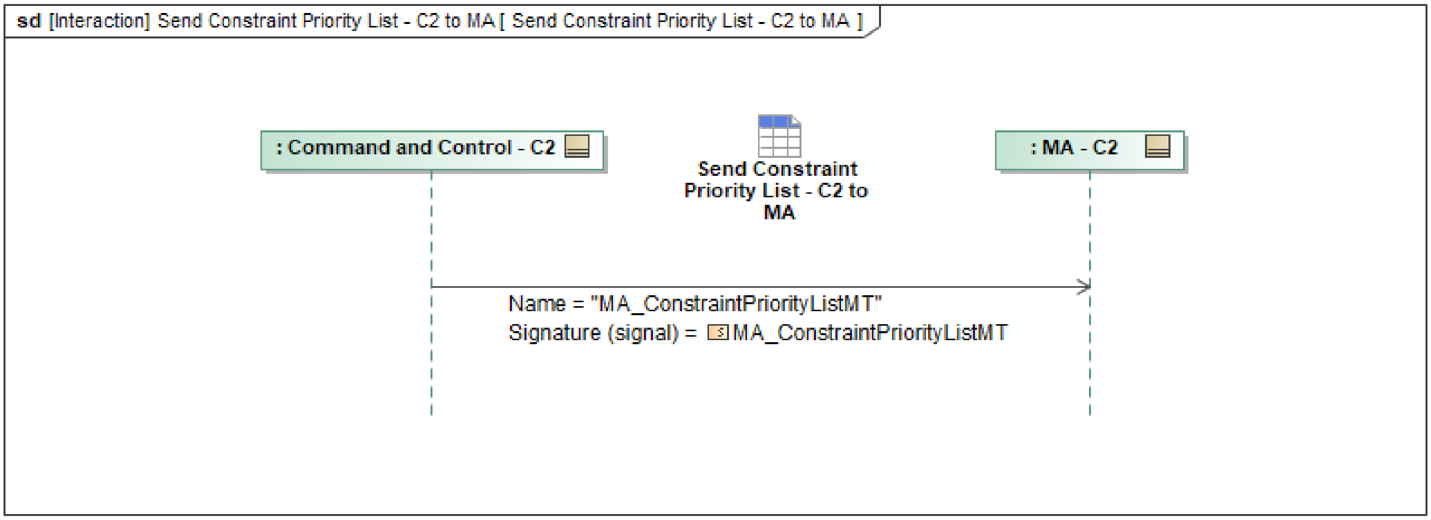


Figure 1-103: Send Constraint Priority List - C2 to MA

Table 1-102: Send Constraint Priority List - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ConstraintPriorityListMT: MA_ConstraintPriorityListMT	ObjectState: ObjectStateEnum ListItem: MA_ConstraintPriorityListItemType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.9 Send Constraint Priority List Activation Data - C2 to MA

This interaction has C2 send *MA_ConstraintPriorityListActivationDataMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

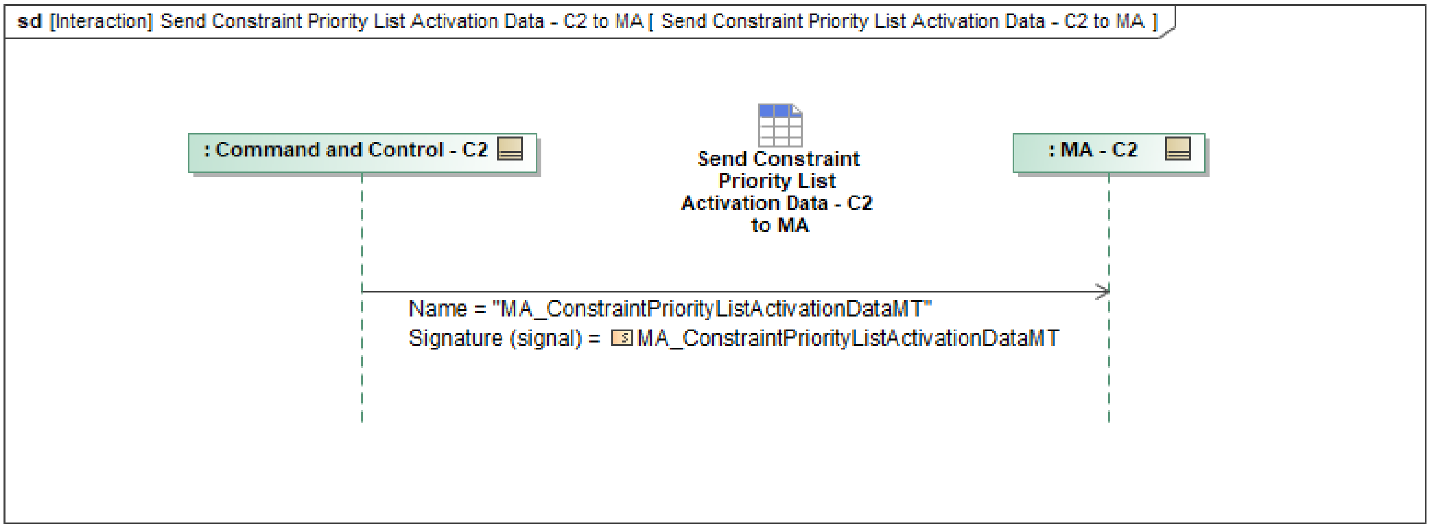


Figure 1-104: Send Constraint Priority List Activation Data - C2 to MA

Table 1-103: Send Constraint Priority List Activation Data - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ConstraintPriorityListActivationDataMT: MA_ConstraintPriorityListActivationDataMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.10 Send Engagement Criteria - C2 to MA

This interaction has C2 send *MA_EngagementCriteriaMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

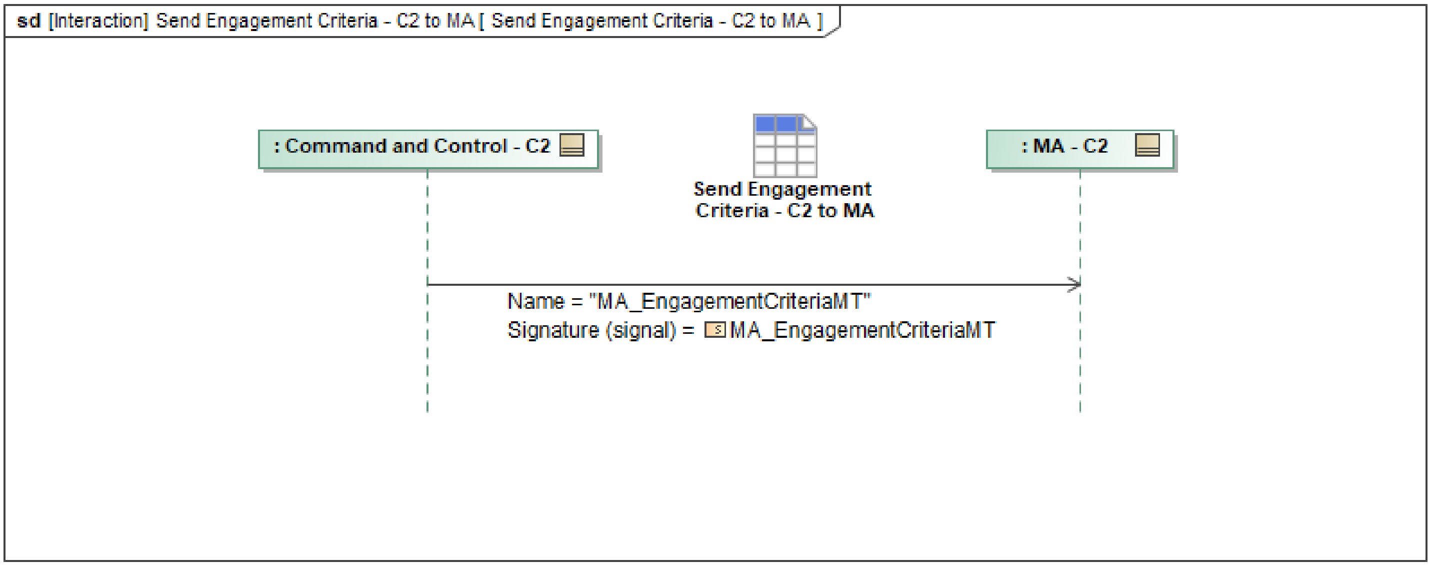


Figure 1-105: Send Engagement Criteria - C2 to MA

Table 1-104: Send Engagement Criteria - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_EngagementCriteriaMT: MA_EngagementCriteriaMT	TargetAuthorityCriteria: MA_AuthorityCriteriaType TargetDesignationCriteria: MA_TargetDesignationCriteriaType TargetCustodyCriteria: MA_TargetCustodyCriteriaType IdMatrixCriteria: MA_IdMatrixCriteriaType TargetDesignationCriteria: MA_TargetDesignationCriteriaType TargetCustodyCriteria: MA_TargetCustodyCriteriaType IdMatrixCriteria: MA_IdMatrixCriteriaType TargetAuthorityCriteria: MA_AuthorityCriteriaType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.11 Send Entity - C2 to MA

This interaction has C2 send *EntityMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

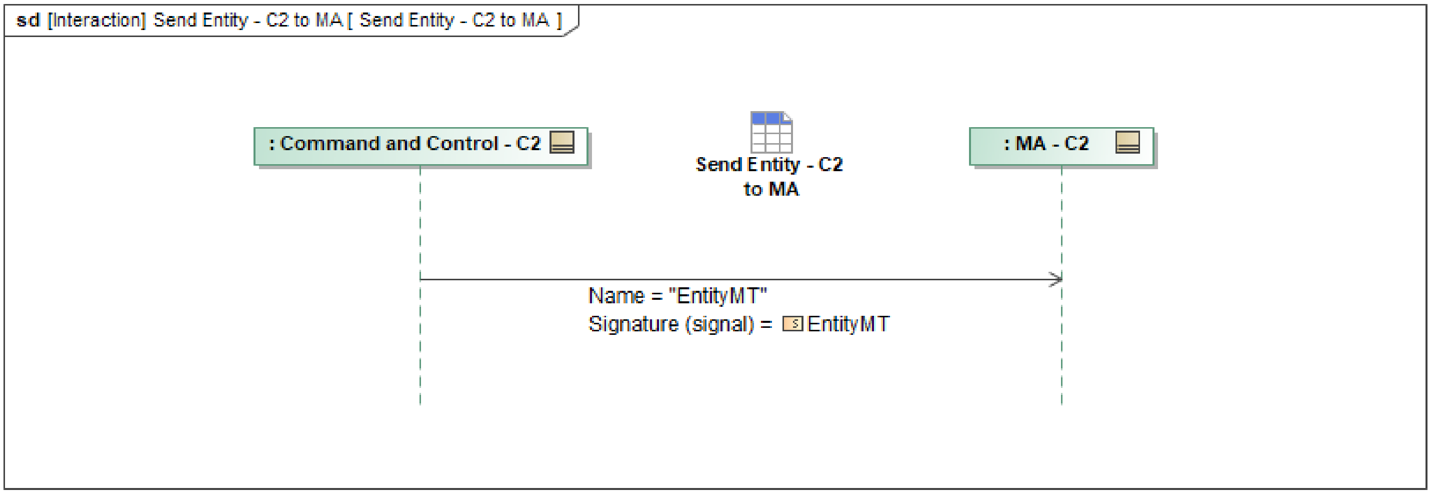


Figure 1-106: Send Entity - C2 to MA

Table 1-105: Send Entity - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
EntityMT: EntityMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.12 Send Execution Priority List - C2 to MA

This interaction has C2 send *MA_ExecutionPriorityListMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

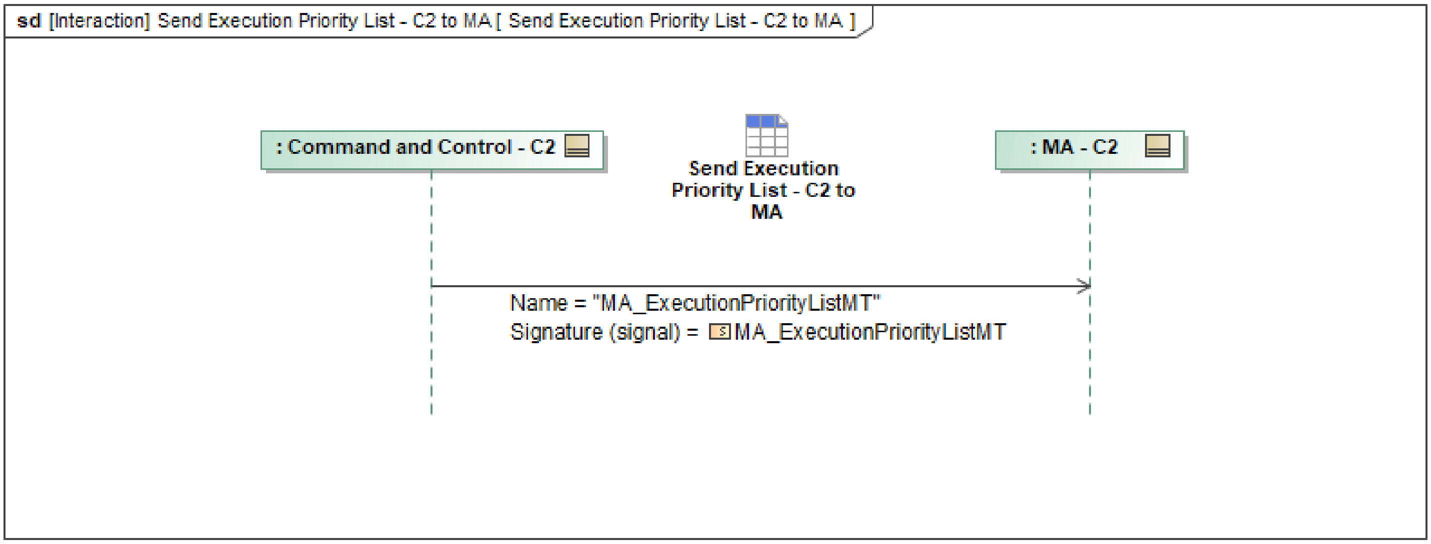


Figure 1-107: Send Execution Priority List - C2 to MA

Table 1-106: Send Execution Priority List - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ExecutionPriorityListMT: MA_ExecutionPriorityListMT	ObjectState: ObjectStateEnum ListItem: MA_ExecutionPriorityListItemType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.13 Send Execution Priority List Activation Data - C2 to MA

This interaction has C2 send *MA_ExecutionPriorityListActivationDataMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

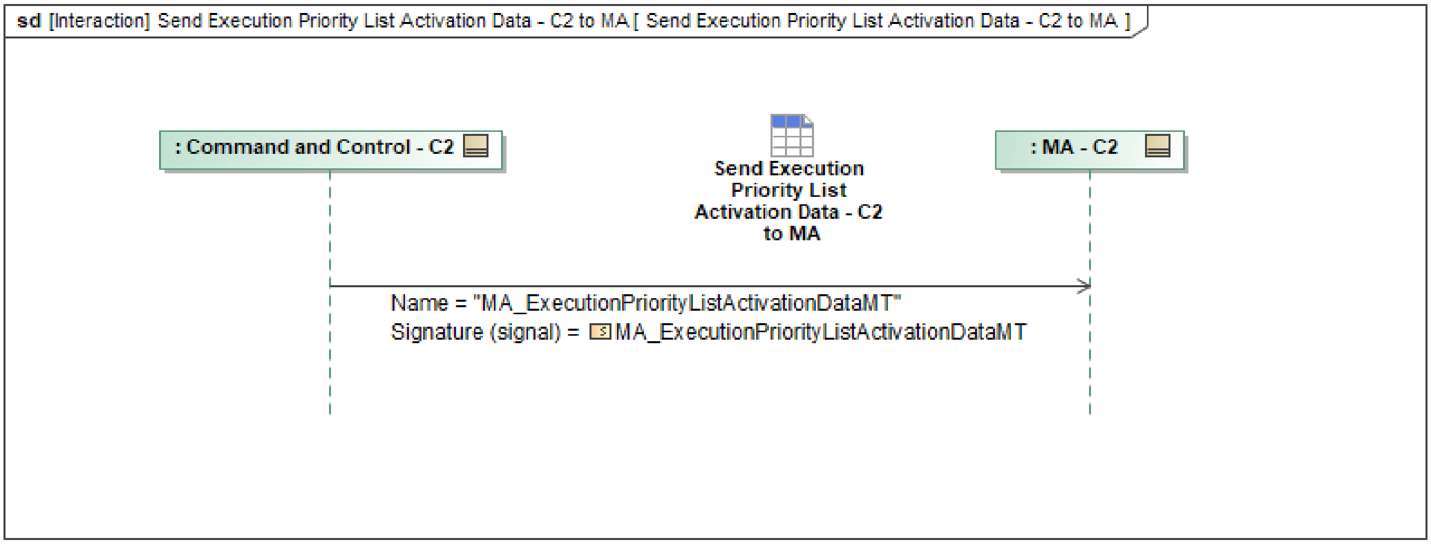


Figure 1-108: Send Execution Priority List Activation Data - C2 to MA

Table 1-107: Send Execution Priority List Activation Data - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ExecutionPriorityListActivationDataMT: MA_ExecutionPriorityListActivationDataMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.14 Send Interoperability Policy - C2 to MA

This interaction has C2 send *MA_InteroperabilityPolicyMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

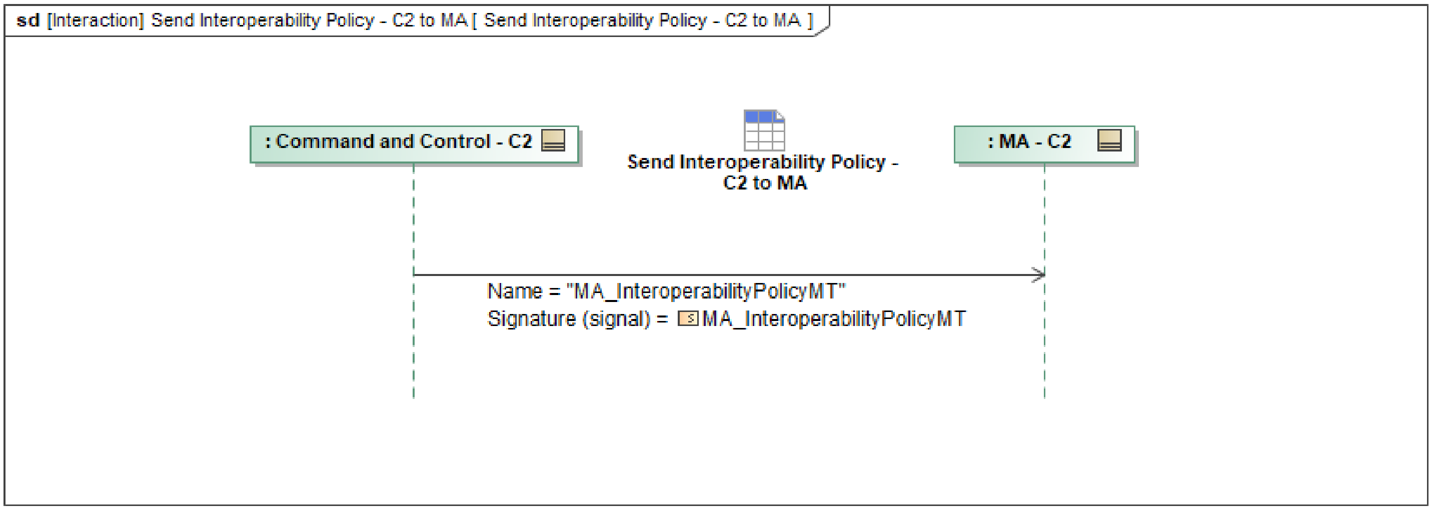


Figure 1-109: Send Interoperability Policy - C2 to MA

Table 1-108: Send Interoperability Policy - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_InteroperabilityPolicyMT: MA_InteroperabilityPolicyMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.15 Send Leadership Metrics - C2 to MA

This interaction has C2 send *MA_LeadershipMetricsMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

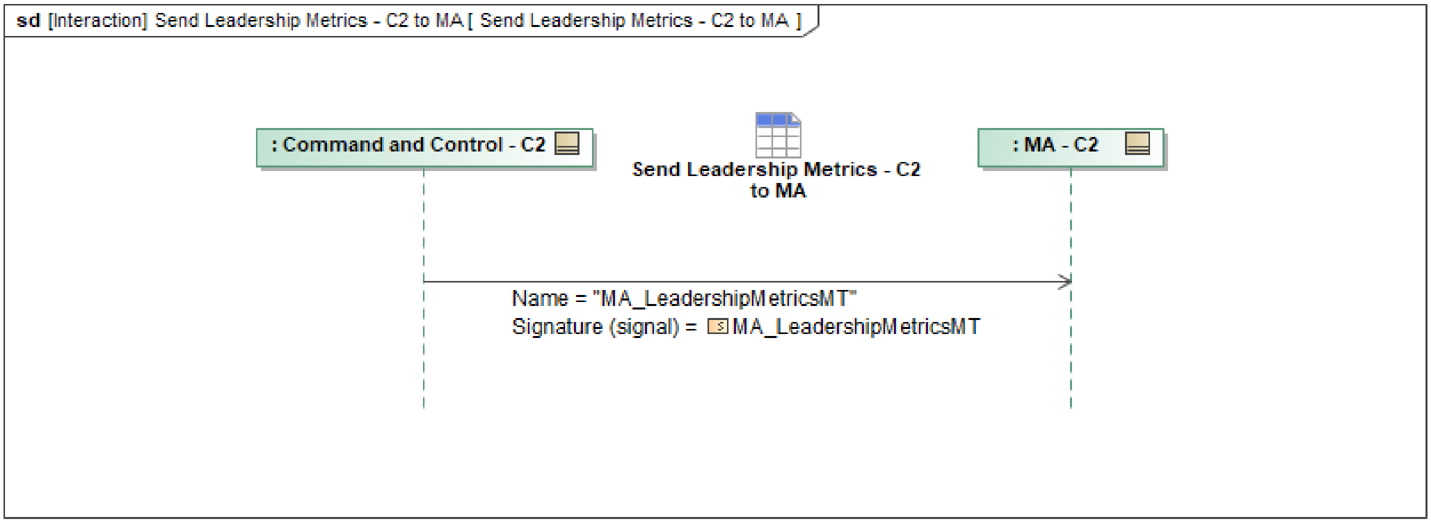


Figure 1-110: Send Leadership Metrics - C2 to MA

Table 1-109: Send Leadership Metrics - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_LeadershipMetricsMT: MA_LeadershipMetricsMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.16 Send Mission Plan - C2 to MA

This interaction has C2 send *MA_MissionPlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

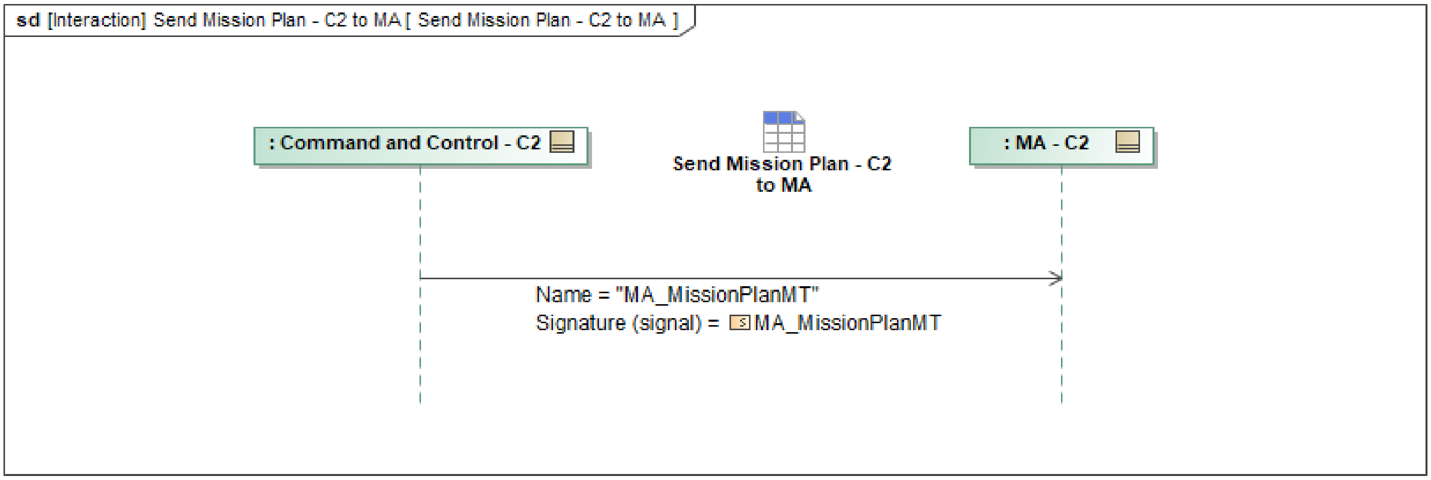


Figure 1-111: Send Mission Plan - C2 to MA

Table 1-110: Send Mission Plan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanMT: MA_MissionPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.17 Send Operator - C2 to MA

This interaction has C2 send *OperatorMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

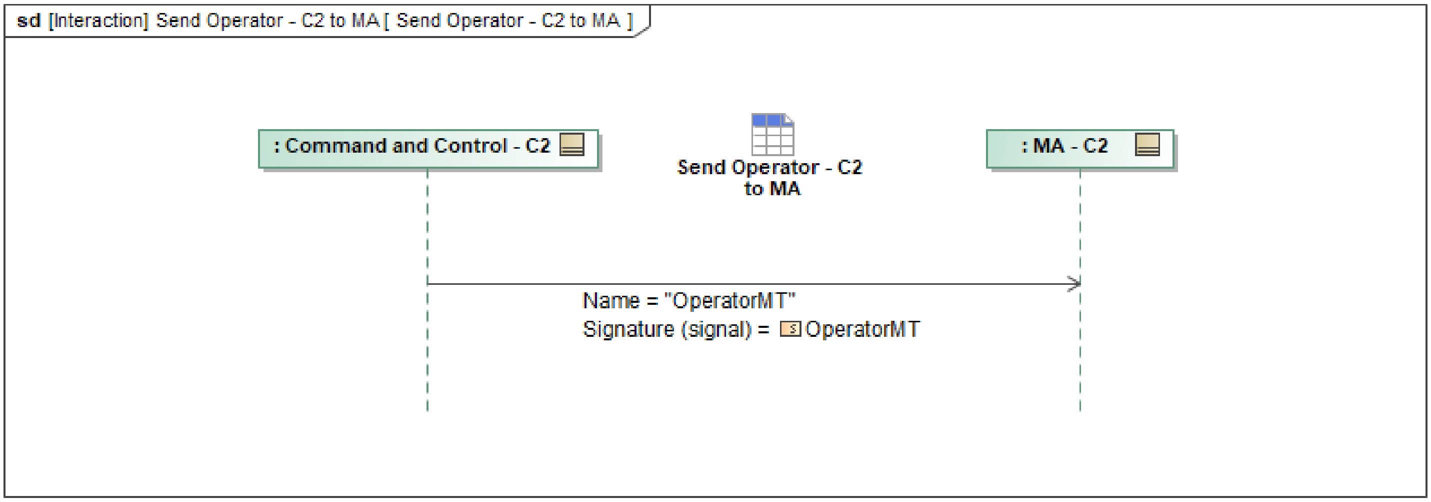


Figure 1-112: Send Operator - C2 to MA

Table 1-111: Send Operator - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
OperatorMT: OperatorMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.18 Send Operator Role - C2 to MA

This interaction has C2 send *MA_OperatorRoleMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

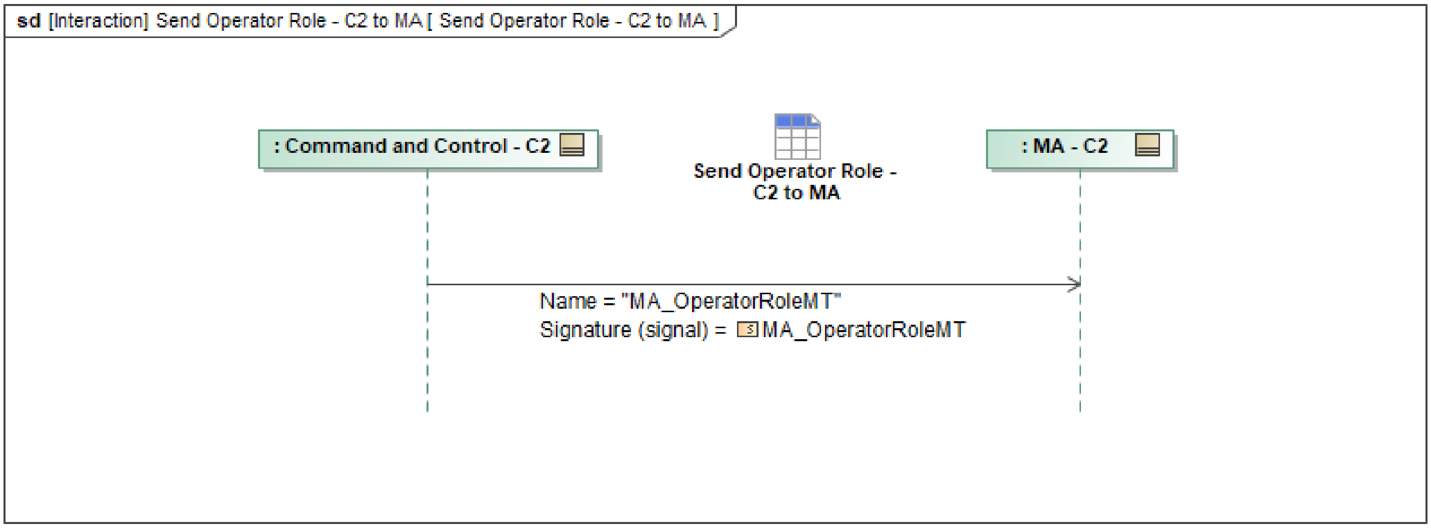


Figure 1-113: Send Operator Role - C2 to MA

Table 1-112: Send Operator Role - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OperatorRoleMT: MA_OperatorRoleMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.19 Send OpLine - C2 to MA

This interaction has C2 send *MA_OpLineMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

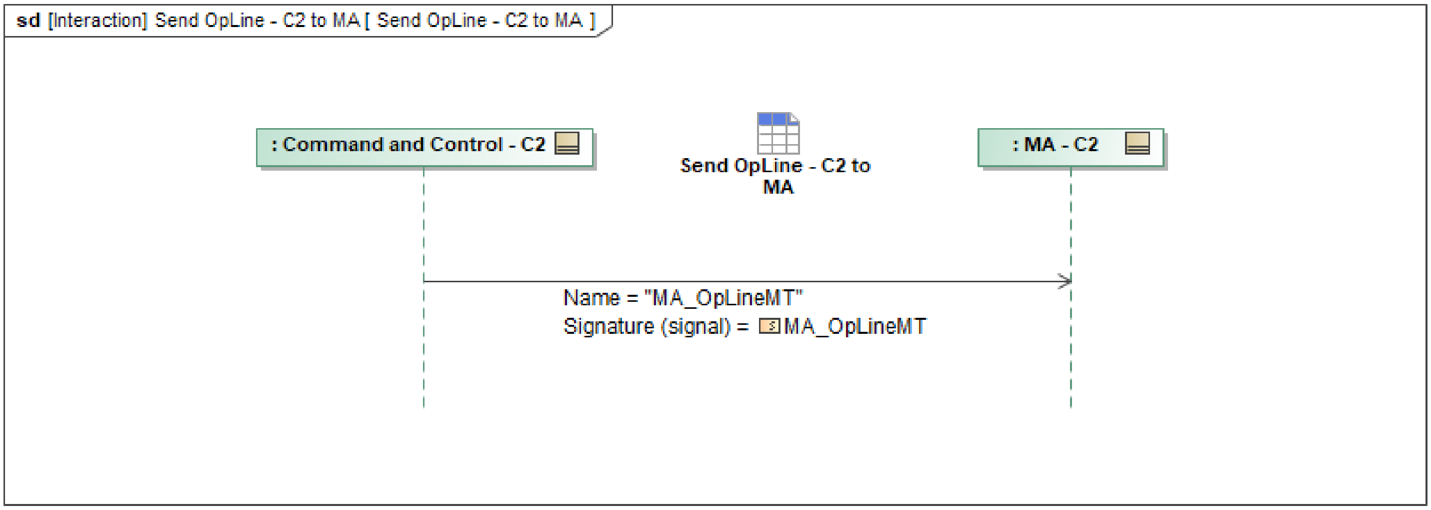


Figure 1-114: Send OpLine - C2 to MA

Table 1-113: Send OpLine - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpLineMT: MA_OpLineMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.20 Send OpPlan - C2 to MA

This interaction has C2 send *MA_OpPlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

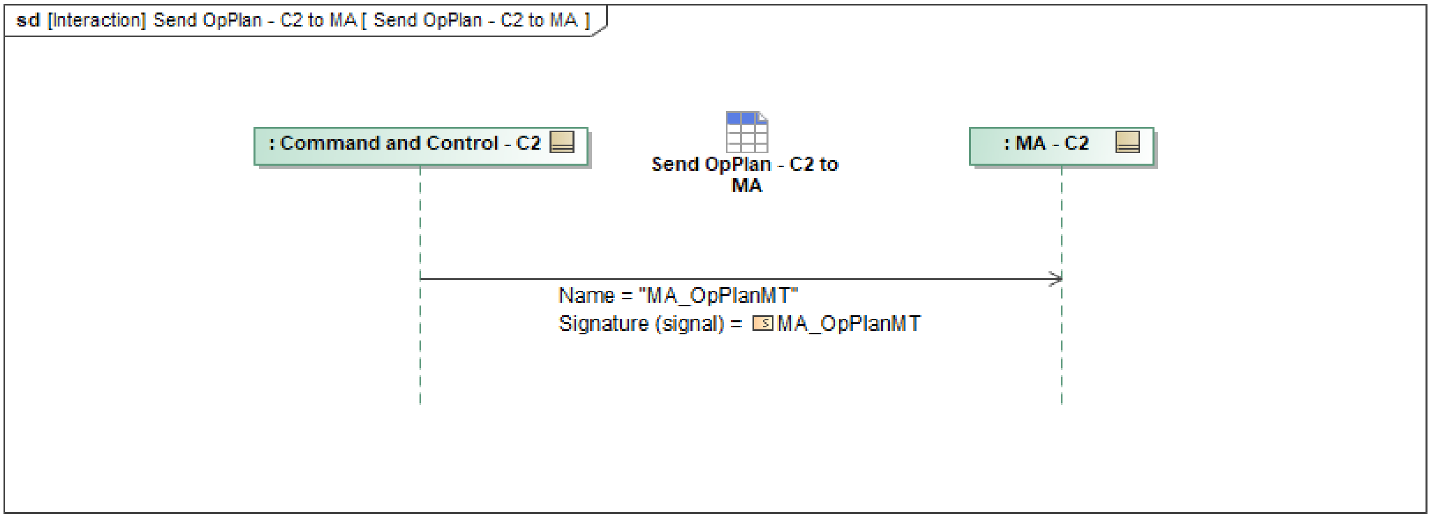


Figure 1-115: Send OpPlan - C2 to MA

Table 1-114: Send OpPlan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpPlanMT: MA_OpPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.21 Send OpPoint - C2 to MA

This interaction has C2 send *MA_OpPointMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

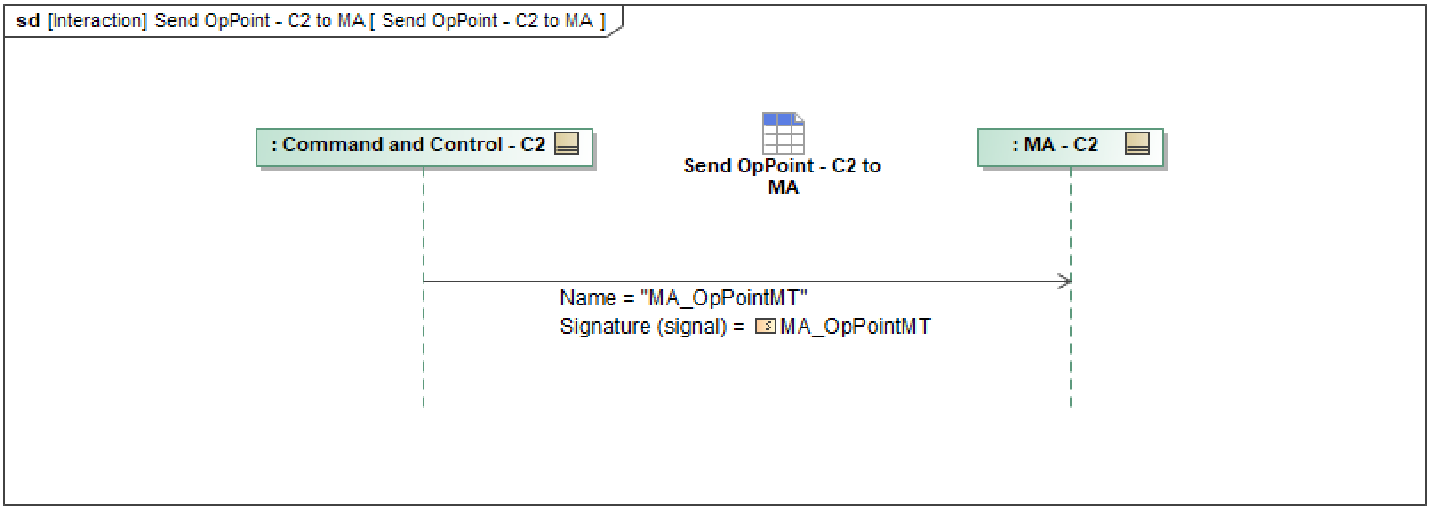


Figure 1-116: Send OpPoint - C2 to MA

Table 1-115: Send OpPoint - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpPointMT: MA_OpPointMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.22 Send OpRouting - C2 to MA

This interaction has C2 send *MA_OpRoutingMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

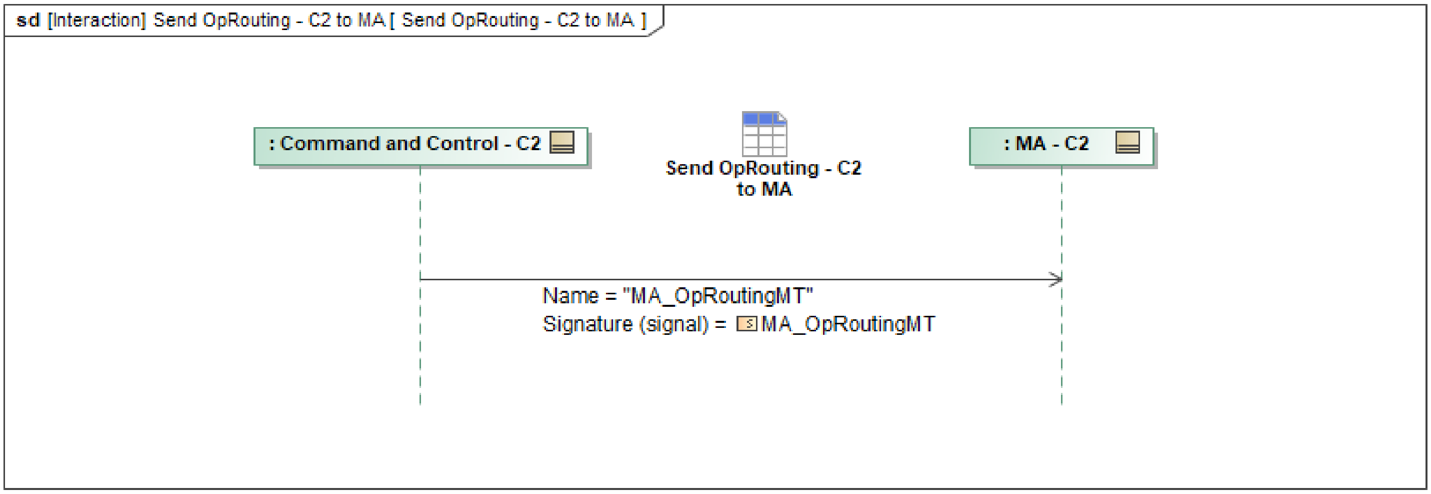


Figure 1-117: Send OpRouting - C2 to MA

Table 1-116: Send OpRouting - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpRoutingMT: MA_OpRoutingMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.23 Send OpVolume - C2 to MA

This interaction has C2 send *MA_OpVolumeMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

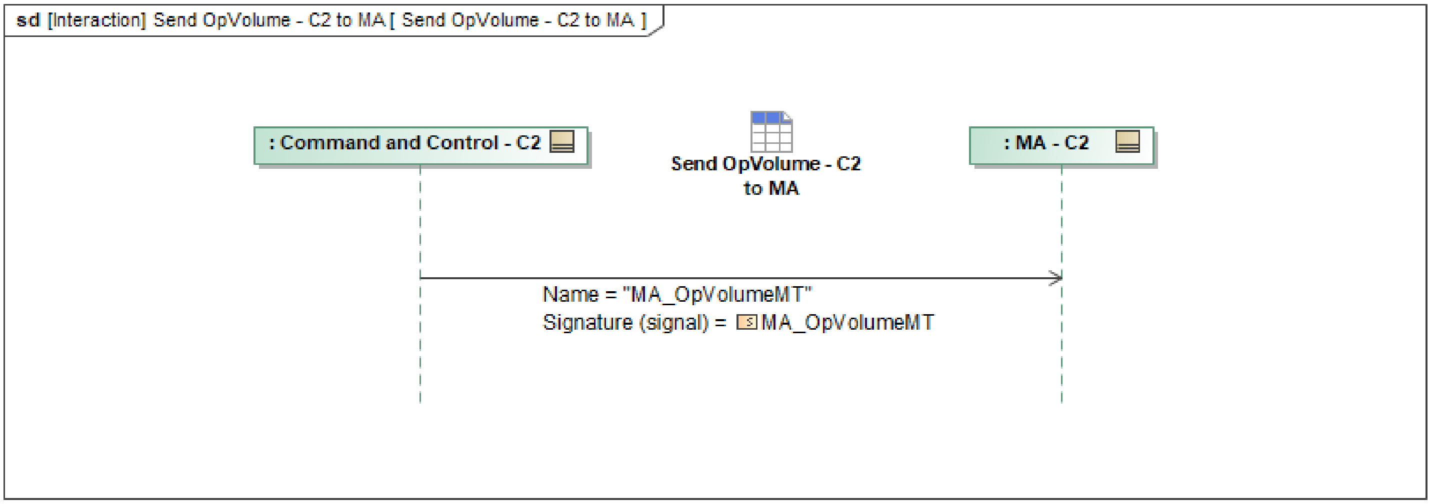


Figure 1-118: Send OpVolume - C2 to MA

Table 1-117: Send OpVolume - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpVolumeMT: MA_OpVolumeMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.24 Send OpZone - C2 to MA

This interaction has C2 send *MA_OpZoneMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

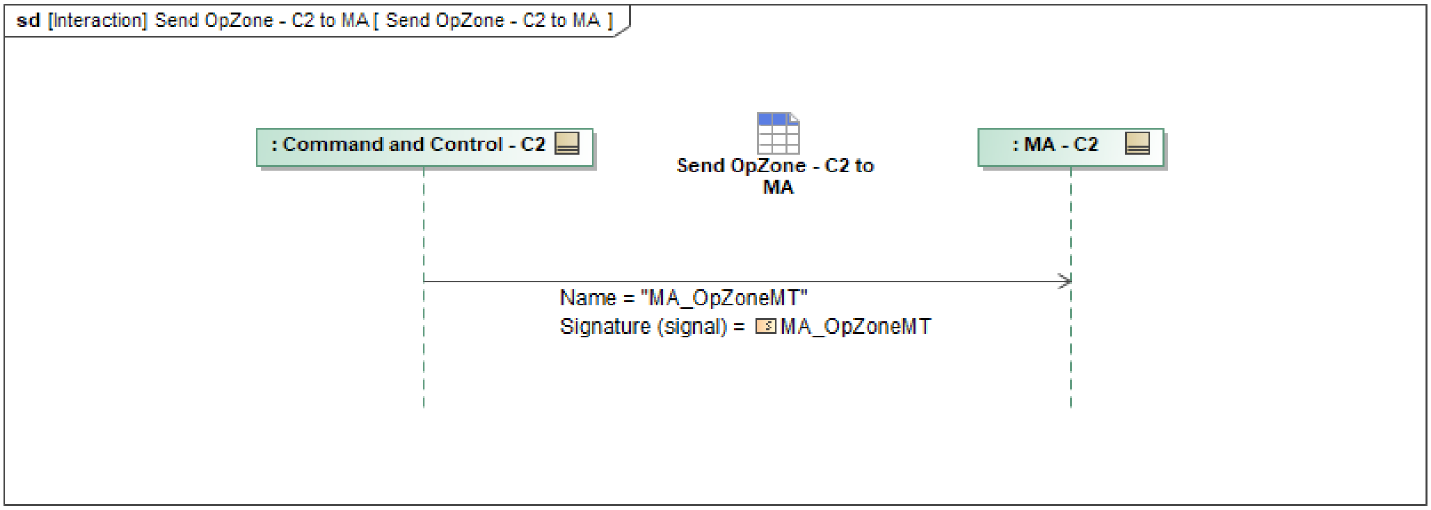


Figure 1-119: Send OpZone - C2 to MA

Table 1-118: Send OpZone - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpZoneMT: MA_OpZoneMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.25 Send Package - C2 to MA

This interaction has C2 send *PackageMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

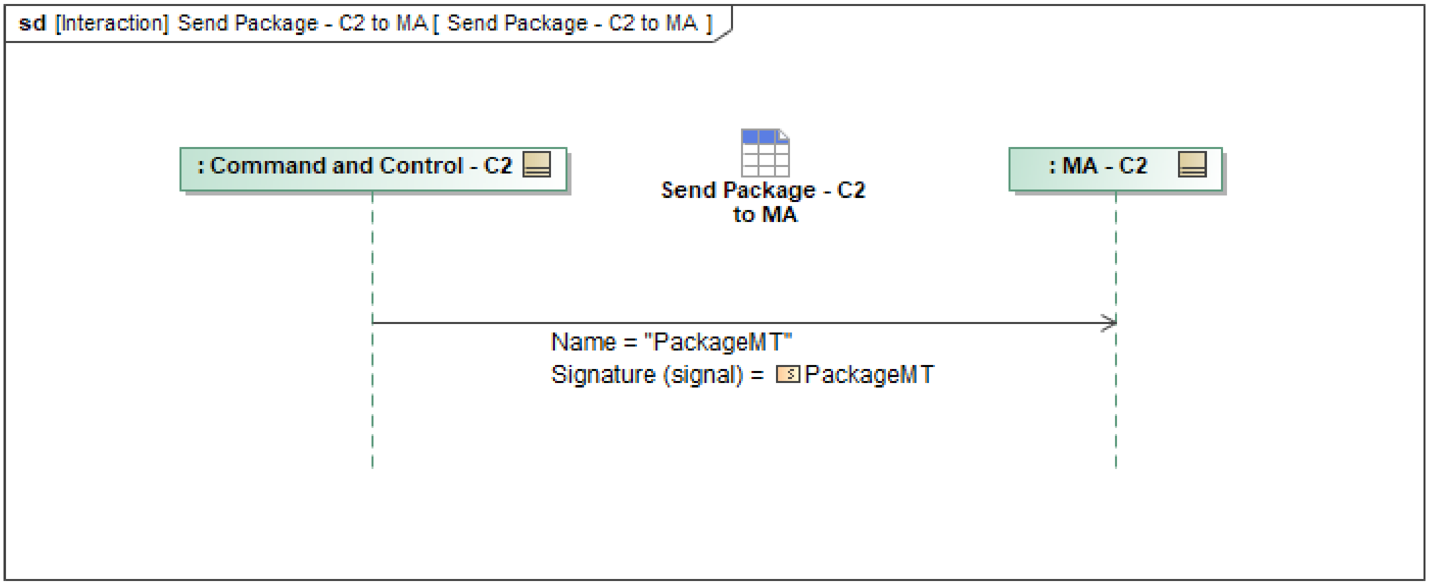


Figure 1-120: Send Package - C2 to MA

Table 1-119: Send Package - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
PackageMT: PackageMT	DataRecordInstanceID: DataRecordInstanceID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.26 Send Response - C2 to MA

This interaction has C2 send *MA_ResponseMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

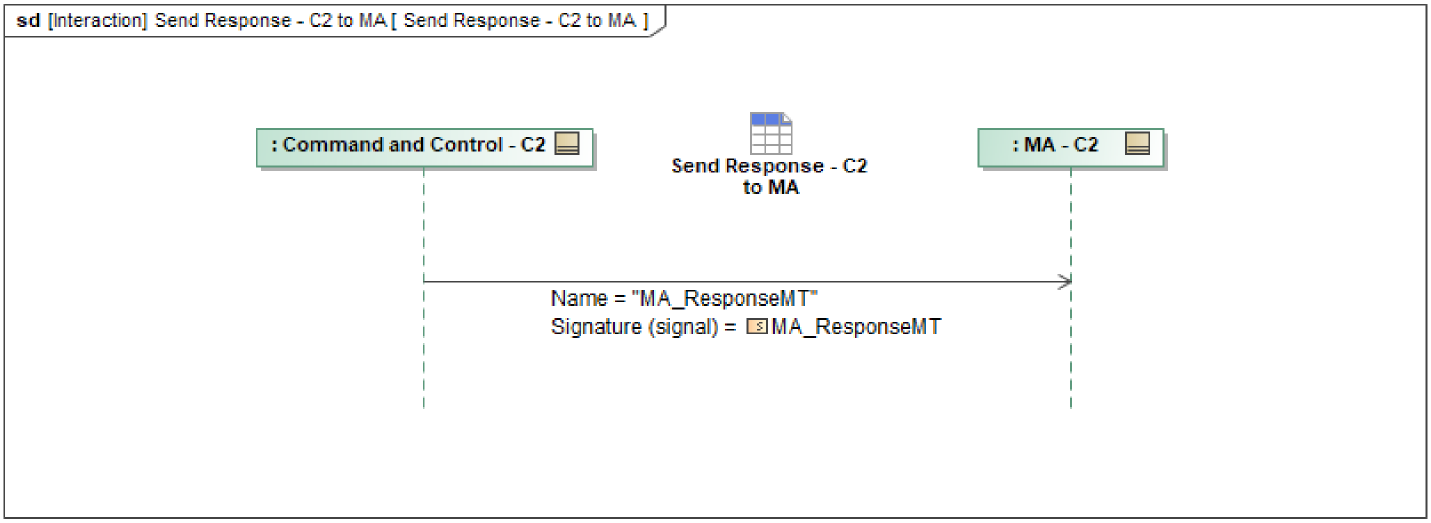


Figure 1-121: Send Response - C2 to MA

Table 1-120: Send Response - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ResponseMT: MA_ResponseMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.27 Send Response Plan - C2 to MA

This interaction has C2 send *ResponsePlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

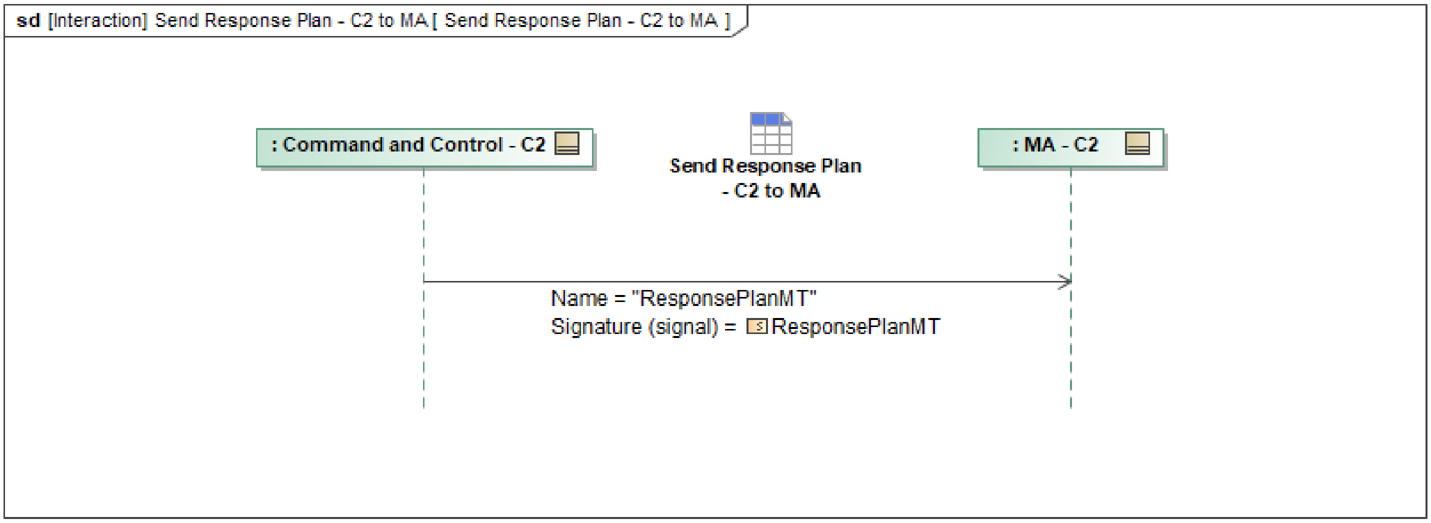


Figure 1-122: Send Response Plan - C2 to MA

Table 1-121: Send Response Plan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
ResponsePlanMT: ResponsePlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.28 Send Route Activity Plan - C2 to MA

This interaction has C2 send *MA_RouteActivityPlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

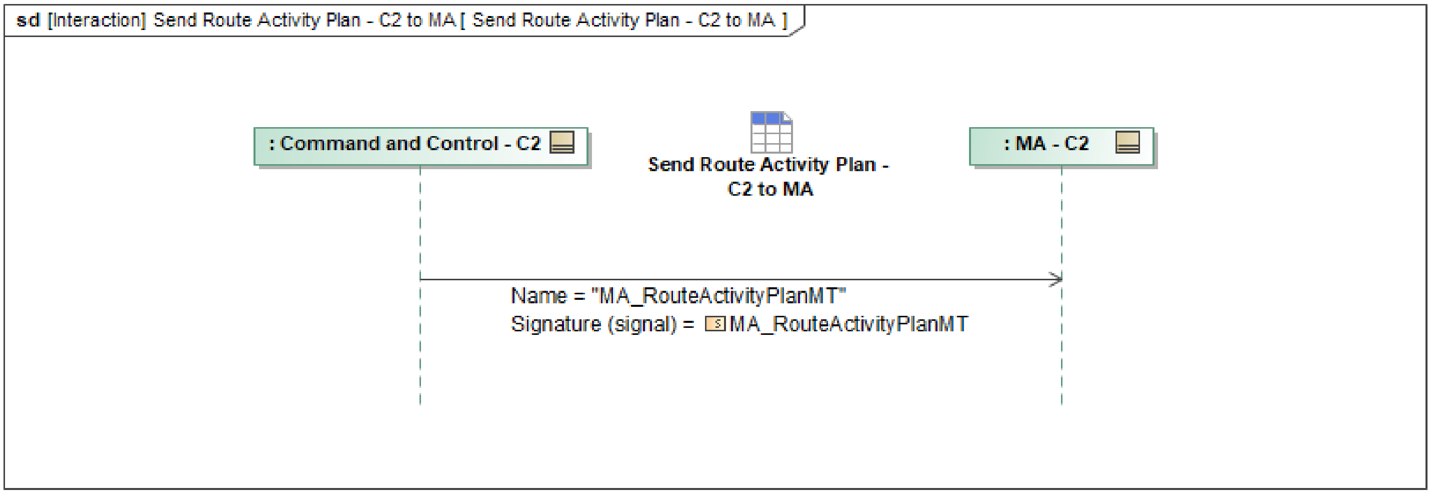


Figure 1-123: Send Route Activity Plan - C2 to MA

Table 1-122: Send Route Activity Plan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RouteActivityPlanMT: MA_RouteActivityPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.29 Send Route Metrics - C2 to MA

This interaction has C2 send *RouteMetricsMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

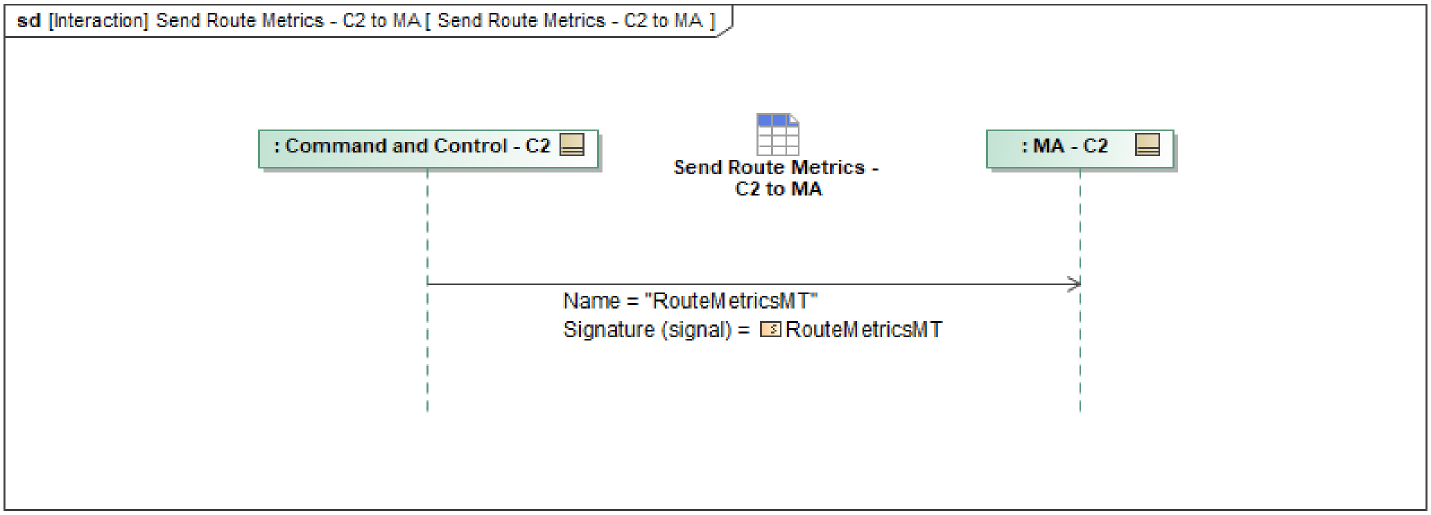


Figure 1-124: Send Route Metrics - C2 to MA

Table 1-123: Send Route Metrics - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
RouteMetricsMT: RouteMetricsMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.30 Send Route Plan - C2 to MA

This interaction has C2 send *MA_RoutePlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

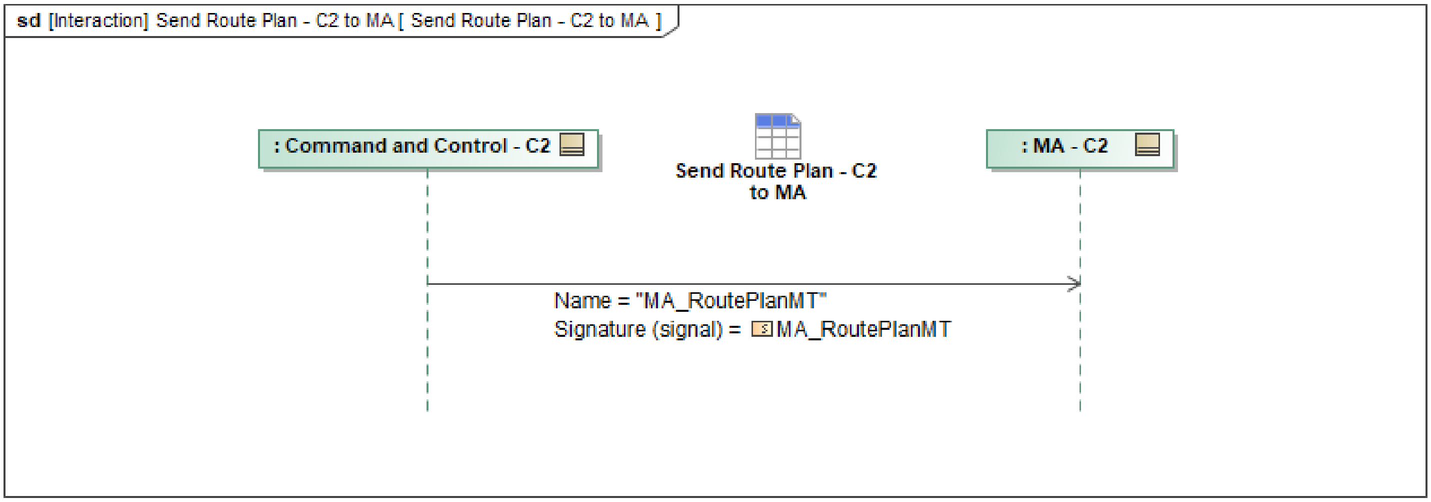


Figure 1-125: Send Route Plan - C2 to MA

Table 1-124: Send Route Plan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RoutePlanMT: MA_RoutePlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.31 Send Task - C2 to MA

This interaction has C2 send *MA_TaskMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

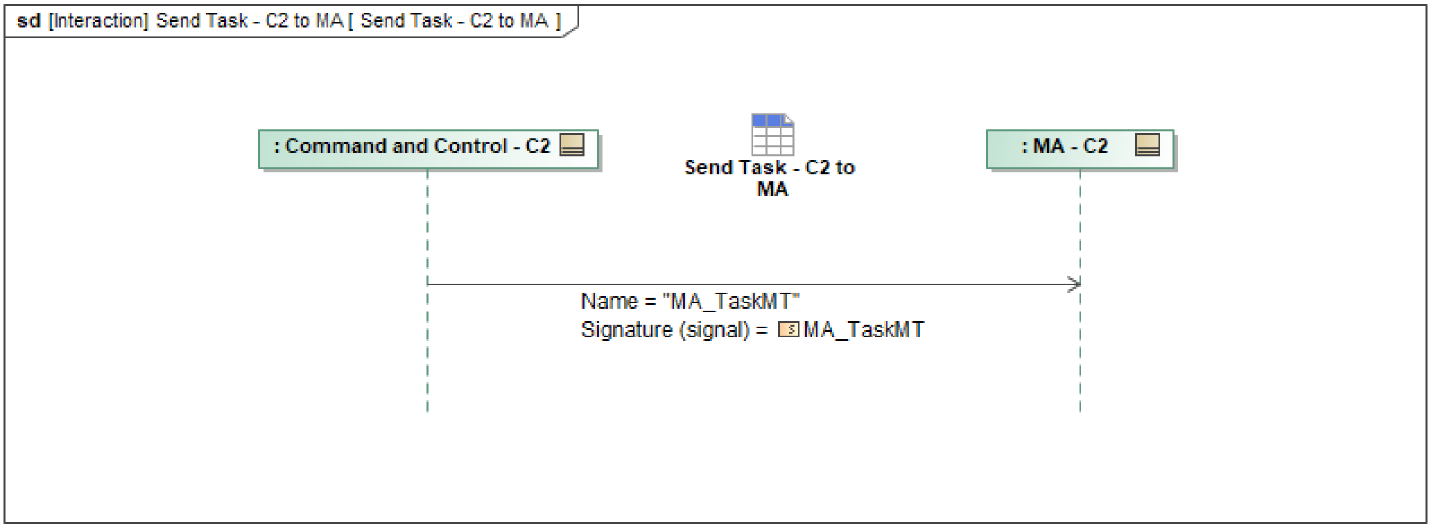


Figure 1-126: Send Task - C2 to MA

Table 1-125: Send Task - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskMT: MA_TaskMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.32 Send Task Plan - C2 to MA

This interaction has C2 send *MA_TaskPlanMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

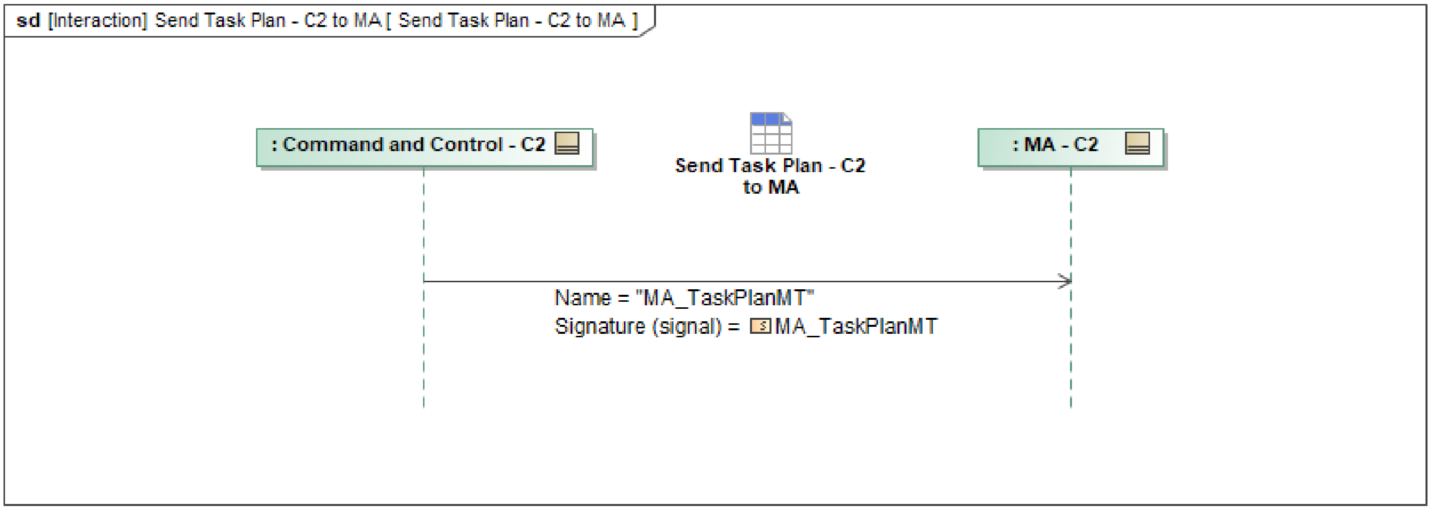


Figure 1-127: Send Task Plan - C2 to MA

Table 1-126: Send Task Plan - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskPlanMT: MA_TaskPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.22.33 Send Weather Dataset - C2 to MA

This interaction has C2 send *WeatherDatasetMT* to MA.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

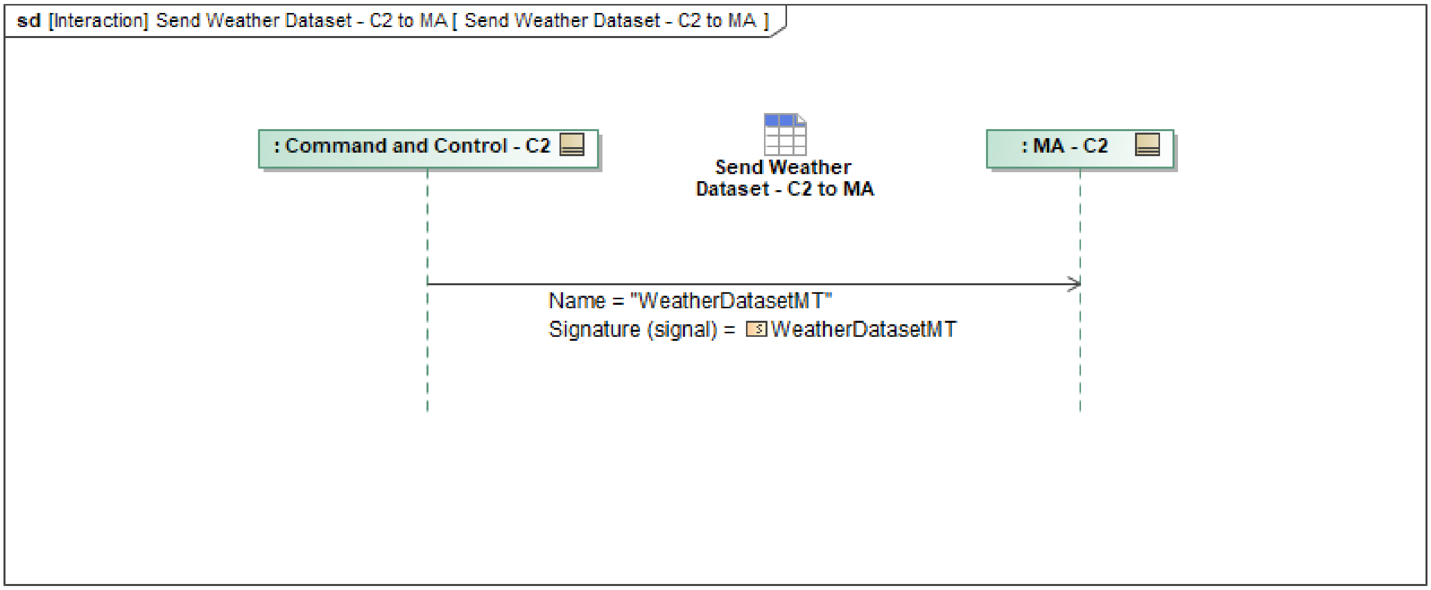


Figure 1-128: Send Weather Dataset - C2 to MA

Table 1-127: Send Weather Dataset - C2 to MA

Message Name (Name:Type)	Required Fields (Name:Type)
WeatherDatasetMT: WeatherDatasetMT	DataRecordInstanceID: DataRecordInstanceID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23 Send Data - MA to C2

1.2.8.23.1 Send Action - MA to C2

This interaction has MA send *MA_ActionMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

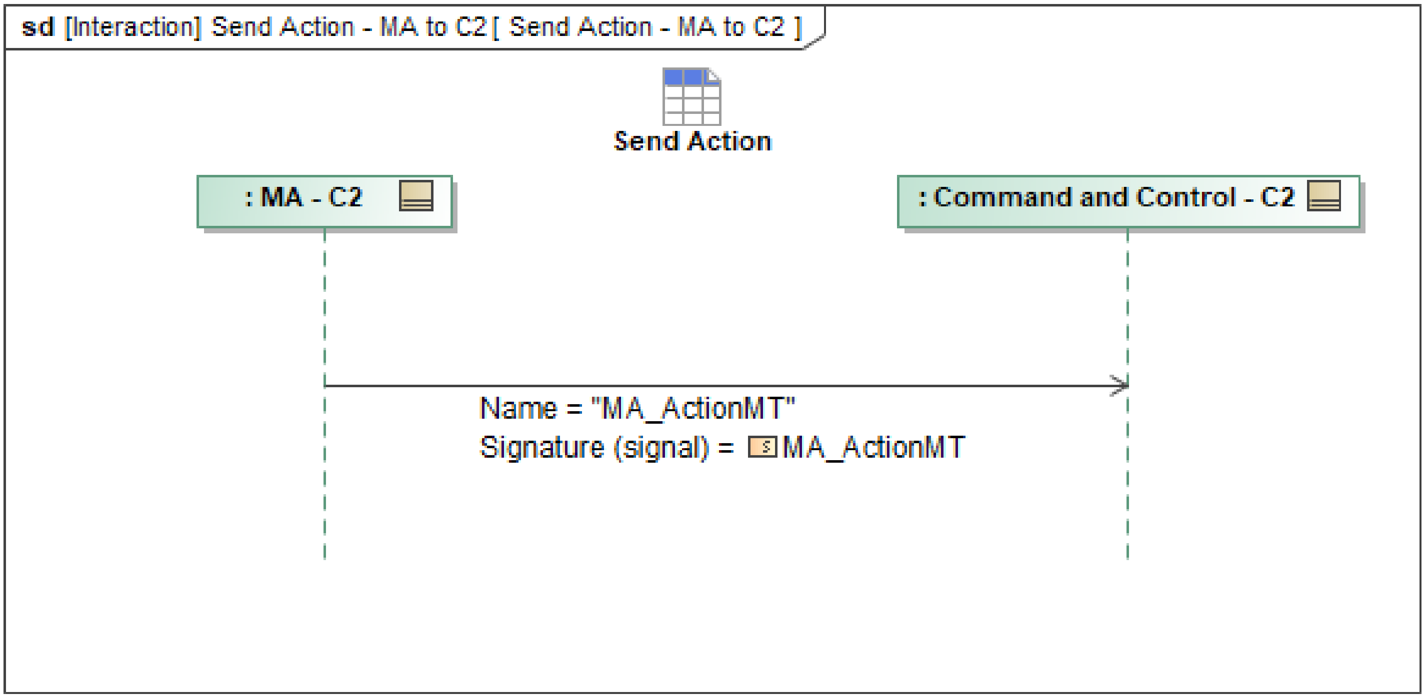


Figure 1-129: Send Action - MA to C2

Table 1-128: Send Action - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionMT: MA_ActionMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.2 Send Action Plan - MA to C2

This interaction has MA send *MA_ActionPlanMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

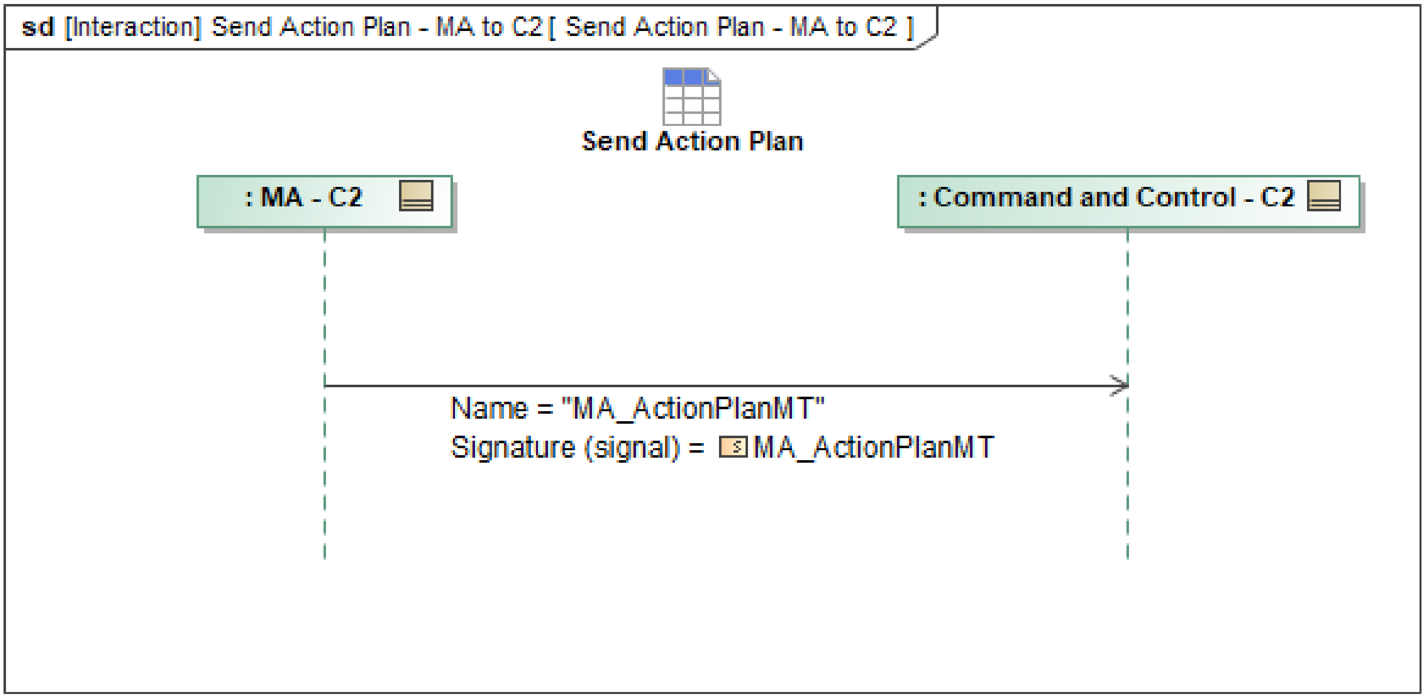


Figure 1-130: Send Action Plan - MA to C2

Table 1-129: Send Action Plan - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionPlanMT: MA_ActionPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.3 Send Activity Plan - MA to C2

This interaction has MA send *MA_ActivityPlanMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

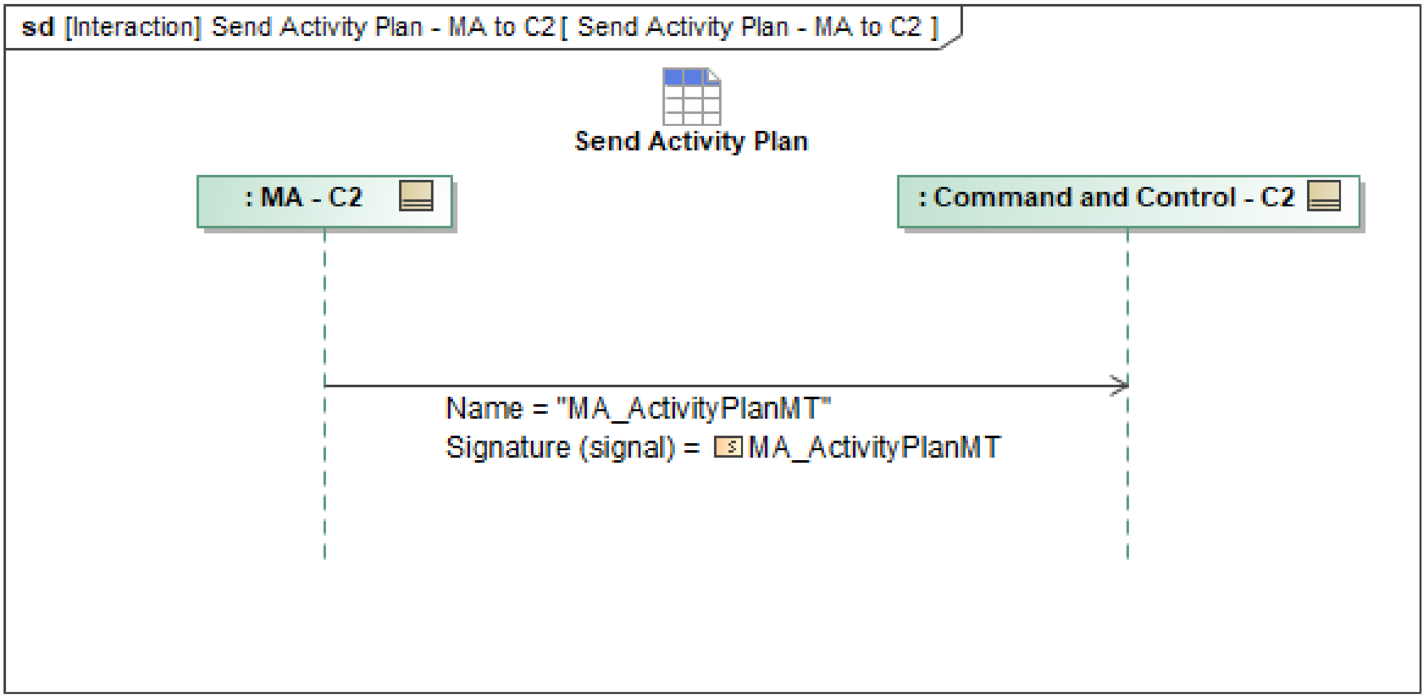


Figure 1-131: Send Activity Plan - MA to C2

Table 1-130: Send Activity Plan - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActivityPlanMT: MA_ActivityPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.4 Send Approval Authority - MA to C2

This interaction has MA send *MA_ApprovalAuthorityMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

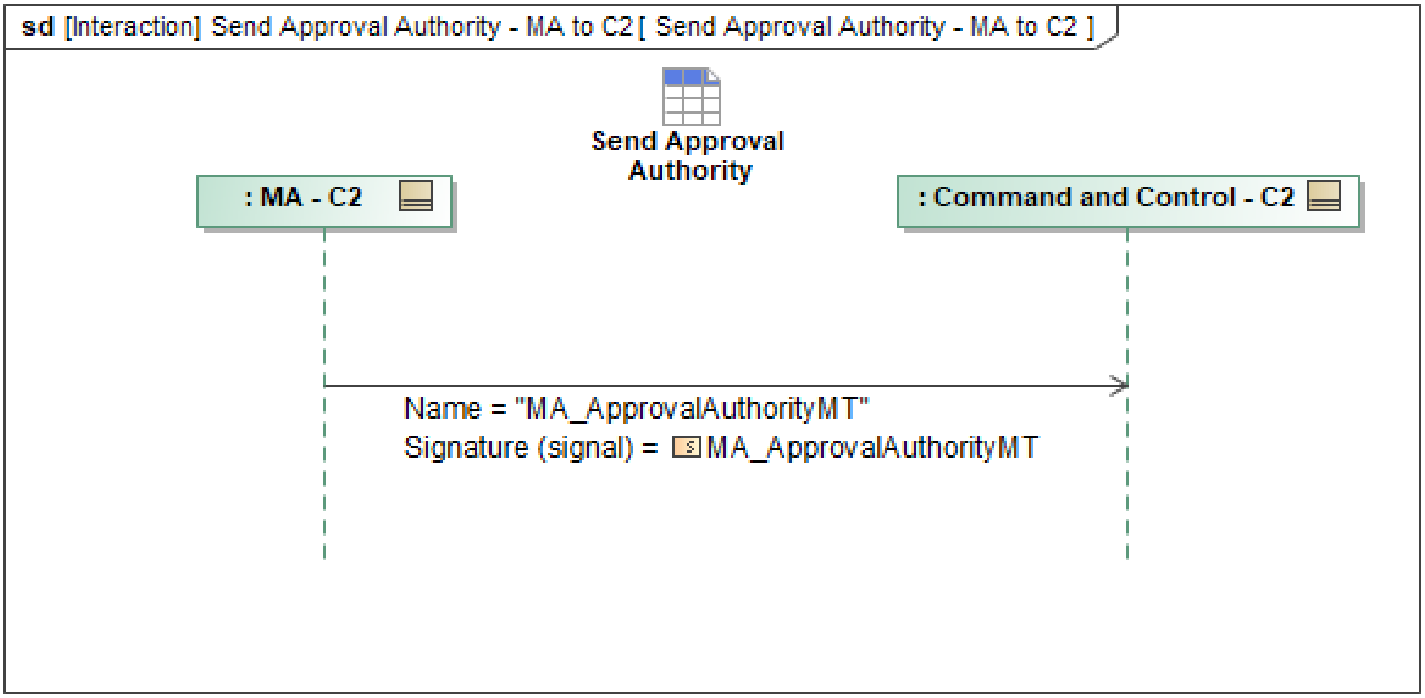


Figure 1-132: Send Approval Authority - MA to C2

Table 1-131: Send Approval Authority - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalAuthorityMT: MA_ApprovalAuthorityMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.5 Send Approval Policy - MA to C2

This interaction has MA send *MA_ApprovalPolicyMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

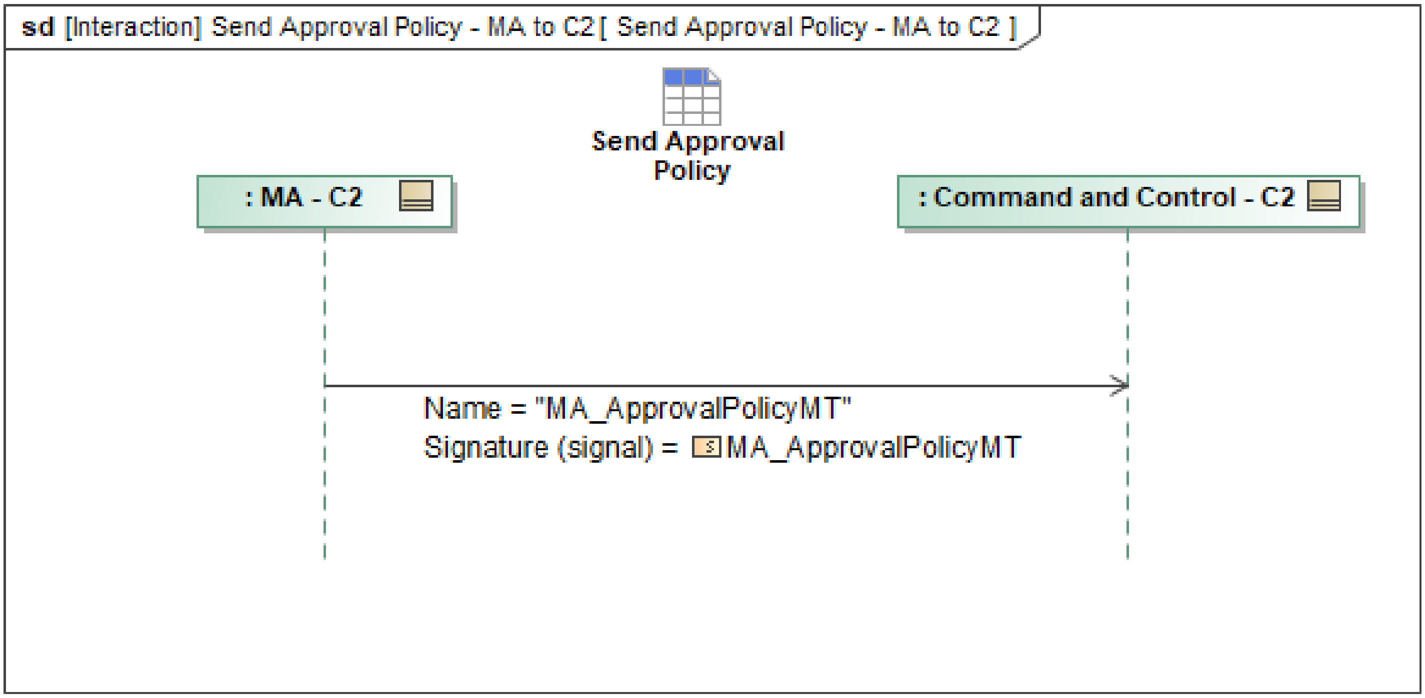


Figure 1-133: Send Approval Policy - MA to C2

Table 1-132: Send Approval Policy - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalPolicyMT: MA_ApprovalPolicyMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.6 Send Capability Set - MA to C2

This interaction has MA send *MA_CapabilitySetMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

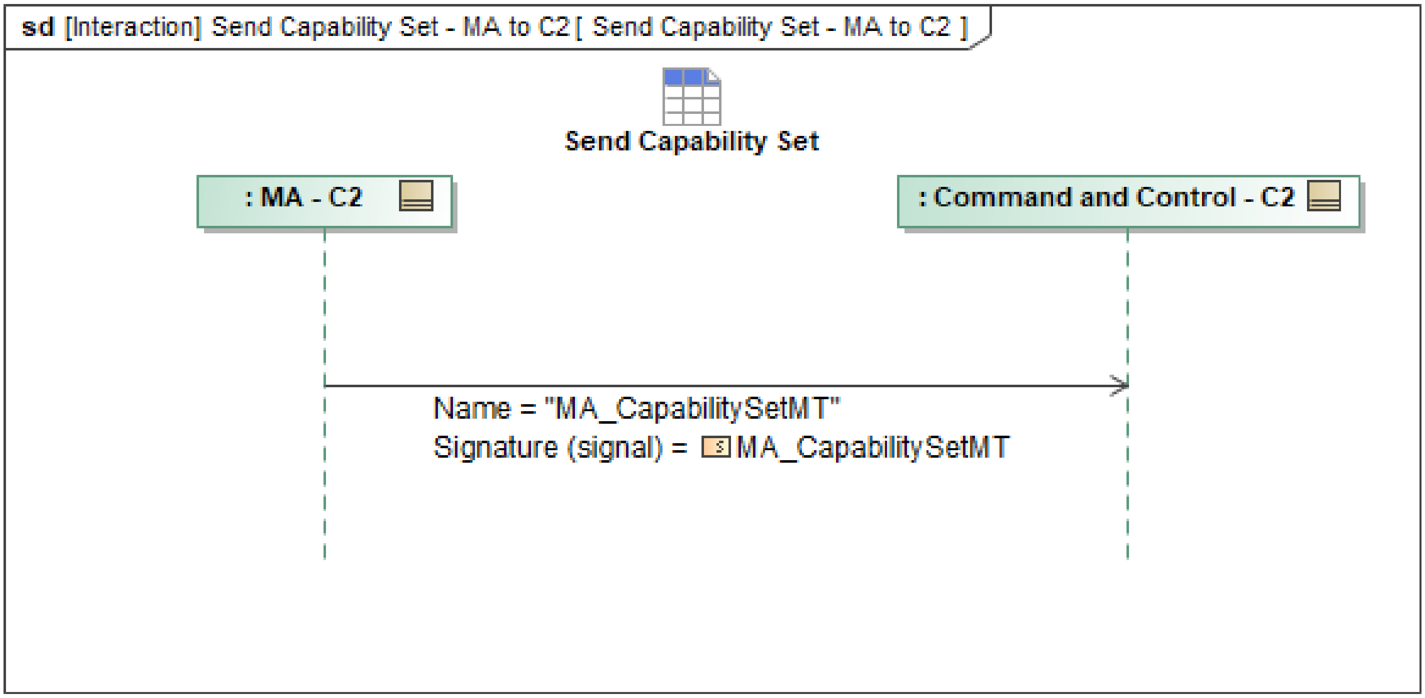


Figure 1-134: Send Capability Set - MA to C2

Table 1-133: Send Capability Set - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_CapabilitySetMT: MA_CapabilitySetMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.7 Send Entity - MA to C2

This interaction has MA send *EntityMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

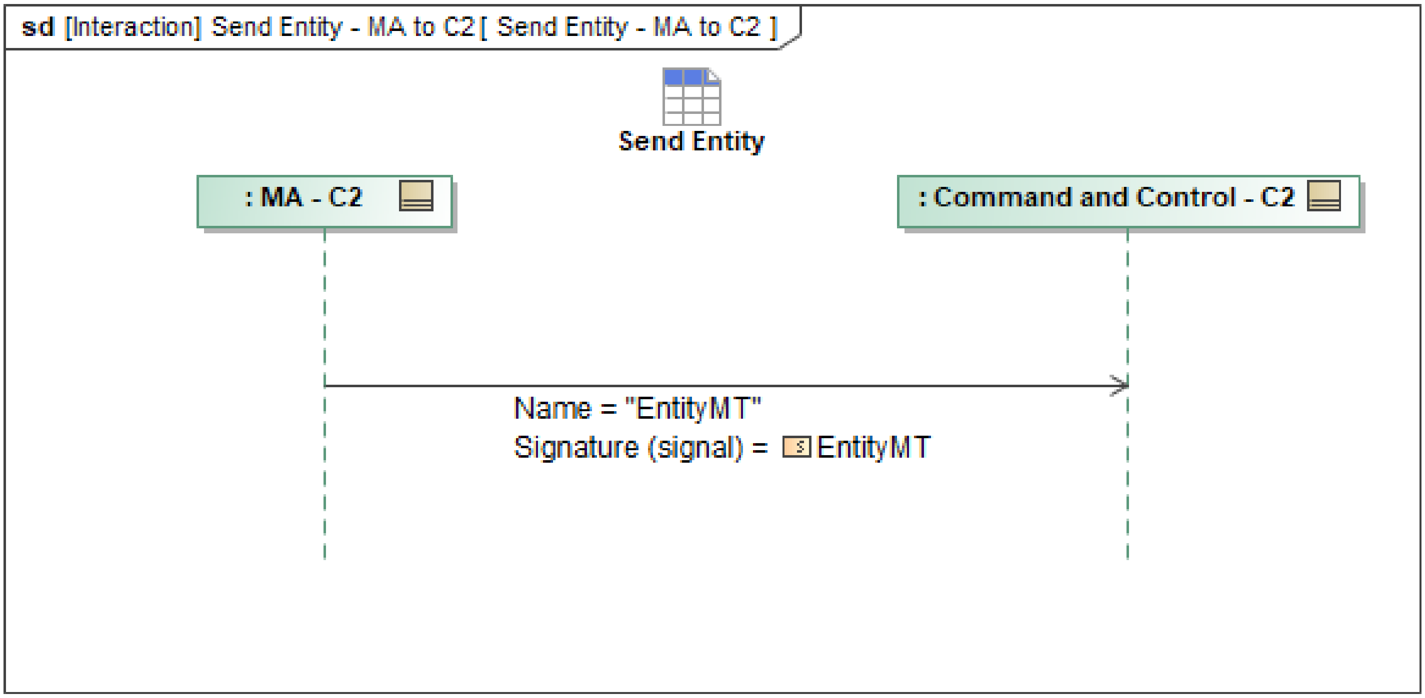


Figure 1-135: Send Entity - MA to C2

Table 1-134: Send Entity - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
EntityMT: EntityMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.8 Send Mission Plan - MA to C2

This interaction has MA send *MA_MissionPlanMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

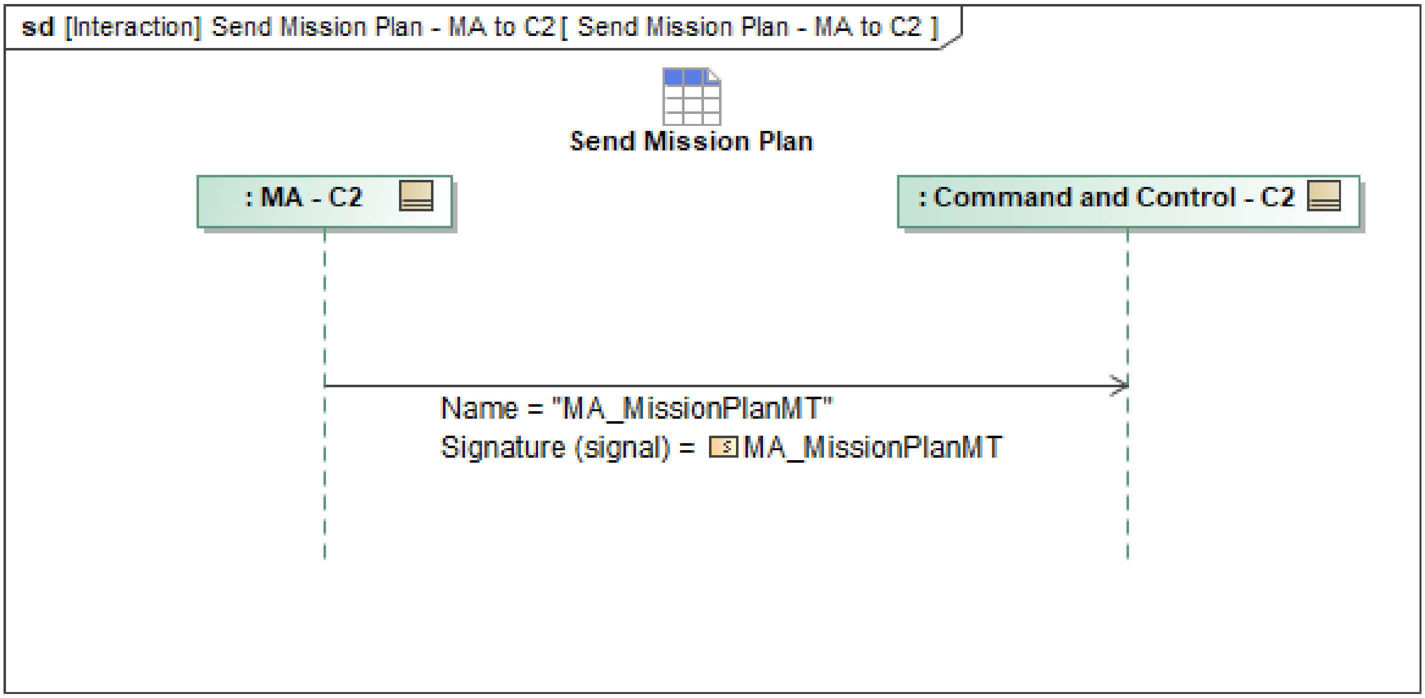


Figure 1-136: Send Mission Plan - MA to C2

Table 1-135: Send Mission Plan - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_MissionPlanMT: MA_MissionPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.9 Send Op Line - MA to C2

This interaction has MA send *MA_OpLineMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

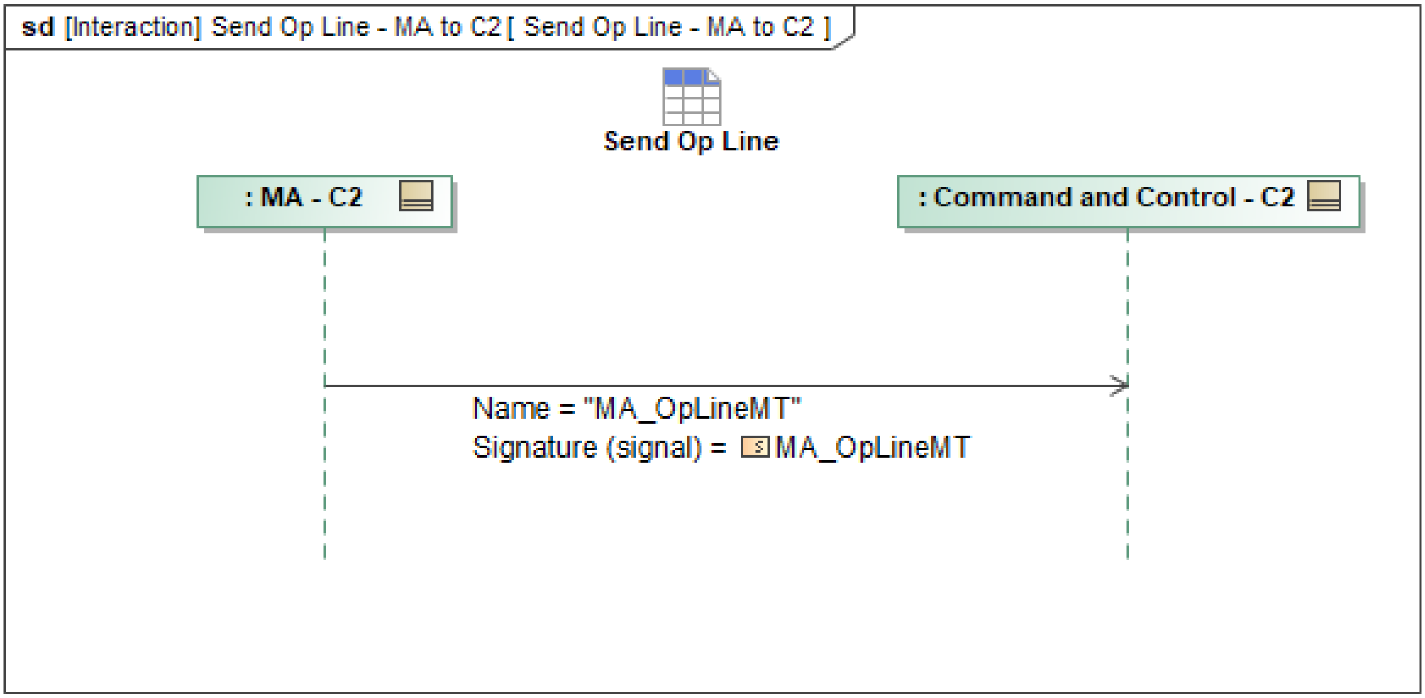


Figure 1-137: Send Op Line - MA to C2

Table 1-136: Send Op Line - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpLineMT: MA_OpLineMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.10 Send Op Routing - MA to C2

This interaction has MA send *MA_OpRoutingMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

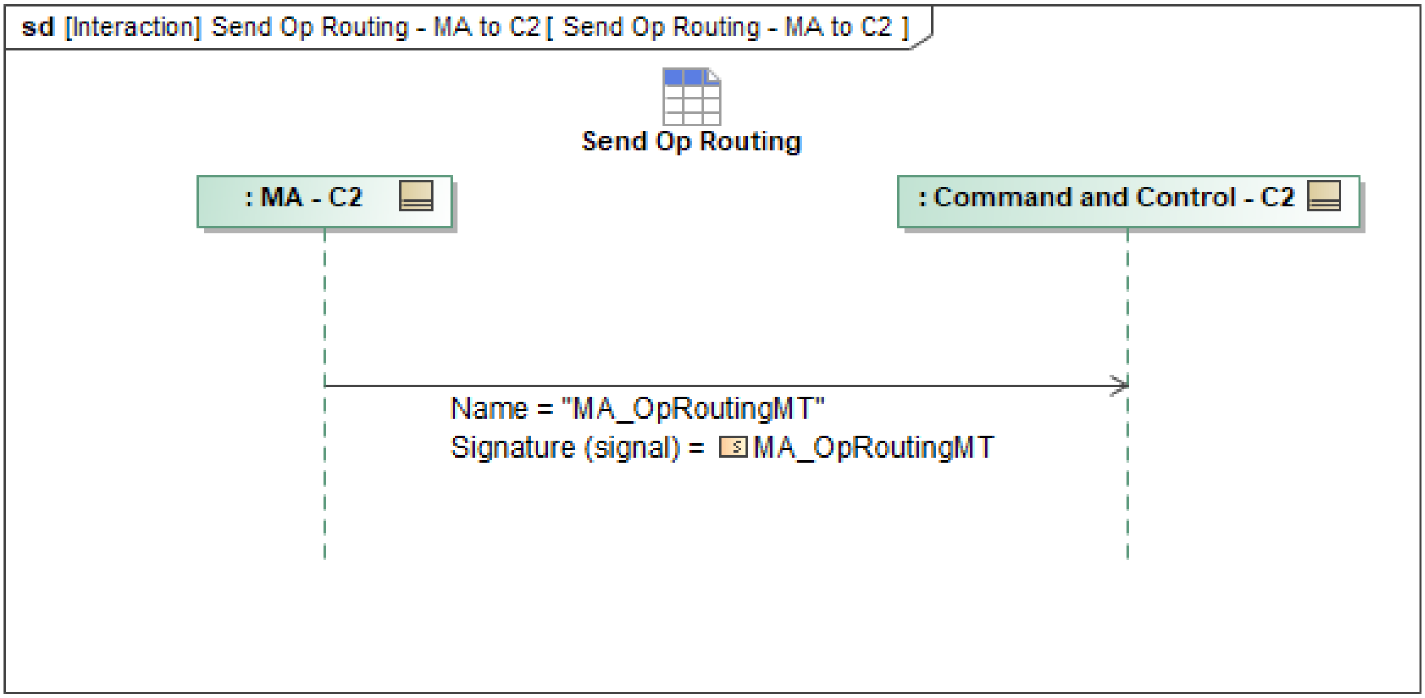


Figure 1-138: Send Op Routing - MA to C2

Table 1-137: Send Op Routing - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpRoutingMT: MA_OpRoutingMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.11 Send Op Volume - MA to C2

This interaction has MA send *MA_OpVolumeMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

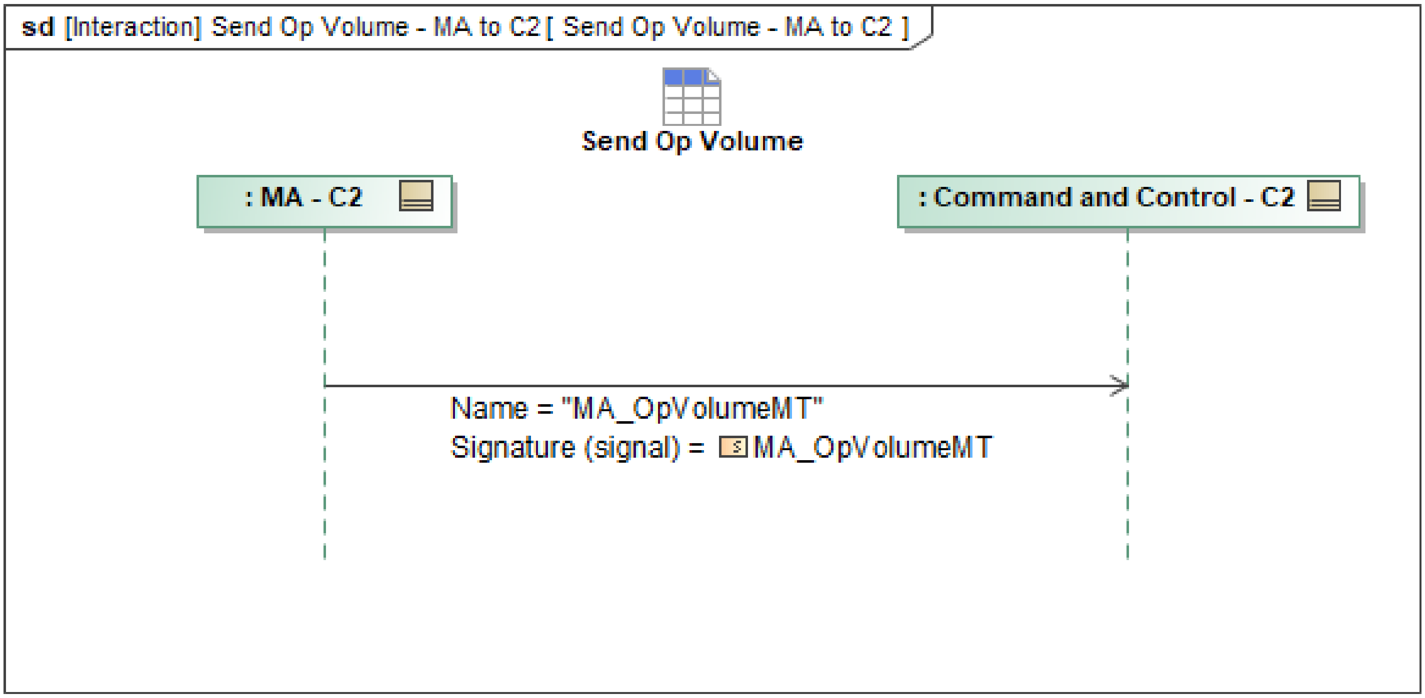


Figure 1-139: Send Op Volume - MA to C2

Table 1-138: Send Op Volume - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpVolumeMT: MA_OpVolumeMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.12 Send Op Zone - MA to C2

This interaction has MA send *MA_OpZoneMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

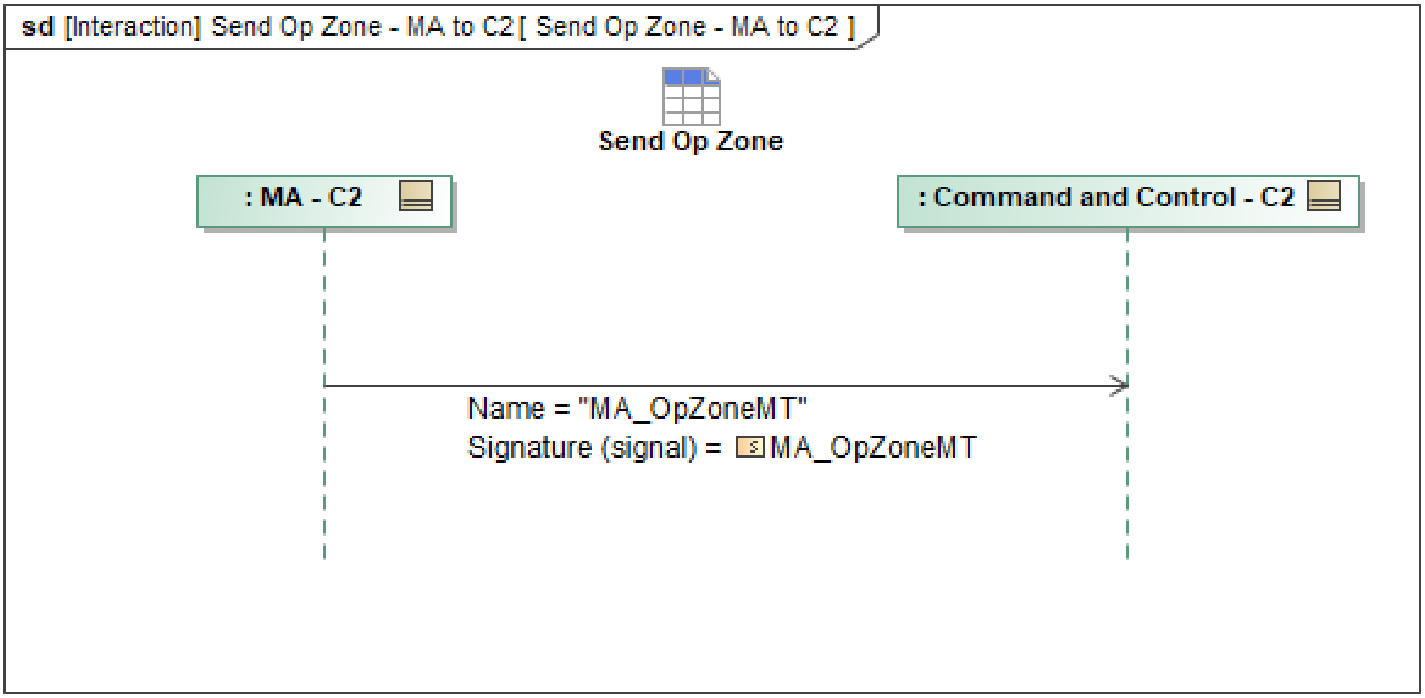


Figure 1-140: Send Op Zone - MA to C2

Table 1-139: Send Op Zone - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OpZoneMT: MA_OpZoneMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.13 Send Operator - MA to C2

This interaction has MA send *OperatorMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

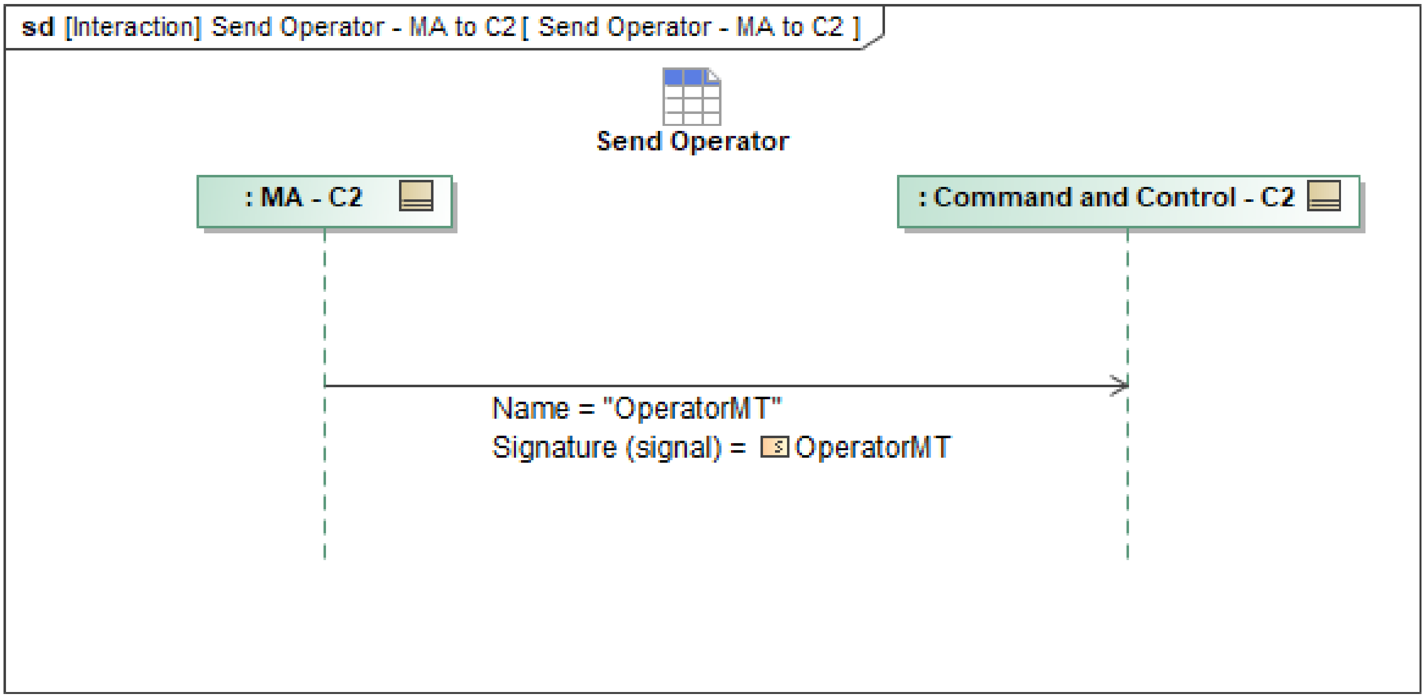


Figure 1-141: Send Operator - MA to C2

Table 1-140: Send Operator - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
OperatorMT: OperatorMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.14 Send Operator Notification - MA to C2

This interaction has MA send *OperatorNotificationMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

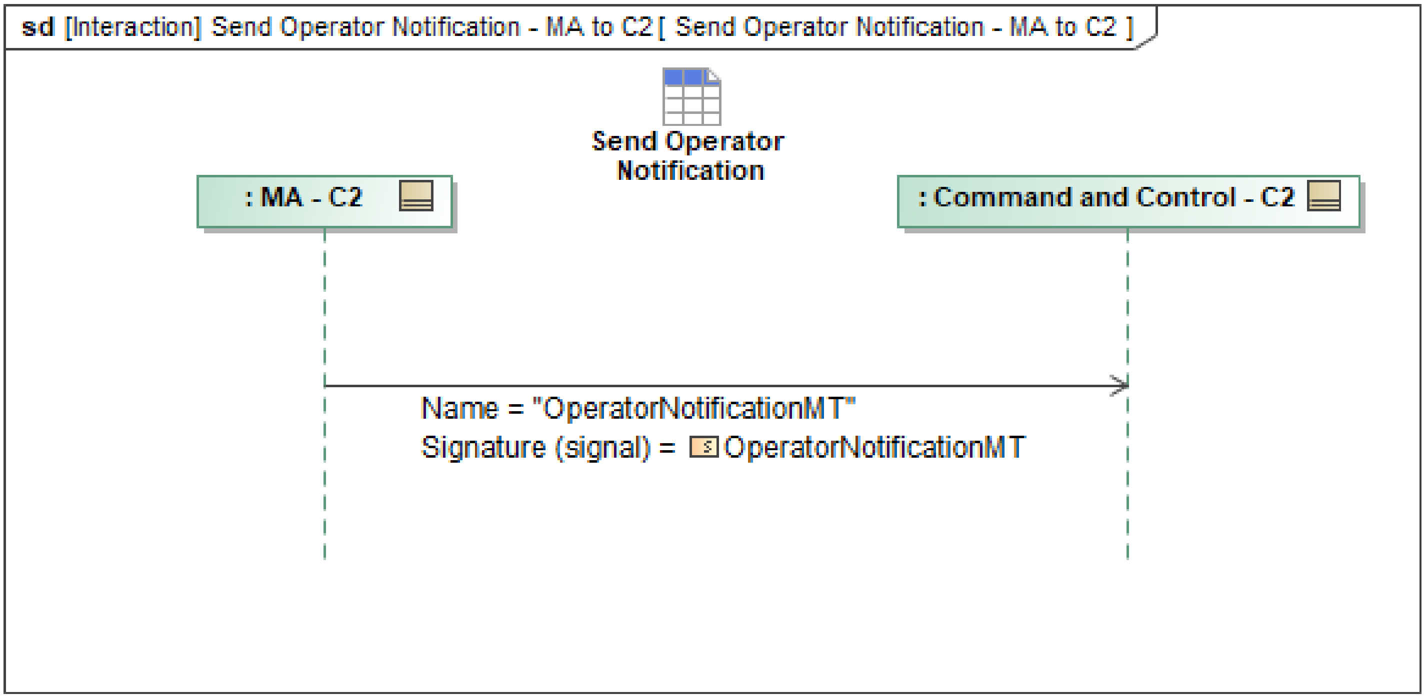


Figure 1-142: Send Operator Notification - MA to C2

Table 1-141: Send Operator Notification - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
OperatorNotificationMT: OperatorNotificationMT	Destinations: OperatorRoleType ObjectState: ObjectStateEnum Source: NotificationSourceType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.15 Send Operator Role - MA to C2

This interaction has MA send *MA_OperatorRoleMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

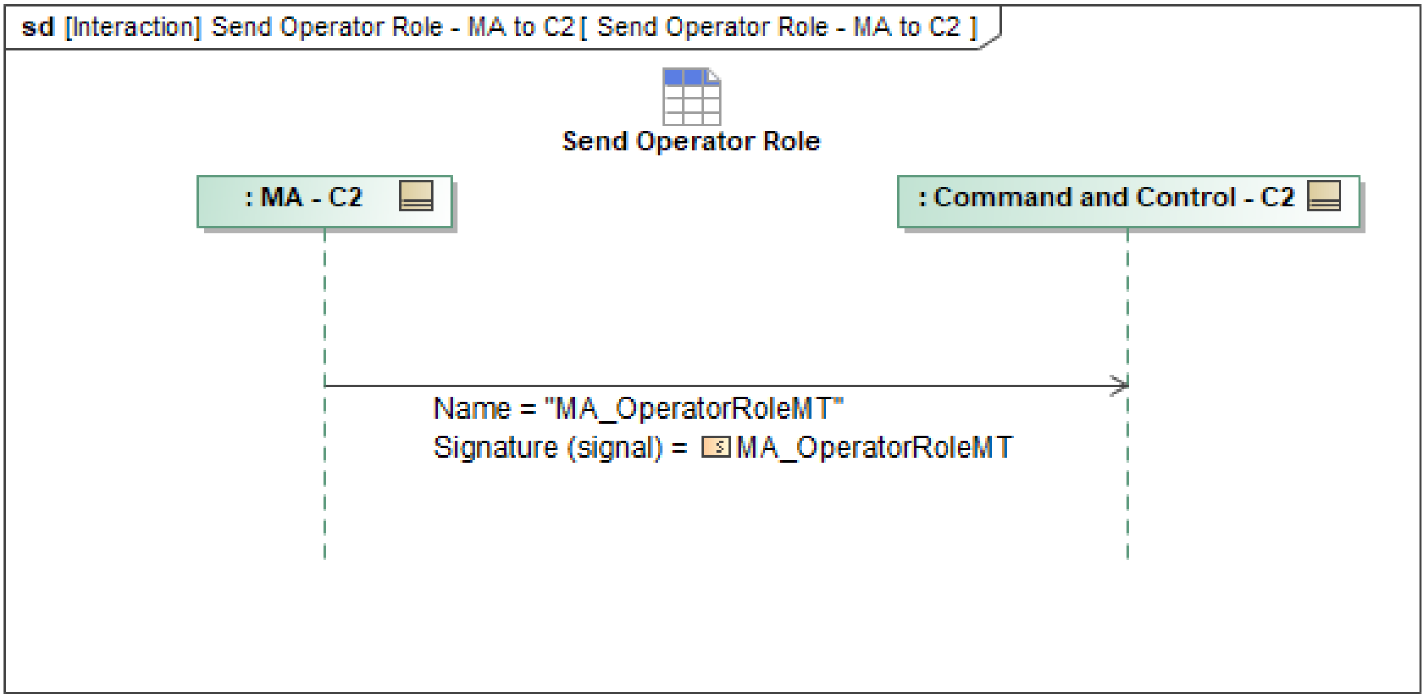


Figure 1-143: Send Operator Role - MA to C2

Table 1-142: Send Operator Role - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OperatorRoleMT: MA_OperatorRoleMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.16 Send Response - MA to C2

This interaction has MA send *MA_ResponseMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

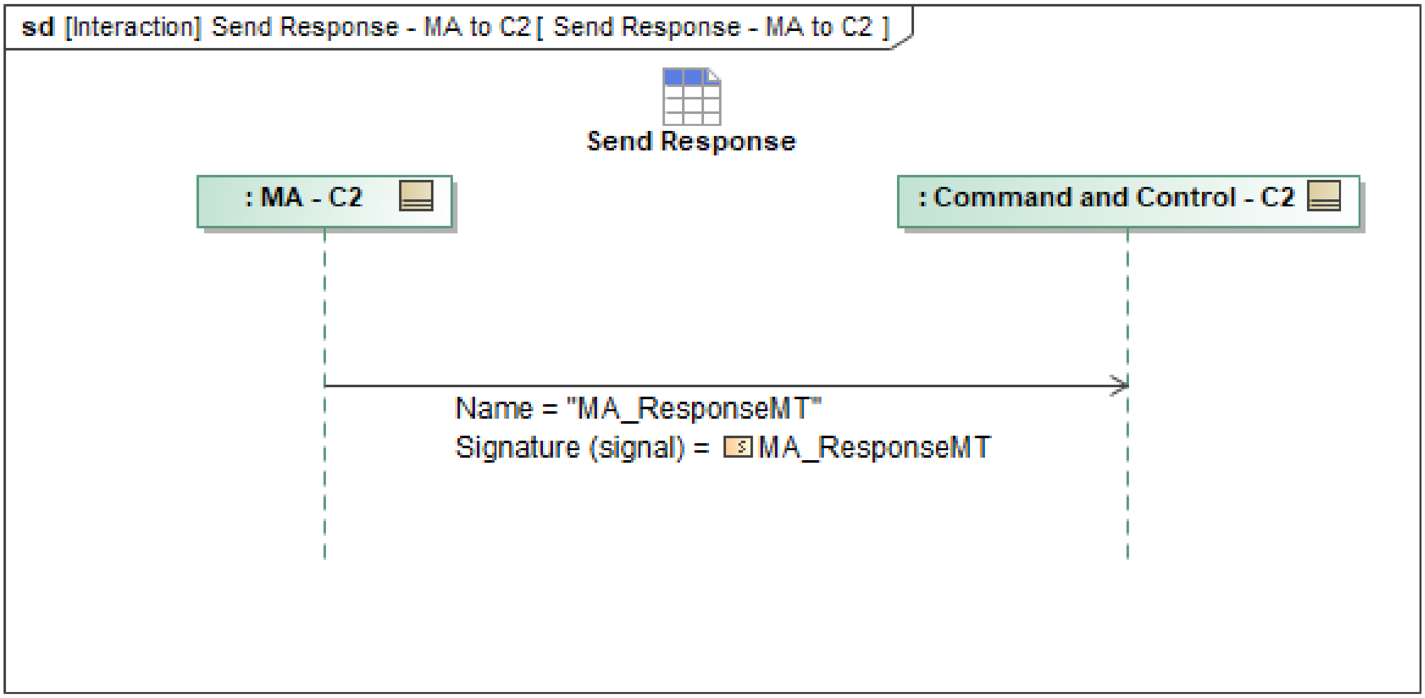


Figure 1-144: Send Response - MA to C2

Table 1-143: Send Response - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ResponseMT: MA_ResponseMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.17 Send Response Plan - MA to C2

This interaction has MA send *ResponsePlanMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

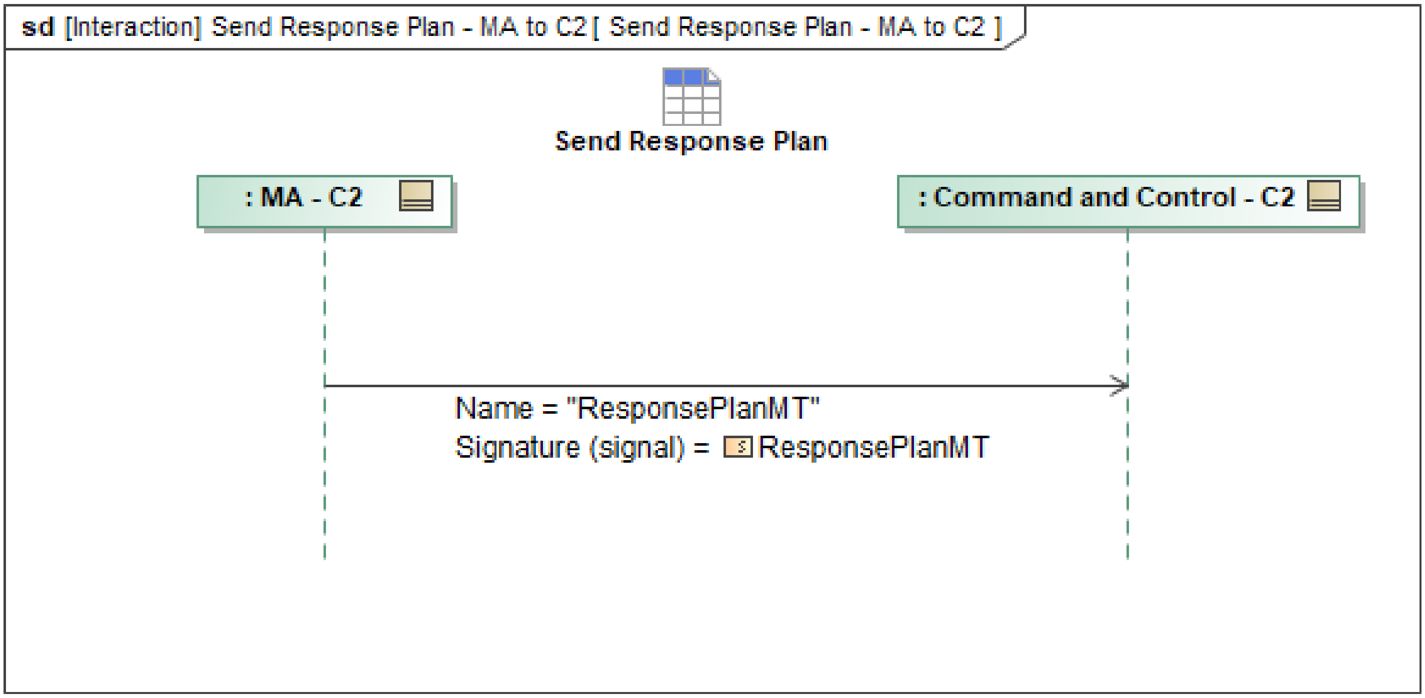


Figure 1-145: Send Response Plan - MA to C2

Table 1-144: Send Response Plan - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
ResponsePlanMT: ResponsePlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.18 Send Route Activity Plan - MA to C2

This interaction has MA send *MA_RouteActivityPlanMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

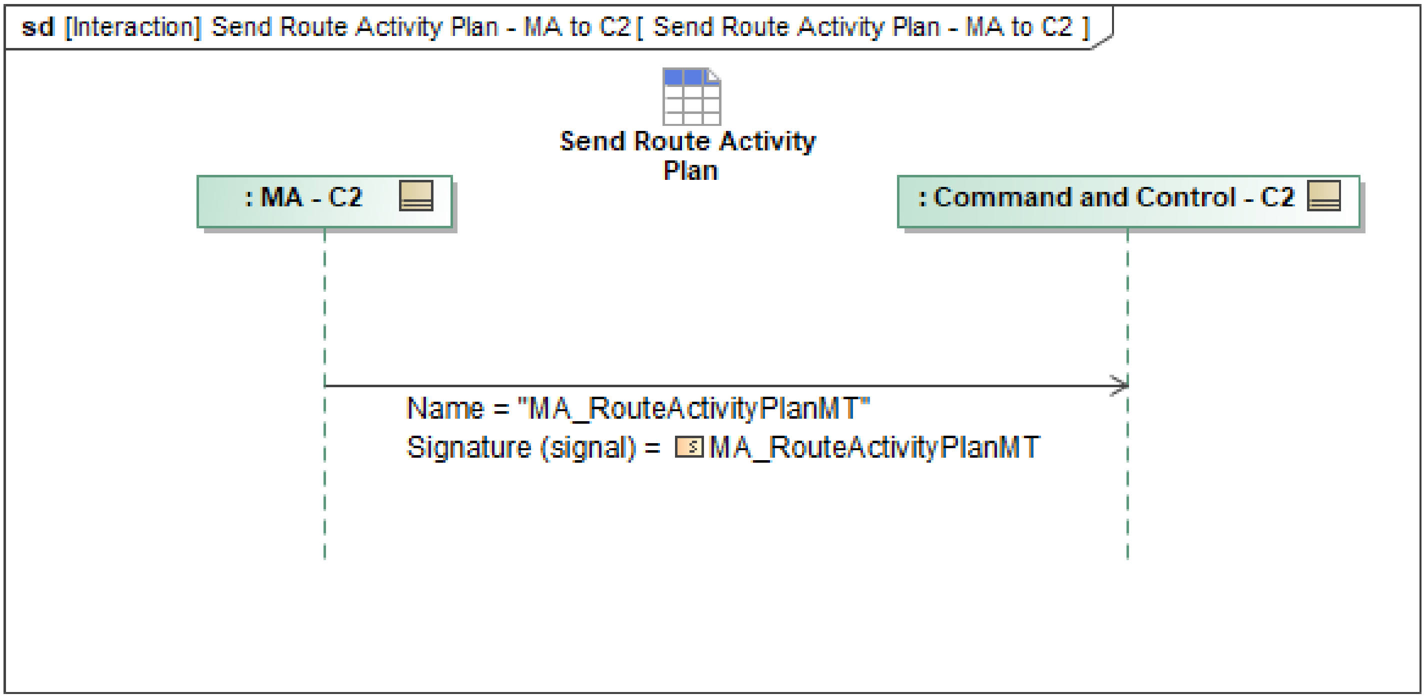


Figure 1-146: Send Route Activity Plan - MA to C2

Table 1-145: Send Route Activity Plan - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RouteActivityPlanMT: MA_RouteActivityPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.19 Send Route Plan - MA to C2

This interaction has MA send *MA_RoutePlanMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

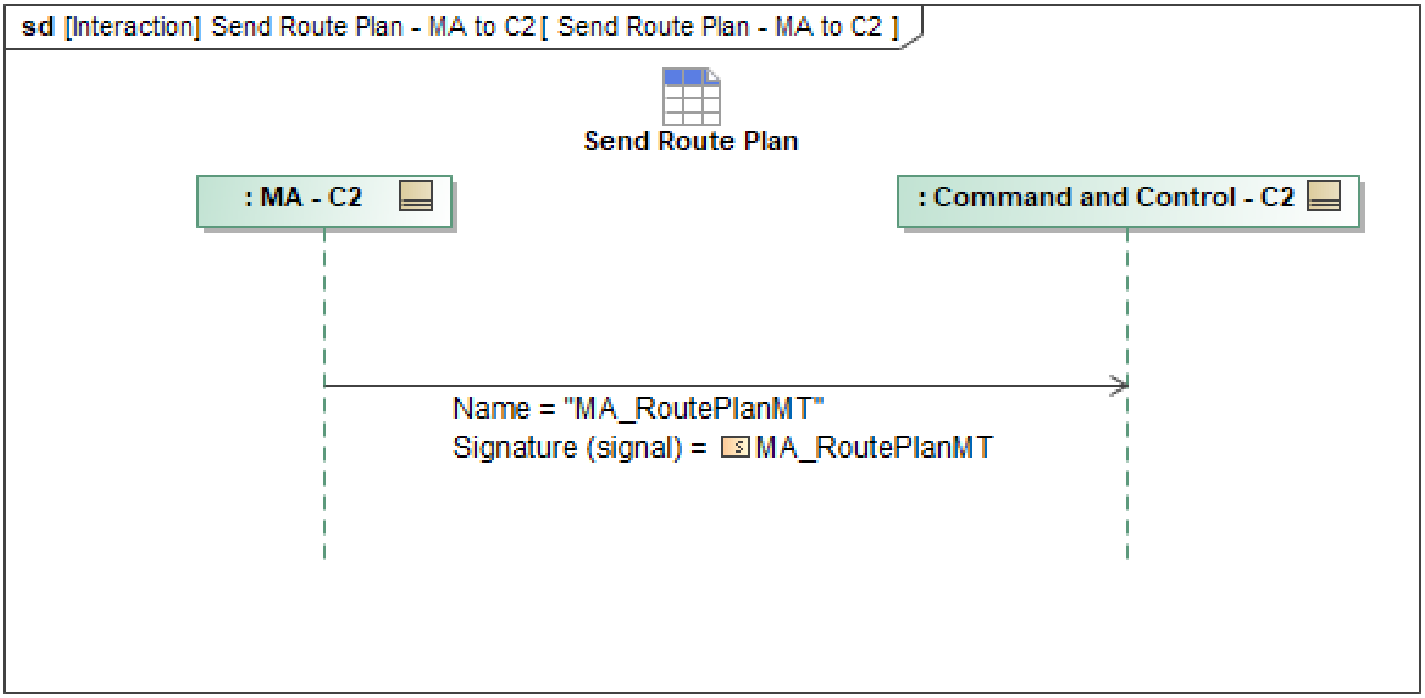


Figure 1-147: Send Route Plan - MA to C2

Table 1-146: Send Route Plan - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_RoutePlanMT: MA_RoutePlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.20 Send Task - MA to C2

This interaction has MA send *MA_TaskMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

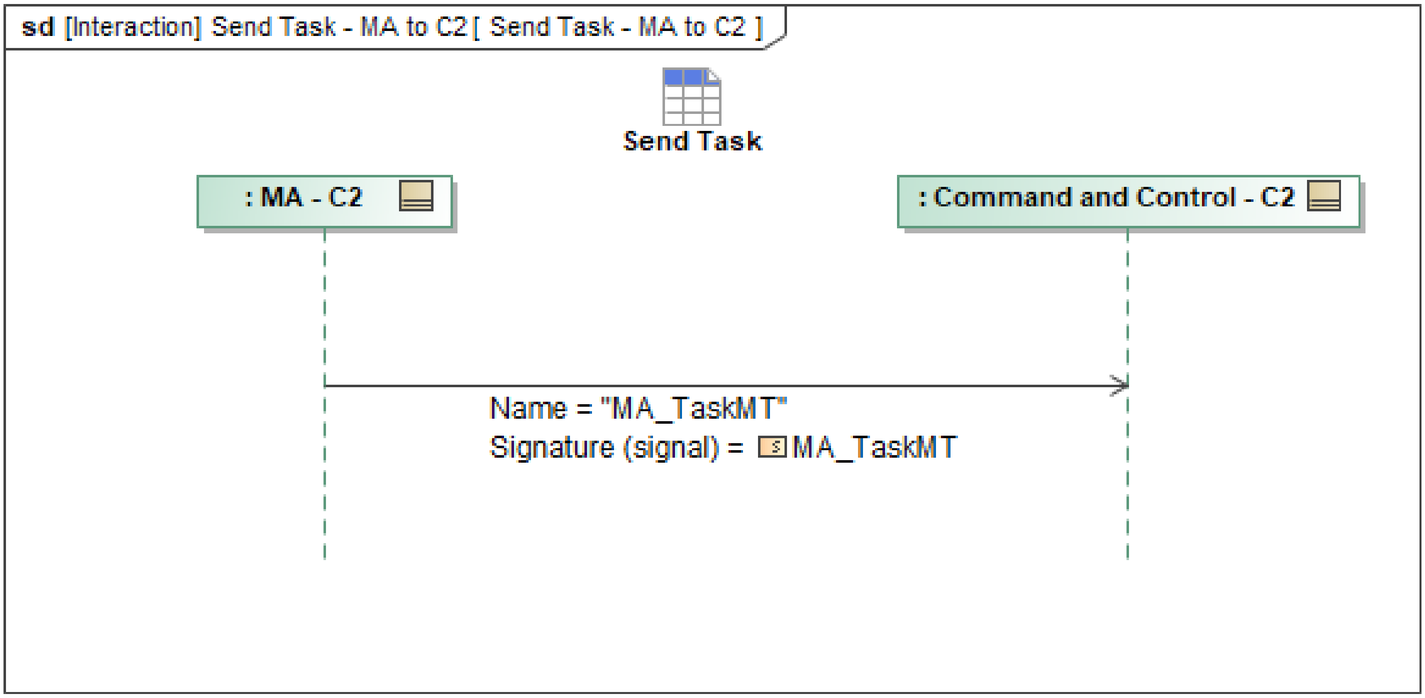


Figure 1-148: Send Task - MA to C2

Table 1-147: Send Task - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskMT: MA_TaskMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.8.23.21 Send Task Plan - MA to C2

This interaction has MA send *MA_TaskPlanMT* to C2.

If the message being sent represents the first instance of the data, the *ObjectState* field is set to *NEW*. When sending an update to revise existing data, *ObjectState* is set to *UPDATED*. Each message is populated with the definition to detail the data being conveyed.

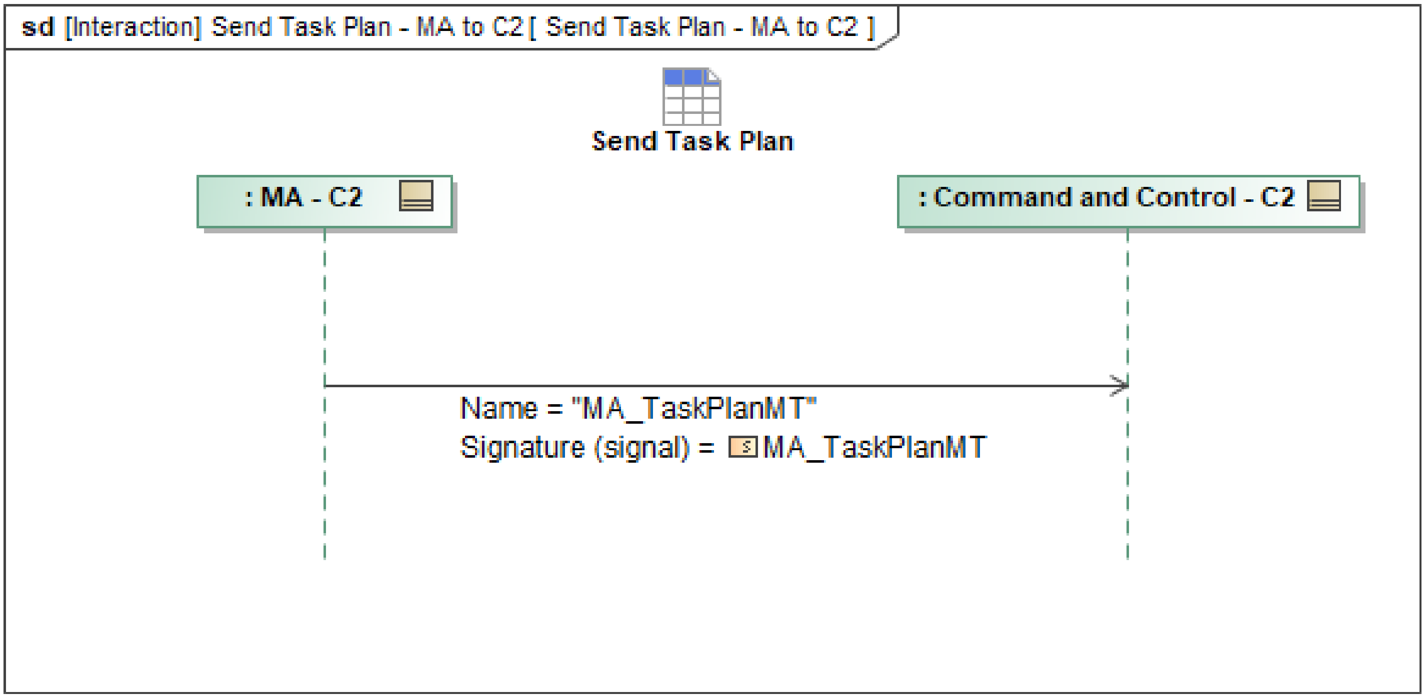


Figure 1-149: Send Task Plan - MA to C2

Table 1-148: Send Task Plan - MA to C2

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskPlanMT: MA_TaskPlanMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9 C2 RBAC Behaviors Package

Overview

The A-GRA defines a configurable role-based access control (RBAC) paradigm. RBAC defines C2 user roles which will have a configurable set of permissions allocated to them. These permissions will govern what messages a C2 user with a role is authorized to send. These roles are defined during mission planning using *MA_OperatorRoleMT*. If needed, RBAC allows for user roles to be generated during Mission Execution by authorized C2 users. It is imperative that any new roles are broadcast on the network so all players are aware of new user roles and their permissions. This message will be the source of truth for user roles, and messages that use an operator role ID will use the operator role IDs present in the *MA_OperatorRoleMT*s defined for the mission. To define the set of permissions a C2 user role has over Mission Autonomy, the *MA_AuthorizationMT* message is used. In order to route approval requests to C2 users with specific roles, Mission Planners and authorized C2 users will use *MA_ApprovalPolicyMT* to configure what types of autonomous actions require an approval, and *MA_ApprovalAuthorityMT* to configure which C2 role is authorized to accept or reject the requested action.

The A-GRA defines a set of five initial roles, with example permissions that can be further customized. The "**Admin**" RBAC role is characterized by full authorization to send messages to mission autonomy. In addition to having full permissions, the "**Admin**" role is the only role that can update the permissions of other roles during a mission. It is recommended that the **Admin** have full control over Mission Autonomy meaning that nothing is unauthorized. The "**QB**" role also has full authorization to command mission autonomy. The key difference between the "**Admin**" and "**QB**" roles is the ability to update the permissions and configuration of other roles. The "**Autonomous Vehicle Controller**" role (**AVC**) is authorized to perform flight tasks but has restrictions on specific weapon capabilities by default. The **AVC** role is designed to be highly configurable to meet mission needs. While all A-GRA roles are configurable, the **AVC** role serves as a generalized C2 role from which more specific roles will be derived. The Role Configuration section provides further details on configuring existing roles or generating new ones. The "**Launch and Recovery**" (**LRE**) role specifies permissions for takeoff and landing procedures. The "**Observer**" role is characterized by being unauthorized to send messages to mission autonomy; it is a read-only role.

When a C2 user starts a C2 application, they log in to the C2 system using their credentials. This authenticates the user and allows the system to retrieve the role data associated with that user. The C2 system (HMI/PVI) then configures itself to present only the options available for that RBAC role. Upon initial connection to the network, Mission Systems authenticates the C2 system and establishes secure communication. Once this process is complete, the C2 user is authorized to send messages to Mission Autonomy according to the permissions allocated to their role. If a C2 user needs to change their role during a mission, they will need to resend the ***Authorize*** sequence of messages declaring the new role. Alternatively, the **Admin** C2 user can change the permissions of the user's current C2 role, or create a new role to meet the specific needs of the mission.

It should also be noted that RBAC is not intended to implement a hierarchy of C2 systems to deconflict commanded tasks. Instead, it is expected that Mission Autonomy will always execute the last received command for prioritizing commands from C2 systems.

1.2.9.1 Admin Create Role

An authorized C2 that has declared as an admin may create an entirely new role not contained within the pre-mission planned RBAC configuration. It is assumed that the C2 admin will have the most up-to-date *MA_OperatorRoleMT*, *MA_AuthorizationMT, MA_ApprovalPolicyMT*, and *MA_ApprovalAuthorityMT* RBAC configuration in order to create a new role and configure it. If the C2 admin does not have the current configuration, they may query for them via the *Query Operator Roles* and *Query Role Permissions* interactions. To create a new role, the C2 admin sends *MA_OperatorRoleMT* with a unique *OperatorRoleID* representing the new role. The C2 admin is then required to configure the new role with an existing set of permissions or with new permissions via the *Admin Update MA Role Request Permissions* and *Admin Update Role Permissions* interactions after sharing the new role to MA. To complete the update, MA shares the updated list of roles to all connected authorized C2 via the Controller Update Broadcast interaction. MA will retain the updated *MA_OperatorRoleMT* along with *MA_ApprovalAuthorityMT, MA_ApprovalPolicyMT,* and *MA_AuthorizationMT* to share to any newly connected C2.

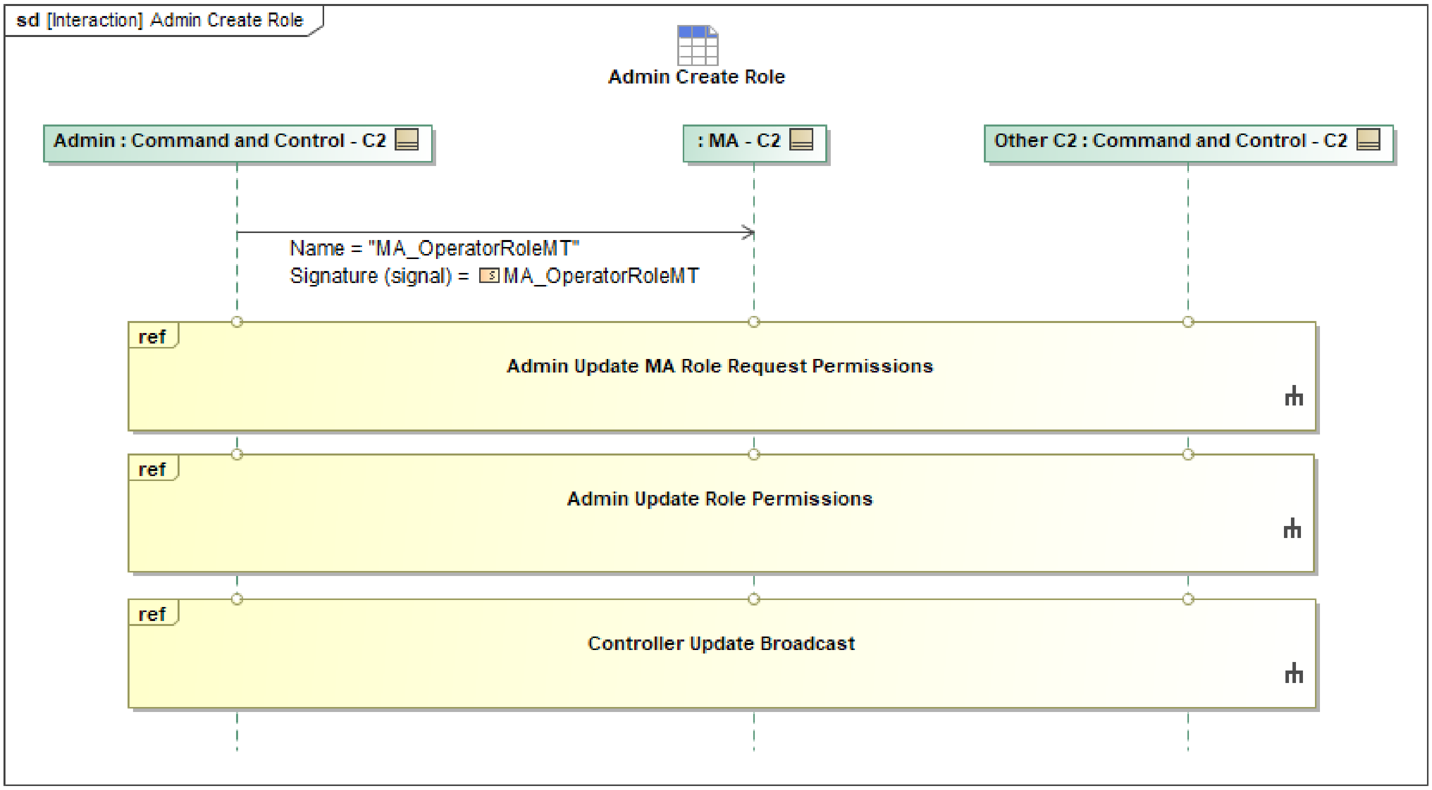


Figure 1-150: Admin Create Role

Table 1-149: Admin Create Role

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OperatorRoleMT: MA_OperatorRoleMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.2 Admin Update MA Role Request Permissions

An authorized C2 that has declared as an admin may update the individual pre-planned approval request permissions, the blanket approval request response permissions, and/or a group of approval request permissions for a given role during Mission Execution. These approval request permissions define the set of message types MA will request approval for in addition to which roles are permitted to respond to the request. It is assumed that the C2 admin will have the most up-to-date *MA_OperatorRoleMT*, *MA_AuthorizationMT, MA_ApprovalPolicyMT*, and *MA_ApprovalAuthorityMT* RBAC configuration in order to update them. If the C2 admin does not have the current configuration, they may query for them via the *Query Operator Roles* and *Query Role Permissions* interactions. The C2 admin will update the specific approval request permissions for a C2 role by sending an updated *MA_ApprovalPolicyMT* containing the complete set of configured approval request permissions for a role to MA. The C2 admin may also update the set of approval request permissions a role has by sending an updated *MA_ApprovalAuthorityMT*.

To update approval request permissions for a given role, the C2 admin would update the previously configured *MA_ApprovalPolicyMT* linked to that role's *OperatorRoleID* via *MA_ApprovalAuthorityMT*.

To configure specific approval request permissions, C2 admin would add a new requirement or update an existing requirement specified within the *RequirementExecution.ByRequirementPolicy.Requirement.RequirementTypes* field and update *RequirementExecution.ByRequirementPolicy.Policy*. The C2 admin would then send the updated *MA_ApprovalPolicyMT* to MA. To update blanket policies for a given role, the C2 admin may similarly configure *RequirementExecution.DefaultReponse* before sending the updated *MA_ApprovalPolicyMT* to MA. Setting a default response to approval request permissions for a role will automatically approve or reject any request sent to that role if it was not already defined in *RequirementExecution.ByRequirementPolicy.Requirement.RequirementTypes*.

To add an existing set of approval request permissions to a role, the C2 admin would send an updated *MA_ApprovalAuthorityMT* linking a role's *OperatorRoleID* to the existing *ApprovalPolicyID*. Similarly, to remove an existing set of approval request permissions that a role already has, the C2 admin would send an updated *MA_ApprovalAuthorityMT* removing the role's *OperatorID* from the existing *ApprovalPolicyID*.

If a set of approval request permissions does not exist, the C2 admin may create a new *MA_ApprovalPolicyMT* with a unique *ApprovalPolicyID*. The C2 admin would then update *MA_ApprovalAuthorityMT* with the new *ApprovalPolicyID* and the *OperatorRoleID* of the role receiving the new set of request permissions. The C2 admin would send both the new *MA_ApprovalPolicyMT* and updated *MA_ApprovalAuthorityMT* to finalize the permission update.

To complete the update MA will share the updated *MA_ApprovalPolicyMT* and/or *MA_ApprovalAuthorityMT* to all other C2 connected. MA will retain the updated *MA_ApprovalPolicyMT* and/or *MA_ApprovalAuthorityMT* to share to any newly connected C2.

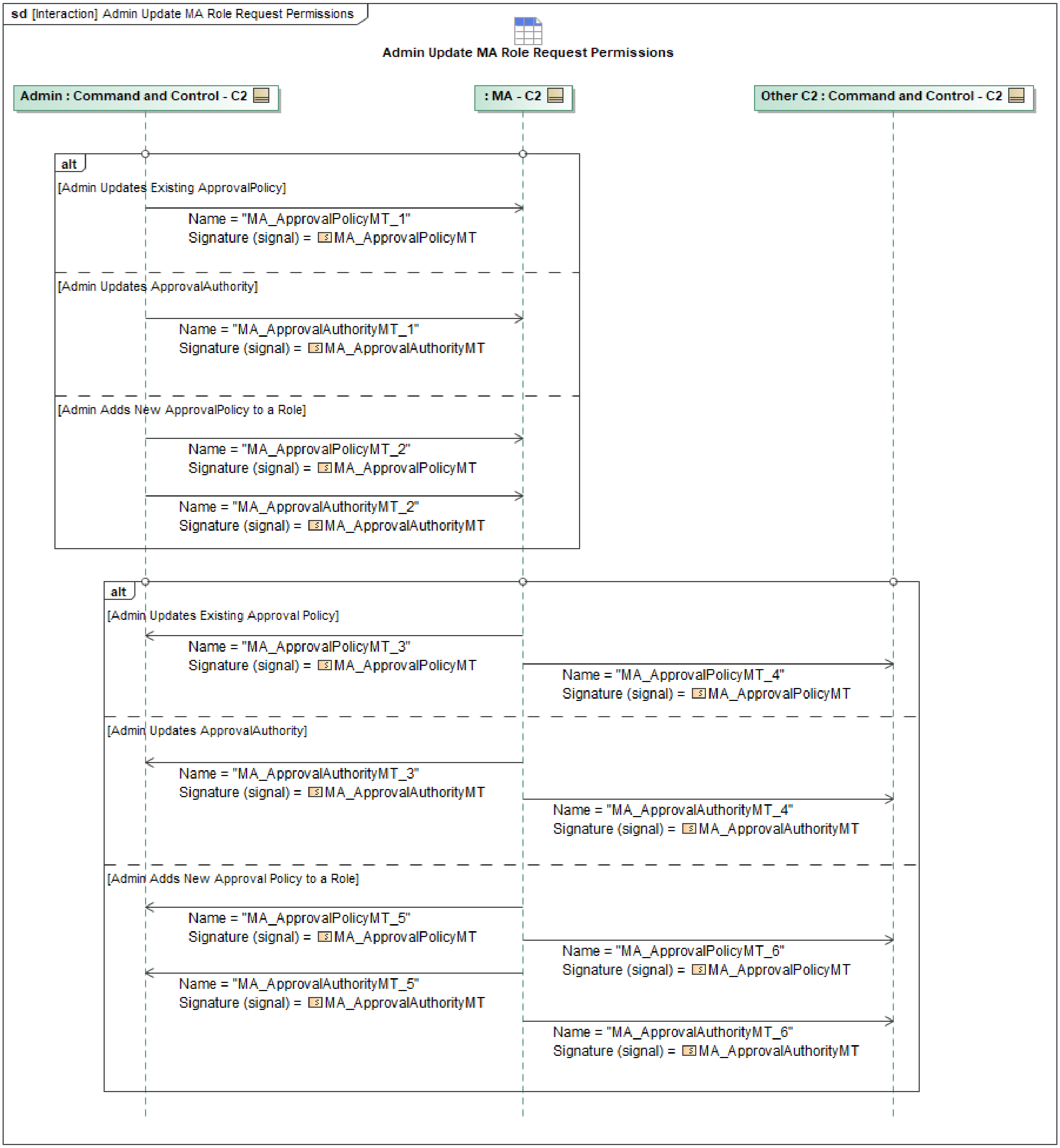


Figure 1-151: Admin Update MA Role Request Permissions

Table 1-150: Admin Update MA Role Request Permissions

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalAuthorityMT_1: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalAuthorityMT_2: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalAuthorityMT_3: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalAuthorityMT_4: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalAuthorityMT_5: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalAuthorityMT_6: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalPolicyMT_1: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType
MA_ApprovalPolicyMT_2: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType
MA_ApprovalPolicyMT_3: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType
MA_ApprovalPolicyMT_4: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType
MA_ApprovalPolicyMT_5: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType
MA_ApprovalPolicyMT_6: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.3 Admin Update Role Permissions

This interaction describes the updating of permissions for roles and the set of messages a role is permitted or not permitted to send. For updating the permissions for specific MA-generated messages that require approval and from which roles, see the Admin Update MA Role Request Permissions interaction. An authorized C2 that has declared as an admin may update the individual pre-planned permissions, the blanket permissions, and/or a group of permissions for a given role during Mission Execution via the *MA_AuthorizationMT* message. These role permissions define the set of message types that a role is either authorized or unauthorized to send via the *AuthorizedExceptFor* and *UnauthorizedExceptFor* fields provided under each C2 message type field. Exceptions to the blanket authorized/unauthorized messages may be specified under each message type field. For more information about *MA_AuthorizationMT*, refer to the C2 RBAC Behaviors Package description.

It is assumed that the C2 admin will have the most up-to-date *MA_OperatorRoleMT*, *MA_AuthorizationMT*, *MA_ApprovalPolicyMT*, and *MA_ApprovalAuthorityMT* RBAC configuration in order to update them. If the C2 admin does not have the current configuration, they may query for them via the *Query Operator Roles* and *Query Role Permissions* interactions. The C2 admin will update specific permissions for a C2 role by sending an updated *MA_AuthorizationMT* containing the complete set of configured permissions for a role to MA. The C2 admin is required to define the default permissions for each message type field within *MA_AuthorizationMT* with the *AuthorizedExceptFor* or *UnauthorizedExceptFor* permissions type. Specific exceptions may be optionally specified underneath the chosen permissions type for each message type. If a set of permissions does not exist for a role (for example, if a new role was created that needs to be configured), the C2 admin may create a new *MA_AuthorizationMT*. To associate the permissions to a new role, the C2 admin will populate *OperatorRole* field under *UserIdentifier* with same *UUID* as that of the new role.

To complete the update MA will share the updated *MA_AuthorizationMT* to all other C2 connected. MA will retain the updated *MA_AuthorizationMT* to share to any newly connected C2.

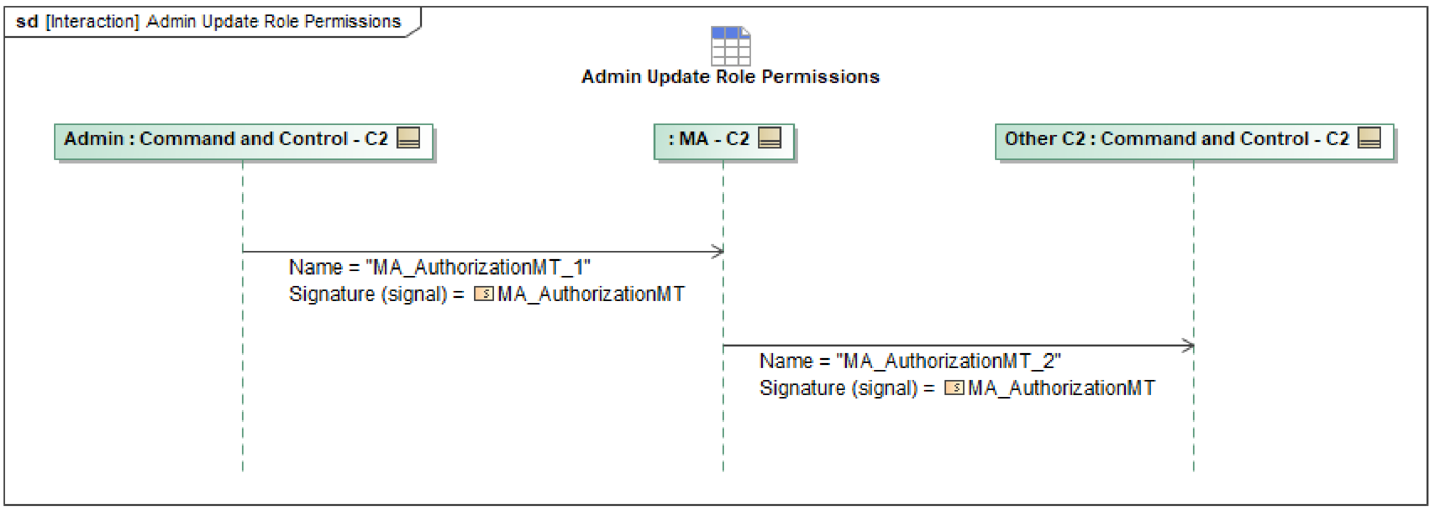


Figure 1-152: Admin Update Role Permissions

Table 1-151: Admin Update Role Permissions

Message Name (Name:Type)	Required Fields (Name:Type)
MA_AuthorizationMT_1: MA_AuthorizationMT	ByRequirementExecution: MA_RequirementAuthorizationType PackageAuthorization: boolean ByPlatformSettingsModification: MA_SettingsModficationAuthorizationType ByPlanActivation: MA_PlanAuthorizationType ByDataActivation: MA_DataActivationAuthorizationType ByDataModification: MA_DataModificationAuthorizationType ByOperatorRequest: MA_OperatorRequestAuthorizationType OperatorRoleID: OperatorRoleID_Type OperatorRoleID: OperatorRoleID_Type PackageAuthorization: boolean
MA_AuthorizationMT_2: MA_AuthorizationMT	ByRequirementExecution: MA_RequirementAuthorizationType ByPlanActivation: MA_PlanAuthorizationType

Message Name (Name:Type)	Required Fields (Name:Type)
	ByPlatformSettingsModification: MA_SettingsModficationAuthorizationType ByDataActivation: MA_DataActivationAuthorizationType PackageAuthorization: boolean OperatorRoleID: OperatorRoleID_Type ByOperatorRequest: MA_OperatorRequestAuthorizationType ByDataModification: MA_DataModificationAuthorizationType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.4 Authorize

After authentication, C2 systems will declare their role to Mission Autonomy using *OperatorMT*. The field *OperatorID* will be filled out with the C2 System's Operator *UUID* and the *DescriptiveLabel* with the operator's callsign. The *OperatorStatus* will be set to *AVAILABLE* to indicate that this C2 system can receive requests from Mission Autonomy. The *OperatorRoleID* will be populated with an *OperatorRole UUID* that is associated with a pre-planned operator role (discussed in section C2 RBAC Behaviors Package). This *OperatorMT* message is used by mission autonomy to keep a mapping of *SystemID*'s or *OperatorID*'s to Operator Roles.

Once Mission Autonomy receives an *OperatorID*, it will advertise which C2 systems have authorized themselves as described in the Controller Update Broadcast sequence by sending an *MA_OperatorRoleMT* with the newly declared C2 system appended to the list in the *SystemID* field. The *MA_OperatorRoleMT* will also include the *UUID* of the updated role in the *OperatorRoleID* field.

If a C2 system wishes to change its declared role during a mission, it can send another *OperatorMT* message to repeat this sequence and update its role as described in the Update Operator Roles section.

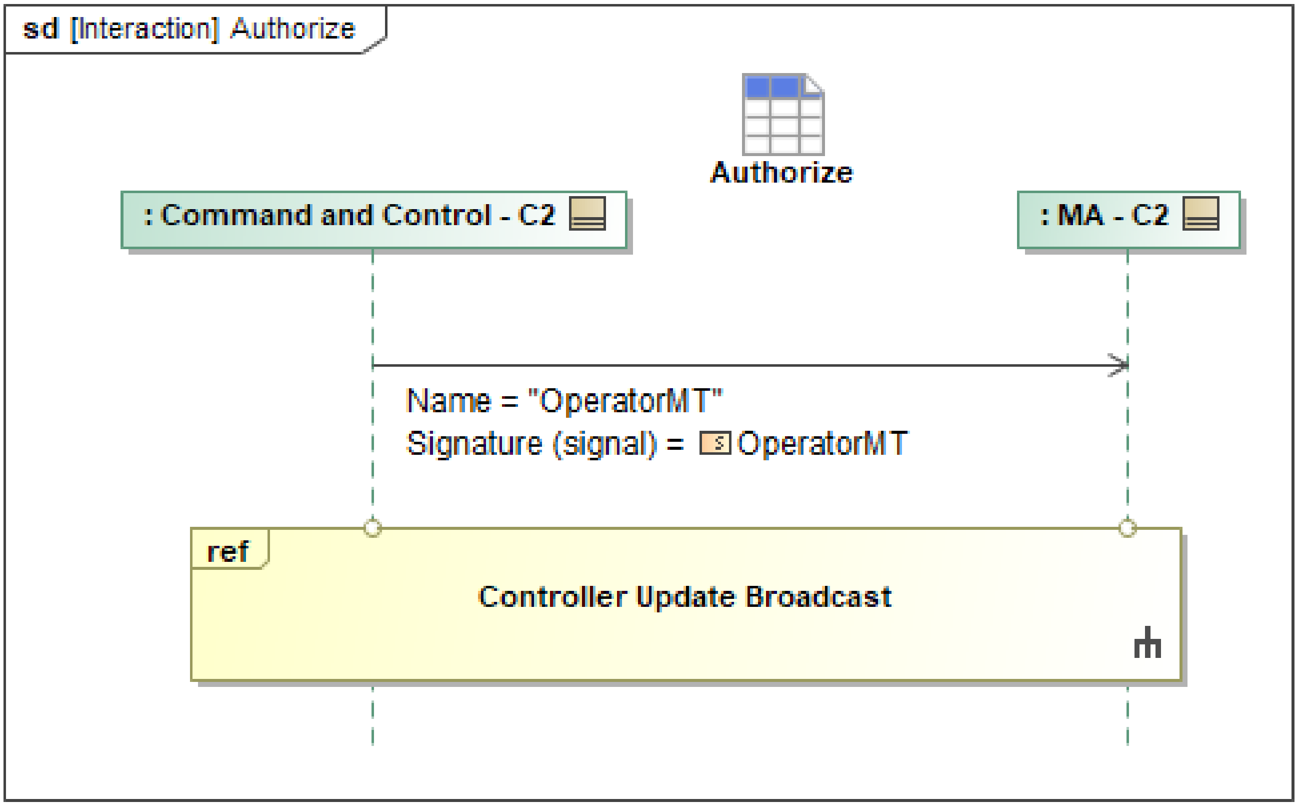


Figure 1-153: Authorize

Table 1-152: Authorize

Message Name (Name:Type)	Required Fields (Name:Type)
OperatorMT: OperatorMT	OperatorRoleID: OperatorRoleID_Type DescriptiveLabel: VisibleString256Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.5 C2 Unauthorized Response

After a message is determined to be unauthorized, MA will send a response message to C2 based on UCI Primitive Type:

- For Data-1 and DataRecord-1 types, a *MA_SystemNotificationMT* message will be sent.
- For the Command-2 types, a *Command-2 Status Message* will be sent.
- For the ActionRequest-2 type, *ActionRequest-2 Status Message* will be sent.
- For the DataRequest-2 type, *DataRequest-2 Status Message* will be sent.

The Command-2, ActionRequest-2 and DataRequest-2 messages are each further specified by a corresponding table (Unauthorized Command-2 Message Pairs , Unauthorized ActionRequest-2 Message Pairs and Unauthorized DataRequest-2 Message Pairs, respectively) in the Model. Each table contains a triggering message, the corresponding unauthorization response message, and the applicable interactions.

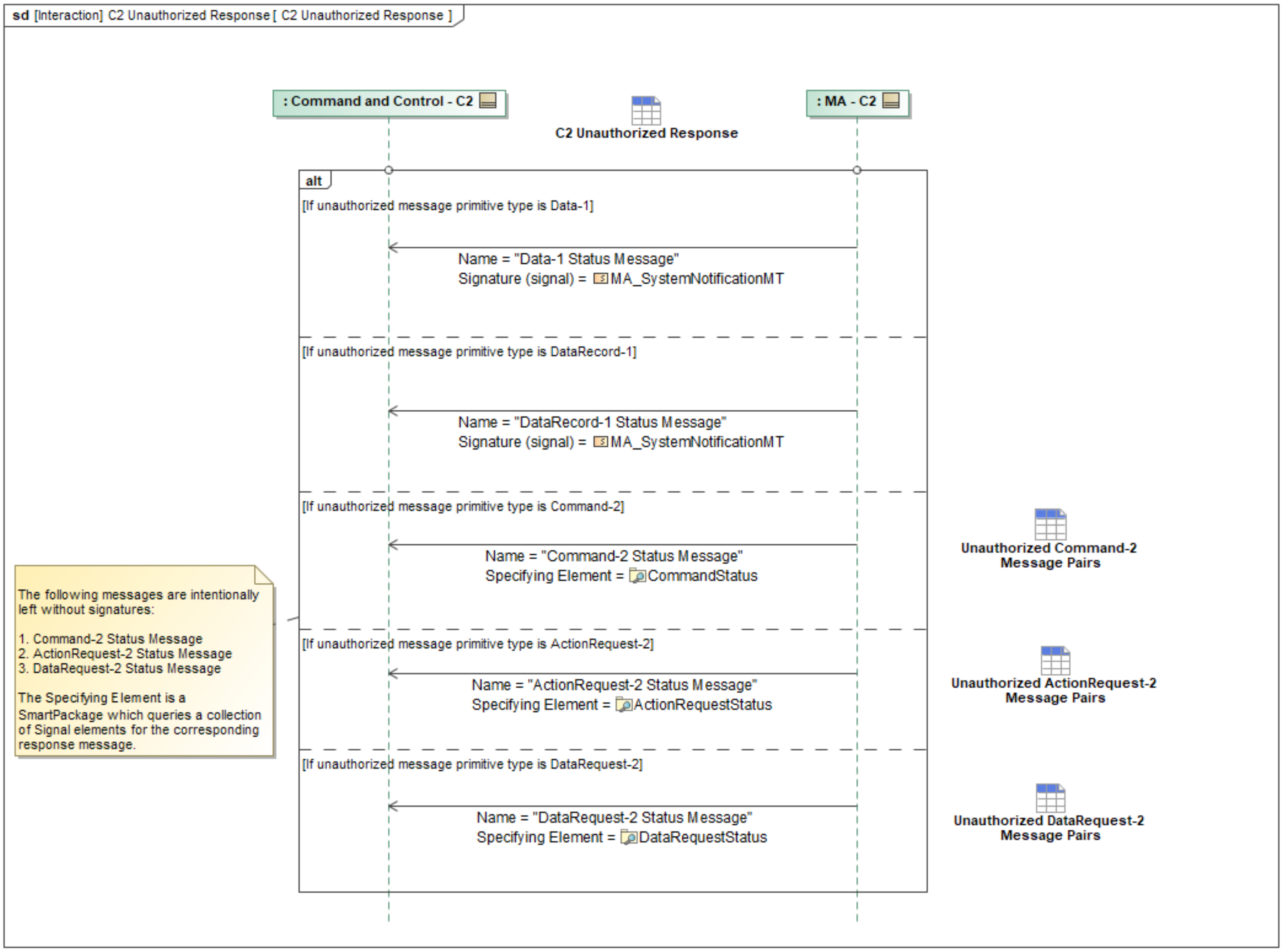


Figure 1-154: C2 Unauthorized Response

Table 1-153: C2 Unauthorized Response

Message Name (Name:Type)	Required Fields (Name:Type)
ActionRequest-2 Status Message:	RequestProcessingStateReason: CannotComplyType
Command-2 Status Message:	CommandProcessingStateReason: CannotComplyType
Data-1 Status Message: MA_SystemNotificationMT	AssociatedMessage: AssociatedMessageType
DataRecord-1 Status Message: MA_SystemNotificationMT	AssociatedMessage: AssociatedMessageType
DataRequest-2 Status Message:	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.6 Controller Update Broadcast

Whenever Mission Autonomy makes updates to the field *SystemID* in *MA_OperatorRoleMT*, it will broadcast its updated *MA_OperatorRoleMT* to other C2 systems. In addition to the *SystemID*, MA will also broadcast operators' callsigns in the *DescriptiveLabel* field for each *SystemID*. See Section C2 RBAC Behaviors Package for more information on what control is permitted per role.

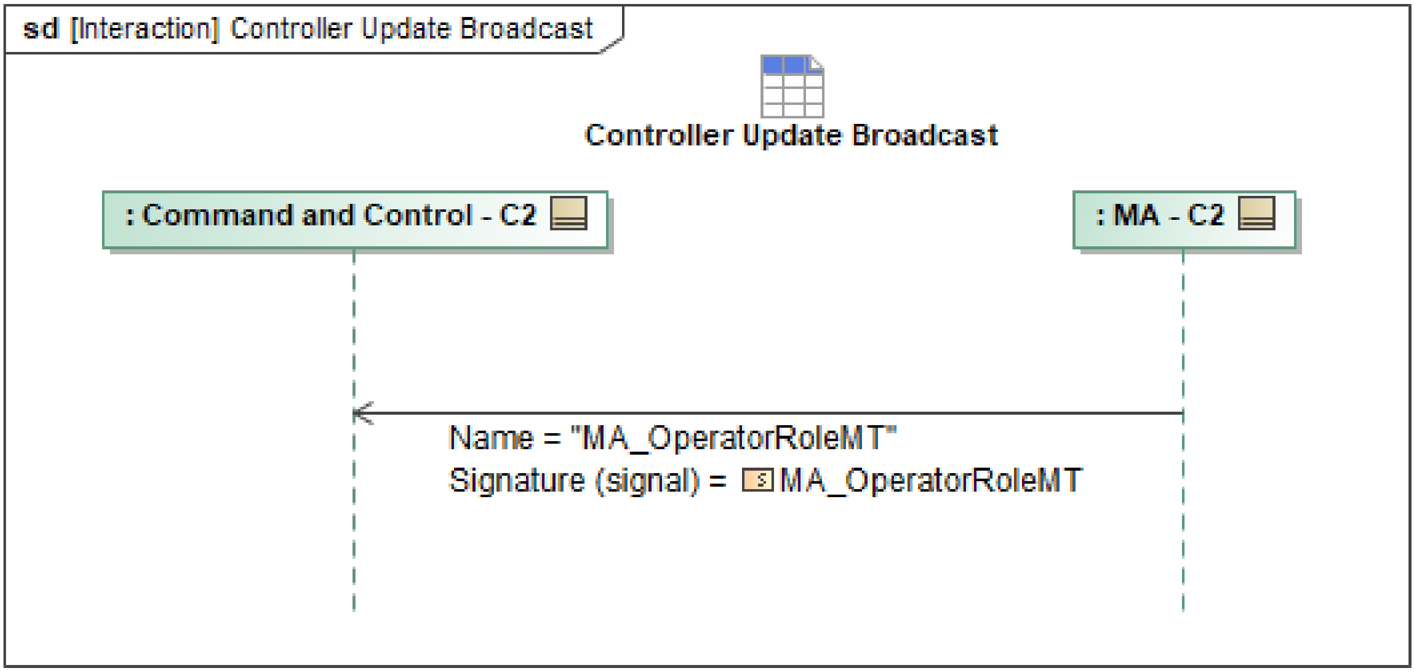


Figure 1-155: Controller Update Broadcast

Table 1-154: Controller Update Broadcast

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OperatorRoleMT: MA_OperatorRoleMT	SystemID: SystemID_Type DescriptiveLabel: VisibleString256Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.7 Query Operator Roles

A C2 system can query the authorized operators and their roles from MA via *QueryDataRequestMT*. *MA_OperatorRoleMT* contains the list of authorized operators, their roles, and their callsigns and are stored or cached in data manager services which may have a database backend. This data is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate *RequestProcessingState* such as “*QUEUED*” and “*PROCESSING*” to indicate additional in-process states. The *QueryDataRequestStatusMT* message with a *RequestProcessingState* set to “*COMPLETED*” and *ResultsInNative* field set to “*FALSE*” contains the requested data. When *ResultsInNative* is set to “*TRUE*”, this causes the underlying data management services to publish the requested data in its original or raw form.

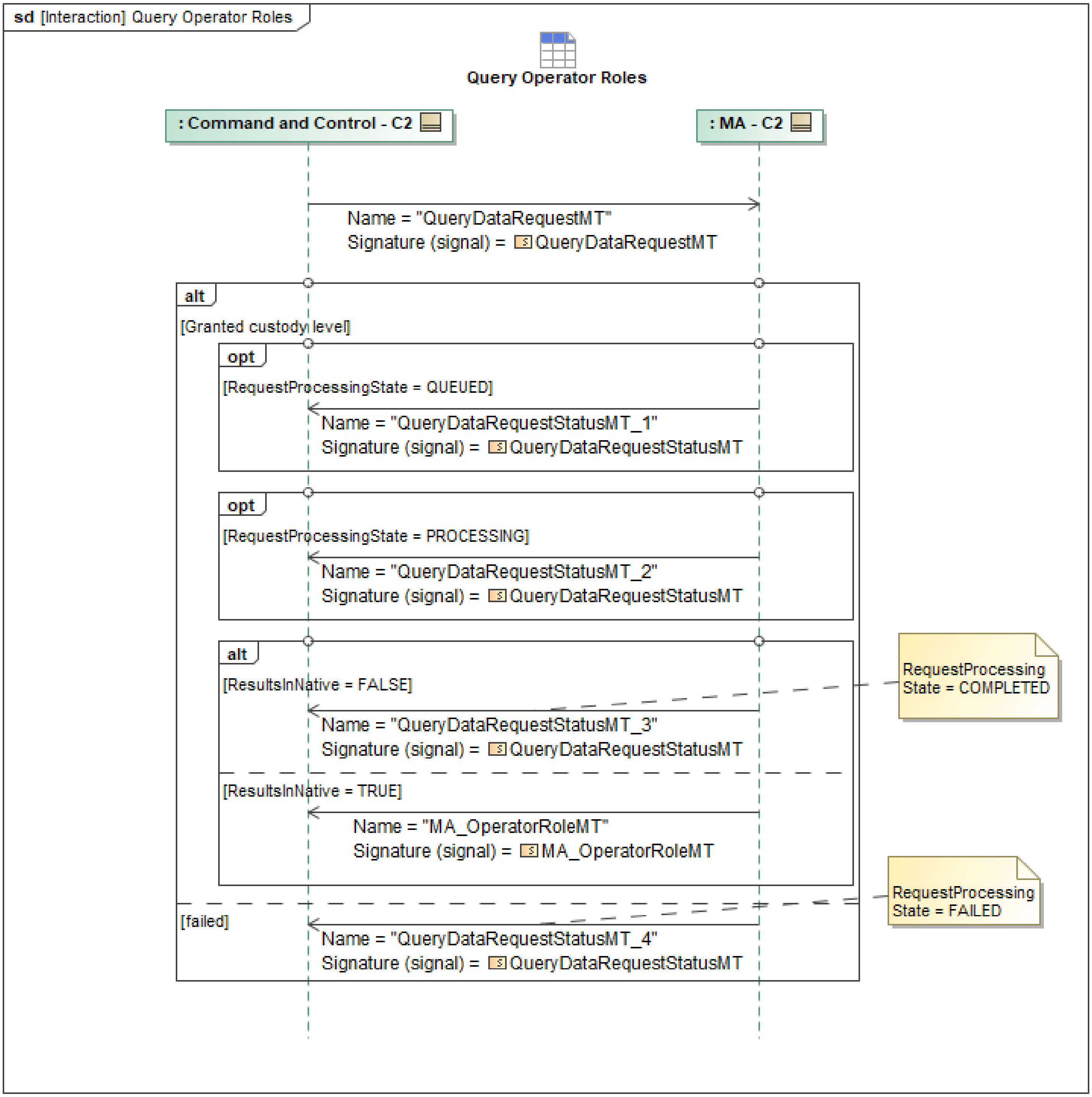


Figure 1-156: Query Operator Roles

Table 1-155: Query Operator Roles

Message Name (Name:Type)	Required Fields (Name:Type)
MA_OperatorRoleMT: MA_OperatorRoleMT	DescriptiveLabel: VisibleString256Type SystemID: SystemID_Type
QueryDataRequestMT: QueryDataRequestMT	ResultsInNative: QueryTopicPairType

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	RequestProcessingState: RequestProcessingStateEnum
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	RequestProcessingState: RequestProcessingStateEnum
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	RequestProcessingState: RequestProcessingStateEnum
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingState: RequestProcessingStateEnum RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.8 Query Role Permissions

A C2 admin may query for the permissions configuration in order update them for a given role. Other authorized roles may query for them for situational awareness. A C2 system may query for the complete RBAC role permissions configuration as well as the approval request permissions configuration via *QueryDataRequestMT*. The C2 has the option to query for specific permissions messages for all RBAC roles, or for the permissions messages for a specific role by using that role's *OperatorRoleID UUID*. *MA_AuthorizationMT* contains the set of permissions for each role. *MA_ApprovalPolicyMT* contains the set of permissions that require approval when generated by MA and is used in conjunction with *MA_ApprovalAuthorityMT* to define which roles MA should route the approval request to. C2 may query for any combination of these messages from MA. These RBAC permission messages are stored or cached in data manager services which may have a database backend. This data is queried via *QueryDataRequestMT* and responded to with one or more *QueryDataRequestStatusMT* messages. Data management services may respond with intermediate *RequestProcessingState* such as "*QUEUED*" and "*PROCESSING*" to indicate additional in-process states. The *QueryDataRequestStatusMT* message with a *RequestProcessingState* set to "*COMPLETED*" and *ResultsInNative* field set to "*FALSE*" contains the requested data. When *ResultsInNative* is set to "*TRUE*", this causes the underlying data management services to publish the requested data in its original or raw form. If *QueryDataRequestMT* does not specify an *OperatorRoleID UUID* via *QueryPET*, MA will share the requested permissions messages for all roles defined within *MA_OperatorRoleMT*. If an *OperatorRoleID UUID* is specified in *QueryPET*, MA will respond with the requested permissions messages for only that role.

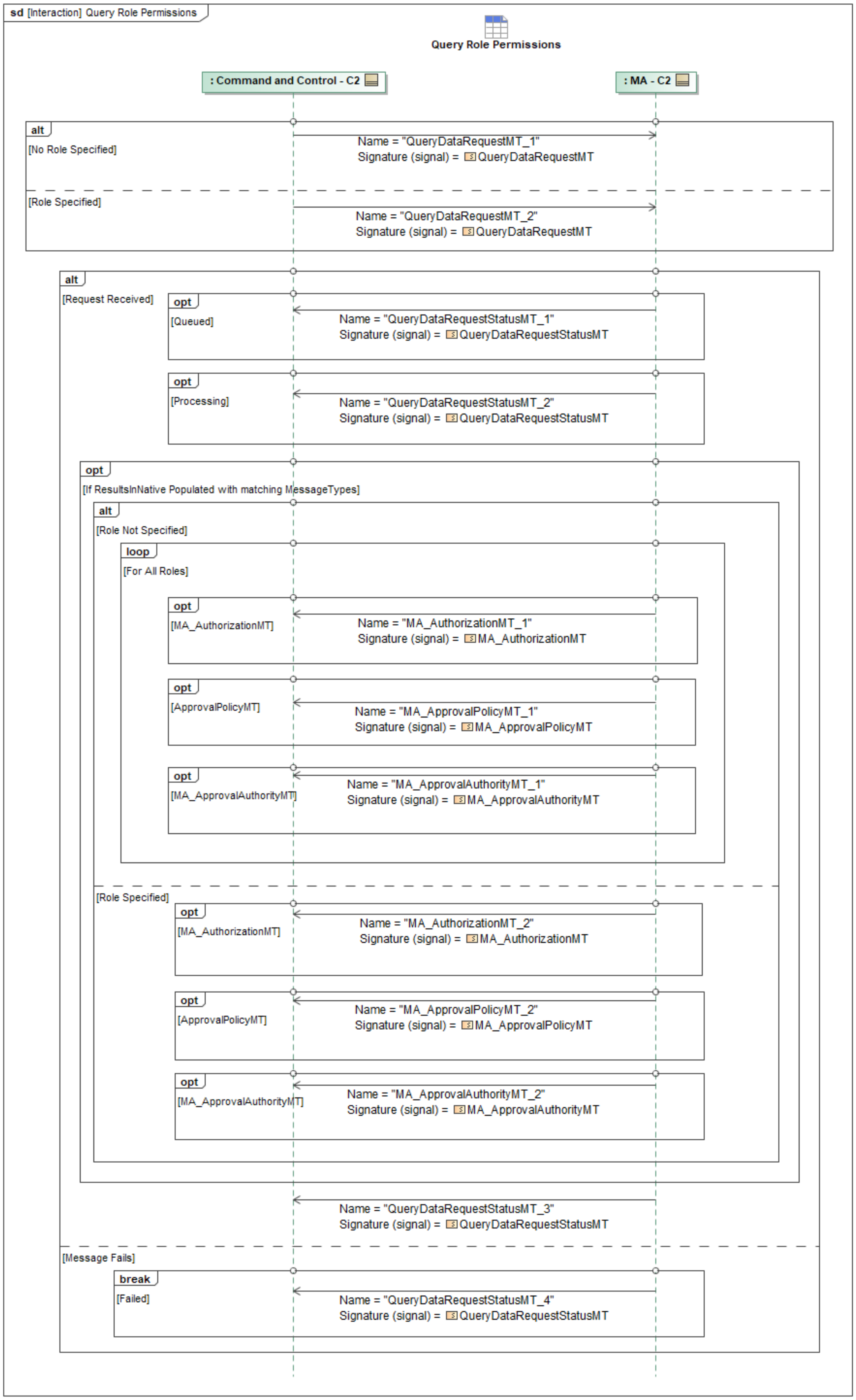


Figure 1-157: Query Role Permissions

Table 1-156: Query Role Permissions

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ApprovalAuthorityMT_1: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalAuthorityMT_2: MA_ApprovalAuthorityMT	SystemID: SystemID_Type
MA_ApprovalPolicyMT_1: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType
MA_ApprovalPolicyMT_2: MA_ApprovalPolicyMT	RequirementExecution: RequirementExecutionPolicyType
MA_AuthorizationMT_1: MA_AuthorizationMT	ByDataActivation: MA_DataActivationAuthorizationType ByPlanActivation: MA_PlanAuthorizationType OperatorRoleID: OperatorRoleID_Type ByDataModification: MA_DataModificationAuthorizationType PackageAuthorization: boolean ByPlatformSettingsModification: MA_SettingsModficationAuthorizationType ByRequirementExecution: MA_RequirementAuthorizationType ByOperatorRequest: MA_OperatorRequestAuthorizationType PackageAuthorization: boolean OperatorRoleID: OperatorRoleID_Type
MA_AuthorizationMT_2: MA_AuthorizationMT	ByPlanActivation: MA_PlanAuthorizationType OperatorRoleID: OperatorRoleID_Type ByDataModification: MA_DataModificationAuthorizationType ByDataActivation: MA_DataActivationAuthorizationType PackageAuthorization: boolean ByRequirementExecution: MA_RequirementAuthorizationType ByOperatorRequest: MA_OperatorRequestAuthorizationType ByPlatformSettingsModification: MA_SettingsModficationAuthorizationType PackageAuthorization: boolean OperatorRoleID: OperatorRoleID_Type
QueryDataRequestMT_1: QueryDataRequestMT	None
QueryDataRequestMT_2: QueryDataRequestMT	Query: QueryPET
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.9.9 RBAC Construct

Table 1-157: RBAC Roles

Roles	Controls
Admin	A C2 system with the **Admin** role will have full permissions to control mission autonomy. An **Admin** C2 system will have additional permissions to edit the existing mission plan to change role configuration during mission execution. This means that an **Admin** C2 system can update an existing role that needs new permissions, or create a totally new role. In the case that a new role is created, any user can switch to the new role via *MA_OperatorRoleMT*. This allows RBAC to remain flexible as a mission evolves. These updated roles will be sent out across the network to update all instances of C2 and Mission Autonomy.
QB	The **QB** Role is meant for C2 Users that have the most tactical awareness. These users have full authorization to command Mission Autonomy, with the exception of role configuration data.
AVC	This role is meant to be the most general of the C2 roles. It allows full flight and sensor permissions but limits weapon capability commands by default. It can be configured or used to create other roles as needed per mission. This is the "template" role. If desired, weapon capabilities can be configured for this role even if they are not authorized by default.
LRE	This role is intended to serve launch and recovery functions.
Observer	A C2 system with the **Observer** role will have no permissions to control mission autonomy. It is purely a listener to Mission Autonomy's data. This Observer C2 is not authorized to send any messages to mission autonomy, and mission autonomy will reject any messages that were sent from an observer C2.

*note for capability permissions, it is assumed that if a *CapabilityTypeEnum* is applicable for a permission, it applies to all enum values that are included in that capability data type. For example, the **AVC** role is authorized to command *"FLIGHT"* capabilities. This means that all *FlightCapabilityEnum* values are authorized. If permissions need to be subdivided further within a specific capability, the *CapabilityTaxonomyType* field can be used.

Role Configuration

In addition to the default roles, the A-GRA defines a process for how to create and configure roles based on a set of UCI configuration messages. In this way, mission planning and Admin C2 systems can ensure that RBAC will always fit the needs of the mission.

Roles are defined and configured through the use of four UCI messages:

MA_OperatorRoleMT: The *MA_OperatorRoleMT* is what defines a role. This is the message that other configuration messages will reference to define role permissions and request routing.

MA_ApprovalPolicyMT: The *MA_ApprovalPolicyMT* message defines what actions Mission Autonomy can take autonomously, and which require approval from C2 to execute the triggered autonomous action.

MA_ApprovalAuthorityMT: the *MA_ApprovalAuthorityMT* is what links *MA_ApprovalPolicyMT* to an approving authority. For RBAC, this message configures what roles Mission Autonomy will send its Approval Requests to.

*MA_Authoriza*tionMT: The *MA_AuthorizationMT* message configures which roles are authorized to task Mission Autonomy with certain message types.

MA_OperatorRoleMT

The *MA_OperatorRoleMT* message defines the user role. This message contains an *OperatorRoleID*, which serves as a unique identifier for the role throughout the mission. Agreement on this ID across all network nodes is required. The *Role* field is intended to reference an external database; however, with Role-Based Access Control (RBAC), the role's permissions are integrated directly into UCI messages. Consequently, the *Role* field's subfields, *Key* and *SystemName*, will be set to empty strings, as they are required by the schema but not utilized by Mission Autonomy. During mission planning, the *SystemID* field will remain empty, as no Command and Control (C2) system has yet declared itself as fulfilling this role. During mission execution, the *SystemID* field will be populated with an active list of C2 nodes that have declared themselves with this role. This list is advertised to other network nodes whenever it is updated.

MA_ApprovalPolicyMT

The *MA_ApprovalPolicyMT* is used to define which autonomous actions require approval. The *RequirementExecution* and *PlanActivation* fields enable the mission plan or an **Admin** C2 system to associate specific UCI requirements or plan types with a UCI Policy. This *Policy* will be set to *"AUTO_APPROVE"* if the UCI requirement or plan is authorized to be executed autonomously and *"AUTO_REJECT"* if the UCI requirement is not authorized to be executed autonomously. If executing this requirement or plan is authorized, but requires C2 approval to execute, then *Policy* will be set to *"REQUIRES_APPROVAL"*. The *RequirementExecution* field includes the subfield "*DefaultResponse*," which allows the *MA_ApprovalPolicy* to specify a default enum (either approve or reject). Exceptions to this default policy can then be supplied through the *ByRequirementPolicy* field. This approach allows for a comprehensive role definition without explicitly associating every UCI requirement enumeration with a Policy.

MA_ApprovalAuthorityMT

The *MA_ApprovalAuthorityMT* message establishes the link between the *OperatorRoleID* and the *ApprovalPolicyID*, letting Mission Autonomy route approval messages generated from a policy to a specific role. The *OperatorRoleID* is specified in the *SystemApprovalPolicy.Approver.ApproverRole.OperatorRoleID* field, while the *ApprovalPolicyID* is specified in the *SystemApprovalPolicy.ApprovalPolicyID* field. The A-GRA expects each role to have a single MA_A*pprovalAuthorityMT* message defining its Approval Policies. This design facilitates minimal updates if a permission needs to be modified for a role during mission execution.

MA_AuthorizationMT

The *MA_AuthorizationMT* message is used to configure the set of permissions allowed to each C2 Operator Role. The *AuthorizationID* will be set with a *UUID*. The *UserIdentifier* choice type will be filled with the *OperatorRoleID* of the role being configured. The *SystemId* field will be set to the *SystemId* of Mission Autonomy instance that this configuration file is for. The *Precedence* field will be set to zero as it is unused by RBAC, but is required by the schema. The fields *ByPlanActivation*, *ByRequirementExecution*, *ByDataModification*, *ByDataActivation*, *ByOperatorRequest*, *ByPackageAuthorization* and *ByPlatformSettingsModification*, and will be configured to set permissions for each role. The following fields are required fields for RBAC.

The *ByPlanActivation* field authorizes the activation of mission plans with the *ByMissionPlan* field, and authorization to activate specific plan types by their enum with the *ByPlanParts* field. Note that this field does not authorize updating or modifying a mission plan or plan parts, only their activation. For example, if a role was authorized to activate a *"RETURN_TO_BASE"* plan, it would have permission to command the activation of a Route Plan with the type *"RETURN_TO_BASE"*. However, this field does not authorize a C2 role to modify the route plan it is activating. That is a separate permission handled by the *ByDataModification* field.

The *ByRequirementExecution* field authorizes the C2 Operator Role to execute *Effects*, *Actions*, *Tasks*, *Capabilities* and *Responses* by their Enum type. This field similarly only gives permissions to command the execution of a UCI requirement, and does not include its modification. For example, if a role was configured to have permission to execute *"KINETIC_ATTACK"* type actions, then they could command any existing *MA_ActionMT*s with the type of *"KINETIC_ATTACK"* with *MA_ActionCommandMT*. This permission does not give the C2 user role authorization to modify the *MA_ActionMT* that it is activating. That is a separate permission handled by the *ByDataModification* field.

The *ByDataModification* field is used to specify what types of data this C2 user role is allowed to Create, Modify, and Delete. Note that this field does not include activating or deactivating data type messages like *MA_OpZone* or *MA_EngagementCriteriaEngagement*, that is handled by the *ByDataActivation* field.

The *ByDataActivation* field controls which types of data messages can be activated by a C2 user uole, broken out by message type. An example of this is authorization to command *MA_EngagementCriteriaCommand*. Note that this field does not imply permission to modify these messages, just activate them.

The *ByOperatorRequest* field is used to indicate which types of Requests the C2 role is authorized to task Mission Autonomy with. An example of this is the *AssessmentRequestMT* message.

The *PackageAuthorization* field configures if this role is authorized to create new teams, modify existing teams and dissolve teams.

The *BySettingsCommandAuthorization* field gives the C2 user role permissions to modify the existing settings on Mission Autonomy. This allows for changing default settings on the different capabilities and services that Mission Autonomy has access to.

Role Definition

In order to define a role, Mission Autonomy will have an *MA_OperatorRoleMT* message with a *OperatorRoleID* that corresponds to the role. The UUID field will be set to the unique ID for that role. The *DescriptiveLabel* field may also be used to include a human readable text string to indicate the role associated with this UUID, but this is not required.

MA_ApprovalAuthorityMT will have a unique *ApprovalAuthorityID*. The *ApplicableSystemPackageID* choice type field will select "*SystemID*" and set the UUID to this instance of Mission Autonomy. The *Approver.ApproverRole.OperatorRoleID* field should be set to the *UUID* of the role defined in the corresponding *MA_OperatorRoleMT* message. The *RequiredApproval* field should be set to false if C2 nodes with this role should not receive requests from Mission Autonomy (like the observer role). The *ApprovalPolicyID* will be set to the ID of the applicable policy for the role being configured.

MA_ApprovalPolicyMT will have an unique *ApprovalPolicyID*. For the fields *Plans*, *PlanActivation*, and *RequirementExecution* the *Policy* field should be set to *"REQUIRES_APPROVAL"* if it is required that a certain autonomous action (specified in the sibling fields of the policy), needs approval before it can be executed. Further details on how to configure Approval Policys are given in the section

Respond to Approval Request.

The next section will cover how to configure default *MA_AuthorizationMT*'s for each of the default A-GRA role permission sets.

Observer: The *AuthorizationID* will be set to a *UUID*. The *UserIdentifier* choice type will be set to *OperatorRole*, with the ID set to the *UUID* defined in the *MA_OperatorRoleMT* for the **Observer** role. This tells Mission Autonomy that this is the set of permissions authorized for the **Observer** Role. The *Precedence* field will be set to *"0"* as it is not utilized by RBAC, but is required by the schema. Each of the permission fields will chose "*UnauthorizedExceptFor*" from the choice type, and leave the exceptions list empty. The RBAC Permissions Table above outlines how the **Observer** role should be configured.

LRE: The *AuthorizationID* will be set to a *UUID*. The *UserIdentifier* choice type will be set to *OperatorRole*, with the ID set to the *UUID* defined in the *MA_OperatorRoleMT* for the **LRE** role. This tells Mission Autonomy that this is the set of permissions authorized for the **LRE** Role. The *Precedence* field will be set to *"0"* as it is not utilized by RBAC, but is required by the schema. The rest of the message will be configured in accordance with the RBAC permission table above.

AVC: The *AuthorizationID* will be set to a *UUID*. The *UserIdentifier* choice type will be set to *OperatorRole*, with the ID set to the *UUID* defined in the *MA_OperatorRoleMT* for the **AVC** role. This tells Mission Autonomy that this is the set of permissions authorized for the **AVC** Role. The *Precedence* field will be set to *"0"* as it is not utilized by RBAC, but is required by the schema. The rest of the message will be configured in accordance with the RBAC permission table above. If new roles need to be created, the new roles should be extended from the **AVC** role.

QB: The *AuthorizationID* will be set to a *UUID*. The *UserIdentifier* choice type will be set to *OperatorRole*, with the ID set to the *UUID* defined in the *MA_OperatorRoleMT* for the **QB** role. This tells Mission Autonomy that this is the set of permissions authorized for the **QB** Role. The *Precedence* field will be set to *"0"* as it is not utilized by RBAC, but is required by the schema. The rest of the message will be configured in accordance with the RBAC permission table above. The **QB** role will also be the approver for all strike requests generated by Mission Autonomy. To configure this, there will be an Approval Policy with the field *RequirementExecution.ByRequirementPolicy.Policy* set to *"REQUIRES_APPROVAL"* and the sibling *Requirement* field should have *Requirement.Requirement.Task* set to *"STRIKE"*.

Admin: The *AuthorizationID* will be set to a *UUID*. The *UserIdentifier* choice type will be set to *OperatorRole*, with the ID set to the *UUID* defined in the *MA_OperatorRoleMT* for the **Admin** role. This tells Mission Autonomy that this is the set of permissions authorized for the **Admin** Role. The *Precedence* field will be set to *"0"* as it is not utilized by RBAC, but is required by the schema. The rest of the message will be configured in accordance with the RBAC permission table above.

If Mission Autonomy generates a plan, or triggers a response that causes it to activate an existing plan, The *ApprovalRequirementEnum* value of *"REQUEST_APPROVAL"* can be used to route an approval request to the correct role. For example, if mission autonomy reaches bingo fuel, it might try to activate an *"RETURN_TO_BASE"* Route Plan. If it was desired that **QB's** needed to approve activating an RTB plan, then the **QB**'s *MA_ApprovalPolicyMT* would have the *PlanActivation.ByPlanParts.RoutePlan* set to the *"RETURN_TO_BASE"* enumeration. This would indicate that anytime Mission Autonomy tried to activate a *"RETURN_TO_BASE"* route plan that wasn't directly commanded, it should get an approval from a C2 system with the **QB** role. More details on sending and receiving approval requests can be found in the section ***Respond to***

Approval Request. It is important to note that in the context of RBAC, the *"APPROVAL_REQUIRED" ApprovalRequirementEnum* is only used for plan activation commands that originate from inside Mission Autonomy and were not directly commanded by C2. If Mission Autonomy receives a command from C2, it should either execute the command or reject it - it should not ask C2 for permissions to execute.

1.2.10 C2 Sensor/Vehicle Tasking Behaviors Package

1.2.10.1 Request Approval for Task Allocation Plan

This interaction allows a lead to receive approval for an Action Plan if the *ApprovalPolicy* calls for C2 to approve *ActionPlans* of this type. MA sends the plans using the "Send Action Plan - MA to C2" sequence. Optionally, MA can send additional information associated with the *Plans* using the "Synchronize Assessment" sequence. Then the "Respond to Approval Request" sequence will be started by sending an *ApprovalRequestMT* with *ApprovalItem.PlanApproval* set to the ID of the Plan being approved. If approved, the MA may then execute the Plan if activation has been commanded or otherwise triggered.

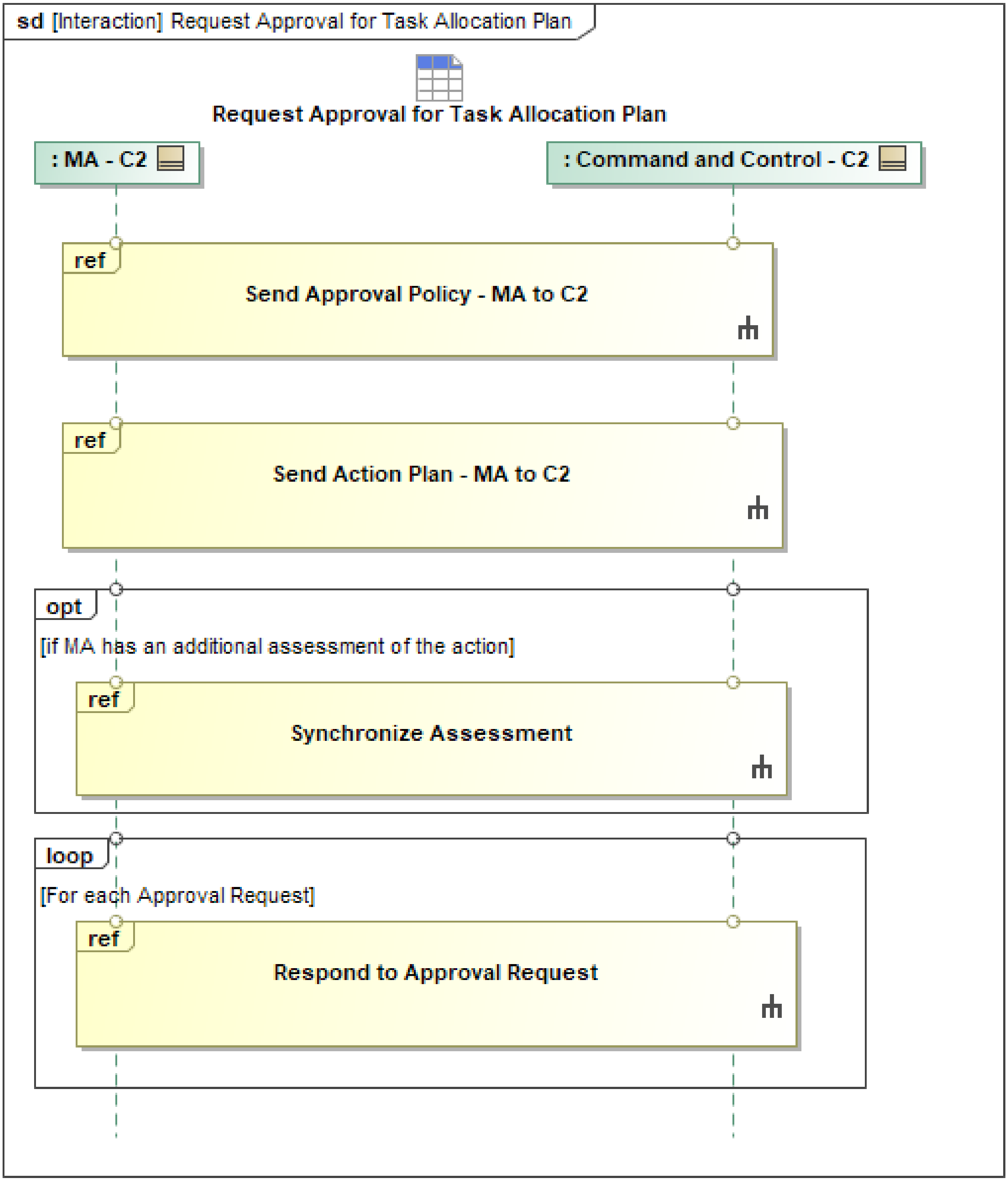


Figure 1-158: Request Approval for Task Allocation Plan

1.2.10.2 Synchronize Assessment

This interaction allows an ACP to send an Assessment to C2. The Assessment can be either initiated by MA or by C2.

To initiate the Assessment from C2, C2 sends a *MA_AssessmentRequestMT*. The required parameters depend on the type of assessment being requested. Optionally MA can send a *MA_AssessmentRequestStatusMT* when the request is *QUEUED* or *PROCESSING*. If the *MA_AssessmentRequestMT* can't be processed, MA responds with a *MA_AssessmentRequestStatusMT* with *RequestProcessingState* == *FAILED*. If the *AssessmentRequest* processing is completed, MA sends a *MA_AssessmentRequestStatusMT* with *RequestProcessingState* == *COMPLETED*. If the *ResultsInAssessmentMessage* field in the *MA_AssessmentRequestMT* is set to *FALSE*, then MA will include the Assessment in the *MA_AssessmentRequestStatusMT.* Otherwise MA will send a separate *MA_AssessmentMT*. To initiate the Assessment from MA, send *MA_AssessmentMT* to C2. The parameters in the Assessment should match the parameters that the ACP used to create the Assessment.

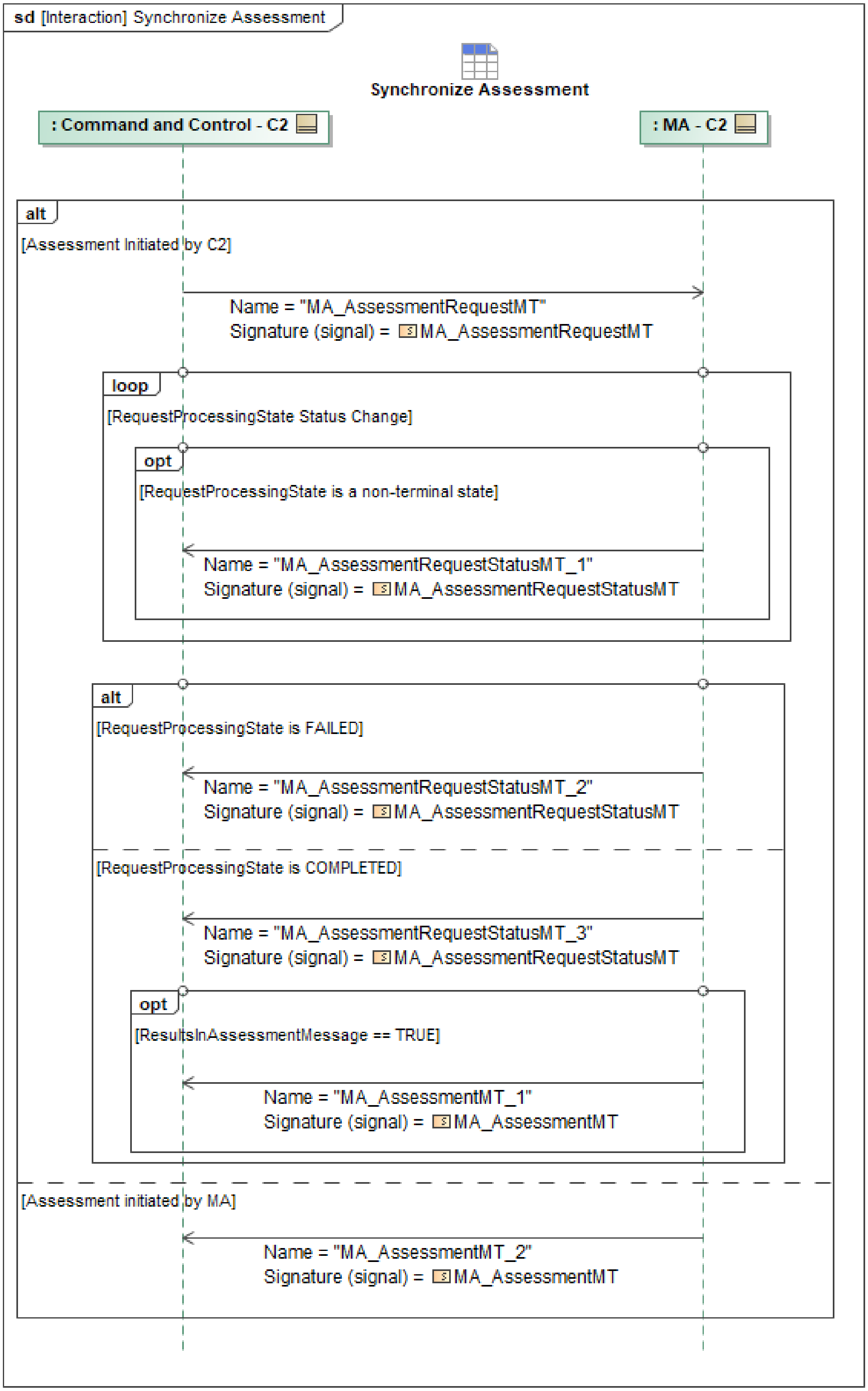


Figure 1-159: Synchronize Assessment

Table 1-158: Synchronize Assessment

Message Name (Name:Type)	Required Fields (Name:Type)
MA_AssessmentMT_1: MA_AssessmentMT	None
MA_AssessmentMT_2: MA_AssessmentMT	None
MA_AssessmentRequestMT: MA_AssessmentRequestMT	None
MA_AssessmentRequestStatusMT_1: MA_AssessmentRequestStatusMT	None
MA_AssessmentRequestStatusMT_2: MA_AssessmentRequestStatusMT	None
MA_AssessmentRequestStatusMT_3: MA_AssessmentRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11 C2 Tactic Behaviors Package

1.2.11.1 Advertise Tactic Capabilities

MA sends *MA_ActionCapabilityMT* periodically to advertise the Tactic capabilities of the platform, as well as the Tactics the platform is currently capable of performing based on system/subsystem status and available stores when MA sends *ActionCapabilityStatusMT* periodically.

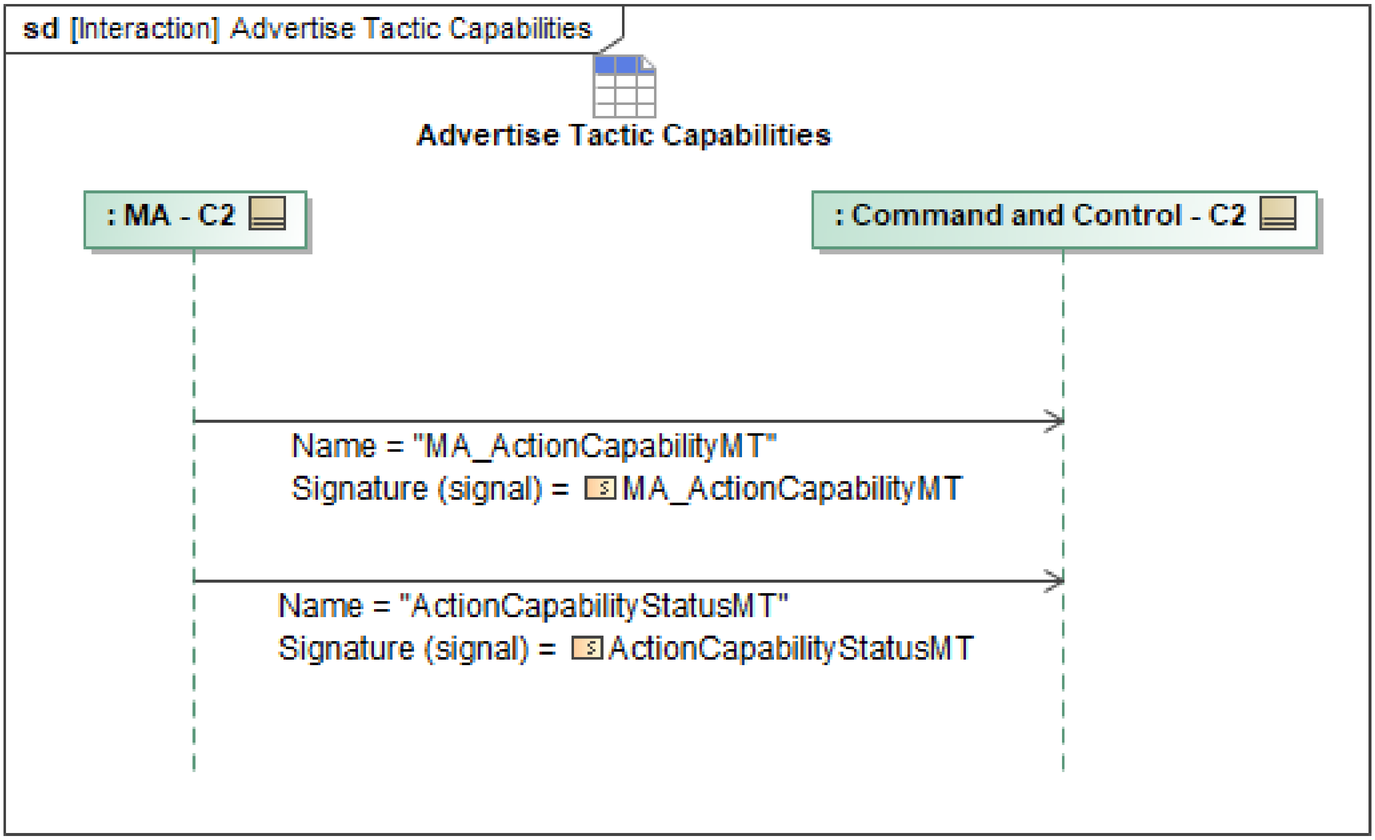


Figure 1-160: Advertise Tactic Capabilities

Table 1-159: Advertise Tactic Capabilities

Message Name (Name:Type)	Required Fields (Name:Type)
ActionCapabilityStatusMT: ActionCapabilityStatusMT	None
MA_ActionCapabilityMT: MA_ActionCapabilityMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.2 Cancel Tactic

C2 can command MA to cancel a pending Tactic (Action), or to cancel an already executing Tactic when possible with *ActionCancelCommandMT*, to which MA will reply with a corresponding *ActionCancelCommandStatusMT_1.*

Canceling a Tactic should cause the platform to revert to a pre-defined fallback state (ex: return-to-base or returning to the original PlannedC2 mission plan) unless a new set of instructions is commanded.

In the case that the *ActionCancelCommandMT* is rejected, the returned *ActionCancelCommandStatusMT_2* message with *CommandProcessingState = REJECTED* must also include the *CommandProcessingStateReason* field to include an explanation why the command was rejected.

As the Tactic is canceled, MA will continue to update C2 on the Tactic's status (see: Receive Tactic Status behavior) until the Tactic and it's sub-Tasks have all been successfully canceled/aborted.

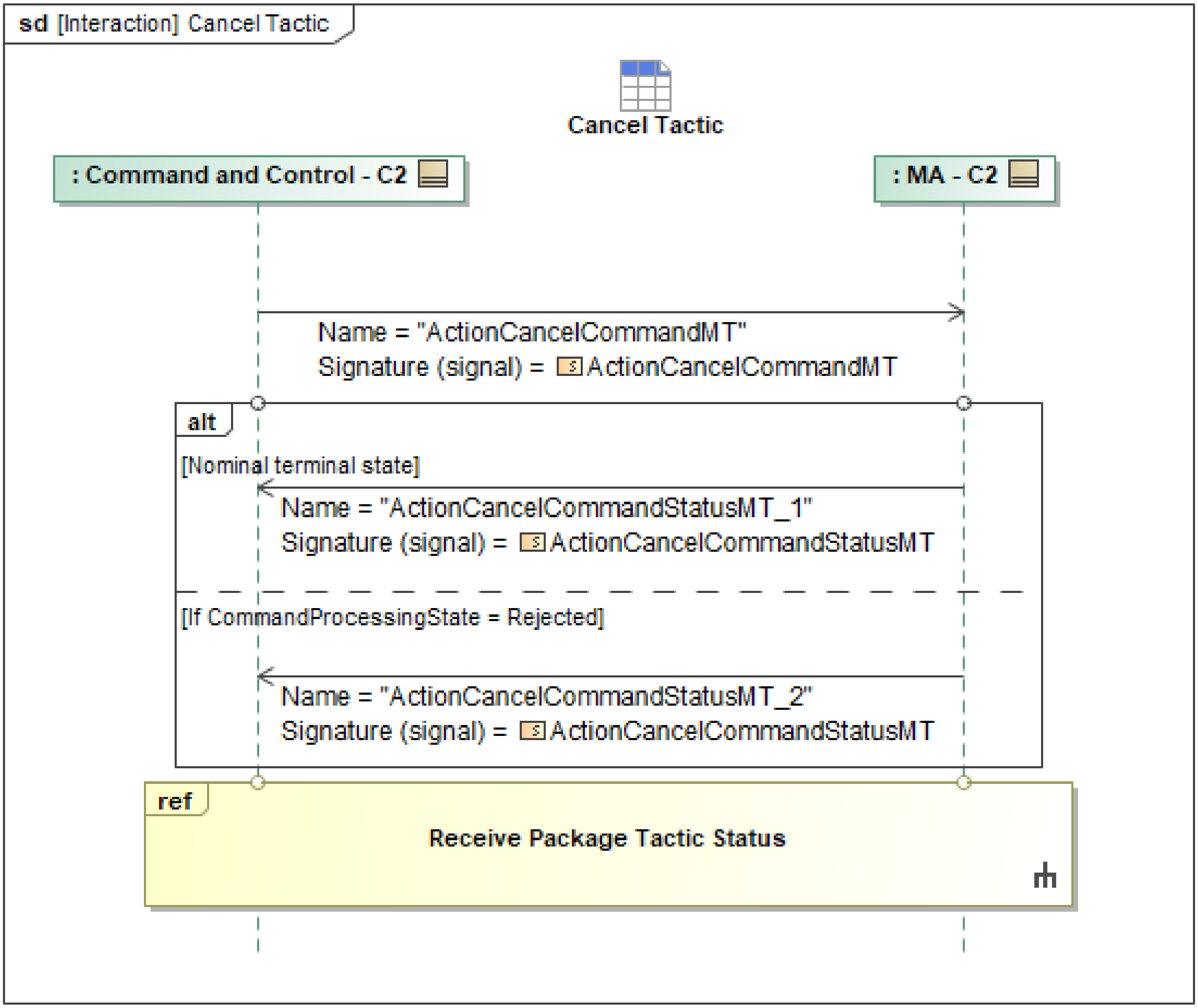


Figure 1-161: Cancel Tactic

Table 1-160: Cancel Tactic

Message Name (Name:Type)	Required Fields (Name:Type)
ActionCancelCommandMT: ActionCancelCommandMT	None
ActionCancelCommandStatusMT_1: ActionCancelCommandStatusMT	None
ActionCancelCommandStatusMT_2: ActionCancelCommandStatusMT	CommandProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.3 Cold

The "Cold" command is a tactic that commands a single ship or package to cancel any active engagements and turn away from the fight. This is specified in the *MA_ActionMT* message under the *ActionType* field, using the *WITHDRAW MA_ActionTypeEnum* value. If updates to Cold Tactic Parameters are needed, the referenced sequence can be used to do so.

C2 will activate the "Cold" command by sending an *MA_ActionCommandMT* message to MA. The *MA_ActionCommandMT* message to start the tactic will only be accepted if MA is intercepting a target. MA will check if there is an ongoing tactic with a designated target. This would happen internally in MA, where MA would check if it were actively performing flight maneuvers towards a target. If MA is engaging a target, then the "Cold" command can be accepted.

MA will respond to C2 with a *MA_ActionCommandStatusMT* message. If the tactic is valid and MA is intercepting a track, the *MA_ActionCommandStatusMT_1* message will be used to confirm receipt of the command. If the message was rejected because MA was not intercepting a track, *MA_ActionCommandStatusMT_2* message will have the CommandProcessingStateReason field populated with the *INIT_CRITERIA_NOT_MET* value.

Once the command is accepted, MA will start sending the tactic status to C2. Any existing engagements will be canceled upon receipt of the *WITHDRAW* action command which will also be appropriately reported back to C2.

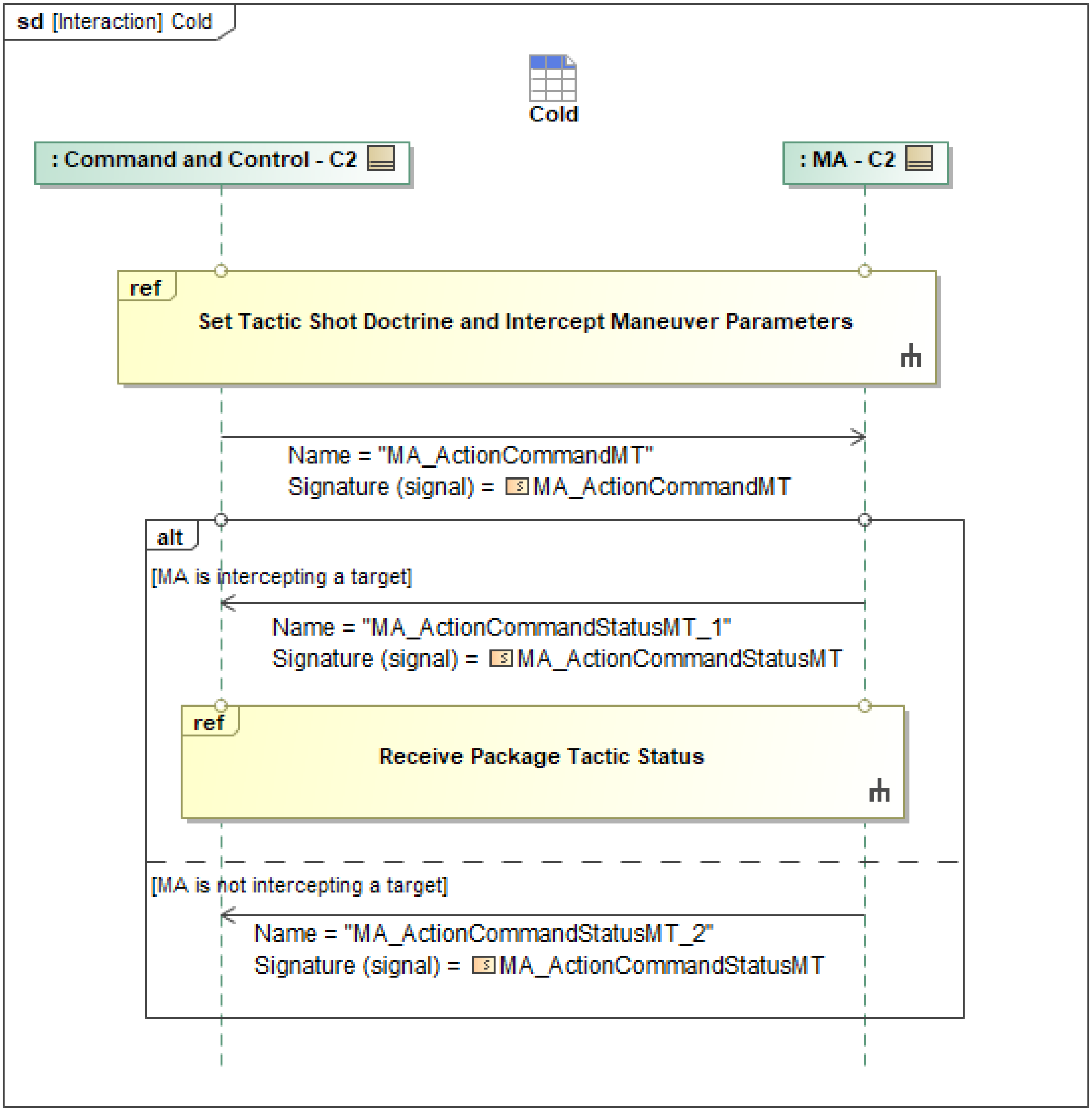


Figure 1-162: Cold

Table 1-161: Cold

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandMT: MA_ActionCommandMT	Capability: MA_ActionCapabilityCommandType ActionID: ActionID_Type

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandStatusMT_1: MA_ActionCommandStatusMT	None
MA_ActionCommandStatusMT_2: MA_ActionCommandStatusMT	CommandProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.4 Command Tactic

C2 will command all tactics, both to individual platforms, and packages, at the action level. MA will then be responsible for breaking down the action into tasks and/or activities. C2 will define the action type and tactical parameters by issuing *MA_ActionMT*. Then, C2 can command the vehicle to begin populating a Tactic Template and begin executing a Tactic by sending an *MA_ActionCommandMT* message to MA, which will then return *MA_ActionCommandStatusMT_1* messages to acknowledge receipt and confirm acceptance of the *MA_ActionCommandMT*.

Additionally, if the *MA_ActionCommand* was addressed to a *PackageID*, the associated *MA_ActionCommandStatus* should reflect the *Package ID*. If partial rejections occur during Lead-to-Follower delegation the Lead ACP must revert to individual messages to report multiple rejecting followers.

Once MA has accepted the command to begin the Tactic, MA will send *MA_ActionCommandStatusMT_2 as a response to C2 and* it will then begin providing periodic status updates to C2 on the status of the Tactic via the "Receive Package Tactic Status behavior". The interaction is the same for individual systems providing a tactic status to C2. If C2 is authorized, but MA cannot accept the command, *MA_ActionCommandStatusMT_3* is the response.

If C2 allocates a requirement to a specific subset of platforms (via discrete *SystemID* allocations rather than *PackageID*), the Lead ACP shall bypass package-level aggregation and return individual status messages for those tasked members.

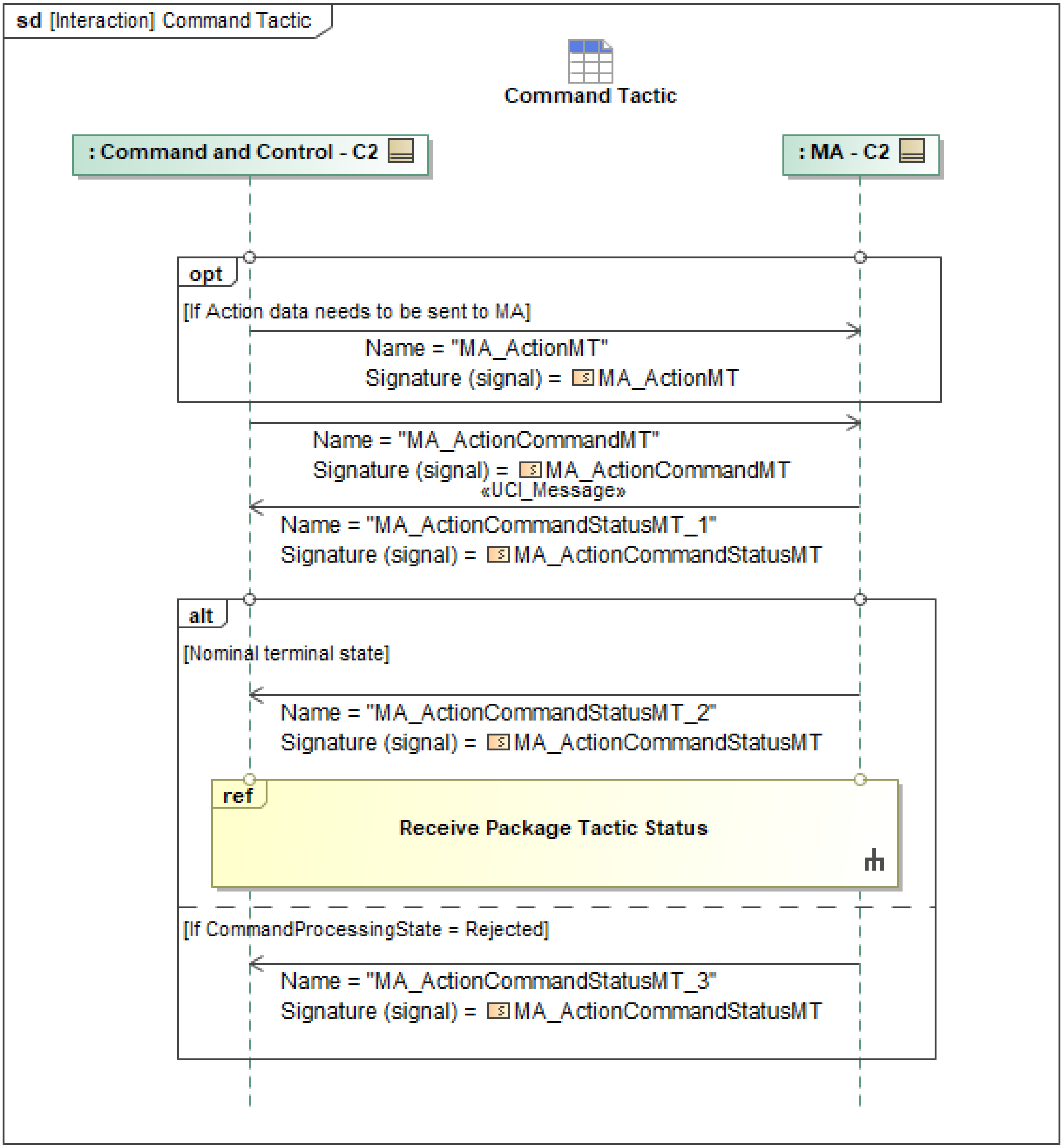


Figure 1-163: Command Tactic

Table 1-162: Command Tactic

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandMT: MA_ActionCommandMT	Capability: MA_ActionCapabilityCommandType
MA_ActionCommandStatusMT_1: MA_ActionCommandStatusMT	None
MA_ActionCommandStatusMT_2: MA_ActionCommandStatusMT	None
MA_ActionCommandStatusMT_3: MA_ActionCommandStatusMT	CommandProcessingStateReason: CannotComplyType
MA_ActionMT: MA_ActionMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.5 Drop

The "Drop" command described in this C2 L1 interaction cancels any existing target pairings and shot allocations between ACPs and tracks. C2 can send the drop command and specify who it applies to via the *AppliesToIDs*. Within this field there are options to specify a *PackageID*, *SystemID*, or an *EmptyType* of "*ALL*" or "*NON*." The Drop of the existing *"TARGET"* Action will apply to all ACPs specified using that field. The Drop command is sent using the *MA_DesignationMT* UCI message. The *DesignationID* field is required to contain a UUID and represents this new designation of a dropped Entity. The *TypeOfDesignation* field should be set to *"DESIGNATED_NO_TARGET"* via the *DesignationEnum.* The *DesignationObjective* field should call out an *MA_ActionTypeEnum* of *"TARGET"*. The *Target* field is where the *TargetType* is specified. This field can refer to an *EntityID,* or any other field contained within *TargetType* that should be dropped from ongoing or planned targeting actions. ACPs will acknowledge the "Drop" designation with an *MA_SystemNotificationMT.* If MA is positively acknowledging the *MA_DesignationMT* to drop track(s), and any ACPs were previously targeting and/or had shots allocated to the now-dropped tracks, ACPs will then send the *MA_ActionStatusMT* to report any ongoing "Target" activities as canceled. The UUID of the previously applicable *ActionID* corresponding to the "*TARGET*" *MA_ActionTypeEnum* within the "Target" command should be set to "*CANCELED*" so ACPs know to stop executing any target actions and designate selected tracks as non-targets.

When tracks are designated as non-targets, mission autonomy will trigger an action to cancel any existing target pairings and shot allocations. This update of targets uses the *RelationshipDesignationMT* message where the *Relationship* field contains an *EngagementStatus* field that contains an *ExternalCommandExecutionStateEnum*. This enum should be set to *"CANCELED"*. The *RelationshipDesignationID* field contains a UUID indicating each individual unique relationship between a *Source* and a *Destination.* In this case of updating C2 with target pairings, or dropped tracks, the sources are ACPs represented by *SystemIDs* and the destinations are Entities represented by *EntityIDs.*

It is assumed that each ACP will then update any applicable MA_*StrikeActivityMTs* to remove each allocated shot, preventing it from firing, and therefore will report this activity by sending MA_*StrikeActivityMT* periodically. Messaging details are explained in the use cases "Receive Package Tactic Status" and "Report Tactic Status - Peer", which cover C2 and Peer to Peer behaviors respectively. These use cases allow Peers to report the status of tactics back to a leader for the leader to then report the entire package's tactic status back to C2.

The *MA_SystemNotificationMT* message is the initial response that provides C2 an acknowledgment that the *MA_DesignationMT* message was received by MA successfully or not.

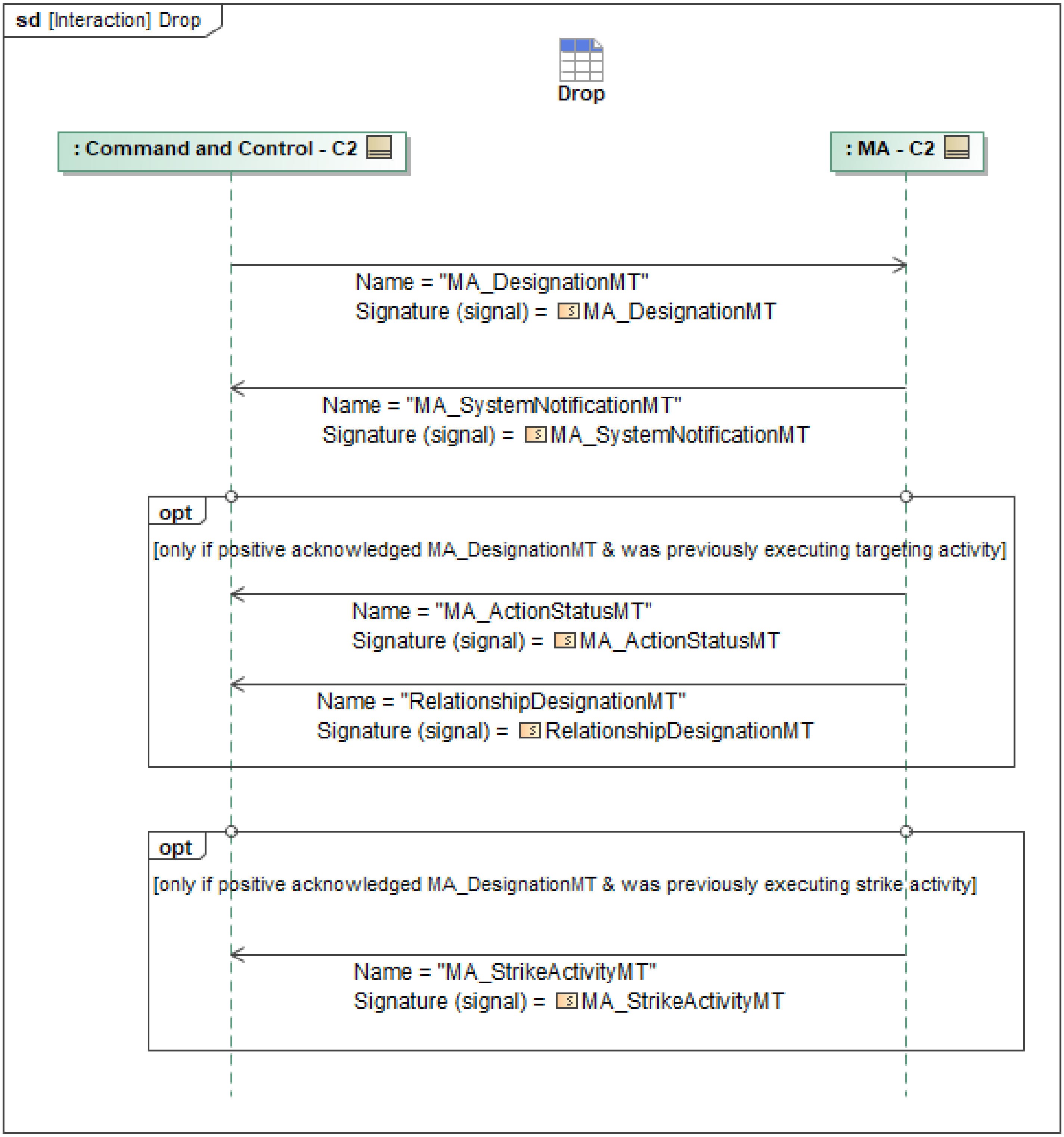


Figure 1-164: Drop

Table 1-163: Drop

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionStatusMT: MA_ActionStatusMT	SystemID: SystemID_Type

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DesignationMT: MA_DesignationMT	DesignationType: DesignationEnum DESIGNATED_NO_TARGET: AppliesToIDs: MA_PackageSystemType DesignationObjective: MA_RequirementTaxonomyType DesignationType: DesignationEnum
MA_StrikeActivityMT: MA_StrikeActivityMT	None
MA_SystemNotificationMT: MA_SystemNotificationMT	AssociatedMessage: AssociatedMessageType
RelationshipDesignationMT: RelationshipDesignationMT	Relationship: RelationshipType Destination: GeoLocatedObjectType EngagementStatus: ExternalCommandExecutionStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.6 Holster

The “Holster” Tactic Command is meant to cancel any existing shot allocations between ACP(s) and tracks. If the ACP(s) are currently targeting and/or intercepting tracks, this command will keep the ACP(s) on the intercept course but remove their allocated shot. This command is sent using the *MA_DesignationMT* with a *DesignationData* field for each track that needs to be holstered. Each *DesignationData* must set the *TypeOfDesignation* field to “*DESIGNATED_NO_TARGET*” and the *Target.EntityID* set to the *EntityID* of the track to be holstered. The *DesignationObjective* field should have an *MA_ActionTypeEnum* of "*TARGET*" informing MA it can still continue its intercept behavior, but should un-allocate shots for the specified *EntityID*. Additionally, the *DesignationID* field must be set with a valid UUID and the *AppliesToIDs* will be set to the targeting System or Package's UUID.

ACP(s) acknowledge the “Holster” designation with the *MA_SystemNotificationMT* message. The *AssociatedMessageType* field will be set to “*MA_DESIGNATION*”. If positively acknowledged (meaning *NotificationDetail* is not included) and any ACP(s) previously had shots allocated, ACP(s) will update any applicable *StrikeActivityMT* to remove each allocated shot. MS would publish *StrikeActivityMT*(s) communicating any updates of current employment of weapons to MA, which MA can forward to C2. The removed shot allocation would be represented by the field *AssignedTarget being* no longer included by MS to MA, and then sent from MA to C2. This prevents the ACP's MS from firing. Updates to a strike activity will be reported via the *StrikeActivityMT* message.

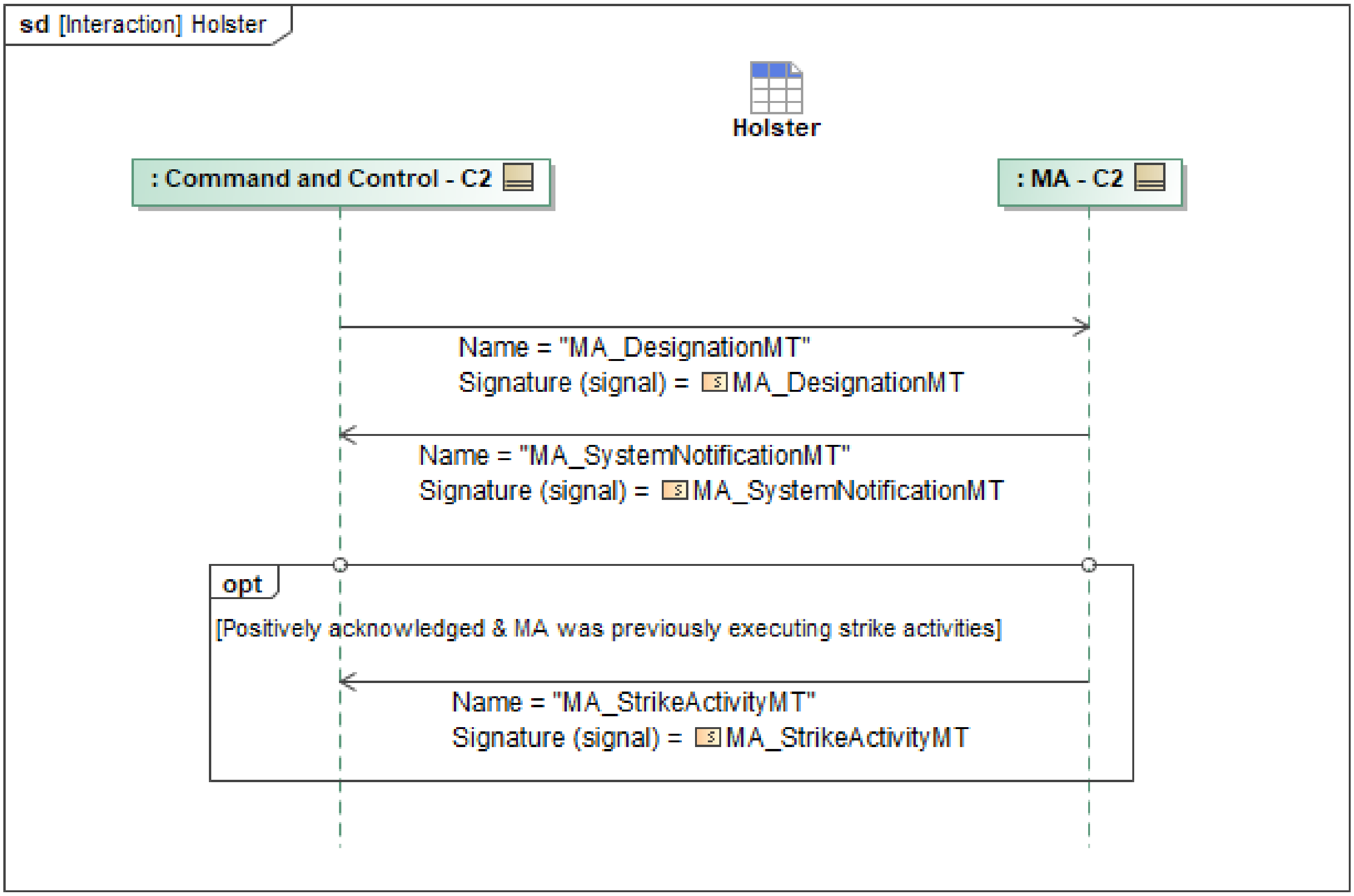


Figure 1-165: Holster

Table 1-164: Holster

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DesignationMT: MA_DesignationMT	TypeOfDesignation: DesignationEnum DESIGNATED_NO_TARGET: AppliesToIDs: MA_PackageSystemType DesignationObjective: MA_RequirementTaxonomyType
MA_StrikeActivityMT: MA_StrikeActivityMT	ConsentState: ConsentEnum SelectedForRelease: boolean
MA_SystemNotificationMT: MA_SystemNotificationMT	AssociatedMessage: AssociatedMessageType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.7 Receive Package Tactic Status

MA will periodically provide updates to C2 on the status of a Tactic while the Tactic is being executed or queued to begin executing. MA sends *ActionActivityMT* periodically to convey periodic updates regarding the ongoing activity executing the tactic. MA sends *MA_ActionStatusMT* periodically to convey the status of the Tactic as a whole, while MA sends *MA_TaskStatusMT* periodically so it can provide more detailed statuses on the respective Tasks that make up the Tactic. This can apply to an entire package, where the lead would communicate all of it's Peers statuses, or just a single ACP. The "Report Tactic Status - Peer" sequence describes how peers will communicate ongoing tactics to the lead, where the lead is responsible for reporting the package tactic status to C2. In the case of a single ship, the single ship would be the only ACP in the package, and the sequence would only be executed one time.

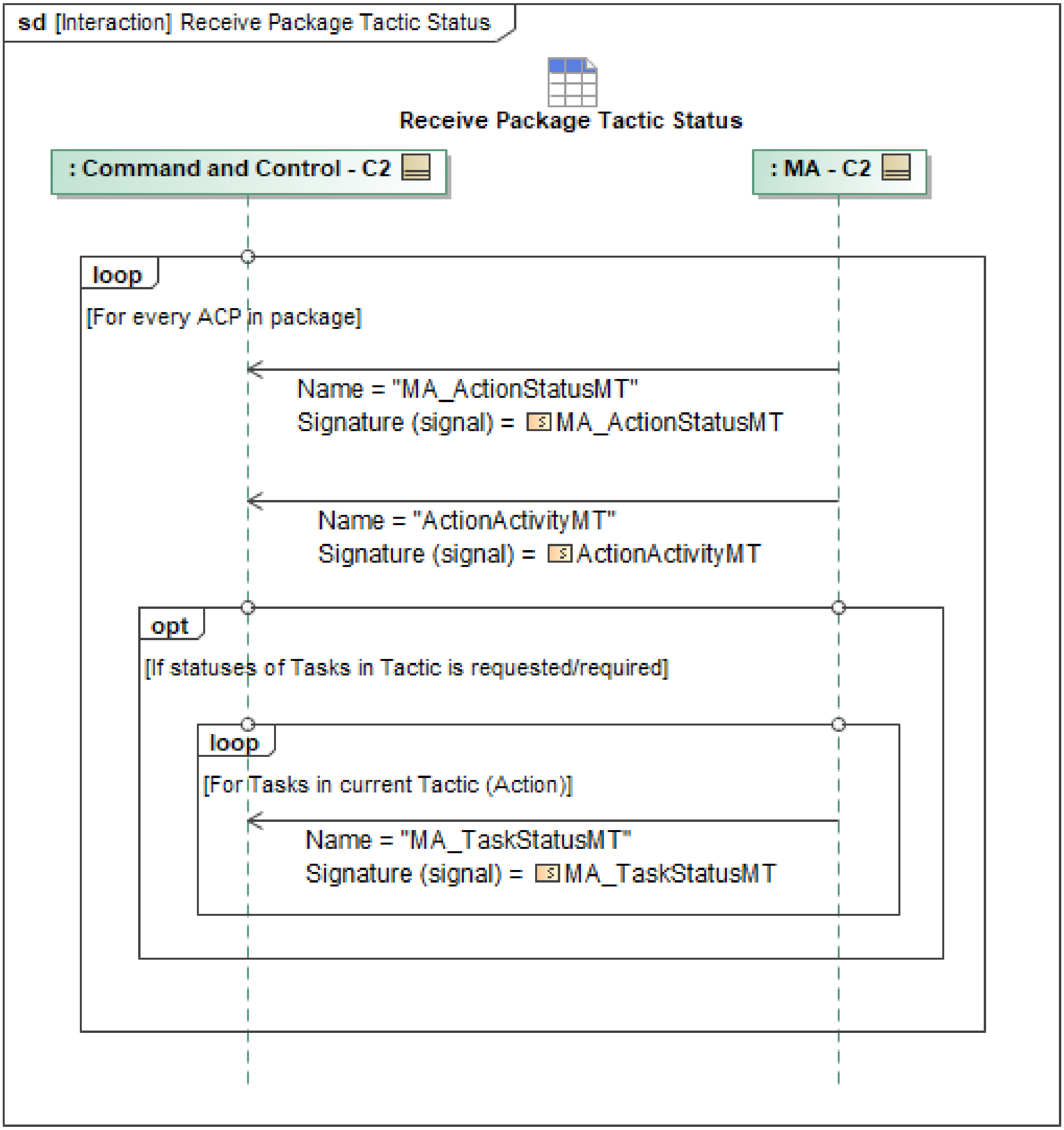


Figure 1-166: Receive Package Tactic Status

Table 1-165: Receive Package Tactic Status

Message Name (Name:Type)	Required Fields (Name:Type)
ActionActivityMT: ActionActivityMT	None
MA_ActionStatusMT: MA_ActionStatusMT	PackageID: PackageID_Type

Message Name (Name:Type)	Required Fields (Name:Type)
	PackageID: PackageID_Type
MA_TaskStatusMT: MA_TaskStatusMT	TaskID: TaskID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.8 Set Tactic Shot Doctrine and Intercept Maneuver Parameters

Updating parameters that may inform a shot doctrine and/or intercept maneuver tactics for Mission Autonomy to ingest is done via an *MA_ActionMT* message. A C2 operator would send an *MA_ActionMT* message to provide this update. This *MA_ActionMT* can also be setup during the mission planning process using the fields discussed below. Tactics will then be employed from C2, at the Action level, using an *MA_ActionCommandMT.* In the *MA_ActionMT*, targeting tactics will utilize the *MA_ActionTypeEnum* of *TARGET* while strikes will utilize *ATTACK_KINETIC*. Other *MA_ActionTypeEnum's* can be used depending on the tactic desired, which are discussed in other C2 behaviors. Other Values related to a shot doctrine live within the *ActionConstraints* field. Specifically, within *ActionConstraints*, the field *EngagementParameters* contains fields to specify *InterceptTactic*, *ShotParameters*, and *EnableAutomaticRecalculation,* the later which specifies whether to enable automatic recalculation during engagement. The *InterceptTactic* is a choice type for C2 to set either a *SkateType* or *BanzaiType.* Within *SkateType*, a *ManeuverOutRange* can be set via a *DistanceType*. Within *BanzaiType* a *TerminationType* can be set which uses the *MA_InterceptTacticTerminationTypeEnum.* The enumeration values available are *BLOW_THROUGH, VELCRO,* or *COLLIDE. ShotParameters contains MissilesPerContact and ShootChoice. The MissilesPerContact field denotes how many missiles should be used per entity contact, specified as an integer. The ShootChoice allows a choice between toggling UseRangeProbabilityOfIntercept or setting ShotParameterDetails. ShotParameterDetails contains UseRangeTurnAndRunOrDMC for C2 to specify an ACP to abide by a RangeTurnAndRun or follow a DigitalManeuveringCueAngle. The Digital Maneuver Cue value can either be specified using the RangeTurnAndRun value or the DigitalManeuveringCueAngle. If RangeTurnAndRun is desired, the RangeTurnAndRun value will specified within MA_WEZ_MT (see "Exchange WEZ" Interaction) or DLZ_MT (see "Exchange DLZ" Interaction) and the UseRangeTurnAndRun field should be populated. If the DigitalManeuveringCueAngle field is used, it should be a value in radians between 0.17452393 and 3.14159265358979323846, set in increments of 0.17452393 radians (which corresponds to 10 to 180 degrees set in increments of 10 degrees). ShotParameterDetails also specify the ShootNLT and ShootNET fields which signify shoot no later than and shoot no earlier than values allow C2 to specify a shot window. These fields are DistanceTypes.*

When commanded via an *MA_ActionCommandMT,* an ACP will follow these tactic parameters when engaging a target.

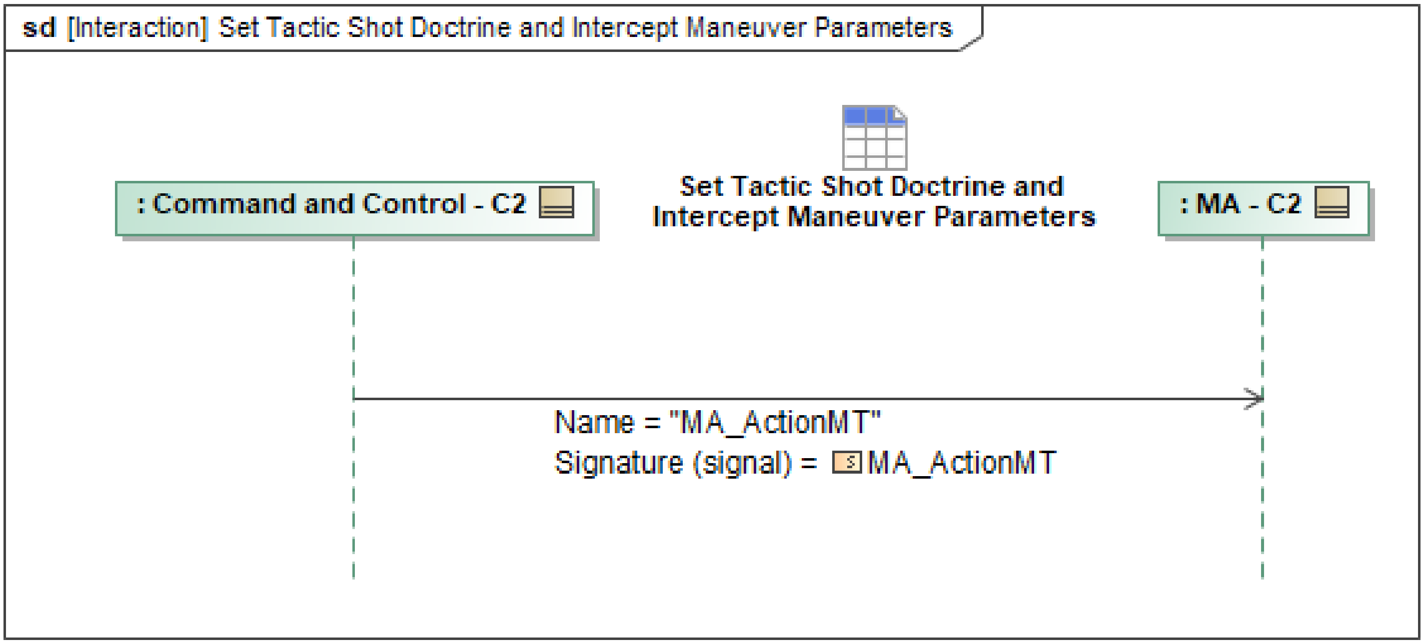


Figure 1-167: Set Tactic Shot Doctrine and Intercept Maneuver Parameters

Table 1-166: Set Tactic Shot Doctrine and Intercept Maneuver Parameters

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionMT: MA_ActionMT	ActionID: ActionID_Type ActionConstraints: MA_RequirementConstraintsType EngagementParameters: MA_EngagementParametersType ActionType: MA_ActionTypeEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.9 Shoot

A "Shoot" tactic commanded from an operator tells one or more ACPs to allocate shots to applicable tracks and to shoot them according to a pre-determined shot doctrine. The "Shoot" tactic can be executed using either a Capability-based or an Activity-based command path. The tactic begins executing when C2 sends an *MA_ActionCommandMT*. In the Capability-based path, C2 references a pre-planned *MA_ActionMT* via the *ActionID* field by *UUID* so MA references the desired shot doctrine parameters. In this case, the *ActionType* of the referenced *MA_ActionMT* needs to be *ATTACK_KINETIC*. The parameters of the tactic are configured during mission planning by setting the fields within *MA_ActionMT*.

If the operator uses the *Activity* path, autonomy advertises an ongoing *ActionActivity* and C2 sends an *MA_ActionCommandMT* whose *ActivityID* matches the advertised *ActivityID* for that ongoing play. In this Activity-based path, updates to the ongoing tactic's shot doctrine and engagement parameters are conveyed via *ChangeShotDoctrine, ChangeActionConstraints, and ChangeAttackKineticParameters*.

If desired, the 5.2.9.8 "Set Tactic Shot Doctrine and Intercept Maneuver Parameters" behavior may still be used to change the underlying *MA_ActionMT* for an ongoing tactic.

If operators employing the *Shoot* tactic wish to use a different parameter value for anything within the shot doctrine, the *ConstraintOverride* field remains available. Contained within this field is the *ShotDoctrineParameters* field. This only needs to contain values if the commanded tactic should follow updated parameters. Fields populated in *ShotDoctrineParameters* override the default values.

For the *Activity Audible* path, omission versus explicit values follows - fields that are omitted from *ChangeShotDoctrine, ChangeActionConstraints, or ChangeAttackKineticParameters* remain at their current effective value for the ongoing play; Audibles do not implicitly reset unspecified parameters. Fields that are present with concrete values replace the current effective value for that parameter for the duration of the ongoing *ActionActivity*, until further update or tactic completion. Where supported, an explicit “use default” indicator may be used to revert a parameter to the value defined in the referenced *MA_ActionMT* for this play.

These overrides apply to the ongoing *ActionActivity* for the duration of that play. Subsequent Audibles referencing the same *ActivityID* may further update these parameters; parameters not mentioned in a given Audible retain their current effective values for that ongoing play. The *MA_ActionCommandMT* (whether using the *ActionID*-initiated baseline or the *ActivityID*-based Audible path) will need to contain information about the entities the Shoot command should be acted on. This information is contained within the *ActionConstraintOverride* field within *Capability* – specifically the *TargetObject*, where *EntityIDs* can be added *ByInstance.*

Additional updates to per-target missile allocation for an ongoing play are expressed via *PerTargetShotDoctrine* structure within *ChangeActionConstraints*, whose cardinality matches the *TargetObject* when used with *ChangeActionConstraints*. When *ChangeActionConstraints*.*TargetObject* and *MissilesPerContact* are provided for an ongoing *ActionActivity*, they are interpreted as an add/modify set for that play: targets newly introduced are added, and existing targets may have their associated missile counts updated. Targets that are part of the ongoing tactic and are not mentioned in *ChangeActionConstraints* remain unaffected by that Audible; removal of targets continues to be expressed via existing “Drop” behaviors rather than by omission from *TargetObject.* Similarly to the previous “Target” tactic, MA (lead) is responsible for allocating and assigning shots within a package of ACPs for n number of targets.

The allocated shot pairings are shared with C2 by a lead MA using the *RelationshipDesignationMT* message, which is covered by the “Report Shot Allocations” sequence. Furthermore, the ongoing statuses of all commanded shots are shared via the use of two behaviors on the Peer and C2 interfaces respectively: “Peer Report Tactic Status” and “Receive Package Tactic Status”.

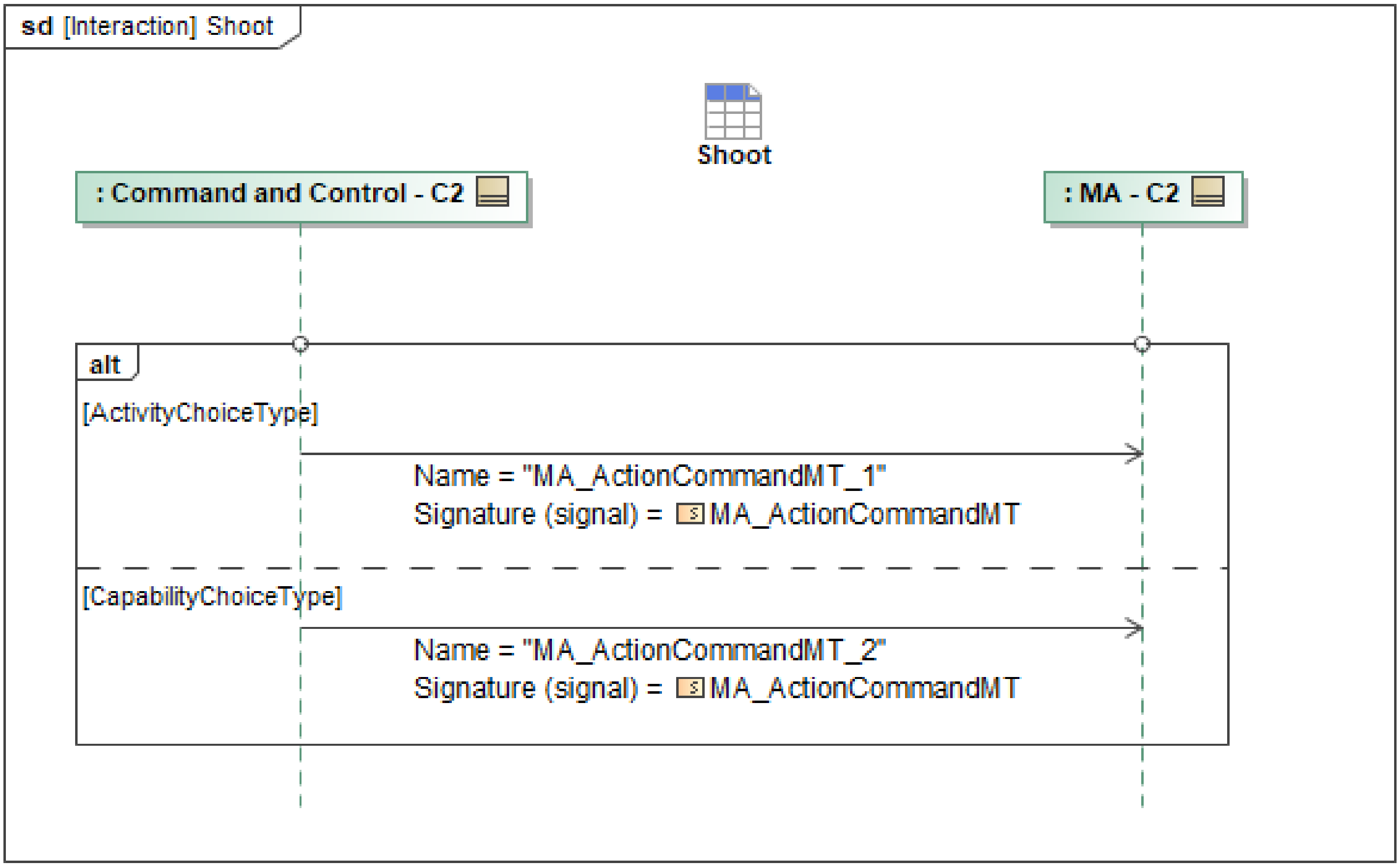


Figure 1-168: Shoot

Table 1-167: Shoot

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandMT_1: MA_ActionCommandMT	TargetObject: IdentityKindInstanceType MissilesPerContact: MA_ObjectShotCountChoiceType Activity: MA_ActivityCommandBaseType ChangeActionConstraints: MA_ActionConstraintType
MA_ActionCommandMT_2: MA_ActionCommandMT	ActionConstraintOverride: MA_ActionConstraintType Capability: MA_ActionCapabilityCommandType TargetObject: IdentityKindInstanceType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.10 Shoot Now

A "Shoot Now" tactic command is very similar to the "Shoot" tactic, but rather than ensuring ACPs abide by specific shot-doctrine shot windows (Shoot NET and Shoot NLT), Shoot Now will not follow a shot window and relaxes release criteria to meet the Range Probable Intercept (RPI) value. The "Shoot Now" tactic is expected to be set up pre-mission as a separate *MA_ActionMT*, with the same

ActionType = ATTACK_KINETIC as the "Shoot" tactic. Operators can specify whatever parameters are desired to align with their operational intent for a "Shoot Now" tactic. However, it would be expected that the *ShootNLT*, *ShootNET*, and *DigitalManeuveringCueAngle* fields are excluded from the *ShotDoctrineParameters* within *ActionConstraints* in this pre-defined *MA_ActionMT*. The "Shoot Now" tactic can be executed using either a Capability-based or an Activity-based command path.

In the Capability-based path, the pre-defined *MA_ActionMT* is referenced by *UUID* via *ActionID* in *MA_ActionCommandMT*, and the pre-planned parameters are applied, including any pre-mission *ShotDoctrineParameters* that define how "Shoot Now" should behave by default.

When "Shoot Now" is used as an audible on an ongoing tactic, autonomy advertises an ongoing *ActionActivity* and C2 sends an *MA_ActionCommandMT* whose *ActivityID* matches the advertised *ActivityID* for that ongoing play. In this Activity-based path, updates to the ongoing tactic's shot doctrine and engagement parameters are conveyed via *ChangeShotDoctrine* where *UseRangeProbabilityOfIntercept* is used to explicitly specify a shot constraint. *ShootNLT*, *ShootNET* and *DigitalManeuveringCueAngle* are still expected to be excluded. *ChangeActionConstraints*, *ChangeAttackKineticParameters*, and *Capability* are not used for this audible. If an operator wishes to override a certain field or value within a pre-defined shot doctrine, this can be done in the same way as for other tactics that utilize an *MA_ActionCommandMT* message. For example, if a "Shoot Now" tactic default includes *MissilesPerContact* = 1, but release of 2 missiles is needed within RPI, the *MissilesPerContact* field within *ConstraintOverride* should specify "2". This will command this tactic only with the overriding constraints that are filled out.

For the Activity-based Audible path, omission versus explicit values follow these rules: fields that are omitted from *ChangeShotDoctrine*, *ChangeActionConstraints*, or *ChangeAttackKineticParameters* remain at their current effective value for the ongoing play; Audibles do not implicitly reset unspecified parameters. Fields that are present with concrete values replace the current effective value for that parameter for the duration of the ongoing *ActionActivity*, until further update or tactic completion. Where supported, an explicit "use default" indicator may be used to revert a parameter to the value defined in the referenced *MA_ActionMT* for this play. "Default shot doctrine" refers to the *ShotDoctrine* parameters as defined on the referenced *MA_ActionMT*, not to the state prior to any given Audible. When using the *ActivityID* path, these overrides apply to the ongoing *ActionActivity* for the duration of that play. Subsequent Audibles referencing the same *ActivityID* may further update these parameters; parameters not mentioned in a given Audible retain their current effective values for that ongoing play. The *MA_ActionCommandMT* message needs to contain information about the applicable targets the "Shoot Now" command applies to as well. This information is contained within the *ActionConstraintOverride* field within Capability – specifically the *TargetObject* field, where *EntityIDs* can be added *ByInstance*.

Additional updates to per-target missile allocation for an ongoing play are expressed via the *MissilePerTarget* within *PerTargetShotDoctrine*, whose cardinality matches the *TargetObject* when used with *ChangeActionConstraints*. When *ChangeActionConstraints*.*TargetObject* and *MissilesPerTarget* are provided for an ongoing *ActionActivity*, they are interpreted as an add/modify set for that play: targets newly introduced are added, and existing targets may have their associated missile counts updated. Targets that are part of the ongoing tactic and are not mentioned in ChangeActionConstraints remain unaffected by that Audible; removal of targets continues to be expressed via existing "Drop" behaviors rather than by omission from *TargetObject*.

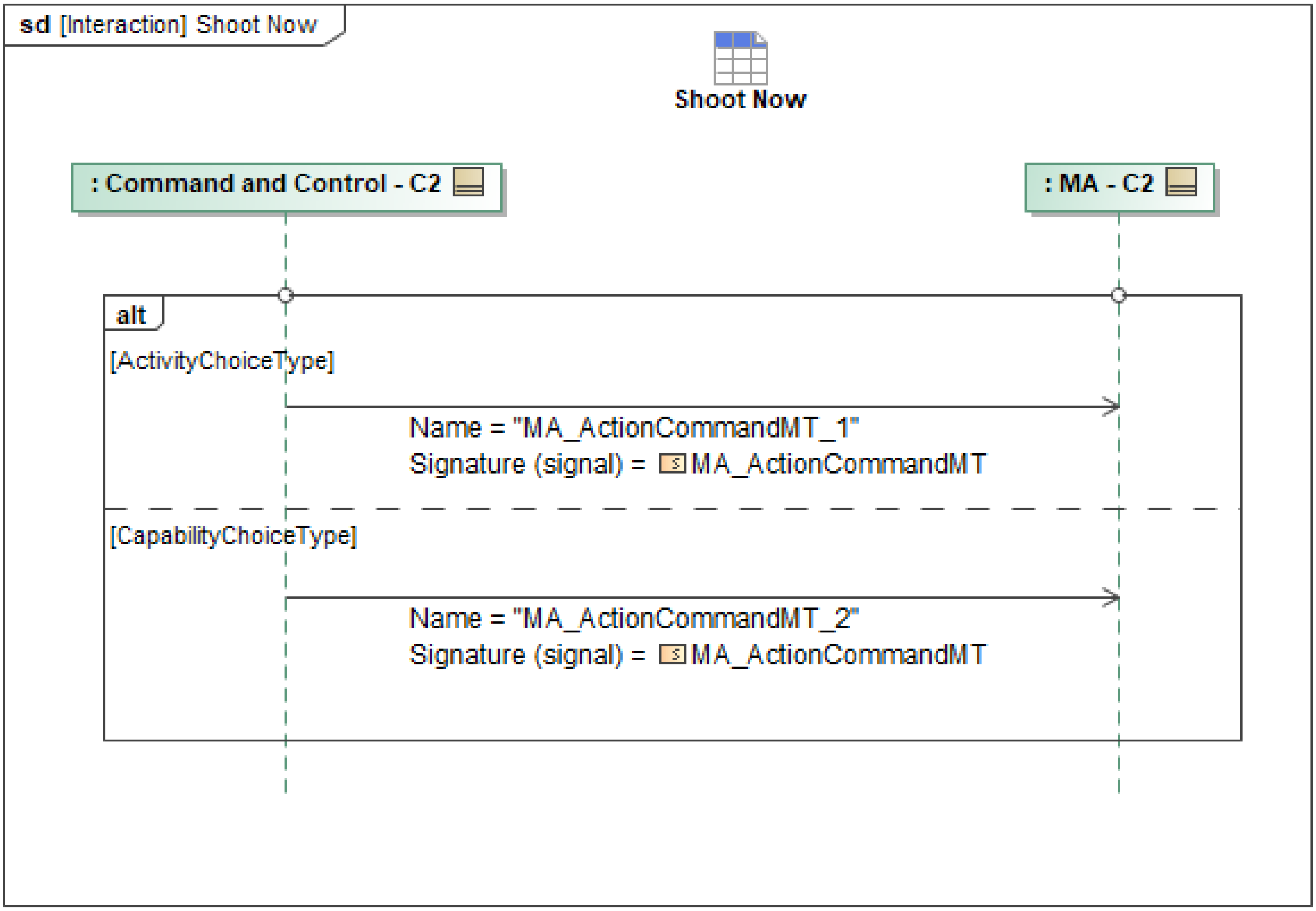


Figure 1-169: Shoot Now

Table 1-168: Shoot Now

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandMT_1: MA_ActionCommandMT	TargetObject: IdentityKindInstanceType MissilesPerContact: MA_ObjectShotCountChoiceType Activity: ActivityCommandBaseType ChangeActionConstraints: MA_ActionConstraintType
MA_ActionCommandMT_2: MA_ActionCommandMT	Capability: MA_ActionCapabilityCommandType ActionConstraintOverride: MA_ActionConstraintType TargetObject: IdentityKindInstanceType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.11.11 Target

The Target tactic enables a C2 operator to command ACP platforms individually or as a package to target 1 to many entities. MA will abide by shot doctrine and intercept maneuver parameters that are configured during mission planning via setting the fields within *MA_ActionMT* or can be modified during mission execution. For more details on updating these parameters see the "Set Tactic Shot Doctrine and Intercept Maneuver Parameters" C2 behavior. Targeting tactics commanded to an entire package result in the specific ACP-target pairings being determined by a lead MA to assign. Upon receipt of a target tactic command, a lead MA determines optimal pairings and makes the target assignment to its peers. This behavior is described in more detail in the Peer Interaction "Assign Target Pairings - Peer". The pre-planned target tactic needs to include an *MA_ActionMT* that specifies an *ActionType* of "*TARGET*". This targeting action will then be broken down into tasks and/or activities and shared amongst the peers to execute the desired intent. For C2 to command a tactic of "Target", the sequence is similar to the commanding of a generic tactic.

As mentioned, the "Target" tactic should be pre-planned in the form of a *MA_ActionMT* message that contains the desired intercept maneuver and shot doctrine parameters as well as specifying the *MA_ActionTypeEnum* to "*TARGET*". To command the tactic, C2 will just send the *MA_ActionCommandMT* message. Specifying the *TargetObjects* is needed and should be done within the *TargetObject* field nested within *ActionConstraintOverride.* One to many *EntityIDs* should be set here using *EntityID* under *ByInstance.* The specific UUIDs of all entities to be targeted is used. If an operator wishes to send a "Target" tactic with one or more of the default shot doctrine parameters overridden for the this tactic, they should be populated within the applicable *EngagementParameters* field*.* This field is nested within the *ConstraintOverride* field in the *MA_ActionCommandMT* message. By sending a target command with *EngagementParameters* within *ConstraintOverride*, that field will override the planned default parameters and MA will fall back to the default values for everything left out of *ConstraintOverride.* The commanded values will only be used for this command, and the defaults will still be active on MA.

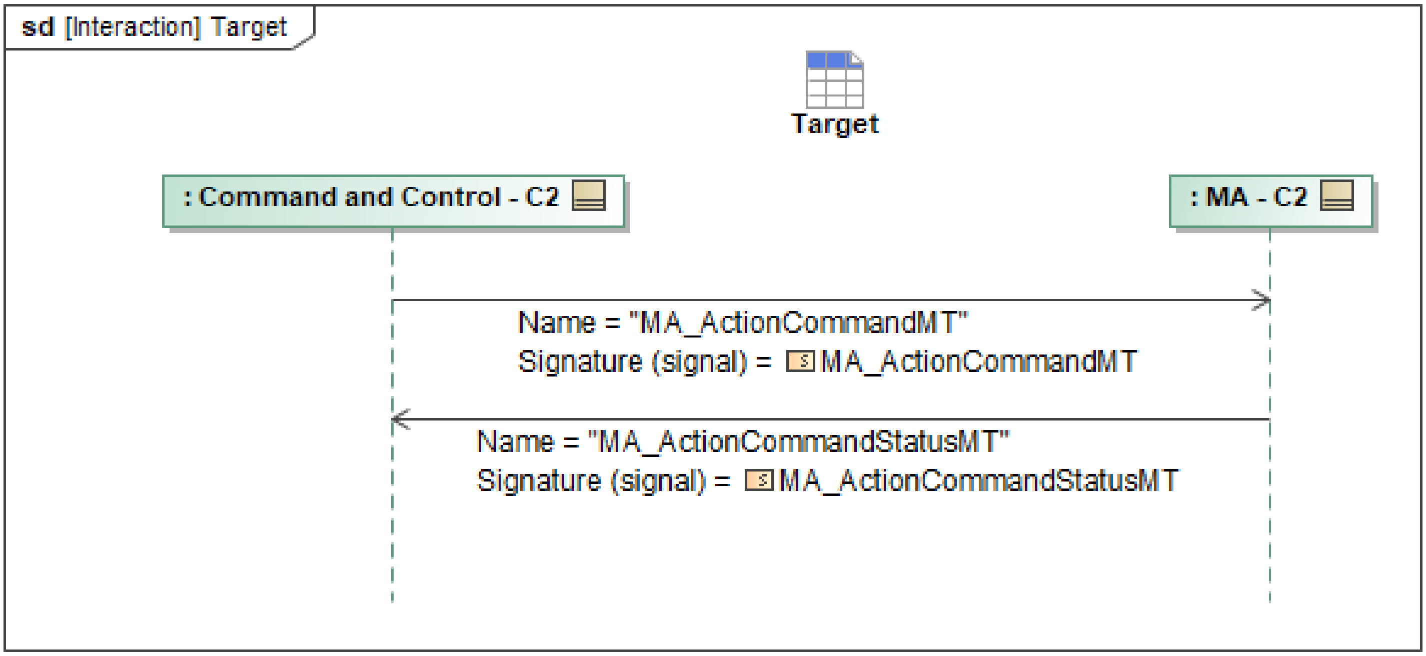


Figure 1-170: Target

Table 1-169: Target

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ActionCommandMT: MA_ActionCommandMT	TargetObject: IdentityKindInstanceType ByInstance: TargetType Capability: MA_ActionCapabilityCommandType ActionConstraintOverride: MA_ActionConstraintType ActionID: ActionID_Type
MA_ActionCommandStatusMT: MA_ActionCommandStatusMT	CommandProcessingState: CommandProcessingStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12 C2 Task Prioritization and Mission Context Behaviors Package

1.2.12.1 C2 Define Global Override

The Command and Control (C2) system can send the global override settings to Mission Autonomy (MA) via *MA_PlatformOverrideCommandMT*. This global settings value will be a speed, altitude, formation, or Identification Friend or Foe (IFF) setting command to be sent to an individual Autonomy Capable Platform (ACP) or to a team lead, and this interaction allows an overriding value for one of those parameters to be set or cleared. If setting a new global override, then the *CommandState* should be set to *NEW*. The specific request (speed, altitude, formation, or IFF override) should be set via the *PlatformOverrideCommand* field. If sending a speed override, the *SpeedOverrideCommand* field should be populated. Likewise, if setting an altitude override, the *AltitudeOverrideCommand* should be used. If setting a global formation command, then the *FormationOverrideCommand* field should be used. Finally, if setting IFF settings, the *IFF_OverrideCommand* field should be used. To cancel an existing global override, the *CommandState* should be set to *CANCEL* and the same *PlatformOverrideCommand* type as was used when sending the override should be populated. The exact value is a design detail left to implementation; the only expectation is that the corresponding field is populated as a mechanism for identifying which override has been cleared. The *Applicability* field indicates the unique identifier of the System or Package intended to have effects applied. The *ExpirationTime* field may be optionally populated when sending a new override command to define the time at which the override will expire; if unpopulated, the command is indefinite until canceled. Likewise, the *ApplicableZoneID* field may be optionally populated to define one or more zones within which the override is applicable; if unpopulated, the command is valid throughout the operating area.

MA sends C2 the *MA_PlatformOverrideCommandStatusMT* message, accepting or rejecting the command. If the command is rejected, the *MA_PlatformOverrideCommandStatusMT* message will specify the *CommandProcessingStateReason* field to provide further details. If the command is accepted, MA will reply with *MA_PlatformOverrideCommandStatusMT* where *CommandProcessingState* is set to *ACCEPTED* and will report the active platform override(s) by setting the *PlatformOverrideStatus* field with the value of each active override setting within *MA_PlatformOverrideStatusMT*.

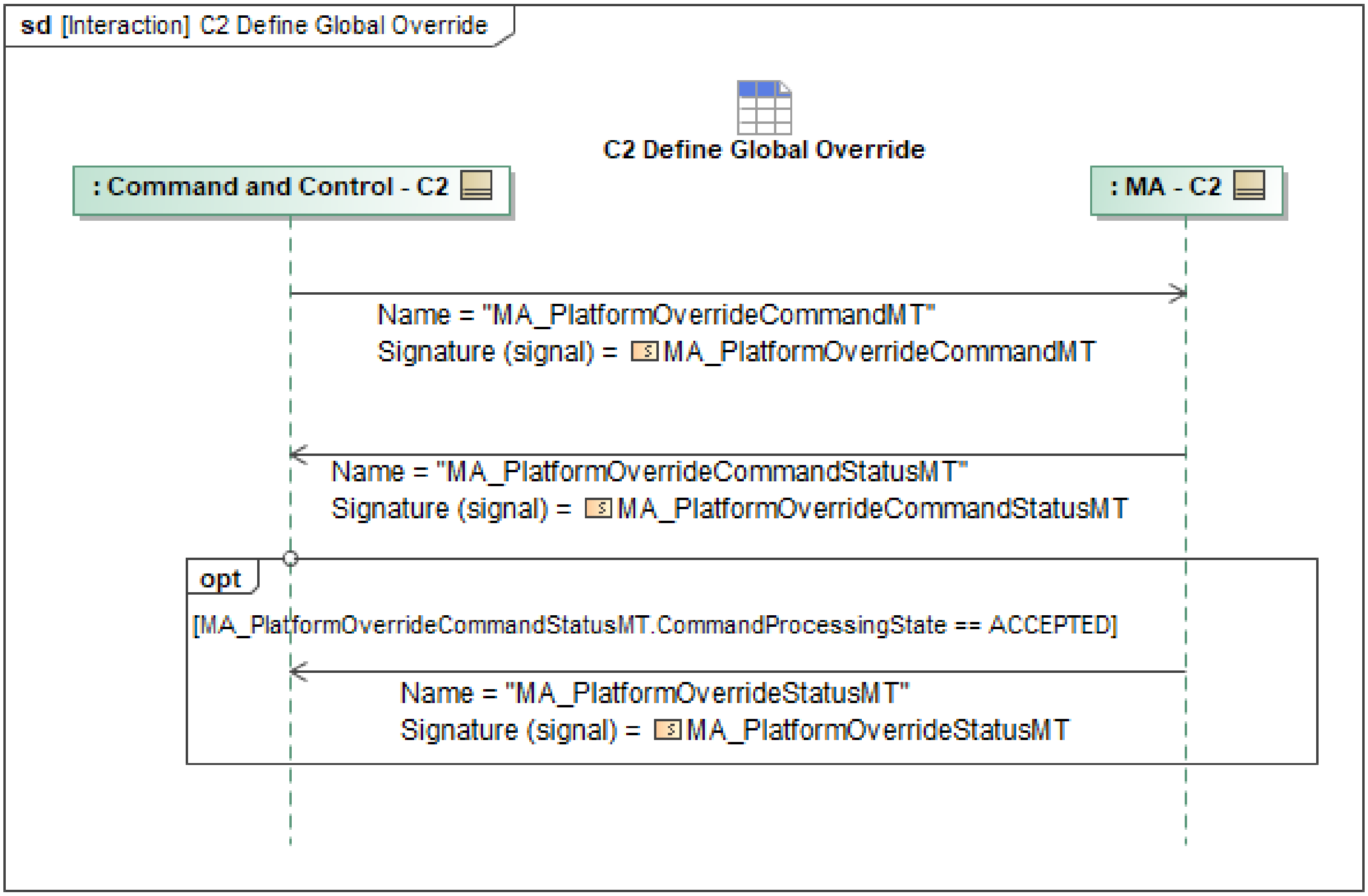


Figure 1-171: C2 Define Global Override

Table 1-170: C2 Define Global Override

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PlatformOverrideCommandMT: MA_PlatformOverrideCommandMT	None
MA_PlatformOverrideCommandStatusMT: MA_PlatformOverrideCommandStatusMT	None
MA_PlatformOverrideStatusMT: MA_PlatformOverrideStatusMT	PlatformOverrideStatus: MA_PlatformDefaultAndOverrideDataType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.2 C2 Acceptable Levels of Risk Behaviors

1.2.12.2.1 Provide Risk Assessment Results

Mission Use Case: ALR

The optional Provide Risk Assessment Results sequence diagram specifies the interaction where an ACP provides C2 with risk assessment results if an updated threat triggers an assessment of the current Route Plan. Upon determining that the current Route Plan exceeds the specified Acceptable

Level of Risk (ALR) setting, the ACP may attempt to generate a new Route Plan through the threat environment which complies with the ALR setting. (If the risk assessment for the current route plan does not exceed the ALR setting, the risk assessment results will not be sent to C2, and this interaction would not apply). After a new Route Plan is generated, the ACP sends *MA_AssessmentMT* to C2 to inform of the risk assessment results, populating the *Data* of the *RouteThreatAssessment* field within the *Assessment* field of this message. The message will have a populated *AnalysisRouteID* to identify the Route Plan for which the assessment applies. If a Route Plan which complies with the ALR setting is not generated or able to be achieved, the *MA_AssessmentMT* message will convey this to C2 by populating the *Achievability* field within *AchievabilityAssessment* as *NOT_ACHIEVABLE*.

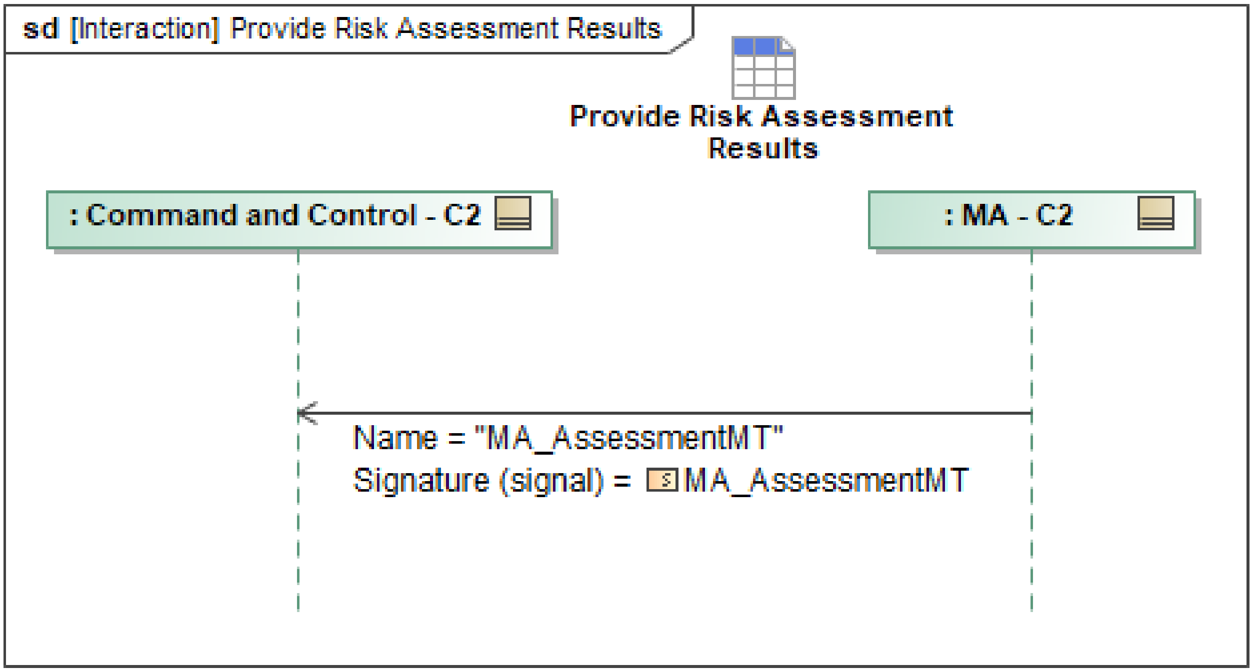


Figure 1-172: Provide Risk Assessment Results

Table 1-171: Provide Risk Assessment Results

Message Name (Name:Type)	Required Fields (Name:Type)
MA_AssessmentMT: MA_AssessmentMT	AnalysisRouteID: AnalysisRouteID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.2.2 Request Route Threat Assessment

Mission Use Case: ALR

The optional Request Route Threat Assessment sequence diagram specifies the interaction where C2 commands an ACP to perform a risk assessment for a Route Plan. This interaction assumes that the Route Plan to be assessed has been previously uploaded to the ACP or that the “Update Route Plan” interaction has occurred where C2 uploads a new or updated Route Plan. C2 will request a risk assessment for a Route Plan, sending *MA_AssessmentRequestMT* with a specified *AnalysisRouteID* and the *Parameters* of the *RouteThreatAssessment* field within the *Category* field populated. MA may optionally report status on the request to C2 using *MA_AssessmentRequestStatusMT_1* to convey the processing state and/or estimated completion information. MA will deliver the risk assessment results with the *MA_AssessmentRequestStatusMT_2* message, populating the *Data* of the *RouteThreatAssessment* field within the *Assessment* field.

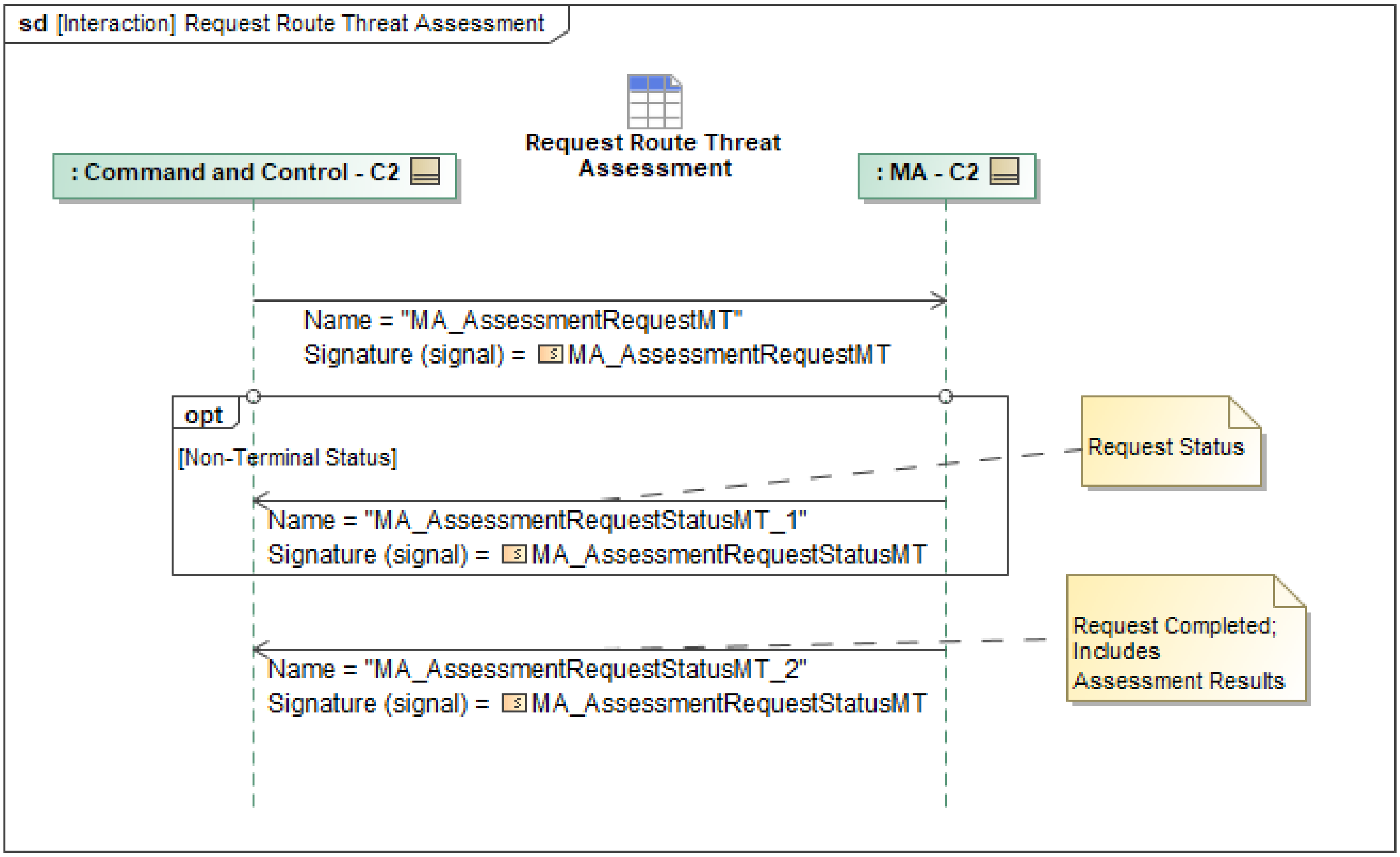


Figure 1-173: Request Route Threat Assessment

Table 1-172: Request Route Threat Assessment

Message Name (Name:Type)	Required Fields (Name:Type)
MA_AssessmentRequestMT: MA_AssessmentRequestMT	AnalysisRouteID: AnalysisRouteID_Type
MA_AssessmentRequestStatusMT_1: MA_AssessmentRequestStatusMT	None
MA_AssessmentRequestStatusMT_2: MA_AssessmentRequestStatusMT	Assessment: AssessmentType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.2.3 Update Acceptable Level of Risk

Mission Use Case: ALR

Overview

The Acceptable Level of Risk (ALR) command message (*MA_AcceptableLevelofRiskCommandMT*) allows an operator to set a specific level of risk the MA is allowed to incur during the planning or execution of a Mission Plan. The *AcceptableRiskSetting* field allows the value to include *HIGH, MEDIUM, LOW*, or *NONE* at the package or vehicle level. This ALR setting applies to all tasks and actions for the vehicle or package. The *SurvivabilityRiskLevelID* field allows an operator to select a specific Survivability Risk Definition for MA, which has previously been defined in the Mission Plan or by C2 using the *SurvivabilityRiskLevelMT* message. The difference between ALR and Survivability Risk Level is that ALR defines the maximum allowed risk the planner is allowed to plan for and the *SurvivabilityRiskLevelMT* describes what *HIGH, MEDIUM,* and *LOW* mean with respect to the *MaximumContinuousExposure* for a given Vulnerability Type. Where *MaximumContinuousExposure* refers to the Composite and Cumulative Exposure Time allowed and the Vulnerability Type includes Acquisition (detection by acquisition radars), Track (exposure to track radars), Intercept (probably of intercept (impact) by a threat), Launch (probably of launch by a threat), and Kill (overall survivability prediction). *SurvivabilityRiskLevelMT* can be set in the mission plan or updated by C2 to define the various levels of risk. If the MA does not have an active ALR, this is the same as having an ALR of *NONE*. The MA (leader or individual ACP) will provide status on the commanded *MA_AcceptableLevelofRiskCommandMT* messages using the *MA_AcceptableLevelofRiskCommandStatusMT* message and will report its currently activated ALR setting and Survivability Risk Level to C2 using *MA_AcceptableLevelOfRiskStatusMT.* The C2 will send messages to a package lead or non-package ACPs. Package leads will distribute the messages to the rest of the follower ACPs.

Mission Execution

The ALR and *SurvivabilityRiskLevelMT* are used to inform tactical decisions and planning during mission execution. Any planner that supports the use of ALR may choose to alter tactics (i.e. produce a different plan) based on the planner's best assessment of the risk involved with executing the plan in the operational environment. Risk assessment is a complicated modeling process and is not currently governed by the A-GRA. During mission execution, if a new *MA_AcceptableLevelofRiskCommandMT* is received/updated during execution, this should trigger a replan if necessary to avoid the current activity exceeding the new ALR setting based on the *SurvivabilityRiskLevelMT* definitions and the operational environment

Sequence Description

The Update Acceptable Level of Risk sequence diagram specifies the interaction where C2 commands updates to the ALR setting for a non-package ACP, an entire package, or specific ACP(s) within a package. First, C2 sends *MA_AcceptableLevelOfRiskCommandMT* to MA, where the *AppliesToID* field can have multiple entries of *PackageID* or *SystemID*. This allows C2 to send a Lead MA the Package ID (or multiple sub-Package IDs) or a set of System IDs. The *MA_AcceptableLevelOfRiskCommandMT* message shall define an *AcceptableRiskSetting*, which allows for the ALR to be set as *HIGH, MEDIUM, LOW*, or *NONE* at the package or vehicle level. It also defines a *SurvivabilityRiskLevelID* which allows for a *RiskSetting* to be populated to describe what *HIGH, MEDIUM*, and *LOW* mean with respect to the *MaximumContinuousExposure* for a given Vulnerability Type. Any new *MA_AcceptableLevelOfRiskCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA_AcceptableLevelOfRiskCommandMT* is sent with the *CommandState* set to *CANCEL,* this will cancel the active command

In response to receiving the command, MA shall send

MA_AcceptableLevelOfRiskCommandStatusMT_1 to C2, with *CommandProcessingState* set to *RECEIVED* to indicate receipt of the command. For this status message, the *AppliesToID* field shall be set to the Package ID if MA is the leader and has received a status message from all members of the Package ID. Otherwise, it shall be set to the System ID(s) and/or sub-Package ID(s) for which the status message applies.

If the MA is a non-package ACP, it shall send *MA_AcceptableLevelOfRiskCommandStatusMT_2* to C2 with the *CommandProcessingState* set to *ACCEPTED* to indicate that its ALR setting has been updated. An ACP will then send *MA_AcceptableLevelOfRiskStatusMT_1* to C2 to provide status on its currently activated ALR setting and Survivability Risk Level, using the *AcceptableRiskSetting* and *SurvivabilityRiskLevelID* fields, and populating its System ID in the *AppliesToID* field. If MA is the lead ACP, then MA shall send its follower ACPs *MA_AcceptableLevelOfRiskCommandMT* (See "Update Peer Acceptable Level of Risk"). After receiving *ACCEPTED* status from all applicable follower ACPs, the MA lead sends C2 *MA_AcceptableLevelOfRiskCommandStatusMT_3* with the *CommandProcessingState* set to *ACCEPTED,* reflecting the *AppliesToID* as needed. The Lead ACP will then send *MA_AcceptableLevelOfRiskStatusMT_2* to C2 to provide status on the package's currently activated ALR setting and Survivability Risk Level, using the *AcceptableRiskSetting* and *SurvivabilityRiskLevelID* fields, and populating the Package ID in the *AppliesToID* field.

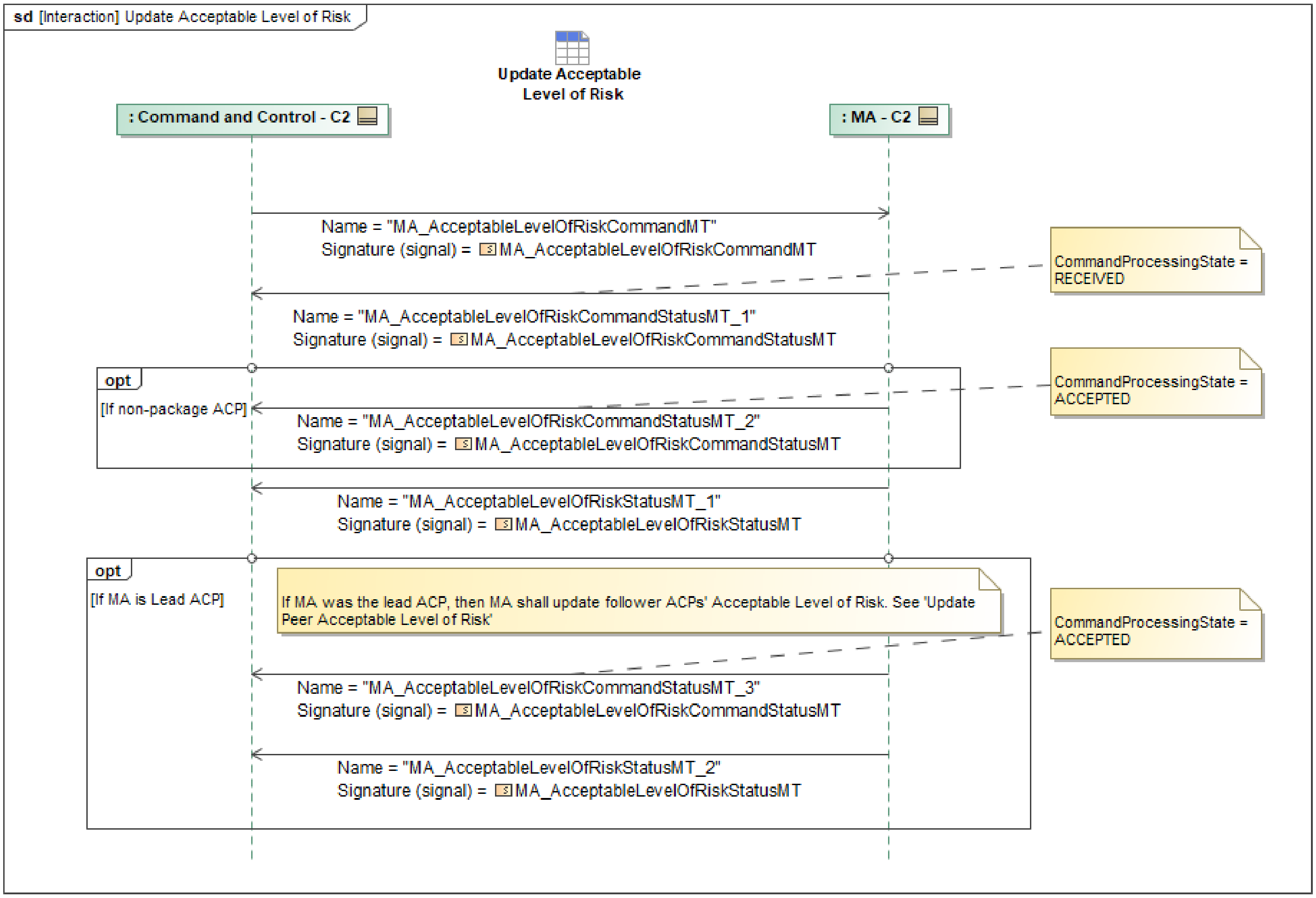


Figure 1-174: Update Acceptable Level of Risk

Table 1-173: Update Acceptable Level of Risk

Message Name (Name:Type)	Required Fields (Name:Type)
MA_AcceptableLevelOfRiskCommandMT: MA_AcceptableLevelOfRiskCommandMT	None
MA_AcceptableLevelOfRiskCommandStatusMT_1: MA_AcceptableLevelOfRiskCommandStatusMT	None
MA_AcceptableLevelOfRiskCommandStatusMT_2: MA_AcceptableLevelOfRiskCommandStatusMT	None
MA_AcceptableLevelOfRiskCommandStatusMT_3: MA_AcceptableLevelOfRiskCommandStatusMT	None
MA_AcceptableLevelOfRiskStatusMT_1: MA_AcceptableLevelOfRiskStatusMT	None
MA_AcceptableLevelOfRiskStatusMT_2: MA_AcceptableLevelOfRiskStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy

Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.2.4 Update Package Survivability Risk Definition

Mission Use Case: ALR

The optional Update Package Survivability Risk Definition sequence diagram specifies the interaction where C2 commands the Lead ACP to update the package members' Survivability Risk Definition data. First, C2 sends the Lead ACP the new Survivability Risk Definition data, and the Lead ACP confirms it has received the updated data, as shown in the referenced "Update Survivability Risk Definition" sequence diagram. Then, the Lead ACP updates the applicable follower ACPs with the data, which may be the entire package or a subset of followers. (See "Update Peer Survivability Risk Definition"). Once the Lead ACP has received a response from all followers, the Lead ACP will send *MA_SystemNotificationMT* to C2 to inform that the entire package has updated the Survivability Risk Definition. *MA_SystemNotificationMT* contains *SystemSubjectID*, *SystemPerspective*, and *AssociatedMessage* fields. The *SystemSubjectID* shall be set to the Package ID, the *SystemPerspective* enumeration shall be set to *RESPONSE_SUBJECT,* and the *AssociatedMessage* shall define the received message's *MessageTypeEnum* as *MA_SURVIVABILITY_RISK_LEVEL* with *AssociatedID* set to the *SurvivabilityRiskLevelID*.

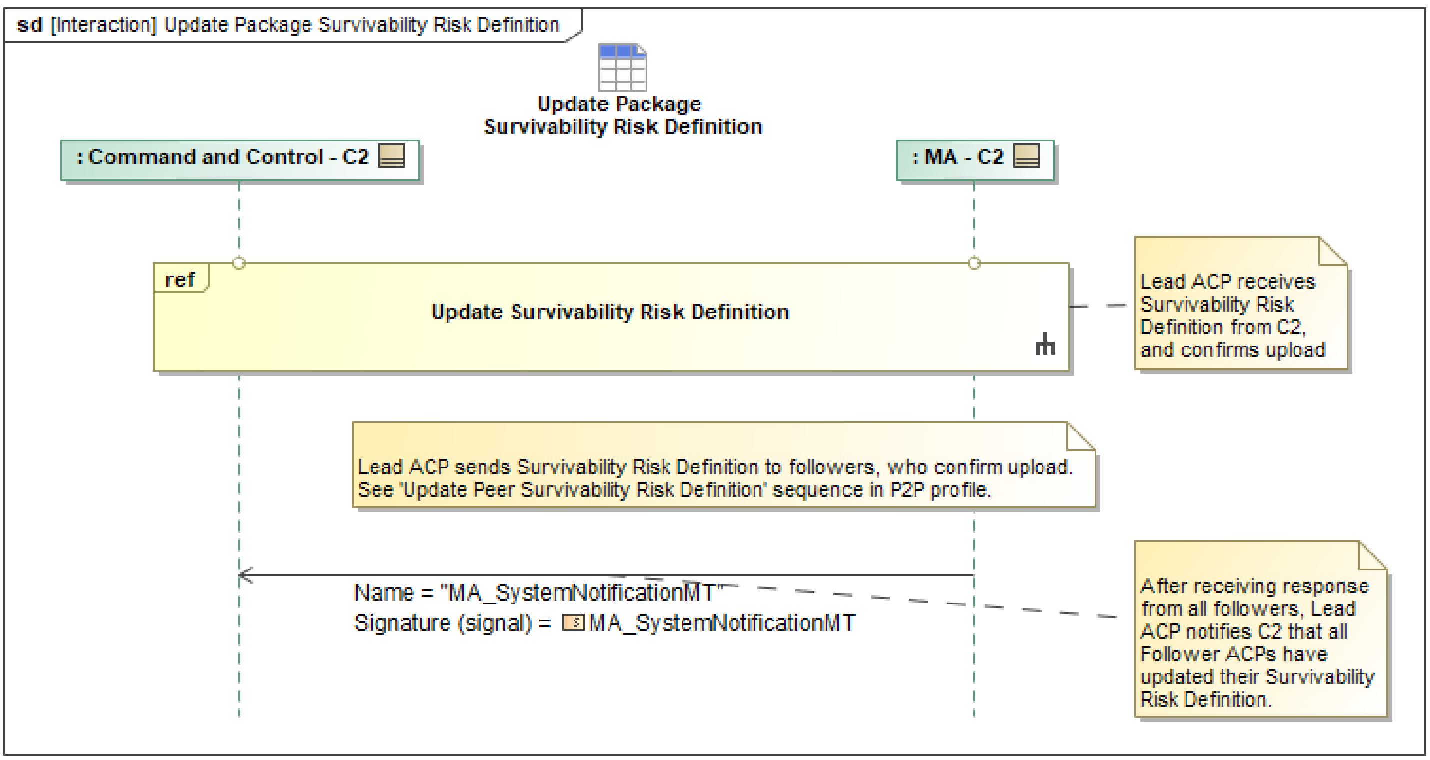


Figure 1-175: Update Package Survivability Risk Definition

Table 1-174: Update Package Survivability Risk Definition

Message Name (Name:Type)	Required Fields (Name:Type)
MA_SystemNotificationMT: MA_SystemNotificationMT	AssociatedMessage: AssociatedMessageType AssociatedMessage: AssociatedMessageType

Note: The fields above are required for this use case in addition to what is specified in Autonomy

Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.2.5 Update Survivability Risk Definition

Mission Use Case: ALR

The optional Update Survivability Risk Definition sequence diagram specifies the interaction where C2 sends an updated Survivability Risk Definition to a Lead ACP or non-package ACP. C2 sends *SurvivabilityRiskLevelMT* with the *ObjectState* field set to *UPDATED*. The message has a *SurvivabilityRiskLevelID* field to uniquely identify the message and a *RiskSetting* field, which describes what the different risk settings (*HIGH, MEDIUM, LOW, NONE*) mean with respect to the *MaximumContinuousExposure* for a given Vulnerability Type. The *AppliesToSystemID* can be left blank, since the *MA_AcceptableLevelOfRiskCommandMT* will need to be used to activate a given Survivability Risk Definition and will specify the applicable System or Package IDs.

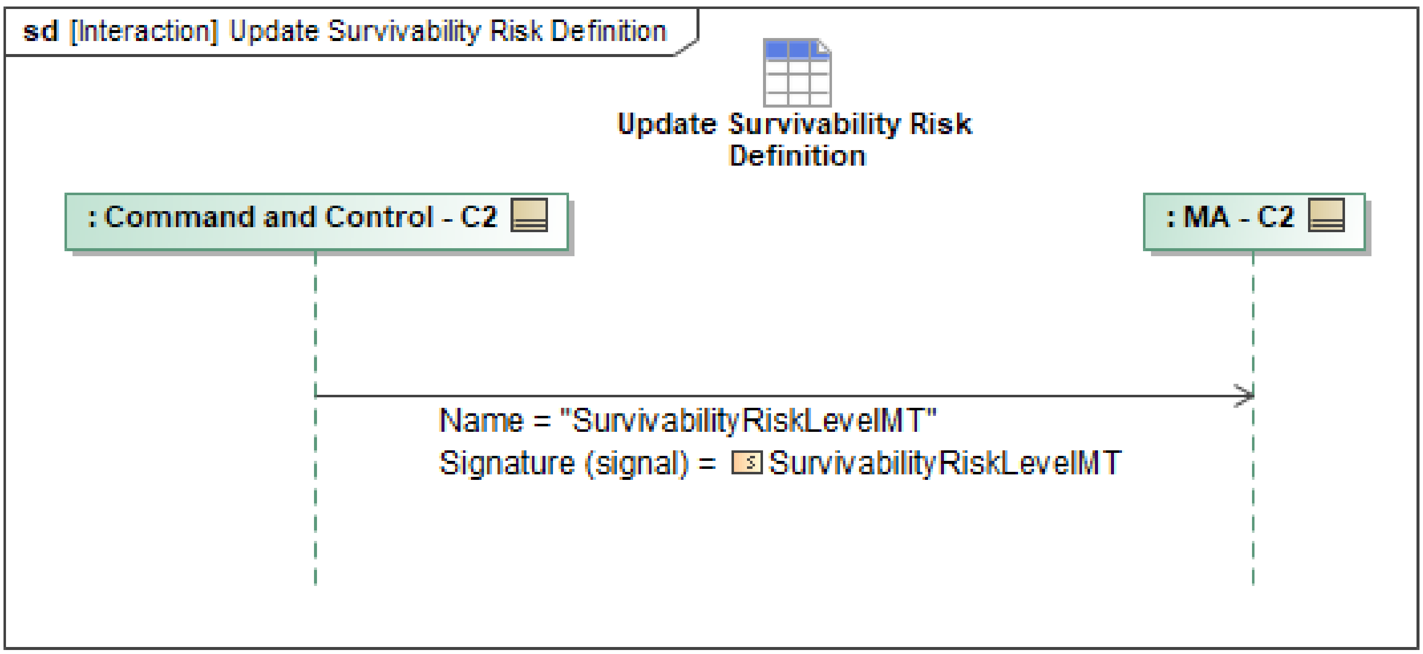


Figure 1-176: Update Survivability Risk Definition

Table 1-175: Update Survivability Risk Definition

Message Name (Name:Type)	Required Fields (Name:Type)
SurvivabilityRiskLevelMT: SurvivabilityRiskLevelMT	ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.3 C2 Constraint Priority List Behaviors

1.2.12.3.1 Activate Constraint Priority List

Overview

The Constraint Priority List (CPL), *MA_ConstraintPriorityListMT,* allows an operator to set default values for the *Priority* field in *OpZoneMT, OpVolumeMT,* and *OpLineMT* (further referred to as geozones) messages based on type. If the *Priority* field in those pre-planned messages are empty and the type is in the CPL, the *Priority* value from the CPL shall be used. A *Priority* value of zero (0) is the highest priority value. If the geozone message *Priority* field is empty and the geozone type is not in the CPL, the *Priority* field shall be set to the *DefaultPriority* from the CPL. If the *DefaultPriority* field is empty, the value would be blank or the highest priority ("above" 0). The *DefaultPriority* and *ListItem* fields are optional and if they are left blank, this indicates that only directly set *Priority* values in the geozones will be utilized and there is no default priority value. The C2 will send messages related to the CPL to a package lead or non-package ACPs. Package leads will distribute the messages to the rest of the follower ACPs. The *MA_ConstraintPriorityListCommandMT* message will identify which *MA_ConstraintPriorityListActivationDataMT* should be active. The *MA_ConstraintPriorityListActivationDataMT* defines a default CPL and CPLs associated with *OpZones* or *OpVolumes*. Only a single CPL shall be active at any given time. When the ACP is in an *OpZones* or *OpVolumes* associated with engagement criteria, it should be activated; otherwise, the default CPL would be activated.

Mission Execution

Prior to planning or execution, the ActivationData, CPLs, and associated geozones will be available for use. During planning, if there is an active non-empty *MA_ConstraintPriorityListMT*, the planner will assign the *Priority* values to the geozones as the types that are listed in the CPL, if they were not previously set. The planner will use the *Priority* values in the geozones to make decisions on routes and other items as needed. During mission execution, if there is a change to the active *MA_ConstraintPriorityListMT* message, due to a new geozone being entered or C2 sending a new/updated *MA_ConstraintPriorityListCommandMT* message, MA should trigger a replan. A replan is not necessary if it is determined that the updated *MA_ConstraintPriorityListMT* will not affect the current plan by no active geozones changing priority value. A *MA_ConstraintPriorityListActivationStatusMT* message will be sent upon the transition to a different *MA_ConstraintPriorityListMT.*

Sequence Description

The Activate Constraint Priority List sequence diagram below specifies the interaction of C2 commanding MA to activate one of its stored CPL activation schema. First C2 sends a *MA_ConstraintPriorityListCommandMT* message to MA. The *AppliesToID* field can have multiple entries of *PackageID* or *SystemID*. This allows a C2 to send a Lead MA the Package ID (or multiple sub Package IDs) or a set of System IDs. The *MA_ConstraintPriorityListCommandMT* message shall specify the *ConstraintPriorityListActivationDataID* field containing the ID of the Constraint Priority List Activation Data message (or schema) that C2 wishes to be active. Any new *MA_ConstraintPriorityListCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA_ConstraintPriorityListCommandMT* is sent with the *CommandState* set to *CANCEL,* this will cancel the active command and CPL activation schema.

In response to receiving the command, MA shall send *MA_ConstraintPriorityListCommandStatusMT_1* to C2. The *CommandProcessingState* will be set to *RECEIVED* to indicate receipt of the command. For this status message and all other status messages (*MA_ConstraintPriorityListCommandStatus* & *MA_ConstraintPriorityListActivationStatus*), the *AppliesToID* field shall be set to the Package ID if MA is the leader and has received a status message from all members of the Package ID, otherwise it shall be set to System ID(s) and/or sub Package ID(s) that the status message applies to.

The MA shall then use the *ConstraintPriorityListActivationDataID* to query its onboard MP to retrieve the *MA_ConstraintPriorityListActivationDataMT* (see "Query Constraint Priority List Activation Data"). Then it will use the *MA_ConstraintPriorityListActivationDataMT* to query for all the CPL messages (see "Query Constraint Priority List"). If the MA is a non-package ACP, it shall send *MA_ConstraintPriorityListCommandStatusMT_2* to C2 with the *CommandProcessingState* set to *ACCEPTED*.

The MA shall use the ACP Location and the *MA_ConstraintPriorityListActivationDataMT* to determine which CPL ID should be activated (default or geozone based). Once doing so, MA will send *MA_ConstraintPriorityListActivationStatusMT_1* to C2 containing the *ConstraintPriorityListActivationDataID* and the *ActiveConstraintPriorityListID*. Optionally, if MA is the lead ACP, then MA shall message its follower ACPs with the *MA_ConstraintPriorityListCommandMT* (See "Activate Peer Constraint Priority List"). After receiving the respective responses from the follower ACPs, MA (the lead ACP) would send C2 the *MA_ConstraintPriorityListCommandStatusMT_3* with the *CommandProcessingState* set to *ACCEPTED* and the *MA_ConstraintPriorityListActivationStatusMT_2* both reflecting the *AppliesToID* as needed.

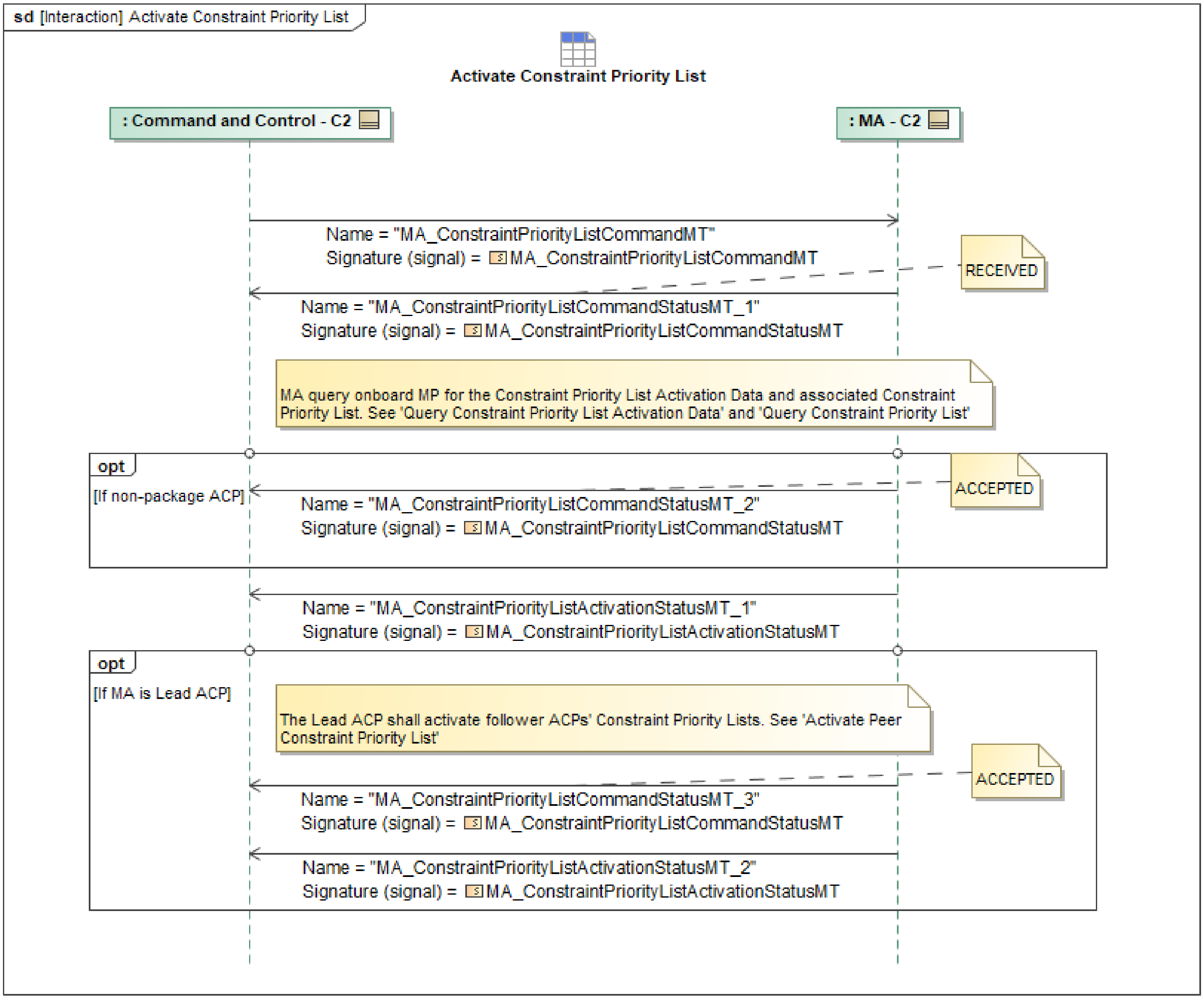


Figure 1-177: Activate Constraint Priority List

Table 1-176: Activate Constraint Priority List

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ConstraintPriorityListActivationStatusMT_1: MA_ConstraintPriorityListActivationStatusMT	None
MA_ConstraintPriorityListActivationStatusMT_2: MA_ConstraintPriorityListActivationStatusMT	None
MA_ConstraintPriorityListCommandMT: MA_ConstraintPriorityListCommandMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ConstraintPriorityListCommandStatusMT_1: MA_ConstraintPriorityListCommandStatusMT	None
MA_ConstraintPriorityListCommandStatusMT_2: MA_ConstraintPriorityListCommandStatusMT	None
MA_ConstraintPriorityListCommandStatusMT_3: MA_ConstraintPriorityListCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.3.2 Update Constraint Priority List Activation Status

The Update Constraint Priority List Activation Status sequence diagram specifies the MA notifying C2 of a new active CPL after entering a new geozone. Prior to this interaction if this is the Lead ACP, all follower ACPs shall message the lead ACP their activation statuses. If this is the Lead ACP, it should wait for all package members to respond. MA then sends C2 *MA_ConstraintPriorityListActivationStatusMT* containing the *ConstraintPriorityListActivationDataID* and the *ActiveConstraintPriorityListID*. The *AppliesToID* shall be set to the Package ID if it is the lead and if all follower status have been received, or it shall be set to System ID(s) and/or sub Package ID(s) that the status message applies to.

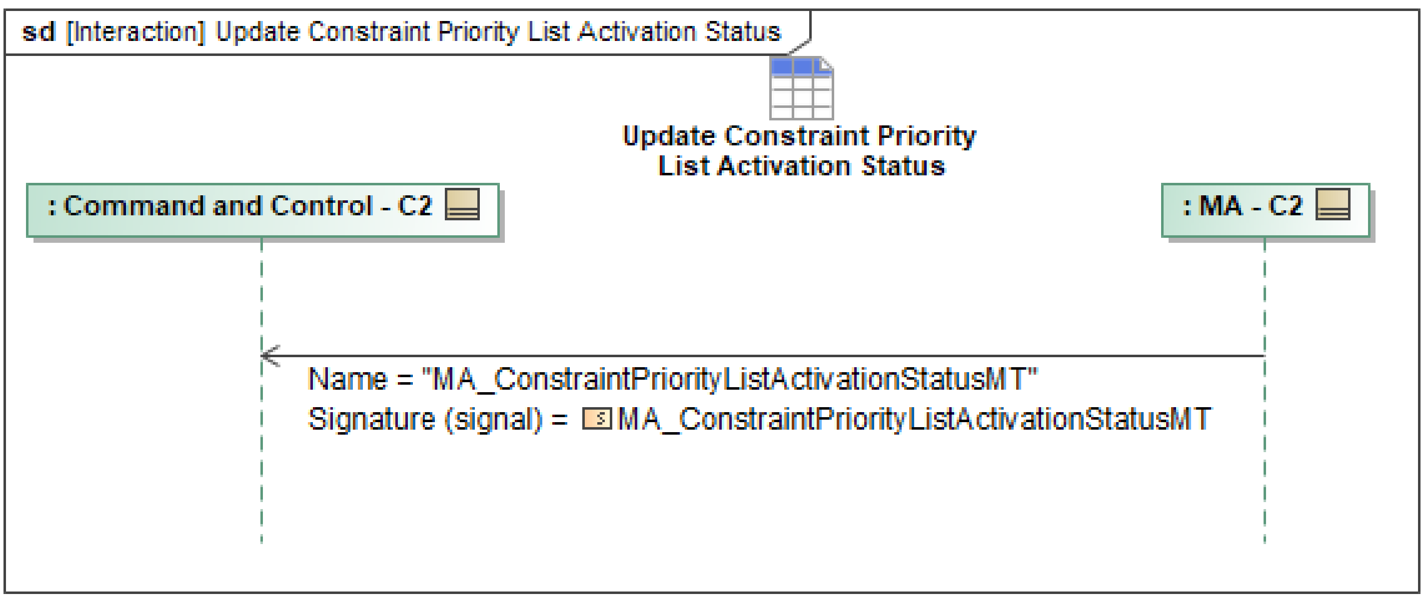


Figure 1-178: Update Constraint Priority List Activation Status

Table 1-177: Update Constraint Priority List Activation Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ConstraintPriorityListActivationStatusMT: MA_ConstraintPriorityListActivationStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.4 C2 Execution Priority List Behaviors

1.2.12.4.1 Activate Execution Priority List

Overview

The Execution Priority List (EPL), *MA_ExecutionPriorityListMT*, allows an operator to set default values for the *Rank* field in the *TaskMT* and *ActionMT* (further referred to as task) messages based on the type. If the *Rank* field in those messages is empty and the type is in the EPL, the *Rank* from the EPL will be used. A *Rank→ Priority* value of zero (0) is the highest priority value, and an empty field is the lowest. If the planned task message *Rank* field is empty and the task type is not in the EPL, the *Rank* field shall be left empty and have the lowest priority. The C2 will send messages to a package lead or non-package ACPs. Package leads will distribute the messages to the rest of the follower ACPs. The *MA_ExecutionPriorityListCommandMT* message will identify which *MA_ExecutionPriorityListActivationDataMT* should be active. The *MA_ExecutionPriorityListActivationDataMT* defines a default EPL and EPLs associated with *OpZones* or *OpVolumes*. Only a single EPL shall be active at any given time. When the ACP is in an *OpZones* or *OpVolumes* associated with an EPL, it should be activated otherwise the default EPL would be activated.

Mission Execution

Prior to planning or execution, the ActivationData, EPLs, and associated geozones will be available for use. During planning, if there is an active non-empty *MA_ExecutionPriorityListMT*, the planner will assign the *Rank* values to the tasks as the types that are listed in the EPL, if they were not previously set. Child tasks shall inherit the *Rank* value associated with a parent task. The *Rank* value does not explicitly specify the execution order of the tasks. During dynamic re-planning, if a team or an ACP is assigned a collection of tasks to execute, the resulting dynamic planning process should prioritize those tasks and actions based on their priority values. During mission execution, if there is a change to the active *MA_ExecutionPriorityListMT* message, due to a new geozone being entered or C2 sending a new/updated *MA_ExecutionPriorityListCommandMT* message, MA should trigger a replan. A replan is not necessary if it is determined that the updated *MA_ExecutionPriorityListMT* will not affect the active plan by no active tasks changing rank value or the changes do not affect the task sequence order. A *MA_ExecutionPriorityListCommandStatusMT* message will be sent upon the transition to a different *MA_ExecutionPriorityListMT.*

Sequence Description

The Activate Execution Priority List (EPL) sequence diagram below specifies the interaction of C2 commanding MA to activate one of its stored EPL activation schema. First C2 sends a *MA_ExecutionPriorityListCommandMT* message to MA. The *AppliesToID* field can have multiple entries of *PackageID* or *SystemID*. This allows a C2 to send a Lead MA the Package ID (or multiple sub Package IDs) or a set of System IDs. The *MA_ExecutionPriorityListCommandMT* message shall specify the *ExecutionPriorityListActivationDataID* field containing the ID of the Execution Priority List Activation Data message (or schema) that C2 wishes to be active. Any new *MA_ExecutionPriorityListCommandMT* will override any previously sent message, even if the *CommandID* has changed. When a *MA_ExecutionPriorityListCommandMT* is sent with the *CommandState* set to *CANCEL,* this will cancel the active command and EPL activation schema.

In response to receiving the command, MA shall send *MA_ExecutionPriorityListCommandStatusMT_1* to C2. The *CommandProcessingState* will be set to *RECEIVED* to indicate receipt of the command. For this status message and all other status messages. (*MA_ExecutionPriorityListCommandStatus* & *MA_ExecutionPriorityListActivationStatus*), the *AppliesToID* field shall be set to the Package ID if MA is the leader and has received a status message from all members of the Package ID, otherwise it shall be set to System ID(s) and/or sub Package ID(s) that the status message applies to

The MA shall then use the *ExecutionPriorityListActivationDataID* to query its onboard MP to retrieve the *MA_ExecutionPriorityListActivationDataMT* (see "Query Execution Priority List Activation Data"). Then it will use the *MA_ExecutionPriorityListActivationDataMT* to query for all the EPL messages (see "Query Execution Priority List"). If the MA is a non-package ACP, it shall send *MA_ExecutionPriorityListCommandStatusMT_2* to C2 with the *CommandProcessingState* set to *ACCEPTED.*

The MA shall use the ACP Location and the *MA_ExecutionPriorityListActivationDataMT* to determine which EPL ID should be activated (default or geozone based). Once doing so, MA will send *MA_ExecutionPriorityListActivationStatusMT_1* to C2 containing the *ExecutionPriorityListActivationDataID* and the *ActiveExecutionPriorityListID*. Optionally, if MA is the lead ACP, then MA shall message its follower ACPs with the *MA_ExecutionPriorityListCommandMT* (See "Activate Peer Execution Priority List"). After receiving the respective responses from the follower ACPs, MA (the lead ACP) would send C2 the *MA_ExecutionPriorityListCommandStatusMT_3* with the *CommandProcessingState* set to *ACCEPTED* and the *MA_ExecutionPriorityListActivationStatusMT_2* both reflecting the *AppliesToID* as needed.

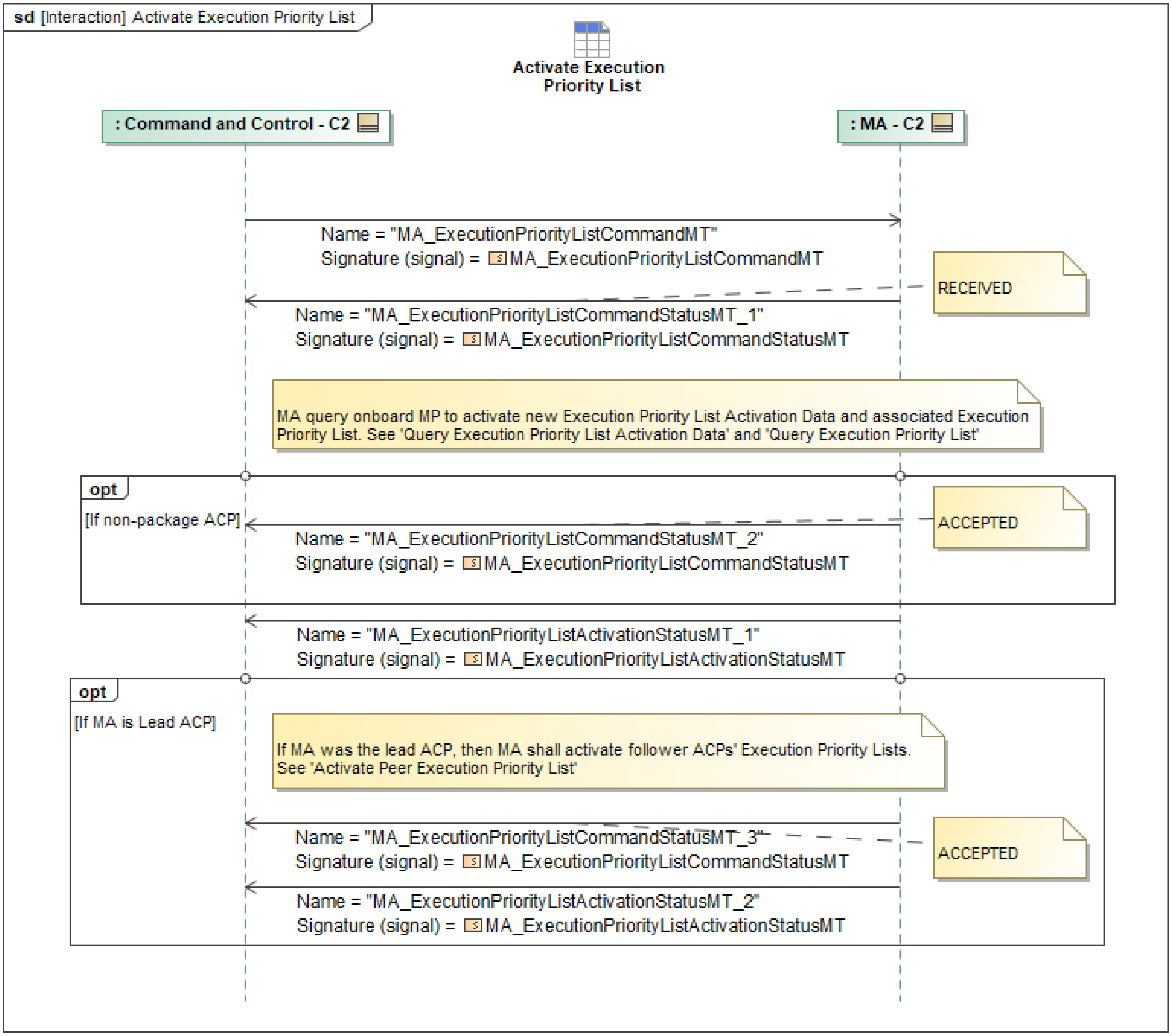


Figure 1-179: Activate Execution Priority List

Table 1-178: Activate Execution Priority List

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ExecutionPriorityListActivationStatusMT_1 : MA_ExecutionPriorityListActivationStatusMT	None
MA_ExecutionPriorityListActivationStatusMT_2 : MA_ExecutionPriorityListActivationStatusMT	None
MA_ExecutionPriorityListCommandMT: MA_ExecutionPriorityListCommandMT	None
MA_ExecutionPriorityListCommandStatusMT_1 : MA_ExecutionPriorityListCommandStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ExecutionPriorityListCommandStatusMT_2 : MA_ExecutionPriorityListCommandStatusMT	None
MA_ExecutionPriorityListCommandStatusMT_3 : MA_ExecutionPriorityListCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.4.2 Update Execution Priority List Activation Status

The Update Execution Priority List Activation Status sequence diagram specifies the MA notifying C2 of a new active EPL after entering a new geozone. Prior to this interaction if this is the Lead ACP, all follower ACPs shall message the lead ACP their activation statuses. If this is the Lead ACP, it should wait for all package members to respond. MA then sends C2 *MA_ExecutionPriorityListActivationStatusMT* containing the *ExecutionPriorityListActivationDataID* and the *ActiveExecutionPriorityListID*. The *AppliesToID* shall be set to the Package ID if it is the lead and if all follower status have been received, or it shall be set to System ID(s) and/or sub Package ID(s) that the status message applies to.

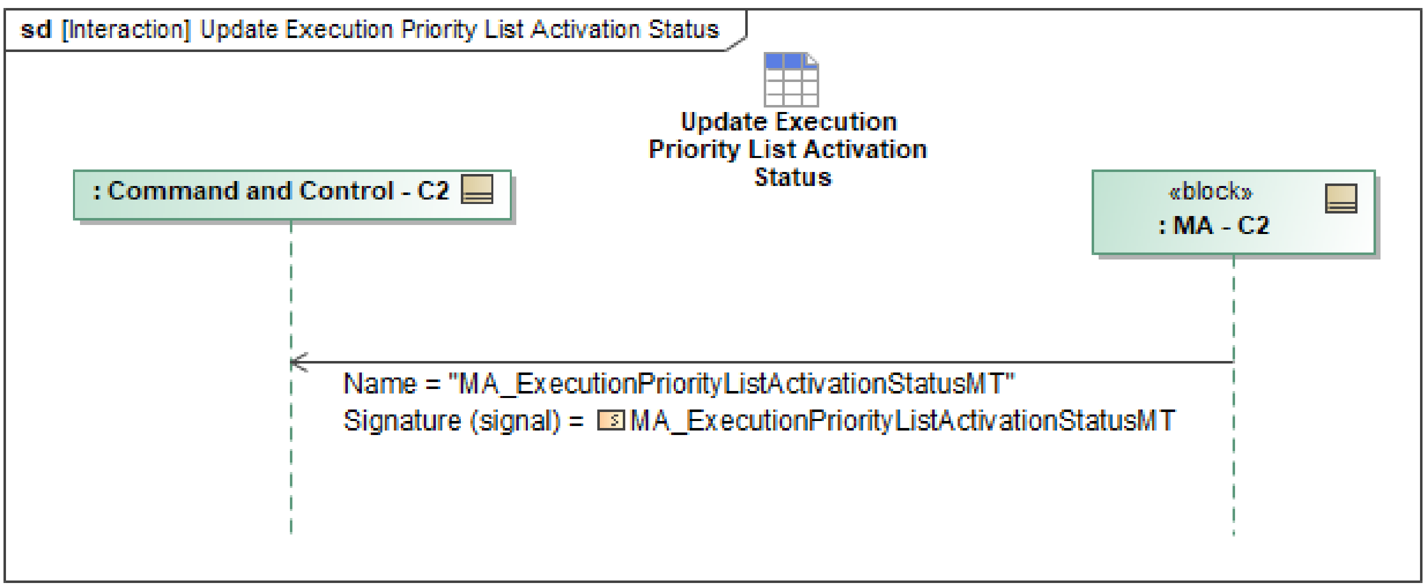


Figure 1-180: Update Execution Priority List Activation Status

Table 1-179: Update Execution Priority List Activation Status

Message Name (Name:Type)	Required Fields (Name:Type)
MA_ExecutionPriorityListActivationStatusMT: MA_ExecutionPriorityListActivationStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.12.5 C2 Ignore Zone Constraints Behaviors

1.2.12.5.1 Activate Ignore Zone Constraints

Overview

The Ignore Zone Constraints command message (*MA_IgnoreConstraintCommandMT*) allows an operator to specify individual tasks to ignore specific geozones. The task(s) are specified in the *ReferenceType* field and will be one of the following: Effect, Action, Task, Response, or Capability Command. The *ListOfConstraintsToIgnore* field allows setting one or more OpZone or OpVolume ID to be ignored. The *IgnoreAllConstraints* field allows the operator to override any and all constraints as an attempt to accomplish a task. Multiple command messages can be active at the same time and each can be updated or canceled as needed. The *MA_IgnoreConstraintCommandStatusMT* message will be used to provide status of the active commands. The C2 will send messages to a package lead or non-package ACPs. Package leads will distribute the messages to the rest of the follower ACPs.

Mission Execution

During planning, if there is one or more active *MA_IgnoreConstraintCommandMT* message, the planner will ignore the OpZone and/or OpVolumes when planning for the tasks specified in the *ReferenceType* field. During mission execution, if a new or updated *MA_IgnoreConstraintCommandMT* message is received with an active task specified in the *ReferenceType* field, this shall trigger a replan to occur.

Sequence Description

The Activate Ignore Zone Constraints sequence diagram below specifies the interaction of C2 commanding MA to ignore a zone constraint for specified task(s). Note that the intent of this sequence only covers OpZone and OpVolume constraints. First, C2 sends a *MA_IgnoreConstraintCommandMT* message to MA. The *AppliesToID* field can have multiple entries of *PackageID* or *SystemID*. This allows a C2 to send a Lead MA the Package ID (or multiple sub Package IDs) or a set of System IDs. The *MA_IgnoreConstraintCommandMT* message shall specify either a list of zone constraints to ignore (*ListOfConstraintsToIgnore*), or an indicator that all constraints should be ignored (*IgnoreAllConstraints*). It also contains the ID(s) of the task(s) the command applies to (*ReferenceType)*. Multiple commands can be active on the MA and each should have a unique *CommandID*. For new commands, the *CommandState* shall be set to *NEW*. For C2 updates to active commands, the *CommandState* shall be set to *UPDATE* and the *CommandID* shall match that of the active message being updated. To cancel an active command, the *CommandState* shall be set to *CANCEL* with *CommandID* set to that of the message being cancelled.

In response to receiving the command, MA shall send *MA_IgnoreConstraintCommandStatusMT_1* to C2. The *CommandProcessingState* will be set to *RECEIVED* to indicate receipt of the command. For this status message the *AppliesToID* field shall be set to the Package ID if MA is the leader and has received a status message from all members of the Package ID. Otherwise, it shall be set to the System ID(s) and/or sub Package ID(s) for which the status message applies.

The MA shall perform the command within MA to ignore the specified constraints for the designated task(s). If the MA is a non-package ACP, it shall send the C2 *MA_IgnoreConstraintCommandStatusMT_2* with *CommandProcessingState* set to *ACCEPTED* to indicate it has completed the command. If MA is the lead ACP, then MA shall message its follower ACPs with the *MA_IgnoreConstraintCommandMT* (See "Activate Peer Ignore Zone Constraints"). After receiving the respective responses from the follower ACPs, MA (the lead ACP) sends C2 the

MA_IgnoreConstraintCommandStatusMT_3 with the *CommandProcessingState* set to *ACCEPTED*, reflecting the *AppliesToID* as needed.

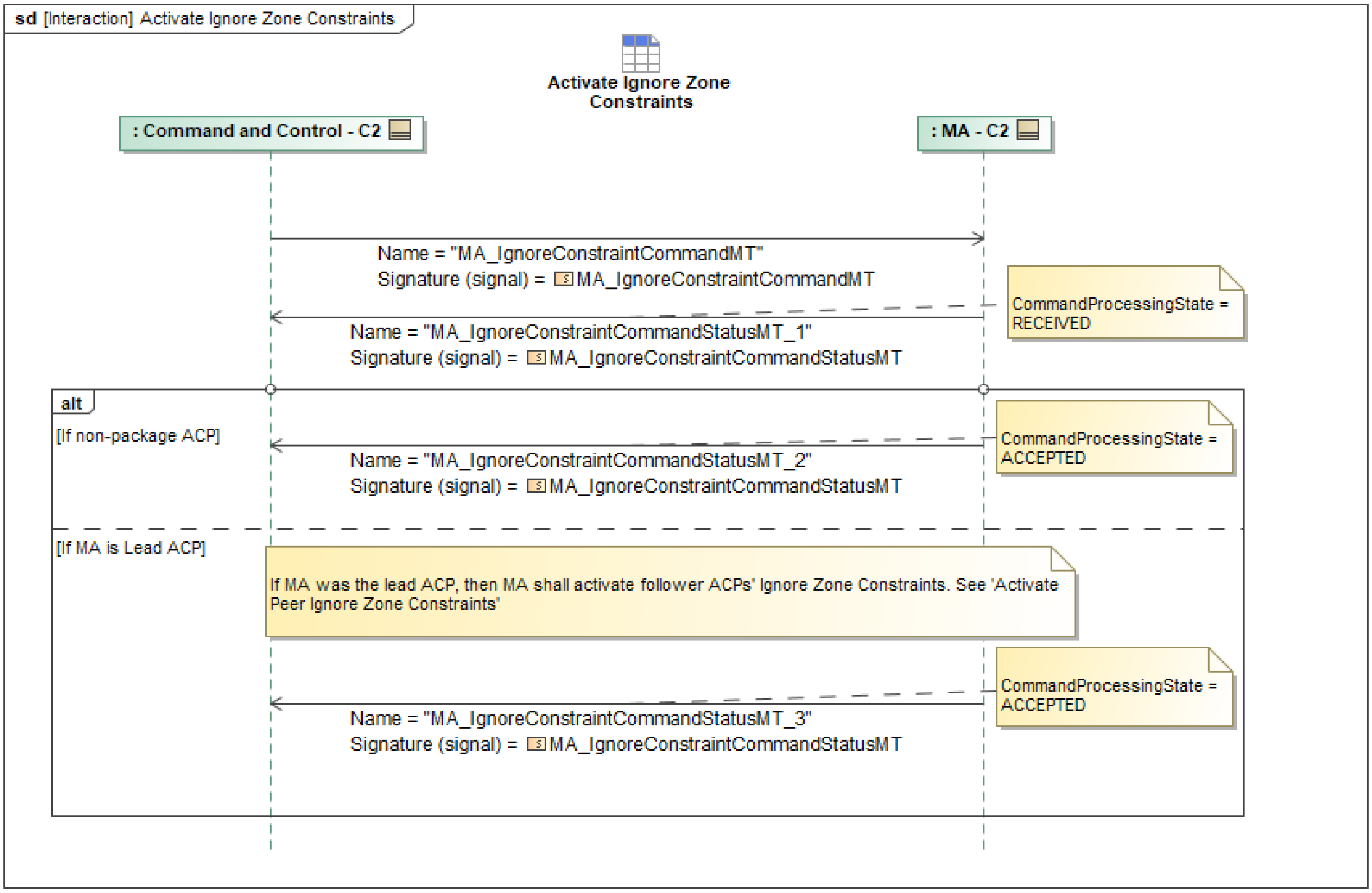


Figure 1-181: Activate Ignore Zone Constraints

Table 1-180: Activate Ignore Zone Constraints

Message Name (Name:Type)	Required Fields (Name:Type)
MA_IgnoreConstraintCommandMT: MA_IgnoreConstraintCommandMT	ListOfConstraintsToIgnore: MA_RegionChoiceType IgnoreAllConstraints: boolean
MA_IgnoreConstraintCommandStatusMT_1: MA_IgnoreConstraintCommandStatusMT	None
MA_IgnoreConstraintCommandStatusMT_2: MA_IgnoreConstraintCommandStatusMT	None
MA_IgnoreConstraintCommandStatusMT_3: MA_IgnoreConstraintCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13 C2 Teaming Behaviors Package

A “Task” is a defined behavior that can be executed by multiple collaborating ACPs as a Joint System

Task (e.g. Altitude Stacked Marshall), or by a single ACP as a Single System Task (e.g. single vehicle tasks, and Escort assigned to a single formation). A formation is defined and commanded as a task, using *MA_TaskMT* and *MA_TaskCommandMT*. A task may either be directly commanded to invoke immediate execution or be used within a task plan for future execution when the plan is activated. The C2 system invokes a task in two stages. If the task has not yet been uploaded, then it is first uploaded as shown in the Uploading a Task sequence diagram. Then, the task can either be “assigned”, where C2 directly assigns systems, or it can be “delegated”, where the autonomy determines which systems are assigned at execution. Therefore, a task can be assigned using a plan, assigned directly, delegated using a plan, or delegated directly. If multiple C2 systems task the same package or its package members, the C2 RBAC levels and hierarchy dictate the higher priority tasking to execute. For package dissemination, C2 can send tasks, updates, and data to any ACP within a package. If the receiving ACP is not the package leader, the package partner will forward the appropriate messages to the package leader.

1.2.13.1 Changing a Package Leader

The Changing a Package Leader sequence diagram specifies the interaction enabling C2 to reassign the package leader or request a package leader election. If C2 is reassigning the leader, C2 shall send *MA_PackageManagementCommandMT_1* to the Initial Package Leader with the *ChangeType* field set to *UPDATE_PACKAGE_LEADER*. C2 shall specify the new leader with the *PackageLeaderID* field. Otherwise, if C2 is requesting a package leader election, C2 shall send *MA_PackageManagementCommandMT_2* to the Initial Package Leader with the *ChangeType* field set to *REQUEST_LEADER_ELECTION*. C2 shall specify the leader election method with the *PackageLeaderElectionMethod* field (If method is unspecified, the Bully Leader Election will be used). In response to receiving a *MA_PackageManagementCommandMT* message, the Initial Package Leader shall send C2 *MA_PackageManagementCommandStatusMT* confirming or rejecting the package change command. Once the new lead has been established, the New Package Leader shall send package status to C2 according to the "Receive Package Status" sequence. Lastly, the New Package Leader shall send *MA_PackagePartnerStatusReportMT_1*, and the Initial Package Leader (the now previous Package Leader) shall send *MA_PackagePartnerStatusReportMT_2.* These messages are sent to reflect any updates made during the package change.

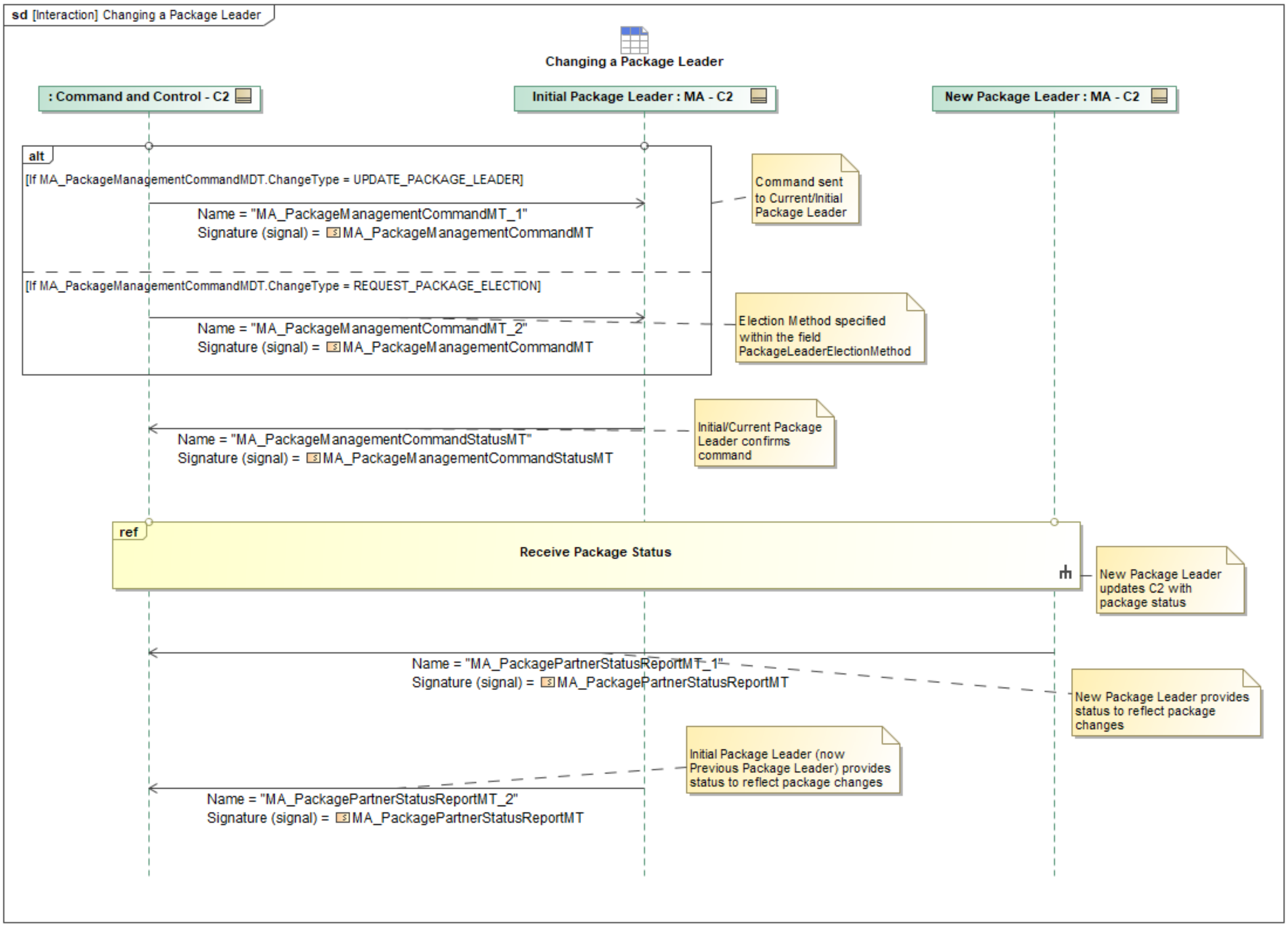


Figure 1-182: Changing a Package Leader

Table 1-181: Changing a Package Leader

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PackageManagementCommandMT_1: MA_PackageManagementCommandMT	PackageLeaderID: SystemID_Type
MA_PackageManagementCommandMT_2: MA_PackageManagementCommandMT	None
MA_PackageManagementCommandStatusMT: MA_PackageManagementCommandStatusMT	None
MA_PackagePartnerStatusReportMT_1: MA_PackagePartnerStatusReportMT	PackageLeaderID: SystemID_Type
MA_PackagePartnerStatusReportMT_2: MA_PackagePartnerStatusReportMT	PackageLeaderID: SystemID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13.2 Command Team/Platform via Task

MA_TaskMT contains different tasks that a system can perform against a specific entity or geographical location. *MA_TaskPlanMT* is the collection of these different Tasks for a team of ACPs and can be optionally sent to the ACP package to assist with commanding coordinated tasking. To command a team via tasking, C2 provides *MA_TaskMT* (and optionally *MA_TaskPlanMT)* messages with the specific TaskType field identified to dictate the task that should be performed. If sending the *MA_TaskPlanMT* message, the *PlanCommandID* field can be used to specify the planning command ID used to generate the Task Plan. C2 then sends an *MA_TaskCommandMT* to trigger immediate execution of the task, specifying the *TaskID* and *RequiredAllocation*. If C2 doesn't specify assigned assets and delegates to the ACP Lead to select package members, then *RequiredAllocation* should specify *PackageID*, *PlatformType*, and *PlatformTypeCategory*. If C2 desires to specify which package members are assigned the tasks, then *RequiredAllocation* should include *SystemID* of the ACPs requested. If the receiving MA system is not the Package lead, the Package partner will forward *MA_TaskMT* and *MA_TaskCommandMT* to the Package lead.

MA responds with a *MA_TaskCommandStatusMT* that tells C2 whether the command has been *ACCEPTED* or *REJECTED*. MA will then provide *MA_TaskStatusMT* (and *TaskPlanStatusMT* when a Task Plan was used) that describe the status of the *Task* (and *TaskPlan* when applicable*)*, respectively.

Additionally, if the *MA_TaskCommand* was addressed to a *PackageID*, the associated *MA_TaskCommandStatus* should reflect the *Package ID*. If partial rejections occur during Lead-to-Follower delegation the Lead ACP must revert to individual messages to report multiple rejecting followers.

If C2 allocates a requirement to a specific subset of platforms (via discrete *SystemID* allocations rather than *PackageID*), the Lead ACP shall bypass package-level aggregation and return individual status messages for those tasked members.

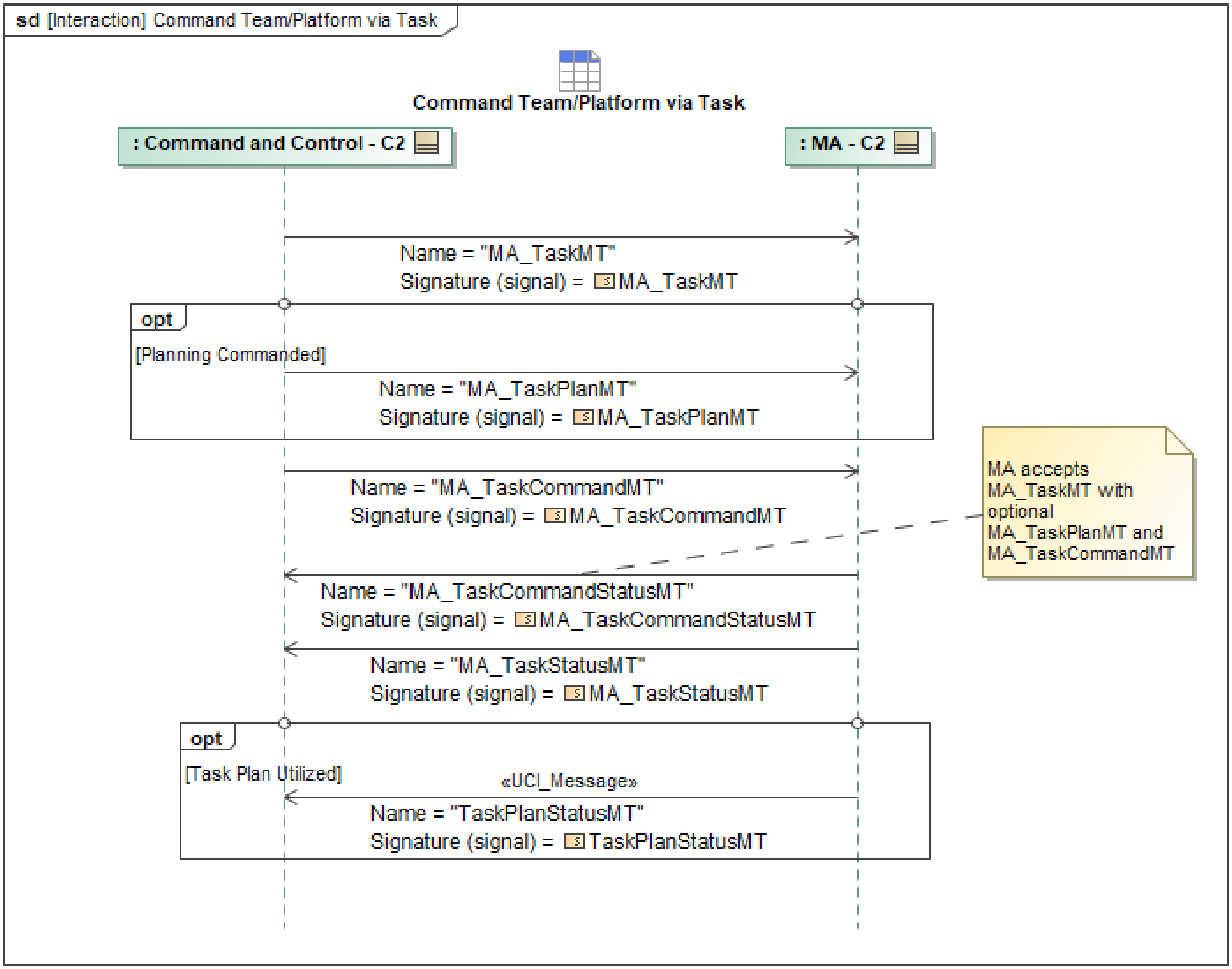


Figure 1-183: Command Team/Platform via Task

Table 1-182: Command Team/Platform via Task

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskCommandMT: MA_TaskCommandMT	RequiredAllocation: MA_RequirementAllocationConstraintType
MA_TaskCommandStatusMT: MA_TaskCommandStatusMT	None
MA_TaskMT: MA_TaskMT	None
MA_TaskPlanMT: MA_TaskPlanMT	None
MA_TaskStatusMT: MA_TaskStatusMT	Metrics: MetricsType
TaskPlanStatusMT: TaskPlanStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13.3 Creating or Changing a Package

The Creating or Changing a Package sequence diagram specifies the interaction enabling C2 to command ACP(s) to form a new package, join an existing package, or leave an existing package. Message *MA_PackageManagementCommandMT* is sent from the C2 system to each Commanded MA ACP(s) (1 or *n* ACPs, where *n* is the total number of assets). This message allows one or more Commanded MA ACP(s) to do one of the following based on specifying the *ChangeType* field:

- Form a package (*FORM_NEW_PACKAGE*)
- Join a package (*JOIN_PACKAGE*)
- Leave a package (LEAVE_PACKAGE)

The *MA_PackageManagementCommandMT* message may optionally include a *PackageLeaderID* in order to designate the leader when forming a package. The message may also include a list of *ConstraintIDs* to apply to the package. In response to receiving *MA_PackageManagementCommandMT*, each Commanded MA that received a message sends C2 a response *MA_PackageManagementCommandStatusMT* confirming or rejecting the *MA_PackageManagementCommandMT*. If the leader was not designated by C2, all associated ACPs would execute the specified *PackageLeaderElectionMethod* to determine which ACP will be the leader. Once the lead has been established, the Package Leader shall send package status to C2 according to the "Receive Package Status" sequence. The Package Leader will send *MA_PackagePartnerStatusReportMT* as a single message containing the status for all ACPs using the *SystemStatus* field to combine individual members.

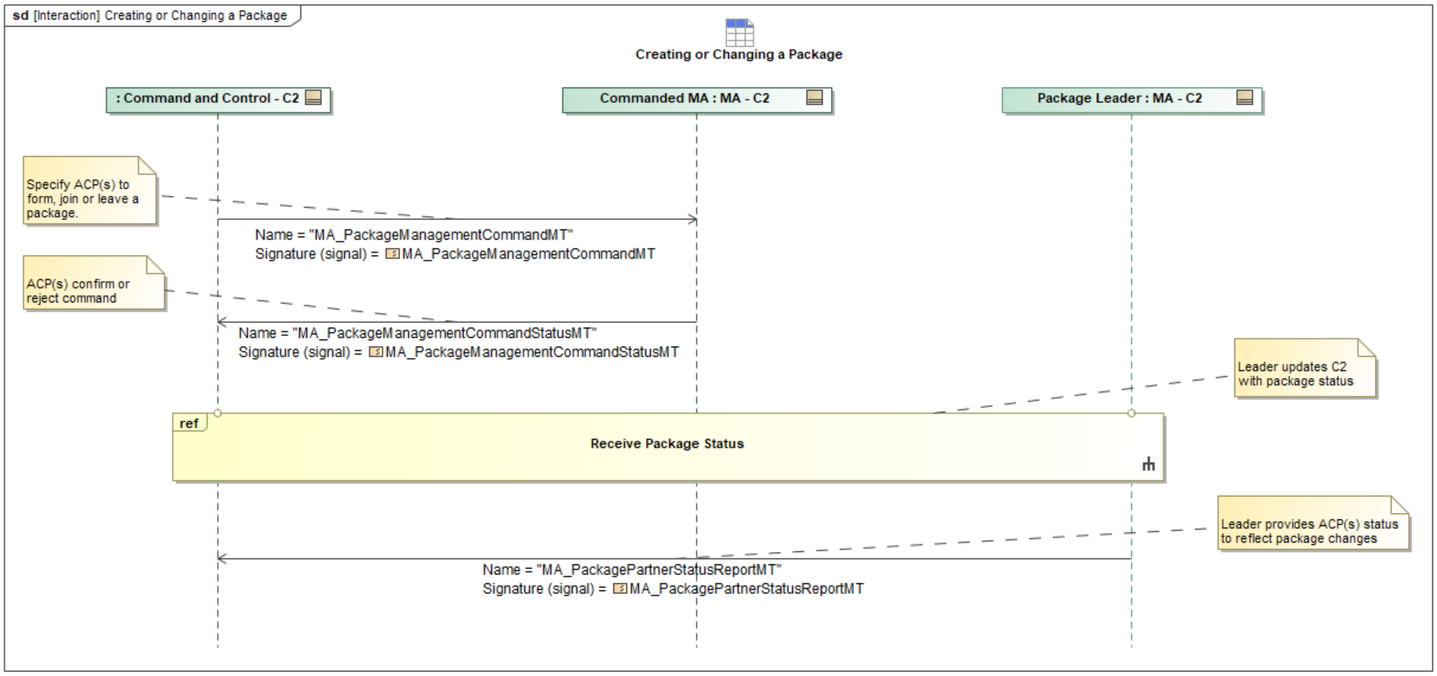


Figure 1-184: Creating or Changing a Package

Table 1-183: Creating or Changing a Package

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PackageManagementCommandMT: MA_PackageManagementCommandMT	None
MA_PackageManagementCommandStatusMT: MA_PackageManagementCommandStatusMT	None
MA_PackagePartnerStatusReportMT: MA_PackagePartnerStatusReportMT	PackageLeaderID: SystemID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13.4 Distribute C2 Track Data to Facilitate Global Fusion and Update COP

This interaction describes how C2 will distribute tracks via *EntityMT* data to MA in order to support global fusion and contribute to the development and maintenance of the common operating picture (COP). C2 will provide any C2 Source tracks and any External source tracks to MA via *EntityMT,* which can then be utilized by Fusion and COP services. *ObservationMeasurementReportMT* will be sent alongside *EntityMT* from MA to C2 to support any Fusion and COP functionality C2 wants to perform.

Within the *ObservationMeasurementReportMT* message, there are two optional paths to different sets of *SubsystemIDs* and *SystemIDs:*

1) *ObservationMeasurementReportMDT* → *ObservationMeasurement* → *MeasurementSource* → *ElementDetails* → *SourceElementIdentifier* → *SystemID* + *SubsystemID*

- These messages provide the system/subsystem that's the source of the OMR's kinematics

2) *ObservationMeasurementReportMDT → ObservationMeasurement → MeasurementSource → SourceIdentity → CapabilityID + SystemID + SubsystemID*

- These messages provide the system/subsystem and capability for the source of identity in the OMR (it's expected this will usually be the same system/subsystem as the kinematics) A client would want to know the capability type of the provided entity ID, and may have to subscribe to additional messages to understand the type (e.g. [Capability]StatusMT messages).

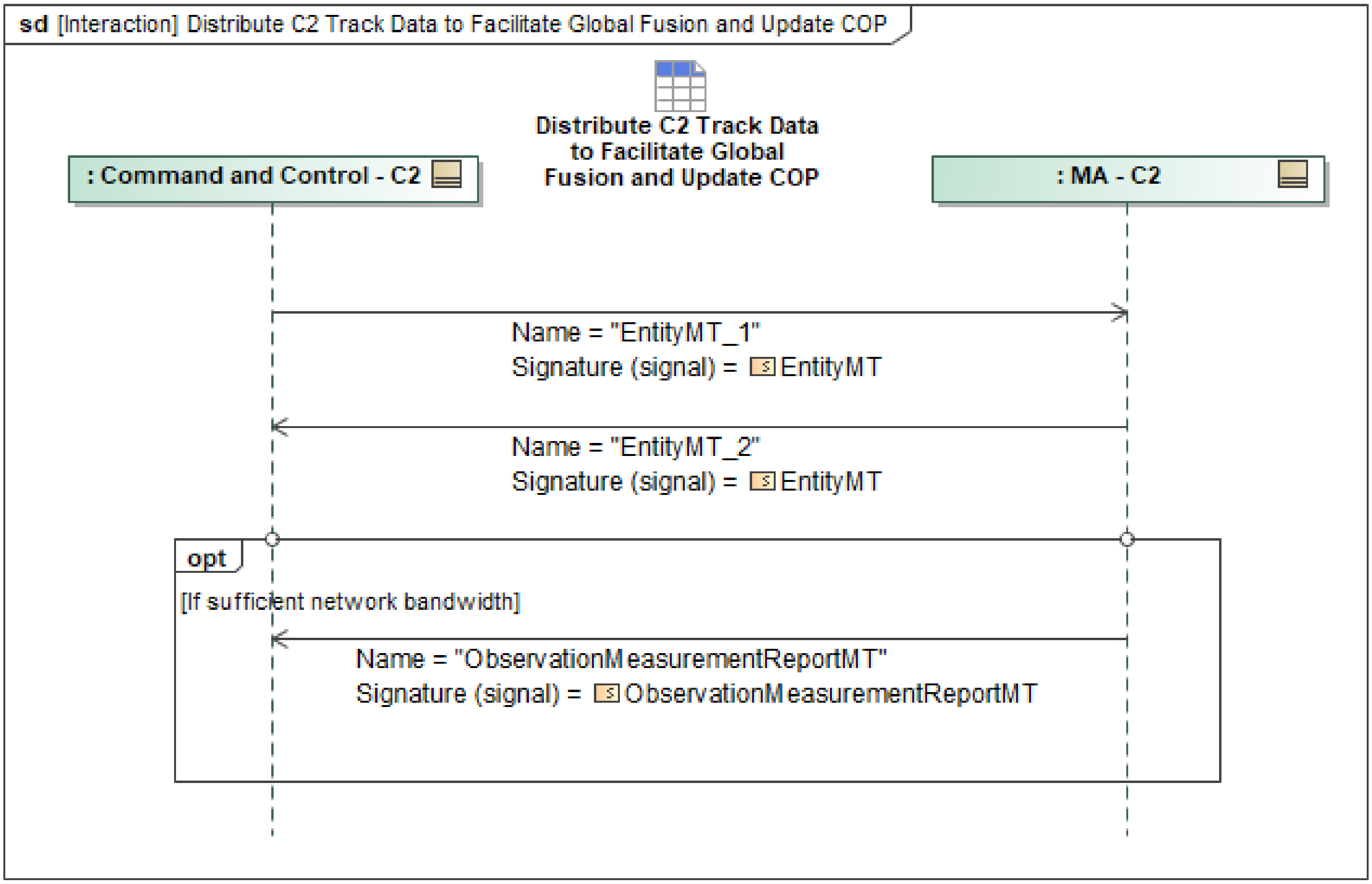


Figure 1-185: Distribute C2 Track Data to Facilitate Global Fusion and Update COP

Table 1-184: Distribute C2 Track Data to Facilitate Global Fusion and Update COP

Message Name (Name:Type)	Required Fields (Name:Type)
EntityMT_1: EntityMT	None
EntityMT_2: EntityMT	SourceTypeIdentifier: EntitySourceIdentifierType MeasurementID: MeasurementID_Type
ObservationMeasurementReportMT: ObservationMeasurementReportMT	MeasurementSource: ObservationMeasurementSourceType ElementDetails: ElementDetailsType SourceElementIdentifier: SourceKinematicsElementIdentityType SystemID: SystemID_Type SubsystemID: SubsystemID_Type

Message Name (Name:Type)	Required Fields (Name:Type)
	SourceIdentity: MeasurementSourceIdentityType CapabilityID: CapabilityID_Type SystemID: SystemID_Type SubsystemID: SubsystemID_Type MeasurementKinematics: MeasurementKinematicsChoiceType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13.5 Receive Package Status

This interaction specifies the periodic package status messages which are sent from the Package Leader MA to C2 systems. MA sends *PackageMT* periodically and advertises the fully formed package, with all package member System IDs included in the *PackagePartner* field. MA sends *PackageStatusMT* periodically and contains status information for the package and all package members, including their state in the *PartnerState* field and capabilities in the *PartnerCapabilityStatus* field.

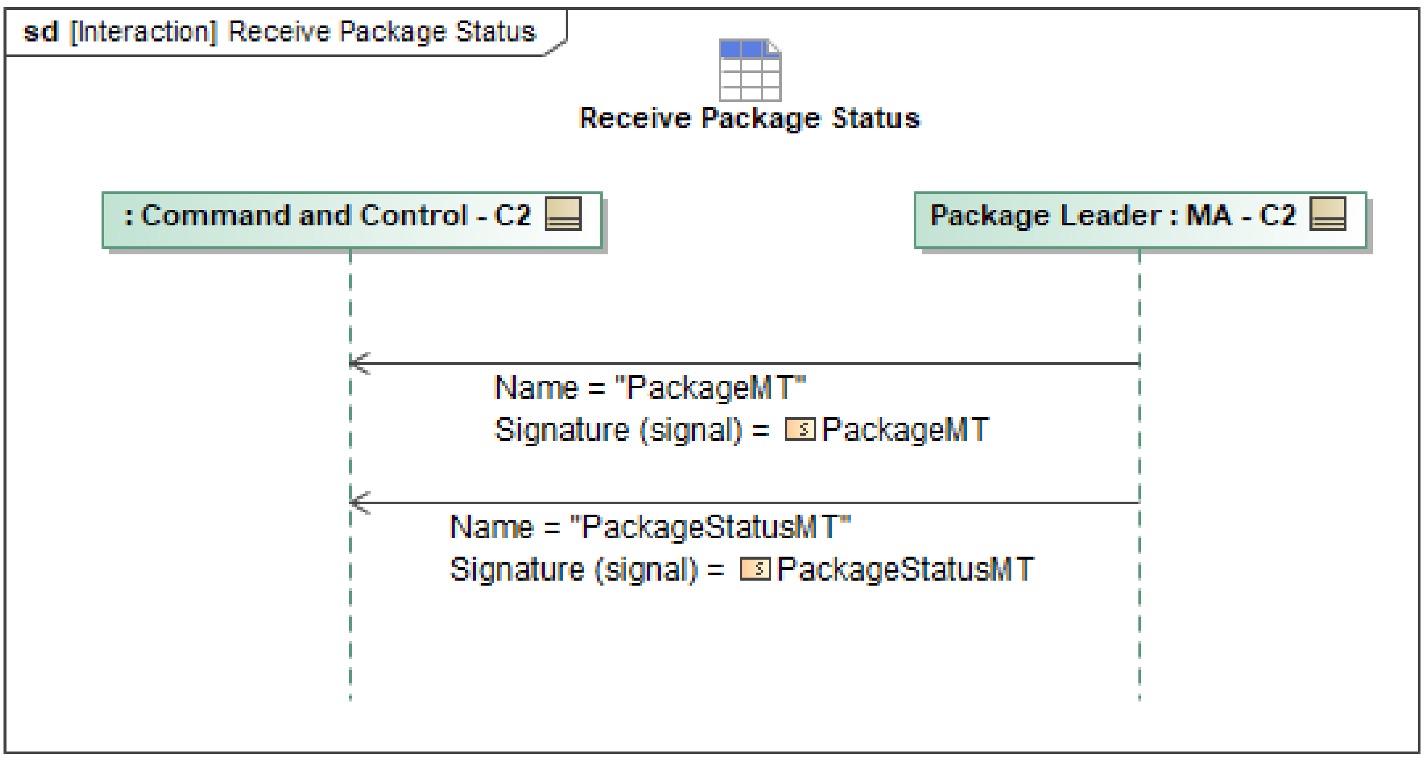


Figure 1-186: Receive Package Status

Table 1-185: Receive Package Status

Message Name (Name:Type)	Required Fields (Name:Type)
PackageMT: PackageMT	LeadSystemID: SystemID_Type
PackageStatusMT: PackageStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.13.6 Uploading a Task

To upload any task, the message *MA_TaskMT* is sent containing task parameters and a *TaskID*. No ACK is required for task messages issued. Optional constraint message(s) may be sent linking constraint(s) to the task. The example in the Uploading a Task sequence diagram works for any task type.

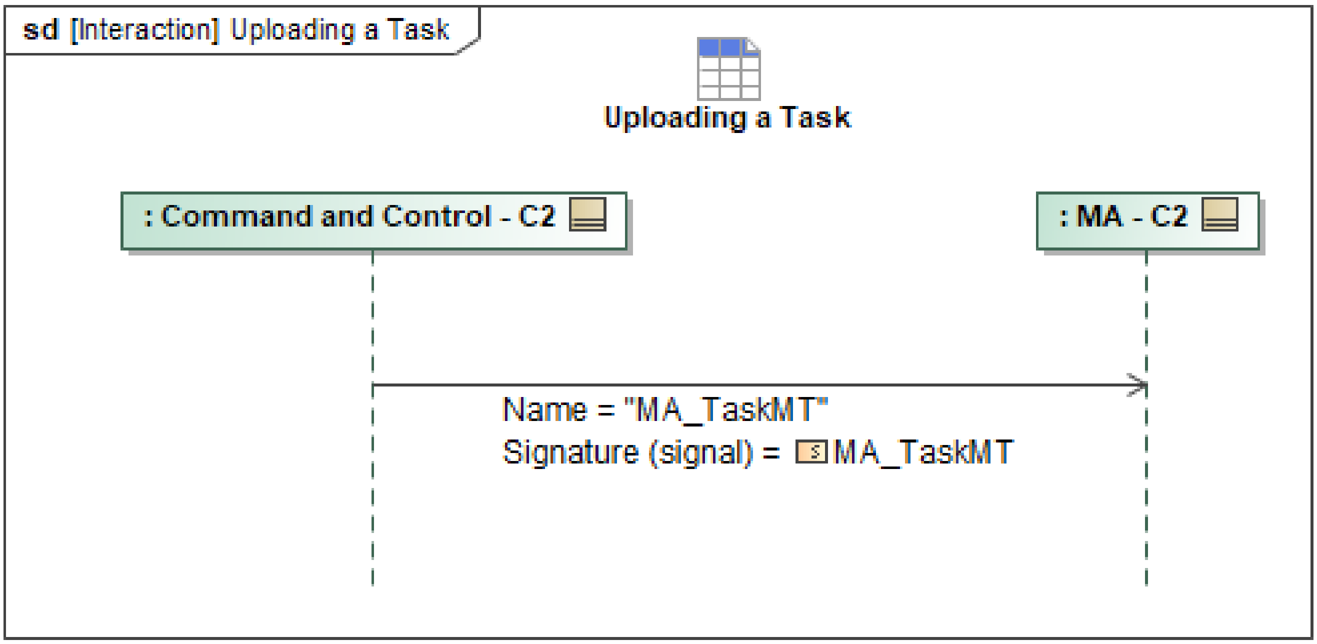


Figure 1-187: Uploading a Task

Table 1-186: Uploading a Task

Message Name (Name:Type)	Required Fields (Name:Type)
MA_TaskMT: MA_TaskMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.14 C2 Threat Assessment Package

1.2.14.1 C2 Threat Assessment

The Threat Assessment sequence diagram specifies the interactions where an ACP provides C2 with vehicle threat assessment results. C2 may request an ACP to perform vehicle threat assessment(s) for specific Entity ID's or all known red Entity ID's by sending *MA_AssessmentRequest* with respective V*ehicleThreatAssessmentRequestType* message. C2 can request the assessment be performed once or periodically at a defined rate by setting the *RequestFrequencyType* to PERIODIC and providing a value in *RequestFrequencyPeriod*. C2 may also request what capability MA will use to perform the vehicle threat assessment (i.e. onboard sensors), however if C2 wants MA to use historical tracks provided by track management function then the *ThreatCapability* enumeration will be *Other_Unspecified_Detection*. The ACP will respond, once or periodically, with a *MA_AssessmentRequestStatusMT* and, if the request specified *ResultsInAssessmentMessage* to be FALSE, a *MA_AssessmentMT.* If periodic assessments were requested, C2 will send *MA_AssessmentRequestMT* with *RequestFrequencyType* set to STOP. MA will respond with the corresponding *MA_AssessmentRequestStatusMT* to indicate request complete.

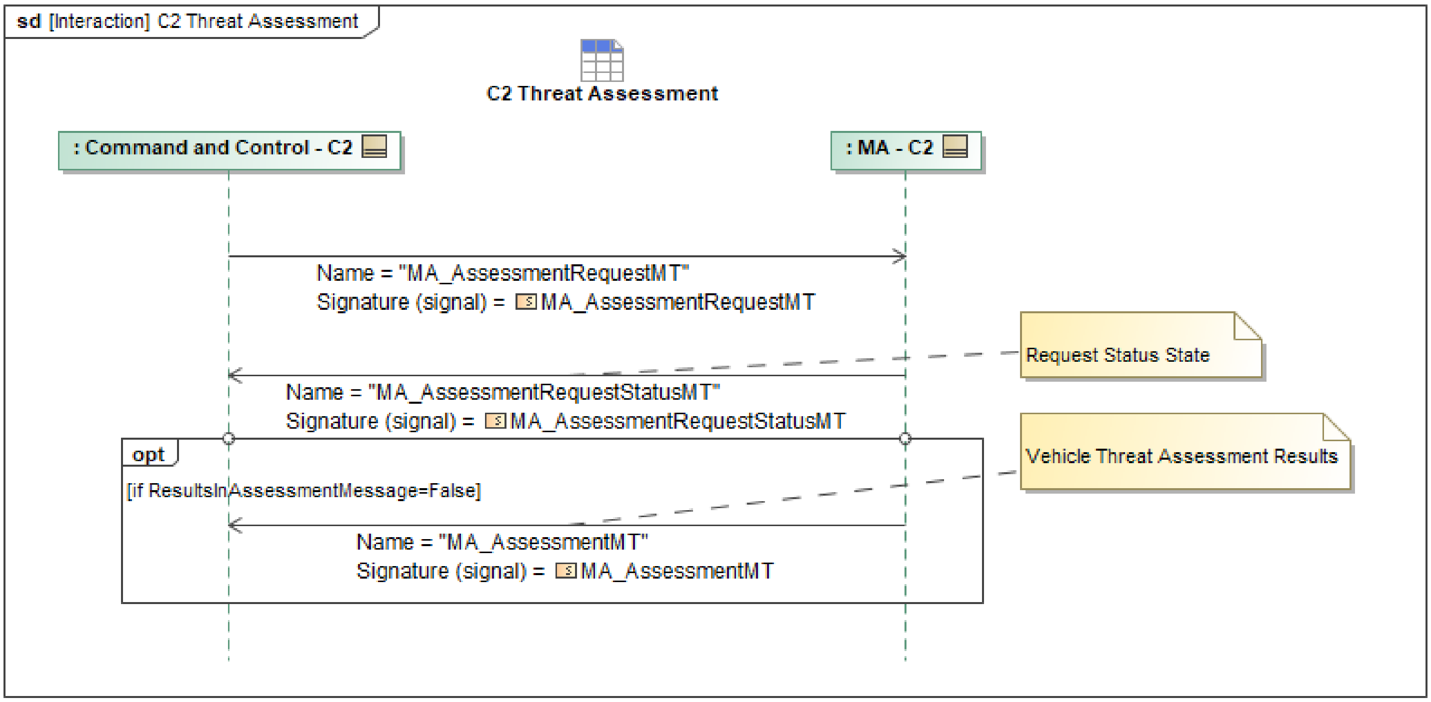


Figure 1-188: C2 Threat Assessment

Table 1-187: C2 Threat Assessment

Message Name (Name:Type)	Required Fields (Name:Type)
MA_AssessmentMT: MA_AssessmentMT	VehicleThreatAssessment: VehicleThreatAssessmentType ObjectState: ObjectStateEnum AssessmentRequestID: RequestID_Type
MA_AssessmentRequestMT: MA_AssessmentRequestMT	VehicleThreatAssessment: VehicleThreatAssessmentRequestType
MA_AssessmentRequestStatusMT: MA_AssessmentRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15 C2 Weapon Employment Behaviors Package

1.2.15.1 DLZ Compound Convergence

DLZ Compound Convergence is a fusion of multiple DLZ from team members within a flight group. The fusion of DLZ enables C2 to generate a composite view of flight group's weapon engagement capabilities to assess tasking for the best shot. Once ACPs are producing a DLZ in support of strike task, MA sends DLZ_MT messages to C2 periodically.

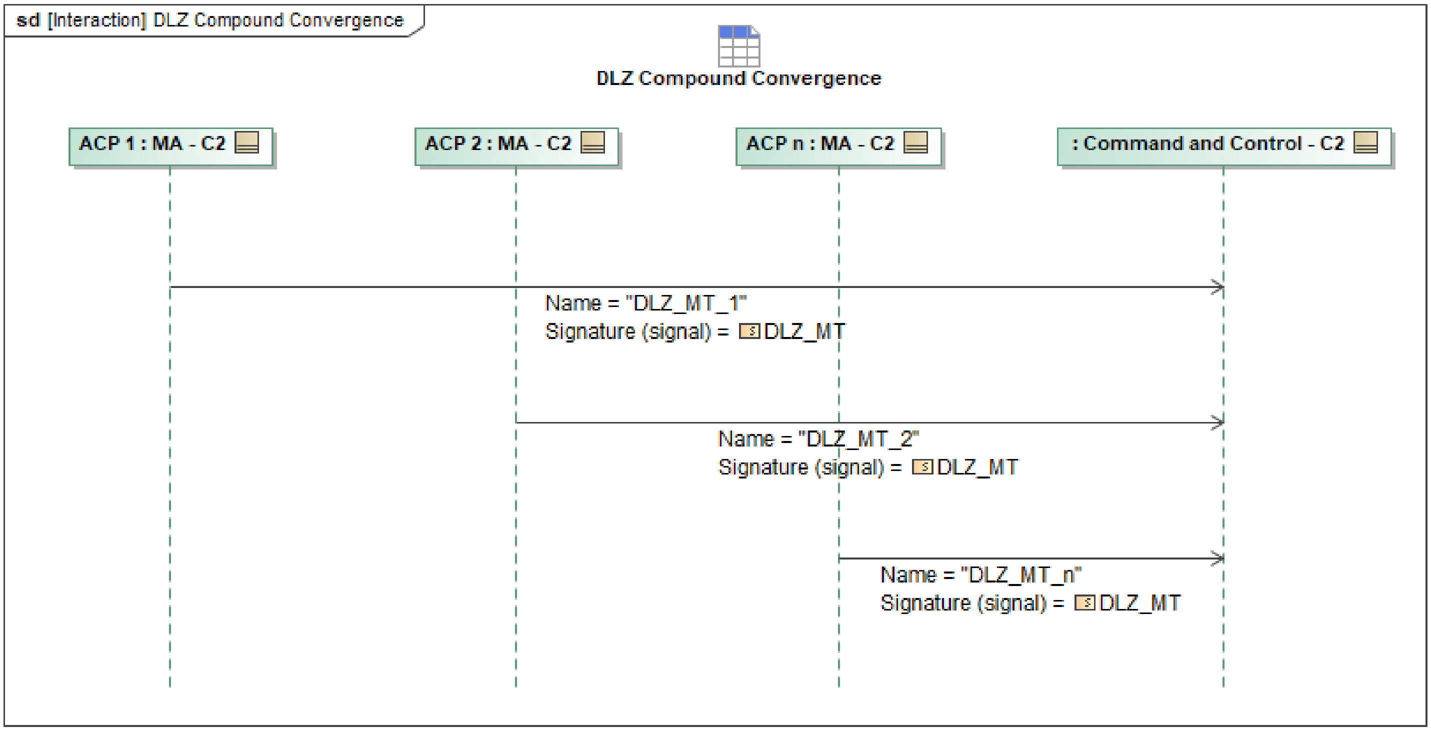


Figure 1-189: DLZ Compound Convergence

Table 1-188: DLZ Compound Convergence

Message Name (Name:Type)	Required Fields (Name:Type)
DLZ_MT_1: DLZ_MT	None
DLZ_MT_2: DLZ_MT	None
DLZ_MT_n: DLZ_MT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.2 Exchange Shot Queue Prioritization

Shot Queue Prioritization List is a list of mission targets ranked by assessing multiple factors to determine which target entities should be engaged first. The shot queue will utilize *MA_PrioritizationList* with a *ListType* of *Mission_Target_List (enum)*. The list requires subjects (*EntityID*) to be assigned a priority, where the lowest assigned value is the highest priority level. The message also includes an optional field, *ProbabilityOfKill,* that may be included with each assessed subject entity. MA (Single Ship or Multi-Ship Lead) may send an updated shot queue prioritization list to C2 based on the active engagement criteria.

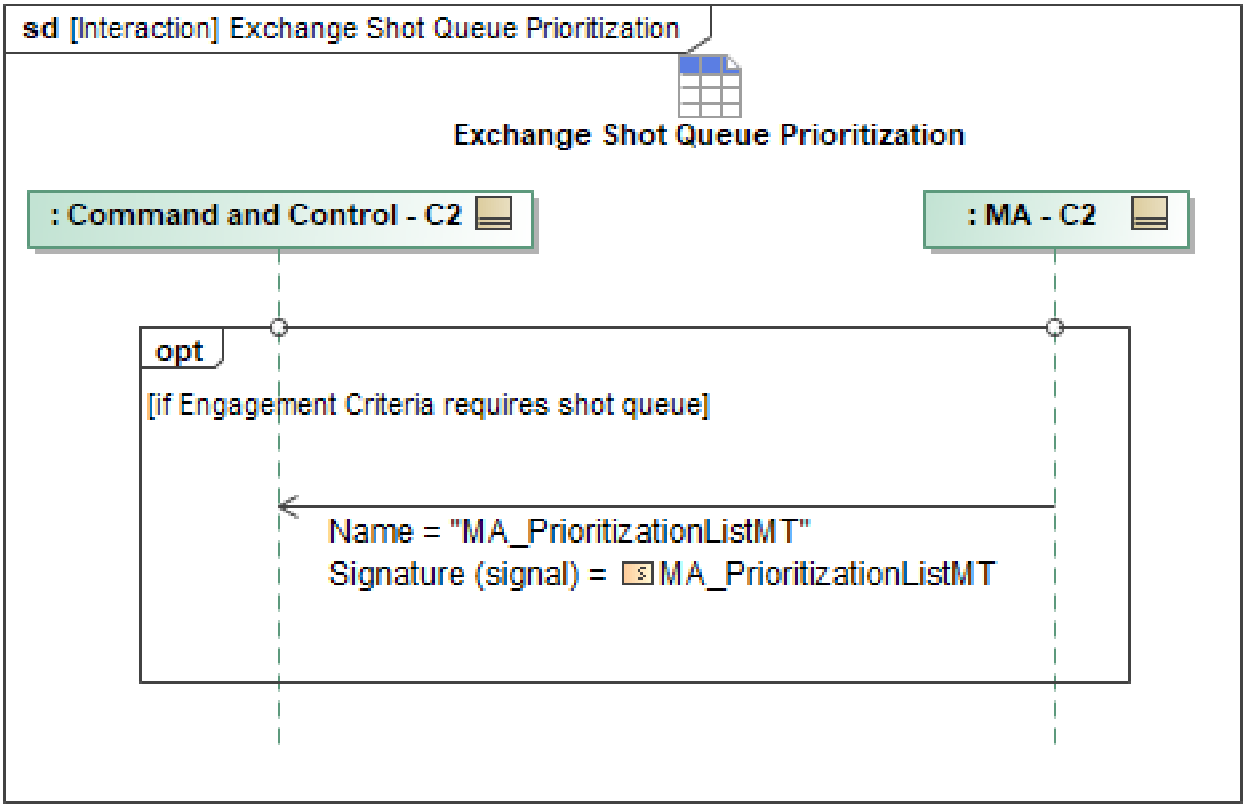


Figure 1-190: Exchange Shot Queue Prioritization

Table 1-189: Exchange Shot Queue Prioritization

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PrioritizationListMT: MA_PrioritizationListMT	MISSION_TARGET_LIST: ObjectState: ObjectStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.3 Mission Store Status Exchange

Once mission store status is received through the MS Interface, MA sends *StrikeCapabilityStatusMT* periodically to C2 for each store capability ID regardless of the store presence. MA will also send *MA_StrikeCapabilityMT* periodically. C2 will utilize *StrikeCapabilityStatusMT and MA_StrikeCapabilityMT* to maintain persistent awareness of the store state, loaded mission data, and mission store parameters in accordance with the selected Mission Data File (MDF). In the event that MA has an unexpected or undesirable state reported, MA is then required to update *StrikeCapabilityStatus.MessageData.CapabilityStatus.AvailabilityInfo.AvailabilityReason.*

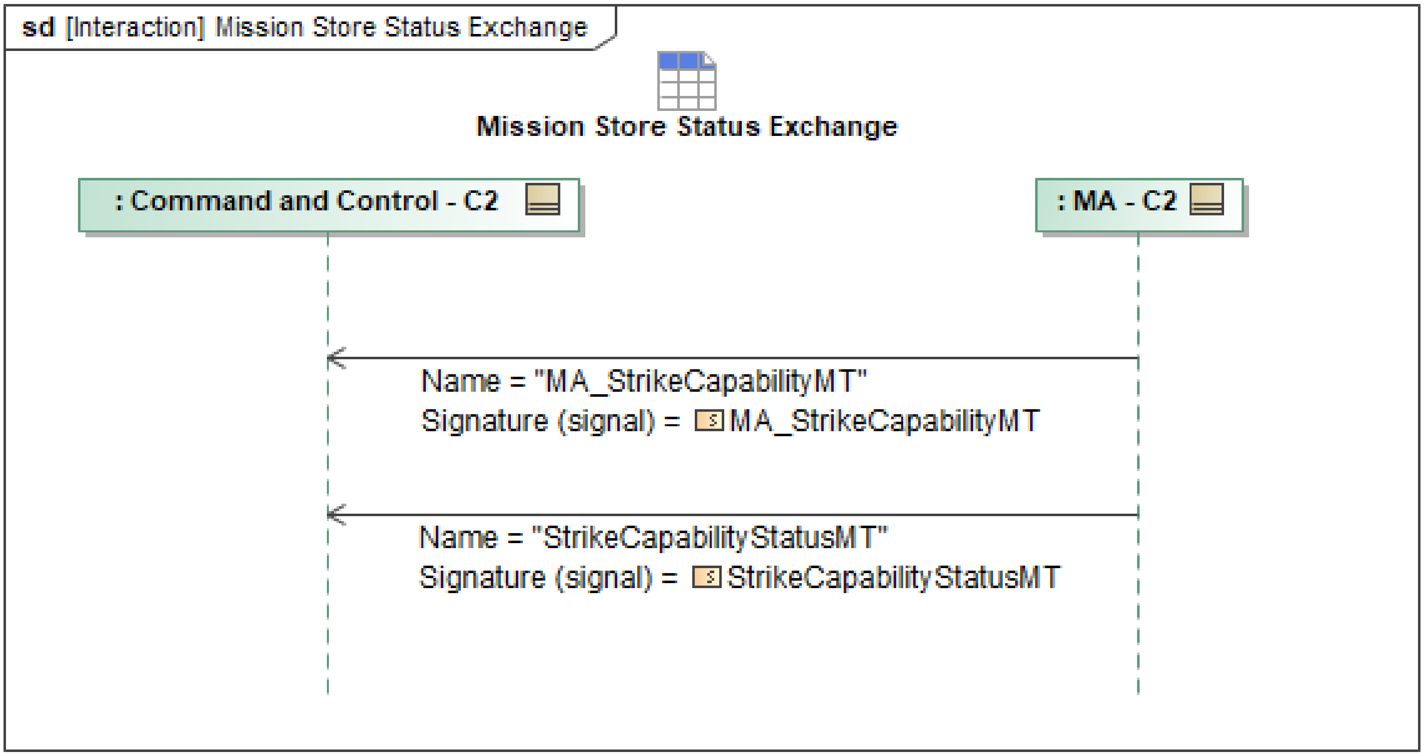


Figure 1-191: Mission Store Status Exchange

Table 1-190: Mission Store Status Exchange

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	None
StrikeCapabilityStatusMT: StrikeCapabilityStatusMT	StoreStatus: StoreStatusType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.4 Query Entity Track History

The Track Management functionality for multi-ship/single-ship weapon employment is to maintain target tracks historical data along with entity predictions data communicated from MA to C2 when requested using a data query request message. C2 queries the MA node to get the historical and prediction track data. The track data may be used by other functions such as sending external track reports, stimulating alerts/notifications if rule sets are triggered, and updating the track display, history, and attribute data. MA will be getting Entity data from the COP, so the Track Management Data Fusion engine will take the COP data and its sensor data to build track history. Track propagation service takes the track state updates and predicts the extrapolated position (Kinematics) of an entity in the battlespace and maintains these as a database within the MA.

The query for entity and propagation data from C2 to MA will be using the existing UCI message : *QueryDataRequest*.

- *ResultsInNative* field needs *MessageType* & *ReceiverAddress* for querying *EntityPropagation* historical data from C2 to MA. This would not serve the purpose since it doesn't specify *EntityId*(s) that need to be queried.

- *QueryMessage* field *MessageType* & *Query* PET (Polymorphic Extension Type) for querying *Entity* and *EntityPropagation* historical data from C2 to MA.

To create a query, *QueryDataRequest* UCI message uses *QueryMessage field*. For example, *QueryDataRequestMT* (starting from the query element) shows querying for *EntityID* from the historical Entity messages stored by MA's Entity Management Service. The query can be combined for querying both *Entity* & *EntityPropagation* messages (historical) in the *QueryType for entityid*.

C2 receives *QueryDataRequestStatusMT_1 response from the MA node with intermediate response of RequestProcessingState field set to QUEUED, followed by QueryDataRequestStatusMT_2 response with RequestProcessingState set to PROCESSING.* Upon successful query for the entity track management or propagation data, MA would receive *QueryDataRequestStatusMT_3* message with the entity database along with the *RequestProcessingState* set to *COMPLETED.* A failure response can result in a *QueryDataRequestStatusMT_4* with *RequestProcessingState* set to *FAILED* or *REJECTED* or *CANCELED* from *MA* node to C2*.* For any non-compliance related issues, *QueryDataRequestStatusMT_4* response is sent with *RequestProcessingStateReason* set to appropriate non-compliance code.

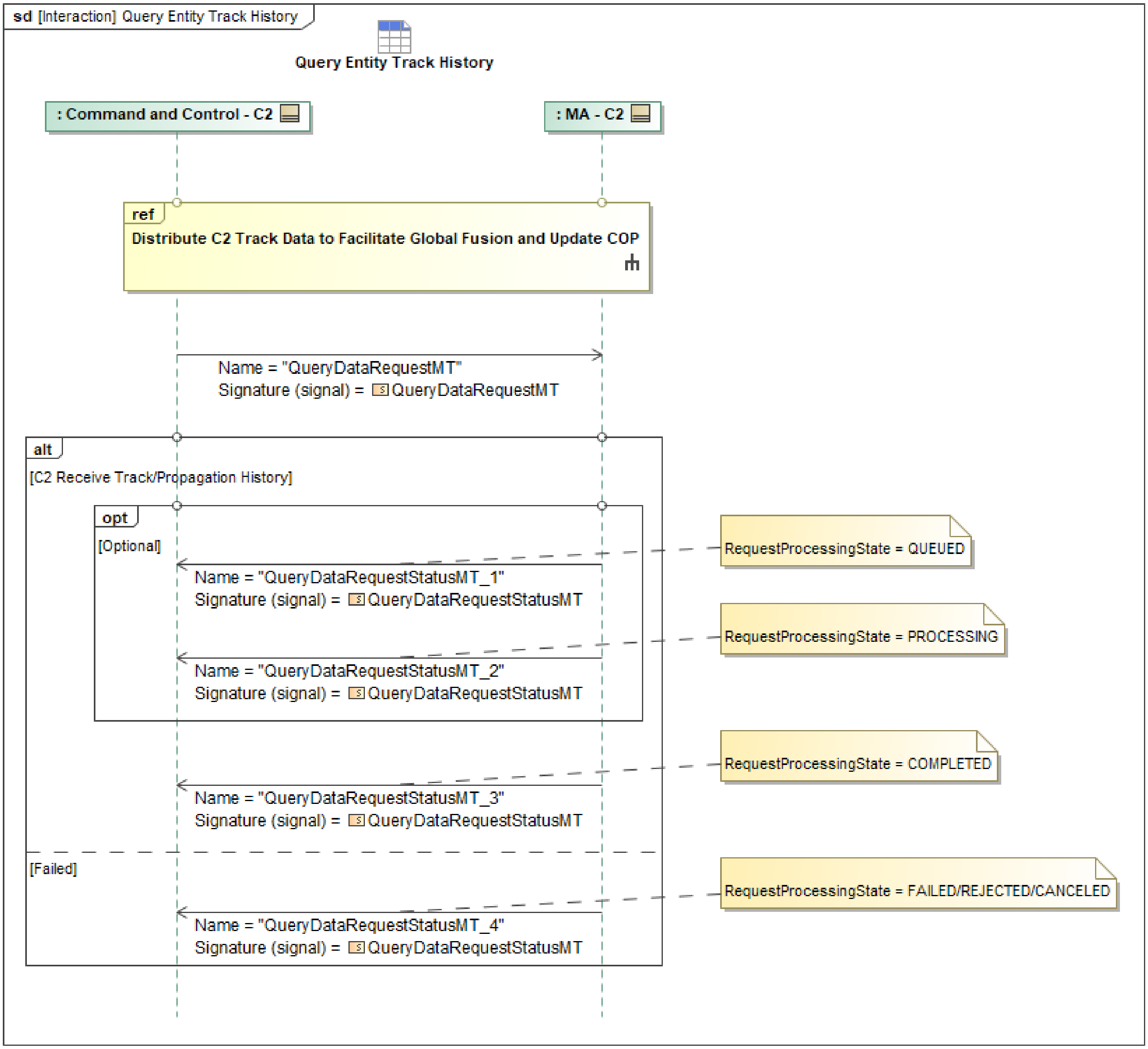


Figure 1-192: Query Entity Track History

Table 1-191: Query Entity Track History

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestMT: QueryDataRequestMT	QueryMessage: QueryMessageType
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.5 Query Shot Queue Prioritization

Shot Queue Prioritization List is a list of mission targets ranked by assessing multiple factors to determine which target entities should be engaged first. The shot queue will utilize *MA_PrioritizationListMT* with a *ListType* of *MISSION_TARGET_LIST enumeration*. The list requires subjects (EntityID) to be assigned a priority, where the lowest assigned value is the highest priority level. The message also includes an optional field, *ProbabilityOfKill*, that may be included with each assessed subject entity. C2 may query (*QueryDataRequestMT*) an ACP (Single Ship, Multi-Ship Lead, or Multi-Ship Peer) to request *MA_PrioritizationListMT* in its native form. MA will respond with *QueryDataRequestStatusMT* to provide state changes (Queued "*QueryDataRequestStatusMT_1*", Processing "*QueryDataRequestStatusMT_2*", and Completed "*QueryDataRequestStatusMT_3*") in the processing of the request as depicted in the sequence diagram. MA will send *MA_PrioritizationListMT* result in native form to the requesting *SystemID.*

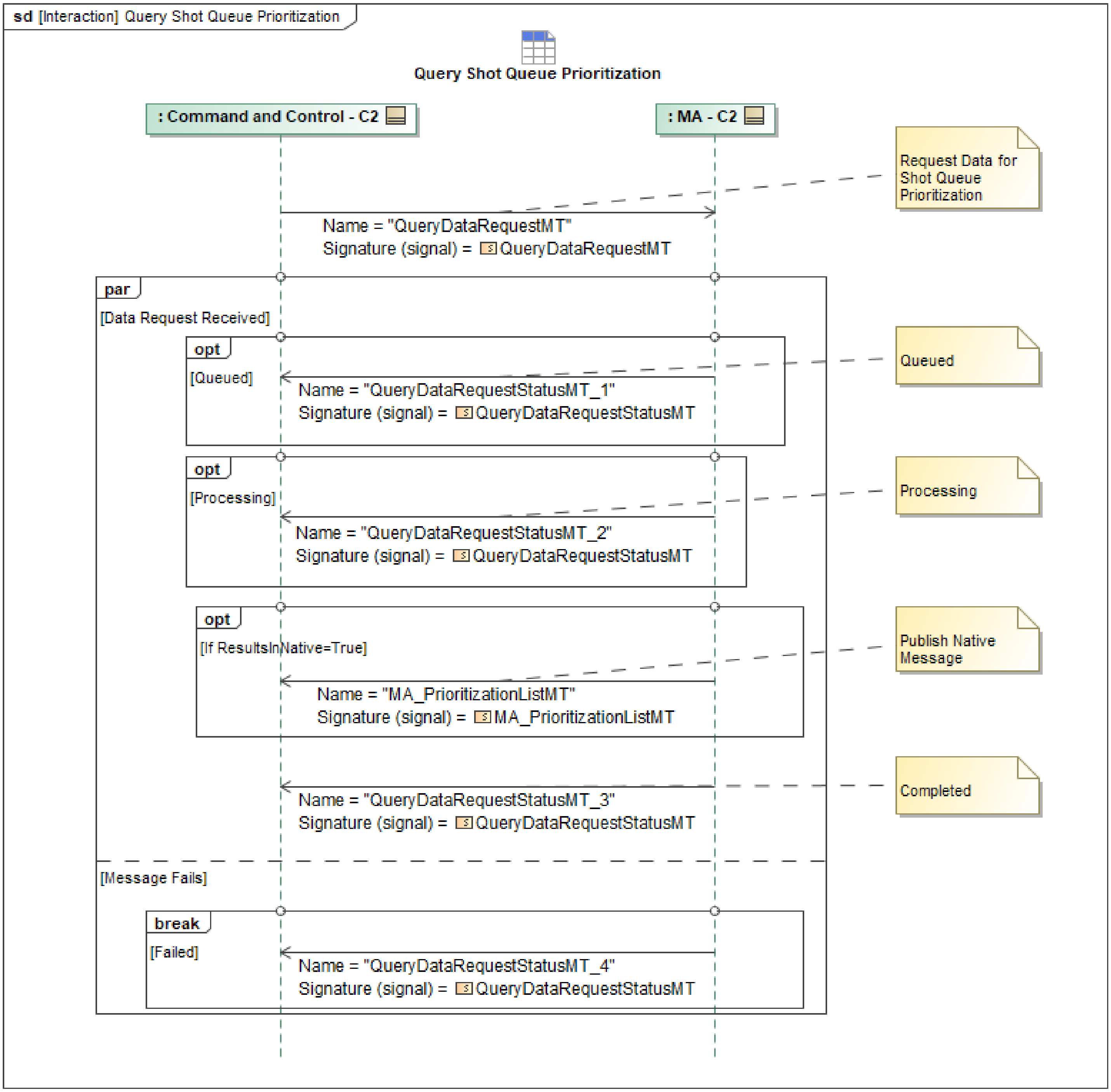


Figure 1-193: Query Shot Queue Prioritization

Table 1-192: Query Shot Queue Prioritization

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PrioritizationListMT: MA_PrioritizationListMT	ObjectState: ObjectStateEnum MISSION_TARGET_LIST:
QueryDataRequestMT: QueryDataRequestMT	None

Message Name (Name:Type)	Required Fields (Name:Type)
QueryDataRequestStatusMT_1: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_2: QueryDataRequestStatusMT	None
QueryDataRequestStatusMT_3: QueryDataRequestStatusMT	Result: QueryResultType
QueryDataRequestStatusMT_4: QueryDataRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.6 Release Consent

In the following sequence, MA enforces the Weapons Authorization Policy by checking its Authorization list, compiled from receiving Weapons Authorization Engagement Criteria messages previously from C2 or MP.

- Once the appropriate "authorizers" have been determined, MA sends a *StrikeConsentRequestMT* to C2 to obtain final authorization to fire
- C2 responds with *StrikeConsentRequestStatusMT*

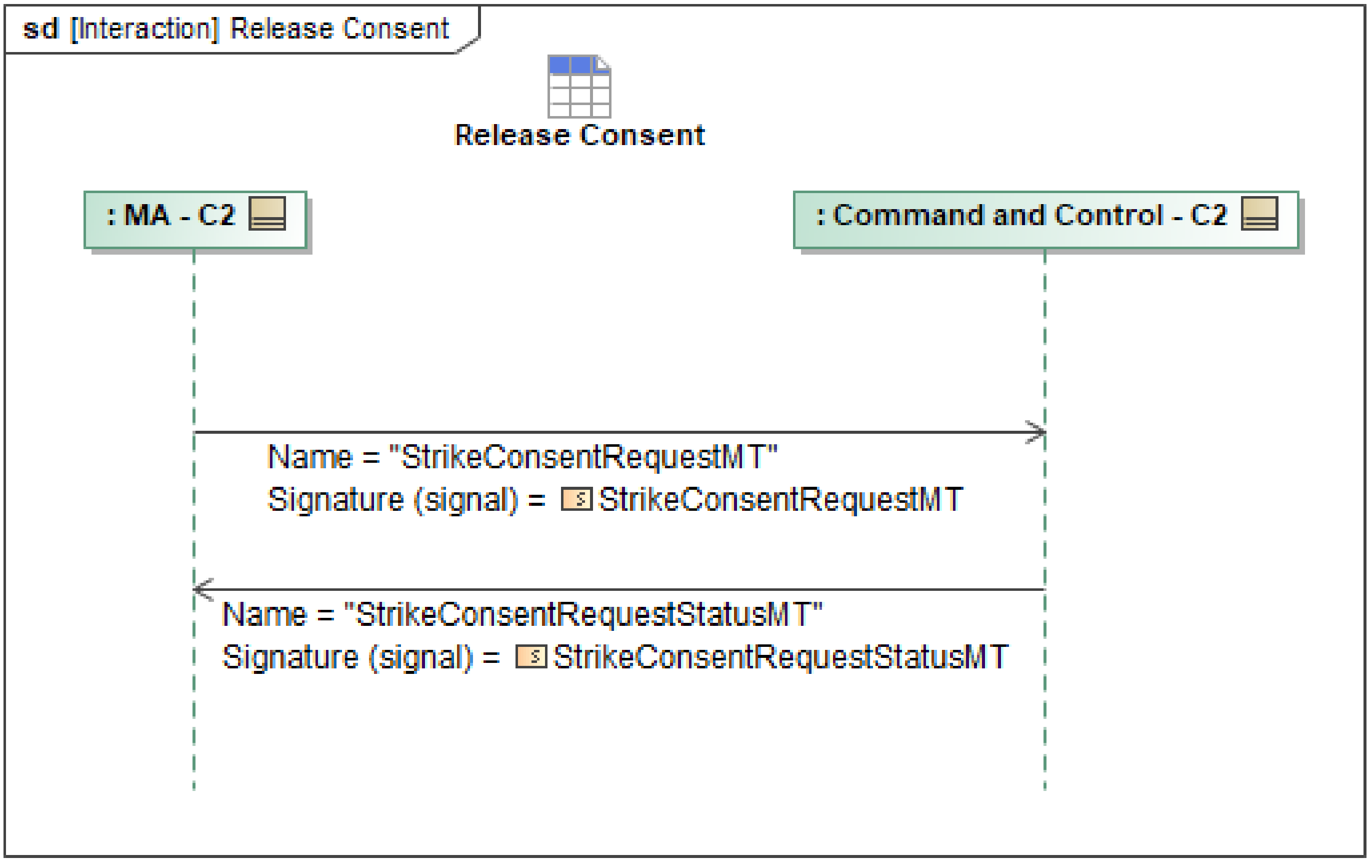


Figure 1-194: Release Consent

Table 1-193: Release Consent

Message Name (Name:Type)	Required Fields (Name:Type)
StrikeConsentRequestMT: StrikeConsentRequestMT	None
StrikeConsentRequestStatusMT: StrikeConsentRequestStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.7 Report Intercept Pairings

The intercept pairings of an engagement describe the individual targets that each ACP will do tactical maneuvers towards. The assigned targets may be a single or group of enemy tracks. Whenever the package lead MA assigns target pairings it will determine the tactical maneuvers employed by each ACP. This sequence describes the messages from MA to C2 to report back that tactical assignment. This would occur after a Lead ACP has assigned the intercept pairings amongst a package a received a positive acknowledgment back from the "Peer Assign Target Pairings" interaction.

The selected intercept pairings should follow the intercept maneuver parameters included in the engagement criteria by C2 prior to an engagement described in more detail in section 4.2.2.8.5 Update Intercept Parameters. If C2 commands a targeting action to MA and intercept pairings have been calculated, MA will return the *RelationshipDesignationMT* message back to inform C2 of the assigned intercept pairings for each ACP to target. In *RelationshipDesignationMT*, required fields for this sequence are the *Source* field set to the *SystemID* of the ACP, the *Destination* field set to the *EntityID* of the track being targeted, and the *Relationship.EnagementStatus* field set to *ENGAGING* to denote that the target is being intercepted.

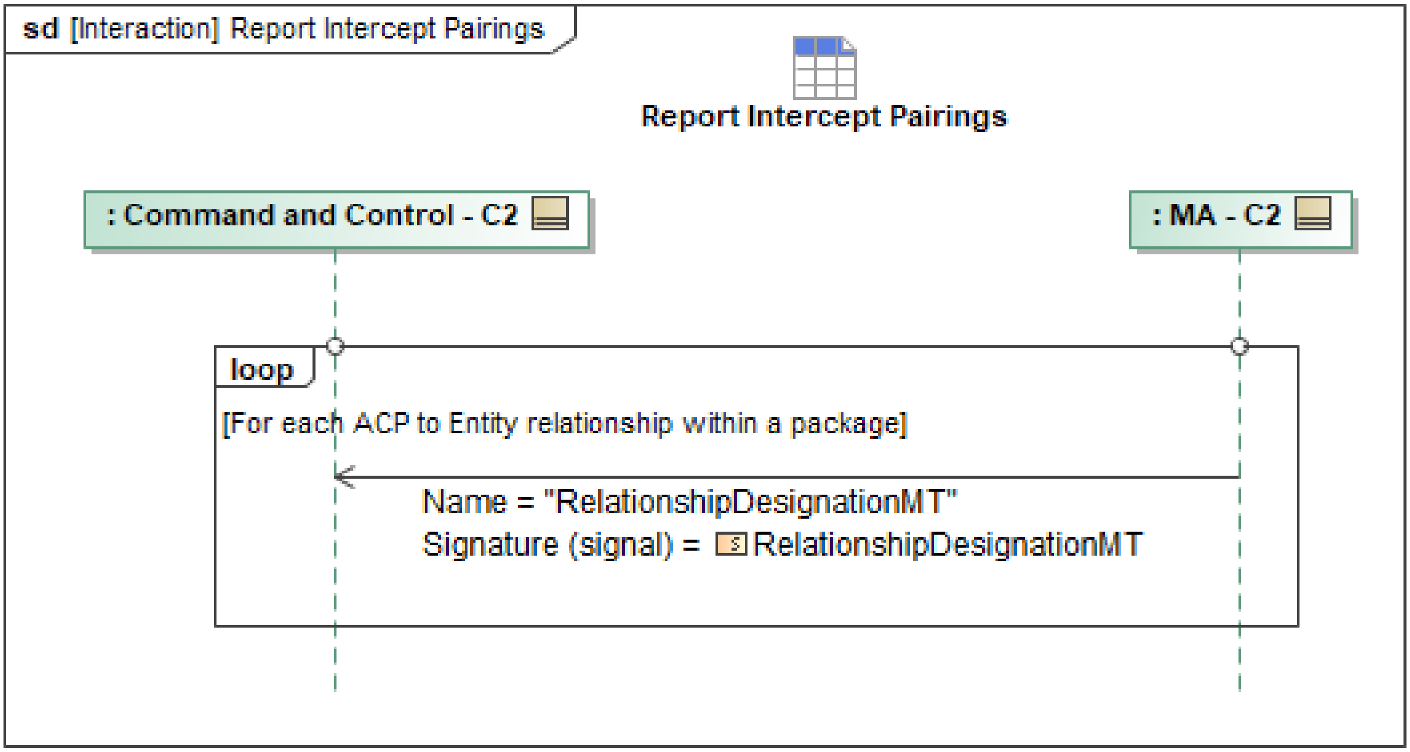


Figure 1-195: Report Intercept Pairings

Table 1-194: Report Intercept Pairings

Message Name (Name:Type)	Required Fields (Name:Type)
RelationshipDesignationMT: RelationshipDesignationMT	Relationship: RelationshipType Destination: GeoLocatedObjectType EngagementStatus: ExternalCommandExecutionStateEnum

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.8 Report Shot Allocations

An ACP's shot allocation(s) describe targeting or shooting tactics commanded by C2 that have resulted in MA allocating one or more shots to a hostile target. The shots may be allocated to an individual target, or across a group of tracks. This sequence describes how a single or Lead MA reports allocated shots of an individual ACP, or multiple ACPs as part of a package via the team lead. These Shot Allocations will follow a shot doctrine which can be set pre-mission via fields within the *MA_ActionMT* message or updated by C2 using the "Set Tactic Shot Doctrine and Intercept Maneuver Parameters". Additionally, if C2 commands a shoot tactic, new shot doctrine parameters can be sent along with this tactic. See the "Shoot" tactic L1 behavior. If C2 commands 1 or more

ACPs to target or shoot any number of tracks, the Lead MA will calculate optimal target pairings along with allocating the shots across an entire package. Upon positive acknowledgement by Peers to a Lead MA is when this sequence of Reporting Shot Allocations to C2 is used by the Lead.

The *RelationshipDesignationMT* contains the needed information to convey the information to C2. For each allocated shot, MA should specify a *Source* and a *Destination,* which are both *GeoLocatedObjectTypes.* For the case of reporting shot allocations to C2, the *Source* field should contain a *SystemID,* representing the ACP with a shot allocated to a target. The *Destination* field should contain an *EntityID* which would represent the target being fired at. If the same ACP has allocated shots to multiple targets, multiple *RelationshipDesignationMT* messages should be sent to C2, denoting the same ACP via *SystemID* within the *Source* field, and different *EntityIDs* within the *Destination* field. Each individual shot allocation will also garner their own unique UUID sent in the *RelationshipDesignationID* field. This relationship denotes an ACP has a *WEAPON_ASSIGNED* for the *Destination* field. The *EngagementStatus* field is where the state of the relationship is conveyed to C2. Using the *ExternalCommandExecutionStateEnum,* MA should use the enumeration of *WEAPON_ASSIGNED* to denote the shots have been allocated towards a target. If an ACP has already reached target lock, an enum value of *LOCKED_ON* can be used to better describe the allocated shots status.

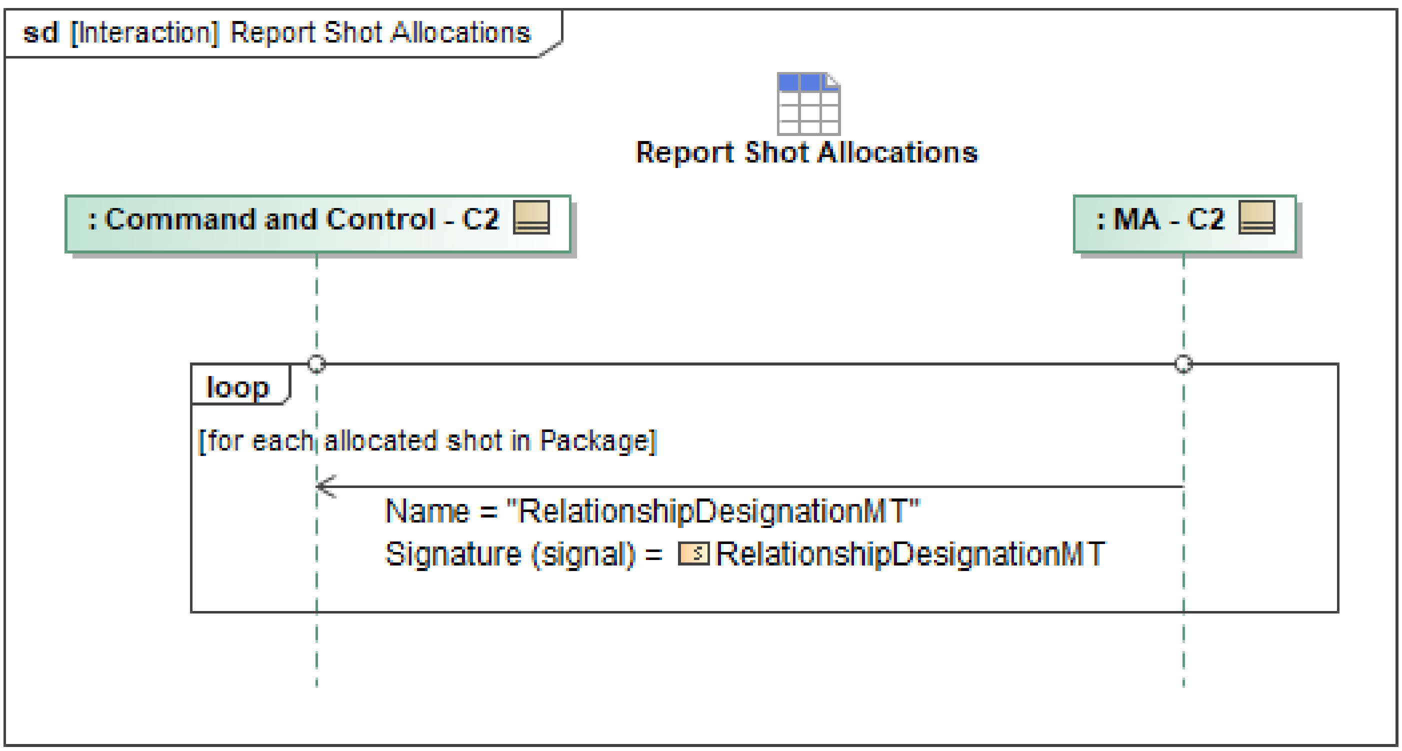


Figure 1-196: Report Shot Allocations

Table 1-195: Report Shot Allocations

Message Name (Name:Type)	Required Fields (Name:Type)
RelationshipDesignationMT: RelationshipDesignationMT	EngagementStatus: ExternalCommandExecutionStateEnum Destination: GeoLocatedObjectType WEAPON_ASSIGNED:

Message Name (Name:Type)	Required Fields (Name:Type)
	Relationship: RelationshipType EntityID: EntityID_Type

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.9 Store Inventory and Descriptions Exchange

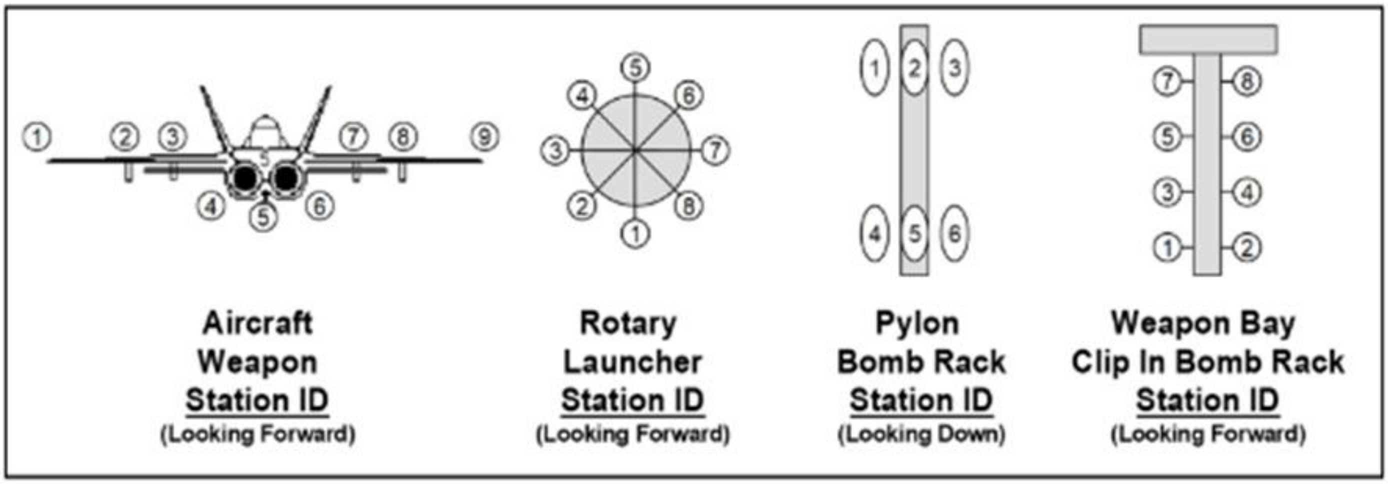


Figure 6.01R.13.00-1 - Examples of Station ID [01R/13/00...07]

Figure 1-197: Examples of Station ID

The supporting weapon subsystems (e.g. Stores Management System, Weapons Resource Manager) within MS are required to publish store inventory data (e.g. installed mission stores and carriages) to MA using *MA_StrikeCapabilityMT*. MA will monitor and forward the periodic *MA_StrikeCapabilityMT* message to determine the inventory of the installed store types, station locations and installed capabilities. MS will assign a capability ID for each installed mission store. The supporting capability is related to the carriage or rack that the store is attached to. MA can request the weapon subsystem via MS to verify the reported inventory is compatible with the platform using *StoreManagementCommandMT*. C2 can also initiate this request by sending a *StoreManagementCommandMT*, which MA will forward to MS. MA receives a response of *StoreManagementCommandStatusMT* from MS and should forward this to C2. Store compatibility infers the platform has completed weapon system integration testing to include safe separation testing.

This message is used for inventory and mission capability. This message will be updated as required when a tactic is selected, and a Mission Data File (MDF) is loaded. MA can directly forward this information to C2. C2 will remain aware of the mission store inventory and descriptions for overall mission effectiveness by receiving *StoreManagementCommandStatusMT*'s from MA, originating from MS.

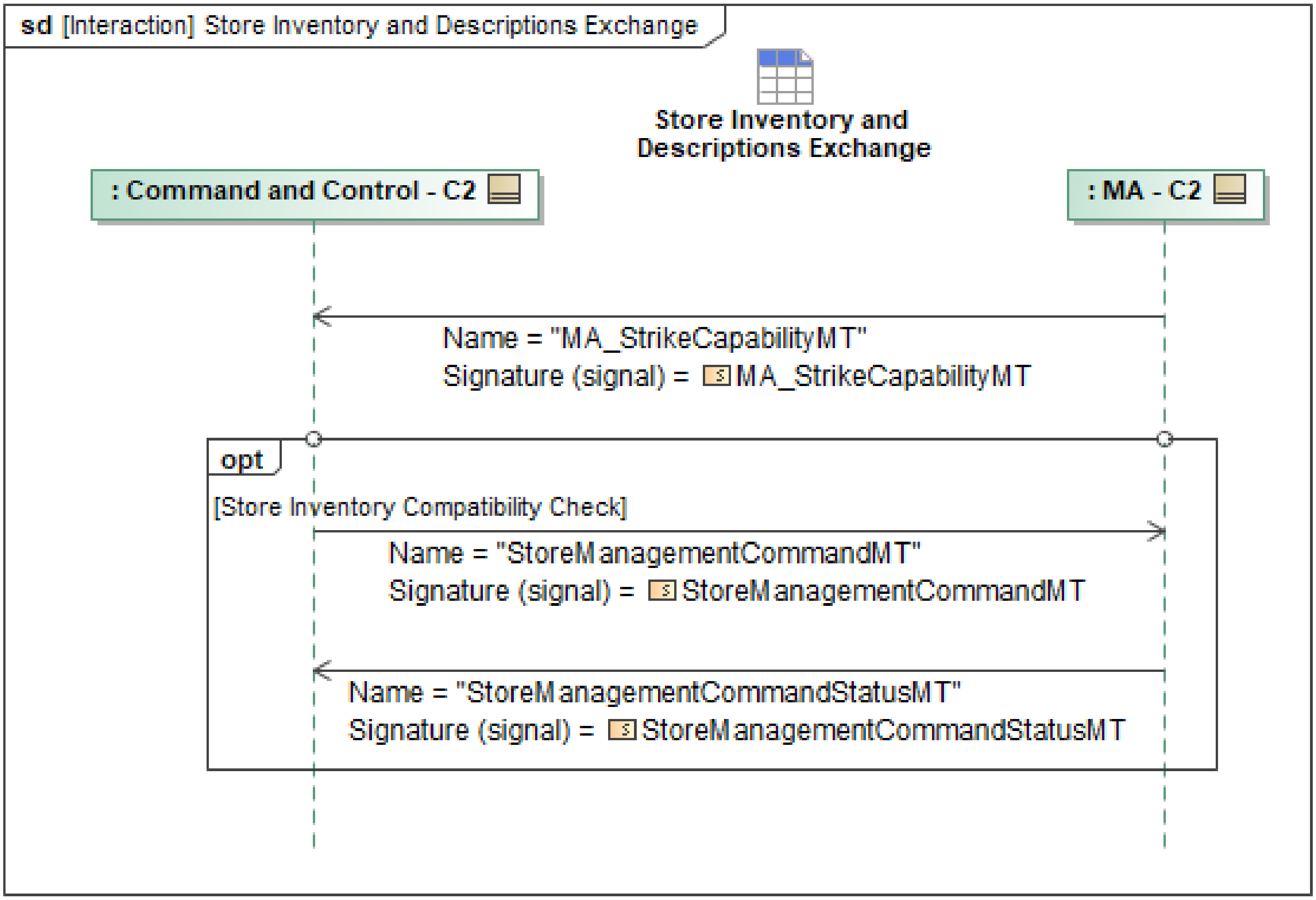


Figure 1-198: Store Inventory and Descriptions Exchange

Table 1-196: Store Inventory and Descriptions Exchange

Message Name (Name:Type)	Required Fields (Name:Type)
MA_StrikeCapabilityMT: MA_StrikeCapabilityMT	Mnemonic: VisibleString256Type StoreTypeVariant: ForeignKeyType Powerable: boolean LoadoutMnemonic: VisibleString256Type Carriage: StoreCarriageCapabilityType
StoreManagementCommandMT: StoreManagementCommandMT	None
StoreManagementCommandStatusMT: StoreManagementCommandStatusMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.10 Target Designation

The Target Designation sequence diagram specifies the interaction where C2 designates one or more specific entities as a target. C2 sends *MA_DesignationMT* to MA, populating one or more *DesignationData* for the systems to receive a target designation. If an entity is to be designated as a target, the *DesignationEnum* within *TypeOfDesignation* will be set to *DESIGNATED_TARGET*. The *AppliesToID* field will specify the System ID or Package ID which shall be assigned to the target. The use of *PackageID* allows a Lead ACP to delegate the target assignment, whereas *SystemID* allows for direct assignment to a specific ACP. The optional *DesignationObjective* field allows for specification of a Requirement that the ACP should use against the target. To acknowledge receipt of the message, MA responds with the *MA_SystemNotificationMT* message, referencing the *SystemSubjectID*, *SystemPerspective*, and *AssociatedMessage* fields. The *SystemSubjectID* shall be set to the MA's System ID, the *SystemPerspective* enumeration shall be set to *RESPONSE_SUBJECT*, and *AssociatedMessage* shall define the received message's *MessageTypeEnum* as *DESIGNATION* and the *AssociatedID* shall be set to the *DesignationID*.

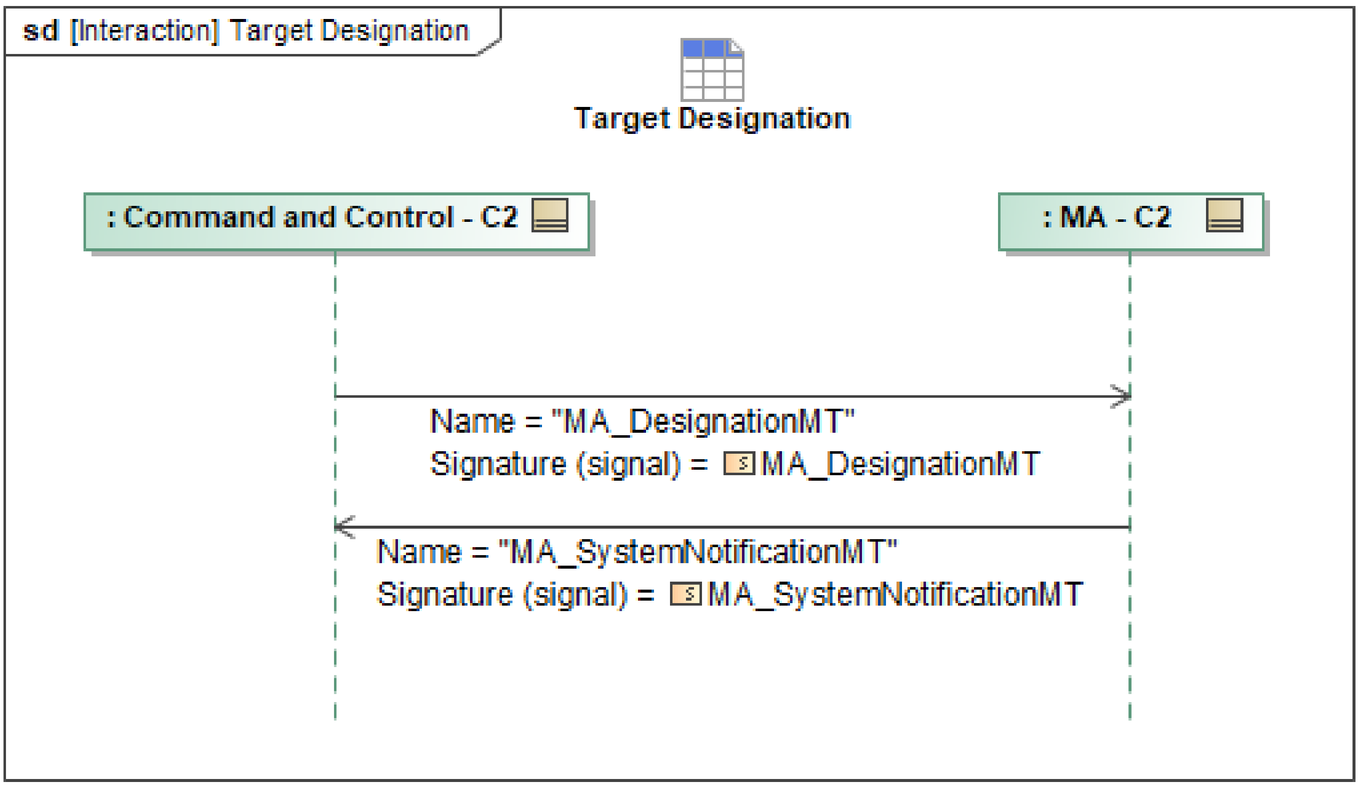


Figure 1-199: Target Designation

Table 1-197: Target Designation

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DesignationMT: MA_DesignationMT	None
MA_SystemNotificationMT: MA_SystemNotificationMT	AssociatedMessage: AssociatedMessageType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.11 Target Designation Request

The Target Designation Request sequence diagram specifies the interaction that happens after MA identifies one or more targets and sends a *MA_DesignationRequestMT* to C2 to be Designated. The Target Designation Criteria can be used to identify Targets for engagement, and MA would send *MA_DesignationRequestMT* to an authorized C2 that is determined using the Target Authority Criteria. The *MA_DesignationRequestMT* contains one or more *DesignationData* that contains the target designation information. The *DesignationID* field is a specific identifier for that specific target. The Target field can be set to many items, but for an engagement, the Target Designation Criteria is only applicable to *EntityMTs*. The *TypeOfDesignation* field would likely be set to *DESIGNATED_TARGET* but could also be set to *DESIGNATED_NO_TARGET* if the service determines the target should no longer be a target. The *AppliesToID* field will specify the System ID or Package ID which shall be assigned to the target. The use of *PackageID* allows a Lead ACP to delegate the target assignment, whereas *SystemID* allows for direct assignment to a specific ACP. The optional *DesignationObjective* field allows for specification of a Requirement that the ACP should use against the target. For an engagement, these Requirement types would match what is in the *ApprovalPolicyMT* that is linked to the Engagement Criteria Target Authority. C2 can then use the information from the *DesignationData* field to populate a *MA_DesignationMT* message and send it back to the MA. C2 can update the information as needed. C2 will then send a *DesignationRequestStatusMT* populating the *RequestID* field with the same ID as was in the request. The *RequestProcessingState* field will be filled out as needed with the main options being *COMPLETED* (if a *MA_DesignationMT* was sent) or *REJECTED* (if C2 did not send a *MA_DesignationMT* message). The *DesignationDataStatus* field will be populated with the relevant target data from the request message. Including the associated *DesignationID* and the *RequestProcessingState* with similar options as stated above. There is an optional *RequestProcessingStateReason* field that can be filled out if desired.

If C2 is unable to complete the request it will ignore the *MA_DesignationRequestMT* command and reply with *MA_DesignationRequestStatusMT_4* where *RequestProcessingState* is set to "*REJECTED*". If the request is rejected because C2 does not have the request information, the *Reason* field within *RequestProcessingStateReason* will be set to "*UNKNOWN_ID"*. If the request is rejected because the missing requested information cannot be transmitted via UIC, the *Reason* field will be set instead to "*INVALID_INPUT_PARAMETER"*.

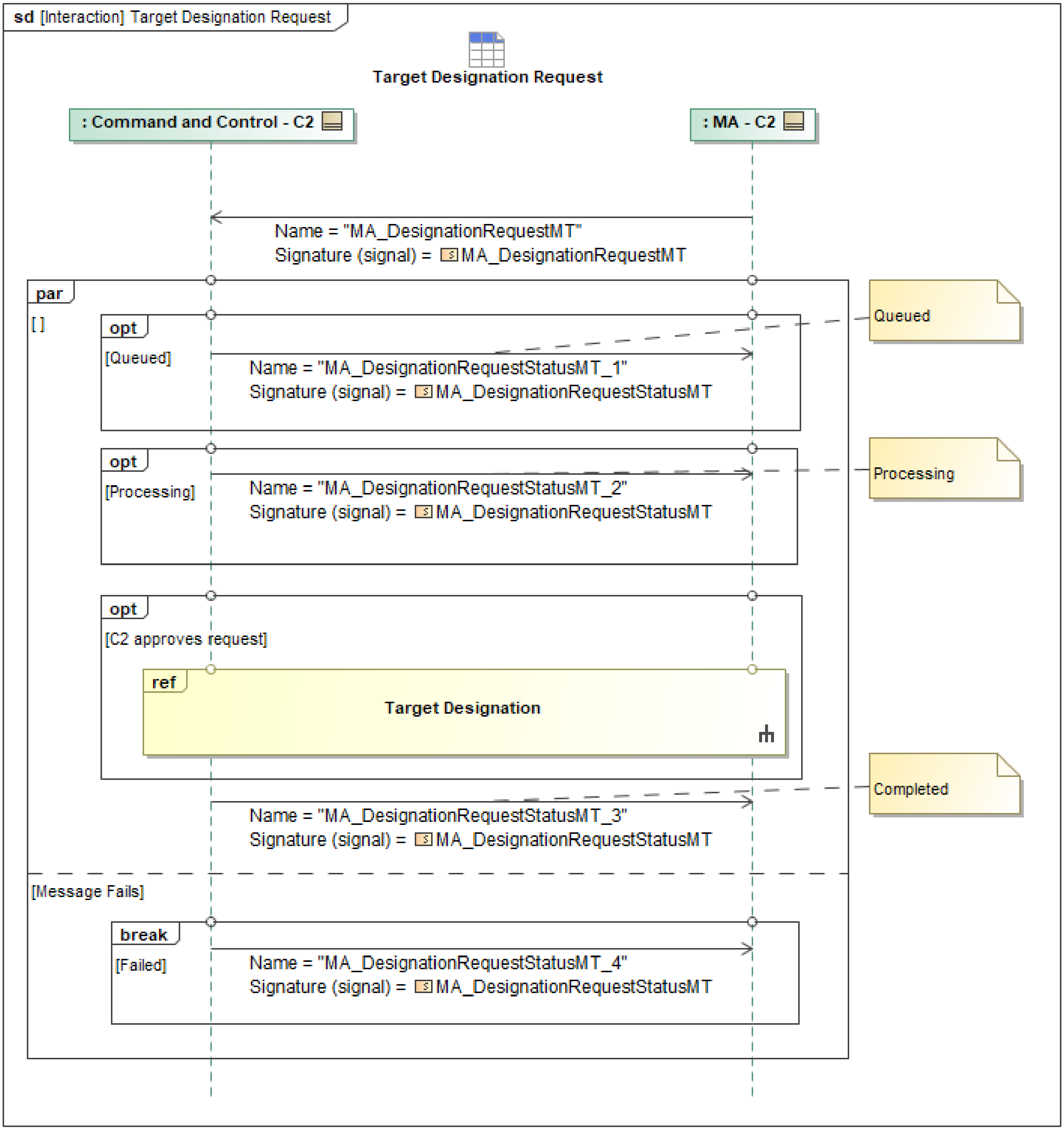


Figure 1-200: Target Designation Request

Table 1-198: Target Designation Request

Message Name (Name:Type)	Required Fields (Name:Type)
MA_DesignationRequestMT: MA_DesignationRequestMT	None
MA_DesignationRequestStatusMT_1: MA_DesignationRequestStatusMT	None
MA_DesignationRequestStatusMT_2: MA_DesignationRequestStatusMT	None
MA_DesignationRequestStatusMT_3: MA_DesignationRequestStatusMT	None
MA_DesignationRequestStatusMT_4: MA_DesignationRequestStatusMT	RequestProcessingStateReason: CannotComplyType

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.12 Threat Prioritization

Mission Use Case: ALR

The Threat Prioritization List ranks entities by assessing the movements and behaviors to determine threat levels to a respective position, whether it is the ACP ownship or another platform. Threat Prioritization will utilize *MA_PrioritizationListMT* with a *ListType* of *Threat_Target_Prioritization_List* enumeration. The list requires subjects (*EntityID*) to be assigned a priority, where the lowest assigned value is the highest threat level.

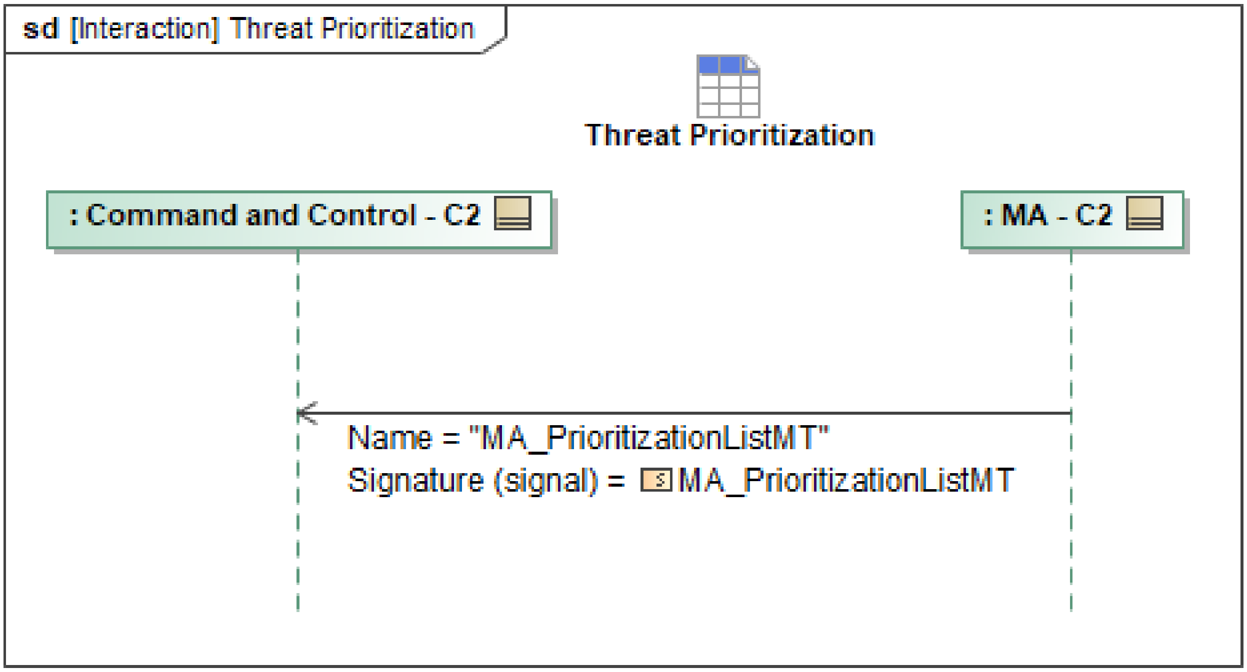


Figure 1-201: Threat Prioritization

Table 1-199: Threat Prioritization

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PrioritizationListMT: MA_PrioritizationListMT	EntityID: EntityID_Type THREAT_TARGET_PRIORITIZATION_LIST:

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.13 Update Shot Queue Prioritization

Shot Queue Prioritization List is a list of mission targets ranked by assessing multiple factors to determine which target entities should be engaged first. The shot queue will utilize *MA_PrioritizationListMT* with a ListType of *Mission_Target_List (enum)*. The list requires subjects (EntityID) to be assigned a priority, where the lowest assigned value is the highest priority level. The message also includes an optional field, *ProbabilityOfKill*, that may be included with each assessed subject entity. C2 may send an updated shot queue prioritization list to an ACP (Single Ship, Multi-Ship Lead, or Multi-Ship Peer). Once an updated *MA_PrioritizationListMT* is received, MA will respond to the originating System with an *MA_SystemNotificationMT*.

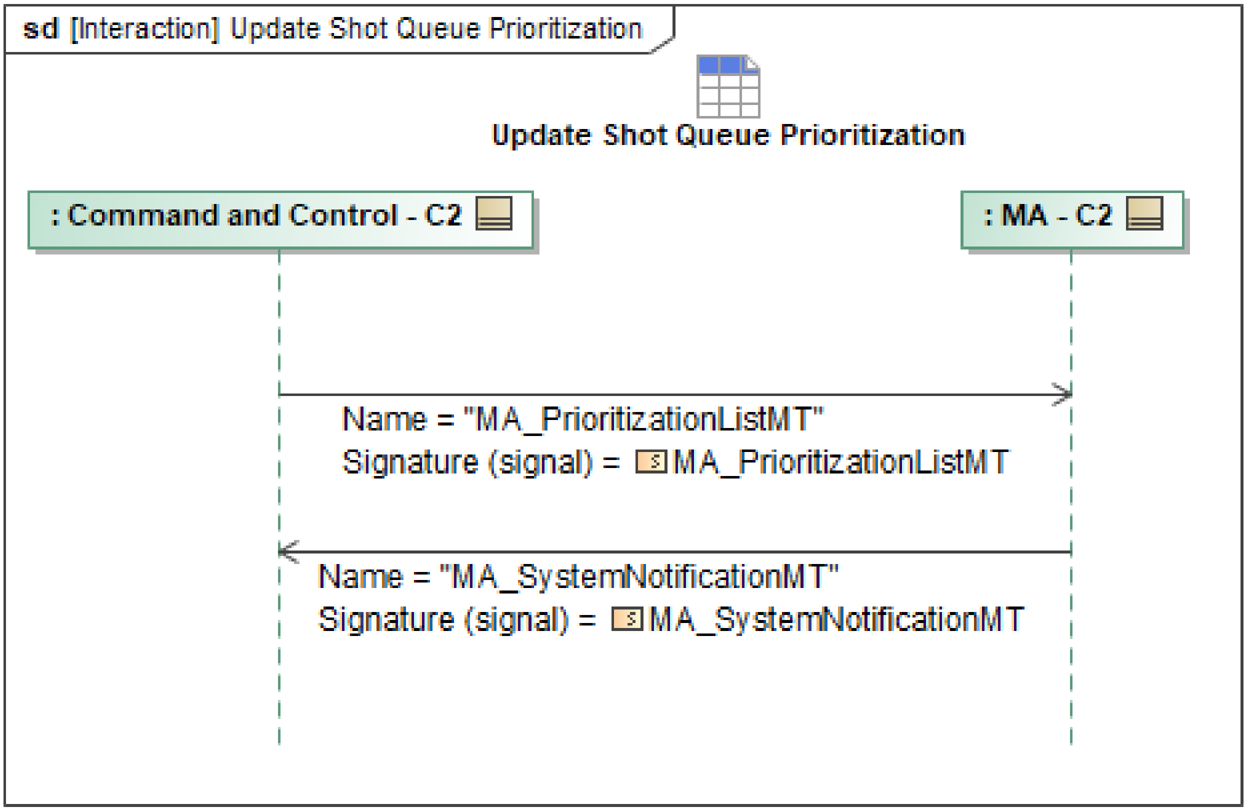


Figure 1-202: Update Shot Queue Prioritization

Table 1-200: Update Shot Queue Prioritization

Message Name (Name:Type)	Required Fields (Name:Type)
MA_PrioritizationListMT: MA_PrioritizationListMT	MISSION_TARGET_LIST: ObjectState: ObjectStateEnum
MA_SystemNotificationMT: MA_SystemNotificationMT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.2.15.14 WEZ Compound Convergence

WEZ Compound Convergence is a fusion of multiple WEZ from team members within a flight group. The fusion of WEZ enables C2 to generate a composite view of flight group's weapon engagement capabilities to assess tasking for the best shot. Once ACPs are producing a WEZ in support of strike task, MA will send *MA_WEZ_MT* periodically to C2.

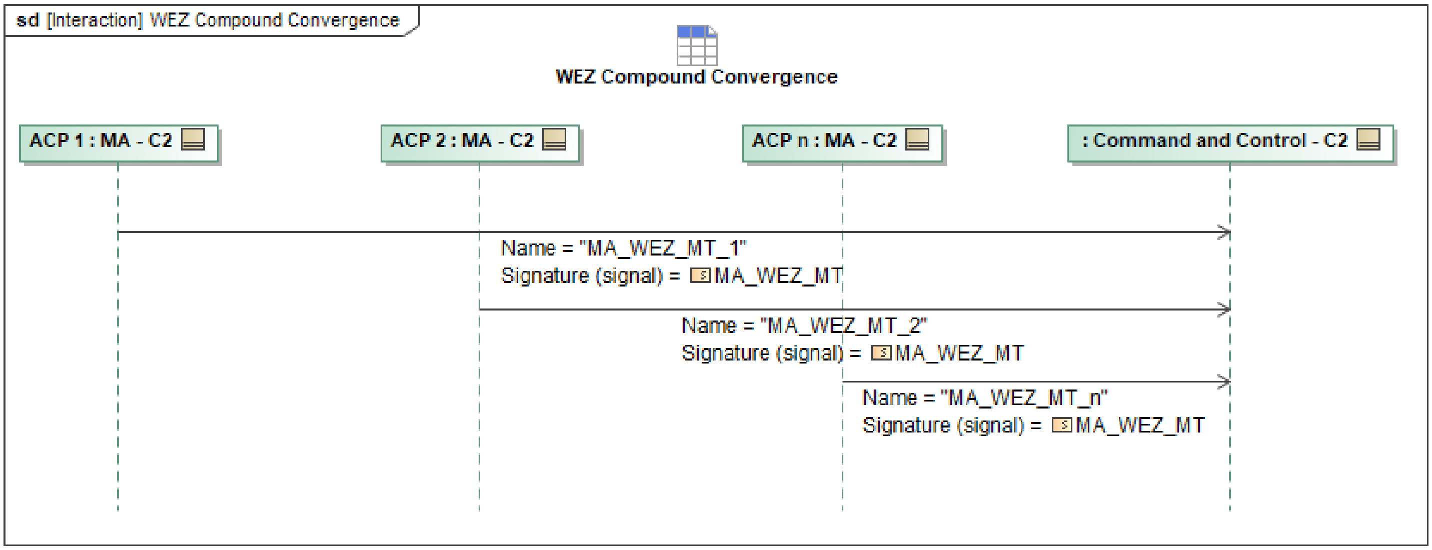


Figure 1-203: WEZ Compound Convergence

Table 1-201: WEZ Compound Convergence

Message Name (Name:Type)	Required Fields (Name:Type)
MA_WEZ_MT_1: MA_WEZ_MT	None
MA_WEZ_MT_2: MA_WEZ_MT	None
MA_WEZ_MT_n: MA_WEZ_MT	None

Note: The fields above are required for this use case in addition to what is specified in Autonomy Schema. The exhaustive list of all fields are in the Autonomy Schema. (See Message Set Tables section)

1.3 Interface Compliance

The Interface Compliance table contains the compliance requirements for the L1 C2 interface.

Table 1-202: C2 Interface Compliance Requirements

ID	Requirement
MA-L1-001	A Level 1 C2 interface shall support the C2 Interface Minimum Message Set defined in the C2 MMS table in the C2 feature profile (A-GRA location: 1 Problem Domain > 3 Exchange Items > Minimum Message Set (MMS)) for all messages in Core Mission Use Case sequences and all non-Core Mission Use Case sequences required for platform specific implementations.
MA-L1-002	A Level 1 C2 interface shall implement all required messaging sequences in the C2 feature profile (A-GRA location: 1 Problem Domain > 2 White Box > 3 Functional Analysis > L1 Functional Analysis > L1 Interface Behaviors > C2 Interface Behavior), marked Core and all non-Core Mission Use Case sequences that apply to the platform specific implementation.
MA-L1-017	Upon receipt of a Malformed Message (a message missing expected fields (i.e., required by the behavior but not the UCI schema) or with invalid context-dependent information within the scope of that individual message) from the C2 interface, MA shall respond to the error by notifying C2 of the malformed message and requesting the corrected message according to the Malformed Message Response (C2) sequence.
MA-L1-021	Upon receipt of any message from the C2 interface that is unauthorized according to C2's role and the RBAC configuration, MA shall respond in accordance with the C2 Unauthorization Response sequence, selecting the response message that corresponds to C2's unauthorized message the three tables (Unauthorized ActionRequest-2 Message Pairs, Unauthorized Command-2 Message Pairs, Unauthorized DataRequest-2 Message Pairs). The tables include primitive type, triggering message, and the specific unauthorization response message.
MA-L1-023	A Level 1 C2 interface shall follow state transitions for CommandProcessingState and RequestProcessingState in accordance with OAC-SPC-001 Rev C, Section 5.1.1.1 and 5.1.2.1. Representations of non-terminal statuses are optional and should align with the definitions and usage prescribed by the UCI Schema Style and Design Specification, ensuring consistent message state handling.

1.3.1 C2 MMS

The Minimum Message Set referenced in the Interface Compliance Requirements is shown in this section. For an implementor to be compliant, all messages shown in the table shall be supported. The "Direction" column specifies the message direction relative to the C2 interface. The "MUC" column specifies the message's Mission Use Case. The message and its respective interaction(s) are only required for the particular Mission Use Case listed.
Mission Use Cases (MUCs) identify specific implementation needs for different ACP solutions. These include a Core set of behaviors that all A-GRA-compliant systems must meet, as well as select use cases such as Carrier-based or Electronic Support Measure-capable. MUCs are considered to be L0 as they describe the use of an ACP, not necessarily a single L1 system. For L1 interface compliance, all implementations must meet the interactions delegated to the Core MUC and all interactions delegated to other MUCs relevant to their implementation. Compliance to interactions not delegated to MUCs do not count towards A-GRA compliance.

Table 1-203: C2 MMS

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	**ActionActivityMT**	C2	in	Core
UCI 2.3	**ActionCancelCommandMT**	C2	out	Core
UCI 2.3	**ActionCancelCommandStatusMT**	C2	in	Core
UCI 2.3	**ActionCapabilityStatusMT**	C2	in	Core
UCI 2.3	**ActionPlanStatusMT**	C2	in	Core
UCI 2.3	**ActivityPlanApprovalStatusMT**	C2	in	Core
UCI 2.3	**ActivityPlanExecutionStatusMT**	C2	in	Core
UCI 2.3	**ActivityPlanStatusMT**	C2	in	Core
UCI 2.3	**AirfieldReportMT**	C2	inout	Core
UCI 2.3	**AirSampleActivityMT**	C2	in	Core
UCI 2.3	**AirSampleCommandMT**	C2	out	Core
UCI 2.3	**AirSampleCommandStatusMT**	C2	in	Core
UCI 2.3	**AirSampleSettingsCommandMT**	C2	out	Core
UCI 2.3	**AirSampleSettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**AMTI_ActivityMT**	C2	in	Core
UCI 2.3	**AMTI_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**AMTI_CommandMT**	C2	out	Core
UCI 2.3	**AMTI_CommandStatusMT**	C2	in	Core
UCI 2.3	**AMTI_SettingsCommandMT**	C2	out	Core
UCI 2.3	**AMTI_SettingsCommandStatusMT**	C2	in	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	**AO_ActivityMT**	C2	in	Core
UCI 2.3	**AO_CapabilityMT**	C2	in	Core
UCI 2.3	**AO_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**AO_CommandMT**	C2	out	Core
UCI 2.3	**AO_CommandStatusMT**	C2	in	Core
UCI 2.3	**AO_SettingsCommandMT**	C2	out	Core
UCI 2.3	**AO_SettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**CargoDeliveryCapabilityMT**	C2	in	Core
UCI 2.3	**CargoDeliveryCapabilityStatusMT**	C2	in	Core
UCI 2.3	**COMINT_ActivityMT**	C2	in	Core
UCI 2.3	**COMINT_CapabilityMT**	C2	in	Core
UCI 2.3	**COMINT_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**COMINT_CommandMT**	C2	out	Core
UCI 2.3	**COMINT_CommandStatusMT**	C2	in	Core
UCI 2.3	**COMINT_SettingsCommandMT**	C2	out	Core
UCI 2.3	**COMINT_SettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**CommPointingCommandMT**	C2	out	Core
UCI 2.3	**CommPointingCommandStatusMT**	C2	in	Core
UCI 2.3	**CommRelayActivityMT**	C2	in	Core
UCI 2.3	**CommRelayCommandMT**	C2	out	Core
UCI 2.3	**CommRelayCommandStatusMT**	C2	in	Core
UCI 2.3	**CommRelaySettingsCommandMT**	C2	out	Core
UCI 2.3	**CommRelaySettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**CommTerminalCapabilityMT**	C2	in	Core
UCI 2.3	**CommTerminalCapabilityStatusMT**	C2	in	Core
UCI 2.3	**CommTerminalCommandMT**	C2	out	Core
UCI 2.3	**CommTerminalCommandStatusMT**	C2	in	Core
UCI 2.3	**CommTerminalPlanActivationCommandMT**	C2	out	Core
UCI 2.3	**CommTerminalPlanActivationCommandStatusMT**	C2	in	Core
UCI 2.3	**CommTerminalSettingsCommandMT**	C2	out	Core
UCI 2.3	**CommTerminalSettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**ComponentStatusMT**	C2	in	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	**ControlRequestMT**	C2	out	Core
UCI 2.3	**ControlRequestStatusMT**	C2	in	Core
UCI 2.3	**ControlStatusMT**	C2	in	Core
UCI 2.3	**DLZ_MT**	C2	in	Core
UCI 2.3	**DMPI_CancelCommandMT**	C2	out	Core
UCI 2.3	**DMPI_CancelCommandStatusMT**	C2	in	Core
UCI 2.3	**DoorCommandMT**	C2	out	Core
UCI 2.3	**DoorCommandStatusMT**	C2	in	Core
UCI 2.3	**EA_ActivityMT**	C2	in	Core
UCI 2.3	**EA_CapabilityMT**	C2	in	Core
UCI 2.3	**EA_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**EA_CommandMT**	C2	out	Core
UCI 2.3	**EA_CommandStatusMT**	C2	in	Core
UCI 2.3	**EffectCapabilityStatusMT**	C2	in	Core
UCI 2.3	**EntityMT**	C2	inout	Core
UCI 2.3	**ESM_ActivityMT**	C2	in	Core
UCI 2.3	**ESM_CapabilityMT**	C2	in	Core
UCI 2.3	**ESM_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**ESM_CommandMT**	C2	out	Core
UCI 2.3	**ESM_CommandStatusMT**	C2	in	Core
UCI 2.3	**ESM_SettingsCommandMT**	C2	out	Core
UCI 2.3	**ESM_SettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**GatewayCapabilityMT**	C2	in	Additional Comms Capabilities
UCI 2.3	**GatewayCapabilityStatusMT**	C2	in	Additional Comms Capabilities
UCI 2.3	**IFF_CapabilityMT**	C2	in	Core
UCI 2.3	**IFF_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**IFF_CommandMT**	C2	out	Core
UCI 2.3	**IFF_CommandStatusMT**	C2	in	Core
UCI 2.3	**IFF_SettingsCommandMT**	C2	out	Core
UCI 2.3	**IFF_SettingsCommandStatusMT**	C2	in	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
A-GRA Custom Extension	**MA_AcceptableLevelOfRiskCommandMT**	C2	out	ALR
A-GRA Custom Extension	**MA_AcceptableLevelOfRiskCommandStatusMT**	C2	in	ALR
A-GRA Custom Extension	**MA_AcceptableLevelOfRiskStatusMT**	C2	in	ALR
A-GRA Custom Extension	**MA_ActionCapabilityMT**	C2	in	Core
A-GRA Custom Extension	**MA_ActionCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_ActionCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ActionMT**	C2	inout	Core
A-GRA Custom Extension	**MA_ActionPlanMT**	C2	inout	Core
A-GRA Custom Extension	**MA_ActionStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ActivityPlanMT**	C2	inout	Core
A-GRA Custom Extension	**MA_AMTI_CapabilityMT**	C2	in	Core
A-GRA Custom Extension	**MA_ApprovalAuthorityMT**	C2	inout	Core
A-GRA Custom Extension	**MA_ApprovalPolicyMT**	C2	inout	Core
A-GRA Custom Extension	**MA_ApprovalRequestMT**	C2	in	Core
A-GRA Custom Extension	**MA_ApprovalRequestStatusMT**	C2	out	Core
A-GRA Custom Extension	**MA_AssessmentMT**	C2	in	Core, ALR
A-GRA Custom Extension	**MA_AssessmentRequestMT**	C2	out	Core, ALR
A-GRA Custom Extension	**MA_AssessmentRequestStatusMT**	C2	in	Core, ALR
A-GRA Custom Extension	**MA_AuthorizationMT**	C2	inout	Core
A-GRA Custom Extension	**MA_CAP_ExecutionStatusReportMT**	C2	in	Core
A-GRA Custom Extension	**MA_CapabilitySetMT**	C2	inout	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
A-GRA Custom Extension	**MA_CarrierReportMT**	C2	out	Carrier
A-GRA Custom Extension	**MA_CarrierStateMT**	C2	out	Carrier
A-GRA Custom Extension	**MA_ConstraintPriorityListActivationDataMT**	C2	out	Core
A-GRA Custom Extension	**MA_ConstraintPriorityListActivationStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ConstraintPriorityListCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_ConstraintPriorityListCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ConstraintPriorityListMT**	C2	out	Core
A-GRA Custom Extension	**MA_COP_ConfigurationSettingsCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_COP_ConfigurationSettingsCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_COP_ConfigurationSettingsMT**	C2	in	Core
A-GRA Custom Extension	**MA_COP_ConfigurationSettingsRequestMT**	C2	out	Core
A-GRA Custom Extension	**MA_COP_ConfigurationSettingsRequestStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_DataDisseminationPlanActivationCommandMT**	C2	out	Additional Comms Capabilities
A-GRA Custom Extension	**MA_DataDisseminationPlanActivationCommandStatusMT**	C2	in	Additional Comms Capabilities
A-GRA Custom Extension	**MA_DataDisseminationPlanActivationStatusMT**	C2	in	Additional Comms Capabilities
A-GRA Custom Extension	**MA_DataDisseminationPlanMT**	C2	out	Additional Comms Capabilities
A-GRA Custom Extension	**MA_DataDisseminationPlanOverrideRequestMT**	C2	out	Additional Comms Capabilities
A-GRA Custom Extension	**MA_DataDisseminationPlanOverrideRequestStatusMT**	C2	in	Additional Comms Capabilities
A-GRA Custom	**MA_DesignationMT**	C2	out	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
Extension				
A-GRA Custom Extension	**MA_DesignationRequestMT**	C2	in	Core
A-GRA Custom Extension	**MA_DesignationRequestStatusMT**	C2	out	Core
A-GRA Custom Extension	**MA_EffectCapabilityMT**	C2	in	Core
A-GRA Custom Extension	**MA_EngagementCriteriaCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_EngagementCriteriaCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_EngagementCriteriaMT**	C2	out	Core
A-GRA Custom Extension	**MA_EngagementCriteriaStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ExecutionPriorityListActivationDataMT**	C2	out	Core
A-GRA Custom Extension	**MA_ExecutionPriorityListActivationStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ExecutionPriorityListCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_ExecutionPriorityListCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ExecutionPriorityListMT**	C2	out	Core
A-GRA Custom Extension	**MA_FlightActivityMT**	C2	in	Core
A-GRA Custom Extension	**MA_FlightCapabilityMT**	C2	inout	Core
A-GRA Custom Extension	**MA_FlightCapabilityStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_FlightCommandMT**	C2	out	Core, Carrier
A-GRA Custom Extension	**MA_FlightCommandStatusMT**	C2	in	Core, Carrier
A-GRA Custom Extension	**MA_GatewayActivityMT**	C2	inout	Additional Comms Capabilities
A-GRA Custom Extension	**MA_IgnoreConstraintCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_IgnoreConstraintCommandStatusMT**	C2	in	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
A-GRA Custom Extension	**MA_InteroperabilityPolicyMT**	C2	out	Core
A-GRA Custom Extension	**MA_LeadershipMetricsMT**	C2	out	Core
A-GRA Custom Extension	**MA_MessageAcknowledgmentConfigurationMT**	C2	inout	Core
A-GRA Custom Extension	**MA_MessageAcknowledgmentConfigurationRequestMT**	C2	inout	Core
A-GRA Custom Extension	**MA_MessageAcknowledgmentConfigurationRequestStatusMT**	C2	inout	Core
A-GRA Custom Extension	**MA_MissionPlanActivationCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_MissionPlanActivationCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_MissionPlanActivationStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_MissionPlanCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_MissionPlanCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_MissionPlanExecutionStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_MissionPlanMT**	C2	inout	Core
A-GRA Custom Extension	**MA_OpCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_OpCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_OperatorRoleMT**	C2	inout	Core
A-GRA Custom Extension	**MA_OpLineMT**	C2	inout	Core
A-GRA Custom Extension	**MA_OpPlanMT**	C2	out	Core
A-GRA Custom Extension	**MA_OpPointMT**	C2	inout	Core
A-GRA Custom Extension	**MA_OpRoutingMT**	C2	inout	Core
A-GRA Custom Extension	**MA_OpStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_OpVolumeMT**	C2	inout	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
A-GRA Custom Extension	**MA_OpZoneMT**	C2	inout	Core
A-GRA Custom Extension	**MA_PackageManagementCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_PackageManagementCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_PackagePartnerStatusReportMT**	C2	in	Core
A-GRA Custom Extension	**MA_PlanModificationRequestMT**	C2	in	Core
A-GRA Custom Extension	**MA_PlanModificationRequestStatusMT**	C2	out	Core
A-GRA Custom Extension	**MA_PlanningFunctionMT**	C2	in	Core
A-GRA Custom Extension	**MA_PlanningFunctionSettingsCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_PlanningFunctionSettingsCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_PlanningFunctionStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_PlatformOverrideCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_PlatformOverrideCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_PlatformOverrideStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_PositionReportDetailedMT**	C2	in	Core, Carrier
A-GRA Custom Extension	**MA_PositionReportMT**	C2	in	Core, Carrier
A-GRA Custom Extension	**MA_PrioritizationListMT**	C2	inout	ALR, Core
A-GRA Custom Extension	**MA_ResponseActivityMT**	C2	in	Core
A-GRA Custom Extension	**MA_ResponseCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_ResponseCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_ResponseMT**	C2	inout	Core
A-GRA Custom Extension	**MA_ResponsePlanCommandMT**	C2	out	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
A-GRA Custom Extension	**MA_ResponsePlanCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_RouteActivityPlanMT**	C2	inout	Core
A-GRA Custom Extension	**MA_RoutePlanMT**	C2	inout	Core, Carrier
A-GRA Custom Extension	**MA_StrikeActivityMT**	C2	in	Core
A-GRA Custom Extension	**MA_StrikeCapabilityMT**	C2	in	Core
A-GRA Custom Extension	**MA_SystemManagementRequestMT**	C2	out	Core, Carrier
A-GRA Custom Extension	**MA_SystemManagementRequestStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_SystemNotificationMT**	C2	inout	Core, Carrier, ALR
A-GRA Custom Extension	**MA_SystemsNeededRequestMT**	C2	in	Core
A-GRA Custom Extension	**MA_SystemsNeededRequestStatusMT**	C2	out	Core
A-GRA Custom Extension	**MA_TaskCommandMT**	C2	out	Core
A-GRA Custom Extension	**MA_TaskCommandStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_TaskMT**	C2	inout	Core, Carrier
A-GRA Custom Extension	**MA_TaskPlanMT**	C2	inout	Core, Carrier
A-GRA Custom Extension	**MA_TaskStatusMT**	C2	in	Core, Carrier
A-GRA Custom Extension	**MA_WEZ_MT**	C2	in	Core
UCI 2.3	**MissionContingencyAlertMT**	C2	in	Core
UCI 2.3	**MissionPlanApprovalStatusMT**	C2	in	Core
UCI 2.3	**MissionPlanStatusMT**	C2	in	Core
A-GRA Custom Extension	**MA_NavigationReportMT**	C2	in	Core, Carrier
UCI 2.3	**ObservationMeasurementReportMT**	C2	in	Core
UCI 2.3	**OpaqueCapabilityMT**	C2	in	Core
UCI 2.3	**OpaqueCapabilityStatusMT**	C2	in	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	**OperatorMT**	C2	inout	Core
UCI 2.3	**OperatorNotificationAckMT**	C2	out	Carrier
UCI 2.3	**OperatorNotificationMT**	C2	in	Carrier, Core
UCI 2.3	**OpNotificationMT**	C2	in	Core
UCI 2.3	**PackageMT**	C2	inout	Core, Carrier
UCI 2.3	**PackageStatusMT**	C2	in	Core
UCI 2.3	**PO_ActivityMT**	C2	in	Core
UCI 2.3	**PO_CapabilityMT**	C2	in	Core
UCI 2.3	**PO_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**PO_CommandMT**	C2	out	Core
UCI 2.3	**PO_CommandStatusMT**	C2	in	Core
UCI 2.3	**PO_SettingsCommandMT**	C2	out	Core
UCI 2.3	**PO_SettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**QueryDataRequestMT**	C2	inout	Core
UCI 2.3	**QueryDataRequestStatusMT**	C2	inout	Core
UCI 2.3	**RadarAltimeterCapabilityMT**	C2	in	Core
UCI 2.3	**RadarAltimeterCapabilityStatusMT**	C2	in	Core
UCI 2.3	**RadarAltimeterCommandMT**	C2	out	Core
UCI 2.3	**RadarAltimeterCommandStatusMT**	C2	in	Core
UCI 2.3	**RadarAltimeterSettingsCommandMT**	C2	out	Core
UCI 2.3	**RadarAltimeterSettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**ReferenceFrameMT**	C2	in	Carrier
UCI 2.3	**RefuelCapabilityMT**	C2	in	Core
UCI 2.3	**RefuelCapabilityStatusMT**	C2	in	Core
UCI 2.3	**RefuelCommandMT**	C2	out	Core
UCI 2.3	**RefuelCommandStatusMT**	C2	in	Core
UCI 2.3	**RefuelSettingsCommandMT**	C2	out	Core
UCI 2.3	**RefuelSettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**RelationshipDesignationMT**	C2	in	Core
UCI 2.3	**ResponsePlanApprovalStatusMT**	C2	in	Core
UCI 2.3	**ResponsePlanExecutionStatusMT**	C2	in	Core
UCI 2.3	**ResponsePlanMT**	C2	inout	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	**RouteActivityPlanExecutionStatusMT**	C2	in	Core
UCI 2.3	**RouteActivityPlanStatusMT**	C2	in	Core
UCI 2.3	**RouteMetricsMT**	C2	inout	Core
UCI 2.3	**RouteMetricsRequestMT**	C2	out	Core
UCI 2.3	**RouteMetricsRequestStatusMT**	C2	in	Core
UCI 2.3	**RouteModificationRequestMT**	C2	in	Core
UCI 2.3	**RouteModificationRequestStatusMT**	C2	out	Core
UCI 2.3	**RoutePlanApprovalStatusMT**	C2	in	Core
UCI 2.3	**RoutePlanExecutionStatusMT**	C2	in	Core
UCI 2.3	**RoutePlanStatusMT**	C2	in	Core
UCI 2.3	**SAR_ActivityMT**	C2	in	Core
UCI 2.3	**SAR_CapabilityMT**	C2	in	Core
UCI 2.3	**SAR_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**SAR_CommandMT**	C2	out	Core
UCI 2.3	**SAR_CommandStatusMT**	C2	in	Core
UCI 2.3	**SAR_SettingsCommandMT**	C2	out	Core
UCI 2.3	**SAR_SettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**ServiceConfigurationChangeRequestMT**	C2	out	Carrier
UCI 2.3	**SMTI_ActivityMT**	C2	in	Core
UCI 2.3	**SMTI_CapabilityMT**	C2	in	Core
UCI 2.3	**SMTI_CapabilityStatusMT**	C2	in	Core
UCI 2.3	**SMTI_CommandMT**	C2	out	Core
UCI 2.3	**SMTI_CommandStatusMT**	C2	in	Core
UCI 2.3	**SMTI_SettingsCommandMT**	C2	out	Core
UCI 2.3	**SMTI_SettingsCommandStatusMT**	C2	in	Core
UCI 2.3	**StoreManagementCommandMT**	C2	out	Core
UCI 2.3	**StoreManagementCommandStatusMT**	C2	in	Core
UCI 2.3	**StrikeCapabilityStatusMT**	C2	in	Core
UCI 2.3	**StrikeCommandMT**	C2	out	Core
UCI 2.3	**StrikeCommandStatusMT**	C2	in	Core
UCI 2.3	**StrikeConsentRequestMT**	C2	in	Core
UCI 2.3	**StrikeConsentRequestStatusMT**	C2	out	Core

Message Type	Message Name	Interface	Direction	Applicable MUCs
UCI 2.3	**SubsystemStatusMT**	C2	in	Core
UCI 2.3	**SurvivabilityRiskLevelMT**	C2	out	ALR
UCI 2.3	**SystemDeploymentCapabilityMT**	C2	in	Core
UCI 2.3	**SystemDeploymentCapabilityStatusMT**	C2	in	Core
UCI 2.3	**SystemStatusMT**	C2	inout	Core
UCI 2.3	**TacticalOrderCapabilityMT**	C2	in	Core
UCI 2.3	**TacticalOrderCapabilityStatusMT**	C2	in	Core
UCI 2.3	**TaskCancelCommandMT**	C2	out	Core
UCI 2.3	**TaskCancelCommandStatusMT**	C2	in	Core
UCI 2.3	**TaskPlanApprovalStatusMT**	C2	in	Core
UCI 2.3	**TaskPlanExecutionStatusMT**	C2	in	Core
UCI 2.3	**TaskPlanStatusMT**	C2	in	Core
UCI 2.3	**TurretCommandMT**	C2	out	Core
UCI 2.3	**TurretCommandStatusMT**	C2	in	Core
UCI 2.3	**WeatherDatasetMT**	C2	out	Core
UCI 2.3	**WeatherMT**	C2	in	Core
UCI 2.3	**WeatherRadarActivityMT**	C2	in	Core
UCI 2.3	**WeatherRadarCapabilityMT**	C2	in	Core
UCI 2.3	**WeatherRadarCapabilityStatusMT**	C2	in	Core
UCI 2.3	**WeatherRadarCommandMT**	C2	out	Core
UCI 2.3	**WeatherRadarCommandStatusMT**	C2	in	Core
UCI 2.3	**WeatherRadarSettingsCommandMT**	C2	out	Core
UCI 2.3	**WeatherRadarSettingsCommandStatusMT**	C2	in	Core

1.3.2 C2 UCI Extensions

The tables in the following sections provide the specifications for extended UCI 2.3 messages utilized by the A-GRA to communicate Autonomy L1 interface interactions. The tables include both extended messages and message types. Each table includes the message name, the functional description, and fields contained within. Top-level messages include images of the nested message structure. For a better view of the diagram and the message structure, it is strongly recommended to look at the schema in a XML viewer tool.

1.3.2.1MA_FlightCapabilityMT UCI Extension

MA_FlightCapabilityMT	
Description	Extended to indicate the type of flight capability via the FlightCapabilityEnum and provide per-

	control mode performance parameters. See the XSD File for more info
Ext. Fields	HSA_CSA_PerformanceProfile : MA_FlightControlModesPerformanceProfileType WaypointFollowingPerformanceProfile : MA_FlightControlModesPerformanceProfileType CurveFollowingPerformanceProfile : MA_FlightControlModesPerformanceProfileType AccelerationLimit : MA_BodyReferenceAccelerationType AccelerationLimitValue : MA_AccelerationLimitPairType BodyReferenceOrientationRate : MA_BodyReferenceOrientationRateType OrientationRateLimits : MA_BodyReferenceOrientationRateType AirspeedPair : PathSegmentSpeedValueType

Table A-1-204: C2 UCI Extensions MA_FlightCapabilityMT UCI Extension

HSA_CSA_PerformanceProfile		
Description	Indicates performance profile for HSA_CSA flight control mode.	
Base Type	MA_FlightControlModesPerformanceProfileType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MinAirspeed	MA_AirspeedLimitType	Repeated
Description	Minimum allowable airspeed given as a function of altitude.	
MaxAirspeed	MA_AirspeedLimitType	Repeated
Description	Maximum allowable airspeed given as a function of altitude.	
BestEnduranceAirspeed	MA_AirspeedLimitType	Repeated
Description	Airspeed of maximum endurance given as a function of altitude.	
BestRangeAirspeed	MA_AirspeedLimitType	Repeated
Description	Airspeed of maximum range given as a function of altitude.	
MinAltitude	AltitudeReferenceType	Optional
Description	Minimum allowable altitude.	
MaxAltitude	AltitudeReferenceType	Optional
Description	Maximum allowable altitude.	
MinAccelerationLimits	MA_AccelerationLimitsType	Repeated
Description	The vehicle's acceleration limits in the current operating domain of the system as a function of time. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.	
ExcessPowerOrAcceleration	MA_ExcessPowerOrAccelerationType	Optional
Description	Provides either specific excess power or paired acceleration and climb rate data for	

	the platform.	
MaxOrientationLimits	MA_OrientationLimitType	Repeated
Description	Euler Angle Sequence describing the maximum orientation of the vehicle in the order yaw, pitch, roll as a function of airspeed, altitude, and weight.	
MaxOrientationRateLimits	MA_OrientationRateLimitsType	Repeated
Description	Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.	
MaxTurnRate	AngleRateType	Optional
Description	Maximum allowable turn rate in radians/sec.	
MaxDescentRate	MA_SpeedType	Repeated
Description	Maximum allowable descent rate as a function of airspeed, altitude, and weight.	
MaxDeceleration	MA_AccelerationLimitsType	Repeated
Description	Maximum allowable deceleration limits as a function of airspeed, altitude, and weight.	
FuelBurnRate	MA_FuelBurnRateType	Repeated
Description	Fuel burn rate as a function of airspeed, altitude, and weight.	

Table A-1-205: C2 UCI Extensions MA_FlightCapabilityMT - HSA_CSA_PerformanceProfile UCI Extension Field

WaypointFollowingPerformanceProfile		
Description	Indicates performance profile for Waypoint Following flight control mode.	
Base Type	MA_FlightControlModesPerformanceProfileType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MinAirspeed	MA_AirspeedLimitType	Repeated
Description	Minimum allowable airspeed given as a function of altitude.	
MaxAirspeed	MA_AirspeedLimitType	Repeated
Description	Maximum allowable airspeed given as a function of altitude.	
BestEnduranceAirspeed	MA_AirspeedLimitType	Repeated
Description	Airspeed of maximum endurance given as a function of altitude.	
BestRangeAirspeed	MA_AirspeedLimitType	Repeated
Description	Airspeed of maximum range given as a function of altitude.	
MinAltitude	AltitudeReferenceType	Optional
Description	Minimum allowable altitude.	
MaxAltitude	AltitudeReferenceType	Optional
Description	Maximum allowable altitude.	

MinAccelerationLimits	MA_AccelerationLimitsType	Repeated
Description	The vehicle's acceleration limits in the current operating domain of the system as a function of time. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.	
ExcessPowerOrAcceleration	MA_ExcessPowerOrAccelerationType	Optional
Description	Provides either specific excess power or paired acceleration and climb rate data for the platform.	
MaxOrientationLimits	MA_OrientationLimitType	Repeated
Description	Euler Angle Sequence describing the maximum orientation of the vehicle in the order yaw, pitch, roll as a function of airspeed, altitude, and weight.	
MaxOrientationRateLimits	MA_OrientationRateLimitsType	Repeated
Description	Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.	
MaxTurnRate	AngleRateType	Optional
Description	Maximum allowable turn rate in radians/sec.	
MaxDescentRate	MA_SpeedType	Repeated
Description	Maximum allowable descent rate as a function of airspeed, altitude, and weight.	
MaxDeceleration	MA_AccelerationLimitsType	Repeated
Description	Maximum allowable deceleration limits as a function of airspeed, altitude, and weight.	
FuelBurnRate	MA_FuelBurnRateType	Repeated
Description	Fuel burn rate as a function of airspeed, altitude, and weight.	

Table A-1-206: C2 UCI Extensions MA_FlightCapabilityMT - WaypointFollowingPerformanceProfile UCI Extension Field

CurveFollowingPerformanceProfile		
Description	Indicates performance profile for Curve Following flight control mode.	
Base Type	MA_FlightControlModesPerformanceProfileType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MinAirspeed	MA_AirspeedLimitType	Repeated
Description	Minimum allowable airspeed given as a function of altitude.	
MaxAirspeed	MA_AirspeedLimitType	Repeated
Description	Maximum allowable airspeed given as a function of altitude.	
BestEnduranceAirspeed	MA_AirspeedLimitType	Repeated
Description	Airspeed of maximum endurance given as a function of altitude.	

BestRangeAirspeed	MA_AirspeedLimitType	Repeated
Description	Airspeed of maximum range given as a function of altitude.	
MinAltitude	AltitudeReferenceType	Optional
Description	Minimum allowable altitude.	
MaxAltitude	AltitudeReferenceType	Optional
Description	Maximum allowable altitude.	
MinAccelerationLimits	MA_AccelerationLimitsType	Repeated
Description	The vehicle's acceleration limits in the current operating domain of the system as a function of time. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.	
ExcessPowerOrAcceleration	MA_ExcessPowerOrAccelerationType	Optional
Description	Provides either specific excess power or paired acceleration and climb rate data for the platform.	
MaxOrientationLimits	MA_OrientationLimitType	Repeated
Description	Euler Angle Sequence describing the maximum orientation of the vehicle in the order yaw, pitch, roll as a function of airspeed, altitude, and weight.	
MaxOrientationRateLimits	MA_OrientationRateLimitsType	Repeated
Description	Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle. Flight control modes such as HSA and Waypoint Following should only use the first time index, while Curve Following may utilize an unbounded number of time indices to form dynamically feasible trajectories.	
MaxTurnRate	AngleRateType	Optional
Description	Maximum allowable turn rate in radians/sec.	
MaxDescentRate	MA_SpeedType	Repeated
Description	Maximum allowable descent rate as a function of airspeed, altitude, and weight.	
MaxDeceleration	MA_AccelerationLimitsType	Repeated
Description	Maximum allowable deceleration limits as a function of airspeed, altitude, and weight.	
FuelBurnRate	MA_FuelBurnRateType	Repeated
Description	Fuel burn rate as a function of airspeed, altitude, and weight.	

Table A-1-207: C2 UCI Extensions MA_FlightCapabilityMT - CurveFollowingPerformanceProfile UCI Extension Field

AccelerationLimit		
Description	Specified acceleration limits in body reference or NED coordinate frame.	
Base Type	MA_BodyReferenceAccelerationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

X_Accel	AccelerationType	Required
Description	Body-referenced acceleration component along the longitudinal (x or roll) axis, measured in meters/sec^2	
Y_Accel	AccelerationType	Required
Description	Body-referenced acceleration component along the lateral (y or pitch) axis, measured in meters/sec^2	
Z_Accel	AccelerationType	Required
Description	Body-referenced acceleration component along the vertical (z or yaw) axis, measured in meters/sec^2.	
Timestamp	DateTimeType	Optional
Description	Indicates the time when the acceleration data was measured or last known to be correct. The element can also indicate the time at which a future velocity of a System, Entity, etc. will be achieved.	

Table A-1-208: C2 UCI Extensions MA_FlightCapabilityMT - AccelerationLimit UCI Extension Field

AccelerationLimitValue		
Description	Value for the specified acceleration limit, provided either as a set of body referenced orientation rates, or a unitless Mach value.	
Base Type	MA_AccelerationLimitPairType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
BodyReferenceOrientationRate	MA_BodyReferenceOrientationRateType	Required
Description	Body reference orientation rates to be paired with the specified acceleration limit.	
MachValue	MachType	Required
Description	Unitless Mach value to be paired with the specified acceleration limit.	

Table A-1-209: C2 UCI Extensions MA_FlightCapabilityMT - AccelerationLimitValue UCI Extension Field

BodyReferenceOrientationRate		
Description	Body reference orientation rates to be paired with the specified acceleration limit.	
Base Type	MA_BodyReferenceOrientationRateType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RollRate	AngleRateType	Required
Description	Body-referenced attitude rate component along the longitudinal (x or roll) axis, measured in radians/sec.	
PitchRate	AngleRateType	Required
Description	Body-referenced attitude rate component along the lateral (y or pitch) axis, measured	

	in radians/sec.	
YawRate	AngleRateType	Required
Description	Body-referenced attitude rate component along the vertical (z or yaw) axis, measured in radians/sec.	
Timestamp	DateTimeType	Optional
Description	Indicates the time when the velocity data was measured or last known to be correct. The element can also indicate the time at which a future velocity of a System, Entity, etc. will be achieved.	

Table A-1-210: C2 UCI Extensions MA_FlightCapabilityMT - BodyReferenceOrientationRate UCI Extension Field

OrientationRateLimits		
Description	Specified orientation rate limits in body reference or NED coordinate frame.	
Base Type	MA_BodyReferenceOrientationRateType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RollRate	AngleRateType	Required
Description	Body-referenced attitude rate component along the longitudinal (x or roll) axis, measured in radians/sec.	
PitchRate	AngleRateType	Required
Description	Body-referenced attitude rate component along the lateral (y or pitch) axis, measured in radians/sec.	
YawRate	AngleRateType	Required
Description	Body-referenced attitude rate component along the vertical (z or yaw) axis, measured in radians/sec.	
Timestamp	DateTimeType	Optional
Description	Indicates the time when the velocity data was measured or last known to be correct. The element can also indicate the time at which a future velocity of a System, Entity, etc. will be achieved.	

Table A-1-211: C2 UCI Extensions MA_FlightCapabilityMT - OrientationRateLimits UCI Extension Field

AirspeedPair		
Description	Airspeed pair value for associated orientation rate limit, specified in body reference or NED coordinate frame.	
Base Type	PathSegmentSpeedValueType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	SpeedType	Required
Description	Indicates the speed along the PathSegment at the EndPoint.	

Reference	SpeedReferenceEnum	Required
Description	Indicates whether the given vehicle speed is reference to ground speed or true air speed.	

Table A-1-212: C2 UCI Extensions MA_FlightCapabilityMT - AirspeedPair UCI Extension Field

1.3.2.2MA_FlightCommandMT UCI Extension

MA_FlightCommandMT	
Description	Message extension included to invoke a flight capability to perform a new flight activity or change and existing flight activity. See the XSD File for more info
Ext. Fields	Command : MA_FlightCommandType Capability : MA_FlightCapabilityCommandType HSA_CSA : MA_HSA_CSA_Type WaypointFollowing : MA_WaypointFollowingType CurveFollowing : MA_CurveControlType Speed : PathSegmentSpeedType Heading : MA_DirectionReferenceType SpeedChoice : MA_HSA_SpeedChoiceType SpeedValue : PathSegmentSpeedValueType CurveFitMethod : MA_CurveFitMethodEnum Reference : MA_HeadingReferenceEnum ControlPoint : MA_CurveFollowingControlPointType RelativeAcceleration : MA_AccelerationChoiceType RelativeVelocity : MA_VelocityChoiceType OrientationRate : MA_OrientationRateChoiceType Direction : MA_DirectionChoiceType Course : MA_DirectionReferenceType

Table A-1-213: C2 UCI Extensions MA_FlightCommandMT UCI Extension

Command	
Description	Indicates an instance of a Flight Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate independent Capability CommandStatus message.
Base Type	MA_FlightCommandType
Message Structure	

Field Name	Field Type	Field Usage
Capability	MA_FlightCapabilityCommandType	Required
Description	Indicates a new invocation of a Flight Capability. Generally, if accepted, the command will result in one or more new Flight Activities being created and reported via the Flight Activity message. The request/response interaction terminates as soon as the command is accepted or rejected; this element is not used to interact with an "active" command. Updates and/or additional interaction with the resulting Activity (after the command is accepted) is accomplished via the sibling Activity element.	
Activity	MA_FlightActivityCommandType	Required
Description	Indicates a command to modify an existing Flight Activity (which was previously reported via the Flight Activity message and was marked as "interactive"). The request/response interaction terminates as soon as the modification is accepted or rejected. The modifications are reflected in subsequent Flight Activity messages.	

Table A-1-214: C2 UCI Extensions MA_FlightCommandMT - Command UCI Extension Field

Capability		
Description	Indicates a new invocation of a Flight Capability. Generally, if accepted, the command will result in one or more new Flight Activities being created and reported via the Flight Activity message. The request/response interaction terminates as soon as the command is accepted or rejected; this element is not used to interact with an "active" command. Updates and/or additional interaction with the resulting Activity (after the command is accepted) is accomplished via the sibling Activity element.	
Base Type	MA_FlightCapabilityCommandType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Loiter	MA_LoiterType	Optional
Description	Indicates the parameters for a loiter.	
MustFly	MustFlyType	Optional
Description	Indicates the parameters for a must fly.	
Formation	MA_FormationType	Repeated
Description	Indicates the relative positions for the entities in the formation.	
FlightControlMode	MA_FlightControlModesChoiceType	Optional
Description	See annotations in child elements and messages/elements that use this type for details.	
AltitudeStackedMarshall	MA_AltitudeStackedMarshallType	Optional
Description	Indicates kinematics for an altitude stacked marshal.	
Departure	MA_DepartureType	Optional
Description	This element identifies the parameters for a departure/take-off task.	
Recovery	MA_RecoveryType	Optional

Description	This element identifies the parameters for a recovery/land task.	
RoutePlanIntercept	MA_RoutePlanInterceptType	Optional
Description	This element defines the parameters for a Route intercept	
CapabilityID	CapabilityID_Type	Required
Description	Indicates the Capability instance being commanded.	
Ranking	CapabilityCommandRankingType	Required
Description	Indicates the command's rank relative to other commands and rank-related overrides and tailoring.	
TemporalConstraints	CapabilityCommandTemporalConstraintsType	Optional
Description	Indicates temporal constraints for the command. If omitted, the command should be scheduled immediately while satisfying other applicable constraints.	
OverrideRejection	boolean	Optional
Description	Indicates that the Command shall be performed even though it was previously rejected.	
Traceability	TraceabilityType	Optional
Description	C2 sources work at different levels of detail and abstraction, progressing from what to do to how specifically to do it. Inputs from abstract tasking sources and Tasks can be decomposed into discrete, actionable Tasks and/or Capability commands. This element provides traceability to the external tasking or Task from which this Capability command was derived.	
Classification	SecurityInformationType	Optional
Description	This element provides the classification label that the Subsystem should place on messages and other data resulting from the Command.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-215: C2 UCI Extensions MA_FlightCommandMT - Capability UCI Extension Field

HSA_CSA		
Description	This message shall be used to provide the ability to command a new flight vector to the Platform.	
Base Type	MA_HSA_CSA_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Altitude	AltitudeReferenceType	Optional
Description	Altitude hold value to be achieved and reference to altitude measurement source.	

Speed	PathSegmentSpeedType	Optional
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Direction	MA_DirectionChoiceType	Optional
Description	Defines heading or course type and reference to hold	

Table A-1-216: C2 UCI Extensions MA_FlightCommandMT - HSA_CSA UCI Extension Field

WaypointFollowing		
Description	This message shall be used to provide the ability to send a new waypoint path or segment for the Platform to follow.	
Base Type	MA_WaypointFollowingType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Route	MA_RouteType	Required
Description	Indicates the Paths and Segments that make up the Route to be flown.	

Table A-1-217: C2 UCI Extensions MA_FlightCommandMT - WaypointFollowing UCI Extension Field

CurveFollowing		
Description	Indicates the parameters for curve following control.	
Base Type	MA_CurveControlType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CurveSegments	MA_NURBS_PointType	Repeated
Description	Specifies the curve segments as a set of NURBS points.	
AppendCurve	EmptyType	Optional
Description	Specifies this curve will be appended to a previous curve command. This is only valid if the vehicle is currently executing a curve.	
EndOfCurveBehavior	MA_EndOfCurveBehaviorEnum	Optional
Description	Specifies the vehicle behavior at the completion of the curve.	

Table A-1-218: C2 UCI Extensions MA_FlightCommandMT - CurveFollowing UCI Extension Field

Speed	
Description	Defines speed type (reference frame) for all speed related fields in this message.
Base Type	PathSegmentSpeedType
Message Structure	

Field Name	Field Type	Field Usage
SpeedChoice	PathSegmentSpeedChoiceType	Optional
Description	Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number.	
SpeedOptimization	SpeedOptimizationEnum	Optional
Description	Indicates that the System will vary its speed autonomously according to the specified optimization algorithm.	

Table A-1-219: C2 UCI Extensions MA_FlightCommandMT - Speed UCI Extension Field

Heading		
Description	Defines heading type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-220: C2 UCI Extensions MA_FlightCommandMT - Heading UCI Extension Field

SpeedChoice		
Description	Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number.	
Base Type	MA_HSA_SpeedChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SpeedValue	PathSegmentSpeedValueType	Required
Description	Indicates speed in meters per second (m/s).	
MachValue	MachType	Required
Description	Indicates the speed reference for the associated speed value.	

Table A-1-221: C2 UCI Extensions MA_FlightCommandMT - SpeedChoice UCI Extension Field

SpeedValue	
Description	Indicates speed in meters per second (m/s).

Base Type	PathSegmentSpeedValueType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	SpeedType	Required
Description	Indicates the speed along the PathSegment at the EndPoint.	
Reference	SpeedReferenceEnum	Required
Description	Indicates whether the given vehicle speed is reference to ground speed or true air speed.	

Table A-1-222: C2 UCI Extensions MA_FlightCommandMT - SpeedValue UCI Extension Field

CurveFitMethod	
Description	Method through which a curve should be fitted to the discretized points in ControlPoint.
Base Type	MA_CurveFitMethodEnum
Message Structure	
Enumeration Literal	**Description**
B_SPLINE	B-Spline (basis spline) fit.
BEZIER	Bezier curve fit.
CATMULL_ROM	Catmull-Rom spline fit.
NATURAL_CUBIC	Natural Cubic curve fit.

Table A-1-223: C2 UCI Extensions MA_FlightCommandMT - CurveFitMethod UCI Extension Field

Reference	
Description	Reference to type of heading or course given.
Base Type	MA_HeadingReferenceEnum
Message Structure	
Enumeration Literal	**Description**
TRUE_NORTH	Heading value according to Geographic North.
MAGNETIC_NORTH	Heading value according to location where the planet's magnetic field points vertically downward.

Table A-1-224: C2 UCI Extensions MA_FlightCommandMT - Reference UCI Extension Field

ControlPoint	
Description	Control Points for the desired discretized curve or spline path.
Base Type	MA_CurveFollowingControlPointType
Message Structure	

Field Name	Field Type	Field Usage
Position	PointChoice4D_Type	Required
Description	Physical location of the referenced item in geospatial coordinates. The use of the timestamp in this context represents the time the positional data was last measured. Position data may be specified as either a geospatial point or relative to a separately defined reference frame.	
PositionUncertainty	UncertaintyType	Optional
Description	Uncertainty of the physical location of the referenced item.	
DomainVelocity	Velocity3D_Type	Optional
Description	Vehicle's velocity in the current operating domain of the system. Air vehicle components are in True Air Speed.	
GroundVelocity	Velocity2D_Type	Optional
Description	Representation of the vehicle's ground velocity as directional speed components.	
RelativeVelocity	MA_VelocityChoiceType	Optional
Description	Represents the vehicle's velocity, in units of m/s, in the separately defined reference frame referenced in the sibling Position element. This element should only be used when the RelativePoint choice is selected and the seperately defined reference frame is itself moving.	
DomainAcceleration	Acceleration3D_Type	Optional
Description	The vehicle's acceleration in the current operating domain of the system.	
Orientation	OrientationType	Optional
Description	Euler Angle Sequence describing the orientation of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.	
OrientationRate	MA_OrientationRateChoiceType	Optional
Description	Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.	
RelativeAcceleration	MA_AccelerationChoiceType	Optional
Description		

Table A-1-225: C2 UCI Extensions MA_FlightCommandMT - ControlPoint UCI Extension Field

RelativeAcceleration		
Description		
Base Type	MA_AccelerationChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
BodyReferenceAcceleration	MA_BodyReferenceAccelerationType	Required
Description	Body reference accelerations.	

NED_Acceleration	Acceleration3D_Type	Required
Description	Inertial, NED coordinate frame accelerations.	

Table A-1-226: C2 UCI Extensions MA_FlightCommandMT - RelativeAcceleration UCI Extension Field

RelativeVelocity		
Description	Represents the vehicle's velocity, in units of m/s, in the separately defined reference frame referenced in the sibling Position element. This element should only be used when the RelativePoint choice is selected and the seperately defined reference frame is itself moving.	
Base Type	MA_VelocityChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
BodyReferenceVelocity	MA_BodyReferenceVelocityType	Required
Description	Velocity in body reference frame.	
NED_Velocity	Velocity3D_Type	Required
Description	Velocity in North East Down reference frame.	

Table A-1-227: C2 UCI Extensions MA_FlightCommandMT - RelativeVelocity UCI Extension Field

OrientationRate		
Description	Euler Angle Sequence describing the rate of orientation change of the vehicle in the order yaw, pitch, roll. The angles are given in a locally level, North-East-Down coordinate system centered on the vehicle.	
Base Type	MA_OrientationRateChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
BodyReferenceOrientationRate	MA_BodyReferenceOrientationRateType	Required
Description	Orientation rate in body reference.	
NED_OrientationRate	OrientationRateType	Required
Description	Orientation rate in inertial reference.	

Table A-1-228: C2 UCI Extensions MA_FlightCommandMT - OrientationRate UCI Extension Field

Direction		
Description	Defines heading or course type and reference to hold	
Base Type	MA_DirectionChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

Heading	MA_DirectionReferenceType	Optional
Description	Defines heading type and reference to hold.	
Course	MA_DirectionReferenceType	Optional
Description	Defines course type and reference to hold.	

Table A-1-229: C2 UCI Extensions MA_FlightCommandMT - Direction UCI Extension Field

Course		
Description	Defines course type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-230: C2 UCI Extensions MA_FlightCommandMT - Course UCI Extension Field

1.3.2.3MA_FlightCommandStatusMT UCI Extension

MA_FlightCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_FlightCommandStatusMDT CannotComplyDetails : MA_CannotComplyDetailsType Severity : MA_FaultSeverityEnum

Table A-1-231: C2 UCI Extensions MA_FlightCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_FlightCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CannotComplyDetails	MA_CannotComplyDetailsType	Optional
Description	Provides detail for why FA cannot comply with the flight command.	

Activity	ResultingActivityType	Repeated
Description	Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-232: C2 UCI Extensions MA_FlightCommandStatusMT - MessageData UCI Extension Field

CannotComplyDetails		
Description	Provides detail for why FA cannot comply with the flight command.	
Base Type	MA_CannotComplyDetailsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ValidationResult	MA_ValidationResultEnum	Repeated
Description	In case of flight command rejection, provides the reason for Flight Autonomy rejection. Further details can be provided by Flight Autonomy through the ValidationResultReason. List size for this element is based on "Select All That Apply" condition.	
ValidationResultReason	MA_ValidationResultReasonType	Repeated
Description	In case of flight command rejection, provides additional context for Flight Autonomy rejection.	

Table A-1-233: C2 UCI Extensions MA_FlightCommandStatusMT - CannotComplyDetails UCI Extension Field

Severity	
Description	The severity of the fault. This enumeration is modified for Mission Autonomy adaptation.
Base Type	MA_FaultSeverityEnum
Message Structure	
Enumeration Literal	**Description**

NOMINAL	A fault that carries the nominal level indicates that fault does not impact the current mission but may indicate something that should be checked in maintenance. Faults should be tagged as nominal if they are targeting maintenance. An example might be air filter hasn't been changed in 5000 hours or queue length reached 60% of max.
ADVISORY	A fault that is important but is not a Caution or Warning; this can be a configuration change, a condition that is contributing to a degraded aircraft or subsystem operation, or an anomaly that is being automatically controlled or corrected by a higher level system.
CAUTION	A fault that is indicative of a condition that, without a timely and correct response, could result in equipment damage or failure of mission.
WARNING	A fault that is indicative of a condition requiring immediate response or attention to prevent loss of aircraft, or failure of mission.
FAILED	A fault that carries a failed level indicates the failure of a component which may preclude the use of a capability. Faults should be tagged with failed if the component state will be marked faulted. Failed components may or may not prevent the completion of the mission.

Table A-1-234: C2 UCI Extensions MA_FlightCommandStatusMT - Severity UCI Extension Field

1.3.2.4MA_FlightActivityMT UCI Extension

MA_FlightActivityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	VehicleCommandState : Activity : MA_FlightActivityType Speed : PathSegmentSpeedValueType Heading : MA_DirectionReferenceType

Table A-1-235: C2 UCI Extensions MA_FlightActivityMT UCI Extension

VehicleCommandState		
Description		
Base Type		
Message Structure		
Field Name	**Field Type**	**Field Usage**

Table A-1-236: C2 UCI Extensions MA_FlightActivityMT - VehicleCommandState UCI Extension Field

Activity

Description	Indicates the dynamic operational characteristics of an Activity of a System's Flight Capability.	
Base Type	MA_FlightActivityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActualEndPoint	MA_EndPointType	Repeated
Description	Indicates end points for the Flight activity. The current end point (the end point the System is currently flying to) is listed first followed by subsequent end points if known. If the Flight Activity was planned and the actual Activity is following the plan, the RouteActivityPlan* messages provide details relative to the plan. For planned Flight Activities against moving targets the actual, dynamically calculated kinematic details can be provided with this element.	
ActualCompletionTime	DateTimeType	Optional
Description	This field specifies the actual time when the Flight Activity completed.	
VehicleCommandState	MA_VehicleCommandStateType	Required
Description	Indicates the currently commanded parameters per the Flight Autonomy's execution of the request FlightCommand.	
CurveFollowingControl	MA_CurveFollowingControlType	Optional
Description	Indicates the curve following parameters, validity and status.	
RoutePlanInterceptStatus	MA_RoutePlanInterceptStatusType	Optional
Description	Status of the executing RoutePlan Intercept.	
ActivityID	ActivityID_Type	Required
Description	Indicates the unique ID of the Activity.	
CapabilityID	CapabilityID_Type	Repeated
Description	Indicates a Capability which the Activity is an instance of. Generally an Activity is an instance of a single Capability. An Activity can be associated with multiple Capabilities when, for example, a Subsystem has distinct Capabilities for automated and manual. In this case, if a manual command is issued while an automated Activity that achieves the desired result already exists, a single Activity represents both Capabilities.	
Interactive	boolean	Required
Description	Indicates whether or not the Activity can be edited/modified via a [Capability]Command...Activity message.	
ActivityState	ActivityStateEnum	Required
Description	Indicates the current state of the Activity its state machine. See enumeration annotations for further details.	
Basis	ActivityBasisEnum	Optional
Description	Indicates the basis of the Activity. See enumerated type annotations for further details. If omitted, a basis of ACTUAL should be assumed.	
ActivityRank	ComparableRankingType	Optional
Description	Indicates the rank of the Activity amongst all other Activities of the Capability and/or Subsystem. This isn't necessarily the same as the Rank associated with the sibling ActivitySource. This element is meant to convey the Subsystem determined relative	

	ranking of all Activities of a Capability category (ESM, EA, etc.). For Subsystems with multiple Capability categories this element is meant to convey relative ranking across all Capability categories. In addition to the rank of the ActivitySource, an Activity's rank can be influenced by Capability priority (see CapabilitySettings), RF_ControlCommands and other factors that influence the Subsystem's internal scheduling and resource allocation.	
ActivityReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling ActivityState element. This element is only expected when the Activity is in a potentially undesirable state such as constrained, failed or completed with less than optimum results.	
Source	ActivitySourceType	Repeated
Description	Indicates an originating "source" of the Activity; the command or other item which invoked this specific Activity of a Capability. A single Activity can fully or partially satisfy Activity sources.	
EstimatedStartTime	DateTimeType	Optional
Description	Indicates the estimated time when the Activity started or is expected to start.	
EstimatedCompletionTime	DateTimeType	Optional
Description	Indicates the estimated time when the Activity will complete.	
EstimatedPercentComplete	PercentType	Optional
Description	Indicates the current estimated percent complete of the Activity.	

Table A-1-237: C2 UCI Extensions MA_FlightActivityMT - Activity UCI Extension Field

Speed		
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Base Type	PathSegmentSpeedValueType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	SpeedType	Required
Description	Indicates the speed along the PathSegment at the EndPoint.	
Reference	SpeedReferenceEnum	Required
Description	Indicates whether the given vehicle speed is reference to ground speed or true air speed.	

Table A-1-238: C2 UCI Extensions MA_FlightActivityMT - Speed UCI Extension Field

Heading	
Description	Intended heading of the aircraft.
Base Type	MA_DirectionReferenceType

Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-239: C2 UCI Extensions MA_FlightActivityMT - Heading UCI Extension Field

1.3.2.5MA_OpVolumeMT UCI Extension

MA_OpVolumeMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_OpVolumeMDT

Table A-1-240: C2 UCI Extensions MA_OpVolumeMT UCI Extension

MessageData		
Description	This element represents the details regarding the operational volume.	
Base Type	MA_OpVolumeMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpVolumeID	OpVolumeID_Type	Required
Description	This element represents the unique identifier of the OpVolume associated with this message.	
Category	MA_OpZoneCategoryEnum	Required
Description	This element indicates the type of volume.	
CategoryUniqueData	MA_OpZoneCategoryType	Optional
Description	Contains parameters unique to certain OpZoneCategoryEnum values.	
Volume	MA_OpVolumeChoiceType	Required
Description	This element represents the characteristics of the operational volume.	
SituationalAwareness	boolean	Required
Description	Volume used only for situational awareness. Not used for planning purposes.	
OpDescription	OpDescriptionType	Optional
Description	A human-readable descriptive name with free text remarks.	
MissionTraceability	MissionTraceabilityType	Optional

Description	Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc.	
Source	CreationSourceEnum	Optional
Description	Indicates the source of the information.	
Schedule	ScheduleType	Optional
Description	This element represents a date/time interval associated with the operational item during which the item is in effect.	
AssociatedTime	TimeFunctionType	Repeated
Description	Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options.	
DataLinkIdentifier	DataLinkIdentifierPET	Repeated
Description	List of data link ID. Multiple data link IDs can be reported for the same network type.	
Priority	unsignedInt	Optional
Description	This field represents the priority to be considered when choosing between multiple available points.	
QualifyingTags	QualifyingTagsType	Optional
Description	Provides identity qualifications to augment identification and associated processing of the OpPoint.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	One or more systems to which this item applies. If omitted, item applies to all systems.	
SystemScheduleOverride	SystemScheduleStateType	Repeated
Description	System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution.	

Table A-1-241: C2 UCI Extensions MA_OpVolumeMT - MessageData UCI Extension Field

1.3.2.6MA_TaskMT UCI Extension

MA_TaskMT	
Description	This message specifies a Task that a System will perform against a specific entity or geographical location. Tasks include flight, collection, strike, electronic attack, directed energy, laser designation, cargo delivery and communication relay. They are abstracted to a level of C2 and autonomy derived from UCI operational vignettes. Tasks enable services, operators and other actors to do automated, dynamic, machine-to-machine allocation of Tasks to Systems and subsequent mission planning. If necessary, authorization of Users to Task Systems can be controlled using the Authorization message. See the XSD File for more info
Ext. Fields	TaskType : MA_TaskType TaskPlurality : MA_TaskPluralityEnum

	ConstraintID : MA_ConstraintID_Type HSA_CSA : MA_HSA_CSA_Type WaypointFollowing : MA_WaypointFollowingType CurveFollowing : MA_CurveControlType Speed : PathSegmentSpeedType Heading : MA_DirectionReferenceType SpeedChoice : MA_HSA_SpeedChoiceType SpeedValue : PathSegmentSpeedValueType CurveFitMethod : MA_CurveFitMethodEnum

Table A-1-242: C2 UCI Extensions MA_TaskMT UCI Extension

TaskType		
Description	Identifies the type of this Task instance.	
Base Type	MA_TaskType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AirSample	AirSampleTaskType	Required
Description	Air sample includes direct sampling of the air (SAMPLE) and remote sensing with spectral analysis (SPECTROMETER) with the intent of detecting NBC events.	
AMTI	AMTI_TaskType	Required
Description	Indicates a Task to collect Air Moving Target Indicator (AMTI) data.	
AO	AO_TaskType	Required
Description	Indicates a Task to perform an optical emission such as laser designation.	
CAP	MA_CAP_TaskType	Required
Description	Indicates a Task to perform Combat Air Patrol.	
CargoDelivery	CargoDeliveryTaskType	Required
Description	Indicates a Task to transfer cargo between locations.	
COMINT	COMINT_TaskType	Required
Description	Indicates a Task to provide COMINT.	
CommRelay	CommRelayTaskType	Required
Description	Indicates a Task to provide communications relay support.	
CounterSpace	CounterSpaceTaskType	Required
Description	Indicates a Task to employ a CounterSpace capability.	
Escort	MA_EscortTaskType	Required
Description	Indicates a Task to escort another Entity or System.	
EA	EA_TaskType	Required

Description	Indicates a Task to provide electronic attack support to another System. It guides/constrains the EA System by specifying where it should fly, what it should protect and what the threat is.	
ESM	ESM_TaskType	Required
Description	Indicates a Task to collect ESM data.	
Flight	MA_FlightTaskType	Required
Description	Indicates a Task to effect the flight path/plan of the System.	
Jettison	MA_JettisonTaskType	Required
Description	Indicates a Task to a jettisonable weapon from the system.	
OrbitChange	OrbitChangeTaskType	Required
Description	Indicates a task to perform an orbit change via a spacecraft maneuver.	
OrbitalSurveillance	OrbitalSurveillanceTaskType	Required
Description	Indicates an Orbital Surveillance Task.	
OrbitalSurveillanceSensor	OrbitalSurveillanceSensorTaskType	Required
Description	Indicates a task to perform orbital surveillance sensor tasking.	
PO	PO_TaskType	Required
Description	Indicates a Task to collect Passive Optical data, imagery and video as well as perform PO search and track capabilities.	
Refuel	RefuelTaskType	Required
Description	Indicates a Task for one System to refuel another.	
SAR	SAR_TaskType	Required
Description	Indicates a Task to collect a Synthetic Aperture Radar (SAR) image.	
SMTI	SMTI_TaskType	Required
Description	Indicates a Task to collect Moving Target Indicator (MTI) data.	
Strike	StrikeTaskType	Required
Description	Indicates a Task to kinetically attack/strike, with a weapon that can be released from the System.	
SystemDeployment	SystemDeploymentTaskType	Required
Description	Indicates a task to perform a deployment or release of a system at a specified location.	
TacticalOrder	TacticalOrderTaskType	Required
Description	Indicates a task to perform a tactical order.	
WeatherRadar	WeatherRadarTaskType	Required
Description	Indicates a task to collect weather radar data.	

Table A-1-243: C2 UCI Extensions MA_TaskMT - TaskType UCI Extension Field

TaskPlurality	
Description	Indicates whether the task is meant to be executed by a single entity or as a joint operation. A join operation implies forming a group prior to execution. A

	single entity could be a single system or a single pre-formed group.	
Base Type	MA_TaskPluralityEnum	
Message Structure		
Enumeration Literal	**Description**	
UNSPECIFIED	Indicates the task plurality is unspecified, and should be defaulted or determined by the receiving system.	
JOINT	Indicates the task is to be executed as a joint task. A group will be formed when this task is assigned.	
SINGLE_ENTITY	Indicates the task is to be executed by a single entity. A single entity could be a single system or a single pre-existing group.	

Table A-1-244: C2 UCI Extensions MA_TaskMT - TaskPlurality UCI Extension Field

ConstraintID		
Description	Indicates any Constraints that apply to the Task (e.g FormationConstraint). A FormationConstraint implies that if a group is formed to execute this task, the referenced Constraint(s) should be applied to that group.	
Base Type	MA_ConstraintID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-245: C2 UCI Extensions MA_TaskMT - ConstraintID UCI Extension Field

HSA_CSA		
Description	This message shall be used to provide the ability to command a new flight vector to the Platform.	
Base Type	MA_HSA_CSA_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Altitude	AltitudeReferenceType	Optional
Description	Altitude hold value to be achieved and reference to altitude measurement source.	
Speed	PathSegmentSpeedType	Optional
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Direction	MA_DirectionChoiceType	Optional

Description	Defines heading or course type and reference to hold

Table A-1-246: C2 UCI Extensions MA_TaskMT - HSA_CSA UCI Extension Field

WaypointFollowing		
Description	This message shall be used to provide the ability to send a new waypoint path or segment for the Platform to follow.	
Base Type	MA_WaypointFollowingType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Route	MA_RouteType	Required
Description	Indicates the Paths and Segments that make up the Route to be flown.	

Table A-1-247: C2 UCI Extensions MA_TaskMT - WaypointFollowing UCI Extension Field

CurveFollowing		
Description	Indicates the parameters for curve following control.	
Base Type	MA_CurveControlType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CurveSegments	MA_NURBS_PointType	Repeated
Description	Specifies the curve segments as a set of NURBS points.	
AppendCurve	EmptyType	Optional
Description	Specifies this curve will be appended to a previous curve command. This is only valid if the vehicle is currently executing a curve.	
EndOfCurveBehavior	MA_EndOfCurveBehaviorEnum	Optional
Description	Specifies the vehicle behavior at the completion of the curve.	

Table A-1-248: C2 UCI Extensions MA_TaskMT - CurveFollowing UCI Extension Field

Speed		
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Base Type	PathSegmentSpeedType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SpeedChoice	PathSegmentSpeedChoiceType	Optional
Description	Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach	

	number.	
SpeedOptimization	SpeedOptimizationEnum	Optional
Description	Indicates that the System will vary its speed autonomously according to the specified optimization algorithm.	

Table A-1-249: C2 UCI Extensions MA_TaskMT - Speed UCI Extension Field

Heading		
Description	Defines heading type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-250: C2 UCI Extensions MA_TaskMT - Heading UCI Extension Field

SpeedChoice		
Description	Indicates the speed that the vehicle shall traverse the path segment. Speed can be represented as referenced speed (e.g., ground speed in m/s) or as a unitless Mach number.	
Base Type	MA_HSA_SpeedChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SpeedValue	PathSegmentSpeedValueType	Required
Description	Indicates speed in meters per second (m/s).	
MachValue	MachType	Required
Description	Indicates the speed reference for the associated speed value.	

Table A-1-251: C2 UCI Extensions MA_TaskMT - SpeedChoice UCI Extension Field

SpeedValue		
Description	Indicates speed in meters per second (m/s).	
Base Type	PathSegmentSpeedValueType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

Value	SpeedType	Required
Description	Indicates the speed along the PathSegment at the EndPoint.	
Reference	SpeedReferenceEnum	Required
Description	Indicates whether the given vehicle speed is reference to ground speed or true air speed.	

Table A-1-252: C2 UCI Extensions MA_TaskMT - SpeedValue UCI Extension Field

CurveFitMethod	
Description	Method through which a curve should be fitted to the discretized points in ControlPoint.
Base Type	MA_CurveFitMethodEnum
Message Structure	
Enumeration Literal	**Description**
B_SPLINE	B-Spline (basis spline) fit.
BEZIER	Bezier curve fit.
CATMULL_ROM	Catmull-Rom spline fit.
NATURAL_CUBIC	Natural Cubic curve fit.

Table A-1-253: C2 UCI Extensions MA_TaskMT - CurveFitMethod UCI Extension Field

1.3.2.7MA_MissionPlanActivationCommandStatusMT UCI Extension

MA_MissionPlanActivationCommandStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	Plans : MA_PlansReferenceType MessageData : MA_MissionPlanActivationCommandStatusMDT ActivationCommandByState : MA_PlanActivationStatusType

Table A-1-254: C2 UCI Extensions MA_MissionPlanActivationCommandStatusMT UCI Extension

Plans		
Description	Indicates the Plan or Plans in the sibling CommandState.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Repeated

Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-255: C2 UCI Extensions MA_MissionPlanActivationCommandStatusMT - Plans UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanActivationCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	MissionPlanActivationCommandID_Type	Required
Description	The ID of the MissionPlanActivationCommand this status pertains to.	
ActivationStatus	PlanCommandStatusType	Required
Description	Indicates the status of the MissionPlanActivationCommand.	
ActivationCommandByState	MA_PlanActivationStatusType	Repeated
Description	Indicates plans in a given commanded activation state. List size allows for each available PlanActivationCommandState to be represented.	

Table A-1-256: C2 UCI Extensions MA_MissionPlanActivationCommandStatusMT - MessageData UCI

Extension Field

ActivationCommandByState		
Description	Indicates plans in a given commanded activation state. List size allows for each available PlanActivationCommandState to be represented.	
Base Type	MA_PlanActivationStatusType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanActivationCommandState	PlanActivationStateEnum	Required
Description	Indicates an activation command state that one or more *Plans is in.	
Plans	MA_PlansReferenceType	Required
Description	Indicates the Plan or Plans in the sibling CommandState.	

Table A-1-257: C2 UCI Extensions MA_MissionPlanActivationCommandStatusMT - ActivationCommandByState UCI Extension Field

1.3.2.8MA_MissionPlanMT UCI Extension

MA_MissionPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_MissionPlanMDT

Table A-1-258: C2 UCI Extensions MA_MissionPlanMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Required
Description	Indicates a unique ID assigned to the MissionPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
MissionPlanCommandID	MissionPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which this MissionPlan originated from.	
Applicability	MA_PlanApplicabilityType	Required

Description	Indicates the System or Systems which have a direct relationship with the Mission Plan. See annotations for the underlying type and its child elements for details.	
Window	DateTimeRangeType	Optional
Description	The time window for which the Plan is applicable.	
SubPlans	MA_PlansReferenceBaseType	Optional
Description	Indicates the set of *Plans which are included as part of the MissionPlan. The included SubPlans share the same Applicability as the MissionPlan.	
ExecutionSequence	ExecutionSequenceType	Optional
Description	Indicates the MissionPlan's flow of execution when there are multiple kinematic and action plans. The kinematic and action plans are combined into sets and given unique IDs. An execution flow may be created by linking *ExecutionPlanSets of different types (e.g. OrbitExecutionPlanSet and RouteExecutionPlanSet).	
ParentMissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a MissionPlan that is the parent of this MissionPlan.	
SubordinateMissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a MissionPlan that is subordinate to this MissionPlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	MA_MissionPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	
Remarks	RemarksType	Optional
Description	Indicates information about the mission plan from an operator intended for display. Both short and longer detailed information can be provided.	

Table A-1-259: C2 UCI Extensions MA_MissionPlanMT - MessageData UCI Extension Field

1.3.2.9MA_PositionReportMT UCI Extension

MA_PositionReportMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_PositionReportMDT

Table A-1-260: C2 UCI Extensions MA_PositionReportMT UCI Extension

MessageData

Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PositionReportMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Required
Description	The unique ID of the System corresponding to this PositionReport.	
DisplayName	VisibleString256Type	Optional
Description	This is the free-text, generally human-readable, name of the System or Resource corresponding to this PositionReport. This name isn't necessarily unique.	
Source	SystemSourceEnum	Required
Description	Indicates the source of the data in sibling elements. See enumeration annotations for further details.	
CurrentOperatingDomain	EnvironmentEnum	Required
Description	This enumerated element represents the domain/environment in which the system is currently operating.	
InertialState	MA_InertialStateType	Required
Description	This element represents the inertial state of the system or resource.	
WanderAngle	AngleType	Optional
Description	The angle by which the Inertial Navigation System (INS) rotates the geodetic frame about the vertical axis of the North-East-Down (NED) coordinate frame to avoid the introduction of a singularity at the North pole when it is calculating vehicle location. A rotation from true North towards East is reported as a positive angle.	
MagneticHeading	AngleType	Optional
Description	Indicates the heading in relationship to magnetic North.	

Table A-1-261: C2 UCI Extensions MA_PositionReportMT - MessageData UCI Extension Field

1.3.2.10 MA_TaskPlanMT UCI Extension

MA_TaskPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_TaskPlanMDT

Table A-1-262: C2 UCI Extensions MA_TaskPlanMT UCI Extension

MessageData	
Description	Indicates the data being conveyed by the message.

Base Type	MA_TaskPlanMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskPlanID	TaskPlanID_Type	Required
Description	Indicates a unique ID assigned to the TaskPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	TaskPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which the TaskPlan originated from.	
Plan	MA_TaskPlanType	Required
Description	Contains the details of the task allocation plan.	
ForPlanningUseOnly	Boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	MA_TaskPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-263: C2 UCI Extensions MA_TaskPlanMT - MessageData UCI Extension Field

1.3.2.11 MA_PackagePartnerStatusReportMT UCI Extension

MA_PackagePartnerStatusReportMT	
Description	This message reports the resulting status of a package change for a PackageManagementCommand for individual entities in a package. See the XSD File for more info
Ext. Fields	MessageData : MA_PackagePartnerStatusReportMDT

Table A-1-264: C2 UCI Extensions MA_PackagePartnerStatusReportMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PackagePartnerStatusReportMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**

CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command that formed the Package specified in the sibling PackageID element.	
PackageID	PackageID_Type	Required
Description	Indicates the unique ID of the package that this message is reporting a change status for.	
PackageLeaderID	SystemID_Type	Optional
Description	Indicates the unique ID of the system that represents the Package lead.	
PackageStatus	RequestProcessingStateEnum	Required
Description	The status of the Package as a whole.	
ConstraintID	MA_ConstraintID_Type	Repeated
Description	Indicates any constraints that were applied to the original PackageManagementCommand, indicated in the sibling CommandID element.	
SystemStatus	MA_PackagePartnerStatusType	Repeated
Description	The status for an individual system within the package.	
StaticLeadershipFitnessScore	FloatMinMaxType	Optional
Description	Indicates the static leadership fitness score for the partners in the package.	
DynamicLeadershipFitnessScore	MA_PartnerLeadershipFitnessScoreType	Optional
Description	Indicates the dynamic leadership fitness score and determination type for a partner in the package.	

Table A-1-265: C2 UCI Extensions MA_PackagePartnerStatusReportMT - MessageData UCI Extension Field

1.3.2.12 MA_ApprovalRequestMT UCI Extension

MA_ApprovalRequestMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	ActivationDetails : MA_MissionPlanActivationDetailsType OpPlanID : MA_OpPlanID_Type MissionPlanActivationApproval : MA_MissionPlanActivationCommandType ApprovalReferences : ApprovalRequestPolicyReferenceType ApprovalItem : ApprovalRequestItemReferenceType ByExecutionPlanSet : MA_ExecutionPlanSetActivationType MessageData : ApprovalRequestMDT BySubPlan : MA_MissionPlanSubplanActivationType OpPlan : MA_OpPlanActivationType

Table A-1-266: C2 UCI Extensions MA_ApprovalRequestMT UCI Extension

ActivationDetails		
Description	Indicates the commanded activation details.	
Base Type	MA_MissionPlanActivationDetailsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ByMissionPlan	MissionPlanActivationType	Required
Description	Indicates simultaneous activation of all sub-*Plans (RoutePlan, RouteActivityPlan for example) of a MissionPlan into the same activation state.	
BySubPlan	MA_MissionPlanSubplanActivationType	Required
Description	Indicates activation by sub-*Plan (RoutePlan or OrbitPlan for example) of the MissionPlan, with potentially different states for each.	
ByExecutionPlanSet	MA_ExecutionPlanSetActivationType	Required
Description	Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state.	

Table A-1-267: C2 UCI Extensions MA_ApprovalRequestMT - ActivationDetails UCI Extension Field

OpPlanID		
Description	Indicates the unique ID of the TaskPlan to perform the sibling activation command on.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-268: C2 UCI Extensions MA_ApprovalRequestMT - OpPlanID UCI Extension Field

MissionPlanActivationApproval	
Description	Indicates the MissionPlanActivationCommand details that are under review for approval. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to

	transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance. This transition would be subject to approval.	
Base Type	MA_MissionPlanActivationCommandType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Required
Description	Indicates the unique ID of the MissionPlan whose commanded activation state or states are summarized in the sibling element.	
ActivationDetails	MA_MissionPlanActivationDetailsType	Required
Description	Indicates the commanded activation details.	

Table A-1-269: C2 UCI Extensions MA_ApprovalRequestMT - MissionPlanActivationApproval UCI Extension Field

ApprovalReferences		
Description	Indicates the policy that resulted in the ApprovalRequest along with references to the data under review.	
Base Type	ApprovalRequestPolicyReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApprovalPolicyID	ApprovalPolicyID_Type	Required
Description	Indicates the unique ID of the ApprovalPolicy that resulted in this ApprovalRequest.	
ApprovalItem	ApprovalRequestItemReferenceType	Required
Description	Indicates the ID of the specific item that is being requested for approval.	

Table A-1-270: C2 UCI Extensions MA_ApprovalRequestMT - ApprovalReferences UCI Extension Field

ApprovalItem		
Description	Indicates the ID of the specific item that is being requested for approval.	
Base Type	ApprovalRequestItemReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanApproval	PlanReferenceID_ChoiceType	Required
Description	Indicates the unique ID of the *Plan that is under review for approval.	
RequirementExecutionApproval	ApprovalRequestItemType	Required
Description	Indicates the Requirement, including any associated DMPIs, that is under review for approval to execute.	
MissionPlanActivationApproval	MissionPlanActivationCommandType	Repeated

Description	Indicates the MissionPlanActivationCommand details that are under review for approval. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance. This transition would be subject to approval.

Table A-1-271: C2 UCI Extensions MA_ApprovalRequestMT - ApprovalItem UCI Extension Field

ByExecutionPlanSet		
Description	Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state.	
Base Type	MA_ExecutionPlanSetActivationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutionPlanSetID	ExecutionPlanSetID_Type	Required
Description	Indicates the unique ID of the execution plan set.	
CommandType	PlanActivationCommandEnum	Required
Description	Indicates the activation to perform on the associated execution plan set.	

Table A-1-272: C2 UCI Extensions MA_ApprovalRequestMT - ByExecutionPlanSet UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	ApprovalRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Approver	OperatorRoleType	Required
Description	An Operator Role or non-Operator identifier that is allowed to approve the included approval policy and approval item.	
ApprovalReferences	ApprovalRequestPolicyReferenceType	Required
Description	Indicates the policy that resulted in the ApprovalRequest along with references to the data under review.	
RespondBy	DateTimeType	Required
Description	If a response is not received from this approval authority by the given time, the approval policy's default response will be used. This field should be driven based on the timeout specified within the approval policy for approvals of this type.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required

Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.

Table A-1-273: C2 UCI Extensions MA_ApprovalRequestMT - MessageData UCI Extension Field

BySubPlan		
Description	Indicates activation by sub-*Plan (RoutePlan or OrbitPlan for example) of the MissionPlan, with potentially different states for each.	
Base Type	MA_MissionPlanSubplanActivationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskPlan	TaskPlanActivationType	Repeated
Description	Indicates commanded details for activation of a TaskPlan.	
RoutePlan	MA_RoutePlanActivationType	Repeated
Description	Indicates commanded details for activation of a RoutePlan.	
RouteActivityPlan	RouteActivityPlanActivationType	Repeated
Description	Indicates commanded details for activation of a RouteActivityPlan.	
OrbitPlan	OrbitPlanActivationType	Repeated
Description	Indicates commanded details for activation of an OrbitPlan.	
OrbitActivityPlan	OrbitActivityPlanActivationType	Repeated
Description	Indicates commanded details for activation of an OrbitActivityPlan.	
ActivityPlan	ActivityPlanActivationType	Repeated
Description	Indicates commanded details for activation of an ActivityPlan.	
EffectPlan	EffectPlanActivationType	Repeated
Description	Indicates commanded details for activation of an EffectPlan.	
ActionPlan	ActionPlanActivationType	Repeated
Description	Indicates commanded details for activation of an ActionPlan.	
ResponsePlan	ResponsePlanActivationType	Repeated
Description	Indicates commanded details for activation of a ResponsePlan.	
OpPlan	MA_OpPlanActivationType	Repeated
Description	Indicates commanded details for activation of an OpPlan.	

Table A-1-274: C2 UCI Extensions MA_ApprovalRequestMT - BySubPlan UCI Extension Field

OpPlan	
Description	Indicates commanded details for activation of an OpPlan.
Base Type	MA_OpPlanActivationType
Message Structure	

Field Name	Field Type	Field Usage
OpPlanID	MA_OpPlanID_Type	Required
Description	Indicates the unique ID of the TaskPlan to perform the sibling activation command on.	
CommandType	PlanActivationCommandEnum	Required
Description	Indicates the activation to perform on the associated TaskPlan.	

Table A-1-275: C2 UCI Extensions MA_ApprovalRequestMT - OpPlan UCI Extension Field

1.3.2.13 MA_ResponseActivityMT UCI Extension

MA_ResponseActivityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	RequirementInstantiation : MA_RequirementTemplateInstantiationType ActivatedPlans : MA_PlansReferenceType Activity : MA_ResponseActivityType MessageData : MA_ResponseActivityMDT ResultingPlans : MA_PlansReferenceType OpPlanID : MA_OpPlanID_Type

Table A-1-276: C2 UCI Extensions MA_ResponseActivityMT UCI Extension

RequirementInstantiation		
Description	Indicates an independent set of Requirement or Requirements and *Plans that were instantiated for the Response. Subsequent progress in planning and/or execution is given in [Requirement]Plan, [Requirement]Status, MissionPlan*, etc. messages.	
Base Type	MA_RequirementTemplateInstantiationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequirementID	RequirementInstanceID_ChoiceType	Repeated
Description	Indicates the unique ID of a Requirement that was instantiated as a result of the Response. Subsequent progress in planning and/or execution of the Requirement should be given in [Requirement]Plan, [Requirement]Status, etc. messages.	
ResultingPlans	MA_PlansReferenceType	Optional
Description	Indicates *Plans that were updated or created to include the sibling Requirements.	

Table A-1-277: C2 UCI Extensions MA_ResponseActivityMT - RequirementInstantiation UCI Extension Field

ActivatedPlans

Description	Indicates *Plans that were activated for the Response.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-278: C2 UCI Extensions MA_ResponseActivityMT - ActivatedPlans UCI Extension Field

Activity		
Description	Indicates the dynamic operational characteristics of an Activity of an Response Capability.	
Base Type	MA_ResponseActivityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
InstantiationTime	DateTimeType	Required
Description	Indicates the time at which the Response was instantiated.	

OptionIndex	unsignedInt	Optional
Description	Indicates the specific Option within the Response that was instantiated.	
RequirementInstantiation	MA_RequirementTemplateInstantiationType	Repeated
Description	Indicates an independent set of Requirement or Requirements and *Plans that were instantiated for the Response. Subsequent progress in planning and/or execution is given in [Requirement]Plan, [Requirement]Status, MissionPlan*, etc. messages.	
ActivatedPlans	MA_PlansReferenceType	Optional
Description	Indicates *Plans that were activated for the Response.	
AlertID	MissionContingencyAlertID_Type	Optional
Description	Indicates a MissionContingencyAlert that was generated for the Response.	
ActivityID	ActivityID_Type	Required
Description	Indicates the unique ID of the Activity.	
CapabilityID	CapabilityID_Type	Repeated
Description	Indicates a Capability which the Activity is an instance of. Generally an Activity is an instance of a single Capability. An Activity can be associated with multiple Capabilities when, for example, a Subsystem has distinct Capabilities for automated and manual. In this case, if a manual command is issued while an automated Activity that achieves the desired result already exists, a single Activity represents both Capabilities.	
Interactive	boolean	Required
Description	Indicates whether or not the Activity can be edited/modified via a [Capability]Command...Activity message.	
ActivityState	ActivityStateEnum	Required
Description	Indicates the current state of the Activity its state machine. See enumeration annotations for further details.	
Basis	ActivityBasisEnum	Optional
Description	Indicates the basis of the Activity. See enumerated type annotations for further details. If omitted, a basis of ACTUAL should be assumed.	
ActivityRank	ComparableRankingType	Optional
Description	Indicates the rank of the Activity amongst all other Activities of the Capability and/or Subsystem. This isn't necessarily the same as the Rank associated with the sibling ActivitySource. This element is meant to convey the Subsystem determined relative ranking of all Activities of a Capability category (ESM, EA, etc.). For Subsystems with multiple Capability categories this element is meant to convey relative ranking across all Capability categories. In addition to the rank of the ActivitySource, an Activity's rank can be influenced by Capability priority (see CapabilitySettings), RF_ControlCommands and other factors that influence the Subsystem's internal scheduling and resource allocation.	
ActivityReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling ActivityState element. This element is only expected when the Activity is in a potentially undesirable state such as constrained, failed or completed with less than optimum results.	
Source	ActivitySourceType	Repeated

Description	Indicates an originating "source" of the Activity; the command or other item which invoked this specific Activity of a Capability. A single Activity can fully or partially satisfy Activity sources.	
EstimatedStartTime	DateTimeType	Optional
Description	Indicates the estimated time when the Activity started or is expected to start.	
EstimatedCompletionTime	DateTimeType	Optional
Description	Indicates the estimated time when the Activity will complete.	
EstimatedPercentComplete	PercentType	Optional
Description	Indicates the current estimated percent complete of the Activity.	

Table A-1-279: C2 UCI Extensions MA_ResponseActivityMT - Activity UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ResponseActivityMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Required
Description	Indicates the System which hosts the Response Capabilities whose Activities are given in this message.	
Activity	MA_ResponseActivityType	Repeated
Description	Indicates the dynamic operational characteristics of an Activity of an Response Capability.	

Table A-1-280: C2 UCI Extensions MA_ResponseActivityMT - MessageData UCI Extension Field

ResultingPlans		
Description	Indicates *Plans that were updated or created to include the sibling Requirements.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	

OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-281: C2 UCI Extensions MA_ResponseActivityMT - ResultingPlans UCI Extension Field

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-282: C2 UCI Extensions MA_ResponseActivityMT - OpPlanID UCI Extension Field

1.3.2.14 MA_PackageManagementCommandMT UCI Extension

MA_PackageManagementCommandMT

Description	This message is used to command one or more entities to form, join or leave a Package. See the XSD File for more info
Ext. Fields	MessageData : MA_PackageManagementCommandMDT

Table A-1-283: C2 UCI Extensions MA_PackageManagementCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PackageManagementCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PackageID	PackageID_Type	Require d
Description	Indicates the unique identifier of the Package to form, join or leave.	
ParentPackageID	PackageID_Type	Optional
Description	If the sibling PackageID element is a sub-package, indicates the unique identifier of the parent package.	
ChangeType	MA_PackageChangeEnum	Require d
Description	Indicates the type of package change to perform.	
SystemID	SystemID_Type	Repeate d
Description	Indicates the System(s) commanded to perform a package change.	
ConstraintID	MA_ConstraintID_Type	Repeate d
Description	Indicates any Constraints that may apply to this package (e.g FormationConstraint).	
PackageLeaderID	SystemID_Type	Optional
Description	Indicates the System commanded to assume package leader role. Allows for the explicit specification of a leader without needing MA to conduct leader election.	
PackagePartnerMetrics	MA_PartnerMetricsType	Repeate d
Description	Indicates the leadership fitness metrics for the partners in the package.	
PackageLeaderElectionMethod	Integer	Require d

Description	Specifies which election method/algorithm the package will utilize for electing a Package Leader. Implementors are expected to set a default election method. If blank, the default election method will be used. Integer values and corresponding election methods are as follows: 0 - Bully Election Method. Supports Dynamic Fitness Scores and cases where Leadership Fitness Scores need to be shared as part of an election process. 1- Static Fitness Score Election Method. Supports nomination of a leader assuming leadership fitness scores remain static once they have been defined by MP/C2. 2 - Maximum Consensus Election Method. Supports utilizing consensus algortihms to reach team consensus on the leader while preserving communication bandwidth. 3 - Raft Leader Election Method. Supports leader election using the Raft consensus algorithm, which ensures a distributed and fault-tolerant approach to electing a leader.	
PackageLeadershipFitnessScoreDetermination Method	MA_LeadershipFitnessScoreDeterminationMethodChoiceType	Optional
Description	This field specifies the chosen fitness score determination method to be used for fitness scoring for the package.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-284: C2 UCI Extensions MA_PackageManagementCommandMT - MessageData UCI Extension Field

1.3.2.15 MA_PackageManagementCommandStatusMT UCI Extension

MA_PackageManagementCommandStatusMT	
Description	This message is sent in response to a PackageManagementCommand and indicates if the command was accepted and if not why it was rejected. The resulting status of the Package Change is reported in PackagePartnerStatusReport for each system in the original command. See the XSD File for more info

Ext. Fields	MessageData : MA_PackageManagementCommandStatusMDT

Table A-1-285: C2 UCI Extensions MA_PackageManagementCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PackageManagementCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-286: C2 UCI Extensions MA_PackageManagementCommandStatusMT - MessageData UCI Extension Field

1.3.2.16 MA_ApprovalRequestStatusMT UCI Extension

MA_ApprovalRequestStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_ApprovalRequestStatusMDT

Table A-1-287: C2 UCI Extensions MA_ApprovalRequestStatusMT UCI Extension

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.
Base Type	MA_ApprovalRequestStatusMDT

Message Structure		
Field Name	**Field Type**	**Field Usage**
ApprovalRequestProcessingState	ApprovalStatusEnum	Required
Description	The ApprovalState indicates the current state of the approval for an approval request.	
ApprovalRequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling ApprovalRequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	
OperatorID	OperatorID_Type	Optional
Description	Indicates the Operator that provided the approval/rejection, if available. This element can be beneficial when systems require two-person integrity.	
ApprovedDMPI_ID	DMPI_ID_Type	Repeated
Description	If the ApprovalRequest was for DMPI approval then this element contains the list of approved DMPIs. If any DMPIs are approved, the sibling ApprovalState will indicate that the request is approved.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-288: C2 UCI Extensions MA_ApprovalRequestStatusMT - MessageData UCI Extension Field

1.3.2.17 MA_TaskCommandStatusMT UCI Extension

MA_TaskCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_TaskCommandStatusMDT

Table A-1-289: C2 UCI Extensions MA_TaskCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_TaskCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Activity	ResultingActivityType	Repeated
Description	Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-290: C2 UCI Extensions MA_TaskCommandStatusMT - MessageData UCI Extension Field

1.3.2.18 MA_MissionPlanExecutionStatusMT UCI Extension

MA_MissionPlanExecutionStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	ExecutionPlanSetStatus : MA_ExecutionPlanSetExecutionStateType PlanExecutionStatus : MA_MissionPlanExecutionStateType MessageData : MA_MissionPlanExecutionStatusMDT

Table A-1-291: C2 UCI Extensions MA_MissionPlanExecutionStatusMT UCI Extension

ExecutionPlanSetStatus	
Description	Indicates the execution state of an execution plan set of the MissionPlan indicated by the sibling MissionPlanID element.
Base Type	MA_ExecutionPlanSetExecutionStateType

Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutionPlanSetID	ExecutionPlanSetID_Type	Required
Description	Indicates the unique ID of the execution plan set.	
ExecutionPlanSetExecutionState	PlanExecutionStateEnum	Required
Description	The current state of the execution plan set.	

Table A-1-292: C2 UCI Extensions MA_MissionPlanExecutionStatusMT - ExecutionPlanSetStatus UCI Extension Field

PlanExecutionStatus		
Description	Indicates the execution state of a MissionPlan of the System indicated by the sibling SystemID element.	
Base Type	MA_MissionPlanExecutionStateType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Required
Description	The ID of the MissionPlan this status pertains to.	
PlanExecutionState	PlanExecutionStateEnum	Required
Description	The current state of the plan.	
ExecutionPlanSetStatus	MA_ExecutionPlanSetExecutionStateType	Repeated
Description	Indicates the execution state of an execution plan set of the MissionPlan indicated by the sibling MissionPlanID element.	

Table A-1-293: C2 UCI Extensions MA_MissionPlanExecutionStatusMT - PlanExecutionStatus UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanExecutionStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Required
Description	The unique ID of the System whose Plan is being statused.	
Source	SystemSourceEnum	Required
Description	Indicates the source of the data in sibling elements. See enumeration annotations for further details.	
PlanExecutionStatus	MA_MissionPlanExecutionStateType	Repeated
Description	Indicates the execution state of a MissionPlan of the System indicated by the sibling	

	SystemID element.

Table A-1-294: C2 UCI Extensions MA_MissionPlanExecutionStatusMT - MessageData UCI Extension Field

1.3.2.19 MA_StrikeCapabilityMT UCI Extension

MA_StrikeCapabilityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_StrikeCapabilityMDT

Table A-1-295: C2 UCI Extensions MA_StrikeCapabilityMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_StrikeCapabilityMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Capability	MA_StrikeCapabilityType	Repeated
Description	Indicates a Strike Capability provided by the Subsystem.	
Carriage	StoreCarriageCapabilityType	Repeated
Description	Optional list of carriage related to the StrikeCapability.	
Verification	StoreVerificationStatusType	Optional
Description	This is a rollup of all verification of each capability in this message.	
LoadoutMnemonic	VisibleString256Type	Optional
Description	Mnemonic representation of the store loadout that the reported strike capabilities and carriage support capabilities are derived from.	
AvailableMDF	MDF_Type	Repeated
Description	Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability.	

Table A-1-296: C2 UCI Extensions MA_StrikeCapabilityMT - MessageData UCI Extension Field

1.3.2.20 MA_LeadershipMetricsMT UCI Extension

MA_LeadershipMetricsMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_LeadershipMetricsMDT

Table A-1-297: C2 UCI Extensions MA_LeadershipMetricsMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_LeadershipMetricsMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PackagePartnerMetrics	MA_PartnerMetricsType	Repeated
Description	Indicates the leadership fitness metrics for the partners in the package.	
PackageLeaderElectionMethod	Integer	Required
Description	Specifies which election method/algorithm the package will utilize for electing a Package Leader. Implementors are expected to set a default election method. If blank, the default election method will be used. Integer values and corresponding election methods are as follows: 0 - Bully Election Method. Supports Dynamic Fitness Scores and cases where Leadership Fitness Scores need to be shared as part of an election process. 1- Static Fitness Score Election Method. Supports nomination of a leader assuming leadership fitness scores remain static once they have been defined by MP/C2. 2 - Maximum Consensus Election Method. Supports utilizing consensus algortihms to reach team consensus on the leader while preserving communication bandwidth. 3 - Raft Leader Election Method. Supports leader election using the Raft consensus algorithm, which ensures a distributed and fault-tolerant approach to electing a leader.	

Table A-1-298: C2 UCI Extensions MA_LeadershipMetricsMT - MessageData UCI Extension Field

1.3.2.21 MA_WEZ_MT UCI Extension

MA_WEZ_MT	
Description	This message represents a single Weapon Engagement Zone (WEZ). The WEZ represents a

	less dynamic capability than the DLZ for air to air engagements and is not reliant on target state data to be produced. See the XSD File for more info
Ext. Fields	MessageData : MA_WEZ_MDT WEZ_ID : MA_WEZ_ID_Type WEZ_Data : MA_WEZ_DataType RangeTurnAndClose : MA_WeaponRangeType RangeMinimum : MA_WeaponRangeType RangeTurnAndRun : MA_WeaponRangeType RangeProbabilityOfIntercept : MA_WeaponRangeType RangeMaxGuidance : MA_WeaponRangeType RangeOptimal : MA_WeaponRangeType RangeMaxAero : MA_WeaponRangeType

Table A-1-299: C2 UCI Extensions MA_WEZ_MT UCI Extension

MessageData		
Description	This represents the characteristics that identify and describe a single Weapon Engagement Zone (WEZ).	
Base Type	MA_WEZ_MDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
WEZ_ID	MA_WEZ_ID_Type	Required
Description	This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).	
WEZ_RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
CapabilityID	CapabilityID_Type	Optional
Description	Indicates the weapon Capability instance that this WEZ is associated with.	
GeneratedByWeapon	boolean	Optional
Description	This element indicates if the WEZ was generated using payload algorithms.	
WEZ_Data	MA_WEZ_DataType	Required
Description	Indicates WEZ range calculations for air-to-air engagement assessments.	

Table A-1-300: C2 UCI Extensions MA_WEZ_MT - MessageData UCI Extension Field

WEZ_ID	
Description	This is an ID to uniquely identify the Weapon Engagement Zone (WEZ).

Base Type	MA_WEZ_ID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-301: C2 UCI Extensions MA_WEZ_MT - WEZ_ID UCI Extension Field

WEZ_Data		
Description	Indicates WEZ range calculations for air-to-air engagement assessments.	
Base Type	MA_WEZ_DataType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RangeTurnAndClose	MA_WeaponRangeType	Repeated
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-close maneuver RangeTurnAndClose is not applicable to ballistic trajectory targets.	
RangeMinimum	MA_WeaponRangeType	Repeated
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeTurnAndRun	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed)	

	3.Target turn-and-run maneuver RangeTurnAndRun is not applicable to ballistic trajectory targets.	
RangeProbabilityOfIntercept	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeMaxGuidance	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeOptimal	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
RangeMaxAero	MA_WeaponRangeType	Repeated
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	

TimeToRelease	TimeType	Required
Description	The seconds time to activate for the proposed launch store type.	
TimeOfActivation	TimeType	Optional
Description	The time of weapon seeker activation, assuming no platform guidance or weapon errors, for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No post-launch target maneuvers	

Table A-1-302: C2 UCI Extensions MA_WEZ_MT - WEZ_Data UCI Extension Field

RangeTurnAndClose		
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-close maneuver RangeTurnAndClose is not applicable to ballistic trajectory targets.	
Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-303: C2 UCI Extensions MA_WEZ_MT - RangeTurnAndClose UCI Extension Field

RangeMinimum	
Description	Launch slant range at which the weapon will intercept the target and arm its fuze, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.

Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-304: C2 UCI Extensions MA_WEZ_MT - RangeMinimum UCI Extension Field

RangeTurnAndRun		
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.Target turn-and-run maneuver RangeTurnAndRun is not applicable to ballistic trajectory targets.	
Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-305: C2 UCI Extensions MA_WEZ_MT - RangeTurnAndRun UCI Extension Field

RangeProbabilityOfIntercept	
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers Note: ballistic trajectory velocity changes are not considered maneuvers.
Base Type	MA_WeaponRangeType

Message Structure		
Field Name	**Field Type**	**Field Usage**
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-306: C2 UCI Extensions MA_WEZ_MT - RangeProbabilityOfIntercept UCI Extension Field

RangeMaxGuidance		
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Current steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-307: C2 UCI Extensions MA_WEZ_MT - RangeMaxGuidance UCI Extension Field

RangeOptimal	
Description	Launch range at which the weapon will intercept the target with high energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.
Base Type	MA_WeaponRangeType
Message Structure	

Field Name	Field Type	Field Usage
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-308: C2 UCI Extensions MA_WEZ_MT - RangeOptimal UCI Extension Field

RangeMaxAero		
Description	Launch range at which the weapon will intercept the target with nominal energy, assuming no platform guidance or weapon guidance errors for: 1.Optimal steering 2.Current shooter kinematic conditions (e.g., altitude/airspeed) 3.No target maneuvers, Note: ballistic trajectory velocity changes are not considered maneuvers.	
Base Type	MA_WeaponRangeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Range	DistanceType	Required
Description	Indicates the slant range of the weapon.	
Azimuth	AngleType	Required
Description	Indicates the azimuth the slant range pertains to.	

Table A-1-309: C2 UCI Extensions MA_WEZ_MT - RangeMaxAero UCI Extension Field

1.3.2.22 MA_TaskCommandMT UCI Extension

MA_TaskCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_TaskCommandMDT

Table A-1-310: C2 UCI Extensions MA_TaskCommandMT UCI Extension

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.

Base Type	MA_TaskCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Command	MA_TaskCommandType	Repeated
Description	Indicates an instance of a Task Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message.	

Table A-1-311: C2 UCI Extensions MA_TaskCommandMT - MessageData UCI Extension Field

1.3.2.23 MA_OpZoneMT UCI Extension

MA_OpZoneMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_OpZoneMDT Category : MA_OpZoneCategoryEnum CategoryUniqueData : MA_OpZoneCategoryType SpeedLimits : MA_OpZoneSpeedLimitsType VehicleConfiguration : MA_VehicleCommandDataType Zone : MA_OpZoneChoiceType GeometryReference : MA_ReferentialOpZoneType ReferenceObjectsFilter : MA_GeoLocatedObjectFilterType OpPointFilter : MA_OpPointFilterType OpLineFilter : MA_OpLineFilterType OpLineCategory : MA_OpLineCategoryEnum OpZoneFilter : MA_OpZoneFilterType OpZoneCategory : MA_OpZoneCategoryEnum OpVolumeFilter : MA_OpVolumeFilterType OpVolumeCategory : MA_OpZoneCategoryEnum DMPI_Filter : MA_DMPI_FilterType SignalReportFilter : MA_SignalReportFilterType OpVolumeFilter : MA_OpVolumeFilterType OpLineFilter : MA_OpLineFilterType OpLineCategory : MA_OpLineCategoryEnum SpeedLimits : MA_OpZoneSpeedLimitsType

	SignalReportFilter : MA_SignalReportFilterType OpPointFilter : MA_OpPointFilterType VehicleConfiguration : MA_VehicleCommandDataType DMPI_Filter : MA_DMPI_FilterType Zone : MA_OpZoneChoiceType Category : MA_OpZoneCategoryEnum CategoryUniqueData : MA_OpZoneCategoryType ReferenceObjectsFilter : MA_GeoLocatedObjectFilterType OpVolumeCategory : MA_OpZoneCategoryEnum GeometryReference : MA_ReferentialOpZoneType OpZoneCategory : MA_OpZoneCategoryEnum OpZoneFilter : MA_OpZoneFilterType

Table A-1-312: C2 UCI Extensions MA_OpZoneMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_OpZoneMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpZoneID	OpZoneID_Type	Required
Description	This element represents the unique identifier of the zone associated with this message.	
Category	MA_OpZoneCategoryEnum	Required
Description	This element indicates the type of zone.	
CategoryUniqueData	MA_OpZoneCategoryType	Optional
Description	Contains parameters unique to certain OpZoneCategoryEnum values.	
Zone	MA_OpZoneChoiceType	Required
Description	This element represents the geospatial and temporal characteristics of the OpZone.	
SituationalAwareness	boolean	Required
Description	Zone used only for situational awareness. Not used for planning purposes.	
OpDescription	OpDescriptionType	Optional
Description	A human-readable descriptive name with free text remarks.	
MissionTraceability	MissionTraceabilityType	Optional
Description	Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc.	
Source	CreationSourceEnum	Optional

Description	Indicates the source of the information.	
Schedule	ScheduleType	Optional
Description	This element represents a date/time interval associated with the operational item during which the item is in effect.	
AssociatedTime	TimeFunctionType	Repeated
Description	Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options.	
DataLinkIdentifier	DataLinkIdentifierPET	Repeated
Description	List of data link ID. Multiple data link IDs can be reported for the same network type.	
Priority	unsignedInt	Optional
Description	This field represents the priority to be considered when choosing between multiple available points.	
QualifyingTags	QualifyingTagsType	Optional
Description	Provides identity qualifications to augment identification and associated processing of the OpPoint.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	One or more systems to which this item applies. If omitted, item applies to all systems.	
SystemScheduleOverride	SystemScheduleStateType	Repeated
Description	System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution.	

Table A-1-313: C2 UCI Extensions MA_OpZoneMT - MessageData UCI Extension Field

Category	
Description	This element indicates the type of zone.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.

DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.

NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-314: C2 UCI Extensions MA_OpZoneMT - Category UCI Extension Field

CategoryUniqueData		
Description	Contains parameters unique to certain OpZoneCategoryEnum values.	
Base Type	MA_OpZoneCategoryType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ConstrainedEntryExit	ConstrainedEntryExitType	Required
Description	Defines boundaries which applicable Systems can only enter and exit through defined edges.	
FilterArea	OpZoneFilterAreaPET	Repeated

Description	Indicates that the OpZone can be a zone filter type.	
Jamming	OpZoneJammingType	Required
Description	Indicates that the OpZone is a jamming control zone.	
KeepIn	IngressEgressType	Required
Description	Defines boundaries to which applicable Systems must stay inside.	
MissileLaunchPoint	OpZoneMissileDataType	Required
Description	Data defining a missile type, related track, and source of launch position.	
NoFire	OpZoneNoFireType	Required
Description	Defines areas where strike impact is restricted. Does not restrict the launch of weapons.	
NoFly	OpZoneNoFlyType	Required
Description	Defines area where flight is restricted. Equivalent to MIL-STD-6016 restricted zone.	
SpeedLimits	MA_OpZoneSpeedLimitsType	Required
Description	Defines area where speeds are restricted.	
VehicleConfiguration	MA_VehicleCommandDataType	Required
Description	Defines vehicle configuration parameters that should change based on the planned location of a vehicle.	
WeaponRestriction	OpZoneWeaponRestrictionType	Required
Description	Set of restricted weapons that cannot be used against a target type and or in a zone.	
WeatherConditions	OpZoneWeatherType	Required
Description	Defines area of weather conditions with potential of mission impact.	

Table A-1-315: C2 UCI Extensions MA_OpZoneMT - CategoryUniqueData UCI Extension Field

SpeedLimits		
Description	Defines area where speeds are restricted.	
Base Type	MA_OpZoneSpeedLimitsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MinSpeedLimit	PathSegmentSpeedValueType	Optional
Description	Indicates the minimum speed limit in meters per second within a zone/region.	
MaxSpeedLimit	PathSegmentSpeedValueType	Optional
Description	Indicates the maximum speed limit in meters per second within a zone/region.	

Table A-1-316: C2 UCI Extensions MA_OpZoneMT - SpeedLimits UCI Extension Field

VehicleConfiguration	
Description	Defines vehicle configuration parameters that should change based on the

	planned location of a vehicle.	
Base Type	MA_VehicleCommandDataType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
VehicleAction	VehicleActionEnum	Optional
Description	Defines vehicle actions that can be done along a path segment of a route.	
IFF_Settings	IFF_Type	Optional
Description	If present the IFF settings for this vehicle will be changed along this path segment.	
SurvivabilityMode	VehicleSurvivabilityModeEnum	Optional
Description	Specifies that the vehicle should put itself into a more survivable mode. Different vehicles will have specific behaviors that reduce the signature of the vehicle (doors closed, minimal use of flaps, etc.) If this element is not present then it is assumed that the aircraft being configured has no survivability mode.	
LostCommTimeout	DurationType	Optional
Description	Defines the lost communication contingency timeout. When this timeout value is exceeded, the AV considers that lost communication has occurred and begins to perform contingency measures.	
LOS	boolean	Optional
Description	Use Line Of Sight (LOS) communication as backup contingency.	
LossOfLinkProcessing	VehicleLossOfLinkProcessingEnum	Optional
Description	Specifies whether the vehicle should invoke special Loss of Link processing. This could be turned off if the vehicle is going to be in a known and accepted lost communication situation.	
CommAllocationAction	CommAllocationActionType	Optional
Description	Indicates a change of communication allocation.	
QNH_Setting	PressureType	Optional
Description	Represents the barometric altimeter setting for correct MSL reading, commonly known as QNH, requested. Expressed in kilopascals (kPa).	

Table A-1-317: C2 UCI Extensions MA_OpZoneMT - VehicleConfiguration UCI Extension Field

Zone		
Description	This element represents the geospatial and temporal characteristics of the OpZone.	
Base Type	MA_OpZoneChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
GeometrySpecification	OpZoneType	Required
Description	Defines an OpZone geometry	

GeometryReference	MA_ReferentialOpZoneType	Required
Description	Defines an OpZone geometry via reference to other filtered objects	

Table A-1-318: C2 UCI Extensions MA_OpZoneMT - Zone UCI Extension Field

GeometryReference		
Description	Defines an OpZone geometry via reference to other filtered objects	
Base Type	MA_ReferentialOpZoneType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ReferenceObjectsFilter	MA_GeoLocatedObjectFilterType	Repeated
Description	The geometry of all objects returned by this filter will be used for the OpZone being defined.	
HorizontalBuffer	DistanceType	Required
Description	The area of the resulting OpZone is the area of the referenced object, expanded by this horizontal buffer along the plane orthogonal to the NED frame's down direction	
MinAltitudeBuffer	DistanceType	Required
Description	The min altitude of the resulting OpZone is the min altitude of the referenced object, decreased by this min altitude buffer	
MaxAltitudeBuffer	DistanceType	Required
Description	The max altitude of the resulting OpZone is the max altitude of the referenced object, increased by this max altitude buffer	

Table A-1-319: C2 UCI Extensions MA_OpZoneMT - GeometryReference UCI Extension Field

ReferenceObjectsFilter		
Description	The geometry of all objects returned by this filter will be used for the OpZone being defined.	
Base Type	MA_GeoLocatedObjectFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
EntityFilter	EntityFilterType	Optional
Description	Indicates a name/type of Entity required to match the filter. If omitted, any name/type of Entity satisfies the filter.	
SystemFilter	SystemFilterType	Optional
Description	Indicates a name/type of System required to match the filter. If omitted, any name/type of System satisfies the filter.	
OpPointFilter	MA_OpPointFilterType	Optional
Description	Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter.	

OpLineFilter	MA_OpLineFilterType	Optional
Description	Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.	
OpZoneFilter	MA_OpZoneFilterType	Optional
Description	Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.	
OpVolumeFilter	MA_OpVolumeFilterType	Optional
Description	Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.	
DMPI_Filter	MA_DMPI_FilterType	Optional
Description	Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter.	
SignalReportFilter	MA_SignalReportFilterType	Optional
Description	Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter.	

Table A-1-320: C2 UCI Extensions MA_OpZoneMT - ReferenceObjectsFilter UCI Extension Field

OpPointFilter		
Description	Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter.	
Base Type	MA_OpPointFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPointID	OpPointID_Type	Repeated
Description	Indicates a specific OpPointID that is required to match the filter. If omitted, any OpPointID satisfies the filter. Multiple instances are logically ORed.	
OpPointCategory	OpPointCategoriesType	Repeated
Description	Indicates an OpPoint category that is required to match the filter. If omitted, any OpPoint category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpPoint must fall within to satisfy the filter. Multiple instances are logically ORed.Indicates that the OpRouting type is applicable. List size for this element is based on there being no category type for OpRouting objects	

Table A-1-321: C2 UCI Extensions MA_OpZoneMT - OpPointFilter UCI Extension Field

OpLineFilter	
Description	Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.

Base Type	MA_OpLineFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpLineID	OpLineID_Type	Repeated
Description	Indicates a specific OpLineID that is required to match the filter. If omitted, any OpLineID satisfies the filter. Multiple instances are logically ORed.	
OpLineCategory	MA_OpLineCategoryEnum	Repeated
Description	Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpLine must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-322: C2 UCI Extensions MA_OpZoneMT - OpLineFilter UCI Extension Field

OpLineCategory	
Description	Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpLineCategoryEnum
Message Structure	
Enumeration Literal	**Description**
GENERAL_LINE	A general line where the function must be described by other procedures.
FORWARD_EDGE_OF_BATTLE_AREA	The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.
GUN_TARGET_LINE	Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.
CORRIDOR	Line with width describing the passageway designated for travel by friendly aircraft.
HOSTILE_BOUNDARY	Line that indicates a limit beyond which exists a hostile environment.
BUFFER_ZONE_BOUNDARY	Line beyond which friendly aircraft usually do not fly in peacetime.
LOW_LEVEL_TRANSIT_ROUTE	Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.
TACTICAL_ACTION_LINE	Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.
FIRE_SUPPORT_COORDINATION_LINE	Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must

	be coordinated with the appropriate ground commander.
FORWARD_LINE_OF_OWN_TROOPS	Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.
LINE_OF_COMMUNICATIONS	Line the route that connects an operating military unit with its supply base.
JETWAY	Line with width that defines areas where commercial aircraft are flying through.
DEPARTURE_RADIAL	Line the defines the departure radial when performing Carrier admin departures.
RECOVERY_RADIAL	Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.
FINAL_BEARING	Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.
BRC	Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.

Table A-1-323: C2 UCI Extensions MA_OpZoneMT - OpLineCategory UCI Extension Field

OpZoneFilter		
Description	Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.	
Base Type	MA_OpZoneFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpZoneID	OpZoneID_Type	Repeated
Description	Indicates a specific OpZoneID that is required to match the filter. If omitted, any OpZoneID satisfies the filter. Multiple instances are logically ORed.	
OpZoneCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpZone must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-324: C2 UCI Extensions MA_OpZoneMT - OpZoneFilter UCI Extension Field

OpZoneCategory	
Description	Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.

Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.

KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-325: C2 UCI Extensions MA_OpZoneMT - OpZoneCategory UCI Extension Field

OpVolumeFilter		
Description	Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.	
Base Type	MA_OpVolumeFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpVolumeID	OpVolumeID_Type	Repeated
Description	Indicates a specific OpVolumeID that is required to match the filter. If omitted, any OpVolumeID satisfies the filter. Multiple instances are logically ORed.	
OpVolumeCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpVolume must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-326: C2 UCI Extensions MA_OpZoneMT - OpVolumeFilter UCI Extension Field

OpVolumeCategory	
Description	Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).

DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.

NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-327: C2 UCI Extensions MA_OpZoneMT - OpVolumeCategory UCI Extension Field

DMPI_Filter		
Description	Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter.	
Base Type	MA_DMPI_FilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DMPI_ID	DMPI_ID_Type	Repeated
Description	Indicates a specific DMPI_ID that is required to match the filter. If omitted, any DMPI_ID satisfies the filter. Multiple instances are logically ORed.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the DMPI must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-328: C2 UCI Extensions MA_OpZoneMT - DMPI_Filter UCI Extension Field

SignalReportFilter		
Description	Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter.	
Base Type	MA_SignalReportFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SignalReportID	SignalReportID_Type	Repeated
Description	Indicates a specific SignalReportID that is required to match the filter. If omitted, any SignalReportID satisfies the filter. Multiple instances are logically ORed.	
SystemID	SystemID_Type	Repeated
Description	Indicates a specific source SystemID that is required to match the filter. If omitted, any source SystemID satisfies the filter. Multiple instances are logically ORed.	
SubsystemID	SubsystemID_Type	Repeated
Description	Indicates a specific source SubsystemID that is required to match the filter. If omitted, any source SubsystemID satisfies the filter. Multiple instances are logically ORed.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the SignalReport must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-329: C2 UCI Extensions MA_OpZoneMT - SignalReportFilter UCI Extension Field

OpVolumeFilter		
Description	Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.	
Base Type	MA_OpVolumeFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpVolumeID	OpVolumeID_Type	Repeated
Description	Indicates a specific OpVolumeID that is required to match the filter. If omitted, any OpVolumeID satisfies the filter. Multiple instances are logically ORed.	
OpVolumeCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpVolume must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-330: C2 UCI Extensions MA_OpZoneMT - OpVolumeFilter UCI Extension Field

OpLineFilter		
Description	Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.	
Base Type	MA_OpLineFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpLineID	OpLineID_Type	Repeated
Description	Indicates a specific OpLineID that is required to match the filter. If omitted, any OpLineID satisfies the filter. Multiple instances are logically ORed.	
OpLineCategory	MA_OpLineCategoryEnum	Repeated
Description	Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpLine must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-331: C2 UCI Extensions MA_OpZoneMT - OpLineFilter UCI Extension Field

OpLineCategory	
Description	Indicates an OpLine category that is required to match the filter. If omitted, any OpLine category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpLineCategoryEnum
Message Structure	
Enumeration Literal	**Description**
GENERAL_LINE	A general line where the function must be described by other procedures.
FORWARD_EDGE_OF_BATTLE_AREA	The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.
GUN_TARGET_LINE	Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.
CORRIDOR	Line with width describing the passageway designated for travel by friendly aircraft.
HOSTILE_BOUNDARY	Line that indicates a limit beyond which exists a hostile environment.
BUFFER_ZONE_BOUNDARY	Line beyond which friendly aircraft usually do not fly in peacetime.
LOW_LEVEL_TRANSIT_ROUTE	Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.
TACTICAL_ACTION_LINE	Line positioned in hostile airspace which may be used in conjunction with

	other criteria for hostile determination.
FIRE_SUPPORT_COORDINATION_LINE	Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander.
FORWARD_LINE_OF_OWN_TROOPS	Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.
LINE_OF_COMMUNICATIONS	Line the route that connects an operating military unit with its supply base.
JETWAY	Line with width that defines areas where commercial aircraft are flying through.
DEPARTURE_RADIAL	Line the defines the departure radial when performing Carrier admin departures.
RECOVERY_RADIAL	Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.
FINAL_BEARING	Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.
BRC	Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.

Table A-1-332: C2 UCI Extensions MA_OpZoneMT - OpLineCategory UCI Extension Field

SpeedLimits		
Description	Defines area where speeds are restricted.	
Base Type	MA_OpZoneSpeedLimitsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MinSpeedLimit	PathSegmentSpeedValueType	Optional
Description	Indicates the minimum speed limit in meters per second within a zone/region.	
MaxSpeedLimit	PathSegmentSpeedValueType	Optional
Description	Indicates the maximum speed limit in meters per second within a zone/region.	

Table A-1-333: C2 UCI Extensions MA_OpZoneMT - SpeedLimits UCI Extension Field

SignalReportFilter		
Description	Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter.	
Base Type	MA_SignalReportFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

SignalReportID	SignalReportID_Type	Repeated
Description	Indicates a specific SignalReportID that is required to match the filter. If omitted, any SignalReportID satisfies the filter. Multiple instances are logically ORed.	
SystemID	SystemID_Type	Repeated
Description	Indicates a specific source SystemID that is required to match the filter. If omitted, any source SystemID satisfies the filter. Multiple instances are logically ORed.	
SubsystemID	SubsystemID_Type	Repeated
Description	Indicates a specific source SubsystemID that is required to match the filter. If omitted, any source SubsystemID satisfies the filter. Multiple instances are logically ORed.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the SignalReport must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-334: C2 UCI Extensions MA_OpZoneMT - SignalReportFilter UCI Extension Field

OpPointFilter		
Description	Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter.	
Base Type	MA_OpPointFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPointID	OpPointID_Type	Repeated
Description	Indicates a specific OpPointID that is required to match the filter. If omitted, any OpPointID satisfies the filter. Multiple instances are logically ORed.	
OpPointCategory	OpPointCategoriesType	Repeated
Description	Indicates an OpPoint category that is required to match the filter. If omitted, any OpPoint category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpPoint must fall within to satisfy the filter. Multiple instances are logically ORed.Indicates that the OpRouting type is applicable. List size for this element is based on there being no category type for OpRouting objects	

Table A-1-335: C2 UCI Extensions MA_OpZoneMT - OpPointFilter UCI Extension Field

VehicleConfiguration		
Description	Defines vehicle configuration parameters that should change based on the planned location of a vehicle.	
Base Type	MA_VehicleCommandDataType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

VehicleAction	VehicleActionEnum	Optional
Description	Defines vehicle actions that can be done along a path segment of a route.	
IFF_Settings	IFF_Type	Optional
Description	If present the IFF settings for this vehicle will be changed along this path segment.	
SurvivabilityMode	VehicleSurvivabilityModeEnum	Optional
Description	Specifies that the vehicle should put itself into a more survivable mode. Different vehicles will have specific behaviors that reduce the signature of the vehicle (doors closed, minimal use of flaps, etc.) If this element is not present then it is assumed that the aircraft being configured has no survivability mode.	
LostCommTimeout	DurationType	Optional
Description	Defines the lost communication contingency timeout. When this timeout value is exceeded, the AV considers that lost communication has occurred and begins to perform contingency measures.	
LOS	boolean	Optional
Description	Use Line Of Sight (LOS) communication as backup contingency.	
LossOfLinkProcessing	VehicleLossOfLinkProcessingEnum	Optional
Description	Specifies whether the vehicle should invoke special Loss of Link processing. This could be turned off if the vehicle is going to be in a known and accepted lost communication situation.	
CommAllocationAction	CommAllocationActionType	Optional
Description	Indicates a change of communication allocation.	
QNH_Setting	PressureType	Optional
Description	Represents the barometric altimeter setting for correct MSL reading, commonly known as QNH, requested. Expressed in kilopascals (kPa).	

Table A-1-336: C2 UCI Extensions MA_OpZoneMT - VehicleConfiguration UCI Extension Field

DMPI_Filter		
Description	Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter.	
Base Type	MA_DMPI_FilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DMPI_ID	DMPI_ID_Type	Repeated
Description	Indicates a specific DMPI_ID that is required to match the filter. If omitted, any DMPI_ID satisfies the filter. Multiple instances are logically ORed.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the DMPI must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-337: C2 UCI Extensions MA_OpZoneMT - DMPI_Filter UCI Extension Field

Zone		
Description	This element represents the geospatial and temporal characteristics of the OpZone.	
Base Type	MA_OpZoneChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
GeometrySpecification	OpZoneType	Required
Description	Defines an OpZone geometry	
GeometryReference	MA_ReferentialOpZoneType	Required
Description	Defines an OpZone geometry via reference to other filtered objects	

Table A-1-338: C2 UCI Extensions MA_OpZoneMT - Zone UCI Extension Field

Category	
Description	This element indicates the type of zone.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational

	purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by

	friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-339: C2 UCI Extensions MA_OpZoneMT - Category UCI Extension Field

CategoryUniqueData		
Description	Contains parameters unique to certain OpZoneCategoryEnum values.	
Base Type	MA_OpZoneCategoryType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ConstrainedEntryExit	ConstrainedEntryExitType	Required
Description	Defines boundaries which applicable Systems can only enter and exit through defined edges.	
FilterArea	OpZoneFilterAreaPET	Repeated
Description	Indicates that the OpZone can be a zone filter type.	
Jamming	OpZoneJammingType	Required
Description	Indicates that the OpZone is a jamming control zone.	
KeepIn	IngressEgressType	Required
Description	Defines boundaries to which applicable Systems must stay inside.	
MissileLaunchPoint	OpZoneMissileDataType	Required
Description	Data defining a missile type, related track, and source of launch position.	
NoFire	OpZoneNoFireType	Required
Description	Defines areas where strike impact is restricted. Does not restrict the launch of weapons.	
NoFly	OpZoneNoFlyType	Required

Description	Defines area where flight is restricted. Equivalent to MIL-STD-6016 restricted zone.	
SpeedLimits	MA_OpZoneSpeedLimitsType	Required
Description	Defines area where speeds are restricted.	
VehicleConfiguration	MA_VehicleCommandDataType	Required
Description	Defines vehicle configuration parameters that should change based on the planned location of a vehicle.	
WeaponRestriction	OpZoneWeaponRestrictionType	Required
Description	Set of restricted weapons that cannot be used against a target type and or in a zone.	
WeatherConditions	OpZoneWeatherType	Required
Description	Defines area of weather conditions with potential of mission impact.	

Table A-1-340: C2 UCI Extensions MA_OpZoneMT - CategoryUniqueData UCI Extension Field

ReferenceObjectsFilter		
Description	The geometry of all objects returned by this filter will be used for the OpZone being defined.	
Base Type	MA_GeoLocatedObjectFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
EntityFilter	EntityFilterType	Optional
Description	Indicates a name/type of Entity required to match the filter. If omitted, any name/type of Entity satisfies the filter.	
SystemFilter	SystemFilterType	Optional
Description	Indicates a name/type of System required to match the filter. If omitted, any name/type of System satisfies the filter.	
OpPointFilter	MA_OpPointFilterType	Optional
Description	Indicates a name/type of OpPoint required to match the filter. If omitted, any name/type of OpPoint satisfies the filter.	
OpLineFilter	MA_OpLineFilterType	Optional
Description	Indicates a name/type of OpLine required to match the filter. If omitted, any name/type of OpLine satisfies the filter.	
OpZoneFilter	MA_OpZoneFilterType	Optional
Description	Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.	
OpVolumeFilter	MA_OpVolumeFilterType	Optional
Description	Indicates a name/type of OpVolume required to match the filter. If omitted, any name/type of OpVolume satisfies the filter.	
DMPI_Filter	MA_DMPI_FilterType	Optional
Description	Indicates a name/type of DMPI required to match the filter. If omitted, any name/type of DMPI satisfies the filter.	

SignalReportFilter	MA_SignalReportFilterType	Optional
Description	Indicates a name/type of SignalReport required to match the filter. If omitted, any name/type of SignalReport satisfies the filter.	

Table A-1-341: C2 UCI Extensions MA_OpZoneMT - ReferenceObjectsFilter UCI Extension Field

OpVolumeCategory	
Description	Indicates an OpVolume category that is required to match the filter. If omitted, any OpVolume category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.

HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.

TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-342: C2 UCI Extensions MA_OpZoneMT - OpVolumeCategory UCI Extension Field

GeometryReference		
Description	Defines an OpZone geometry via reference to other filtered objects	
Base Type	MA_ReferentialOpZoneType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ReferenceObjectsFilter	MA_GeoLocatedObjectFilterType	Repeated
Description	The geometry of all objects returned by this filter will be used for the OpZone being defined.	
HorizontalBuffer	DistanceType	Required
Description	The area of the resulting OpZone is the area of the referenced object, expanded by this horizontal buffer along the plane orthogonal to the NED frame's down direction	
MinAltitudeBuffer	DistanceType	Required
Description	The min altitude of the resulting OpZone is the min altitude of the referenced object, decreased by this min altitude buffer	
MaxAltitudeBuffer	DistanceType	Required
Description	The max altitude of the resulting OpZone is the max altitude of the referenced object, increased by this max altitude buffer	

Table A-1-343: C2 UCI Extensions MA_OpZoneMT - GeometryReference UCI Extension Field

OpZoneCategory	
Description	Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**

AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.

MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-344: C2 UCI Extensions MA_OpZoneMT - OpZoneCategory UCI Extension Field

OpZoneFilter

Description	Indicates a name/type of OpZone required to match the filter. If omitted, any name/type of OpZone satisfies the filter.	
Base Type	MA_OpZoneFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpZoneID	OpZoneID_Type	Repeated
Description	Indicates a specific OpZoneID that is required to match the filter. If omitted, any OpZoneID satisfies the filter. Multiple instances are logically ORed.	
OpZoneCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpZone category that is required to match the filter. If omitted, any OpZone category satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the OpZone must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-345: C2 UCI Extensions MA_OpZoneMT - OpZoneFilter UCI Extension Field

1.3.2.24 MA_NavigationReportMT UCI Extension

MA_NavigationReportMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_NavigationReportMDT

Table A-1-346: C2 UCI Extensions MA_NavigationReportMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_NavigationReportMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Required
Description	Indicates the unique ID of the System corresponding to this NavigationReport.	
Source	SystemSourceEnum	Required
Description	Indicates the source of the data in sibling elements. See enumeration annotations for further details.	
ContingencyLevel	SystemContingencyLevelEnum	Required

Description	Designates the current contingency state of the System. Contingencies can be an indication that the System will/has autonomously modified its means of navigation.	
Endurance	EnduranceType	Required
Description	Indicates the System's remaining endurance. It is expected that at least one of the optional children are populated in this context.	
Playtime	DurationType	Optional
Description	Estimated remaining playtime of operation/mission when taking into account RTB routing, fuel overhead, current fuel usage, current tasking, etc.	
Navigation	NavigationType	Optional
Description	Vehicle navigation method currently employed.	

Table A-1-347: C2 UCI Extensions MA_NavigationReportMT - MessageData UCI Extension Field

1.3.2.25 MA_AcceptableLevelOfRiskCommandMT UCI Extension

MA_AcceptableLevelOfRiskCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_AcceptableLevelOfRiskCommandMDT

Table A-1-348: C2 UCI Extensions MA_AcceptableLevelOfRiskCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AcceptableLevelOfRiskCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AcceptableRiskSetting	MA_AcceptableLevelOfRiskEnum	Required
Description	This is the acceptable level of risk permitted at the package or vehicle level, which applies to all tasks and actions for the vehicle/package while active.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	Identifies the Package ID or ACP System ID for which the acceptable risk parameter applies. If Package ID is used, this includes all package member System IDs, so a member ACP's System ID does not need to be separately included. This value overrides the AppliesToSystemID in the SurvivabilityRiskLevelMDT message.	
SurvivabilityRiskLevelID	SurvivabilityRiskLevelID_Type	Required
Description	The unique identifier for the Survivability Risk Level that should be active as the default setting.	
CommandID	CommandID_Type	Required

Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-349: C2 UCI Extensions MA_AcceptableLevelOfRiskCommandMT - MessageData UCI Extension Field

1.3.2.26 MA_ConstraintPriorityListActivationDataMT UCI Extension

MA_ConstraintPriorityListActivationDataMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ConstraintPriorityListActivationDataMDT

Table A-1-350: C2 UCI Extensions MA_ConstraintPriorityListActivationDataMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ConstraintPriorityListActivationDataMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ConstraintPriorityListActivationDataID	MA_ConstraintPriorityListActivationDataID_Type	Required
Description	The unique identifier for the message.	
DefaultConstraintPriorityListID	MA_ConstraintPriorityListID_Type	Required
Description	Contains the ID of the Constraint Priority List that should be set as the default.	
ListOfApplicableOpConstrainedConstraintPriorityListIDs	MA_ListOfApplicableOpConstrainedConstraintPriorityListIDType	Repeated
Description	When an Package/ACP is within this geozone, the linked Constraint Priority List will be used instead of the default. OpZone and OpVolume IDs can only appear once in this list.	

Table A-1-351: C2 UCI Extensions MA_ConstraintPriorityListActivationDataMT - MessageData UCI Extension Field

1.3.2.27 MA_ConstraintPriorityListCommandMT UCI Extension

MA_ConstraintPriorityListCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ConstraintPriorityListCommandMDT

Table A-1-352: C2 UCI Extensions MA_ConstraintPriorityListCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ConstraintPriorityListCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ConstraintPriorityListActivationDataID	MA_ConstraintPriorityListActivationDataID_Type	Required
Description	This ID of the Activated message that should be active.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-353: C2 UCI Extensions MA_ConstraintPriorityListCommandMT - MessageData UCI Extension Field

1.3.2.28 MA_ConstraintPriorityListMT UCI Extension

MA_ConstraintPriorityListMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ConstraintPriorityListMDT

Table A-1-354: C2 UCI Extensions MA_ConstraintPriorityListMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ConstraintPriorityListMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ConstraintPriorityListID	MA_ConstraintPriorityListID_Type	Required
Description	The unique identifier for the message.	
DefaultPriority	unsignedShort	Optional
Description	This field, if set, is the value that would be assigned to any ConstraintType that is not listed in the ListItem field. If this field is not set, the default value of ConstraintTypes not listed the ListItem field would be blank or the highest priority ("above" 0).	
ListItem	MA_ConstraintPriorityListItemType	Repeated
Description	Identifies a constraint type and it's priority in the Constraint Priority List.	

Table A-1-355: C2 UCI Extensions MA_ConstraintPriorityListMT - MessageData UCI Extension Field

1.3.2.29 MA_ExecutionPriorityListActivationDataMT UCI Extension

MA_ExecutionPriorityListActivationDataMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ExecutionPriorityListActivationDataMDT

Table A-1-356: C2 UCI Extensions MA_ExecutionPriorityListActivationDataMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ExecutionPriorityListActivationDataMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutionPriorityListActivationDataID	MA_ExecutionPriorityListActivationDataID_Type	Required
Description	The unique identifier for the message.	
DefaultExecutionPriorityListID	MA_ExecutionPriorityListID_Type	Required

Description	Contains the ID of the message that should be set as the default.	
ListOfApplicableOpConstrainedExecutionPriorityListIDs	MA_ListOfApplicableOpConstrainedExecutionPriorityListIDType	Repeated
Description	When an ACP is within this geozone, the linked message will be used instead of the default. OpZone and OpVolume IDs can only appear once in this list.	

Table A-1-357: C2 UCI Extensions MA_ExecutionPriorityListActivationDataMT - MessageData UCI Extension Field

1.3.2.30 MA_ExecutionPriorityListCommandMT UCI Extension

MA_ExecutionPriorityListCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ExecutionPriorityListCommandMDT

Table A-1-358: C2 UCI Extensions MA_ExecutionPriorityListCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ExecutionPriorityListCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutionPriorityListActivationDataID	MA_ExecutionPriorityListActivationDataID_Type	Required
Description	The ID of the Activation Data message that should be used.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-359: C2 UCI Extensions MA_ExecutionPriorityListCommandMT - MessageData UCI Extension Field

1.3.2.31 MA_ExecutionPriorityListMT UCI Extension

MA_ExecutionPriorityListMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ExecutionPriorityListMDT

Table A-1-360: C2 UCI Extensions MA_ExecutionPriorityListMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ExecutionPriorityListMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutionPriorityListID	MA_ExecutionPriorityListID_Type	Required
Description	The unique identifier for the message.	
ListItem	MA_ExecutionPriorityListItemType	Repeated
Description	Identifies an item (task or action) and its rank in the Execution Priority List.	

Table A-1-361: C2 UCI Extensions MA_ExecutionPriorityListMT - MessageData UCI Extension Field

1.3.2.32 MA_IgnoreConstraintCommandMT UCI Extension

MA_IgnoreConstraintCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_IgnoreConstraintCommandMDT

Table A-1-362: C2 UCI Extensions MA_IgnoreConstraintCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_IgnoreConstraintCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ReferenceType	RequirementsReferenceType	Required

Description	ID of the task(s) that the command applies to.	
ListOfConstraintsToIgnore	MA_RegionChoiceType	Repeated
Description	Indicates the constraint(s) being overridden.	
IgnoreAllConstraints	boolean	Optional
Description	If True, ignores all constraints.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-363: C2 UCI Extensions MA_IgnoreConstraintCommandMT - MessageData UCI Extension Field

1.3.2.33 MA_EngagementCriteriaCommandMT UCI Extension

MA_EngagementCriteriaCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_EngagementCriteriaCommandMDT

Table A-1-364: C2 UCI Extensions MA_EngagementCriteriaCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_EngagementCriteriaCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
EngagementCriteriaID	MA_EngagementCriteriaID_Type	Required
Description	Contains the ID of the Engagement Criteria message that should be active (for CommandState NEW or Update) or deactivated (for CommandState CANCEL).	
ApplicableRegion	MA_RegionChoiceType	Repeated
Description	The ID(s) of an OpZone/OpVolume or specified zone(s) that defines where this Engagement Criteria is applicable. If this field is not empty, an ACP and the Target must be within this region for the ACP to fire and the weapon to strike the intended	

	target. The MA shall only start an engagement for targets inside of the Region(s).	
TimeValid	DateTimeRangeType	Optional
Description	The time window for which the Engagement Criteria is valid. If this field is left empty, the Engagement Criteria is valid until the Command is canceled or an updated.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in this field as well.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-365: C2 UCI Extensions MA_EngagementCriteriaCommandMT - MessageData UCI Extension Field

1.3.2.34 MA_EngagementCriteriaMT UCI Extension

MA_EngagementCriteriaMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_EngagementCriteriaMDT

Table A-1-366: C2 UCI Extensions MA_EngagementCriteriaMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_EngagementCriteriaMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
EngagementCriteriaID	MA_EngagementCriteriaID_Type	Required
Description	The unique identifier for the message.	
IdMatrixCriteria	MA_IdMatrixCriteriaType	Optional
Description	These markers are used to identify an Entity as a Friend, Neutral, or Enemy and which enables or confirms the Entity's Standard Identity type to be set to FRIEND, NEUTRAL, SUSPECT or HOSTILE.	
TargetDesignationCriteria	MA_TargetDesignationCriteriaType	Repeated
Description	This provides a list of criteria that when met can trigger a MA_DesignationRequestMT to be sent to a Target Authority who can use the info in this message to send a	

	MA_DesignationMT that will designate an Entity as a Target.	
TargetAuthorityCriteria	MA_AuthorityCriteriaType	Optional
Description	This provides a list of Authorized Operators that are allowed to set an Entity as a Target and to respond to a StrikeConsentRequestMT. An Entity becomes a Target when a Designation message declares that entity to be a Target. This can also be done by an Action or Task specifying an Entity as a Target (TargetType). The associated Approval Policy will specify which Actions or Tasks make an Entity a Target and who is authorized to command them or approve a response that activates them. MA can send a MA_DesignationRequestMT to Operators identified in the Approval Authority. An Authorized operator can also directly send a MA_DesignationMT. An MA can be an authorized approver if specified, but an MA cannot send a MA_DesignationMT for itself. A Lead can send a MA_DesignationMT to a follower if a C2 has already approved the Target.	
TargetCustodyCriteria	MA_TargetCustodyCriteriaType	Repeated
Description	This describes how the Custody of a Target is confirmed and must be held through a Weapons Engagement with that Target.	
StrikeConsentRequired	boolean	Required
Description	Indicates whether (TRUE) or not (FALSE) the Strike Activities requires consent to transition from the ENABLED state to an ACTIVE state. If TRUE, each Strike Activity would be required to send a StrikeConsentRequestMT to an Authorized Operator (defined in the TargetAuthorityCriteria field) and receive a StrikeConsentRequestStatusMT before each Strike Activity can execute the Strike.	

Table A-1-367: C2 UCI Extensions MA_EngagementCriteriaMT - MessageData UCI Extension Field

1.3.2.35 MA_AcceptableLevelOfRiskCommandStatusMT UCI Extension

MA_AcceptableLevelOfRiskCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_AcceptableLevelOfRiskCommandStatusMDT

Table A-1-368: C2 UCI Extensions MA_AcceptableLevelOfRiskCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AcceptableLevelOfRiskCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.	

CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-369: C2 UCI Extensions MA_AcceptableLevelOfRiskCommandStatusMT - MessageData UCI Extension Field

1.3.2.36 MA_ConstraintPriorityListActivationStatusMT UCI Extension

MA_ConstraintPriorityListActivationStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ConstraintPriorityListActivationStatusMDT

Table A-1-370: C2 UCI Extensions MA_ConstraintPriorityListActivationStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ConstraintPriorityListActivationStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ConstraintPriorityListActivationDataID	MA_ConstraintPriorityListActivationDataID_Type	Required
Description	The unique identifier for the CPL Activation Data message.	
ActiveConstraintPriorityListID	MA_ConstraintPriorityListID_Type	Required
Description	The unique identifier for the active Constraint Priority List.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to respond for the Package or individual ACPs as needed.	

Table A-1-371: C2 UCI Extensions MA_ConstraintPriorityListActivationStatusMT - MessageData UCI Extension Field

1.3.2.37 MA_ConstraintPriorityListCommandStatusMT UCI Extension

MA_ConstraintPriorityListCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ConstraintPriorityListCommandStatusMDT

Table A-1-372: C2 UCI Extensions MA_ConstraintPriorityListCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ConstraintPriorityListCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-373: C2 UCI Extensions MA_ConstraintPriorityListCommandStatusMT - MessageData UCI Extension Field

1.3.2.38 MA_ExecutionPriorityListActivationStatusMT UCI Extension

MA_ExecutionPriorityListActivationStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details.

	See the XSD File for more info
Ext. Fields	MessageData : MA_ExecutionPriorityListActivationStatusMDT

Table A-1-374: C2 UCI Extensions MA_ExecutionPriorityListActivationStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ExecutionPriorityListActivationStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutionPriorityListActivationDataID	MA_ExecutionPriorityListActivationDataID_Type	Required
Description	The unique identifier for the EPL Activation Data message.	
ActiveExecutionPriorityListID	MA_ExecutionPriorityListID_Type	Required
Description	The unique identifier for the active EPL message.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to respond for the Package or individual ACPs as needed.	

Table A-1-375: C2 UCI Extensions MA_ExecutionPriorityListActivationStatusMT - MessageData UCI Extension Field

1.3.2.39 MA_ExecutionPriorityListCommandStatusMT UCI Extension

MA_ExecutionPriorityListCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ExecutionPriorityListCommandStatusMDT

Table A-1-376: C2 UCI Extensions MA_ExecutionPriorityListCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ExecutionPriorityListCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to	

	respond with ACCEPTED for the Package or individual ACPs as needed.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-377: C2 UCI Extensions MA_ExecutionPriorityListCommandStatusMT - MessageData UCI Extension Field

1.3.2.40 MA_IgnoreConstraintCommandStatusMT UCI Extension

MA_IgnoreConstraintCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_IgnoreConstraintCommandStatusMDT

Table A-1-378: C2 UCI Extensions MA_IgnoreConstraintCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_IgnoreConstraintCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state	

	machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-379: C2 UCI Extensions MA_IgnoreConstraintCommandStatusMT - MessageData UCI Extension Field

1.3.2.41 MA_EngagementCriteriaStatusMT UCI Extension

MA_EngagementCriteriaStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_EngagementCriteriaStatusMDT

Table A-1-380: C2 UCI Extensions MA_EngagementCriteriaStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_EngagementCriteriaStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
EngagementCriteriaID	MA_EngagementCriteriaID_Type	Optional
Description	Contains the ID of the Engagement Criteria message that is currently active. If this field is empty, that means there is no active Engagement Criteria and Engagement is not authorized.	
ApplicableRegion	MA_RegionChoiceType	Repeated
Description	The ID(s) of an OpZone/OpVolume or specified zone(s) that defines where the Engagement Criteria is applicable. If this field is not empty, the ACP must be in this region when it fires, and the calculated weapon impact location of the Target must be within this region. The MA shall only start an engagement for targets inside of the Region(s).	
TimeValid	DateTimeRangeType	Optional
Description	The time window for which the Engagement Criteria is valid. If this field is left empty, the Engagement Criteria is valid until the Command is canceled or an updated.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. If an ACP is a member of a package, the ACPs SystemID does not need to be in the field as well.	

Table A-1-381: C2 UCI Extensions MA_EngagementCriteriaStatusMT - MessageData UCI Extension Field

1.3.2.42 MA_EngagementCriteriaCommandStatusMT UCI Extension

MA_EngagementCriteriaCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_EngagementCriteriaCommandStatusMDT

Table A-1-382: C2 UCI Extensions MA_EngagementCriteriaCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_EngagementCriteriaCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual ACPs as needed.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-383: C2 UCI Extensions MA_EngagementCriteriaCommandStatusMT - MessageData UCI Extension Field

1.3.2.43 MA_SystemNotificationMT UCI Extension

MA_SystemNotificationMT	
Description	See annotations in child elements and messages/elements that use this type for details.

	See the XSD File for more info
Ext. Fields	MessageData : MA_SystemNotificationMDT NotificationDetail : CannotComplyType NotificationDetail : CannotComplyType

Table A-1-384: C2 UCI Extensions MA_SystemNotificationMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_SystemNotificationMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemSubjectIDs	MA_PackageSystemType	Required
Description	Indicates the subject of the Notification, the thing which the Notification is about.	
SystemPerspective	NotificationPerspectiveEnum	Repeated
Description	Indicates a perspective (characteristic, behavior and/or change) of the Subject that is the basis of the Notification. List size for this element is based on "Select All That Apply" condition.	
AssociatedMessage	AssociatedMessageType	Repeated
Description	Indicates a message that is associated with the Notification.	
NotificationDetail	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful	
NotificationID	NotificationID_Type	Required
Description	Indicates the unique ID of the Notification.	
NotificationState	NotificationStateEnum	Required
Description	Indicates the current state within an enumerated state machine that characterizes the life cycle of a notification.	
Timestamp	DateTimeType	Required
Description	The time at which the Notification was originally triggered (ex: event time).	
Source	NotificationSourceType	Required
Description	Indicates the originating source of the Notification.	
Severity	NotificationSeverityEnum	Required
Description	Indicates the overall severity of the notification. Informational notifications are meant to provide informational context of events. Indicates the existence of a hazardous condition requiring immediate action to prevent loss of life, equipment damage, or abortion of the mission. Critical notifications indicate the need for immediate attention.	

AppliesTo	SubjectType	Repeated
Description	Indicates an Asset that the Notification applies to and/or is affected by. Optional when the Notification characterizes the Subject without regard for Assets it might affect. Omission of this element doesn't necessarily mean the Notification isn't applicable to an Asset.	
NotificationNarrative	VisibleString1024Type	Optional
Description	Human readable message for display to an operator.	

Table A-1-385: C2 UCI Extensions MA_SystemNotificationMT - MessageData UCI Extension Field

NotificationDetail		
Description	Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful	
Base Type	CannotComplyType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Reason	CannotComplyEnum	Required
Description	Indicates the reason why an action (Task, [Capability]Command, [SupportingCapability]Command, [Capability]Activity, etc.) was rejected, interrupted, unallocated, failed and/or generally can't be completed satisfactorily. See enumeration annotations for further details.	
Description	VisibleString1024Type	Optional
Description	Indicates a human-readable text description of the reason for the "can't comply".	
AssociatedID	ID_Type	Optional
Description	Indicates an ID corresponding to another object that is associated with or the cause of the "can't comply".	

Table A-1-386: C2 UCI Extensions MA_SystemNotificationMT - NotificationDetail UCI Extension Field

NotificationDetail		
Description	Indicates details regarding the reason for and/or a contributing factor to the unsuccessful processing of the sibling AssociatedMessage element. This element is only expected when the Associated Message processing is not successful	
Base Type	CannotComplyType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Reason	CannotComplyEnum	Required
Description	Indicates the reason why an action (Task, [Capability]Command, [SupportingCapability]Command, [Capability]Activity, etc.) was rejected, interrupted,	

	unallocated, failed and/or generally can't be completed satisfactorily. See enumeration annotations for further details.	
Description	VisibleString1024Type	Optional
Description	Indicates a human-readable text description of the reason for the "can't comply".	
AssociatedID	ID_Type	Optional
Description	Indicates an ID corresponding to another object that is associated with or the cause of the "can't comply".	

Table A-1-387: C2 UCI Extensions MA_SystemNotificationMT - NotificationDetail UCI Extension Field

1.3.2.44 MA_EffectCapabilityMT UCI Extension

MA_EffectCapabilityMT	
Description	This message represents the Effect category of UCI Capability. See the XSD File for more info
Ext. Fields	AssociatedTaskType : TaskTypeEnum

Table A-1-388: C2 UCI Extensions MA_EffectCapabilityMT UCI Extension

AssociatedTaskType	
Description	Indicates a Task type associated with the Effect. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.
Base Type	TaskTypeEnum
Message Structure	
Enumeration Literal	**Description**
AIR_SAMPLE	Indicates the task is to sample and analyze the air around the vehicle.
AMTI	Indicates the task is to track air targets using an Airborne Moving Target Indicator.
AO	Indicates the task is to employ an Active Optical capability.
CARGO_DELIVERY	Indicates the task is to deliver cargo.
COMINT	Indicates the task is to employ a communications intelligence capability.
COMM_RELAY	Indicates the task is to relay communications from other battlespace participants.
COUNTER_SPACE	Indicates the task is to employ a Counter Space Capability.
EA	Indicates the task is to employ an electronic attack capability.
ESM	Indicates the task is to employ an Electronic Support Measure.
FLIGHT	Indicates the task is to fly.
ORBIT_CHANGE	Indicates a task to perform an orbit change via a spacecraft maneuver.

ORBITAL_SURVEILLANCE	Indicates a task to perform orbital surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates a task to perform orbital surveillance via sensor collection capabilities.
PO	Indicates the task is to employ a Passive Optical capability.
REFUEL	Indicates the task is to refuel another System or Entity.
SAR	Indicates the task is to employ a Synthetic Aperture Radar capability.
SMTI	Indicates the task is to track surface targets using a Surface Moving Target Indicator.
STRIKE	Indicates the task is to strike a target.
SYSTEM_DEPLOYMENT	Indicates the task is to deploy a system.
TACTICAL_ORDER	Indicates the task is to perform a tactical order.
WEATHER_RADAR	Indicates the task is to collect weather data.

Table A-1-389: C2 UCI Extensions MA_EffectCapabilityMT - AssociatedTaskType UCI Extension Field

1.3.2.45 MA_ActionCapabilityMT UCI Extension

MA_ActionCapabilityMT	
Description	This message represents the Action category of UCI Capability. See the XSD File for more info
Ext. Fields	AssociatedTaskType : MA_TaskTypeEnum Capability : MA_ActionCapabilityType MessageData : MA_ActionCapabilityMDT

Table A-1-390: C2 UCI Extensions MA_ActionCapabilityMT UCI Extension

AssociatedTaskType	
Description	Indicates a Task type associated with the Action. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_TaskTypeEnum
Message Structure	
Enumeration Literal	**Description**
AIR_SAMPLE	Indicates the task is to sample and analyze the air around the vehicle.
AMTI	Indicates the task is to track air targets using an Airborne Moving Target Indicator.
AO	Indicates the task is to employ an Active Optical capability.
CARGO_DELIVERY	Indicates the task is to deliver cargo.
CAP	Indicates the task is to execute a Combat Air Patrol.

COMINT	Indicates the task is to employ a communications intelligence capability.
COMM_RELAY	Indicates the task is to relay communications from other battlespace participants.
COUNTER_SPACE	Indicates the task is to employ a Counter Space Capability.
EA	Indicates the task is to employ an electronic attack capability.
ESCORT	Indicates the task is to escort another entity.
ESM	Indicates the task is to employ an Electronic Support Measure.
FLIGHT	Indicates the task is to fly.
ORBIT_CHANGE	Indicates a task to perform an orbit change via a spacecraft maneuver.
ORBITAL_SURVEILLANCE	Indicates a task to perform orbital surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates a task to perform orbital surveillance via sensor collection capabilities.
PO	Indicates the task is to employ a Passive Optical capability.
REFUEL	Indicates the task is to refuel another System or Entity.
SAR	Indicates the task is to employ a Synthetic Aperture Radar capability.
SMTI	Indicates the task is to track surface targets using a Surface Moving Target Indicator.
STRIKE	Indicates the task is to strike a target.
SYSTEM_DEPLOYMENT	Indicates the task is to deploy a system.
TACTICAL_ORDER	Indicates the task is to perform a tactical order.
WEATHER_RADAR	Indicates the task is to collect weather data.

Table A-1-391: C2 UCI Extensions MA_ActionCapabilityMT - AssociatedTaskType UCI Extension Field

Capability		
Description	Indicates an installed Action Capability, its configurable characteristics and its controllability.	
Base Type	MA_ActionCapabilityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityType	MA_ActionTypeEnum	Required
Description	Indicates the type of Action. See enumerated type annotations for further details.	
AssociatedTaskType	MA_TaskTypeEnum	Repeated
Description	Indicates a Task type associated with the Action. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.	
AssociatedCapabilityType	CapabilityTypeEnum	Repeated
Description	Indicates a Capability type associated with the Action. See enumerated type annotations for further details. List size for this element is based on "Select All That	

	Apply" condition.	
CapabilityOptions	ActionCapabilityOptionsType	Required
Description	Indicates the Capability's support for select optional elements in its accepted control interface messages. Support for these elements is expected to vary across Capability implementations but must be known in order to adapt and configure upstream Systems and Services which use the Capability.	
MessageOutput	ActionMessageOutputsEnum	Repeated
Description	Indicates a message that is an output of the Capability. See enumerated type annotations for details. List size for this element is based on "Select All That Apply" condition.	
AcceptedInterface	CapabilityControlInterfacesEnum	Repeated
Description	Indicates an accepted control interface for the Capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.	
CommsRequired	boolean	Optional
Description	Indicates whether (TRUE) or not (FALSE) communications are required to use the Capability. Omitted is equivalent to FALSE. For example, an unmanned system with no onboard storage would require communications to disseminate the outputs of the Capability.	
CapabilityID	CapabilityID_Type	Required
Description	Indicates the unique ID of this Capability. This ID is used to invoke the Capability via Command messages, report its activity via Activity messages, etc.	
DomainPairing	EnvironmentPairingEnum	Repeated
Description	Indicates the domain/environment which the Capability is employed from and the "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.	
ProductOutput	ProductOutputType	Repeated
Description	Indicates a Product that can be an output of the Capability.	
PackageOperation	PackageOperationEnum	Optional
Description	Indicates if and how the operation of the Capability can be coordinated with a partner Capability of another System in a multi-System Package. Omitting this element is equivalent to the enumerate NO_PACKAGE_OPERATION. The actual Package configuration including per-Capability partnering between Systems is given in the PackageStatus message. For example, a System that can perform cooperative geolocation could advertise with this element in the ESM_Capability message. A System that can navigate in a swarm with other Systems could advertise with this element in the FlightCapability message.	

Table A-1-392: C2 UCI Extensions MA_ActionCapabilityMT - Capability UCI Extension Field

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.
Base Type	MA_ActionCapabilityMDT

Message Structure		
Field Name	**Field Type**	**Field Usage**
Capability	MA_ActionCapabilityType	Repeated
Description	Indicates an installed Action Capability, its configurable characteristics and its controllability.	
AvailableMDF	MDF_Type	Repeated
Description	Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability.	

Table A-1-393: C2 UCI Extensions MA_ActionCapabilityMT - MessageData UCI Extension Field

1.3.2.46 MA_AMTI_CapabilityMT UCI Extension

MA_AMTI_CapabilityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_AMTI_CapabilityMDT Capability : MA_AMTI_CapabilityType

Table A-1-394: C2 UCI Extensions MA_AMTI_CapabilityMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AMTI_CapabilityMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Capability	MA_AMTI_CapabilityType	Repeated
Description	Indicates an installed AMTI Capability, its configurable characteristics and its controllability.	
AvailableMDF	MDF_Type	Repeated
Description	Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising	

	their availability and activating them is. Their content is abstracted by the UCI concept of Capability.

Table A-1-395: C2 UCI Extensions MA_AMTI_CapabilityMT - MessageData UCI Extension Field

Capability		
Description	Indicates an installed AMTI Capability, its configurable characteristics and its controllability.	
Base Type	MA_AMTI_CapabilityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MT_IIRS_Level	NIIRS_Type	Repeated
Description	Indicates a Moving Target Indication Interpretability Rating Scale level supported by the Capability. See annotations of the underlying type for further details.	
ComponentFieldOfRegard	ComponentFieldOfRegardType	Repeated
Description	Fields of Regard for each component in the capability.	
CapabilityType	AMTI_CapabilityEnum	Required
Description	Indicates the Capability's first tier type in the AMTI taxonomy. For AMTI, the first tier is an indication of the physical area covered. See enumerated type annotations for further details.	
SubCapabilityType	AMTI_SubCapabilityEnum	Repeated
Description	Indicates a SubCapability of the AMTI Capability; the second tier in the taxonomy of AMTI. For AMTI, the second tier is waveform category. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition. The selection of AUTO_SUBCAPABILITY_SELECT still requires the evaluation of all sibling attributes by the Capability.	
CapabilityOptions	AMTI_CapabilityOptionsType	Required
Description	Indicates the Capability's support for select optional elements in its accepted control interface messages. Support for these elements is expected to vary across Capability implementations but must be known in order to adapt and configure upstream Systems and Services which use the Capability.	
SupportedFrequencies	FrequencyRangeType	Repeated
Description	Indicates a frequency range used by the Capability.	
MessageOutput	AMTI_MessageOutputsEnum	Repeated
Description	Indicates a message that is an output of the Capability. See enumerated type annotations for details. List size for this element is based on "Select All That Apply" condition.	
AcceptedInterface	CapabilityControlInterfacesEnum	Repeated
Description	Indicates an accepted control interface for the Capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.	
CommsRequired	boolean	Optional

Description	Indicates whether (TRUE) or not (FALSE) communications are required to use the Capability. Omitted is equivalent to FALSE. For example, an unmanned system with no onboard storage would require communications to disseminate the outputs of the Capability.	
CapabilityID	CapabilityID_Type	Required
Description	Indicates the unique ID of this Capability. This ID is used to invoke the Capability via Command messages, report its activity via Activity messages, etc.	
DomainPairing	EnvironmentPairingEnum	Repeated
Description	Indicates the domain/environment which the Capability is employed from and the "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.	
ProductOutput	ProductOutputType	Repeated
Description	Indicates a Product that can be an output of the Capability.	
PackageOperation	PackageOperationEnum	Optional
Description	Indicates if and how the operation of the Capability can be coordinated with a partner Capability of another System in a multi-System Package. Omitting this element is equivalent to the enumerate NO_PACKAGE_OPERATION. The actual Package configuration including per-Capability partnering between Systems is given in the PackageStatus message. For example, a System that can perform cooperative geolocation could advertise with this element in the ESM_Capability message. A System that can navigate in a swarm with other Systems could advertise with this element in the FlightCapability message.	

Table A-1-396: C2 UCI Extensions MA_AMTI_CapabilityMT - Capability UCI Extension Field

1.3.2.47 MA_SystemManagementRequestMT UCI Extension

MA_SystemManagementRequestMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_SystemManagementRequestMDT RequestType : MA_SystemManagementRequestType VehicleSettings : MA_VehicleCommandDataType

Table A-1-397: C2 UCI Extensions MA_SystemManagementRequestMT UCI Extension

MessageData		
Description	This contains the message data.	
Base Type	MA_SystemManagementRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**

AuthorityType	EntityManagementRequestAuthorityEnum	Optional
Description	Indicates whether this SystemManagementRequest is a request to change data or an order that must be followed. Defaults to REQUEST if not filled.	
RequestSource	RequestSourceEnum	Optional
Description	This element indicates the type of service requesting the change. Defaults to AUTOMATIC_INTERNAL if not filled.	
ReferenceSystemID	SystemID_Type	Required
Description	Indicates the system to be acted upon by this message.	
RequestType	MA_SystemManagementRequestType	Repeated
Description	Indicates the type of request to perform on a system.	
UpdateFromControllingUnit	boolean	Optional
Description	Indicates if this SystemManagementRequest came from the system controlling the referenced system. This indicates a higher confidence in the new data.	
Source	SystemMessageIdentifierType	Optional
Description	Indicates the system initiating this request and allows the requesting Service, Activity, or Network to be specified.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-398: C2 UCI Extensions MA_SystemManagementRequestMT - MessageData UCI Extension Field

RequestType		
Description	Indicates the type of request to perform on a system.	
Base Type	MA_SystemManagementRequestType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SetMode	MessageModeEnum	Required
Description	Indicates a request to modify the mode of a system.	
SetIdentity	SystemIdentityType	Required
Description	Indicates a request to set the identity of a system.	
SetLink16Metadata	Link16MetadataType	Required
Description	Indicates a request to modify the Link16Metadata of a system.	
SetVoiceControl	VoiceControlType	Required
Description	Set the voice control frequency of a system.	
SetSensorEntityReporting	boolean	Required

Description	When TRUE, indicates an order to report all locally derived sensor, signal, track or Entity data. When FALSE, indicates an order to stop reporting.	
VehicleSettings	MA_VehicleCommandDataType	Required
Description	Indicates a request to modify vehicle settings.	

Table A-1-399: C2 UCI Extensions MA_SystemManagementRequestMT - RequestType UCI Extension Field

VehicleSettings		
Description	Indicates a request to modify vehicle settings.	
Base Type	MA_VehicleCommandDataType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
VehicleAction	VehicleActionEnum	Optional
Description	Defines vehicle actions that can be done along a path segment of a route.	
IFF_Settings	IFF_Type	Optional
Description	If present the IFF settings for this vehicle will be changed along this path segment.	
SurvivabilityMode	VehicleSurvivabilityModeEnum	Optional
Description	Specifies that the vehicle should put itself into a more survivable mode. Different vehicles will have specific behaviors that reduce the signature of the vehicle (doors closed, minimal use of flaps, etc.) If this element is not present then it is assumed that the aircraft being configured has no survivability mode.	
LostCommTimeout	DurationType	Optional
Description	Defines the lost communication contingency timeout. When this timeout value is exceeded, the AV considers that lost communication has occurred and begins to perform contingency measures.	
LOS	boolean	Optional
Description	Use Line Of Sight (LOS) communication as backup contingency.	
LossOfLinkProcessing	VehicleLossOfLinkProcessingEnum	Optional
Description	Specifies whether the vehicle should invoke special Loss of Link processing. This could be turned off if the vehicle is going to be in a known and accepted lost communication situation.	
CommAllocationAction	CommAllocationActionType	Optional
Description	Indicates a change of communication allocation.	
QNH_Setting	PressureType	Optional
Description	Represents the barometric altimeter setting for correct MSL reading, commonly known as QNH, requested. Expressed in kilopascals (kPa).	

Table A-1-400: C2 UCI Extensions MA_SystemManagementRequestMT - VehicleSettings UCI Extension Field

1.3.2.48 MA_SystemManagementRequestStatusMT UCI Extension

MA_SystemManagementRequestStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_SystemManagementRequestStatusMDT

Table A-1-401: C2 UCI Extensions MA_SystemManagementRequestStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_SystemManagementRequestStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Required
Description	Indicates the ID of the system responding to the SystemManagementRequest.	
ServiceID	ServiceID_Type	Required
Description	Indicates the ID of the service responding to the SystemManagementRequest.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-402: C2 UCI Extensions MA_SystemManagementRequestStatusMT - MessageData UCI Extension Field

1.3.2.49 MA_RoutePlanMT UCI Extension

MA_RoutePlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info

Ext. Fields	MessageData : MA_RoutePlanMDT

Table A-1-403: C2 UCI Extensions MA_RoutePlanMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_RoutePlanMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RoutePlanID	RoutePlanID_Type	Required
Description	Indicates a unique ID assigned to the RoutePlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	RoutePlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which the RoutePlan originated from.	
Plan	MA_RoutePlanType	Required
Description	Contains the details of the route plan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	RoutePlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-404: C2 UCI Extensions MA_RoutePlanMT - MessageData UCI Extension Field

1.3.2.50 MA_ActionMT UCI Extension

MA_ActionMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ActionMDT

Table A-1-405: C2 UCI Extensions MA_ActionMT UCI Extension

MessageData	
Description	See annotations in child elements and messages/elements that use this type

	for details.	
Base Type	MA_ActionMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActionID	ActionID_Type	Required
Description	Indicates a unique ID assigned to the Action. Minor updates to the Action can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new instance with a different child UUID value.	
ActionType	MA_ActionTypeEnum	Required
Description	Indicates the type of Action; the type of behavior to be performed when the Action is executed. An Action is a representation of an operational planning sentence/statement whose most complex form is: 1. Do [ActionType] within [ActionConstraints] to [TargetObject] within [TargetObjectConstraint] to/on/against/toward [SecondaryObject] Many other forms are also possible, including: 2. do [ActionType] 3. do [ActionType] to [TargetObject] 4. do [ActionType] to [TargetObject] doing [TargetObjectConstraint] to [SecondaryObject].	
ActionConstraints	MA_RequirementConstraintsType	Optional
Description	Indicates constraints on the Action. The constraint type used here is common to all Requirement types: Effect, Action, Task, [Capability]Command, Response. Requirement constraints can be 1) given in a [Requirement] message, 2) overridden as a planning input in a [Requirement]PlanCommand message, 3) further refined/overridden during plan creation and reported in a [Requirement]Plan message to inform subsequent planning or execution. See sibling ActionType element annotation for an overall description of Action structure.	
ActionGuidance	RequirementGuidanceType	Optional
Description	Indicates guidance on planning and C2 of the Action.	
TargetObject	IdentityKindInstanceType	Repeated
Description	Indicates the target of the Action; the object the Action will be done to or for. If omitted, the Action places no explicit limitation on which TargetObject it can be done to. See sibling ActionType element annotation for an overall description of Action structure. Note that the term "target" is used in a general sense. For some Requirement types, a more suitable term might be "subject", "point", "location", etc. For example, the "target" of a COVER Action could be a blue/friendly System.	
TargetObjectConstraints	RequirementTargetConstraintsType	Optional
Description	Indicates constraints "on" the TargetObject that must be satisfied before the Action can be done to or for it. See sibling ActionType element annotation for an overall	

	description of Effect structure.	
SecondaryObject	IdentityKindInstanceType	Repeated
Description	Indicates a thing the sibling TargetObject must act on/against/toward to make the Action applicable. See sibling ActionType element annotation for an overall description of Action structure.	
Metadata	RequirementMetadataType	Optional
Description	Indicates traceability and other metadata associated with the Action.	

Table A-1-406: C2 UCI Extensions MA_ActionMT - MessageData UCI Extension Field

1.3.2.51 MA_COP_ConfigurationSettingsCommandStatusMT UCI Extension

MA_COP_ConfigurationSettingsCommandStatusMT	
Description	Command status message in response to MA_COP_ConfigurationSettingsCommand.UCI_PRIMITIVE: Command-2. See the XSD File for more info
Ext. Fields	MessageData : MA_COP_ConfigurationSettingsCommandStatusMDT

Table A-1-407: C2 UCI Extensions MA_COP_ConfigurationSettingsCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_COP_ConfigurationSettingsCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-408: C2 UCI Extensions MA_COP_ConfigurationSettingsCommandStatusMT - MessageData UCI Extension Field

1.3.2.52 MA_COP_ConfigurationSettingsRequestStatusMT UCI Extension

MA_COP_ConfigurationSettingsRequestStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_COP_ConfigurationSettingsRequestStatusMDT

Table A-1-409: C2 UCI Extensions MA_COP_ConfigurationSettingsRequestStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_COP_ConfigurationSettingsRequestStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ConfigurationSettings	MA_COP_ConfigurationSettingsMDT	Optional
Description	The configuration settings	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-410: C2 UCI Extensions MA_COP_ConfigurationSettingsRequestStatusMT - MessageData UCI Extension Field

1.3.2.53 MA_COP_ConfigurationSettingsCommandMT UCI Extension

MA_COP_ConfigurationSettingsCommandMT	
Description	Command message to adjust the configuration settings for the COP among ACPs.UCI_PRIMITIVE: Command-2. See the XSD File for more info
Ext. Fields	MessageData : MA_COP_ConfigurationSettingsCommandMDT

Table A-1-411: C2 UCI Extensions MA_COP_ConfigurationSettingsCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_COP_ConfigurationSettingsCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApplicableToIDs	MA_PackageSystemType	Required
Description	Indicates the ID of the Package or System being configured.	
COP_LeaderID	SystemID_Type	Repeated
Description	Indicates a prioritized list of unique SystemIDs for leaders for the COP. If not specified, no change to leader requested	
RequestAuthority	MA_COP_ConfigurationAuthorityEnum	Required
Description	Indicates the priority of the request being made. Generally C2 will set this to ORDER and the COP lead will set this to REQUEST	
Configuration	MA_COP_ConfigurationType	Repeated
Description	A list of configuration items.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-412: C2 UCI Extensions MA_COP_ConfigurationSettingsCommandMT - MessageData UCI Extension Field

1.3.2.54 MA_COP_ConfigurationSettingsRequestMT UCI Extension

MA_COP_ConfigurationSettingsRequestMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_COP_ConfigurationSettingsRequestMDT

Table A-1-413: C2 UCI Extensions MA_COP_ConfigurationSettingsRequestMT UCI Extension

MessageData

Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_COP_ConfigurationSettingsRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApplicableToIDs	MA_PackageSystemType	Repeated
Description	Indicates the ID of the Package or System that should respond to the request.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-414: C2 UCI Extensions MA_COP_ConfigurationSettingsRequestMT - MessageData UCI Extension Field

1.3.2.55 MA_DesignationRequestMT UCI Extension

MA_DesignationRequestMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_DesignationRequestMDT

Table A-1-415: C2 UCI Extensions MA_DesignationRequestMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DesignationRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DesignationData	MA_DesignationDataType	Repeated
Description	Indicates the data for one or more designations.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-416: C2 UCI Extensions MA_DesignationRequestMT - MessageData UCI Extension Field

1.3.2.56 MA_DesignationMT UCI Extension

MA_DesignationMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_DesignationMDT

Table A-1-417: C2 UCI Extensions MA_DesignationMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DesignationMDT	
Message Structure		
Field Name	Field Type	Field Usage
DesignationData	MA_DesignationDataType	Repeated
Description	Indicates the data of one or more designations.	

Table A-1-418: C2 UCI Extensions MA_DesignationMT - MessageData UCI Extension Field

1.3.2.57 MA_DesignationRequestStatusMT UCI Extension

MA_DesignationRequestStatusMT	
Description	Reports the current status of a Designation. See the XSD File for more info
Ext. Fields	MessageData : MA_DesignationRequestStatusMDT

Table A-1-419: C2 UCI Extensions MA_DesignationRequestStatusMT UCI Extension

MessageData		
Description	See the annotation in the associated message for an overall description of the message and this type.	
Base Type	MA_DesignationRequestStatusMDT	
Message Structure		
Field Name	Field Type	Field Usage
DesignationDataStatus	DesignationDataStatusType	Repeated
Description	Indicates the status for one or more designations.	

RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-420: C2 UCI Extensions MA_DesignationRequestStatusMT - MessageData UCI Extension Field

1.3.2.58 MA_AcceptableLevelOfRiskStatusMT UCI Extension

MA_AcceptableLevelOfRiskStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_AcceptableLevelOfRiskStatusMDT

Table A-1-421: C2 UCI Extensions MA_AcceptableLevelOfRiskStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AcceptableLevelOfRiskStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AcceptableRiskSetting	MA_AcceptableLevelOfRiskEnum	Required
Description	This is the active acceptable level of risk permitted at the package or vehicle level, which applies to all tasks and actions for the vehicle/package while active.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	The IDs, Package or ACP, that this message applies to. A lead is able to respond for the Package or individual ACPs as needed.	
SurvivabilityRiskLevelID	SurvivabilityRiskLevelID_Type	Optional
Description	The unique identifier for the active Survivability Risk Level.	

Table A-1-422: C2 UCI Extensions MA_AcceptableLevelOfRiskStatusMT - MessageData UCI Extension Field

1.3.2.59 MA_CapabilitySetMT UCI Extension

MA_CapabilitySetMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info

Ext. Fields	MessageData : MA_CapabilitySetMDT

Table A-1-423: C2 UCI Extensions MA_CapabilitySetMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_CapabilitySetMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilitySetID	MA_CapabilitySetID_Type	Required
Description	Indicates the unique ID of the CapabilitySet.	
CapabilityID	CapabilityID_Type	Repeated
Description	Indicates a Capability that exists in the set.	
ApplicableSystemID	SystemID_Type	Optional
Description	Indicates the System that the CapabilitySet is relevant to.	
MissionPlanID	MissionPlanID_Type	Optional
Description	Indicates the MissionPlan that the CapabilitySet is relevant to.	
MissionID	MissionID_Type	Optional
Description	Indicates the Mission that the CapabilitySet is relevant to.	
DataRecordInstanceID	DataRecordInstanceID_Type	Optional
Description	Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee.	

Table A-1-424: C2 UCI Extensions MA_CapabilitySetMT - MessageData UCI Extension Field

1.3.2.60 MA_CAP_ExecutionStatusReportMT UCI Extension

MA_CAP_ExecutionStatusReportMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_CAP_ExecutionStatusReportMDT

Table A-1-425: C2 UCI Extensions MA_CAP_ExecutionStatusReportMT UCI Extension

MessageData	
Description	See annotations in child elements and messages/elements that use this type

	for details.	
Base Type	MA_CAP_ExecutionStatusReportMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PackageID	PackageID_Type	Optional
Description	The ID of the package executing the CAP.	
PackageRequirementID	RequirementInstanceID_ChoiceType	Repeated
Description	Contains the unique ID(s) of the package-level requirements associated with the ongoing CAP activity. This will typically be a package-level PROTECT_CAP Action and/or a package-level CAP Task.	
AOI_Coverage	PercentType	Optional
Description	Percent of the Area of Interest (AOI) currently covered by any vehicle's sensors.	
PlanMetrics	MA_CAP_PlanMetrics_Type	Optional
Description	Sensor coverage and time gap metrics of the planned CAP routes over the next racetrack cycle.	
ExecutionStatus	MA_CAP_ExecutionStatus_Type	Repeated
Description	Contains a list of CAP statuses from vehicles executing the CAP.	

Table A-1-426: C2 UCI Extensions MA_CAP_ExecutionStatusReportMT - MessageData UCI Extension Field

1.3.2.61 MA_SystemsNeededRequestMT UCI Extension

MA_SystemsNeededRequestMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_SystemsNeededRequestMDT

Table A-1-427: C2 UCI Extensions MA_SystemsNeededRequestMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_SystemsNeededRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Candidate	MA_SystemCandidateChoiceType	Required
Description	Indicates a list of possible systems that could be used to solve the RequiredTasks.	
ProposedRequirements	RequirementAllocationCommandType	Required

Description	Indicates a proposed Requirement for the System's needed analysis request.	
AssociationConstraint	RequirementAssociationConstraintType	Repeated
Description	Indicates an association between Requirements that is a constraint on how they are allocated and planned.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-428: C2 UCI Extensions MA_SystemsNeededRequestMT - MessageData UCI Extension Field

1.3.2.62 MA_SystemsNeededRequestStatusMT UCI Extension

MA_SystemsNeededRequestStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_SystemsNeededRequestStatusMDT

Table A-1-429: C2 UCI Extensions MA_SystemsNeededRequestStatusMT UCI Extension

MessageData		
Description	Indicates the minimum number of systems needed to complete a desired set of tasks.	
Base Type	MA_SystemsNeededRequestStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
IncompletePlanning	IncompleteProcessingType	Repeated
Description	Indicates planning items that failed to complete. It is used to indicate specific details about why something failed to plan.	
Option	SystemsNeededOptionType	Repeated
Description	Provides results of the command that was performed. Indicates a specific option for satisfying the tasks specified.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	

RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-430: C2 UCI Extensions MA_SystemsNeededRequestStatusMT - MessageData UCI Extension Field

1.3.2.63 MA_MissionPlanActivationCommandMT UCI Extension

MA_MissionPlanActivationCommandMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	OpPlanID : MA_OpPlanID_Type ApplicableToIDs : MA_PackageSystemType ByExecutionPlanSet : MA_ExecutionPlanSetActivationType MessageData : MA_MissionPlanActivationCommandMDT ActivationDetails : MA_MissionPlanActivationDetailsType Applicability : MA_PlanApplicabilityType OpPlan : MA_OpPlanActivationType Command : MA_MissionPlanActivationCommandType

Table A-1-431: C2 UCI Extensions MA_MissionPlanActivationCommandMT UCI Extension

OpPlanID		
Description	Indicates the unique ID of the TaskPlan to perform the sibling activation command on.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional

Description	Human readable text label with content meaningful to human operators.

Table A-1-432: C2 UCI Extensions MA_MissionPlanActivationCommandMT - OpPlanID UCI Extension Field

ApplicableToIDs		
Description	Indicates the unique ID of the System (e.g. aircraft, ground-based sensor, ship, satellite, site, etc.) or Package whose Capabilities, performance, position, etc. the plan is based on. If the plan is a fully detailed one that can be activated and executed, this element indicates the System intended/responsible for executing the plan and producing the intended effects. See annotations in sibling element PlannedForID which provide additional context.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	
PackageID	PackageID_Type	Required
Description	The ID of a Package.	
SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-433: C2 UCI Extensions MA_MissionPlanActivationCommandMT - ApplicableToIDs UCI Extension Field

ByExecutionPlanSet		
Description	Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state.	
Base Type	MA_ExecutionPlanSetActivationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutionPlanSetID	ExecutionPlanSetID_Type	Required
Description	Indicates the unique ID of the execution plan set.	
CommandType	PlanActivationCommandEnum	Required
Description	Indicates the activation to perform on the associated execution plan set.	

Table A-1-434: C2 UCI Extensions MA_MissionPlanActivationCommandMT - ByExecutionPlanSet UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanActivationCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	MissionPlanActivationCommandID_Type	Required
Description	The ID of the Mission Plan Activation Command. Used to reference the command when providing status.	
ApprovalManagementCommandID	CommandID_Type	Repeated
Description	Indicates the command(s) that initiated the Activation Approvals for the *Plans identified in the sibling Command element. The approval status can be found by locating the corresponding MissionPlanActivationApprovalStatus with the ID identified here. The ActivationCommand identified in the sibling Command\ActivationDetails should be verified against the ActivationType that is identified in the associated MissionPlanActivationApprovalCommand.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	
Command	MA_MissionPlanActivationCommandType	Repeated
Description	Indicates command or commands to change the activation state of a MissionPlan and/or it subplans or subordinate plans. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance.	
Applicability	MA_PlanApplicabilityType	Optional
Description	Indicates the System or Systems who are being commanded to activate a specified plan.	

Table A-1-435: C2 UCI Extensions MA_MissionPlanActivationCommandMT - MessageData UCI Extension Field

ActivationDetails		
Description	Indicates the commanded activation details.	
Base Type	MA_MissionPlanActivationDetailsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ByMissionPlan	MissionPlanActivationType	Required

Description	Indicates simultaneous activation of all sub-*Plans (RoutePlan, RouteActivityPlan for example) of a MissionPlan into the same activation state.	
BySubPlan	MA_MissionPlanSubplanActivationType	Required
Description	Indicates activation by sub-*Plan (RoutePlan or OrbitPlan for example) of the MissionPlan, with potentially different states for each.	
ByExecutionPlanSet	MA_ExecutionPlanSetActivationType	Required
Description	Indicates simultaneous activation of all sub-*Plans (ActivityPlan, RoutePlan for example) of an *ExecutionPlanSet into the same activation state.	

Table A-1-436: C2 UCI Extensions MA_MissionPlanActivationCommandMT - ActivationDetails UCI Extension Field

Applicability		
Description	Indicates the System or Systems who are being commanded to activate a specified plan.	
Base Type	MA_PlanApplicabilityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlannedForID	SystemID_Type	Optional
Description	Indicates the System which is the intended user of the plan. This type can have multiple meanings, but primarily enables multiple systems, possibly at different levels of warfare or C2 hierarchy levels, to maintain independent plans for the same System identified in the sibling ApplicableToID element. For instance, during planning at the operational level of warfare, a non-tactical planning service may create and simulate execution of plans internally. Such plans would identify the operational level System as the PlannedFor System; they would never be activated or executed by a real world tactical System. Alternatively, the operational level System could have a tactical planning service that is functionally capable and responsible for creating fully detailed plans that can be activated and executed by real world tactical Systems. Such a tactical planning service could be used even if the operational level System does not have control authority over the tactical System at the time the plan is created. Such plans would identify the tactical level System as the PlannedFor System. As a final example, a plan which is intended to be used by the same System which produces its intended effects would have a common SystemID between the sibling PlannedForID and ApplicableToID elements. This element does not replace existing approval policies, authorization or control status and therefore does not automatically imply that the PlannedFor System has any kind of authority over the receiving system.	
ApplicableToIDs	MA_PackageSystemType	Required

Description	Indicates the unique ID of the System (e.g. aircraft, ground-based sensor, ship, satellite, site, etc.) or Package whose Capabilities, performance, position, etc. the plan is based on. If the plan is a fully detailed one that can be activated and executed, this element indicates the System intended/responsible for executing the plan and producing the intended effects. See annotations in sibling element PlannedForID which provide additional context.

Table A-1-437: C2 UCI Extensions MA_MissionPlanActivationCommandMT - Applicability UCI Extension Field

OpPlan		
Description	Indicates commanded details for activation of an OpPlan.	
Base Type	MA_OpPlanActivationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPlanID	MA_OpPlanID_Type	Required
Description	Indicates the unique ID of the TaskPlan to perform the sibling activation command on.	
CommandType	PlanActivationCommandEnum	Required
Description	Indicates the activation to perform on the associated TaskPlan.	

Table A-1-438: C2 UCI Extensions MA_MissionPlanActivationCommandMT - OpPlan UCI Extension Field

Command		
Description	Indicates command or commands to change the activation state of a MissionPlan and/or it subplans or subordinate plans. If more than one instance of this element is given, each should correspond to a different MissionPlanID. For example, if the intent is to transition from one MissionPlan to another, the new MissionPlan can be activated in one instance and the old MissionPlan can be deactivated in another instance.	
Base Type	MA_MissionPlanActivationCommandType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Required
Description	Indicates the unique ID of the MissionPlan whose commanded activation state or states are summarized in the sibling element.	
ActivationDetails	MA_MissionPlanActivationDetailsType	Required
Description	Indicates the commanded activation details.	

Table A-1-439: C2 UCI Extensions MA_MissionPlanActivationCommandMT - Command UCI Extension Field

1.3.2.64 MA_ApprovalAuthorityMT UCI Extension

MA_ApprovalAuthorityMT

Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ApprovalAuthorityMDT ApplicableIDs : MA_PackageSystemType

Table A-1-440: C2 UCI Extensions MA_ApprovalAuthorityMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ApprovalAuthorityMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApprovalAuthorityID	ApprovalAuthorityID_Type	Required
Description	Indicates associated Approval Authority identifier for applicable system and approval policy.	
ApplicableIDs	MA_PackageSystemType	Repeated
Description	Indicates to which system this Approval Authority applies.	
SystemApprovalPolicy	SystemApprovalPolicyType	Repeated
Description	Indicates an ApprovalPolicy and one or more approval authorities ("approvers").	
DataRecordInstanceID	DataRecordInstanceID_Type	Optional
Description	Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee.	

Table A-1-441: C2 UCI Extensions MA_ApprovalAuthorityMT - MessageData UCI Extension Field

ApplicableIDs		
Description	Indicates to which system this Approval Authority applies.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	

PackageID	PackageID_Type	Required
Description	The ID of a Package.	
SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-442: C2 UCI Extensions MA_ApprovalAuthorityMT - ApplicableIDs UCI Extension Field

1.3.2.65 MA_ResponseMT UCI Extension

MA_ResponseMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ResponseMDT

Table A-1-443: C2 UCI Extensions MA_ResponseMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ResponseMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ResponseID	ResponseID_Type	Required
Description	Indicates the unique ID of the Response. Minor updates to the Response can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new instance with a different child UUID value.	
ResponseType	ResponseTypeEnum	Repeated
Description	Indicates the types of Response options included in the message. List size for this element is based on "Select All That Apply" condition.	
ResponseManagementConstraints	RequirementConstraintsType	Optional
Description	Indicates monitoring constraints on the Response. The constraint type used here is common to all Requirement types: Effect, Action, Task, [Capability]Command, Response. Requirement constraints can be 1) given in a [Requirement] message, 2) overridden as a planning input in a [Requirement]PlanCommand message, 3) further refined/overridden during plan creation and reported in a [Requirement]Plan message to inform subsequent planning or execution. See sibling Option element annotation for an overall description of Response structure. Constraints given here aren't conveyed to instantiated Response options for a particular triggering event. The constraints given here apply to Response management (planning, allocation and trigger monitoring prior to a triggering	

	event). Constraints to apply to instantiated Response options, if any, are given elsewhere in the message.	
ResponseManagementGuidance	RequirementGuidanceType	Optional
Description	Indicates guidance on planning and C2 of the Response.	
ResponsePlurality	MA_TaskPluralityEnum	Optional
Description	Indicates whether the response applies a single entity or as a joint condition. A joint condition implies condiseration of the response trigger and activation of the response by all associated entities.	
Option	MA_ResponseOptionDetailsType	Repeated
Description	Indicates an Option of the Response. Responses are designed to be flexible. A Response is a set of prioritized Options to address an anticipated triggering event, events or event sequence in the battlespace. Responses are allocated to Systems via a ResponsePlan. When a ResponsePlan is activated, monitoring for the triggering events in the Options begins. When a triggering event happens, the applicable Option(s) are evaluated.	
RequirementsTemplate	MA_RequirementsTemplateType	Repeated
Description	The sibling Option element associates triggering events with a desired Response one of which is to use this element, a Requirements template, to generate or change one or more Requirements messages/objects (Effect, Action, Task, [Capability]Command). An instance of this element is a RequirementsTemplate; it indicates the types of Requirements to generate or update and their details. Each RequirementsTemplate has a unique ID and therefore can be used for multiple Options.	
Metadata	RequirementMetadataType	Optional
Description	Indicates traceability and other metadata associated with the Response.	

Table A-1-444: C2 UCI Extensions MA_ResponseMT - MessageData UCI Extension Field

1.3.2.66 MA_ActionPlanMT UCI Extension

MA_ActionPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ActionPlanMDT

Table A-1-445: C2 UCI Extensions MA_ActionPlanMT UCI Extension

MessageData	
Description	Indicates the data being conveyed by the message.
Base Type	MA_ActionPlanMDT

Message Structure		
Field Name	**Field Type**	**Field Usage**
ActionPlanID	ActionPlanID_Type	Required
Description	Indicates a unique ID assigned to the ActionPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	ActionPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which this ActionPlan originated from.	
Plan	MA_ActionPlanType	Required
Description	Indicates the details of the ActionPlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	ActionPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-446: C2 UCI Extensions MA_ActionPlanMT - MessageData UCI Extension Field

1.3.2.67 MA_ActivityPlanMT UCI Extension

MA_ActivityPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ActivityPlanMDT

Table A-1-447: C2 UCI Extensions MA_ActivityPlanMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ActivityPlanMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActivityPlanID	ActivityPlanID_Type	Required
Description	Indicates a unique ID assigned to the ActivityPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant	

	updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	ActivityPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which this ActivityPlan originated from.	
Plan	MA_ActivityPlanType	Required
Description	Indicates the details of the ActivityPlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is a used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	ActivityPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-448: C2 UCI Extensions MA_ActivityPlanMT - MessageData UCI Extension Field

1.3.2.68 MA_RouteActivityPlanMT UCI Extension

MA_RouteActivityPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_RouteActivityPlanMDT

Table A-1-449: C2 UCI Extensions MA_RouteActivityPlanMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_RouteActivityPlanMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RouteActivityPlanID	RouteActivityPlanID_Type	Required
Description	Indicates a unique ID assigned to the RouteActivityPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	RouteActivityPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which the RouteActivityPlan originated from.	
Plan	MA_RouteActivityPlanType	Required

Description	Indicates the details of the RouteActivityPlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
PlanInputs	RouteActivityPlanInputsType	Optional
Description	Indicates inputs that were given in the *PlanCommand (or equivalent automated replan) to create the Plan.	

Table A-1-450: C2 UCI Extensions MA_RouteActivityPlanMT - MessageData UCI Extension Field

1.3.2.69 MA_AssessmentRequestMT UCI Extension

MA_AssessmentRequestMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_AssessmentRequestMDT Category : MA_AssessmentRequestType

Table A-1-451: C2 UCI Extensions MA_AssessmentRequestMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AssessmentRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AnalysisRouteID	AnalysisRouteID_Type	Optional
Description	This element indicates the AnalysisRoute to use when performing the Assessment specified in sibling Category.	
RequestFrequencyType	RepeatEnum	Required
Description	Indicates whether this request for assessment is a request for a single reply, or a request for periodic assessments with a corresponding request to cease periodic assessments.	
RequestFrequencyPeriod	DurationType	Optional
Description	If requesting periodic assessments, this field specifies the amount of time that should elapse between the periodic assessments, i.e. the repetition time interval of the assessment.	

ResultsInAssessmentMessage	boolean	Required
Description	If TRUE this element specifies that the response to this message should be an AssessmentRequest and an Assessment message. If FALSE the assessment data that is generated should be bundled inside the response. In some cases there isn't a native message and the results can only be reported via AssessmentRequestStatus.	
Category	MA_AssessmentRequestType	Required
Description	Indicates the assessment category for requested assessment.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-452: C2 UCI Extensions MA_AssessmentRequestMT - MessageData UCI Extension Field

Category		
Description	Indicates the assessment category for requested assessment.	
Base Type	MA_AssessmentRequestType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommPointingPlan	CommPointingPlanRequestType	Required
Description	This element is used to specify that communication requests should be assessed, and a Pointing Plan should be generated based on the requests. The native message is CommPointingPlan.	
CapabilityUtilization	CapabilityUtilizationRequestType	Required
Description	This assessment type is utilized to assess predicted Capability utilization along a mission planned route.	
RouteDeconfliction	RouteDeconflictionRequestType	Required
Description	This element is used to specify that an assessment of the conflicts along a route is requested.	
RouteVulnerabilityMetrics	RouteVulnerabilityMetricsRequestType	Required
Description	This element is used to specify that detection metrics need to be recomputed along a route due to threats. The data generated for detection metrics are different from threat assessments because detection metrics break down the exposure numbers to a much greater detail in order to generate a route than a threat assessment will.	
RouteThreatAssessment	ThreatAssessmentRequestType	Required
Description	This element is used to specify that an assessment of the threat exposure along a route is requested.	
TargetMobility	TargetMobilityRequestType	Required
Description	This element is used to specify that an assessment of the possible zone an Entity could have moved within some given period of time.	

VehicleThreatAssessment	VehicleThreatAssessmentRequestType	Required
Description	This element is used to specify that a threat assessment is being requested for the current position of a vehicle.	
ThreatNominationAssessment	ThreatNominationAssessmentRequestType	Required
Description	This element is used to specify that a threat nomination assessment is being requested for the given mission plans and entities.	
AchievabilityAssessment	AchievabilityAssessmentRequestPET	Required
Description	This element is used to specify that achievability assessment is being requested for the given type of AchievabilityRequest.	
CoverageAssessment	MA_CoverageAssessmentRequestType	Required
Description	This element is used to specify that assessment is being requested for sensor coverage.	

Table A-1-453: C2 UCI Extensions MA_AssessmentRequestMT - Category UCI Extension Field

1.3.2.70 MA_AssessmentRequestStatusMT UCI Extension

MA_AssessmentRequestStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	Assessment : MA_AssessmentType MessageData : MA_AssessmentRequestStatusMDT

Table A-1-454: C2 UCI Extensions MA_AssessmentRequestStatusMT UCI Extension

Assessment		
Description	This element indicates the results of the assessment. It is optional based on whether the assessment request specified that the response should result in native message.	
Base Type	MA_AssessmentType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommPointingPlan	CommPointingPlanAssessmentType	Required
Description	This element defines the response to a communications pointing plan assessment request.	
CapabilityUtilization	CapabilityUtilizationAssessmentType	Required
Description	This assessment type is utilized to assess predicted capability utilization along a mission planned route.	
RouteDeconfliction	RouteDeconflictionAssessmentType	Required

Description	This element defines the response to a route deconfliction assessment request.	
RouteVulnerabilityMetrics	RouteVulnerabilityMetricsAssessmentType	Required
Description	Indicates the results of assessment of vulnerability along a route.	
RouteThreatAssessment	RouteThreatAssessmentType	Required
Description	Indicates the results of assessment of threats along a route.	
TargetMobility	TargetMobilityAssessmentType	Required
Description	This element defines the response to a target mobility assessment request.	
VehicleThreatAssessment	VehicleThreatAssessmentType	Required
Description	This element defines the response to a vehicle threat assessment request.	
ThreatNominationAssessment	ThreatNominationAssessmentType	Required
Description	This element defines the response to a threat nomination assessment.	
AchievabilityAssessment	AchievabilityAssessmentPET	Required
Description	This element defines the response to an achievability assessment.	
CoverageAssessment	MA_CoverageAssessmentType	Required
Description	This element defines the response to a coverage assessment.	

Table A-1-455: C2 UCI Extensions MA_AssessmentRequestStatusMT - Assessment UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AssessmentRequestStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CompletionStatus	CompletionStatusType	Optional
Description	Intermediate status information for an assessment. Typically populated when sending periodic statuses of PROCESSING, but can also store start and finish times for this assessment.	
Assessment	MA_AssessmentType	Optional
Description	This element indicates the results of the assessment. It is optional based on whether the assessment request specified that the response should result in native message.	
IncompleteAssessment	IncompleteProcessingType	Repeated
Description	This element indicates any assessments that failed to complete. It is used to indicate details about why a specific assessment target failed to process.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required

Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-456: C2 UCI Extensions MA_AssessmentRequestStatusMT - MessageData UCI Extension Field

1.3.2.71 MA_AssessmentMT UCI Extension

MA_AssessmentMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_AssessmentMDT Assessment : MA_AssessmentType

Table A-1-457: C2 UCI Extensions MA_AssessmentMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AssessmentMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AssessmentID	AssessmentID_Type	Required
Description	This element references the assessment ID.	
AssessmentRequestID	RequestID_Type	Optional
Description	This element is required when the message is generated in response to an AssessmentRequest. It is omitted when the message is generated unsolicited.	
AnalysisRouteID	AnalysisRouteID_Type	Optional
Description	This element indicates the AnalysisRoute that was used to perform the assessment.	
Assessment	MA_AssessmentType	Required
Description	This element indicates the results of the assessment.	

Table A-1-458: C2 UCI Extensions MA_AssessmentMT - MessageData UCI Extension Field

Assessment

Description	This element indicates the results of the assessment.	
Base Type	MA_AssessmentType	
Message Structure		
Field Name	Field Type	Field Usage
CommPointingPlan	CommPointingPlanAssessmentType	Required
Description	This element defines the response to a communications pointing plan assessment request.	
CapabilityUtilization	CapabilityUtilizationAssessmentType	Required
Description	This assessment type is utilized to assess predicted capability utilization along a mission planned route.	
RouteDeconfliction	RouteDeconflictionAssessmentType	Required
Description	This element defines the response to a route deconfliction assessment request.	
RouteVulnerabilityMetrics	RouteVulnerabilityMetricsAssessmentType	Required
Description	Indicates the results of assessment of vulnerability along a route.	
RouteThreatAssessment	RouteThreatAssessmentType	Required
Description	Indicates the results of assessment of threats along a route.	
TargetMobility	TargetMobilityAssessmentType	Required
Description	This element defines the response to a target mobility assessment request.	
VehicleThreatAssessment	VehicleThreatAssessmentType	Required
Description	This element defines the response to a vehicle threat assessment request.	
ThreatNominationAssessment	ThreatNominationAssessmentType	Required
Description	This element defines the response to a threat nomination assessment.	
AchievabilityAssessment	AchievabilityAssessmentPET	Required
Description	This element defines the response to an achievability assessment.	
CoverageAssessment	MA_CoverageAssessmentType	Required
Description	This element defines the response to a coverage assessment.	

Table A-1-459: C2 UCI Extensions MA_AssessmentMT - Assessment UCI Extension Field

1.3.2.72 MA_FlightCapabilityStatusMT UCI Extension

MA_FlightCapabilityStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	Capability : MA_FlightCapabilityType MessageData : MA_FlightCapabilityMDT

Table A-1-460: C2 UCI Extensions MA_FlightCapabilityStatusMT UCI Extension

Capability		
Description	Provides a list of the specific Flight Capabilities as defined by the FlightCapabilityEnum.	
Base Type	MA_FlightCapabilityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityType	MA_FlightCapabilityEnum	Repeated
Description	Indicates the type of Flight capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.	
FlightCapabilityPerformanceProfile	MA_FlightCapabilityPerformanceProfileType	Optional
Description	Provides per-control mode performance parameters. Enables autonomy commands to properly conform to envelope limits or limit summary values before sending a FlightCommand to the vehicle.	
AcceptedInterface	CapabilityControlInterfacesEnum	Repeated
Description	Indicates an accepted control interface for the Capability. See enumeration annotations for further details. List size for this element is based on "Select All That Apply" condition.	
CommsRequired	boolean	Optional
Description	Indicates whether (TRUE) or not (FALSE) communications are required to use the Capability. Omitted is equivalent to FALSE. For example, an unmanned system with no onboard storage would require communications to disseminate the outputs of the Capability.	
CapabilityID	CapabilityID_Type	Required
Description	Indicates the unique ID of this Capability. This ID is used to invoke the Capability via Command messages, report its activity via Activity messages, etc.	
DomainPairing	EnvironmentPairingEnum	Repeated
Description	Indicates the domain/environment which the Capability is employed from and the "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.	
ProductOutput	ProductOutputType	Repeated
Description	Indicates a Product that can be an output of the Capability.	
PackageOperation	PackageOperationEnum	Optional
Description	Indicates if and how the operation of the Capability can be coordinated with a partner Capability of another System in a multi-System Package. Omitting this element is equivalent to the enumerate NO_PACKAGE_OPERATION. The actual Package configuration including per-Capability partnering between Systems is given in the PackageStatus message. For example, a System that can perform cooperative geolocation could advertise with this element in the ESM_Capability message. A System that can navigate in a swarm with other Systems could advertise with this element in the FlightCapability message.	

Table A-1-461: C2 UCI Extensions MA_FlightCapabilityStatusMT - Capability UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_FlightCapabilityMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Capability	MA_FlightCapabilityType	Repeated
Description	Provides a list of the specific Flight Capabilities as defined by the FlightCapabilityEnum.	
AvailableMDF	MDF_Type	Repeated
Description	Indicates a Mission Data File (MDF) that is available for use. All MDFs currently available for use should be listed. An MDF is a collection of data, organized into files, that is used to tailor, configure or otherwise affect the behavior of a Capability. MDFs can vary from mission to mission or even by phases of a single mission. They are generally developed in pre-mission planning and are unique for each Subsystem model. Their content is not standardized by UCI, but the mechanism for advertising their availability and activating them is. Their content is abstracted by the UCI concept of Capability.	

Table A-1-462: C2 UCI Extensions MA_FlightCapabilityStatusMT - MessageData UCI Extension Field

1.3.2.73 MA_COP_ConfigurationSettingsMT UCI Extension

MA_COP_ConfigurationSettingsMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_COP_ConfigurationSettingsMDT

Table A-1-463: C2 UCI Extensions MA_COP_ConfigurationSettingsMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_COP_ConfigurationSettingsMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApplicableToIDs	MA_PackageSystemType	Required
Description	Indicates the ID of the Package or System with the COP settings.	

COP_LeaderID	SystemID_Type	Optional
Description	Indicates the unique SystemID of the current leader of the COP.	
COP_LeaderBackupID	SystemID_Type	Repeated
Description	Indicates the unique SystemIDs of the backup leaders of the COP.	
COP_FusionCapable	boolean	Required
Description	Indicates if the COP is capable of fusing tracks.	
Configuration	MA_COP_ConfigurationType	Repeated
Description	A list of the configuration items.	

Table A-1-464: C2 UCI Extensions MA_COP_ConfigurationSettingsMT - MessageData UCI Extension Field

1.3.2.74 MA_StrikeActivityMT UCI Extension

MA_StrikeActivityMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	ActiveWeaponHSA_CSA : MA_HSA_CSA_Type Direction : MA_DirectionChoiceType MunitionActive : MA_MunitionActiveType MessageData : MA_StrikeActivityMDT ActiveWeaponState : MA_ActiveWeaponStateEnum Activity : MA_StrikeActivityType Course : MA_DirectionReferenceType Heading : MA_DirectionReferenceType Reference : MA_HeadingReferenceEnum

Table A-1-465: C2 UCI Extensions MA_StrikeActivityMT UCI Extension

ActiveWeaponHSA_CSA		
Description	Provides the heading/course, speed, and altitude of the munition while the Activity is ACTIVE_*.	
Base Type	MA_HSA_CSA_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Altitude	AltitudeReferenceType	Optional
Description	Altitude hold value to be achieved and reference to altitude measurement source.	

Speed	PathSegmentSpeedType	Optional
Description	Defines speed type (reference frame) for all speed related fields in this message.	
Direction	MA_DirectionChoiceType	Optional
Description	Defines heading or course type and reference to hold	

Table A-1-466: C2 UCI Extensions MA_StrikeActivityMT - ActiveWeaponHSA_CSA UCI Extension Field

Direction		
Description	Defines heading or course type and reference to hold	
Base Type	MA_DirectionChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Heading	MA_DirectionReferenceType	Optional
Description	Defines heading type and reference to hold.	
Course	MA_DirectionReferenceType	Optional
Description	Defines course type and reference to hold.	

Table A-1-467: C2 UCI Extensions MA_StrikeActivityMT - Direction UCI Extension Field

MunitionActive		
Description	Data regarding the munition while it is active or in flight.	
Base Type	MA_MunitionActiveType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActiveWeaponState	MA_ActiveWeaponStateEnum	Repeated
Description	Provides further munition state data detail if the Activity is ENABLED, ACTIVE_* or COMPLETED as reported by the ActivityState. This message can be omitted if ActivityState is DISABLED, FAILED or DELETED. List size for this element is based on "Select All That Apply" condition.	
ActiveWeaponHSA_CSA	MA_HSA_CSA_Type	Optional
Description	Provides the heading/course, speed, and altitude of the munition while the Activity is ACTIVE_*.	

Table A-1-468: C2 UCI Extensions MA_StrikeActivityMT - MunitionActive UCI Extension Field

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.
Base Type	MA_StrikeActivityMDT

Message Structure		
Field Name	**Field Type**	**Field Usage**
SubsystemID	SubsystemID_Type	Required
Description	Indicates the Subsystem which hosts the Strike Capabilities whose Activities are given in this message.	
Activity	MA_StrikeActivityType	Repeated
Description	Indicates the dynamic operational characteristics of an Activity of a Strike Capability.	

Table A-1-469: C2 UCI Extensions MA_StrikeActivityMT - MessageData UCI Extension Field

ActiveWeaponState	
Description	Provides further munition state data detail if the Activity is ENABLED, ACTIVE_* or COMPLETED as reported by the ActivityState. This message can be omitted if ActivityState is DISABLED, FAILED or DELETED. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_ActiveWeaponStateEnum
Message Structure	
Enumeration Literal	**Description**
TARGETED	Indicates that a target has been selected and the munition has not been released yet. Report with the ENABLED state generated by ActivityStateEnum.
REQUESTED	Indicates the munition has been requested for release. Report with the ENABLED state generated by ActivityStateEnum.
SEEKER_INACTIVE	Indicates the munition has been released but its local seeker has not been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.
SEEKER_ACTIVE	Indicates the munition has been released and its local seeker has been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.
FUZE_INACTIVE	Indicates the munition has been released but its fuze has not been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.
FUZE_ACTIVE	Indicates the munition has been released and its fuze has been activated. Report with the ACTIVE_* state generated by ActivityStateEnum.
NOT_FULLY_AUTONOMOUS	Indicates the munition has been released not fully autonomous and thus not able to guide without datalink or target destination. Report with the ACTIVE_* state generated by ActivityStateEnum.
FULLY_AUTONOMOUS	Indicates the munition has been released and is fully autonomous and able guide without datalink or target destination. Report with the ACTIVE_* state generated by ActivityStateEnum.
DEFEATED	Indicates the munition has been released and but it has been out maneuvered by the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.
TIMED_OUT	Indicates the munition flight has ended, and it did not intercept the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.
SUCCESS	Indicates the munition flight has ended, and it intercepted the assigned target. Report with the COMPLETED state generated by ActivityStateEnum.

Table A-1-470: C2 UCI Extensions MA_StrikeActivityMT - ActiveWeaponState UCI Extension Field

Activity		
Description	Indicates the dynamic operational characteristics of an Activity of a Strike Capability.	
Base Type	MA_StrikeActivityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SelectedForKeyLoad	boolean	Required
Description	Key Load indicates the store has been put into a state where it is ready to receive cryptographic keys.	
WeaponArmed	boolean	Required
Description	When true, the weapon is in the armed state.	
ReadyForRelease	boolean	Required
Description	The store has been prepared for launch such that when released the store will function properly.	
SelectedForRelease	boolean	Required
Description	When Master Arm, Release Consent, and any LAR or vehicle Attitude parameters enforced by the carriage are all met, this store will be released.	
SelectedForJettison	boolean	Required
Description	Jettison indicates an ejection from the host system without powering or preparing the delivery vehicle. The store will be in an inert state. When Master Arm, Release Consent, and any LAR or vehicle attitude parameters enforced by the carriage are all met, this store will be jettisoned.	
AssignedTarget	GeoLocatedObjectType	Optional
Description	Indicates the Entity target assigned to the StrikeCapability.	
AO_Code	AO_CodeType	Optional
Description	Indicates the current/set AO/laser PRF and/or PIM code.	
LAR_ID	LAR_ID_Type	Optional
Description	Indicates the ID of the associated LAR and/or LAR_Report associated with the Strike capability.	
ConsentRequired	boolean	Optional
Description	Indicates whether (TRUE) or not (FALSE) the Activity requires consent to transition from the ENABLED state to an ACTIVE state. This element is required and shall be TRUE for Activities associated with "mother may I" Capabilities.	
ConsentState	ConsentEnum	Optional
Description	Indicates the state of consent for the Activity. This element is required for Activities which require Consent.	
KineticFootprint	EnduranceFootprintType	Optional
Description	The estimated footprint of where the store may fall if released from the current position.	

TargetLockAcquired	boolean	Optional
Description	Indicated whether (TRUE) or not (FALSE) a lock on target has been acquired.	
MunitionActive	MA_MunitionActiveType	Optional
Description	Data regarding the munition while it is active or in flight.	
TimeToRelease	DurationType	Optional
Description	Time interval before strike capability is released.	

Table A-1-471: C2 UCI Extensions MA_StrikeActivityMT - Activity UCI Extension Field

Course		
Description	Defines course type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-472: C2 UCI Extensions MA_StrikeActivityMT - Course UCI Extension Field

Heading		
Description	Defines heading type and reference to hold.	
Base Type	MA_DirectionReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Value	AngleType	Required
Description	Indicates the heading or course value to hold.	
Reference	MA_HeadingReferenceEnum	Required
Description	Reference to type of heading or course given.	

Table A-1-473: C2 UCI Extensions MA_StrikeActivityMT - Heading UCI Extension Field

Reference	
Description	Reference to type of heading or course given.
Base Type	MA_HeadingReferenceEnum
Message Structure	
Enumeration Literal	**Description**

TRUE_NORTH	Heading value according to Geographic North.
MAGNETIC_NORTH	Heading value according to location where the planet's magnetic field points vertically downward.

Table A-1-474: C2 UCI Extensions MA_StrikeActivityMT - Reference UCI Extension Field

1.3.2.75 MA_OperatorRoleMT UCI Extension

MA_OperatorRoleMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	SystemID : SystemID_Type

Table A-1-475: C2 UCI Extensions MA_OperatorRoleMT UCI Extension

SystemID		
Description	Indicates the SystemID that the OperatorRole belongs to. If not populated, it is assumed to be the SystemID specified in the MessageHeader.	
Base Type	SystemID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-476: C2 UCI Extensions MA_OperatorRoleMT - SystemID UCI Extension Field

1.3.2.76 MA_OpPlanMT UCI Extension

MA_OpPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_OpPlanMDT

Table A-1-477: C2 UCI Extensions MA_OpPlanMT UCI Extension

MessageData

Description	Indicates the data being conveyed by the message.	
Base Type	MA_OpPlanMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPlanID	MA_OpPlanID_Type	Required
Description	Indicates a unique ID assigned to the OpPlan. Minor updates to the Plan can be indicated by an update of the value in the child Version element. More significant updates, replacements, etc. might warrant a new Plan with a different child UUID value.	
PlanCommandID	MA_OpPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which the OpPlan originated from.	
Plan	MA_OpPlanType	Required
Description	Contains the details of the Op* allocation plan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	

Table A-1-478: C2 UCI Extensions MA_OpPlanMT - MessageData UCI Extension Field

1.3.2.77 MA_OpStatusMT UCI Extension

MA_OpStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_OpStatusMDT

Table A-1-479: C2 UCI Extensions MA_OpStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_OpStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ExecutingPackageSystemIDs	MA_PackageSystemType	Required
Description	Identifies the Package ID or ACP System ID whose interpretation of the Op* status is reflected in this message. If Package ID is used, this includes all package member	

	System IDs, so a member ACP's System ID does not need to be separately included.	
CurrentState	OpStateEnum	Required
Description	The state of the schedule (active or inactive) for which the defined schedule timeline applies.	
OpID	OpID_ChoiceType	Required
Description	Indicates the Op* being commanded. Leaving this blank makes the command apply to all Op* IDs	
CommandID	CommandID_Type	Repeated
Description	Indicates the unique ID of the Op* command resulting in this status.	

Table A-1-480: C2 UCI Extensions MA_OpStatusMT - MessageData UCI Extension Field

1.3.2.78 MA_OpCommandMT UCI Extension

MA_OpCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_OpCommandMDT

Table A-1-481: C2 UCI Extensions MA_OpCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_OpCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Command	MA_OpCommandType	Repeated
Description	Indicates an instance of an Op* command. An Op* command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Op* CommandStatus message.	

Table A-1-482: C2 UCI Extensions MA_OpCommandMT - MessageData UCI Extension Field

1.3.2.79 MA_InteroperabilityPolicyMT UCI Extension

MA_InteroperabilityPolicyMT	
Description	See the annotation in the associated message for an overall description of the message and this type.

	See the XSD File for more info
Ext. Fields	MessageData : MA_InteroperabilityPolicyMDT

Table A-1-483: C2 UCI Extensions MA_InteroperabilityPolicyMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_InteroperabilityPolicyMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
InteroperabilityPolicyID	MA_InteroperabilityPolicyID_Type	Required
Description	Indicates the unique ID of the interoperability policy.	
OpPolicy	MA_OpPolicyType	Repeated
Description	Indicates a set of interoperability policies for the handling of OpPoints, OpLines, OpVolumes, etc.	
TimedZone	TimedZoneType	Optional
Description	Indicates the geospatial zone and optionally time/date span or spans when the policy is active. For Op* types, the policy is applicable to 1) the time of Op* instances that lies within the time/date span and 2) the area of Op* instances which falls within the zone. This element supersedes the sibling Expires element.	
Expires	DateTimeType	Optional
Description	Indicates the time and date when the policy expires. The policy is applicable to Op* instances during their existence prior to this time. When the sibling TimedZone element is used, the time spans specified there supersede this element.	
MissionTraceability	MissionTraceabilityType	Optional
Description	Allows the creator of the interoperability policy to specify the source or motivation for this InteroperabilityPolicy such as ATO, ACO, etc.	
DataRecordInstanceID	DataRecordInstanceID_Type	Optional
Description	Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee.	

Table A-1-484: C2 UCI Extensions MA_InteroperabilityPolicyMT - MessageData UCI Extension Field

1.3.2.80 MA_AuthorizationMT UCI Extension

MA_AuthorizationMT	
Description	See annotations in child elements and messages/elements that use this type for details.

	See the XSD File for more info
Ext. Fields	UnauthorizedExceptFor : MA_RequirementAuthorizationPermissionsType CapabilityTaxonomy : MA_CapabilityTaxonomyType Opaque : MA_OpaqueEnum AuthorizedExceptFor : MA_RequirementAuthorizationPermissionsType CapabilityCommand : MA_CapabilityTypeEnum CapabilitySettingTypes : MA_CapabilityTypeEnum ByDataActivation : MA_DataActivationAuthorizationType CAP : MA_CAP_CapabilityEnum UnauthorizedExceptFor : MA_PlanActivationPermissionsType AuthorizedExceptFor : MA_PlatformSettingsPermissionsType DataTypes : MA_DataActivationTypeEnum AuthorizedExceptFor : MA_OperatorRequestPermissionsType CommTerminal : MA_CommTerminalEnum ByRequirementExecution : MA_RequirementAuthorizationType ByDataModification : MA_DataModificationAuthorizationType ActivityPlan : MA_ActivityPlanPartsType MessageData : MA_AuthorizationMDT AuthorizedArea : MA_AuthorizedZoneChoiceType AuthorizedExceptFor : MA_PlanActivationPermissionsType ByPlanActivation : MA_PlanAuthorizationType AuthorizedExceptFor : MA_DataModificationPermissionsType ByPlatformSettingsModification : MA_SettingsModficationAuthorizationType UnauthorizedExceptFor : MA_DataModificationPermissionsType UnauthorizedExceptFor : MA_OperatorRequestPermissionsType RadarAltimeter : MA_RadarAltimeterEnum RequestTypes : MA_OperatorRequestTypeEnum ByPlanParts : MA_PlanPartsType AdditionalPlatformSettingTypes : MA_PlatformSettingPermissionsTypeEnum IFF : MA_IFF_Enum RequirementTypes : MA_RequirementTaxonomyDetailedType UnauthorizedExceptFor : MA_PlatformSettingsPermissionsType UserIdentifier : MA_UserType ByOperatorRequest : MA_OperatorRequestAuthorizationType

Table A-1-485: C2 UCI Extensions MA_AuthorizationMT UCI Extension

UnauthorizedExceptFor

Description	This element indicates that all requirement types are NOT authorized to be executed with the exception of the enumeration values listed in the permissionsType	
Base Type	MA_RequirementAuthorizationPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequirementTypes	MA_RequirementTaxonomyDetailedType	Optional
Description	Indicates the set of Requirements that are authorized or unauthorized to be commanded depending on the parent MA_RequirementAuthorizationType	
AdditionalCapabilityTypes	SupportCapabilityTypeEnum	Repeated
Description	Indicates the set of additional support capabilities that are authorized or unauthorized to be commanded depending on the parent MA_RequirementAuthorizationType. List size for this element is based on "Select All That Apply" condition	

Table A-1-486: C2 UCI Extensions MA_AuthorizationMT - UnauthorizedExceptFor UCI Extension Field

CapabilityTaxonomy		
Description	Indicates specific Capability details/modes that are applicable.	
Base Type	MA_CapabilityTaxonomyType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Action	MA_ActionTypeEnum	Repeated
Description	Indicates an Action Capability. List size for this element is based on "Select All That Apply" condition.	
AirSample	AirSampleCapabilityEnum	Repeated
Description	Indicates an AirSample Capability. List size for this element is based on "Select All That Apply" condition.	
AMTI	AMTI_SpecificDataType	Repeated
Description	Indicates an AMTI Capability and any optional SubCapabilities.	
AO	AO_CapabilityEnum	Repeated
Description	Indicates an AO Capability. List size for this element is based on "Select All That Apply" condition.	
CAP	MA_CAP_CapabilityEnum	Repeated
Description	Indicates a Combat Air Patrol Capability.	
CargoDelivery	CargoDeliverySpecificDataType	Repeated
Description	Indicates a CargoDelivery Capability and any optional SubCapabilities.	
COMINT	COMINT_SpecificDataType	Repeated
Description	Indicates a COMINT Capability and any optional SubCapabilities.	
CommRelay	CommCapabilityEnum	Repeated

Description	Indicates a CommRelay Capability. List size for this element is based on "Select All That Apply" condition.	
CommTerminal	MA_CommTerminalEnum	Optional
Description	Indicates a CommTerminal Capability. List size for this element is based on "Select All That Apply" condition.	
CounterSpace	CS_CapabilityEnum	Optional
Description	Indicates a CounterSpace Capability.	
EA	CapabilityInitiationEnum	Repeated
Description	Indicates an EA Capability. List size for this element is based on "Select All That Apply" condition.	
Effect	EffectTypeEnum	Repeated
Description	Indicates an Effect Capability. List size for this element is based on "Select All That Apply" condition.	
ESM	ESM_SpecificDataType	Repeated
Description	Indicates an ESM Capability and any optional SubCapabilities.	
Flight	FlightCapabilityEnum	Repeated
Description	Indicates a Flight Capability. List size for this element is based on "Select All That Apply" condition.	
IFF	MA_IFF_Enum	Optional
Description	Indicates an IFF Capability. List size for this element is based on "Select All That Apply" condition.	
Opaque	MA_OpaqueEnum	Optional
Description	Indicates an Opaque Capability. List size for this element is based on "Select All That Apply" condition.	
OrbitChange	OrbitChangeCapabilityEnum	Repeated
Description	Indicates an OrbitChange Capability. List size for this element is based on "Select All That Apply" condition.	
OrbitalSurveillance	OrbitalSurveillanceSpecificDataType	Repeated
Description	Indicates an OrbitalSurveillance Capability and any optional SubCapabilities.	
OrbitalSurveillanceSensor	OrbitalSurveillanceSensorCapabilityEnum	Repeated
Description	Indicates an OrbitalSurveillanceSensor Capability. List size for this element is based on "Select All That Apply" condition.	
PO	PO_CapabilityEnum	Repeated
Description	Indicates a PO Capability. List size for this element is based on "Select All That Apply" condition.	
RadarAltimeter	MA_RadarAltimeterEnum	Optional
Description	Indicates a Radar Altimeter Capability. List size for this element is based on "Select All That Apply" condition.	
Refuel	RefuelCapabilityEnum	Repeated
Description	Indicates a Refuel Capability. List size for this element is based on "Select All That	

	Apply" condition.	
Response	ResponseTypeEnum	Repeated
Description	Indicates a Response Capability. List size for this element is based on "Select All That Apply" condition.	
SAR	SAR_SpecificDataType	Repeated
Description	Indicates a SAR Capability and any optional SubCapabilities.	
SMTI	SMTI_SpecificDataType	Repeated
Description	Indicates a SMTI Capability and any optional SubCapabilities.	
Strike	StoreType	Repeated
Description	Indicates a Strike Capability.	
SystemDeployment	SystemDeploymentEnum	Repeated
Description	Indicates a SystemDeployment Capability. List size for this element is based on "Select All That Apply" condition.	
TacticalOrder	TacticalOrderCapabilityEnum	Repeated
Description	Indicates a TacticalOrder Capability. List size for this element is based on "Select All That Apply" condition.	
WeatherRadar	WeatherRadarCapabilityEnum	Optional
Description	Indicates a WeatherRadar Capability.	

Table A-1-487: C2 UCI Extensions MA_AuthorizationMT - CapabilityTaxonomy UCI Extension Field

Opaque	
Description	Indicates an Opaque Capability. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpaqueEnum
Message Structure	
Enumeration Literal	**Description**
OPAQUE	Indicates a Opaque Capability.

Table A-1-488: C2 UCI Extensions MA_AuthorizationMT - Opaque UCI Extension Field

AuthorizedExceptFor		
Description	This element indicates that all requirement types are authorized to be executed with the exception of the enumeration values listed in the permissionsType	
Base Type	MA_RequirementAuthorizationPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequirementTypes	MA_RequirementTaxonomyDetailedType	Optional
Description	Indicates the set of Requirements that are authorized or unauthorized to be	

	commanded depending on the parent MA_RequirementAuthorizationType	
AdditionalCapabilityTypes	SupportCapabilityTypeEnum	Repeated
Description	Indicates the set of additional support capabilities that are authorized or unauthorized to be commanded depending on the parent MA_RequirementAuthorizationType. List size for this element is based on "Select All That Apply" condition	

Table A-1-489: C2 UCI Extensions MA_AuthorizationMT - AuthorizedExceptFor UCI Extension Field

CapabilityCommand	
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_CapabilityTypeEnum
Message Structure	
Enumeration Literal	**Description**
ACTION	Indicates an action capability.
AIR_SAMPLE	Indicates an air sampling capability.
AMTI	Indicates an airborne moving target indication (AMTI) capability.
AO	Indicates an active optical capability.
CAP	Indicates a cargo delivery capability.
CARGO_DELIVERY	Indicates a cargo delivery capability.
COMINT	Indicates a Communications Intelligence (COMINT) capability.
COMM_RELAY	Indicates a communications relay capability.
COMM_TERMINAL	Indicates a communications terminal capability
COUNTER_SPACE	Indicates a Counter Space Capability.
EA	Indicates an electronic attack (EA) capability.
EFFECT	Indicates an effect capability.
ESM	Indicates an electronic support measures (ESM) capability.
FLIGHT	Indicates a capability for waypoint-based flight tasks/commands.
IFF	Indicates an Identification Friend or Foe (IFF) capability.
OPAQUE	Indicates a Opaque Capability
ORBIT_CHANGE	Indicates this space system has the ability to change orbits.
ORBITAL_SURVEILLANCE	Indicates a task to perform Orbital Surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates an Orbital Surveillance Sensor Capability.
PO	Indicates a passive optical capability.
RADAR_ALTIMETER	Indicates a Radar Altimeter capability
REFUEL	Indicates a capability to refuel another asset.
RESPONSE	Indicates an response capability.

SAR	Indicates a synthetic aperture radar (SAR) capability.
SMTI	Indicates a surface moving target indication (SMTI) capability.
STRIKE	Indicates a strike capability.
SYSTEM_DEPLOYMENT	Indicates a deployment capability.
TACTICAL_ORDER	Indicates a tactical order capability.
TASK	Indicates a task capability.
WEATHER_RADAR	Indicates a weather radar capability.

Table A-1-490: C2 UCI Extensions MA_AuthorizationMT - CapabilityCommand UCI Extension Field

CapabilitySettingTypes	
Description	List size for this element is based on "Select All That Apply" condition
Base Type	MA_CapabilityTypeEnum
Message Structure	
Enumeration Literal	**Description**
ACTION	Indicates an action capability.
AIR_SAMPLE	Indicates an air sampling capability.
AMTI	Indicates an airborne moving target indication (AMTI) capability.
AO	Indicates an active optical capability.
CAP	Indicates a cargo delivery capability.
CARGO_DELIVERY	Indicates a cargo delivery capability.
COMINT	Indicates a Communications Intelligence (COMINT) capability.
COMM_RELAY	Indicates a communications relay capability.
COMM_TERMINAL	Indicates a communications terminal capability
COUNTER_SPACE	Indicates a Counter Space Capability.
EA	Indicates an electronic attack (EA) capability.
EFFECT	Indicates an effect capability.
ESM	Indicates an electronic support measures (ESM) capability.
FLIGHT	Indicates a capability for waypoint-based flight tasks/commands.
IFF	Indicates an Identification Friend or Foe (IFF) capability.
OPAQUE	Indicates a Opaque Capability
ORBIT_CHANGE	Indicates this space system has the ability to change orbits.
ORBITAL_SURVEILLANCE	Indicates a task to perform Orbital Surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates an Orbital Surveillance Sensor Capability.
PO	Indicates a passive optical capability.
RADAR_ALTIMETER	Indicates a Radar Altimeter capability
REFUEL	Indicates a capability to refuel another asset.

RESPONSE	Indicates an response capability.
SAR	Indicates a synthetic aperture radar (SAR) capability.
SMTI	Indicates a surface moving target indication (SMTI) capability.
STRIKE	Indicates a strike capability.
SYSTEM_DEPLOYMENT	Indicates a deployment capability.
TACTICAL_ORDER	Indicates a tactical order capability.
TASK	Indicates a task capability.
WEATHER_RADAR	Indicates a weather radar capability.

Table A-1-491: C2 UCI Extensions MA_AuthorizationMT - CapabilitySettingTypes UCI Extension Field

ByDataActivation		
Description	Indicates which types of data are authorized to be activated.	
Base Type	MA_DataActivationAuthorizationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizedExceptFor	MA_DataActivationPermissionsType	Required
Description	This element indicates that all data types are authorized to be activated with the exception of the enumeration values listed in the permissionType.	
UnauthorizedExceptFor	MA_DataActivationPermissionsType	Required
Description	This element indicates that all data types are NOT authorized to be activated with the exception of the enumeration values listed in the permissionType.	

Table A-1-492: C2 UCI Extensions MA_AuthorizationMT - ByDataActivation UCI Extension Field

CAP	
Description	Indicates a Combat Air Patrol Capability.
Base Type	MA_CAP_CapabilityEnum
Message Structure	
Enumeration Literal	**Description**
COMBAT_AIR_PATROL	Indicates a CAP capability

Table A-1-493: C2 UCI Extensions MA_AuthorizationMT - CAP UCI Extension Field

UnauthorizedExceptFor	
Description	This element indicates that all plan types are NOT authorized to be activated with the exception of the enumeration values listed in the permissionsType.
Base Type	MA_PlanActivationPermissionsType
Message Structure	

Field Name	Field Type	Field Usage
ByMissionPlan	MissionPlanActivationSettingType	Repeated
Description	Indicates the set of ByMissionPlan activation commands that are authorized. List size allows for identifying those MissionPlans whose subordinate MissionPlans are either included in the activation (TRUE) or not (FALSE) via the CommandSubordinatePlans setting.	
ByPlanParts	MA_PlanPartsType	Optional
Description	Indicates which *Plan parts are authorized to be activated.	

Table A-1-494: C2 UCI Extensions MA_AuthorizationMT - UnauthorizedExceptFor UCI Extension Field

AuthorizedExceptFor		
Description	This element indicates that all vehicle setting types are authorized to be modified with the exception of the enumeration values listed in the permissionType.	
Base Type	MA_PlatformSettingsPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilitySettingTypes	MA_CapabilityTypeEnum	Repeated
Description	List size for this element is based on "Select All That Apply" condition	
AdditionalPlatformSettingTypes	MA_PlatformSettingPermissionsTypeEnum	Repeated
Description	List size for this element is based on "Select All That Apply" condition	

Table A-1-495: C2 UCI Extensions MA_AuthorizationMT - AuthorizedExceptFor UCI Extension Field

DataTypes	
Description	Indicates the set of data type messages that are authorized or unauthorized to be activated depending on the parent MA_DataActivationAuthorizationType. List size for this element is based on "Select All That Apply" condition
Base Type	MA_DataActivationTypeEnum
Message Structure	
Enumeration Literal	**Description**
COMM_POINTING_COMMAND	See MT type for annotation.
MA_CONSTRAINT_PRIORITY_LIST_COMMAND	See MT type for annotation.
MA_COP_CONFIGURATION_SETTINGS_COMMAND	See MT type for annotation.
MA_ENGAGEMENT_CRITERIA_COMMAND	See MT type for annotation.
MA_EXECUTION_PRIORITY_LIST_COMMAND	See MT type for annotation.
MA_IGNORE_CONSTRAINT_COMMAND	See MT type for annotation.

Table A-1-496: C2 UCI Extensions MA_AuthorizationMT - DataTypes UCI Extension Field

AuthorizedExceptFor		
Description	This element indicates that all request types are authorized to be sent with the exception of the enumeration values listed in the permissionType.	
Base Type	MA_OperatorRequestPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequestTypes	MA_OperatorRequestTypeEnum	Repeated
Description	Indicates the set of Request Messages that C2 users can send to MA. List size for this element is based on "Select All That Apply" condition	

Table A-1-497: C2 UCI Extensions MA_AuthorizationMT - AuthorizedExceptFor UCI Extension Field

CommTerminal	
Description	Indicates a CommTerminal Capability. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_CommTerminalEnum
Message Structure	
Enumeration Literal	**Description**
COMM_TERMINAL	Indicates a Comm Terminal Capability.

Table A-1-498: C2 UCI Extensions MA_AuthorizationMT - CommTerminal UCI Extension Field

ByRequirementExecution		
Description	Indicates which Requirement (Effect, Action, Task, [Capability]Command, Response) execution/performance related activities are authorized.	
Base Type	MA_RequirementAuthorizationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizedExceptFor	MA_RequirementAuthorizationPermissionsType	Required
Description	This element indicates that all requirement types are authorized to be executed with the exception of the enumeration values listed in the permissionsType	
UnauthorizedExceptFor	MA_RequirementAuthorizationPermissionsType	Required
Description	This element indicates that all requirement types are NOT authorized to be executed with the exception of the enumeration values listed in the permissionsType	

Table A-1-499: C2 UCI Extensions MA_AuthorizationMT - ByRequirementExecution UCI Extension Field

ByDataModification		
Description	Indicates which types of data messages are authorized to be created, modified or deleted.	
Base Type	MA_DataModificationAuthorizationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizedExceptFor	MA_DataModificationPermissionsType	Required
Description	This element indicates that all data types are authorized to be created, modified or deleted with the exception of the enumeration values listed in the permissionType.	
UnauthorizedExceptFor	MA_DataModificationPermissionsType	Required
Description	This element indicates that all data types are NOT authorized to be created, modified or deleted with the exception of the enumeration values listed in the permissionType	

Table A-1-500: C2 UCI Extensions MA_AuthorizationMT - ByDataModification UCI Extension Field

ActivityPlan		
Description	Indicates applicable ActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	

TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-501: C2 UCI Extensions MA_AuthorizationMT - ActivityPlan UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_AuthorizationMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizationID	AuthorizationID_Type	Required
Description	Indicates the unique ID of the authorization.	
UserIdentifier	MA_UserType	Required
Description	Indicates the unique ID of the User being granted authorization to task. User types are SystemID, OperatorRoleID, or UserIdentifier	
SystemID	SystemID_Type	Required
Description	Indicates the unique ID of the System which the User, given in the sibling UserIdentifier element, is authorized to task.	
Precedence	unsignedInt	Required
Description	This element indicates the precedence level at which the User is authorized to task the System. The larger the number, the higher the precedence. Any Tasks or Commands created by the User must not exceed this precedence level.	
ByPlanActivation	MA_PlanAuthorizationType	Repeated
Description	Indicates authorization to send Plan Activation commands.	
ByRequirementExecution	MA_RequirementAuthorizationType	Optional
Description	Indicates which Requirement (Effect, Action, Task, [Capability]Command, Response) execution/performance related activities are authorized.	
ByDataModification	MA_DataModificationAuthorizationType	Optional
Description	Indicates which types of data messages are authorized to be created, modified or deleted.	
ByDataActivation	MA_DataActivationAuthorizationType	Optional
Description	Indicates which types of data are authorized to be activated.	
ByOperatorRequest	MA_OperatorRequestAuthorizationType	Optional
Description	Indicates which types of request messages are authorized to be sent.	

PackageAuthorization	boolean	Optional
Description	This element indicates authorization to create, modify and dissolve teams.	
ByPlatformSettingsModification	MA_SettingsModficationAuthorizationType	Optional
Description	This element indicates authorization to send settings commands.	
AuthorizedTime	DateTimeRangeType	Optional
Description	This element indicates the time range during which the Authorization applies. More specifically, this is the time range in which the User must create/submit Tasks in order for them to be allocated to the specified System. If omitted, the Authorization is applicable until removed.	
AuthorizedArea	MA_AuthorizedZoneChoiceType	Optional
Description	This choice type indicates the area over which the authorization applies. It contains either a zone defined in this message or the ID of an existing OpZone	
DataRecordInstanceID	DataRecordInstanceID_Type	Optional
Description	Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee.	

Table A-1-502: C2 UCI Extensions MA_AuthorizationMT - MessageData UCI Extension Field

AuthorizedArea		
Description	This choice type indicates the area over which the authorization applies. It contains either a zone defined in this message or the ID of an existing OpZone	
Base Type	MA_AuthorizedZoneChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizedZone	ZoneType	Required
Description	This element indicates the area over which the authorization applies. More specifically, this is the geospatial zone in which the target of Tasks created by the User must fall in order for them to be allocated to the specified System. If omitted, the Authorization is applicable to all areas.	
AuthorizedOpZoneID	OpZoneID_Type	Required
Description	This element indicates the area defined by an existing OpZone over which the authorization applies. More specifically, this is the geospatial zone in which the target of Tasks created by the User must fall in order for them to be allocated to the specified System. If omitted, the Authorization is applicable to all areas.	

Table A-1-503: C2 UCI Extensions MA_AuthorizationMT - AuthorizedArea UCI Extension Field

AuthorizedExceptFor	
Description	This element indicates that all plan types are authorized to be activated with the exception of the enumeration values listed in the permissionsType.

Base Type	MA_PlanActivationPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ByMissionPlan	MissionPlanActivationSettingType	Repeated
Description	Indicates the set of ByMissionPlan activation commands that are authorized. List size allows for identifying those MissionPlans whose subordinate MissionPlans are either included in the activation (TRUE) or not (FALSE) via the CommandSubordinatePlans setting.	
ByPlanParts	MA_PlanPartsType	Optional
Description	Indicates which *Plan parts are authorized to be activated.	

Table A-1-504: C2 UCI Extensions MA_AuthorizationMT - AuthorizedExceptFor UCI Extension Field

ByPlanActivation		
Description	Indicates authorization to send Plan Activation commands.	
Base Type	MA_PlanAuthorizationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizedExceptFor	MA_PlanActivationPermissionsType	Required
Description	This element indicates that all plan types are authorized to be activated with the exception of the enumeration values listed in the permissionsType.	
UnauthorizedExceptFor	MA_PlanActivationPermissionsType	Required
Description	This element indicates that all plan types are NOT authorized to be activated with the exception of the enumeration values listed in the permissionsType.	

Table A-1-505: C2 UCI Extensions MA_AuthorizationMT - ByPlanActivation UCI Extension Field

AuthorizedExceptFor		
Description	This element indicates that all data types are authorized to be created, modified or deleted with the exception of the enumeration values listed in the permissionType.	
Base Type	MA_DataModificationPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DataTypes	MA_DataMessageTypeEnum	Repeated
Description	Indicates the set of data type messages that are authorized or unauthorized to be edited, created or deleted depending on the parent MA_DataModificationAuthorizationType.	

Table A-1-506: C2 UCI Extensions MA_AuthorizationMT - AuthorizedExceptFor UCI Extension Field

ByPlatformSettingsModification		
Description	This element indicates authorization to send settings commands.	
Base Type	MA_SettingsModficationAuthorizationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizedExceptFor	MA_PlatformSettingsPermissionsType	Required
Description	This element indicates that all vehicle setting types are authorized to be modified with the exception of the enumeration values listed in the permissionType.	
UnauthorizedExceptFor	MA_PlatformSettingsPermissionsType	Required
Description	This element indicates that all vehicle setting types are unauthorized to be modified with the exception of the enumeration values listed in the permissionType.	

Table A-1-507: C2 UCI Extensions MA_AuthorizationMT - ByPlatformSettingsModification UCI Extension Field

UnauthorizedExceptFor		
Description	This element indicates that all data types are NOT authorized to be created, modified or deleted with the exception of the enumeration values listed in the permissionType	
Base Type	MA_DataModificationPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DataTypes	MA_DataMessageTypeEnum	Repeated
Description	Indicates the set of data type messages that are authorized or unauthorized to be edited, created or deleted depending on the parent MA_DataModificationAuthorizationType.	

Table A-1-508: C2 UCI Extensions MA_AuthorizationMT - UnauthorizedExceptFor UCI Extension Field

UnauthorizedExceptFor		
Description	This element indicates that all request types are NOT authorized to be sent with the exception of the enumeration values listed in the permissionType.	
Base Type	MA_OperatorRequestPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequestTypes	MA_OperatorRequestTypeEnum	Repeated
Description	Indicates the set of Request Messages that C2 users can send to MA. List size for this element is based on "Select All That Apply" condition	

Table A-1-509: C2 UCI Extensions MA_AuthorizationMT - UnauthorizedExceptFor UCI Extension Field

RadarAltimeter	
Description	Indicates a Radar Altimeter Capability. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_RadarAltimeterEnum
Message Structure	
Enumeration Literal	Description
RADAR_ALTIMETER	Indicates a Radar Altimeter Capability.

Table A-1-510: C2 UCI Extensions MA_AuthorizationMT - RadarAltimeter UCI Extension Field

RequestTypes	
Description	Indicates the set of Request Messages that C2 users can send to MA. List size for this element is based on "Select All That Apply" condition
Base Type	MA_OperatorRequestTypeEnum
Message Structure	
Enumeration Literal	Description
DATA_RECORD_MANAGEMENT_REQUEST	See MT type for annotation.
MA_ASSESSMENT_REQUEST	See MT type for annotation.
MA_CONTROL_REQUEST	See MT type for annotation.
QUERY_DATA_REQUEST	See MT type for annotation.

Table A-1-511: C2 UCI Extensions MA_AuthorizationMT - RequestTypes UCI Extension Field

ByPlanParts		
Description	Indicates which *Plan parts are authorized to be activated.	
Base Type	MA_PlanPartsType	
Message Structure		
Field Name	Field Type	Field Usage
ActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	
OrbitActivityPlan	MA_ActivityPlanPartsType	Optional

Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-512: C2 UCI Extensions MA_AuthorizationMT - ByPlanParts UCI Extension Field

AdditionalPlatformSettingTypes	
Description	List size for this element is based on "Select All That Apply" condition
Base Type	MA_PlatformSettingPermissionsTypeEnum
Message Structure	
Enumeration Literal	**Description**
MA_ACCEPTABLE_LEVEL_OF_RISK_COMMAND	See MT type for annotation.
MA_SYSTEM_MANAGEMENT_REQUEST	See MT type for annotation.
PLANNING_FUNCTION_SETTINGS_COMMAND	See MT type for annotation.

Table A-1-513: C2 UCI Extensions MA_AuthorizationMT - AdditionalPlatformSettingTypes UCI Extension Field

IFF	
Description	Indicates an IFF Capability. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_IFF_Enum
Message Structure	
Enumeration Literal	**Description**
IFF	Indicates an IFF Capability.

Table A-1-514: C2 UCI Extensions MA_AuthorizationMT - IFF UCI Extension Field

RequirementTypes	
Description	Indicates the set of Requirements that are authorized or unauthorized to be

	commanded depending on the parent MA_RequirementAuthorizationType	
Base Type	MA_RequirementTaxonomyDetailedType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityTaxonomy	MA_CapabilityTaxonomyType	Optional
Description	Indicates specific Capability details/modes that are applicable.	
Effect	EffectTypeEnum	Repeated
Description	Indicates Effects applicable via the Effect* message set. List size for this element is based on "Select All That Apply" condition.	
Action	MA_ActionTypeEnum	Repeated
Description	Indicates Actions applicable via the Action* message set. List size for this element is based on "Select All That Apply" condition.	
Task	MA_TaskTypeEnum	Repeated
Description	Indicates Tasks applicable via the Task* message set. List size for this element is based on "Select All That Apply" condition.	
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates CapabilityCommands applicable via a [Capability]Command message set. List size for this element is based on "Select All That Apply" condition.	
Response	ResponseTypeEnum	Repeated
Description	Indicates Responses applicable via the Response* message set. List size for this element is based on "Select All That Apply" condition.	

Table A-1-515: C2 UCI Extensions MA_AuthorizationMT - RequirementTypes UCI Extension Field

UnauthorizedExceptFor		
Description	This element indicates that all vehicle setting types are unauthorized to be modified with the exception of the enumeration values listed in the permissionType.	
Base Type	MA_PlatformSettingsPermissionsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilitySettingTypes	MA_CapabilityTypeEnum	Repeated
Description	List size for this element is based on "Select All That Apply" condition	
AdditionalPlatformSettingTypes	MA_PlatformSettingPermissionsTypeEnum	Repeated
Description	List size for this element is based on "Select All That Apply" condition	

Table A-1-516: C2 UCI Extensions MA_AuthorizationMT - UnauthorizedExceptFor UCI Extension Field

UserIdentifier

Description	Indicates the unique ID of the User being granted authorization to task. User types are SystemID, OperatorRoleID, or UserIdentifier	
Base Type	MA_UserType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OperatorRoleID	OperatorRoleID_Type	Required
Description	This choice type is used in the context of RBAC to specify an operator role to tie a set of authorized permissions to.	
SystemID	SystemID_Type	Required
Description	This choice type is used to specify a system ID to tie a set of authorized permissions to.	
UserIdentifier	UserIdentifierType	Required
Description	This choice type is used to specify a User Identifier to tie a set of authorized permissions to.	

Table A-1-517: C2 UCI Extensions MA_AuthorizationMT - UserIdentifier UCI Extension Field

ByOperatorRequest		
Description	Indicates which types of request messages are authorized to be sent.	
Base Type	MA_OperatorRequestAuthorizationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AuthorizedExceptFor	MA_OperatorRequestPermissionsType	Required
Description	This element indicates that all request types are authorized to be sent with the exception of the enumeration values listed in the permissionType.	
UnauthorizedExceptFor	MA_OperatorRequestPermissionsType	Required
Description	This element indicates that all request types are NOT authorized to be sent with the exception of the enumeration values listed in the permissionType.	

Table A-1-518: C2 UCI Extensions MA_AuthorizationMT - ByOperatorRequest UCI Extension Field

1.3.2.81 MA_OpCommandStatusMT UCI Extension

MA_OpCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_OpCommandStatusMDT

Table A-1-519: C2 UCI Extensions MA_OpCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_OpCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-520: C2 UCI Extensions MA_OpCommandStatusMT - MessageData UCI Extension Field

1.3.2.82 MA_ApprovalPolicyMT UCI Extension

MA_ApprovalPolicyMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	TaskPlan : MA_TaskPlanPartsType OpLineCategory : MA_OpLineCategoryEnum ActivityPlan : MA_ActivityPlanPartsType ApplicablePlanningActivity : MA_PlanPolicyApplicablePlanType OpVolumeCategory : MA_OpZoneCategoryEnum TaskType : MA_TaskTypeEnum OrbitActivityPlan : MA_ActivityPlanPartsType MessageData : MA_ApprovalPolicyMDT OpZoneCategory : MA_OpZoneCategoryEnum ByPlanParts : MA_PlanPartsType CapabilityCommand : MA_CapabilityTypeEnum PlanParts : MA_PlanPartsType

	Plans : MA_PlanPolicyType RouteActivityPlan : MA_ActivityPlanPartsType OpPlan : MA_OpPlanPartsType PlanActivation : MA_PlanActivationPolicyType

Table A-1-521: C2 UCI Extensions MA_ApprovalPolicyMT UCI Extension

TaskPlan		
Description	Indicates applicable TaskPlan parts.	
Base Type	MA_TaskPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskType	MA_TaskTypeEnum	Repeated
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	

Table A-1-522: C2 UCI Extensions MA_ApprovalPolicyMT - TaskPlan UCI Extension Field

OpLineCategory	
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpLineCategoryEnum
Message Structure	
Enumeration Literal	**Description**
GENERAL_LINE	A general line where the function must be described by other procedures.
FORWARD_EDGE_OF_BATTLE_AREA	The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.
GUN_TARGET_LINE	Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.
CORRIDOR	Line with width describing the passageway designated for travel by friendly aircraft.
HOSTILE_BOUNDARY	Line that indicates a limit beyond which exists a hostile environment.
BUFFER_ZONE_BOUNDARY	Line beyond which friendly aircraft usually do not fly in peacetime.
LOW_LEVEL_TRANSIT_ROUTE	Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.
TACTICAL_ACTION_LINE	Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.
FIRE_SUPPORT_COORDINATION_LINE	Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current

	tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander.
FORWARD_LINE_OF_OWN_TROOPS	Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.
LINE_OF_COMMUNICATIONS	Line the route that connects an operating military unit with its supply base.
JETWAY	Line with width that defines areas where commercial aircraft are flying through.
DEPARTURE_RADIAL	Line the defines the departure radial when performing Carrier admin departures.
RECOVERY_RADIAL	Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.
FINAL_BEARING	Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.
BRC	Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.

Table A-1-523: C2 UCI Extensions MA_ApprovalPolicyMT - OpLineCategory UCI Extension Field

ActivityPlan		
Description	Indicates applicable ActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	

ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-524: C2 UCI Extensions MA_ApprovalPolicyMT - ActivityPlan UCI Extension Field

ApplicablePlanningActivity		
Description	Indicates the set of planning activities whose generated plans are subject to the sibling *Policy elements.	
Base Type	MA_PlanPolicyApplicablePlanType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanType	PlanTypeEnum	Required
Description	Indicates the PlanType that is generated by the planning activity, which is applicable to the associated *Policy elements.	
PlanParts	MA_PlanPartsType	Optional
Description	Indicates the parts of the sibling PlanType that are generated by the planning activity, which is applicable to the associated *Policy elements. If omitted, it indicates all PlanParts are applicable.	
ConstrainingPlans	AutonomousPlanningConstrainingPlansType	Optional
Description	Indicates a set of constraining plans (applicable to same system than the one planned for) that are used in the planning activity, which is applicable to the associated *Policy elements.	
OtherSystemConstrainingPlans	AutonomousPlanningOtherSystemConstrainingPlansType	Repeated
Description	Indicates a set of constraining plans (applicable to different system than the one planned for) that are used in the planning activity, which is applicable to the associated *Policy elements.	

Table A-1-525: C2 UCI Extensions MA_ApprovalPolicyMT - ApplicablePlanningActivity UCI Extension Field

OpVolumeCategory	
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	

Enumeration Literal	Description
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area

	designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-526: C2 UCI Extensions MA_ApprovalPolicyMT - OpVolumeCategory UCI Extension Field

TaskType	
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_TaskTypeEnum
Message Structure	
Enumeration Literal	Description
AIR_SAMPLE	Indicates the task is to sample and analyze the air around the vehicle.
AMTI	Indicates the task is to track air targets using an Airborne Moving Target Indicator.
AO	Indicates the task is to employ an Active Optical capability.
CARGO_DELIVERY	Indicates the task is to deliver cargo.
CAP	Indicates the task is to execute a Combat Air Patrol.
COMINT	Indicates the task is to employ a communications intelligence capability.
COMM_RELAY	Indicates the task is to relay communications from other battlespace participants.
COUNTER_SPACE	Indicates the task is to employ a Counter Space Capability.
EA	Indicates the task is to employ an electronic attack capability.
ESCORT	Indicates the task is to escort another entity.
ESM	Indicates the task is to employ an Electronic Support Measure.
FLIGHT	Indicates the task is to fly.
ORBIT_CHANGE	Indicates a task to perform an orbit change via a spacecraft maneuver.
ORBITAL_SURVEILLANCE	Indicates a task to perform orbital surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates a task to perform orbital surveillance via sensor collection capabilities.
PO	Indicates the task is to employ a Passive Optical capability.
REFUEL	Indicates the task is to refuel another System or Entity.
SAR	Indicates the task is to employ a Synthetic Aperture Radar capability.
SMTI	Indicates the task is to track surface targets using a Surface Moving Target Indicator.
STRIKE	Indicates the task is to strike a target.
SYSTEM_DEPLOYMENT	Indicates the task is to deploy a system.
TACTICAL_ORDER	Indicates the task is to perform a tactical order.
WEATHER_RADAR	Indicates the task is to collect weather data.

Table A-1-527: C2 UCI Extensions MA_ApprovalPolicyMT - TaskType UCI Extension Field

OrbitActivityPlan	
Description	Indicates applicable OrbitActivityPlan parts.

Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-528: C2 UCI Extensions MA_ApprovalPolicyMT - OrbitActivityPlan UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ApprovalPolicyMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApprovalPolicyID	ApprovalPolicyID_Type	Required

Description	Indicates the unique ID of the approval policy.	
Plans	MA_PlanPolicyType	Repeated
Description	Indicates a set of approval policies for the results of planning activities.	
PlanActivation	MA_PlanActivationPolicyType	Repeated
Description	Indicates approval policies which govern Plan Activation commands.	
RequirementExecution	RequirementExecutionPolicyType	Optional
Description	Indicates which Requirement (Effect, Action, Task, [Capability]Command, Response) execution/performance related activities require approval.	
TimedZone	TimedZoneType	Optional
Description	Indicates the geospatial zone and optionally time/date span or spans when the policy is active. For task allocation, the policy is applicable to allocations that 1) complete within the time/date span and 2) include Task or Tasks whose target falls within the zone, regardless of when the Task will be executed. For route planning, the policy is applicable to route plans with segments that fall within the zone during the time/date span, regardless of when the route plan was completed or the target of Tasks associated with the segments. For task execution, the policy is applicable to Tasks that 1) are scheduled to complete within the time/date span and 2) have a target which falls within the zone. This element supersedes the sibling Expires element.	
Expires	DateTimeType	Optional
Description	Indicates the time and date when the policy expires. The policy is applicable to task allocation and route planning activities completed prior to this time. The policy is applicable to task execution activities initiated prior to this time. When the sibling TimedZone element is used, the time spans specified there supersede this element.	
MissionTraceability	MissionTraceabilityType	Optional
Description	Allows the creator of the approval policy to specify the source or motivation for this ApprovalPolicy such as ATO, ACO, etc.	
DataRecordInstanceID	DataRecordInstanceID_Type	Optional
Description	Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee.	

Table A-1-529: C2 UCI Extensions MA_ApprovalPolicyMT - MessageData UCI Extension Field

OpZoneCategory	
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.

CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.

MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-530: C2 UCI Extensions MA_ApprovalPolicyMT - OpZoneCategory UCI Extension Field

ByPlanParts	
Description	Indicates *Plan parts which the Plan to be activated must contain in order for the Policy to be applicable. If omitted, the Policy is applicable regardless of *Plan parts.

Base Type	MA_PlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	
OrbitActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-531: C2 UCI Extensions MA_ApprovalPolicyMT - ByPlanParts UCI Extension Field

CapabilityCommand	
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_CapabilityTypeEnum
Message Structure	
Enumeration Literal	**Description**
ACTION	Indicates an action capability.
AIR_SAMPLE	Indicates an air sampling capability.
AMTI	Indicates an airborne moving target indication (AMTI) capability.
AO	Indicates an active optical capability.

CAP	Indicates a cargo delivery capability.
CARGO_DELIVERY	Indicates a cargo delivery capability.
COMINT	Indicates a Communications Intelligence (COMINT) capability.
COMM_RELAY	Indicates a communications relay capability.
COMM_TERMINAL	Indicates a communications terminal capability
COUNTER_SPACE	Indicates a Counter Space Capability.
EA	Indicates an electronic attack (EA) capability.
EFFECT	Indicates an effect capability.
ESM	Indicates an electronic support measures (ESM) capability.
FLIGHT	Indicates a capability for waypoint-based flight tasks/commands.
IFF	Indicates an Identification Friend or Foe (IFF) capability.
OPAQUE	Indicates a Opaque Capability
ORBIT_CHANGE	Indicates this space system has the ability to change orbits.
ORBITAL_SURVEILLANCE	Indicates a task to perform Orbital Surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates an Orbital Surveillance Sensor Capability.
PO	Indicates a passive optical capability.
RADAR_ALTIMETER	Indicates a Radar Altimeter capability
REFUEL	Indicates a capability to refuel another asset.
RESPONSE	Indicates an response capability.
SAR	Indicates a synthetic aperture radar (SAR) capability.
SMTI	Indicates a surface moving target indication (SMTI) capability.
STRIKE	Indicates a strike capability.
SYSTEM_DEPLOYMENT	Indicates a deployment capability.
TACTICAL_ORDER	Indicates a tactical order capability.
TASK	Indicates a task capability.
WEATHER_RADAR	Indicates a weather radar capability.

Table A-1-532: C2 UCI Extensions MA_ApprovalPolicyMT - CapabilityCommand UCI Extension Field

PlanParts		
Description	Indicates the parts of the sibling PlanType that are generated by the planning activity, which is applicable to the associated *Policy elements. If omitted, it indicates all PlanParts are applicable.	
Base Type	MA_PlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActivityPlan	MA_ActivityPlanPartsType	Optional

Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	
OrbitActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-533: C2 UCI Extensions MA_ApprovalPolicyMT - PlanParts UCI Extension Field

Plans		
Description	Indicates a set of approval policies for the results of planning activities.	
Base Type	MA_PlanPolicyType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApplicablePlanningActivity	MA_PlanPolicyApplicablePlanType	Repeated
Description	Indicates the set of planning activities whose generated plans are subject to the sibling *Policy elements.	
DefaultPolicy	ApprovalPolicyBaseType	Optional
Description	Indicates the default approval policy for the sibling ApplicablePlanningActivity. The default policy can be overridden in specific conditions as indicated by the ByResultPolicy or ByTriggerPolicy siblings of this element. For example, this element could indicate that approval is required by default for all TaskPlans and then use ByResultPolicy or ByTriggerPolicy siblings to specify conditions when task allocation approval is not required. Alternately, this element could indicate that approval is not required by default and then use ByResultPolicy or ByTriggerPolicy siblings to specify conditions when task allocation approval is required. Creators of this message are expected to carefully avoid creating logically inconsistent ByResultPolicy and ByTriggerPolicy instances and therefore no guidance is given on deconflicting illogical	

	instances. For example, an ApprovalPolicy that indicates approval is not required by default with conflicting ByResultPolicy or ByTriggerPolicy overrides of 1) AUTO_REJECT for DROPPED_TASKS and 2) AUTO_APPROVE for operator triggered DROPPED_TASKS would be considered illogical.	
ByResultPolicy	ByResultPolicyType	Repeated
Description	Indicates approval policies when planning activity results match certain conditions. Conditions specified here are considered to be overrides to the default planning activity approval policy given in the sibling Default element. For example, if the default policy indicates task allocation approval is never required, this element could override the default and require approval whenever a task allocation results in previously allocated Task or Tasks being dropped. See the annotations in the sibling DefaultPolicy element for an overview of how the siblings relate to each other. If multiple instances are given, each should be a different Result instance as indicated by the child Result element.	
ByTriggerPolicy	ByTriggerPolicyType	Repeated
Description	Indicates approval policies for re-plan activities initiated by triggered events, overriding the default approval policy for the planning activity given in the sibling DefaultPolicy element. Additionally, this element allows trigger-specific policies for a specific planning activities. For example, if the default policy indicates route planning approval is never required, this element could override, requiring approval for route plans affecting all Systems triggered by new threats if the route re-plan results in dropped tasks. If multiple instances are given, each should be of a different trigger type as indicated by the child EventTrigger element.	

Table A-1-534: C2 UCI Extensions MA_ApprovalPolicyMT - Plans UCI Extension Field

RouteActivityPlan		
Description	Indicates applicable RouteActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	

EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-535: C2 UCI Extensions MA_ApprovalPolicyMT - RouteActivityPlan UCI Extension Field

OpPlan		
Description	Indicates applicable OpPlan parts.	
Base Type	MA_OpPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPointCategory	OpPointCategoriesType	Repeated
Description	Indicates an OpPoint type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpLineCategory	MA_OpLineCategoryEnum	Repeated
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpZoneCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpVolumeCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpRouting	EmptyType	Optional
Description	Indicates an OpRouting is applicable to the associated *Plan. List size for this element is based on there being no category type for OpRouting objects	

Table A-1-536: C2 UCI Extensions MA_ApprovalPolicyMT - OpPlan UCI Extension Field

PlanActivation	
Description	Indicates approval policies which govern Plan Activation commands.
Base Type	MA_PlanActivationPolicyType

Message Structure		
Field Name	**Field Type**	**Field Usage**
ByMissionPlan	MissionPlanActivationSettingType	Repeated
Description	Indicates the set of ByMissionPlan activation commands that are applicable to the sibling Policy element. List size allows for identifying those MissionPlans whose subordinate MissionPlans are either included in the activation (TRUE) or not (FALSE) via the CommandSubordinatePlans setting.	
BySubPlan	SubPlanActivationSettingType	Repeated
Description	Indicates the set of BySubPlan activation commands that are applicable to the sibling Policy element.	
ByPlanParts	MA_PlanPartsType	Optional
Description	Indicates *Plan parts which the Plan to be activated must contain in order for the Policy to be applicable. If omitted, the Policy is applicable regardless of *Plan parts.	
Policy	ApprovalPolicyBaseType	Optional
Description	Indicates the approval policy for the sibling applicable activation activities.	

Table A-1-537: C2 UCI Extensions MA_ApprovalPolicyMT - PlanActivation UCI Extension Field

1.3.2.83 MA_ActionCommandMT UCI Extension

MA_ActionCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	ActionConstraintOverride : MA_ActionConstraintType RequiredAllocation : MA_RequirementAllocationConstraintType System : MA_SystemFilterType PreferredAllocation : MA_RequirementAllocationConstraintType Analytic : MA_AnalyticConstraintsType Designation : MA_DesignationFilterType AssessedObject : MA_AssetFilterType Characteristics : MA_EntityComparativeType Allocation : MA_AllocationChoiceType ConstraintOverride : MA_RequirementConstraintsType Allocation : MA_RequirementAllocationParametersType AccessAssessmentThreshold : MA_AccessAssessmentFilterType Command : MA_ActionCommandType ExcludedRequirementTypes : MA_RequirementTaxonomyType Kinematic : MA_RequirementKinematicConstraintsType EngagementParameters : MA_EngagementParametersType

	Subject : MA_AssetFilterType MessageData : MA_ActionCommandMDT AllowedRequirementTypes : MA_RequirementTaxonomyType Entity : MA_EntityFilterType

Table A-1-538: C2 UCI Extensions MA_ActionCommandMT UCI Extension

ActionConstraintOverride		
Description	Indicates refined/overridden values of constraints for the Action. Added to allow similar override options in direct commanding as is available in planning.	
Base Type	MA_ActionConstraintType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TargetObject	IdentityKindInstanceType	Repeated
Description	Indicates the target of the Action; the object the Action will be done to or for. If omitted, the Action places no explicit limitation on which TargetObject it can be done to. See sibling ActionType element annotation for an overall description of Action structure. Note that the term "target" is used in a general sense. For some Requirement types, a more suitable term might be "subject", "point", "location", etc. For example, the "target" of a COVER Action could be a blue/friendly System.	
TargetObjectConstraints	RequirementTargetConstraintsType	Optional
Description	Indicates constraints "on" the TargetObject that must be satisfied before the Action can be done to or for it. See sibling ActionType element annotation for an overall description of Effect structure.	
SecondaryObject	IdentityKindInstanceType	Repeated
Description	Indicates a thing the sibling TargetObject must act on/against/toward to make the Action applicable. See sibling ActionType element annotation for an overall description of Action structure.	
PerTargetEngagementParameters	MA_EngagementParametersOrDefaultChoiceType	Repeated
Description	Indicates a command to change the engagement parameters corresponding to an ongoing tactic. This field is meant to be used on a per target basis. Setting this field overrides the engagement parameters sent in MA_Action, and Activity modification by ChangeEngagement, and any previous audibles that modified MA_EngagementParameters for the sibling TargetObject.	

Table A-1-539: C2 UCI Extensions MA_ActionCommandMT - ActionConstraintOverride UCI Extension Field

RequiredAllocation	
Description	Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) MUST be allocated to during the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process.

	If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated System must perform the Requirement. This element allows the Requirement creator to force an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system.	
Base Type	MA_RequirementAllocationConstraintType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Allocation	MA_AllocationChoiceType	Required
Description	Indicates the type of Allocation, either by direct assignment of SystemIDs or by PackageID definition for delegation.	
Identity	IdentityType	Repeated
Description	Indicates a kind/type or instance of platform/system that is the target of the allocation constraint. For example, this element could be used to allocate a Requirement to any fighter aircraft platform, or a specific kind of fighter aircraft, or a specific tail number of aircraft, or an aircraft with an aircrew member with a certain voice call sign.	
CapabilityID	CapabilityID_Type	Repeated
Description	Indicates the CapabilityID of the Capability that is the target of the allocation constraint. If multiple required Capabilities are given: 1) allocation to any one of the Capabilities is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the multiple required Capabilities is acceptable, 3) there is no expectation that each Capability must perform the Requirement.	

Table A-1-540: C2 UCI Extensions MA_ActionCommandMT - RequiredAllocation UCI Extension Field

System		
Description	Indicates a filter on characteristics of a System asset. The asset filter is satisfied if a match is found in this element OR the sibling Entity element.	
Base Type	MA_SystemFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Characteristics	MA_SystemComparativeType	Repeated
Description	This element indicates a System filter requiring a variable logical comparator test between a value in a System-associated message and the value given here. Multiple instances are logically ANDed.	
SystemID	SystemID_Type	Repeated
Description	Indicates a specific SystemID that is required to match the filter. If omitted, any SystemID satisfies the filter. Multiple instances are logically ORed.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the System must fall within to satisfy the clause. Multiple instances	

	are logically ORed.	
OrbitalFilter	OrbitalFiltersQueryType	Repeated
Description	Indicates a zone the System must fall within to satisfy the clause. Multiple instances are logically ORed.	

Table A-1-541: C2 UCI Extensions MA_ActionCommandMT - System UCI Extension Field

PreferredAllocation		
Description	Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) should be allocated to during the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process. If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated System must perform the Requirement. This element allows the Requirement creator to suggest an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system.	
Base Type	MA_RequirementAllocationConstraintType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Allocation	MA_AllocationChoiceType	Required
Description	Indicates the type of Allocation, either by direct assignment of SystemIDs or by PackageID definition for delegation.	
Identity	IdentityType	Repeated
Description	Indicates a kind/type or instance of platform/system that is the target of the allocation constraint. For example, this element could be used to allocate a Requirement to any fighter aircraft platform, or a specific kind of fighter aircraft, or a specific tail number of aircraft, or an aircraft with an aircrew member with a certain voice call sign.	
CapabilityID	CapabilityID_Type	Repeated
Description	Indicates the CapabilityID of the Capability that is the target of the allocation constraint. If multiple required Capabilities are given: 1) allocation to any one of the Capabilities is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the multiple required Capabilities is acceptable, 3) there is no expectation that each Capability must perform the Requirement.	

Table A-1-542: C2 UCI Extensions MA_ActionCommandMT - PreferredAllocation UCI Extension Field

Analytic	
Description	Indicates a constraint in the form of an analytic measure that the Requirement must satisfy when planned/executed.

Base Type	MA_AnalyticConstraintsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpConstraint	OpConstraintWeightingType	Repeated
Description	Indicates an operational constraint to achieve. The multiplicity is based on the list of available OpConstraintEnums.	
RiskSetting	SurvivabilityRiskSettingType	Repeated
Description	Identifies the Risk Level constraint for each provided RiskPriortization (max of 4).	
AccessAssessmentThreshold	MA_AccessAssessmentFilterType	Optional
Description	Indicates access assessment analysis event thresholds.	
RF_TaskPerformance	RF_TaskPerformanceType	Optional
Description	Mission specific RF performance objective.	

Table A-1-543: C2 UCI Extensions MA_ActionCommandMT - Analytic UCI Extension Field

Designation		
Description	Indicates a type of Entity designation, established by the Designation message, required to satisfy the filter.	
Base Type	MA_DesignationFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DesignationType	DesignationEnum	Repeated
Description	Indicates a type of designation required to match the filter. List size for this element is based on "Select All That Apply" condition.	
DesignationObjective	MA_RequirementTaxonomyType	Optional
Description	Indicates objective or objectives that the designation must include to match the filter. When omitted, the designation objective is irrelevant for the filter.	

Table A-1-544: C2 UCI Extensions MA_ActionCommandMT - Designation UCI Extension Field

AssessedObject		
Description	Indicates Assessed Objects of the AccessAssessment that are required to match the filter. If omitted, any Assessed Object satisfies the filter.	
Base Type	MA_AssetFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
System	MA_SystemFilterType	Optional
Description	Indicates a filter on characteristics of a System asset. The asset filter is satisfied if a match is found in this element OR the sibling Entity element.	

Entity	MA_EntityFilterType	Optional
Description	Indicates a filter on characteristics of an Entity asset. The asset filter is satisfied if a match is found in this element OR the sibling System element.	

Table A-1-545: C2 UCI Extensions MA_ActionCommandMT - AssessedObject UCI Extension Field

Characteristics		
Description	This element indicates an Entity filter requiring a variable logical comparator test between a value in an Entity-associated message and the value given here. Multiple instances are logically ANDed.	
Base Type	MA_EntityComparativeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Characteristic	MA_EntityCharacteristicType	Required
Description	Indicates a characteristic/element value to be used to compare to the corresponding value of the Entity or related message.	
Comparator	EqualityExpressionEnum	Required
Description	Indicates the logical comparator to use when comparing the target (Entity or related message) Characteristic to the Response Characteristic. The logical expression is: (target Characteristic) (Comparator) (Response Characteristic). When the logical expression is TRUE then the parent TargetClause instance is TRUE.	

Table A-1-546: C2 UCI Extensions MA_ActionCommandMT - Characteristics UCI Extension Field

Allocation		
Description	Indicates the type of Allocation, either by direct assignment of SystemIDs or by PackageID definition for delegation.	
Base Type	MA_AllocationChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Repeated
Description	Indicates the SystemID(s) of the System that is the target of the allocation constraint.	
PackageID	PackageID_Type	Required
Description	Indicates the unique ID of the package for delegation of Systems.	

Table A-1-547: C2 UCI Extensions MA_ActionCommandMT - Allocation UCI Extension Field

ConstraintOverride	
Description	Indicates refined/overridden values of constraints for the Action. Added to allow similar override options in direct commanding as is available in planning.

Base Type	MA_RequirementConstraintsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Rank	ComparableRankingType	Optional
Description	Indicates a constraint in the form of the Rank of the associated Requirement (Effect, Action, Task, [Capability]Command or Response).	
InterruptLowerRank	boolean	Optional
Description	Indicates whether a) an in-work/planned item or items of a lower rank should be interrupted/replanned in order to address this item (TRUE) or b) this item should be simply queued/planned according to rank without interrupting in-work/planned items (FALSE). For example, a Task being added to an existing TaskPlan (TaskPlanCommand constrained by an existing TaskPlan) would potentially cause unlocked Tasks of lower rank in the existing TaskPlan to be unallocated if 1) this element indicated TRUE and 2) the lower ranked Tasks prevented the higher ranked Task from being allocated.	
Allocation	MA_RequirementAllocationParametersType	Optional
Description	Indicates allocation related constraints which can vary by Requirement at the discretion of the creator.	
Timing	RequirementTimingType	Repeated
Description	Indicates a temporal constraint on the Requirement. Allows for defining for each timing type (RequirementTimingEnum).	
Kinematic	MA_RequirementKinematicConstraintsType	Repeated
Description	Indicates a constraint in the form of kinematic position of Requirement actor or actors. When more than one is given they are logically ANDed.	
Dependency	RequirementDependencyType	Repeated
Description	Indicates a constraint in the form of a temporal dependency between Requirements; between Effect, Action, Task, [Capability]Command or Commands or Response. If more than one is given, they are logically ANDed. For example, a pre-strike image collection (PO Task) might be required several minutes prior to a strike (Strike Task) to determine combat ID and target coordinates. A post-strike image collection (PO Task) might be required several minutes after the strike to inform battle damage assessment. The dependency between the three Tasks can be established with this element. Assuming the Strike Task is the most difficult to plan/coordinate, the other two Tasks would likely each have dependencies on the Strike Task in order to orchestrate the desired "look-shoot-look" sequence.	
MaxProductDisseminationClassificationLevel	DataProductClassificationLevelType	Optional
Description	Indicates the maximum allowed classification level of the networks	

	that the allocated system can disseminate products to for this task. The allocated system must have a dissemination route for data at this classification or lower. For example, when trying to collect data to hand-off to an unclassified system or network, the allocated system has to be able to generate products at no higher than the specified level and have the authority to disseminate on a network at no higher than the specified classification level.	
Analytic	MA_AnalyticConstraintsType	Optional
Description	Indicates a constraint in the form of an analytic measure that the Requirement must satisfy when planned/executed.	
AcceptableClassificationLevel	SecurityInformationType	Optional
Description	Indicates a constraint on the highest acceptable classification level at which the Response can be planned and executed.	
CommsRequired	boolean	Optional
Description	Indicates whether or not communications are required when executing the Requirement. In this context, "required" means the creator or a Requirement considers communications as necessary for C2 and/or Product dissemination related to this Requirement, independent of the type of vehicle that it is allocated to. This element may result in consideration of communications allocations during Requirement allocation.	
AllowedRequirementTypes	MA_RequirementTaxonomyType	Optional
Description	Indicates other types of Requirements that can be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise associated with the subject Requirement.	
ExcludedRequirementTypes	MA_RequirementTaxonomyType	Optional
Description	Indicates other types of Requirements that cannot be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise be associated with the subject Requirement.	
ConstraintsNarrative	VisibleString1024Type	Optional
Description	Free-text explanation of sibling constraints, additional extraordinary constraints, etc.	
AllowedDomainPairing	EnvironmentPairingEnum	Repeated
Description	Indicates an allowed domain/environment which the Requirement can be employed from and allowed "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.	
ExcludedDomainPairing	EnvironmentPairingEnum	Repeated
Description	Indicates an excluded domain/environment which the Requirement can't be employed from an excluded "target" domain/environment. See enumerated type annotations for further details. List size for this element is based on "Select All That Apply" condition.	

EngagementParameters	MA_EngagementParametersType	Optional
Description	Provides engagement parameters that this requirement should adhere to.	

Table A-1-548: C2 UCI Extensions MA_ActionCommandMT - ConstraintOverride UCI Extension Field

Allocation		
Description	Indicates allocation related constraints which can vary by Requirement at the discretion of the creator.	
Base Type	MA_RequirementAllocationParametersType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequiredAllocation	MA_RequirementAllocationConstraintType	Optional
Description	Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) MUST be allocated to during the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process. If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated System must perform the Requirement. This element allows the Requirement creator to force an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system.	
PreferredAllocation	MA_RequirementAllocationConstraintType	Optional
Description	Indicates the system and optionally Capability that the Requirement (Effect, Action, Task, [Capability]Command, Response) should be allocated to during the allocation process. The system can be an explicit UCI System or implied UCI System (by Identity, hosted Capability, kinematics, etc.) in which case the associated UCI System must be determined as part of the allocation process. If multiple child elements are given: 1) allocation to any one of the associated Systems is the minimum expectation, 2) decomposition of the Requirement into multiple child Requirements allocated across the associated Systems is acceptable, 3) there is no expectation that each associated System must perform the Requirement. This element allows the Requirement creator to suggest an allocation to a specific system at the risk of 1) a less than optimal overall solution or 2) failure of the Requirement being allocated despite availability of another suitable system.	

Table A-1-549: C2 UCI Extensions MA_ActionCommandMT - Allocation UCI Extension Field

AccessAssessmentThreshold		
Description	Indicates access assessment analysis event thresholds.	
Base Type	MA_AccessAssessmentFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

AssessmentName	VisibleString32Type	Repeated
Description	Indicates a name/type of AccessAssessment algorithm required to match the filter. If omitted, any name/type of AccessAssessment algorithm satisfies the filter.	
AssessedCapabilities	CapabilityTaxonomyUniversalType	Optional
Description	Indicates assessed Capabilities of the AccessAssessment that are required to match the filter. If omitted, any assessed Capability satisfies the filter.	
Subject	MA_AssetFilterType	Optional
Description	Indicates Subjects of the AccessAssessment that are required to match the filter. If omitted, any Subject satisfies the filter.	
AssessedObject	MA_AssetFilterType	Optional
Description	Indicates Assessed Objects of the AccessAssessment that are required to match the filter. If omitted, any Assessed Object satisfies the filter.	
AccessEvent	AccessEventFilterType	Repeated
Description	Indicates Access Event details necessary to match the filter. If omitted, any Access Event satisfies the filter. Multiple instances are logically ORed.	

Table A-1-550: C2 UCI Extensions MA_ActionCommandMT - AccessAssessmentThreshold UCI Extension Field

Command		
Description	Indicates an instance of an Action Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message.	
Base Type	MA_ActionCommandType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Capability	MA_ActionCapabilityCommandType	Required
Description	Indicates a new invocation of an Action Capability. Generally, if accepted, the command will result in one or more new Action Activities being created and reported via the ActionActivity message. The request/response interaction terminates as soon as the command is accepted or rejected; this element is not used to interact with an "active" command. Updates and/or additional interaction with the resulting Activity (after the command is accepted) is accomplished via the sibling Activity element.	
Activity	MA_ActivityCommandBaseType	Required
Description	Indicates a command to modify an existing Action Activity (which was previously reported via the ActionActivity message and was marked as "interactive"). The request/response interaction terminates as soon as the modification is accepted or rejected. The modifications are reflected in subsequent ActionActivity messages.	

Table A-1-551: C2 UCI Extensions MA_ActionCommandMT - Command UCI Extension Field

ExcludedRequirementTypes

Description	Indicates other types of Requirements that cannot be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise be associated with the subject Requirement.	
Base Type	MA_RequirementTaxonomyType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Effect	EffectTypeEnum	Repeated
Description	Indicates Effects applicable via the Effect* message set. List size for this element is based on "Select All That Apply" condition.	
Action	MA_ActionTypeEnum	Repeated
Description	Indicates Actions applicable via the Action* message set. List size for this element is based on "Select All That Apply" condition.	
Task	MA_TaskTypeEnum	Repeated
Description	Indicates Tasks applicable via the Task* message set. List size for this element is based on "Select All That Apply" condition.	
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates CapabilityCommands applicable via a [Capability]Command message set. List size for this element is based on "Select All That Apply" condition.	
Response	ResponseTypeEnum	Repeated
Description	Indicates Responses applicable via the Response* message set. List size for this element is based on "Select All That Apply" condition.	

Table A-1-552: C2 UCI Extensions MA_ActionCommandMT - ExcludedRequirementTypes UCI Extension Field

Kinematic		
Description	Indicates a constraint in the form of kinematic position of Requirement actor or actors. When more than one is given they are logically ANDed.	
Base Type	MA_RequirementKinematicConstraintsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ApplicableObject	RequirementObjectEnum	Required
Description	Indicates which Requirement actor the sibling kinematic constraints are applicable to.	
Geo	GeoFiltersQueryType	Repeated
Description	This element gives a geographical filter to the query.	
Orbital	OrbitalFiltersQueryType	Repeated
Description	This element gives an orbital filter to the query.	
LookAngleConstraints	AnglePairType	Optional
Description	The angle from true north to the line-of-sight vector from the platform to the target tangent plane.	

Table A-1-553: C2 UCI Extensions MA_ActionCommandMT - Kinematic UCI Extension Field

EngagementParameters		
Description	Provides engagement parameters that this requirement should adhere to.	
Base Type	MA_EngagementParametersType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
InterceptTactic	MA_InterceptTacticChoiceType	Optional
Description	The type of intercept tactic to be employed. Used in combination with shot parameters to define tactics like Skate or Banzai.	
ShotParameters	MA_ShotParametersType	Optional
Description	Provides shot parameters that this requirement should adhere to.	
EnableTargetRecalculation	boolean	Optional
Description	This field specifies that MA should recalculate optimal targets during the engagement.	

Table A-1-554: C2 UCI Extensions MA_ActionCommandMT - EngagementParameters UCI Extension Field

Subject		
Description	Indicates Subjects of the AccessAssessment that are required to match the filter. If omitted, any Subject satisfies the filter.	
Base Type	MA_AssetFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
System	MA_SystemFilterType	Optional
Description	Indicates a filter on characteristics of a System asset. The asset filter is satisfied if a match is found in this element OR the sibling Entity element.	
Entity	MA_EntityFilterType	Optional
Description	Indicates a filter on characteristics of an Entity asset. The asset filter is satisfied if a match is found in this element OR the sibling System element.	

Table A-1-555: C2 UCI Extensions MA_ActionCommandMT - Subject UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ActionCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**

Command	MA_ActionCommandType	Repeated
Description	Indicates an instance of an Action Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message.	

Table A-1-556: C2 UCI Extensions MA_ActionCommandMT - MessageData UCI Extension Field

AllowedRequirementTypes		
Description	Indicates other types of Requirements that can be decomposed from, used to plan, used to implement, inherited from traced parents or otherwise associated with the subject Requirement.	
Base Type	MA_RequirementTaxonomyType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Effect	EffectTypeEnum	Repeated
Description	Indicates Effects applicable via the Effect* message set. List size for this element is based on "Select All That Apply" condition.	
Action	MA_ActionTypeEnum	Repeated
Description	Indicates Actions applicable via the Action* message set. List size for this element is based on "Select All That Apply" condition.	
Task	MA_TaskTypeEnum	Repeated
Description	Indicates Tasks applicable via the Task* message set. List size for this element is based on "Select All That Apply" condition.	
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates CapabilityCommands applicable via a [Capability]Command message set. List size for this element is based on "Select All That Apply" condition.	
Response	ResponseTypeEnum	Repeated
Description	Indicates Responses applicable via the Response* message set. List size for this element is based on "Select All That Apply" condition.	

Table A-1-557: C2 UCI Extensions MA_ActionCommandMT - AllowedRequirementTypes UCI Extension Field

Entity		
Description	Indicates a filter on characteristics of an Entity asset. The asset filter is satisfied if a match is found in this element OR the sibling System element.	
Base Type	MA_EntityFilterType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Designation	MA_DesignationFilterType	Optional
Description	Indicates a type of Entity designation, established by the Designation message,	

	required to satisfy the filter.	
Characteristics	MA_EntityComparativeType	Repeated
Description	This element indicates an Entity filter requiring a variable logical comparator test between a value in an Entity-associated message and the value given here. Multiple instances are logically ANDed.	
EntityID	EntityID_Type	Repeated
Description	Indicates a specific EntityID that is required to match the filter. If omitted, any EntityID satisfies the filter. Multiple instances are logically ORed.	
Source	EntitySourceEnum	Repeated
Description	Indicates an Entity source that is required to match the filter. If omitted, any Entity source satisfies the filter. Multiple instances are logically ORed. List size for this element is based on "Select All That Apply" condition.	
GeoFilter	GeoFiltersQueryType	Repeated
Description	Indicates a zone the Entity must fall within to satisfy the filter. Multiple instances are logically ORed.	
OrbitalFilter	OrbitalFiltersQueryType	Repeated
Description	Indicates a zone the Entity must fall within to satisfy the filter. Multiple instances are logically ORed.	

Table A-1-558: C2 UCI Extensions MA_ActionCommandMT - Entity UCI Extension Field

1.3.2.84 MA_ActionCommandStatusMT UCI Extension

MA_ActionCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ActionCommandStatusMDT

Table A-1-559: C2 UCI Extensions MA_ActionCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ActionCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Activity	ResultingActivityType	Repeated
Description	Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.	
CommandID	CommandID_Type	Required

Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-560: C2 UCI Extensions MA_ActionCommandStatusMT - MessageData UCI Extension Field

1.3.2.85 MA_MissionPlanActivationStatusMT UCI Extension

MA_MissionPlanActivationStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	OpPlanID : MA_OpPlanID_Type MessageData : MA_MissionPlanActivationStatusMDT Plans : MA_PlansReferenceBaseType SubPlanActivationState : MA_PlanActivationStateType

Table A-1-561: C2 UCI Extensions MA_MissionPlanActivationStatusMT UCI Extension

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional

Description	Human readable text label with content meaningful to human operators.

Table A-1-562: C2 UCI Extensions MA_MissionPlanActivationStatusMT - OpPlanID UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanActivationStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Required
Description	The ID of the MissionPlan this status pertains to.	
PlanActivationState	PlanActivationStateEnum	Required
Description	Indicates the activation state of the MissionPlan.	
SubPlanActivationState	MA_PlanActivationStateType	Repeated
Description	Indicates the activation state of a SubPlan or SubPlans of the associated MissionPlan. An instance of this element should include all of the SubPlans in a given state; the cardinality is based in the number of states in the associated activation state enumerated type.	

Table A-1-563: C2 UCI Extensions MA_MissionPlanActivationStatusMT - MessageData UCI Extension Field

Plans		
Description	Indicates the Plan or Plans in the sibling ActivationState.	
Base Type	MA_PlansReferenceBaseType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	

EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-564: C2 UCI Extensions MA_MissionPlanActivationStatusMT - Plans UCI Extension Field

SubPlanActivationState		
Description	Indicates the activation state of a SubPlan or SubPlans of the associated MissionPlan. An instance of this element should include all of the SubPlans in a given state; the cardinality is based in the number of states in the associated activation state enumerated type.	
Base Type	MA_PlanActivationStateType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActivationState	PlanActivationStateEnum	Required
Description	Indicates an activation state that one or more *Plans is in.	
Plans	MA_PlansReferenceBaseType	Required
Description	Indicates the Plan or Plans in the sibling ActivationState.	

Table A-1-565: C2 UCI Extensions MA_MissionPlanActivationStatusMT - SubPlanActivationState UCI Extension Field

1.3.2.86 MA_ActionStatusMT UCI Extension

MA_ActionStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	Traceability : MA_RequirementStatusTraceabilityType Plans : MA_PlansReferenceType OpPlanID : MA_OpPlanID_Type MessageData : MA_ActionStatusMDT ExecutingSystemOrPackage : MA_PackageSystemType

Table A-1-566: C2 UCI Extensions MA_ActionStatusMT UCI Extension

Traceability		
Description	Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture. For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.	
Base Type	MA_RequirementStatusTraceabilityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Plans	MA_PlansReferenceType	Optional
Description	Indicates *Plans the Requirement is associated with.	
PlannedActivityID	PlannedActivityID_Type	Repeated
Description	Indicates a specific Planned Activity the Requirement is associated with.	
CapabilityID	CapabilityID_Type	Repeated
Description	Identifies a specific Capability the Requirement is associated with.	
CommandID	CommandID_Type	Repeated
Description	Indicates the ID of an associated [Capability]Command.	
ActivityID	ActivityID_Type	Repeated
Description	Indicates the ID of an associated actual/executing/executed Activity.	

Table A-1-567: C2 UCI Extensions MA_ActionStatusMT - Traceability UCI Extension Field

Plans		
Description	Indicates *Plans the Requirement is associated with.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	

OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-568: C2 UCI Extensions MA_ActionStatusMT - Plans UCI Extension Field

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-569: C2 UCI Extensions MA_ActionStatusMT - OpPlanID UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ActionStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActionID	ActionID_Type	Required
Description	Indicates the unique ID of the Action whose execution status is given by sibling elements.	
ExecutingSystemOrPackage	MA_PackageSystemType	Required
Description	Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.	
ExecutionState	RequirementExecutionStateEnum	Required
Description	Identifies the state of the Requirement.	
ExecutionStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling State element. This element is only expected when the State is in a potentially undesirable state such as failed or completed with less than optimum results. It is strongly encouraged for transitions to failed or dropped.	
ActualTiming	RequirementTimingType	Repeated
Description	Indicates actual timing status for the Requirement. Allows for defining for each timing type (RequirementTimingEnum).	
PercentCompleted	PercentType	Optional
Description	Percentage of assigned Requirement completion. The percentage increments until the Requirement has COMPLETED or FAILED. A percent complete of 0 indicates the Requirement has not yet begun execution, and a failed Requirement may have any percentage between 1 - 99.	
Traceability	MA_RequirementStatusTraceabilityType	Optional
Description	Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture. For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.	
Metrics	MetricsType	Optional

Description	Specifies the actual metrics associated with the Requirements.

Table A-1-570: C2 UCI Extensions MA_ActionStatusMT - MessageData UCI Extension Field

ExecutingSystemOrPackage		
Description	Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	
PackageID	PackageID_Type	Required
Description	The ID of a Package.	
SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-571: C2 UCI Extensions MA_ActionStatusMT - ExecutingSystemOrPackage UCI Extension Field

1.3.2.87 MA_CarrierReportMT UCI Extension

MA_CarrierReportMT	
Description	See the annotation in the associated message carrier status. See the XSD File for more info
Ext. Fields	LaunchCoordinates : MA_CarrierRunwayCoordinatesType Runway : MA_CarrierRunwayType CarrierReportID : MA_CarrierReportID_Type Direction : MA_AngleRelativeType Catapult : MA_CatapultType Information : MA_CarrierInformationType CatapultID : MA_CatapultID_Type MessageData : MA_CarrierReportMDT LandingCoordinates : MA_CarrierRunwayCoordinatesType Direction : MA_AngleRelativeType

Table A-1-572: C2 UCI Extensions MA_CarrierReportMT UCI Extension

LaunchCoordinates		
Description	Specifies the start coordinates of the catapult used for Launch.	
Base Type	MA_CarrierRunwayCoordinatesType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Start	Point3D_RelativeType	Required
Description	The waypoint at the start of the runway.	
Threshold	Point3D_RelativeType	Optional
Description	The waypoint at the nearest point of the runway.	
Limit	Point3D_RelativeType	Optional
Description	The waypoint at the furthest point of the runway.	

Table A-1-573: C2 UCI Extensions MA_CarrierReportMT - LaunchCoordinates UCI Extension Field

Runway		
Description	Contains details of the runway used for landing.	
Base Type	MA_CarrierRunwayType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RunwayID	RunwayID_Type	Required
Description	Indicates the unique ID for this Runway. Route paths may reference this runway for takeoff or landing paths.	
Direction	MA_AngleRelativeType	Optional
Description	Indicates offset angle to refence frame in radians.	
AvailableLength	DistanceType	Optional
Description	Indicates available length of runway in meters.	
Glideslope	AngleQuarterType	Optional
Description	Indicates glideslope for runway in radians.	
GCA	EmptyType	Optional
Description	Indicates Ground-Controlled Approach service is a available on runway when field is present.	
ILS	EmptyType	Optional
Description	Indicates Instrument Landing Systems service is a available on runway when field is present.	
ApproachLighting	ApproachLightingEnum	Optional
Description	Indicates available approach lighting on runway according to MIL-STD-6016.	

LandingCoordinates	MA_CarrierRunwayCoordinatesType	Optional
Description	Specifies the coordinates of the runway when used for landings.	
DefaultRunwayUsageDetails	RunwayUsageDetailsType	Optional
Description	Specifies terms for using a runway. These details can be used in the decision process when choosing a runway.	

Table A-1-574: C2 UCI Extensions MA_CarrierReportMT - Runway UCI Extension Field

CarrierReportID		
Description	Indicates a unique ID, assigned to this carrier report message.	
Base Type	MA_CarrierReportID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-575: C2 UCI Extensions MA_CarrierReportMT - CarrierReportID UCI Extension Field

Direction		
Description	Indicates offset angle to refence frame in radians.	
Base Type	MA_AngleRelativeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ReferenceFrameID	ReferenceFrameID_Type	Required
Description	UUID of the reference frame object that this relative angle applies to.	
OffsetAngle	AngleType	Optional
Description	Offset angle relative to the reference frame origin.	

Table A-1-576: C2 UCI Extensions MA_CarrierReportMT - Direction UCI Extension Field

Catapult		
Description	Contains details of each catapult for launching.	
Base Type	MA_CatapultType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

CatapultID	MA_CatapultID_Type	Required
Description	Indicates the unique ID for this catapult Route paths may reference this catapult for launch paths.	
Direction	MA_AngleRelativeType	Optional
Description	Indicates offset angle to refence frame in radians.	
AvailableLength	DistanceType	Optional
Description	Indicates available length of runway in meters.	
LaunchCoordinates	MA_CarrierRunwayCoordinatesType	Optional
Description	Specifies the start coordinates of the catapult used for Launch.	

Table A-1-577: C2 UCI Extensions MA_CarrierReportMT - Catapult UCI Extension Field

Information		
Description	Contains information about an Carrier.	
Base Type	MA_CarrierInformationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CrashService	CrashServiceEnum	Optional
Description	Indicates availability of crash service on Carrier.	
Runway	MA_CarrierRunwayType	Optional
Description	Contains details of the runway used for landing.	
Catapult	MA_CatapultType	Repeated
Description	Contains details of each catapult for launching.	

Table A-1-578: C2 UCI Extensions MA_CarrierReportMT - Information UCI Extension Field

CatapultID		
Description	Indicates the unique ID for this catapult Route paths may reference this catapult for launch paths.	
Base Type	MA_CatapultID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-579: C2 UCI Extensions MA_CarrierReportMT - CatapultID UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_CarrierReportMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CarrierReportID	MA_CarrierReportID_Type	Required
Description	Indicates a unique ID, assigned to this carrier report message.	
CarrierID	SystemID_Type	Required
Description	Indicates a unique ID, assigned to this carrier report message.	
IdentityReferenceID	CarrierReferenceID_ChoiceType	Required
Description	Indicates whether message is self-reported or reported by third party.	
Information	MA_CarrierInformationType	Optional
Description	Contains information about an Carrier.	

Table A-1-580: C2 UCI Extensions MA_CarrierReportMT - MessageData UCI Extension Field

LandingCoordinates		
Description	Specifies the coordinates of the runway when used for landings.	
Base Type	MA_CarrierRunwayCoordinatesType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Start	Point3D_RelativeType	Required
Description	The waypoint at the start of the runway.	
Threshold	Point3D_RelativeType	Optional
Description	The waypoint at the nearest point of the runway.	
Limit	Point3D_RelativeType	Optional
Description	The waypoint at the furthest point of the runway.	

Table A-1-581: C2 UCI Extensions MA_CarrierReportMT - LandingCoordinates UCI Extension Field

Direction		
Description	Indicates offset angle to refence frame in radians.	
Base Type	MA_AngleRelativeType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

ReferenceFrameID	ReferenceFrameID_Type	Required
Description	UUID of the reference frame object that this relative angle applies to.	
OffsetAngle	AngleType	Optional
Description	Offset angle relative to the reference frame origin.	

Table A-1-582: C2 UCI Extensions MA_CarrierReportMT - Direction UCI Extension Field

1.3.2.88 MA_CarrierStateMT UCI Extension

MA_CarrierStateMT	
Description	See the annotation in the associated message carrier state. See the XSD File for more info
Ext. Fields	CarrierStateID : MA_CarrierStateID_Type CatapultID : MA_CatapultID_Type Catapult : MA_CatapultType Runway : MA_CarrierRunwayType Information : MA_CarrierStateInformationType MessageData : MA_CarrierStateMDT

Table A-1-583: C2 UCI Extensions MA_CarrierStateMT UCI Extension

CarrierStateID		
Description	Indicates a unique ID, assigned to this carrier state message.	
Base Type	MA_CarrierStateID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-584: C2 UCI Extensions MA_CarrierStateMT - CarrierStateID UCI Extension Field

CatapultID	
Description	Indicates the unique ID for this catapult Route paths may reference this catapult for launch paths.
Base Type	MA_CatapultID_Type
Message Structure	

Field Name	Field Type	Field Usage
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-585: C2 UCI Extensions MA_CarrierStateMT - CatapultID UCI Extension Field

Catapult		
Description	Contains details of each catapult for launching.	
Base Type	MA_CatapultType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CatapultID	MA_CatapultID_Type	Required
Description	Indicates the unique ID for this catapult Route paths may reference this catapult for launch paths.	
Direction	MA_AngleRelativeType	Optional
Description	Indicates offset angle to refence frame in radians.	
AvailableLength	DistanceType	Optional
Description	Indicates available length of runway in meters.	
LaunchCoordinates	MA_CarrierRunwayCoordinatesType	Optional
Description	Specifies the start coordinates of the catapult used for Launch.	

Table A-1-586: C2 UCI Extensions MA_CarrierStateMT - Catapult UCI Extension Field

Runway		
Description	Contains details of the runway used for landing.	
Base Type	MA_CarrierRunwayType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RunwayID	RunwayID_Type	Required
Description	Indicates the unique ID for this Runway. Route paths may reference this runway for takeoff or landing paths.	
Direction	MA_AngleRelativeType	Optional
Description	Indicates offset angle to refence frame in radians.	
AvailableLength	DistanceType	Optional
Description	Indicates available length of runway in meters.	

Glideslope	AngleQuarterType	Optional
Description	Indicates glideslope for runway in radians.	
GCA	EmptyType	Optional
Description	Indicates Ground-Controlled Approach service is a available on runway when field is present.	
ILS	EmptyType	Optional
Description	Indicates Instrument Landing Systems service is a available on runway when field is present.	
ApproachLighting	ApproachLightingEnum	Optional
Description	Indicates available approach lighting on runway according to MIL-STD-6016.	
LandingCoordinates	MA_CarrierRunwayCoordinatesType	Optional
Description	Specifies the coordinates of the runway when used for landings.	
DefaultRunwayUsageDetails	RunwayUsageDetailsType	Optional
Description	Specifies terms for using a runway. These details can be used in the decision process when choosing a runway.	

Table A-1-587: C2 UCI Extensions MA_CarrierStateMT - Runway UCI Extension Field

Information		
Description	Contains information about an Carrier.	
Base Type	MA_CarrierStateInformationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Operational	EmptyType	Optional
Description	Indicates Carrier is operational if present.	
AirRaidState	AirRaidStateEnum	Optional
Description	Indicated air raid state of Carrier according to MIL-STD-6016.	
Contamination	AirfieldContaminationType	Optional
Description	Contains contamination details about a Carrier.	
SHORADEZ_Active	EmptyType	Optional
Description	Indicates the SHORADEZ (Short Range Air Defense Engagement Zone) is active The zone is known by context.	
Runway	MA_CarrierRunwayStateType	Optional
Description	Contains details of the runway used for landing.	
Catapult	MA_CatapultStateType	Repeated
Description	Contains details of each catapult for launching.	
QNH_Setting	PressureType	Optional
Description	Indicates atmospheric pressure of an Carrier in kilopascals.	

Table A-1-588: C2 UCI Extensions MA_CarrierStateMT - Information UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_CarrierStateMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CarrierStateID	MA_CarrierStateID_Type	Required
Description	Indicates a unique ID, assigned to this carrier state message.	
CarrierID	SystemID_Type	Required
Description	Indicates a unique ID, assigned to this carrier state message.	
IdentityReferenceID	CarrierReferenceID_ChoiceType	Required
Description	Indicates whether message is self-reported or reported by third party.	
ObservationTime	DateTimeType	Required
Description	Indicates the observation time of carrier state.	
Information	MA_CarrierStateInformationType	Optional
Description	Contains information about an Carrier.	
Weather	WeatherAreaDataType	Optional
Description	Contains weather details about a carrier. The wind in the WindData element provides wind information at ground level of the carrier.	
DepartureCase	MA_CaseTypeEnum	Optional
Description	This is the Case which indicates visibility level which impacts the departure routing	
RecoveryCase	MA_CaseTypeEnum	Optional
Description	This is the Case which indicates visibility level which impacts the recovery routing	
DepartureReferenceRadial	AngleType	Optional
Description	This is the Reference Radial used by ACPs departing the carrier.	
CarrierStores	AirfieldStoresPET	Repeated
Description	Contains details about stores available on an Carrier.	

Table A-1-589: C2 UCI Extensions MA_CarrierStateMT - MessageData UCI Extension Field

1.3.2.89 MA_PositionReportDetailedMT UCI Extension

MA_PositionReportDetailedMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info

Ext. Fields	PositionReportData : MA_PositionReportDataType MessageData : MA_PositionReportDetailedMDT

Table A-1-590: C2 UCI Extensions MA_PositionReportDetailedMT UCI Extension

PositionReportData		
Description	If multiple instances are given, each should be of a different navigation solution type as indicated by the child element.	
Base Type	MA_PositionReportDataType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PositionSource	PositionSourceID_ChoiceType	Required
Description	The source of the positional data.	
ComponentID	ComponentID_Type	Optional
Description	The ID of the specific component within the subsystem that is the source of this data. This can be used if multiple redundant INS/GPS units are represented as a single subsystem.	
NavigationSolutionState	NavigationSolutionStateEnum	Required
Description	The navigation solution state indicates what sensor inputs the navigation unit is using to generate its kinematic solution.	
FigureOfMerit	BytePositiveType	Optional
Description	A FOM is typically calculated as the root-sum-square of the estimated errors contributing to the solution accuracy. Example criteria include: a. GPS receiver state (e.g., carrier tracking, code tracking, acquisition) b. Carrier to noise ratio c. Satellite geometry (DOP value) d. Satellite range accuracy (URA value) e. Ionospheric measurement or modeling error f. Receiver aiding used g. Kalman filter error estimates. The resultant FOM is presented as a numerical value from 1 to 9, where 1 indicates the best navigation performance.	
Kinematics	MA_DetailedKinematicsType	Required

Description	The kinematic state of the vehicle.	
KinematicsError	DetailedKinematicsErrorType	Required
Description	Indicates the covariance matrix of the vehicle, containing detailed kinematic errors of the position, velocity, and orientation. See the DetailedKinematicsErrorType element annotations for details.	
SolutionCorrections	NavigationSolutionCorrectionsType	Optional
Description	This element contains the corrections applied by the measurement unit to the reported values.	

Table A-1-591: C2 UCI Extensions MA_PositionReportDetailedMT - PositionReportData UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PositionReportDetailedMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PositionReportData	MA_PositionReportDataType	Repeated
Description	If multiple instances are given, each should be of a different navigation solution type as indicated by the child element.	

Table A-1-592: C2 UCI Extensions MA_PositionReportDetailedMT - MessageData UCI Extension Field

1.3.2.90 MA_PrioritizationListMT UCI Extension

MA_PrioritizationListMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	ListType : MA_PrioritizationListEnum ListItem : MA_PrioritizationListItemType MessageData : MA_PrioritizationListMDT

Table A-1-593: C2 UCI Extensions MA_PrioritizationListMT UCI Extension

ListType	
Description	Indicates the type of prioritization list. For example, battlespace objects can be prioritized differently for targeting vs. protection as indicated by enumerates given in this element.
Base Type	MA_PrioritizationListEnum
Message Structure	

Enumeration Literal	Description
THREAT_TARGET_PRIORITIZATION_LIST	A prioritized list of battlespace objects or areas that are prioritized by an operational or tactical planner. For planning purposes, the ListItem associated with this enumeration could include, for example, an entity, a signal, a point or region.
CRITICAL_ASSET_LIST	A prioritized list of battlespace objects or areas that are prioritized by the joint force commander to be defended with the resources available. Critical assets are generally specific assets of such extraordinary importance that their loss or degradation would have a significant and debilitating effect on operations or the mission. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system or a point.
JIPTL	The Joint Integrated Prioritization Target List (JPITL) is a prioritized list of targets and associated data approved by the Joint Forces Commander or designated representative. An approved JIPTL is the central product of the target development cycle. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity or a system.
JIPTL_NOMINATION	A product of the target development phase is the JPITL nomination list. Targeters determine this list during the objectives, effects, and guidance phase and analyze which targets should be struck (or otherwise affected) to accomplish them. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity or a system.
JOINT_TARGET_LIST	A consolidated list of selected targets considered to have military significance in the combatant commander's area of responsibility. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area.
MISSION_TARGET_LIST	The list of battlespace targets that are considered to have military significance and are required for the successful completion of the mission. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area.
NO_STRIKE_LIST	A list of battlespace objects characterized as protected from the effects of military operations under international law or Engagement Criteria. Attacking these may violate Law of Armed Conflict (LOAC) or Engagement Criteria or interfere with friendly relations with indigenous personnel or governments. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area.
RESTRICTED_TARGET_LIST	A list of targets that have specific restrictions imposed upon them. Actions on restricted targets are prohibited until coordinated and approved by the establishing authority. Targets are restricted because certain types of actions against them may have negative political, cultural, or propaganda implications, or may interfere with projected friendly operations. The Restricted Target List (RTL) is nominated by elements of the joint force and approved by the JFC. Targets on this list may only be struck with JFC or higher approval. Actions taken by an opponent may remove a target from the RTL. For planning purposes, the ListItem associated with this enumeration could typically be, for example, an entity, a system, a point, or area.

DEFENDED_ASSET_LIST	The Defended Asset List (DAL) is a listing of those assets from the critical asset list prioritized by the joint force commander to be defended with the resources available. Critical assets that are reinforced with additional protection capabilities or capabilities from other combat power elements become part of the DAL. It represents what can be protected, by priority. The DAL allows commanders to apply finite protection capabilities to the most valuable assets.
PLATFORM_TARGET_LIST	A list of active entity platform targets that meet custody requirements. This list type is used in support of presenting a shot queue prioritization list with List Item of EntityID in accordance with the mission common operating picture.

Table A-1-594: C2 UCI Extensions MA_PrioritizationListMT - ListType UCI Extension Field

ListItem		
Description	Indicates an item in a prioritization list of the sibling ListType.	
Base Type	MA_PrioritizationListItemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Subject	IdentityKindInstanceType	Required
Description	Indicates the subject of the prioritization list item.	
Priority	PrioritizationType	Repeated
Description	Indicates the priority of the item. If priority for any child PriorityType is given, then omitted PriorityTypes for the sibling Subject have no/zero priority. No duplicates allowed; at most one instance of each of the child PriorityTypes is allowed. If the child PriorityType is not used, only one instance is allowed.	
ProbabilityOfKill	PercentType	Optional
Description	Indicates the probability of kill, in a percentage where a value of 100.0 = 100%, of a selected target with an available capability.	
ObjectCorrelation	ObjectCorrelationType	Optional
Description	Object Correlation contains a description of the correlation of this object with another object as well as whether the correlating system has C2 responsibilities or not, to indicate a rough level of confidence in the information.	

Table A-1-595: C2 UCI Extensions MA_PrioritizationListMT - ListItem UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PrioritizationListMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PrioritizationListID	PrioritizationListID_Type	Required

Description	Indicates the unique ID of the prioritization list.	
ListType	MA_PrioritizationListEnum	Required
Description	Indicates the type of prioritization list. For example, battlespace objects can be prioritized differently for targeting vs. protection as indicated by enumerates given in this element.	
ListItem	MA_PrioritizationListItemType	Repeated
Description	Indicates an item in a prioritization list of the sibling ListType.	
OpOrderTraceability	OrderTraceabilityType	Repeated
Description	Indicates an operational order which this PrioritizationList is traceable to. This does not mean ranking. This element refers to a Planning Order, Commander's Intent, or other Order that came from a higher echelon. The higher echelon is identified by the OrderSourceID.	
OpDescription	OpDescriptionType	Optional
Description	Indicates a human readable description of the purpose of the prioritization list.	
ApplicableSystemID	SystemID_Type	Repeated
Description	One or more Systems to which the prioritization applies. If omitted, the prioritization applies to all Systems.	
ApplicableServiceID	ServiceID_Type	Repeated
Description	One or more Services to which the mission response list applies. If omitted, the settings apply to all Services of the Systems given in the sibling ApplicableSystemID.	
ApplicableZone	ZoneType	Optional
Description	Zones to which the prioritization applies.	
Schedule	ScheduleType	Optional
Description	Scheduled times to which the prioritization applies.	
DataRecordInstanceID	DataRecordInstanceID_Type	Optional
Description	Indicates the ID of this Data Record instance. The DataRecordInstanceID is used as a checkpoint for DataRecordManagementRequests to ensure that the state of the DataRecord (or DataRecords) that are the subject of a DataRecordManagementRequest are coherent between the requestor and the requestee.	

Table A-1-596: C2 UCI Extensions MA_PrioritizationListMT - MessageData UCI Extension Field

1.3.2.91 MA_ResponsePlanCommandStatusMT UCI Extension

MA_ResponsePlanCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ResponsePlanCommandStatusMDT AssociatedPlans : MA_PlanReferenceType AllocationResult : MA_ResponsePlanningResultType

	OpPlanID : MA_OpPlanID_Type

Table A-1-597: C2 UCI Extensions MA_ResponsePlanCommandStatusMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_ResponsePlanCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	ResponsePlanCommandID_Type	Required
Description	The unique ID of the ResponsePlanCommand this status applies to.	
PlanningStatus	PlanCommandStatusType	Required
Description	Indicates the status of the *PlanCommand.	
AllocationResult	MA_ResponsePlanningResultType	Repeated
Description	Indicates allocation results per Response.	
ResultingPlanID	ResponsePlanID_Type	Repeated
Description	This element indicates the unique ID or IDs of any ResponsePlans that were generated.	

Table A-1-598: C2 UCI Extensions MA_ResponsePlanCommandStatusMT - MessageData UCI Extension Field

AssociatedPlans		
Description	Indicates *Plan associated with the sibling PlanningResult element. If the sibling PlanningResult indicates the Requirement was/remained allocated then this element indicates the specific *Plan it is/remains allocated in. If the PlanningResult indicates the Requirement was DEALLOCATED then this element indicates the constraining *Plan it was DEALLOCATED from. This element isn't expected when the PlanningResult indicates UNALLOCATED.	
Base Type	MA_PlanReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Optional
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Optional
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Optional
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Optional

Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Optional
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Optional
Description	Indicates a reference to a RouteActivityPlan.	
CommScheduleAllocationID	CommScheduleAllocationID_Type	Optional
Description	Indicates a reference to a Comms allocation.	
ActivityPlanID	ActivityPlanID_Type	Optional
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Optional
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Optional
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Optional
Description	Indicates a reference to a ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Optional
Description	Indicates a reference to an OpPlan.	

Table A-1-599: C2 UCI Extensions MA_ResponsePlanCommandStatusMT - AssociatedPlans UCI Extension Field

AllocationResult		
Description	Indicates allocation results per Response.	
Base Type	MA_ResponsePlanningResultType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ResponseID	ResponseID_Type	Required
Description	Unique object identifier for the Response whose planning results are given by sibling elements.	
PlanningResult	PlanningResultEnum	Required
Description	Indicates the outcome of attempting to allocate or plan activities for the Requirement.	
SystemID	SystemID_Type	Required
Description	Indicates the SystemID of the System whose *Plan are associated with the sibling PlanningResult element.	
AssociatedPlans	MA_PlanReferenceType	Optional
Description	Indicates *Plan associated with the sibling PlanningResult element. If the sibling PlanningResult indicates the Requirement was/remained allocated then this element indicates the specific *Plan it is/remains allocated in. If the PlanningResult indicates the Requirement was DEALLOCATED then this element indicates the constraining	

	*Plan it was DEALLOCATED from. This element isn't expected when the PlanningResult indicates UNALLOCATED.	
NotAllocatedReason	UnallocatedReasonType	Repeated
Description	Indicates a reason why the Requirement wasn't allocated or reallocated. If the sibling PlanningResult element indicates the Requirement was left not allocated then this element indicates why. Multiple instances of this element can provide the reason for each System that was attempted and even many reasons for each. If the PlanningResult indicates the Requirement was allocated then this element can still be used to indicate reasons for failed attempts to allocate to other Systems or plan, prior to the successful allocation.	

Table A-1-600: C2 UCI Extensions MA_ResponsePlanCommandStatusMT - AllocationResult UCI Extension Field

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-601: C2 UCI Extensions MA_ResponsePlanCommandStatusMT - OpPlanID UCI Extension Field

1.3.2.92 MA_ResponsePlanCommandMT UCI Extension

MA_ResponsePlanCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ResponsePlanCommandMDT

Table A-1-602: C2 UCI Extensions MA_ResponsePlanCommandMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_ResponsePlanCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	ResponsePlanCommandID_Type	Required
Description	The unique ID of the command. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	
ParentPlanCommandID	ResponsePlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which this ResponsePlanCommand originated from.	
Inputs	ResponsePlanInputsType	Required
Description	Indicates data and references to be used to create the ResponsePlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	

Table A-1-603: C2 UCI Extensions MA_ResponsePlanCommandMT - MessageData UCI Extension Field

1.3.2.93 MA_PlanningFunctionSettingsCommandMT UCI Extension

MA_PlanningFunctionSettingsCommandMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	AutonomousPlanningAction : MA_PlanningAllowedEscalationType OpPlan : MA_OpPlanPartsType AutonomousPlanningResponseChoice : MA_MissionPlanningAutonomyResponseChoiceType MissionPlanningAutonomy : MA_MissionPlanningAutonomySettingType OrbitActivityPlan : MA_ActivityPlanPartsType RouteActivityPlan : MA_ActivityPlanPartsType ByResultSetting : MA_MissionPlanningAutonomySettingByResultType PlanningAllowed : MA_PlanningAllowedType

	OutputPlanParts : MA_PlanPartsType AllowedType : MA_PlanningAllowedType AutonomousAction : MA_MissionPlanningByResultAutonomousActionType MessageData : MA_PlanningFunctionSettingsCommandMDT ActivityPlan : MA_ActivityPlanPartsType TaskPlan : MA_TaskPlanPartsType

Table A-1-604: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT UCI Extension

AutonomousPlanningAction		
Description	This element indicates the allowed autonomous mission planning for the ancestor Trigger. It also indicates the expected MissionContingencyAlert for the ancestor Trigger; whenever autonomous mission planning is allowed and triggered, a MissionContingencyAlert is expected.	
Base Type	MA_PlanningAllowedEscalationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AutonomousPlanningActionID	AutonomousPlanningActionID_Type	Required
Description	Assigns a unique ID to an allowed planning action. This ID can be reference in *Plans, alerts, etc. that result when the planning action is triggered.	
ByResultSetting	MA_MissionPlanningAutonomySettingByResultType	Repeated
Description	This element allows control of the escalation of the autonomous replanning. If the autonomous function indicated by the sibling AllowedType leads to a "by result" trigger described here, then additional autonomous planning is allowed as described here. For example, if Tasks are dropped as a result of initial autonomous replanning then this element can be used to enable/disable a secondary "escalated" planning iteration that includes global Task allocation. If omitted, the sibling AllowedType is allowed regardless of result.	
AllowedType	MA_PlanningAllowedType	Required
Description	This element defines the mission planning type that is allowed to be autonomously triggered.	

Table A-1-605: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - AutonomousPlanningAction UCI Extension Field

OpPlan		
Description	Indicates applicable OpPlan parts.	
Base Type	MA_OpPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPointCategory	OpPointCategoriesType	Repeated

Description	Indicates an OpPoint type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpLineCategory	MA_OpLineCategoryEnum	Repeated
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpZoneCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpVolumeCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpRouting	EmptyType	Optional
Description	Indicates an OpRouting is applicable to the associated *Plan. List size for this element is based on there being no category type for OpRouting objects	

Table A-1-606: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - OpPlan UCI Extension Field

AutonomousPlanningResponseChoice		
Description	Indicates the autonomous mission planning type or types allowed for the sibling Trigger.	
Base Type	MA_MissionPlanningAutonomyResponseChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AutonomousPlanningAction	MA_PlanningAllowedEscalationType	Repeated
Description	This element indicates the allowed autonomous mission planning for the ancestor Trigger. It also indicates the expected MissionContingencyAlert for the ancestor Trigger; whenever autonomous mission planning is allowed and triggered, a MissionContingencyAlert is expected.	
AlertOnly	EmptyType	Required
Description	This element indicates that autonomous mission planning isn't allowed for the ancestor Trigger but a MissionContingencyAlert is expected.	

Table A-1-607: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - AutonomousPlanningResponseChoice UCI Extension Field

MissionPlanningAutonomy		
Description	Indicates commanded settings for autonomous mission planning capabilities of the System.	
Base Type	MA_MissionPlanningAutonomySettingType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

Trigger	PlanningByCaseTriggerType	Required
Description	Indicates an allowed trigger for autonomous planning.	
AutonomousPlanningResponseChoice	MA_MissionPlanningAutonomyResponseChoiceType	Required
Description	Indicates the autonomous mission planning type or types allowed for the sibling Trigger.	
PlanningProcessID	PlanningProcessID_Type	Optional
Description	Indicates the unique ID of a planning process which the settings are applicable to. If omitted, the setting is applicable to all processes of the corresponding System.	
MinimumResponseInterval	DurationType	Optional
Description	Indicates the minimum time interval between autonomous responses for the sibling Trigger. In other words, regardless of when and how frequently the Trigger occurs, the sibling Response shouldn't be initiated until the minimum time duration given here after the previous Response.	

Table A-1-608: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - MissionPlanningAutonomy UCI Extension Field

OrbitActivityPlan		
Description	Indicates applicable OrbitActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional

Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-609: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - OrbitActivityPlan UCI Extension Field

RouteActivityPlan		
Description	Indicates applicable RouteActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	

OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-610: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - RouteActivityPlan UCI Extension Field

ByResultSetting		
Description	This element allows control of the escalation of the autonomous replanning. If the autonomous function indicated by the sibling AllowedType leads to a "by result" trigger described here, then additional autonomous planning is allowed as described here. For example, if Tasks are dropped as a result of initial autonomous replanning then this element can be used to enable/disable a secondary "escalated" planning iteration that includes global Task allocation. If omitted, the sibling AllowedType is allowed regardless of result.	
Base Type	MA_MissionPlanningAutonomySettingByResultType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Trigger	PlanningByResultTriggerType	Required
Description	Indicates a trigger based on the results of a previously initiated autonomous mission planning function.	
AutonomousAction	MA_MissionPlanningByResultAutonomousActionType	Required
Description	Indicates the action or actions allowed for the sibling trigger.	

Table A-1-611: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - ByResultSetting UCI Extension Field

PlanningAllowed		
Description	This element defines the mission planning type to be triggered. If multiple instances are given, each should be of a different planning type as indicated by the child element.	
Base Type	MA_PlanningAllowedType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanAheadDuration	DurationType	Required
Description	Required duration in the future when the Plan should start. Existing *Plan will continue during this time. This duration should account for everything that must happen from replan initiation to new plan activation.	
PlanningProcessID	PlanningProcessID_Type	Required
Description	Indicates the unique ID of the planning process whose autonomous invocation is described the sibling elements. Planning process are declared in the ControlStatus message.	
ConstrainingPlans	AutonomousPlanningConstrainingPlansType	Optional

Description	Indicates existing types of *Plans that can be used as constraints when invoking the planning process indicated by the sibling PlanningProcessID. The types of *Plans indicated here should be consistent with those supported by the planning process as indicated in the ControlStatus message.	
OtherSystemConstrainingPlans	AutonomousPlanningOtherSystemConstrainingPlansType	Repeated
Description	Indicates existing plans of another System that constrain planning. One for each descendant ConstraintUsage.	
OutputPlanParts	MA_PlanPartsType	Optional
Description	Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType.	
ResultsInMissionPlan	boolean	Required
Description	In some cases, the commanding service or HMI may not want to receive the planning process summarized in a MissionPlan message. If TRUE, the results will be published in individual *Plan messages and summarized in a published MissionPlan message. If FALSE, the results will be published in individual *Plan messages without a summary MissionPlan message.	
ForPlanningUseOnly	boolean	Required
Description	Indicates the state to use for the corresponding element in the *PlanCommand when invoking the planning process.	

Table A-1-612: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - PlanningAllowed UCI Extension Field

OutputPlanParts		
Description	Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType.	
Base Type	MA_PlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	
OrbitActivityPlan	MA_ActivityPlanPartsType	Optional

Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-613: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - OutputPlanParts UCI Extension Field

AllowedType		
Description	This element defines the mission planning type that is allowed to be autonomously triggered.	
Base Type	MA_PlanningAllowedType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanAheadDuration	DurationType	Required
Description	Required duration in the future when the Plan should start. Existing *Plan will continue during this time. This duration should account for everything that must happen from replan initiation to new plan activation.	
PlanningProcessID	PlanningProcessID_Type	Required
Description	Indicates the unique ID of the planning process whose autonomous invocation is described the sibling elements. Planning process are declared in the ControlStatus message.	
ConstrainingPlans	AutonomousPlanningConstrainingPlansType	Optional
Description	Indicates existing types of *Plans that can be used as constraints when invoking the planning process indicated by the sibling PlanningProcessID. The types of *Plans indicated here should be consistent with those supported by the planning process as indicated in the ControlStatus message.	
OtherSystemConstrainingPlans	AutonomousPlanningOtherSystemConstrainingPlansType	Repeated
Description	Indicates existing plans of another System that constrain planning. One for each descendant ConstraintUsage.	
OutputPlanParts	MA_PlanPartsType	Optional
Description	Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the	

	Plan parts for the sibling PlanningType.	
ResultsInMissionPlan	boolean	Required
Description	In some cases, the commanding service or HMI may not want to receive the planning process summarized in a MissionPlan message. If TRUE, the results will be published in individual *Plan messages and summarized in a published MissionPlan message. If FALSE, the results will be published in individual *Plan messages without a summary MissionPlan message.	
ForPlanningUseOnly	boolean	Required
Description	Indicates the state to use for the corresponding element in the *PlanCommand when invoking the planning process.	

Table A-1-614: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - AllowedType UCI Extension Field

AutonomousAction		
Description	Indicates the action or actions allowed for the sibling trigger.	
Base Type	MA_MissionPlanningByResultAutonomousActionType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanningAllowed	MA_PlanningAllowedType	Repeated
Description	This element defines the mission planning type to be triggered. If multiple instances are given, each should be of a different planning type as indicated by the child element.	
AlertOnly	EmptyType	Required
Description	This element indicates that autonomous mission planning isn't allowed for the ancestor Trigger but a MissionContingencyAlert is expected.	

Table A-1-615: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - AutonomousAction UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the PlanningFunctionSettingsCommand message.	
Base Type	MA_PlanningFunctionSettingsCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Required
Description	Indicates the unique ID of the System whose planning functions settings are being commanded/changed.	
TransitionDroppedToFailed	boolean	Required
Description	This element specifies whether or not dropped Tasks that cannot be reallocated within the System can be automatically transitioned to the FAILED state for higher	

	level interaction.	
PlanningInterface	PlanInterfaceCommandType	Repeated
Description	Indicates commanded settings for planning interfaces. If multiple instances of this element are given, each should be of a different type of *Plan as enumerated by the child PlanType element.	
MissionPlanningAutonomy	MA_MissionPlanningAutonomySettingType	Repeated
Description	Indicates commanded settings for autonomous mission planning capabilities of the System.	
PlanActivationAutonomy	PlanActivationAutonomyType	Optional
Description	Indicates commanded settings for autonomous plan activation capabilities of the System.	
RoutePlanningContingencyPath	ContingencyPathAutonomyType	Repeated
Description	Indicates commanded route planning settings for the autonomous generation of contingency paths along the primary path. If multiple instances are given, each must be of a different path type as indicated by the child element.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-616: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - MessageData UCI Extension Field

ActivityPlan		
Description	Indicates applicable ActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional

Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-617: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - ActivityPlan UCI Extension Field

TaskPlan		
Description	Indicates applicable TaskPlan parts.	
Base Type	MA_TaskPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskType	MA_TaskTypeEnum	Repeated
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	

Table A-1-618: C2 UCI Extensions MA_PlanningFunctionSettingsCommandMT - TaskPlan UCI Extension Field

1.3.2.94 MA_PlanningFunctionSettingsCommandStatusMT UCI Extension

MA_PlanningFunctionSettingsCommandStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_PlanningFunctionSettingsCommandStatusMDT

Table A-1-619: C2 UCI Extensions MA_PlanningFunctionSettingsCommandStatusMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the PlanningFunctionSettingsCommandStatus message.	
Base Type	MA_PlanningFunctionSettingsCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-620: C2 UCI Extensions MA_PlanningFunctionSettingsCommandStatusMT - MessageData UCI Extension Field

1.3.2.95 MA_OpLineMT UCI Extension

MA_OpLineMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	Category : MA_OpLineCategoryEnum AppliesToIDs : MA_PackageSystemType MessageData : MA_OpLineMDT

Table A-1-621: C2 UCI Extensions MA_OpLineMT UCI Extension

Category	
Description	This element indicates the type of operational line.
Base Type	MA_OpLineCategoryEnum
Message Structure	
Enumeration Literal	**Description**

GENERAL_LINE	A general line where the function must be described by other procedures.
FORWARD_EDGE_OF_BATTLE_AREA	The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.
GUN_TARGET_LINE	Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.
CORRIDOR	Line with width describing the passageway designated for travel by friendly aircraft.
HOSTILE_BOUNDARY	Line that indicates a limit beyond which exists a hostile environment.
BUFFER_ZONE_BOUNDARY	Line beyond which friendly aircraft usually do not fly in peacetime.
LOW_LEVEL_TRANSIT_ROUTE	Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.
TACTICAL_ACTION_LINE	Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.
FIRE_SUPPORT_COORDINATION_LINE	Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander.
FORWARD_LINE_OF_OWN_TROOPS	Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.
LINE_OF_COMMUNICATIONS	Line the route that connects an operating military unit with its supply base.
JETWAY	Line with width that defines areas where commercial aircraft are flying through.
DEPARTURE_RADIAL	Line the defines the departure radial when performing Carrier admin departures.
RECOVERY_RADIAL	Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.
FINAL_BEARING	Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.
BRC	Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.

Table A-1-622: C2 UCI Extensions MA_OpLineMT - Category UCI Extension Field

AppliesToIDs		
Description	One or more systems to which this item applies. If omitted, item applies to all systems.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	
PackageID	PackageID_Type	Required
Description	The ID of a Package.	
SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-623: C2 UCI Extensions MA_OpLineMT - AppliesToIDs UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_OpLineMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpLineID	OpLineID_Type	Required
Description	Indicates the unique ID of the OpLine.	
Category	MA_OpLineCategoryEnum	Required
Description	This element indicates the type of operational line.	
Line	OpLineType	Required
Description	Indicates the geometric description of the OpLine. In practice, operational lines can be geometric lines, areas or in some cases volumes.	
OpDescription	OpDescriptionType	Optional
Description	A human-readable descriptive name with free text remarks.	
MissionTraceability	MissionTraceabilityType	Optional
Description	Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc.	
Source	CreationSourceEnum	Optional
Description	Indicates the source of the information.	
Schedule	ScheduleType	Optional
Description	This element represents a date/time interval associated with the operational item during which the item is in effect.	
AssociatedTime	TimeFunctionType	Repeated
Description	Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options.	
DataLinkIdentifier	DataLinkIdentifierPET	Repeated

Description	List of data link ID. Multiple data link IDs can be reported for the same network type.	
Priority	unsignedInt	Optional
Description	This field represents the priority to be considered when choosing between multiple available points.	
QualifyingTags	QualifyingTagsType	Optional
Description	Provides identity qualifications to augment identification and associated processing of the OpPoint.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	One or more systems to which this item applies. If omitted, item applies to all systems.	
SystemScheduleOverride	SystemScheduleStateType	Repeated
Description	System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution.	

Table A-1-624: C2 UCI Extensions MA_OpLineMT - MessageData UCI Extension Field

1.3.2.96 MA_MissionPlanCommandMT UCI Extension

MA_MissionPlanCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	Inputs : MA_MissionPlanInputsType ActivityPlan : MA_ActivityPlanPartsType ProposedRequirements : MA_RequirementAllocationCommandType RouteActivityPlan : MA_ActivityPlanPartsType ProposedResponse : MA_ResponseAllocationType OrbitActivityPlan : MA_ActivityPlanPartsType ProposedTaskPlan : MA_TaskPlanConstraintType ProposedTask : MA_TaskAllocationType MessageData : MA_MissionPlanCommandMDT TaskType : MA_TaskTypeEnum ProposedEffect : MA_EffectAllocationType PlanningCandidate : MA_RequirementPlanningCandidateType ProposedAction : MA_ActionAllocationType OutputPlanType : MA_PlanPartsType ProposedRequirementPlans : MA_RequirementPlanConstraintType CapabilityCommand : MA_CapabilityTypeEnum

Table A-1-625: C2 UCI Extensions MA_MissionPlanCommandMT UCI Extension

Inputs		
Description	Indicates data and references to other messages and their data to be used to create the MissionPlan.	
Base Type	MA_MissionPlanInputsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanningCandidate	MA_RequirementPlanningCandidateType	Repeated
Description	Indicates specific Systems to consider and plan for along with System-specific planning constraints. If omitted, all Tiers and all Systems at and below those that the planning process is applicable to shall be planned. A System is a planning "candidate" because its plan may or may not change as a result of the command; zero to one Plans per *Plan type will potentially be created for each candidate.	
ProposedRequirementPlans	MA_RequirementPlanConstraintType	Optional
Description	Indicates an existing [Requirement]Plan of the System superior to the sibling PlanningCandidates that provides a proposed "requirement deck" to allocate amongst the candidates. This element can be supplemented with additional Requirements from the sibling ProposedRequirements element.	
ProposedRequirements	MA_RequirementAllocationCommandType	Optional
Description	Indicates a proposed Requirement for the planning process. A Requirement given here is added to any Requirements provided in the sibling ProposedRequirementPlans and constraining *Plans of the sibling PlanningCandidates to create the full set of Requirements to be considered by the planning process. A Requirement given here overrides the same Requirement in the sibling ProposedRequirementPlan. An empty list here, combined with no sibling ProposedRequirementPlans is considered a command to deallocate previously allocated Requirements.	
AssociationConstraint	RequirementAssociationConstraintType	Repeated
Description	Indicates an association between Requirements that is a constraint on how they are allocated and planned.	
ResultsInMissionPlan	boolean	Required
Description	In some cases, the commanding service or HMI may not want to receive the planning process summarized in a MissionPlan message. For example, if MissionPlanCommand is being used to produce a combination of plans (such as a RoutePlan and RouteActivityPlan) for a portion of a sortie, with no immediate intent of producing a full mission plan for the whole sortie, then the results could be published without creating a MissionPlan message. If TRUE, the results will be published in individual *Plan messages and summarized in a published MissionPlan message. If FALSE, the results will be published in individual *Plan messages without a summary MissionPlan message.	
OutputPlanType	MA_PlanPartsType	Optional
Description	Indicates a type of Plan that is required as an output of the MissionPlanCommand. When populated, this element must match or be a subset of the *Plans associated with the planning process (sibling PlanningProcessID) as advertised in PlanningFunction*messages. This element is optional when all *Plan outputs of the process are required.	

PlanningProcessID	PlanningProcessID_Type	Required
Description	Indicates the unique ID of the planning process invoked by the *PlanCommand.	
PlanInitiation	PlanInitiationSourceEnum	Required
Description	This element indicates whether the Plan was initiated autonomously from a planning service, was operator initiated or edited, or was created externally.	
PlanningDataSource	PlanningDataSourceEnum	Required
Description	Indicates the category of messages to use as the source for System position, Capability, status, etc. data to use as the basis for the plan.	
ReplanReason	MA_ReplanReasonType	Optional
Description	Indicates the reason for the replan. This element is used during live operations and/or operation with autonomous planning, contingency management, etc.	
EnvironmentOverride	MA_MissionEnvironmentConstraintType	Repeated
Description	Indicates a selective override of an item in the mission environment world state to use/used in the Plan.	
OpConstraint	OpConstraintWeightingType	Repeated
Description	Further constrain a Mission Plan to an operational constraint that indicates how much the constraint should play a role in the planning process. The multiplicity is based on the list of available OpConstraintEnums.	
SpecialInstructionsID	FileLocationID_Type	Optional
Description	Free-text explanation of any extraordinary instructions at the plan level.	

Table A-1-626: C2 UCI Extensions MA_MissionPlanCommandMT - Inputs UCI Extension Field

ActivityPlan		
Description	Indicates applicable ActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	

ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-627: C2 UCI Extensions MA_MissionPlanCommandMT - ActivityPlan UCI Extension Field

ProposedRequirements		
Description	Indicates a proposed Requirement for the planning process. A Requirement given here is added to any Requirements provided in the sibling ProposedRequirementPlans and constraining *Plans of the sibling PlanningCandidates to create the full set of Requirements to be considered by the planning process. A Requirement given here overrides the same Requirement in the sibling ProposedRequirementPlan. An empty list here, combined with no sibling ProposedRequirementPlans is considered a command to deallocate previously allocated Requirements.	
Base Type	MA_RequirementAllocationCommandType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ProposedEffect	MA_EffectAllocationType	Repeated
Description	Indicates a proposed Effect.	
ProposedAction	MA_ActionAllocationType	Repeated
Description	Indicates a proposed Action.	
ProposedTask	MA_TaskAllocationType	Repeated
Description	Indicates a proposed Task.	
ProposedResponse	MA_ResponseAllocationType	Repeated
Description	Indicates a proposed Response.	

Table A-1-628: C2 UCI Extensions MA_MissionPlanCommandMT - ProposedRequirements UCI Extension Field

RouteActivityPlan		
Description	Indicates applicable RouteActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-629: C2 UCI Extensions MA_MissionPlanCommandMT - RouteActivityPlan UCI Extension Field

ProposedResponse		
Description	Indicates a proposed Response.	
Base Type	MA_ResponseAllocationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

ResponseID	ResponseID_Type	Required
Description	Indicates the unique ID of the allocated or to-be allocated Response.	
Locked	boolean	Optional
Description	Indicates whether the Requirement's allocation can be modified/removed in subsequent planning steps and/or replans.	
ConstraintOverride	MA_RequirementConstraintsType	Optional
Description	Indicates refined/overridden values of constraints for the Requirement. Constraints can be refined/overridden in inputs (*PlanCommand) and/or outputs (*Plan) of planning processes. Values given in this element refine/override those in the associated [Requirement] message. Reasons for constraint overrides include refinement based on the allocated System, multi-Requirement coordination, etc. Requirement constraints can be 1) given in a [Requirement] message, 2) overridden as a planning input in a [Requirement]PlanCommand message, 3) further refined/overridden during plan creation and reported in a [Requirement]Plan message to inform subsequent planning or execution.	
OtherOverride	RequirementAllocationDetailsType	Optional
Description	Indicates special/override values for miscellaneous [Requirement] message elements not included in the sibling ConstraintOverride element.	

Table A-1-630: C2 UCI Extensions MA_MissionPlanCommandMT - ProposedResponse UCI Extension Field

OrbitActivityPlan		
Description	Indicates applicable OrbitActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional

Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-631: C2 UCI Extensions MA_MissionPlanCommandMT - OrbitActivityPlan UCI Extension Field

ProposedTaskPlan		
Description	Indicates a proposed TaskPlan.	
Base Type	MA_TaskPlanConstraintType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AllocationPlanID	TaskPlanID_Type	Optional
Description	Indicates the unique ID of the TaskPlan being referenced. When omitted, the current/active TaskPlan for the System is implied.	
ChangeableAllocations	PlanChangeableConstraintsEnum	Required
Description	Indicates which kinds/categories of allocations in the TaskPlan indicated by the sibling AllocationPlanID are considered changeable. The exact definition of "changeable" is left to particular Services and/or Systems to develop according to their requirements. A lenient interpretation would equate "changeable" with "deallocation allowed" thereby giving a System and/or planning Service freedom to deallocate any changeable allocations from the constraining plan in order to optimize a new *Plan for new and/or unchangeable allocations. Those unallocated Tasks would then be dealt with in subsequent planning steps/iterations. With this definition, an "unchangeable" allocation item is "System/vehicle constrained" and therefore the new *Plan being developed should avoid changes that would cause the item to be unallocated. A stricter interpretation might additionally require that any temporal constraints of an unchangeable allocation item from the constraining TaskPlan be unchanged as a result of the new *Plan being developed. In a System of Systems with inter-*Plan temporal dependencies, this stricter interpretation could reduce automated replanning chain reactions.	
TaskType	MA_TaskTypeEnum	Repeated
Description	Indicates a type of Task in the TaskPlan referenced by the sibling AllocationPlanID element that constrains the new *Plan. If any are given, only allocations associated with those Task types are constraints. If omitted, allocations associated with all Task types are constraints. List size for this element is based on "Select All That Apply" condition.	

Table A-1-632: C2 UCI Extensions MA_MissionPlanCommandMT - ProposedTaskPlan UCI Extension Field

ProposedTask		
Description	Indicates a proposed Task.	
Base Type	MA_TaskAllocationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskID	TaskID_Type	Required
Description	Indicates the unique ID of the allocated or to-be allocated Task.	

Table A-1-633: C2 UCI Extensions MA_MissionPlanCommandMT - ProposedTask UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	MissionPlanCommandID_Type	Required
Description	The ID of the Mission Plan Command. Used to reference the command when providing status.	
ParentMissionPlanCommandID	MissionPlanCommandID_ChoiceType	Optional
Description	Indicates the unique ID of the *PlanCommand which this MissionPlanCommand originated from. For example, a MissionPlan sent to a System might have as a parent the MissionPlanCommand sent from the operator/commander one level above to the task allocator/planner at that level.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	
Inputs	MA_MissionPlanInputsType	Required
Description	Indicates data and references to other messages and their data to be used to create the MissionPlan.	
ForPlanningUseOnly	boolean	Required
Description	When TRUE, indicates the *Plan is to be used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	

Table A-1-634: C2 UCI Extensions MA_MissionPlanCommandMT - MessageData UCI Extension Field

TaskType	
Description	Indicates a type of Task in the TaskPlan referenced by the sibling AllocationPlanID element that constrains the new *Plan. If any are

	given, only allocations associated with those Task types are constraints. If omitted, allocations associated with all Task types are constraints. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_TaskTypeEnum
Message Structure	
Enumeration Literal	**Description**
AIR_SAMPLE	Indicates the task is to sample and analyze the air around the vehicle.
AMTI	Indicates the task is to track air targets using an Airborne Moving Target Indicator.
AO	Indicates the task is to employ an Active Optical capability.
CARGO_DELIVERY	Indicates the task is to deliver cargo.
CAP	Indicates the task is to execute a Combat Air Patrol.
COMINT	Indicates the task is to employ a communications intelligence capability.
COMM_RELAY	Indicates the task is to relay communications from other battlespace participants.
COUNTER_SPACE	Indicates the task is to employ a Counter Space Capability.
EA	Indicates the task is to employ an electronic attack capability.
ESCORT	Indicates the task is to escort another entity.
ESM	Indicates the task is to employ an Electronic Support Measure.
FLIGHT	Indicates the task is to fly.
ORBIT_CHANGE	Indicates a task to perform an orbit change via a spacecraft maneuver.
ORBITAL_SURVEILLANCE	Indicates a task to perform orbital surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates a task to perform orbital surveillance via sensor collection capabilities.
PO	Indicates the task is to employ a Passive Optical capability.
REFUEL	Indicates the task is to refuel another System or Entity.
SAR	Indicates the task is to employ a Synthetic Aperture Radar capability.
SMTI	Indicates the task is to track surface targets using a Surface Moving Target Indicator.
STRIKE	Indicates the task is to strike a target.
SYSTEM_DEPLOYMENT	Indicates the task is to deploy a system.
TACTICAL_ORDER	Indicates the task is to perform a tactical order.
WEATHER_RADAR	Indicates the task is to collect weather data.

Table A-1-635: C2 UCI Extensions MA_MissionPlanCommandMT - TaskType UCI Extension Field

ProposedEffect	
Description	Indicates a proposed Effect.

Base Type	MA_EffectAllocationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
EffectID	EffectID_Type	Required
Description	Indicates the unique ID of the allocated or to-be allocated Effect.	
Locked	boolean	Optional
Description	Indicates whether the Requirement's allocation can be modified/removed in subsequent planning steps and/or replans.	
ConstraintOverride	MA_RequirementConstraintsType	Optional
Description	Indicates refined/overridden values of constraints for the Requirement. Constraints can be refined/overridden in inputs (*PlanCommand) and/or outputs (*Plan) of planning processes. Values given in this element refine/override those in the associated [Requirement] message. Reasons for constraint overrides include refinement based on the allocated System, multi-Requirement coordination, etc. Requirement constraints can be 1) given in a [Requirement] message, 2) overridden as a planning input in a [Requirement]PlanCommand message, 3) further refined/overridden during plan creation and reported in a [Requirement]Plan message to inform subsequent planning or execution.	
OtherOverride	RequirementAllocationDetailsType	Optional
Description	Indicates special/override values for miscellaneous [Requirement] message elements not included in the sibling ConstraintOverride element.	

Table A-1-636: C2 UCI Extensions MA_MissionPlanCommandMT - ProposedEffect UCI Extension Field

PlanningCandidate		
Description	Indicates specific Systems to consider and plan for along with System-specific planning constraints. If omitted, all Tiers and all Systems at and below those that the planning process is applicable to shall be planned. A System is a planning "candidate" because its plan may or may not change as a result of the command; zero to one Plans per *Plan type will potentially be created for each candidate.	
Base Type	MA_RequirementPlanningCandidateType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RouteGuideline	PlanningGuidelineType	Optional
Description	This element specifies high-level route guidelines that were or should be used for the plan that are not specified or implied by Requirements.	
OrbitGuideline	OrbitGuidelineType	Optional
Description	This element specifies high-level orbit guidelines that were or should be used for the plan that are not specified or implied Requirements.	
SystemID	SystemID_Type	Required
Description	The SystemID of the System to plan for.	

ConstrainingPlans	MA_ConstrainingPlansType	Optional
Description	Indicates existing plans of the System associated with the sibling SystemID that constrain the *Plan or *PlanCommand.	
OtherSystemConstrainingPlans	MA_OtherSystemConstrainingPlansType	Repeated
Description	Indicates existing plans of another System that constrain the *Plan or *PlanCommand.	

Table A-1-637: C2 UCI Extensions MA_MissionPlanCommandMT - PlanningCandidate UCI Extension Field

ProposedAction		
Description	Indicates a proposed Action.	
Base Type	MA_ActionAllocationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActionID	ActionID_Type	Required
Description	Indicates the unique ID of the allocated or to-be allocated Action.	
Locked	boolean	Optional
Description	Indicates whether the Requirement's allocation can be modified/removed in subsequent planning steps and/or replans.	
ConstraintOverride	MA_RequirementConstraintsType	Optional
Description	Indicates refined/overridden values of constraints for the Requirement. Constraints can be refined/overridden in inputs (*PlanCommand) and/or outputs (*Plan) of planning processes. Values given in this element refine/override those in the associated [Requirement] message. Reasons for constraint overrides include refinement based on the allocated System, multi-Requirement coordination, etc. Requirement constraints can be 1) given in a [Requirement] message, 2) overridden as a planning input in a [Requirement]PlanCommand message, 3) further refined/overridden during plan creation and reported in a [Requirement]Plan message to inform subsequent planning or execution.	
OtherOverride	RequirementAllocationDetailsType	Optional
Description	Indicates special/override values for miscellaneous [Requirement] message elements not included in the sibling ConstraintOverride element.	

Table A-1-638: C2 UCI Extensions MA_MissionPlanCommandMT - ProposedAction UCI Extension Field

OutputPlanType	
Description	Indicates a type of Plan that is required as an output of the MissionPlanCommand. When populated, this element must match or be a subset of the *Plans associated with the planning process (sibling PlanningProcessID) as advertised in PlanningFunction*messages. This element is optional when all *Plan outputs of the process are required.
Base Type	MA_PlanPartsType
Message Structure	

Field Name	Field Type	Field Usage
ActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	
OrbitActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-639: C2 UCI Extensions MA_MissionPlanCommandMT - OutputPlanType UCI Extension Field

ProposedRequirementPlans		
Description	Indicates an existing [Requirement]Plan of the System superior to the sibling PlanningCandidates that provides a proposed "requirement deck" to allocate amongst the candidates. This element can be supplemented with additional Requirements from the sibling ProposedRequirements element.	
Base Type	MA_RequirementPlanConstraintType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ProposedEffectPlan	EffectPlanConstraintType	Optional
Description	Indicates a proposed EffectPlan.	
ProposedActionPlan	ActionPlanConstraintType	Optional
Description	Indicates a proposed ActionPlan.	
ProposedTaskPlan	MA_TaskPlanConstraintType	Optional

Description	Indicates a proposed TaskPlan.	
ProposedResponsePlan	ResponsePlanConstraintType	Optional
Description	Indicates a proposed ResponsePlan.	

Table A-1-640: C2 UCI Extensions MA_MissionPlanCommandMT - ProposedRequirementPlans UCI Extension Field

CapabilityCommand	
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_CapabilityTypeEnum
Message Structure	
Enumeration Literal	**Description**
ACTION	Indicates an action capability.
AIR_SAMPLE	Indicates an air sampling capability.
AMTI	Indicates an airborne moving target indication (AMTI) capability.
AO	Indicates an active optical capability.
CAP	Indicates a cargo delivery capability.
CARGO_DELIVERY	Indicates a cargo delivery capability.
COMINT	Indicates a Communications Intelligence (COMINT) capability.
COMM_RELAY	Indicates a communications relay capability.
COMM_TERMINAL	Indicates a communications terminal capability
COUNTER_SPACE	Indicates a Counter Space Capability.
EA	Indicates an electronic attack (EA) capability.
EFFECT	Indicates an effect capability.
ESM	Indicates an electronic support measures (ESM) capability.
FLIGHT	Indicates a capability for waypoint-based flight tasks/commands.
IFF	Indicates an Identification Friend or Foe (IFF) capability.
OPAQUE	Indicates a Opaque Capability
ORBIT_CHANGE	Indicates this space system has the ability to change orbits.
ORBITAL_SURVEILLANCE	Indicates a task to perform Orbital Surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates an Orbital Surveillance Sensor Capability.
PO	Indicates a passive optical capability.
RADAR_ALTIMETER	Indicates a Radar Altimeter capability
REFUEL	Indicates a capability to refuel another asset.
RESPONSE	Indicates an response capability.
SAR	Indicates a synthetic aperture radar (SAR) capability.

SMTI	Indicates a surface moving target indication (SMTI) capability.
STRIKE	Indicates a strike capability.
SYSTEM_DEPLOYMENT	Indicates a deployment capability.
TACTICAL_ORDER	Indicates a tactical order capability.
TASK	Indicates a task capability.
WEATHER_RADAR	Indicates a weather radar capability.

Table A-1-641: C2 UCI Extensions MA_MissionPlanCommandMT - CapabilityCommand UCI Extension Field

1.3.2.97 MA_MissionPlanCommandStatusMT UCI Extension

MA_MissionPlanCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	AllocationResult : MA_RequirementPlanningResultType ResultingPlanIdentifier : MA_PlansReferenceType OpPlanID : MA_OpPlanID_Type OpPlanID : MA_OpPlanID_Type MessageData : MA_MissionPlanCommandStatusMDT AssociatedPlans : MA_PlanReferenceType

Table A-1-642: C2 UCI Extensions MA_MissionPlanCommandStatusMT UCI Extension

AllocationResult		
Description	Indicates allocation results per Requirement.	
Base Type	MA_RequirementPlanningResultType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequirementID	RequirementInstanceID_ChoiceType	Required
Description	Unique object identifier for the Requirement whose planning results are given by sibling elements.	
PlanningResult	PlanningResultEnum	Required
Description	Indicates the outcome of attempting to allocate or plan activities for the Requirement.	
SystemID	SystemID_Type	Required
Description	Indicates the SystemID of the System whose *Plan are associated with the sibling PlanningResult element.	
AssociatedPlans	MA_PlanReferenceType	Optional
Description	Indicates *Plan associated with the sibling PlanningResult element. If the sibling PlanningResult indicates the Requirement was/remained allocated then this element	

	indicates the specific *Plan it is/remains allocated in. If the PlanningResult indicates the Requirement was DEALLOCATED then this element indicates the constraining *Plan it was DEALLOCATED from. This element isn't expected when the PlanningResult indicates UNALLOCATED.	
NotAllocatedReason	UnallocatedReasonType	Repeated
Description	Indicates a reason why the Requirement wasn't allocated or reallocated. If the sibling PlanningResult element indicates the Requirement was left not allocated then this element indicates why. Multiple instances of this element can provide the reason for each System that was attempted and even many reasons for each. If the PlanningResult indicates the Requirement was allocated then this element can still be used to indicate reasons for failed attempts to allocate to other Systems or plan, prior to the successful allocation.	

Table A-1-643: C2 UCI Extensions MA_MissionPlanCommandStatusMT - AllocationResult UCI Extension Field

ResultingPlanIdentifier		
Description	This element indicates the unique ID or IDs of any *Plans that were generated.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated

Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-644: C2 UCI Extensions MA_MissionPlanCommandStatusMT - ResultingPlanIdentifier UCI Extension Field

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-645: C2 UCI Extensions MA_MissionPlanCommandStatusMT - OpPlanID UCI Extension Field

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	

DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-646: C2 UCI Extensions MA_MissionPlanCommandStatusMT - OpPlanID UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_MissionPlanCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	MissionPlanCommandID_Type	Required
Description	The unique ID of the MissionPlanCommand this status applies to.	
PlanningStatus	PlanCommandStatusType	Required
Description	Indicates the status of the *PlanCommand.	
AllocationResult	MA_RequirementPlanningResultType	Repeated
Description	Indicates allocation results per Requirement.	
ResultingPlanIdentifier	MA_PlansReferenceType	Optional
Description	This element indicates the unique ID or IDs of any *Plans that were generated.	

Table A-1-647: C2 UCI Extensions MA_MissionPlanCommandStatusMT - MessageData UCI Extension Field

AssociatedPlans		
Description	Indicates *Plan associated with the sibling PlanningResult element. If the sibling PlanningResult indicates the Requirement was/remained allocated then this element indicates the specific *Plan it is/remains allocated in. If the PlanningResult indicates the Requirement was DEALLOCATED then this element indicates the constraining *Plan it was DEALLOCATED from. This element isn't expected when the PlanningResult indicates UNALLOCATED.	
Base Type	MA_PlanReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Optional
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Optional
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Optional
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Optional
Description	Indicates a reference to an OrbitActivityPlan.	

RoutePlanID	RoutePlanID_Type	Optional
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Optional
Description	Indicates a reference to a RouteActivityPlan.	
CommScheduleAllocationID	CommScheduleAllocationID_Type	Optional
Description	Indicates a reference to a Comms allocation.	
ActivityPlanID	ActivityPlanID_Type	Optional
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Optional
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Optional
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Optional
Description	Indicates a reference to a ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Optional
Description	Indicates a reference to an OpPlan.	

Table A-1-648: C2 UCI Extensions MA_MissionPlanCommandStatusMT - AssociatedPlans UCI Extension Field

1.3.2.98 MA_PlanningFunctionStatusMT UCI Extension

MA_PlanningFunctionStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	AutonomousPlanningAction : MA_PlanningAllowedEscalationType PlanningAllowed : MA_PlanningAllowedType ActivityPlan : MA_ActivityPlanPartsType TaskPlan : MA_TaskPlanPartsType ByResultSetting : MA_MissionPlanningAutonomySettingByResultType RouteActivityPlan : MA_ActivityPlanPartsType OpZoneCategory : MA_OpZoneCategoryEnum MessageData : MA_PlanningFunctionStatusMDT OpPlan : MA_OpPlanPartsType OpLineCategory : MA_OpLineCategoryEnum AutonomousPlanningResponseChoice : MA_MissionPlanningAutonomyResponseChoiceType OrbitActivityPlan : MA_ActivityPlanPartsType

	TaskType : MA_TaskTypeEnum OutputPlanParts : MA_PlanPartsType AllowedType : MA_PlanningAllowedType OpVolumeCategory : MA_OpZoneCategoryEnum CapabilityCommand : MA_CapabilityTypeEnum AutonomousAction : MA_MissionPlanningByResultAutonomousActionType MissionPlanningAutonomy : MA_MissionPlanningAutonomySettingType

Table A-1-649: C2 UCI Extensions MA_PlanningFunctionStatusMT UCI Extension

AutonomousPlanningAction		
Description	This element indicates the allowed autonomous mission planning for the ancestor Trigger. It also indicates the expected MissionContingencyAlert for the ancestor Trigger; whenever autonomous mission planning is allowed and triggered, a MissionContingencyAlert is expected.	
Base Type	MA_PlanningAllowedEscalationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AutonomousPlanningActionID	AutonomousPlanningActionID_Type	Required
Description	Assigns a unique ID to an allowed planning action. This ID can be reference in *Plans, alerts, etc. that result when the planning action is triggered.	
ByResultSetting	MA_MissionPlanningAutonomySettingByResultType	Repeated
Description	This element allows control of the escalation of the autonomous replanning. If the autonomous function indicated by the sibling AllowedType leads to a "by result" trigger described here, then additional autonomous planning is allowed as described here. For example, if Tasks are dropped as a result of initial autonomous replanning then this element can be used to enable/disable a secondary "escalated" planning iteration that includes global Task allocation. If omitted, the sibling AllowedType is allowed regardless of result.	
AllowedType	MA_PlanningAllowedType	Required
Description	This element defines the mission planning type that is allowed to be autonomously triggered.	

Table A-1-650: C2 UCI Extensions MA_PlanningFunctionStatusMT - AutonomousPlanningAction UCI Extension Field

PlanningAllowed	
Description	This element defines the mission planning type to be triggered. If multiple instances are given, each should be of a different planning type as indicated by the child element.
Base Type	MA_PlanningAllowedType
Message Structure	

Field Name	Field Type	Field Usage
PlanAheadDuration	DurationType	Required
Description	Required duration in the future when the Plan should start. Existing *Plan will continue during this time. This duration should account for everything that must happen from replan initiation to new plan activation.	
PlanningProcessID	PlanningProcessID_Type	Required
Description	Indicates the unique ID of the planning process whose autonomous invocation is described the sibling elements. Planning process are declared in the ControlStatus message.	
ConstrainingPlans	AutonomousPlanningConstrainingPlansType	Optional
Description	Indicates existing types of *Plans that can be used as constraints when invoking the planning process indicated by the sibling PlanningProcessID. The types of *Plans indicated here should be consistent with those supported by the planning process as indicated in the ControlStatus message.	
OtherSystemConstrainingPlans	AutonomousPlanningOtherSystemConstrainingPlansType	Repeated
Description	Indicates existing plans of another System that constrain planning. One for each descendant ConstraintUsage.	
OutputPlanParts	MA_PlanPartsType	Optional
Description	Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType.	
ResultsInMissionPlan	boolean	Required
Description	In some cases, the commanding service or HMI may not want to receive the planning process summarized in a MissionPlan message. If TRUE, the results will be published in individual *Plan messages and summarized in a published MissionPlan message. If FALSE, the results will be published in individual *Plan messages without a summary MissionPlan message.	
ForPlanningUseOnly	boolean	Required
Description	Indicates the state to use for the corresponding element in the *PlanCommand when invoking the planning process.	

Table A-1-651: C2 UCI Extensions MA_PlanningFunctionStatusMT - PlanningAllowed UCI Extension Field

ActivityPlan		
Description	Indicates applicable ActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	

SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-652: C2 UCI Extensions MA_PlanningFunctionStatusMT - ActivityPlan UCI Extension Field

TaskPlan		
Description	Indicates applicable TaskPlan parts.	
Base Type	MA_TaskPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskType	MA_TaskTypeEnum	Repeated
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	

Table A-1-653: C2 UCI Extensions MA_PlanningFunctionStatusMT - TaskPlan UCI Extension Field

ByResultSetting	
Description	This element allows control of the escalation of the autonomous replanning. If the autonomous function indicated by the sibling AllowedType leads to a "by result" trigger described here, then additional autonomous planning is allowed as described here. For example, if Tasks are dropped as a result of initial

	autonomous replanning then this element can be used to enable/disable a secondary "escalated" planning iteration that includes global Task allocation. If omitted, the sibling AllowedType is allowed regardless of result.	
Base Type	MA_MissionPlanningAutonomySettingByResultType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Trigger	PlanningByResultTriggerType	Required
Description	Indicates a trigger based on the results of a previously initiated autonomous mission planning function.	
AutonomousAction	MA_MissionPlanningByResultAutonomousActionType	Required
Description	Indicates the action or actions allowed for the sibling trigger.	

Table A-1-654: C2 UCI Extensions MA_PlanningFunctionStatusMT - ByResultSetting UCI Extension Field

RouteActivityPlan		
Description	Indicates applicable RouteActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional

Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-655: C2 UCI Extensions MA_PlanningFunctionStatusMT - RouteActivityPlan UCI Extension Field

OpZoneCategory	
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate

	ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted

	without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-656: C2 UCI Extensions MA_PlanningFunctionStatusMT - OpZoneCategory UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the PlanningFunctionStatus message.	
Base Type	MA_PlanningFunctionStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SystemID	SystemID_Type	Required
Description	Indicates the unique ID of the System whose planning function status is being reported.	
TransitionDroppedToFailed	boolean	Required
Description	This element specifies whether or not dropped Tasks that cannot be reallocated within the System will be automatically transitioned to the FAILED state for higher level interaction.	
PlanningInterface	PlanningInterfaceType	Repeated
Description	Indicates a currently enabled planning interface of the System. If multiple instances of this element are given, each should be of a different type of *Plan as enumerated by the child PlanType element.	
MissionPlanningAutonomy	MA_MissionPlanningAutonomySettingType	Repeated
Description	Indicates currently enabled autonomous mission planning capabilities of the System.	
PlanActivationAutonomy	PlanActivationAutonomyType	Optional
Description	Indicates current enabled autonomous plan activation capabilities of the System.	
RoutePlanningContingencyPath	ContingencyPathAutonomyType	Repeated
Description	Indicates current route planning settings for the autonomous generation of contingency paths along the primary path. If multiple instances are given, each must be of a different path type as indicated by the child element.	

Table A-1-657: C2 UCI Extensions MA_PlanningFunctionStatusMT - MessageData UCI Extension Field

OpPlan		
Description	Indicates applicable OpPlan parts.	
Base Type	MA_OpPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPointCategory	OpPointCategoriesType	Repeated
Description	Indicates an OpPoint type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpLineCategory	MA_OpLineCategoryEnum	Repeated
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpZoneCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpVolumeCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpRouting	EmptyType	Optional
Description	Indicates an OpRouting is applicable to the associated *Plan. List size for this element is based on there being no category type for OpRouting objects	

Table A-1-658: C2 UCI Extensions MA_PlanningFunctionStatusMT - OpPlan UCI Extension Field

OpLineCategory	
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpLineCategoryEnum
Message Structure	
Enumeration Literal	**Description**
GENERAL_LINE	A general line where the function must be described by other procedures.
FORWARD_EDGE_OF_BATTLE_AREA	The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.
GUN_TARGET_LINE	Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.
CORRIDOR	Line with width describing the passageway designated for travel by friendly aircraft.
HOSTILE_BOUNDARY	Line that indicates a limit beyond which exists a hostile environment.
BUFFER_ZONE_BOUNDARY	Line beyond which friendly aircraft usually do not fly in peacetime.
LOW_LEVEL_TRANSIT_ROUTE	Line describing the center line of a route through the forward missile

	engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.
TACTICAL_ACTION_LINE	Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.
FIRE_SUPPORT_COORDINATION_LINE	Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander.
FORWARD_LINE_OF_OWN_TROOPS	Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.
LINE_OF_COMMUNICATIONS	Line the route that connects an operating military unit with its supply base.
JETWAY	Line with width that defines areas where commercial aircraft are flying through.
DEPARTURE_RADIAL	Line the defines the departure radial when performing Carrier admin departures.
RECOVERY_RADIAL	Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.
FINAL_BEARING	Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.
BRC	Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.

Table A-1-659: C2 UCI Extensions MA_PlanningFunctionStatusMT - OpLineCategory UCI Extension Field

AutonomousPlanningResponseChoice		
Description	Indicates the autonomous mission planning type or types allowed for the sibling Trigger.	
Base Type	MA_MissionPlanningAutonomyResponseChoiceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AutonomousPlanningAction	MA_PlanningAllowedEscalationType	Repeated
Description	This element indicates the allowed autonomous mission planning for the ancestor Trigger. It also indicates the expected MissionContingencyAlert for the ancestor Trigger; whenever autonomous mission planning is allowed and triggered, a MissionContingencyAlert is expected.	
AlertOnly	EmptyType	Required
Description	This element indicates that autonomous mission planning isn't allowed for the ancestor Trigger but a MissionContingencyAlert is expected.	

Table A-1-660: C2 UCI Extensions MA_PlanningFunctionStatusMT - AutonomousPlanningResponseChoice UCI Extension Field

OrbitActivityPlan		
Description	Indicates applicable OrbitActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-661: C2 UCI Extensions MA_PlanningFunctionStatusMT - OrbitActivityPlan UCI Extension Field

TaskType	
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_TaskTypeEnum
Message Structure	

Enumeration Literal	Description
AIR_SAMPLE	Indicates the task is to sample and analyze the air around the vehicle.
AMTI	Indicates the task is to track air targets using an Airborne Moving Target Indicator.
AO	Indicates the task is to employ an Active Optical capability.
CARGO_DELIVERY	Indicates the task is to deliver cargo.
CAP	Indicates the task is to execute a Combat Air Patrol.
COMINT	Indicates the task is to employ a communications intelligence capability.
COMM_RELAY	Indicates the task is to relay communications from other battlespace participants.
COUNTER_SPACE	Indicates the task is to employ a Counter Space Capability.
EA	Indicates the task is to employ an electronic attack capability.
ESCORT	Indicates the task is to escort another entity.
ESM	Indicates the task is to employ an Electronic Support Measure.
FLIGHT	Indicates the task is to fly.
ORBIT_CHANGE	Indicates a task to perform an orbit change via a spacecraft maneuver.
ORBITAL_SURVEILLANCE	Indicates a task to perform orbital surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates a task to perform orbital surveillance via sensor collection capabilities.
PO	Indicates the task is to employ a Passive Optical capability.
REFUEL	Indicates the task is to refuel another System or Entity.
SAR	Indicates the task is to employ a Synthetic Aperture Radar capability.
SMTI	Indicates the task is to track surface targets using a Surface Moving Target Indicator.
STRIKE	Indicates the task is to strike a target.
SYSTEM_DEPLOYMENT	Indicates the task is to deploy a system.
TACTICAL_ORDER	Indicates the task is to perform a tactical order.
WEATHER_RADAR	Indicates the task is to collect weather data.

Table A-1-662: C2 UCI Extensions MA_PlanningFunctionStatusMT - TaskType UCI Extension Field

OutputPlanParts		
Description	Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType.	
Base Type	MA_PlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**

ActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	
OrbitActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-663: C2 UCI Extensions MA_PlanningFunctionStatusMT - OutputPlanParts UCI Extension Field

AllowedType		
Description	This element defines the mission planning type that is allowed to be autonomously triggered.	
Base Type	MA_PlanningAllowedType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanAheadDuration	DurationType	Required
Description	Required duration in the future when the Plan should start. Existing *Plan will continue during this time. This duration should account for everything that must happen from replan initiation to new plan activation.	
PlanningProcessID	PlanningProcessID_Type	Required
Description	Indicates the unique ID of the planning process whose autonomous invocation is described the sibling elements. Planning process are declared in the ControlStatus message.	
ConstrainingPlans	AutonomousPlanningConstrainingPlansType	Optional

Description	Indicates existing types of *Plans that can be used as constraints when invoking the planning process indicated by the sibling PlanningProcessID. The types of *Plans indicated here should be consistent with those supported by the planning process as indicated in the ControlStatus message.	
OtherSystemConstrainingPlans	AutonomousPlanningOtherSystemConstrainingPlansType	Repeated
Description	Indicates existing plans of another System that constrain planning. One for each descendant ConstraintUsage.	
OutputPlanParts	MA_PlanPartsType	Optional
Description	Indicates the specific Plan parts of the sibling PlanningType that the System can produce and include in the output Plan. The child choice in this element should match the sibling PlanningType. When omitted, the System can produce all of the Plan parts for the sibling PlanningType.	
ResultsInMissionPlan	boolean	Required
Description	In some cases, the commanding service or HMI may not want to receive the planning process summarized in a MissionPlan message. If TRUE, the results will be published in individual *Plan messages and summarized in a published MissionPlan message. If FALSE, the results will be published in individual *Plan messages without a summary MissionPlan message.	
ForPlanningUseOnly	boolean	Required
Description	Indicates the state to use for the corresponding element in the *PlanCommand when invoking the planning process.	

Table A-1-664: C2 UCI Extensions MA_PlanningFunctionStatusMT - AllowedType UCI Extension Field

OpVolumeCategory	
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.

EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.

ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-665: C2 UCI Extensions MA_PlanningFunctionStatusMT - OpVolumeCategory UCI Extension Field

CapabilityCommand	
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_CapabilityTypeEnum
Message Structure	
Enumeration Literal	**Description**
ACTION	Indicates an action capability.
AIR_SAMPLE	Indicates an air sampling capability.
AMTI	Indicates an airborne moving target indication (AMTI) capability.
AO	Indicates an active optical capability.
CAP	Indicates a cargo delivery capability.
CARGO_DELIVERY	Indicates a cargo delivery capability.

COMINT	Indicates a Communications Intelligence (COMINT) capability.
COMM_RELAY	Indicates a communications relay capability.
COMM_TERMINAL	Indicates a communications terminal capability
COUNTER_SPACE	Indicates a Counter Space Capability.
EA	Indicates an electronic attack (EA) capability.
EFFECT	Indicates an effect capability.
ESM	Indicates an electronic support measures (ESM) capability.
FLIGHT	Indicates a capability for waypoint-based flight tasks/commands.
IFF	Indicates an Identification Friend or Foe (IFF) capability.
OPAQUE	Indicates a Opaque Capability
ORBIT_CHANGE	Indicates this space system has the ability to change orbits.
ORBITAL_SURVEILLANCE	Indicates a task to perform Orbital Surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates an Orbital Surveillance Sensor Capability.
PO	Indicates a passive optical capability.
RADAR_ALTIMETER	Indicates a Radar Altimeter capability
REFUEL	Indicates a capability to refuel another asset.
RESPONSE	Indicates an response capability.
SAR	Indicates a synthetic aperture radar (SAR) capability.
SMTI	Indicates a surface moving target indication (SMTI) capability.
STRIKE	Indicates a strike capability.
SYSTEM_DEPLOYMENT	Indicates a deployment capability.
TACTICAL_ORDER	Indicates a tactical order capability.
TASK	Indicates a task capability.
WEATHER_RADAR	Indicates a weather radar capability.

Table A-1-666: C2 UCI Extensions MA_PlanningFunctionStatusMT - CapabilityCommand UCI Extension Field

AutonomousAction		
Description	Indicates the action or actions allowed for the sibling trigger.	
Base Type	MA_MissionPlanningByResultAutonomousActionType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanningAllowed	MA_PlanningAllowedType	Repeated
Description	This element defines the mission planning type to be triggered. If multiple instances are given, each should be of a different planning type as indicated by the child element.	
AlertOnly	EmptyType	Required

Description	This element indicates that autonomous mission planning isn't allowed for the ancestor Trigger but a MissionContingencyAlert is expected.

Table A-1-667: C2 UCI Extensions MA_PlanningFunctionStatusMT - AutonomousAction UCI Extension Field

MissionPlanningAutonomy		
Description	Indicates currently enabled autonomous mission planning capabilities of the System.	
Base Type	MA_MissionPlanningAutonomySettingType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Trigger	PlanningByCaseTriggerType	Required
Description	Indicates an allowed trigger for autonomous planning.	
AutonomousPlanningResponseChoice	MA_MissionPlanningAutonomyResponseChoiceType	Required
Description	Indicates the autonomous mission planning type or types allowed for the sibling Trigger.	
PlanningProcessID	PlanningProcessID_Type	Optional
Description	Indicates the unique ID of a planning process which the settings are applicable to. If omitted, the setting is applicable to all processes of the corresponding System.	
MinimumResponseInterval	DurationType	Optional
Description	Indicates the minimum time interval between autonomous responses for the sibling Trigger. In other words, regardless of when and how frequently the Trigger occurs, the sibling Response shouldn't be initiated until the minimum time duration given here after the previous Response.	

Table A-1-668: C2 UCI Extensions MA_PlanningFunctionStatusMT - MissionPlanningAutonomy UCI Extension Field

1.3.2.99 MA_OpPointMT UCI Extension

MA_OpPointMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_OpPointMDT AppliesToIDs : MA_PackageSystemType

Table A-1-669: C2 UCI Extensions MA_OpPointMT UCI Extension

MessageData

Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_OpPointMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Category	OpPointCategoriesType	Required
Description	This element indicates the type of operational point.	
CategoryUniqueData	OpPointCategoriesUniqueDataType	Optional
Description	This element indicates the unique data for the type of operational point.	
OpPointID	OpPointID_Type	Required
Description	This element represents the unique identifier of the point associated with this message.	
PointChoice	OpPointChoiceType	Required
Description	Choice of either geospatial or relative position of the OpPoint.	
SystemConfiguration	SystemConfigurationType	Repeated
Description	Indicates a System which the OpPoint is applicable for and its planned configuration/state upon arriving at the OpPoint. If omitted, the OpPoint applies to all Systems.	
TurnDirection	RelativeDirectionEnum	Optional
Description	This field provides the turn direction for orbit.	
IngressConstraint	AnglePairType	Optional
Description	When present this field constrains the vehicle ingress path toward the OpPoint. The allowable ingress range is defined clockwise from Min to Max.	
SafeAltitude	SafeAltitudeType	Optional
Description	This field specifies a set of safe minimum approach altitudes and an emergency safe altitude. This field also specifies the cylindrical area, defined by a center point and radius, where the altitudes apply. This field defines minimum approach altitude and emergency safe altitude consistent with standard FAA instrument approach chart definitions (using UCI standard data types).	
OpDescription	OpDescriptionType	Optional
Description	A human-readable descriptive name with free text remarks.	
MissionTraceability	MissionTraceabilityType	Optional
Description	Allows the creator of this operational item to specify what the source or motivation for it originated from. This could be ATO, ACO, etc.	
Source	CreationSourceEnum	Optional
Description	Indicates the source of the information.	
Schedule	ScheduleType	Optional
Description	This element represents a date/time interval associated with the operational item during which the item is in effect.	
AssociatedTime	TimeFunctionType	Repeated

Description	Indicates time related events that are associated to the Op data type such as when activated or when operational. The multiplicity is limited to N-1 options where NONE is an event that has no value, thus removed from list of available options.	
DataLinkIdentifier	DataLinkIdentifierPET	Repeated
Description	List of data link ID. Multiple data link IDs can be reported for the same network type.	
Priority	unsignedInt	Optional
Description	This field represents the priority to be considered when choosing between multiple available points.	
QualifyingTags	QualifyingTagsType	Optional
Description	Provides identity qualifications to augment identification and associated processing of the OpPoint.	
AppliesToIDs	MA_PackageSystemType	Repeated
Description	One or more systems to which this item applies. If omitted, item applies to all systems.	
SystemScheduleOverride	SystemScheduleStateType	Repeated
Description	System-specific schedule that will override any part of the main schedule. This allows for system-specific schedule different than main schedule. When adding a SystemScheduleOverride, the child SystemID also must be added to the sibling SystemID list if the sibling list is not empty. The SystemScheduleOverride is independent of time planning. For the system, this override can be done at execution.	

Table A-1-670: C2 UCI Extensions MA_OpPointMT - MessageData UCI Extension Field

AppliesToIDs		
Description	One or more systems to which this item applies. If omitted, item applies to all systems.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	
PackageID	PackageID_Type	Required
Description	The ID of a Package.	
SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-671: C2 UCI Extensions MA_OpPointMT - AppliesToIDs UCI Extension Field

1.3.2.100 MA_OpRoutingMT UCI Extension

MA_OpRoutingMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	ApplicableToIDs : MA_PackageSystemType MessageData : MA_OpRoutingMDT

Table A-1-672: C2 UCI Extensions MA_OpRoutingMT UCI Extension

ApplicableToIDs		
Description	One or more Systems to which the routing constraints apply. If omitted, the constraints apply to all Systems.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	
PackageID	PackageID_Type	Required
Description	The ID of a Package.	
SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-673: C2 UCI Extensions MA_OpRoutingMT - ApplicableToIDs UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_OpRoutingMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpRoutingID	OpRoutingID_Type	Required
Description	The unique identifier of the set of routing constraints during operation associated with this message.	
DefaultBlueSeparation	SeparationParametersType	Optional

Description	Indicates default minimum geospatial separation to be maintained when planning or analyzing routes or manually navigating. The separation is between the planned/navigated route and other routes or conflicting items. This is the default separation. It can be overridden in specific cases by the sibling element.	
DefaultRedSeparation	SeparationParametersType	Optional
Description	Indicates default minimum geospatial separation to be maintained when planning or analyzing routes or manually navigating. The separation is between the planned/navigated route and other routes or conflicting items. This is the default separation. It can be overridden in specific cases by the sibling element.	
SpecificBlueSeparation	SpecificBlueSeparationType	Repeated
Description	Defines parameters to use when deconflicting blue player routes or generating routes to ensure proper separation from other blue players. These parameters apply at the vehicle type level.	
SpecificRedSeparation	SpecificRedSeparationType	Repeated
Description	Defines parameters to use when deconflicting blue player routes or generating routes to ensure proper separation from other red players. These parameters apply at the vehicle type level.	
OpDescription	OpDescriptionType	Optional
Description	Represents the name and description of the set of associated routing constraints.	
ApplicableToIDs	MA_PackageSystemType	Repeated
Description	One or more Systems to which the routing constraints apply. If omitted, the constraints apply to all Systems.	
ApplicableZone	ZoneType	Optional
Description	Indicates the geospatial zone in which the routing constraints apply.	
Schedule	ScheduleType	Optional
Description	Indicates the time schedule during which the routing constraints apply.	
MissionTraceability	MissionTraceabilityType	Optional
Description	Allows the creator of this zone to specify what the source or motivation for this zone originated from. This could be ATO, ACO, etc.	

Table A-1-674: C2 UCI Extensions MA_OpRoutingMT - MessageData UCI Extension Field

1.3.2.101 MA_PlanningFunctionMT UCI Extension

MA_PlanningFunctionMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	Process : MA_MissionPlanProcessDescriptionType RouteActivityPlan : MA_ActivityPlanPartsType ActivityPlan : MA_ActivityPlanPartsType

	OpVolumeCategory : MA_OpZoneCategoryEnum MissionPlan : MA_MissionPlanProcessType MessageData : MA_PlanningFunctionMDT PlanningInterfaceDetails : MA_PlanningInterfaceDetailsType TaskPlan : MA_TaskPlanPartsType OrbitActivityPlan : MA_ActivityPlanPartsType OpZoneCategory : MA_OpZoneCategoryEnum Output : MA_PlanPartsType CapabilityCommand : MA_CapabilityTypeEnum TaskType : MA_TaskTypeEnum OpLineCategory : MA_OpLineCategoryEnum OpPlan : MA_OpPlanPartsType

Table A-1-675: C2 UCI Extensions MA_PlanningFunctionMT UCI Extension

Process		
Description	Indicates details of a MissionPlan-related planning process.	
Base Type	MA_MissionPlanProcessDescriptionType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Output	MA_PlanPartsType	Required
Description	Indicates a type of Plan output by the planning process and the Parts that can be included.	
PlanningProcessID	PlanningProcessID_Type	Required
Description	Indicates the unique ID of a planning process invocable through a *PlanCommand message/interface.	
ValidationSupported	SupportedFunctionEnum	Repeated
Description	Indicates whether this process supports validation via the *PlanValidationCommand messages. List size for this element is based on "Select All That Apply" condition.	
ApplicableTo	PlanningApplicabilitySystemType	Repeated
Description	Indicates a type or instance, of System or Capabilities, that the planning process is applicable to; a System the process can create the specified *Plan type or types for.	
Supports	ApplicabilityType	Repeated
Description	Indicates a type or instance of System, Capabilities or existing *Plans which the planning process can develop support for in the new *Plan. For example, the sibling ApplicableTo element could indicate the planning process can plan electronic attack routes. This element can then indicate the type or instance of Systems, Capabilities or *Plan types that can be supported by the electronic attack route/plan.	
PlanningTriggers	MissionPlanningByCaseTriggerEnum	Repeated
Description	Indicates the set of triggers that the System is able to monitor to autonomously initiate	

	the planning process. List size for this element is based on "Select All That Apply" condition.

Table A-1-676: C2 UCI Extensions MA_PlanningFunctionMT - Process UCI Extension Field

RouteActivityPlan		
Description	Indicates applicable RouteActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-677: C2 UCI Extensions MA_PlanningFunctionMT - RouteActivityPlan UCI Extension Field

ActivityPlan

Description	Indicates applicable ActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional
Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-678: C2 UCI Extensions MA_PlanningFunctionMT - ActivityPlan UCI Extension Field

OpVolumeCategory	
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**

AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.
EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.

MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.
ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-679: C2 UCI Extensions MA_PlanningFunctionMT - OpVolumeCategory UCI Extension Field

MissionPlan

Description	Indicates details of MissionPlan-related planning processes.	
Base Type	MA_MissionPlanProcessType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DefaultPlanningProcessID	PlanningProcessID_Type	Required
Description	Indicates which one of the sibling Process elements is the default.	
Process	MA_MissionPlanProcessDescriptionType	Repeated
Description	Indicates details of a MissionPlan-related planning process.	

Table A-1-680: C2 UCI Extensions MA_PlanningFunctionMT - MissionPlan UCI Extension Field

MessageData		
Description	Indicates the data being conveyed by the PlanningFunction message.	
Base Type	MA_PlanningFunctionMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanningFunctionID	PlanningFunctionID_Type	Required
Description	Indicates a unique ID, assigned to this PlanningFunction.	
SystemID	SystemID_Type	Required
Description	Indicates the unique ID of the System whose planning functions are being reported.	
PlanningInterfaces	PlanningInterfacesType	Repeated
Description	Indicates planning interfaces supported by the System. A supported interface isn't necessarily enabled currently; see the PlanningStatus message for current status/configuration of control interfaces advertised here.	
PlanningInterfaceDetails	MA_PlanningInterfaceDetailsType	Optional
Description	This element is used to describe the details of planning interfaces/messages, specifically *PlanCommand interfaces of a System. It provides discoverability of distinct planning process instances along with the functional scope and output *Plan type or types. A planning interface/process described here can only be invoked when the sibling Interfaces element indicates it is enabled. Planning processes are dynamically discoverable because Systems tend to have widely varying planning functionality. Decomposing planning into discrete processes allows planning functionality to be distributed across a service oriented system of systems. Making planning processes discoverable encourages planning-related services at multiple levels in a C2 hierarchy to adapt to and simultaneously support Systems with different planning functionality and processes.	
PlanActivationAutonomyDetails	SupportedPlanActivationAutonomyType	Optional

Description	Indicates capabilities of the System to perform plan activation autonomously. Plan activations can be performed either at the MissionPlan or SubPlan level.	
ScoringInterfaceDetails	PlanScoringProcessType	Optional
Description	This element is used to describe the details of scoring interfaces/messages summarized in the sibling Interfaces element. It provides discoverability of distinct scoring process instances along with the functional scope. A scoring interface/process described here can only be invoked when the sibling Interfaces ActivationState element indicates it is enabled. Plan Scoring processes are dynamically discoverable because Systems tend to have varying scoring algorithms. Decomposing plan scoring into discrete processes allows scoring functionality to be distributed across a service oriented system of systems.	

Table A-1-681: C2 UCI Extensions MA_PlanningFunctionMT - MessageData UCI Extension Field

PlanningInterfaceDetails		
Description	This element is used to describe the details of planning interfaces/messages, specifically *PlanCommand interfaces of a System. It provides discoverability of distinct planning process instances along with the functional scope and output *Plan type or types. A planning interface/process described here can only be invoked when the sibling Interfaces element indicates it is enabled. Planning processes are dynamically discoverable because Systems tend to have widely varying planning functionality. Decomposing planning into discrete processes allows planning functionality to be distributed across a service oriented system of systems. Making planning processes discoverable encourages planning-related services at multiple levels in a C2 hierarchy to adapt to and simultaneously support Systems with different planning functionality and processes.	
Base Type	MA_PlanningInterfaceDetailsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlan	MA_MissionPlanProcessType	Optional
Description	Indicates details of MissionPlan-related planning processes.	
TaskPlan	TaskPlanProcessType	Optional
Description	Indicates details of TaskPlan-related planning processes.	
RoutePlan	RoutePlanProcessType	Optional
Description	Indicates details of RoutePlan-related planning processes.	
RouteActivityPlan	ActivityPlanProcessType	Optional
Description	Indicates details of RouteActivityPlan-related planning processes.	
ActivityPlan	ActivityPlanProcessType	Optional

Description	Indicates details of ActivityPlan-related planning processes.	
OrbitPlan	OrbitPlanProcessType	Optional
Description	Indicates details of OrbitPlan-related planning processes.	
OrbitActivityPlan	ActivityPlanProcessType	Optional
Description	Indicates details of OrbitActivityPlan-related planning processes.	
EffectPlan	EffectPlanProcessType	Optional
Description	Indicates details of EffectPlan-related planning processes.	
ActionPlanProcess	ActionPlanProcessType	Optional
Description	Indicates details of ActionPlan-related planning processes.	
ResponsePlanProcess	ResponsePlanProcessType	Optional
Description	Indicates details of ResponsePlan-related planning processes.	

Table A-1-682: C2 UCI Extensions MA_PlanningFunctionMT - PlanningInterfaceDetails UCI Extension Field

TaskPlan		
Description	Indicates applicable TaskPlan parts.	
Base Type	MA_TaskPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskType	MA_TaskTypeEnum	Repeated
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	

Table A-1-683: C2 UCI Extensions MA_PlanningFunctionMT - TaskPlan UCI Extension Field

OrbitActivityPlan		
Description	Indicates applicable OrbitActivityPlan parts.	
Base Type	MA_ActivityPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CapabilityCommand	MA_CapabilityTypeEnum	Repeated
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
SupportingCapabilityCommand	SupportCapabilityTypeEnum	Repeated
Description	Indicates an applicable Supporting CapabilityCommand-based planned Activity type; a [*Supporting Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.	
VehicleSettings	EmptyType	Optional

Description	Indicates VehicleSettings are applicable to the *ActivityPlan.	
CommsUsage	EmptyType	Optional
Description	Indicates Comms usage is applicable to the *ActivityPlan.	
ProductTasks	EmptyType	Optional
Description	Indicates Task or Tasks for Product management (ProductDisseminationTask, ProductClassificationTask, etc.) are applicable to the *ActivityPlan.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-684: C2 UCI Extensions MA_PlanningFunctionMT - OrbitActivityPlan UCI Extension Field

OpZoneCategory	
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpZoneCategoryEnum
Message Structure	
Enumeration Literal	**Description**
AIRBORNE_EARLY_WARNING	A point at which AEW aircraft are or will be orbiting or maneuvering.
COMBAT_AIR_PATROL	A point at which CAP aircraft are or will be orbiting or maneuvering.
CONSTRAINED_ENTRY_EXIT	Indicates an area with constrained entries and exits. The additional parameters used with this zone category are defined in the ConstrainedEntryExitType choice in OpZoneCategoryType.
CONTAMINATED	Indicates the zone is contaminated, an area caused by the deposit and/or absorption of radioactive material, or of biological or chemical agents on and by structures, areas, personnel, or objects.
DANGER	A specified area above, below, or within which there may be a potential danger.
DEFENDED_AREA	An area the source tn is capable of defending against ballistic missiles, etc.,(i.e. the source tn is operational with ready weapons and has designated the area for defense).
DIRECTED_SEARCH_AREA	Indicates the zone is a directed search area.
EOB	Indicates the zone is the boundary for an electronic order of battle request.

EXERCISE	An area for military maneuver or simulated wartime operation involving planning, preparation and execution.
FIGHTER_ENGAGE_AOR	Indicates the zone is a fighter area of responsibility (AOR), an airspace of defined dimensions within which the responsibility for engagement normally rests with fighter aircraft.
FILTER_AREA	Indicates a geographic area of interest for the purpose of limiting data. The additional parameters used with this zone category are defined in the OpZoneFilterAreaPET choice in OpZoneCategoryType.
GENERAL_HAZARD	Indicates the zone is a hazard area that doesn't have a specified operational purpose.
GENERAL_STATION	A general zone describing an air station that must be amplified by voice or predetermined procedures.
GROUND_AOR	Indicates the zone is a ground area of responsibility (AOR), an area of ground within which the responsibility for engagement normally rests with the appropriate ground force commander.
HOSTILE_TACTICAL_AREA	Indicates the zone is a hostile tactical area, an area in which hostile tactical forces (to include ground, air, or maritime) are known to be or will be operating.
HOSTILE_WEAPON	Indicates the zone is a hostile weapon zone, an area in which hostile air defense systems are capable of engagement.
IMPACT_POINT	The real or expected point that a device has or will penetrate or contact a surface.
JAMMING	Indicates the zone is a jamming control zone. The additional parameters used with this zone category are defined in the OpZoneJammingType choice in OpZoneCategoryType.
KEEP_IN	Indicates the zone is a boundary within which applicable systems must stay. The additional parameters used with this zone category are defined in the IngressEgressType choice in OpZoneCategoryType.
KILL	Indicates the zone is a kill zone, an air defense system (SOC) area in which friendly fighters may use less restrictive rules of engagement.
MISSILE_ENGAGEMENT	Indicates the zone is a missile engagement zone, a primary engagement area designated for friendly surface to air missiles.
MISSILE_LAUNCH_POINT	A point indicating the position from which a missile has been or will be launched. The additional parameters used with this zone category are defined in the OpZoneMissileLaunchDataType choice in OpZoneCategoryType.
MISSION_AREA	Indicates the zone is an area in which a mission will take place.
NAMED_AREA_OF_INTEREST	An area usually along an enemy mobility corridor where enemy activity (or inactivity) confirms a particular course of action.
NO_COMM	Indicates the zone is an area where communications use is restricted.
NO_FIRE	Indicates the zone is an area where strike impact is restricted; the zone does not restrict the launch of weapons. The additional parameters used with this zone category are defined in the OpZoneNoFireType choice in OpZoneCategoryType.
NO_FLY	Indicates the zone is an area where flight is restricted. The additional parameters used with this zone category are defined in the OpZoneNoFlyType choice in OpZoneCategoryType.
NO_IMAGE	Indicates the zone is an area where image collection is restricted.
ORBIT_FIGURE_EIGHT	A figure eight pattern within a rectangle that the aircraft is or will be tracking.

ORBIT_RACE_TRACK	An oval within a rectangle that the aircraft is or will be tracking.
ORBIT_RANDOM_CLOSED	A rectangle in which the aircraft will track randomly.
REFERENCE	Indicates the zone is an area that doesn't have a specified operational purpose.
SEARCH	An area where a search is being or will be conducted. normally associated with a SAR mission.
SEPARATION	Indicates an update in reservation volume for an individual vehicle.
SHORAD	Indicates the zone is a short range area defense (SHORAD), an area defended by friendly army short range air defense systems.
SPEED_LIMITS	An area where platform speeds are limited. The additional parameters used with this zone category are defined in the OpZoneSpeedLimitType choice in OpZoneCategoryType.
SUBMARINE_PATROL_AREA	A restricted area established to allow submarine operations to be unimpeded by the operation, or possible attack from friendly forces in wartime, or to be conducted without submerged mutual interference in peacetime.
TARGET_AREA_OF_INTEREST	An engagement area usually along an enemy mobility corridor.
VEHICLE_CONFIGURATION	Indicates the zone is a region where specific vehicle configuration parameters should be changed. The additional parameters used with this zone category are defined in the VehicleCommandDataType choice in OpZoneCategoryType.
WEAPON_RESTRICTION	Indicates the zone is an area within which a set of restricted weapons cannot be used against a target type and/or within in the zone. The additional parameters used with this zone category are defined in the OpZoneWeaponRestrictionType choice in OpZoneCategoryType.
WEATHER_CONDITIONS	Indicates the zone is an area within which weather conditions have a potential for mission impact. The additional parameters used with this zone category are defined in the OpZoneWeatherType choice in OpZoneCategoryType.

Table A-1-685: C2 UCI Extensions MA_PlanningFunctionMT - OpZoneCategory UCI Extension Field

Output		
Description	Indicates a type of Plan output by the planning process and the Parts that can be included.	
Base Type	MA_PlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable ActivityPlan parts.	
RoutePlan	RoutePlanPartsType	Optional
Description	Indicates applicable RoutePlan parts.	
RouteActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable RouteActivityPlan parts.	
OrbitPlan	OrbitPlanPartsType	Optional
Description	Indicates applicable OrbitPlan parts.	

OrbitActivityPlan	MA_ActivityPlanPartsType	Optional
Description	Indicates applicable OrbitActivityPlan parts.	
EffectPlan	EffectPlanPartsType	Optional
Description	Indicates applicable EffectPlan parts.	
ActionPlan	ActionPlanPartsType	Optional
Description	Indicates applicable ActionPlan parts.	
TaskPlan	MA_TaskPlanPartsType	Optional
Description	Indicates applicable TaskPlan parts.	
ResponsePlan	ResponsePlanPartsType	Optional
Description	Indicates applicable ResponsePlan parts.	
OpPlan	MA_OpPlanPartsType	Optional
Description	Indicates applicable OpPlan parts.	

Table A-1-686: C2 UCI Extensions MA_PlanningFunctionMT - Output UCI Extension Field

CapabilityCommand	
Description	Indicates an applicable CapabilityCommand-based planned Activity type; a [*Capability]Command type applicable to the *ActivityPlan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_CapabilityTypeEnum
Message Structure	
Enumeration Literal	**Description**
ACTION	Indicates an action capability.
AIR_SAMPLE	Indicates an air sampling capability.
AMTI	Indicates an airborne moving target indication (AMTI) capability.
AO	Indicates an active optical capability.
CAP	Indicates a cargo delivery capability.
CARGO_DELIVERY	Indicates a cargo delivery capability.
COMINT	Indicates a Communications Intelligence (COMINT) capability.
COMM_RELAY	Indicates a communications relay capability.
COMM_TERMINAL	Indicates a communications terminal capability
COUNTER_SPACE	Indicates a Counter Space Capability.
EA	Indicates an electronic attack (EA) capability.
EFFECT	Indicates an effect capability.
ESM	Indicates an electronic support measures (ESM) capability.
FLIGHT	Indicates a capability for waypoint-based flight tasks/commands.
IFF	Indicates an Identification Friend or Foe (IFF) capability.

OPAQUE	Indicates a Opaque Capability
ORBIT_CHANGE	Indicates this space system has the ability to change orbits.
ORBITAL_SURVEILLANCE	Indicates a task to perform Orbital Surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates an Orbital Surveillance Sensor Capability.
PO	Indicates a passive optical capability.
RADAR_ALTIMETER	Indicates a Radar Altimeter capability
REFUEL	Indicates a capability to refuel another asset.
RESPONSE	Indicates an response capability.
SAR	Indicates a synthetic aperture radar (SAR) capability.
SMTI	Indicates a surface moving target indication (SMTI) capability.
STRIKE	Indicates a strike capability.
SYSTEM_DEPLOYMENT	Indicates a deployment capability.
TACTICAL_ORDER	Indicates a tactical order capability.
TASK	Indicates a task capability.
WEATHER_RADAR	Indicates a weather radar capability.

Table A-1-687: C2 UCI Extensions MA_PlanningFunctionMT - CapabilityCommand UCI Extension Field

TaskType	
Description	Indicates a Task type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_TaskTypeEnum
Message Structure	
Enumeration Literal	**Description**
AIR_SAMPLE	Indicates the task is to sample and analyze the air around the vehicle.
AMTI	Indicates the task is to track air targets using an Airborne Moving Target Indicator.
AO	Indicates the task is to employ an Active Optical capability.
CARGO_DELIVERY	Indicates the task is to deliver cargo.
CAP	Indicates the task is to execute a Combat Air Patrol.
COMINT	Indicates the task is to employ a communications intelligence capability.
COMM_RELAY	Indicates the task is to relay communications from other battlespace participants.
COUNTER_SPACE	Indicates the task is to employ a Counter Space Capability.
EA	Indicates the task is to employ an electronic attack capability.
ESCORT	Indicates the task is to escort another entity.
ESM	Indicates the task is to employ an Electronic Support Measure.
FLIGHT	Indicates the task is to fly.

ORBIT_CHANGE	Indicates a task to perform an orbit change via a spacecraft maneuver.
ORBITAL_SURVEILLANCE	Indicates a task to perform orbital surveillance.
ORBITAL_SURVEILLANCE_SENSOR	Indicates a task to perform orbital surveillance via sensor collection capabilities.
PO	Indicates the task is to employ a Passive Optical capability.
REFUEL	Indicates the task is to refuel another System or Entity.
SAR	Indicates the task is to employ a Synthetic Aperture Radar capability.
SMTI	Indicates the task is to track surface targets using a Surface Moving Target Indicator.
STRIKE	Indicates the task is to strike a target.
SYSTEM_DEPLOYMENT	Indicates the task is to deploy a system.
TACTICAL_ORDER	Indicates the task is to perform a tactical order.
WEATHER_RADAR	Indicates the task is to collect weather data.

Table A-1-688: C2 UCI Extensions MA_PlanningFunctionMT - TaskType UCI Extension Field

OpLineCategory	
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.
Base Type	MA_OpLineCategoryEnum
Message Structure	
Enumeration Literal	**Description**
GENERAL_LINE	A general line where the function must be described by other procedures.
FORWARD_EDGE_OF_BATTLE_AREA	The foremost limits of a series of areas in which ground combat units are deployed, excluding the areas in which the covering or screening forces are operating.
GUN_TARGET_LINE	Straight line from the gun or guns to the target. Altitude, if included, is the apex of the projectile.
CORRIDOR	Line with width describing the passageway designated for travel by friendly aircraft.
HOSTILE_BOUNDARY	Line that indicates a limit beyond which exists a hostile environment.
BUFFER_ZONE_BOUNDARY	Line beyond which friendly aircraft usually do not fly in peacetime.
LOW_LEVEL_TRANSIT_ROUTE	Line describing the center line of a route through the forward missile engagement zone that must be flown by friendly aircraft to ensure they are not designated hostile by defense systems.
TACTICAL_ACTION_LINE	Line positioned in hostile airspace which may be used in conjunction with other criteria for hostile determination.
FIRE_SUPPORT_COORDINATION_LINE	Line established by the appropriate ground commander to ensure coordination of fire not under his command but which may affect current tactical operations. Attacks against ground targets behind this line must be coordinated with the appropriate ground commander.

FORWARD_LINE_OF_OWN_TROOPS	Line defining the forward position of own troops within a ground area of responsibility, that does not usually coincide with the forward edge of battle area.
LINE_OF_COMMUNICATIONS	Line the route that connects an operating military unit with its supply base.
JETWAY	Line with width that defines areas where commercial aircraft are flying through.
DEPARTURE_RADIAL	Line the defines the departure radial when performing Carrier admin departures.
RECOVERY_RADIAL	Line the defines the recovery radial that is used to define the recovery marshal. May be used when performing Carrier admin recoveries.
FINAL_BEARING	Line the defines the final bearing when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.
BRC	Base Recovery Course defines the aircraft carrier heading when an aircraft is approaching a carrier. May be used when performing Carrier admin recoveries.

Table A-1-689: C2 UCI Extensions MA_PlanningFunctionMT - OpLineCategory UCI Extension Field

OpPlan		
Description	Indicates applicable OpPlan parts.	
Base Type	MA_OpPlanPartsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
OpPointCategory	OpPointCategoriesType	Repeated
Description	Indicates an OpPoint type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpLineCategory	MA_OpLineCategoryEnum	Repeated
Description	Indicates an OpLine type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpZoneCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpZone type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpVolumeCategory	MA_OpZoneCategoryEnum	Repeated
Description	Indicates an OpVolume type applicable to the associated *Plan. List size for this element is based on "Select All That Apply" condition.	
OpRouting	EmptyType	Optional
Description	Indicates an OpRouting is applicable to the associated *Plan. List size for this element is based on there being no category type for OpRouting objects	

Table A-1-690: C2 UCI Extensions MA_PlanningFunctionMT - OpPlan UCI Extension Field

1.3.2.102 MA_TaskStatusMT UCI Extension

MA_TaskStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	Plans : MA_PlansReferenceType OpPlanID : MA_OpPlanID_Type ExecutingSystemOrPackage : MA_PackageSystemType MessageData : MA_TaskStatusMDT Traceability : MA_RequirementStatusTraceabilityType

Table A-1-691: C2 UCI Extensions MA_TaskStatusMT UCI Extension

Plans		
Description	Indicates *Plans the Requirement is associated with.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	

ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-692: C2 UCI Extensions MA_TaskStatusMT - Plans UCI Extension Field

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-693: C2 UCI Extensions MA_TaskStatusMT - OpPlanID UCI Extension Field

ExecutingSystemOrPackage		
Description	Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.	
Base Type	MA_PackageSystemType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ALL	EmptyType	Required
Description	Indicates applies to ALL entities.	
NONE	EmptyType	Required
Description	Indicates applies to NO entities.	
PackageID	PackageID_Type	Required
Description	The ID of a Package.	

SystemID	SystemID_Type	Required
Description	The ID of an ACP or C2 Operator.	

Table A-1-694: C2 UCI Extensions MA_TaskStatusMT - ExecutingSystemOrPackage UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_TaskStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskID	TaskID_Type	Required
Description	Indicates the unique ID of the Task whose execution status is being reported.	
ExecutingSystemOrPackage	MA_PackageSystemType	Required
Description	Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.	
ExecutionState	RequirementExecutionStateEnum	Required
Description	Identifies the state of the Requirement.	
ExecutionStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling State element. This element is only expected when the State is in a potentially undesirable state such as failed or completed with less than optimum results. It is strongly encouraged for transitions to failed or dropped.	
ActualTiming	RequirementTimingType	Repeated
Description	Indicates actual timing status for the Requirement. Allows for defining for each timing type (RequirementTimingEnum).	
PercentCompleted	PercentType	Optional
Description	Percentage of assigned Requirement completion. The percentage increments until the Requirement has COMPLETED or FAILED. A percent complete of 0 indicates the Requirement has not yet begun execution, and a failed Requirement may have any percentage between 1 - 99.	
Traceability	MA_RequirementStatusTraceabilityType	Optional
Description	Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture. For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically	

	without *Plans or [Capability]Commands.	
Metrics	MetricsType	Optional
Description	Specifies the actual metrics associated with the Requirements.	

Table A-1-695: C2 UCI Extensions MA_TaskStatusMT - MessageData UCI Extension Field

Traceability		
Description	Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture. For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.	
Base Type	MA_RequirementStatusTraceabilityType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Plans	MA_PlansReferenceType	Optional
Description	Indicates *Plans the Requirement is associated with.	
PlannedActivityID	PlannedActivityID_Type	Repeated
Description	Indicates a specific Planned Activity the Requirement is associated with.	
CapabilityID	CapabilityID_Type	Repeated
Description	Identifies a specific Capability the Requirement is associated with.	
CommandID	CommandID_Type	Repeated
Description	Indicates the ID of an associated [Capability]Command.	
ActivityID	ActivityID_Type	Repeated
Description	Indicates the ID of an associated actual/executing/executed Activity.	

Table A-1-696: C2 UCI Extensions MA_TaskStatusMT - Traceability UCI Extension Field

1.3.2.103 MA_PlanModificationRequestMT UCI Extension

MA_PlanModificationRequestMT	
Description	Indicates the contents of the PlanModificationRequest message details, and also includes the message header.

	See the XSD File for more info
Ext. Fields	SubPlansAdditions : MA_PlansReferenceBaseType SubPlansRetractions : MA_PlansReferenceBaseType SubPlansModification : MA_SubPlansModificationType PlanModificationDetails : MA_PlanModificationDetailsType MessageData : MA_PlanModificationRequestMDT Plans : MA_PlansReferenceType OpPlanID : MA_OpPlanID_Type

Table A-1-697: C2 UCI Extensions MA_PlanModificationRequestMT UCI Extension

SubPlansAdditions		
Description	Indicates an additional set of *Plans which are included as part of the MissionPlan. The included SubPlans share the same Applicability as the MissionPlan.	
Base Type	MA_PlansReferenceBaseType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	

OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-698: C2 UCI Extensions MA_PlanModificationRequestMT - SubPlansAdditions UCI Extension Field

SubPlansRetractions		
Description	Indicates the removal of set of *Plans which are included as part of the MissionPlan.	
Base Type	MA_PlansReferenceBaseType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated
Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-699: C2 UCI Extensions MA_PlanModificationRequestMT - SubPlansRetractions UCI Extension Field

SubPlansModification	
Description	Indicates a change in the set of *Plans which are included as part of the MissionPlan. See annotations for the underlying type and its child elements for

	details.	
Base Type	MA_SubPlansModificationType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
SubPlansAdditions	MA_PlansReferenceBaseType	Optional
Description	Indicates an additional set of *Plans which are included as part of the MissionPlan. The included SubPlans share the same Applicability as the MissionPlan.	
SubPlansRetractions	MA_PlansReferenceBaseType	Optional
Description	Indicates the removal of set of *Plans which are included as part of the MissionPlan.	

Table A-1-700: C2 UCI Extensions MA_PlanModificationRequestMT - SubPlansModification UCI Extension Field

PlanModificationDetails		
Description	Specifies details of the plan modification.	
Base Type	MA_PlanModificationDetailsType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
ForPlanningUseOnly	boolean	Optional
Description	When TRUE, indicates the *Plan is a used only for intermediate planning purposes. It will eventually be replaced/superseded by another *Plan or be discarded/abandoned. *Plans with this flag should only be used when explicitly referenced by other messages. These *Plans should never: trigger functionality, be subject to approvals, be activated, etc.	
WindowModification	PlanWindowModificationTypeChoiceType	Optional
Description	Specifies an absolute time Window or alternatively a TimeOffset by which the Plan's, or set of Plans', Windows should be shifted.	
ApplicabilityModification	PlanApplicabilityModificationType	Optional
Description	Indicates a change in the System or Systems which have a direct relationship with the Mission Plan. See annotations for the underlying type and its child elements for details.	
SubPlansModification	MA_SubPlansModificationType	Optional
Description	Indicates a change in the set of *Plans which are included as part of the MissionPlan. See annotations for the underlying type and its child elements for details.	
HierarchyModification	HierarchyModificationType	Optional
Description	Indicates a change in the hierarchy of the set of MissionPlans.	
ExecutionSequenceModification	ExecutionSequenceReplaceOrModifyChoiceType	Optional
Description	Indicates a change in the execution sequence of the set of *Plans.	
RequirementAllocationLock	RequirementAllocationLockDetailsType	Optional

Description	Indicates a requested change in Allocated Requirements to Lock or Unlock.

Table A-1-701: C2 UCI Extensions MA_PlanModificationRequestMT - PlanModificationDetails UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PlanModificationRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Source	PlanInitiationSourceEnum	Required
Description	Source of the plan modification request.	
Plans	MA_PlansReferenceType	Required
Description	Indicates the Plan, or set of Plans, for which this Plan Modification Request is applicable.	
PlanModificationDetails	MA_PlanModificationDetailsType	Required
Description	Specifies details of the plan modification.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message.	
RequestState	RequestStateEnum	Required
Description	The state of the request message data. Indicates if a request message is a NEW request, an UPDATE to a request, or a CANCEL of a request.	

Table A-1-702: C2 UCI Extensions MA_PlanModificationRequestMT - MessageData UCI Extension Field

Plans		
Description	Indicates the Plan, or set of Plans, for which this Plan Modification Request is applicable.	
Base Type	MA_PlansReferenceType	
Message Structure		
Field Name	**Field Type**	**Field Usage**
MissionPlanID	MissionPlanID_Type	Repeated
Description	Indicates a reference to a MissionPlan.	
TaskPlanID	TaskPlanID_Type	Repeated
Description	Indicates a reference to a TaskPlan.	
OrbitPlanID	OrbitPlanID_Type	Repeated
Description	Indicates a reference to an OrbitPlan.	
OrbitActivityPlanID	OrbitActivityPlanID_Type	Repeated

Description	Indicates a reference to an OrbitActivityPlan.	
RoutePlanID	RoutePlanID_Type	Repeated
Description	Indicates a reference to a RoutePlan.	
RouteActivityPlanID	RouteActivityPlanID_Type	Repeated
Description	Indicates a reference to a RouteActivityPlan.	
ActivityPlanID	ActivityPlanID_Type	Repeated
Description	Indicates a reference to an ActivityPlan.	
EffectPlanID	EffectPlanID_Type	Repeated
Description	Indicates a reference to an EffectPlan.	
ActionPlanID	ActionPlanID_Type	Repeated
Description	Indicates a reference to an ActionPlan.	
ResponsePlanID	ResponsePlanID_Type	Repeated
Description	Indicates a reference to an ResponsePlan.	
OpPlanID	MA_OpPlanID_Type	Repeated
Description	Indicates a reference to an OpPlan.	

Table A-1-703: C2 UCI Extensions MA_PlanModificationRequestMT - Plans UCI Extension Field

OpPlanID		
Description	Indicates a reference to an OpPlan.	
Base Type	MA_OpPlanID_Type	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Version	unsignedInt	Optional
Description	Indicates an incremental version of an item in which only minor changes have been made. An example of a change that would require a new UUID and not just a new version would be for changes that affect task allocations or changes that require re-approval.	
UUID	UniversallyUniqueIdentifierType	Required
Description	Universally Unique Identifier (UUID). A UUID is a 128-bit number (32 hexadecimal digits, 16 bytes) that is conformant to a version and variant of IETF RFC 4122.	
DescriptiveLabel	VisibleString256Type	Optional
Description	Human readable text label with content meaningful to human operators.	

Table A-1-704: C2 UCI Extensions MA_PlanModificationRequestMT - OpPlanID UCI Extension Field

1.3.2.104 MA_PlanModificationRequestStatusMT UCI Extension

MA_PlanModificationRequestStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details.

	See the XSD File for more info
Ext. Fields	MessageData : MA_PlanModificationRequestStatusMDT

Table A-1-705: C2 UCI Extensions MA_PlanModificationRequestStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_PlanModificationRequestStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-706: C2 UCI Extensions MA_PlanModificationRequestStatusMT - MessageData UCI Extension Field

1.3.2.105 MA_MessageAcknowledgmentConfigurationRequestMT UCI Extension

MA_MessageAcknowledgmentConfigurationRequestMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_MessageAcknowledgmentConfigurationRequestMDT

Table A-1-707: C2 UCI Extensions MA_MessageAcknowledgmentConfigurationRequestMT UCI Extension

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.
Base Type	MA_MessageAcknowledgmentConfigurationRequestMDT

Message Structure		
Field Name	**Field Type**	**Field Usage**
DefaultGroupMessageFreq	DurationType	Optional
Description	This will override the current DefaultGroupMessageFreq setting.	
DefaultMessagesToAck	MA_AckMessageEnum	Repeated
Description	This will override the current DefaultMessageToAck setting.	
RequestedReceiverSpecificSetting	MA_AckReceiverSpecificConfigType	Optional
Description	This is an individual configuration settings that is being requested to be configured.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-708: C2 UCI Extensions MA_MessageAcknowledgmentConfigurationRequestMT - MessageData UCI Extension Field

1.3.2.106 MA_MessageAcknowledgmentConfigurationMT UCI Extension

MA_MessageAcknowledgmentConfigurationMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_MessageAcknowledgmentConfigurationMDT

Table A-1-709: C2 UCI Extensions MA_MessageAcknowledgmentConfigurationMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_MessageAcknowledgmentConfigurationMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
DefaultGroupMessageFreq	DurationType	Optional

Description	If set, this provides the frequency at which the Grouped Acknowledgment message is set at. If not set there is no grouping.	
DefaultMessagesToAck	MA_AckMessageEnum	Repeated
Description	If set, the list of messages that an Acknowledgment message will be sent for when received. If not set all messages of the type will be responded to.	
ReceiverSpecificSettings	MA_AckReceiverSpecificConfigType	Repeated
Description	This is the list of individual configuration settings that a remote system is configured. Values in this will override the Default settings.	
ConfigurationSupported	boolean	Required
Description	If FALSE, the publisher does not support grouping, they expect acknowledgments for messages covered by this configuration, and will reject all configuration requests. If TRUE, the publisher supports individual configurations, grouping, and the other items in this collection of messages.	

Table A-1-710: C2 UCI Extensions MA_MessageAcknowledgmentConfigurationMT - MessageData UCI Extension Field

1.3.2.107 MA_MessageAcknowledgmentConfigurationRequestStatusMT UCI Extension

MA_MessageAcknowledgmentConfigurationRequestStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_MessageAcknowledgmentConfigurationRequestStatusMDT

Table A-1-711: C2 UCI Extensions MA_MessageAcknowledgmentConfigurationRequestStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_MessageAcknowledgmentConfigurationRequestStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional

Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.

Table A-1-712: C2 UCI Extensions MA_MessageAcknowledgmentConfigurationRequestStatusMT - MessageData UCI Extension Field

1.3.2.108 MA_ResponseCommandMT UCI Extension

MA_ResponseCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ResponseCommandMDT

Table A-1-713: C2 UCI Extensions MA_ResponseCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ResponseCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Command	MA_ResponseCommandType	Repeated
Description	Indicates an instance of an Response Capability command. A Capability command message can have multiple command instances to allow a sequence of related commands in a single coherent message. Each command instance reports status via a separate, independent Capability CommandStatus message.	

Table A-1-714: C2 UCI Extensions MA_ResponseCommandMT - MessageData UCI Extension Field

1.3.2.109 MA_ResponseCommandStatusMT UCI Extension

MA_ResponseCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_ResponseCommandStatusMDT

Table A-1-715: C2 UCI Extensions MA_ResponseCommandStatusMT UCI Extension

MessageData

Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_ResponseCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Activity	ResultingActivityType	Repeated
Description	Indicates an Activity that satisfies the associated Command. This element is required when the sibling State element indicates ACCEPTED.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-716: C2 UCI Extensions MA_ResponseCommandStatusMT - MessageData UCI Extension Field

1.3.2.110 MA_PlatformOverrideCommandStatusMT UCI Extension

MA_PlatformOverrideCommandStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_PlatformOverrideCommandStatusMDT

Table A-1-717: C2 UCI Extensions MA_PlatformOverrideCommandStatusMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_PlatformOverrideCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	

CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-718: C2 UCI Extensions MA_PlatformOverrideCommandStatusMT - MessageData UCI Extension Field

1.3.2.111 MA_PlatformOverrideCommandMT UCI Extension

MA_PlatformOverrideCommandMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_PlatformOverrideCommandMDT

Table A-1-719: C2 UCI Extensions MA_PlatformOverrideCommandMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_PlatformOverrideCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
Applicability	MA_PlanApplicabilityType	Required
Description	Indicates the System or Systems which have a direct relationship with a global command. See annotations for the underlying type and its child elements for details.	
PlatformOverrideCommand	MA_PlatformDefaultAndOverrideDataType	Required
Description	Indicates a request to override a platform setting in existing plans and commands.	
ExpirationTime	DateTimeType	Optional
Description	Optionally indicates a prescribed expiration time for the platform override command. If not populated, then the command is indefinite until commanded off or expires due to the ApplicableZone(s) being exited.	
ApplicableZoneID	OpZoneID_Type	Repeated
Description	Optionally indicates a zone(s) within which the override is applicable. If not populated, then the command is valid throughout the operating area.	

CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-720: C2 UCI Extensions MA_PlatformOverrideCommandMT - MessageData UCI Extension Field

1.3.2.112 MA_GatewayActivityMT UCI Extension

MA_GatewayActivityMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	ObjectState : ObjectStateEnum MessageData : MA_GatewayActivityMDT

Table A-1-721: C2 UCI Extensions MA_GatewayActivityMT UCI Extension

ObjectState	
Description	Indicates the state of the internal software object that the data in the message is based upon. The state of the software object is determined by the publisher. The publisher could be the original/master publisher or an intermediary re-publisher such as a persistence/data manager. This element is not a commanded CRUD operation. It is not an indication of the state of an instance of a message. It is not a mechanism for partial, incremental, or fragmentary publication of a message. It is a signal from a publisher to subscribers. For REMOVED state, in particular, some subscribers will mirror the publisher and treat the message as if the object no longer exists, but others, such as mission recorders, might not (in order to support retaining historical records of activity, for example).
Base Type	ObjectStateEnum
Message Structure	
Enumeration Literal	**Description**
NEW	Indicates the internal software object corresponding to this message currently exists and this message is the first publication corresponding to the object.
UPDATED	Indicates the internal software object corresponding to this message currently exists and has been updated since the last publication of the message; this publication of the message may contain updates.
REMOVED	Indicates the internal software object corresponding to this message no longer exists; there should be no further publications of this message corresponding to the object.

Table A-1-722: C2 UCI Extensions MA_GatewayActivityMT - ObjectState UCI Extension Field

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_GatewayActivityMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
GatewayServiceID	ServiceID_Type	Required
Description	This element represents the ServiceID of the Gateway Service that sent this message.	
Activity	MA_GatewayActivityType	Repeated
Description	Indicates the dynamic operational characteristics of an Activity of a GatewayCapability.	

Table A-1-723: C2 UCI Extensions MA_GatewayActivityMT - MessageData UCI Extension Field

1.3.2.113 MA_DataDisseminationPlanOverrideRequestMT UCI Extension

MA_DataDisseminationPlanOverrideRequestMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanOverrideRequestMDT

Table A-1-724: C2 UCI Extensions MA_DataDisseminationPlanOverrideRequestMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanOverrideRequestMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
AllocatedToSystemIDs	MA_PackageSystemType	Repeated
Description	Indicates the unique ID of the system(s) or package that is the recipient of this command.	
AllocatedToServiceID	ServiceID_Type	Optional
Description	Indicates the unique ID of the service that is the recipient of this command.	
Request	MA_DataDisseminationPlanOverrideRequestType	Repeated
Description	Indicates an instance of a data plan override request. A data plan override request message can have multiple request instances to allow a sequence of related requests in	

	a single coherent message. Each request instance reports status via a separate, independent DataPlanOverrideRequestStatus message.

Table A-1-725: C2 UCI Extensions MA_DataDisseminationPlanOverrideRequestMT - MessageData UCI Extension Field

1.3.2.114 MA_DataDisseminationPlanOverrideRequestStatusMT UCI Extension

MA_DataDisseminationPlanOverrideRequestStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanOverrideRequestStatusMDT

Table A-1-726: C2 UCI Extensions MA_DataDisseminationPlanOverrideRequestStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanOverrideRequestStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	Indicates the unique ID of the DataDisseminationPlan that this override modification status applies to.	
EffectivityActivationStatus	DataPlanEffectivityActivationStatusType	Repeated
Description	This element indicates the status of pre-mission planned Data-Link gateway settings. It is used to indicate details about specific planned effectivity activations.	
OperatorModificationStatus	OperatorGatewayDesignationStatusType	Repeated
Description	This element indicates the status of operator designations. It is used to indicate details about specific designation configurations.	
RequestID	RequestID_Type	Required
Description	Indicates the unique ID of the request message corresponding to this status message.	
RequestProcessingState	RequestProcessingStateEnum	Required
Description	Indicates the status of the request message. Indicates how the receiver of the request has responded to the request message.	
RequestProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling RequestProcessingState element. This element is only expected when the Request is in a potentially undesirable state such as rejected, canceled or	

	accepted with less than optimum results expected.

Table A-1-727: C2 UCI Extensions MA_DataDisseminationPlanOverrideRequestStatusMT - MessageData UCI Extension Field

1.3.2.115 MA_DataDisseminationPlanMT UCI Extension

MA_DataDisseminationPlanMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	ObjectState : ObjectStateEnum MessageData : MA_DataDisseminationPlanMDT

Table A-1-728: C2 UCI Extensions MA_DataDisseminationPlanMT UCI Extension

ObjectState	
Description	Indicates the state of the internal software object that the data in the message is based upon. The state of the software object is determined by the publisher. The publisher could be the original/master publisher or an intermediary re-publisher such as a persistence/data manager. This element is not a commanded CRUD operation. It is not an indication of the state of an instance of a message. It is not a mechanism for partial, incremental, or fragmentary publication of a message. It is a signal from a publisher to subscribers. For REMOVED state, in particular, some subscribers will mirror the publisher and treat the message as if the object no longer exists, but others, such as mission recorders, might not (in order to support retaining historical records of activity, for example).
Base Type	ObjectStateEnum
Message Structure	
Enumeration Literal	**Description**
NEW	Indicates the internal software object corresponding to this message currently exists and this message is the first publication corresponding to the object.
UPDATED	Indicates the internal software object corresponding to this message currently exists and has been updated since the last publication of the message; this publication of the message may contain updates.
REMOVED	Indicates the internal software object corresponding to this message no longer exists; there should be no further publications of this message corresponding to the object.

Table A-1-729: C2 UCI Extensions MA_DataDisseminationPlanMT - ObjectState UCI Extension Field

MessageData	
Description	See annotations in child elements and messages/elements that use this type for details.

Base Type	MA_DataDisseminationPlanMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	Indicates the unique ID of the DataDisseminationPlan.	
AssociatedPlanID	MA_DataDisseminationPlanID_Type	Repeated
Description	Indicates the unique ID of a DataDisseminationPlan that is a parent to this one.	
Applicability	MA_PlanApplicabilityType	Repeated
Description	Indicates the System or Systems which have a direct relationship with the DataDisseminationPlan. See annotations for the underlying type and its child elements for details. If Applicability is not defined then it indicates this DataDisseminationPlan applies to ALL systems.	
IER	MA_CommIER_Planned_Type	Repeated
Description	Indicates a planned Information Exchange Requirement (IER). An IER can reflect the information needs of local or remote Systems or third party Systems coordinating on behalf of other Systems. The planned assignment of an IER to comms resources is given in the sibling Effectivity element.	
IER_DesiredService	MA_CommIER_DesiredServicesType	Repeated
Description	Defines the desired service constraints for an IER (Information Exchange Requirement), allowing multi-constrained solutions. All child elements are optional; when omitted software should apply no explicit bound for that metric.	
RateGroup	MA_CommRateGroupType	Repeated
Description	Indicates a RateGroup that can be associated with an IER to configure its transmit rate.	
Effectivity	MA_CommDataDisseminationEffectivityType	Repeated
Description	Indicates a DataDisseminationPlan "effectivity set" of items that are used together.	
Vector	MA_CommVectorPlannedType	Repeated
Description	Indicates a planned Vector. A vector describes a logical connection between nodes initiating on a specific NetworkLink (comms resource).	
DistributionGroup	MA_DistributionGroupType	Repeated
Description	Indicates a grouping of tiles or node groups and associated ip address.	
DistributionGroupMembership	MA_DistributionGroupMembershipType	Repeated
Description	Indicates a system's membership to a multicast group.	
Gateway	MA_GatewayReferenceType	Repeated
Description	Indicates a planned Gateway local to the system for which this DataDisseminationPlan is applicable to, with a summary of its Capability to be used for dissemination of IERs in the DataDisseminationPlan.	
Trigger	MA_CommTriggerType	Repeated
Description	Signifies available triggers which would automatically invoke this DataDisseminationPlan or parts within this DataDisseminationPlan. These Triggers that can be associated with activation of an Effectivity by using the TriggerID.	

Table A-1-730: C2 UCI Extensions MA_DataDisseminationPlanMT - MessageData UCI Extension Field

1.3.2.116 MA_DataDisseminationPlanActivationCommandMT UCI Extension

MA_DataDisseminationPlanActivationCommandMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanActivationCommandMDT

Table A-1-731: C2 UCI Extensions MA_DataDisseminationPlanActivationCommandMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanActivationCommandMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	The DataDisseminationPlan that this command pertains to.	
CommandedState	PlanActivationCommandEnum	Required
Description	Indicates the activation to perform on the associated DataDisseminationPlan. See enumerate annotations for further details.	
Applicability	MA_PackageSystemType	Repeated
Description	The System or Package IDs, that this message applies to. A lead is able to respond with ACCEPTED for the Package or individual Systems as needed.	
EffectivityActivation	MA_DataDisseminationPlanEffectivityActivationType	Repeated
Description	Indicates an Effectivity of the DataDisseminationPlan to activate.	
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the Command corresponding to this message. Used to reference the command when providing status.	
CommandState	CommandStateEnum	Required
Description	The state of the command message data. Indicates if a command message is a NEW command, an UPDATE to an existing command, or a CANCEL of an existing command.	

Table A-1-732: C2 UCI Extensions MA_DataDisseminationPlanActivationCommandMT - MessageData UCI Extension Field

1.3.2.117 MA_DataDisseminationPlanActivationCommandStatusMT UCI Extension

MA_DataDisseminationPlanActivationCommandStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanActivationCommandStatusMDT

Table A-1-733: C2 UCI Extensions MA_DataDisseminationPlanActivationCommandStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanActivationCommandStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
CommandID	CommandID_Type	Required
Description	Indicates the unique ID of the command corresponding to this status message.	
CommandProcessingState	CommandProcessingStateEnum	Required
Description	Indicates the state of the command. The command state machine assumes that the Command and CommandStatus messaging interaction is only used to communicate the desired Command to the Subsystem or Service. The state machine and messaging terminates as soon as the host accepts or rejects the Command. See enumeration annotations for further details.	
CommandProcessingStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling CommandProcessingState element. This element is only expected when the Command is in a potentially undesirable state such as rejected, canceled or accepted with less than optimum results expected.	

Table A-1-734: C2 UCI Extensions MA_DataDisseminationPlanActivationCommandStatusMT - MessageData UCI Extension Field

1.3.2.118 MA_DataDisseminationPlanActivationStatusMT UCI Extension

MA_DataDisseminationPlanActivationStatusMT	
Description	See the annotation in the associated message for an overall description of the message and this type. See the XSD File for more info
Ext. Fields	MessageData : MA_DataDisseminationPlanActivationStatusMDT

Table A-1-735: C2 UCI Extensions MA_DataDisseminationPlanActivationStatusMT UCI Extension

MessageData		
Description	See annotations in child elements and messages/elements that use this type for details.	
Base Type	MA_DataDisseminationPlanActivationStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**
PlanID	MA_DataDisseminationPlanID_Type	Required
Description	The ID of the DataDisseminationPlan this status pertains to.	
ActivationState	PlanActivationStateEnum	Required
Description	Indicates the activation state of the associated DataDisseminationPlan.	
StateReason	MA_CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling ActivationState element.	
EffectivityActivationState	MA_DataDisseminationPlanEffectivityStateType	Repeated
Description	Indicates the activation state of an Effectivity of the associated DataDisseminationPlan.	
OverrideModificationState	MA_OverriderModificationStatusType	Repeated
Description	This element indicates the status of operator designations. It is used to indicate details about specific designation configurations.	

Table A-1-736: C2 UCI Extensions MA_DataDisseminationPlanActivationStatusMT - MessageData UCI Extension Field

1.3.2.119 MA_PlatformOverrideStatusMT UCI Extension

MA_PlatformOverrideStatusMT	
Description	See annotations in child elements and messages/elements that use this type for details. See the XSD File for more info
Ext. Fields	MessageData : MA_PlatformOverrideStatusMDT

Table A-1-737: C2 UCI Extensions MA_PlatformOverrideStatusMT UCI Extension

MessageData		
Description	Indicates the data being conveyed by the message.	
Base Type	MA_PlatformOverrideStatusMDT	
Message Structure		
Field Name	**Field Type**	**Field Usage**

PlatformOverrideStatus	MA_PlatformDefaultAndOverrideDataType	Repeated
Description	Indicates the current platform override setting(s) applied.	
ActiveOverrides	boolean	Required
Description	When TRUE, indicates one or more overrides are active. When FALSE, indicates no overrides are active.	
ExecutingSystemOrPackage	MA_PackageSystemType	Required
Description	Indicates the ID of the System or package executing the Requirement; the System or package whose status on execution of the Requirement is reported in sibling elements. *Plans and Requirements must be managed such that only a single System or package executes a given Requirement.	
ExecutionState	RequirementExecutionStateEnum	Required
Description	Identifies the state of the Requirement.	
ExecutionStateReason	CannotComplyType	Optional
Description	Indicates details regarding the reason for and/or a contributing factor to the sibling State element. This element is only expected when the State is in a potentially undesirable state such as failed or completed with less than optimum results. It is strongly encouraged for transitions to failed or dropped.	
ActualTiming	RequirementTimingType	Repeated
Description	Indicates actual timing status for the Requirement. Allows for defining for each timing type (RequirementTimingEnum).	
PercentCompleted	PercentType	Optional
Description	Percentage of assigned Requirement completion. The percentage increments until the Requirement has COMPLETED or FAILED. A percent complete of 0 indicates the Requirement has not yet begun execution, and a failed Requirement may have any percentage between 1 - 99.	
Traceability	MA_RequirementStatusTraceabilityType	Optional
Description	Indicates traceability to the sources and implementation of the Requirement. During the life cycle of execution of a Requirement, the traceability can change/update. Traceability can vary depending on a given System's architecture. For example, traceability to only a CommandID is reasonable if the architecture includes [Capability]Commands without *Plans or [Capability]Activities. Both CommandID and ActivityID are reasonable for architectures where a single [Capability]Command could generate one or multiple [Capability]Activities, e.g. a SAR command could result in multiple activities to cover the area. ActivityID only is reasonable for architectures where [Capability]Activities can originate automatically without *Plans or [Capability]Commands.	
Metrics	MetricsType	Optional
Description	Specifies the actual metrics associated with the Requirements.	

Table A-1-738: C2 UCI Extensions MA_PlatformOverrideStatusMT - MessageData UCI Extension Field